



Merge PACS™

8.2

ADMINISTRATION MANUAL

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INDICATIONS FOR USE:

Merge PACS™ is a Picture Archiving and Communication System (PACS) for multi-modality (CT, MR, PT, US, MG, BTO, CR, DR/DX, NM, XA, RF, secondary capture (SC), and other DICOM-compliant modalities) image processing and display, diagnostic reading and reporting, communication, printing, and storage of medical imaging studies and other patient data. Intended clinical users include radiologists, orthopedic and other surgeons, referring physicians, technologists, and other qualified medical professionals.

Data can be received directly from acquisition modalities, CAD systems, and other image processing systems, or indirectly via importing. Data that is not DICOM-compliant, such as photos, can be converted to DICOM format by Merge PACS.

Merge PACS provides image manipulation tools to enable users to view and compare images such as: linking, MPR, MIP, 3D image fusion/ registration of CT, MR, and PET; as well as CVR (Color Volume Rendering), measurements (linear distances, angles, areas, SUV, etc.), and annotations (for example, outline and label regions of interest, label spinal vertebrae).

The Real Time Worklist (RTWL) displays the real-time status of radiology activity and provides customizable workflow management capabilities. Communication of critical results is facilitated and documented through optional, configurable components.

The Patient Dashboard provides a composite view of patient data, both imaging and non-imaging. Multi-tier patient identity matching provides a comprehensive view even when dealing with multiple disparate patient identities.

Order and report information generated by the HIS/RIS and report creation systems are received and displayed via the transmission of HL7 messaging.

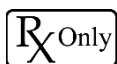
Lossless (reversible) and lossy (irreversible) image compression are supported for viewing, storage and communication. Merge PACS displays full fidelity DICOM images for use in the diagnostic interpretation of mammography using MG or BTO images. Thick slab MIP presentation can be applied to BTO images.

Lossy compressed images and digitized screen film images must not be used for primary diagnosis of mammography studies, and only display monitors that have regulatory clearance for mammography interpretation should be used for the interpretation of mammography studies.

CAUTION: Not for diagnostic use on a mobile device such as a smart phone or a tablet.

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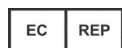
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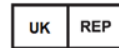
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Symbols Glossary:

Symbol	Title
	Part or Catalog number
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	Country of Manufacture
	Date of Manufacture
	Authorized Representative in the European Community
	Authorized Representative in the United Kingdom
	CE Mark with Notified Body
	Prescription Use Only
	Importer
	Medical Device
	Consult Instructions for Use

The full symbols glossary can be viewed at <https://www.ibm.com/support/pages/node/6262453>.

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Part	Date	Revision	Description
PAX-49666	08/25/2021	1	Initial Release for 8.2
	04/01/2022	2	Update for 8.2.1

The latest version of this document can be found at <https://mergecustomer.force.com/mergeusercommunity/login>.

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Preface

The *Merge PACS 8.2 Admin Manual* describes in detail Merge PACS 8.2 and its features and provides procedures for administering the system.

Purpose

This manual is intended to support all the needs of System Administrators and System Managers responsible for the operations of Merge PACS at a customer site.

Audience

This manual is primarily intended for the on-site PACS Administrator responsible for the daily oversight of the system.

Everyone providing maintenance support to Merge PACS must have a thorough working knowledge of the Windows 2019 operating system (or the Windows 2012 R2 or 2016 operating system, as applicable, for upgraded systems), and this manual assumes that its primary audience has such knowledge.

Organization

The chapters and appendices in this manual are generally organized as follows:

Chapter/Appendix	Description
Chapters 1 through 8	Introduce and describe in detail Merge PACS, its features, and its components.
Chapter 9	Describes the Merge PACS Management Pages, which are the primary tool for monitoring, reconfiguring, and troubleshooting Merge PACS.
Chapter 10	Describes the Advanced Workflow Management Pages.
Chapter 11	Describes the Templated User Management feature.
Chapter 12	Describes the Manage Users page.
Chapter 13	Describes how to create and manage worklists for use with Realtime Worklist and Realtime Study List.
Chapter 14	Contains procedures for managing Hanging Protocols and custom user rules.
Chapter 15	Describes how to configure the Launch and Control feature.
Chapter 16	Contains procedures for custom-configuring the Merge PACS Workstation.
Chapter 17	Contains procedures for maintaining Merge PACS.
Appendices A through F	Contain supplementary information about optional features and quick reference information.

The following table describes each of the chapters and appendices in this manual in greater detail:

Chapter/Appendix	Description
1. Introduction	Introduces Merge PACS, describing the major components and how they work together, as well as various optional features. Chapters 2 through 7 describe each of these components in greater detail.
2. Merge PACS Server	Describes the Merge PACS server, discussing in detail the following components that it comprises: • Services • Background processes • Directories and files • User accounts • System variables
3. Amicas Service	Describes the Amicas service and the individual background processes that this service launches.
4. Amicas Messaging Service	Describes the Amicas Messaging Service and the individual background processes that this service launches.
5. Security Overview	Describes the options available for applying security to the images stored on Merge PACS.
6. System Databases	Describes the DB2 databases that Merge PACS uses, their contents, functions, and the tools and processes for maintaining them.
7. Supported Character Sets	Describes how Merge PACS handles searches performed with accented characters such as those found in the Latin1 character set.
8. Merge PACS Client Applications	Describes the various Merge PACS clients used for viewing patient images.
9. Merge PACS Management Pages	Describes the administration and management tools available from the Merge PACS Management Pages and the tasks these tools perform.
10. Advanced Workflow Management	Describes how to use the configure the various Advanced Workflow options from the Advanced Workflow Management page as well as from the Settings dialog within the Merge PACS Workstation. These options allow you to support the use of auto-assignment of exams and the Launch and Control feature.
11. Templated User Management	Describes the procedures for setting up templates that can be used to provide privileges and default preferences to user groups as well as to the entire site.
12. Managing Users	Describes the Manage Users page, which allows you to view the general privileges and preferences currently set for individual users within the system, remove any user-level overrides for one or more preferences and/or edit a user's preferences on the fly. You can also configure the Advanced Workflow information for the user, including the user's licensure regions, specialties, practice, reader location, etc.

Chapter/Appendix	Description
13. Creating and Managing Worklists	Describes the procedures for creating and managing individual worklists to be used with RealTime Worklist and RealTime Study List.
14. Managing Hanging Protocols and User Rules	Contains procedures for managing hanging protocols and user rules (both those that can be used to control the behavior of Hanging Protocols and those that control other features within the Merge PACS Workstation).
15. Procedures for Configuring Launch and Control	Describes the steps required to configure Merge PACS to use the Launch and Control feature that allows you to set up a macro that can be used to determine which available applications (including the Merge PACS Viewer and any XML-integrated 3 rd -party viewers) should be launched when users click the “View Study” worklist action icon under various conditions (e.g., the type of study, the specific workstation or facility where the study is being read, etc.)
16. Custom Configuring the Merge PACS Workstation	Describes various procedures for customizing the Merge PACS Workstation on a site-wide basis.
17. Procedures for Maintaining Merge PACS	Contains procedures for maintaining the system once it is set up. These procedures include doing the following tasks: • General system maintenance and troubleshooting • Editing patient and study information • Maintaining the system's DB2 databases.
A. Privileges, Preferences and Configurations	Describes the various privileges and preferences that can be included in templates.
B. Standardized Uptake Values (SUV)	Provides detailed information regarding the units that can be displayed within the Workstation when using the Probe , ROI , Region Analysis Area and Region Analysis Volume tools with PET images.
C. Creating a COM+ Application	Contains procedures for creating a COM+ application that will facilitate the communication between a 64-bit Viewer and a 32-bit PS360 client installation.
D. Mapping of RTWL filters to database tables, HL7 fields and DICOM attributes	Describes the mapping of RTWL filters to database tables, HL7 fields and DICOM attributes.
E. Creating New Users for OpenLDAP	Describes how to create new users within OpenLDAP for use with OrthoPACS
F. Creating an X-keys Macro to Simulate Localized Bounce	Describes how to create an X-keys macro on a workstation with an X-keys USB keyboard to provide a Localized Bounce feature for use with mammo tomosynthesis multi-frame images.

Chapter 1. Introduction

1.1. What Is Merge PACS?

Merge PACS™ is a standards-based, flexible, and scalable platform for medical image management over the Intranet/Internet. The hardware supporting Merge PACS applications is off-the-shelf technology, allowing your facility to either reuse existing hardware or to purchase new hardware direct from the manufacturer without additional hidden costs.

Merge PACS can be used as a fully functional Web-based Picture Archiving and Communication Systems (PACS), as an image distribution add-on to an existing PACS, or for Teleradiology. In each case, Merge PACS goes beyond traditional PACS.

Merge PACS addresses the full spectrum of image management requirements, ranging from entry-level Teleradiology scenarios to full-fledged, multi-site enterprise PACS and Web-based image distribution deployments. All of these offerings are based upon systems that deliver the value you expect, without the equipment, implementation and overhead that have typically been required with traditional PACS.

In addition to flexibility, Merge PACS has been conceived, designed and built according to Internet application standards, combining the following characteristics:

- Universal Internet access
- Efficient and flexible Web server technology
- JPEG 2000 lossy image compression for enterprise distribution of images
- JPEG 2000 lossless image compression for primary interpretation of images
- Uncompressed DICOM image storage for medical/legal archiving

Note that Merge PACS can either be configured to run in **Integrated** mode (*i.e.*, using Merge iConnect® Enterprise Archive for image storage) or **Standalone** mode (*i.e.*, using Merge PACS for image storage). This is set during initial installation and cannot be changed afterwards.

1.1.1. The Merge PACS Server

The Merge PACS Server is the backbone of the Merge PACS system. Modalities send DICOM data into this Server, which are then stored for a configurable number of days (typically seven) to permit data validation. The Merge PACS Server also uses Merge's Forever Priors™ technology to store the compressed image data for an indefinite period of time, allowing "one-click" access to previous studies on demand.

1.1.2. Merge PACS Workstation

The Merge PACS Workstation is a full-featured radiology workstation that uses lossless image compression and is designed as a tool for primary interpretation of patient images. Study status information is shared among multiple Merge PACS Workstations. The Merge PACS Workstation allows radiologists to view all previous studies for a patient via Merge's Forever Priors™ technology.

The Merge PACS Workstation consists of the following main components, each of which is described in greater detail below:

- **RealTime Worklist™**
- **RealTime Study List**
- **Patient Record**
- **Merge PACS Viewer**
- **Quality Control Editor**

Note that many of the features and components of the Merge PACS Workstation, such as RealTime Worklist, inherent 3D support, the ability to create hanging protocols, spine labeling, the use of lossless vs lossy image compression, etc., are made available to the users based on their login privileges. Users with “advanced” privileges (typically radiologists) will be able to make use of the full range of features, whereas other users (e.g., referring physicians) will only have access to a subset of those features.

The Merge PACS Workstation can be accessed by pointing a web browser to the following URL:

`http[s]://<MERGE_PACS_SERVER>/merge-viewer`

For detailed information on Merge PACS Workstation functionality, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.1.3. RealTime Worklist™

RealTime Worklist is an optional component that provides a powerful “front end” to the Merge PACS Workstation that allows users to view and manage a list of patient studies that have been assigned to them. Because the Worklist information is stored on a centralized Merge PACS Server and not on individual machines, users can view their individual study lists regardless of which Workstation they log into.

RealTime Worklist is designed to work in conjunction with a hospital's HIS/RIS, populating individual worklists based on orders sent from the HIS/RIS to the Merge PACS Server via the HIS/RIS Integration component. Note, however, that studies that have no associated orders will still be displayed in a worklist.

RealTime Worklist can also be configured to work with the optional Merge RadStream™ Acuity component. With Merge RadStream Acuity enabled, studies can be assigned an “acuity score” that can be used to determine priority within individual worklists. Acuity can be based, in part, on the answers to questions inputted by a Technologist, but the final acuity score is determined automatically by the system according to a complex algorithm.

For a more detailed description on Worklist functionality, refer to Chapter 13 below, and the *Merge PACS 8.2 Workstation User's Guide*.

For information on creating and managing individual worklists, refer to Chapter 13 below.

1.1.4. RealTime Study List

RealTime Study List is a combined interface that lets users view and manage RTWL worklists and Teaching Worklists on the same page, as well as create and save special Quick Filter Worklists on the fly. Note that a Workstation can either have RealTime Study List or RealTime Worklist enabled, but not both at the same time.

For a more detailed description on RealTime Study List functionality, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.1.5. Patient Record

The Merge PACS Patient Record is part of the Merge PACS Workstation and provides a simple, yet powerful, interface for accessing prior exams. When an exam is opened for viewing within the Merge PACS Viewer, described below, the Patient Record displays Series Navigation thumbnails for that exam as well as any prior exams for the same patient. These Series Navigation thumbnails can be used to open the study into the Merge PACS Viewer or into a separate pop-up viewport window.

For a more detailed description on Patient Record functionality, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.1.6. Merge PACS Viewer

The Merge PACS Viewer is part of the Merge PACS Workstation and is the primary application for viewing and manipulating patient images.

Note that many of the features of the Merge PACS Viewer, such as inherent 3D support, the ability to create hanging protocols, spine labeling, the use of lossless vs lossy image compression, etc., are made available to the users based on their login privileges. Users with “advanced” privileges (typically radiologists) will be able to make use of the full range of features, whereas other users (e.g., referring physicians) will only have access to a subset of those features.

For a more detailed description on Merge PACS Viewer functionality, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.1.7. Quality Control Editor

The Quality Control Editor (QC Editor) can be accessed by authorized users from within various components of the Merge PACS Workstation and lets those users perform a wide variety of QC operations.

For more information on the QC Editor, refer to subsection 8.2.5 below and the *Merge PACS 8.2 Quality Control Editor Users Guide*.

1.1.8. HIS/RIS Integration

The HIS/RIS Integration element, actually a part of the Merge PACS Server, permits Merge PACS to convert information from a Hospital Information System (HIS) or Radiology Information System (RIS) into DICOM format, enabling the generation of Modality Work Lists (MWL), and eliminating the need for error-prone manual entry of data at the modality. The HIS/RIS Integration element also sends order information to the RealTime Worklist and allows HL7 Reports to be integrated into the Merge PACS Workstation. Finally, the HIS/RIS Integration element includes an Outbound HL7 Interface that can be used to send Order Request messages and various status messages from the Merge PACS Server back to the HIS/RIS, if so desired.

If the Outbound HL7 Interface is configured to send out system-generated orders, by default the orders are sent after the entire study has been imported. If desired, however, the Outbound HL7 Interface can be configured to send the system-generated orders as soon as the first image is received.

Note that in cases where a single Merge PACS Server is receiving orders from multiple facilities, the Server can be configured so that a MWL request from a modality in one facility will only get the orders sent from that facility. The procedure to do this involves creating individual MWL rules in the form of SQL statements. Each rule can apply to one or more modality, based on AETitle, which allows a separate rule to be created for each facility.

For detailed information on how Merge PACS integrates with HL7 systems, refer to the *Merge PACS 8.2 HL7 Interface Specification*.

1.1.9. Embedded Viewer

The Embedded Viewer is the Merge PACS Viewer toolset embedded into an organization's Electronic Medical Record (EMR).

For detailed information on how Merge PACS integrates with an EMR, refer to the *Merge PACS 8.2 DICOM Conformance Statement* and the *Interface Specification for Amicas Embedded Web Viewer*.

1.1.10. Merge PACS Teleradiology

The optional Merge PACS Teleradiology feature permits sites to move studies from location to location to be read, and either outsource or sell reading services to other organizations. When Merge PACS is configured to run in **Standalone** mode, teleradiology includes **Auto-Routing** for the transmission of uncompressed DICOM images and **Telemedicine** for the transmission of compressed (lossy or lossless) images. In both **Standalone** and **Integrated** mode, Telemedicine can also be configured to transmit annotations, comments, status, spine labels, study presentations, study worklist associations and/or key images.

1.1.11. Watch™

Watch is an optional feature that provides powerful tools for monitoring the health of Merge PACS. In addition, Watch provides the ability to obtain detailed analysis of system events to enhance performance and do preventative maintenance, as well as a variety of business decision support tools.

For information about using Watch, refer to the *Merge PACS 8.2 Watch User Guide*.

1.1.12. Enterprise CD Writer

Enterprise CD Writer is an optional feature that adds the ability to send patient studies to a shared CD or DVD writing device. Depending on how the device itself is configured, these can be just the compressed or uncompressed images or both the compressed and uncompressed images. In addition, auto-writing rules can be created to automatically send studies from one or more import devices (e.g., a modality or another Merge PACS Server) to the shared CD/DVD writing device.

NOTE: Enterprise CD Writer is supported for use with **Windows 2012 R2** as long as Sorna **Vertex** software is used. As an alternative (regardless of the operating system), if Merge PACS is configured to run in **Standalone** mode, a shared CD/DVD writing device (such as Sorna Vertex) can be configured as a DICOM Device, as described in subsection 9.13.2 below, and studies can be sent to it manually either via the Send Studies page, as described in subsection 9.9.3 below, or routed to it from the Merge PACS Workstation. If Merge PACS is configured to run in **Integrated** mode, a shared CD/DVD writing device can be configured as a DICOM Device on the paired iCEA Server and studies can be routed to it from the Merge PACS Workstation.

NOTE: The viewer burnt onto the CD/DVD is configurable by Sorna Personnel; if you want the Merge PACS Viewer to be included, contact Sorna to copy this to the system.

For information about using the Enterprise CD Writer component, refer to Section 9.10 below.

1.1.13. Third-party Application Synchronization

The Merge PACS Workstation can be integrated with third-party applications such as dictation or document management programs in the following two ways:

Type of Integration	Description
Direct API-based Integration	For use with a number of third-party dictation/report applications such as Dictaphone's PowerScribe™ and Agfa's TalkStation™. Integration is initiated by the user from within the Merge PACS Workstation via the 3rd-party Application Synchronization button, and clicking on that button will launch the application prepopulated with patient and study information. Note that with regard the configuration of user and group privileges, this type of integration is referred to as VoiceLink™.

XML File-based Integration

For use with dictation/report applications (such as Epic and SoftMed), document management applications (such as OnBase), and any other application that supports XML file-based integration. Merge PACS integrates with these applications by writing patient and study information to an XML file that can be read by the application. Integration is initiated either automatically when a study is launched or else manually via the 3rd-party Application Synchronization button, depending on how an individual Workstation is configured. Note that some applications can be configured to launch automatically when the XML file is created, whereas others will need to be manually launched after the XML file is created.

For certain applications, such as **PowerScribe 360** and **Epic Hyperspace**, Merge PACS can also be configured to provide **Bi-directional XML Integration**, which lets Merge PACS **receive** inbound XML messages and act on those messages. For example, if an XML file for a study is received where “response event = dictated” and “response event = signed”, the study state can be automatically changed to PRELIMINARY. Or if an XML file for a study is received where “response event = signed,” the study state can be automatically changed to READ.

In addition, a URL is provided that the Dictation/Document Software vendor may use to send a status update notifying Merge PACS that the user has finished a dictation. Once the external application has sent a call to this URL, Merge PACS will process and update the study status to DICTATED. This will save the Radiologist time by removing the need to click the appropriate status update in the Merge PACS Workstation.

Regardless of which type of integration is being used, only one study within the Workstation can be the focus of the third-party application at a time (although a single study can be the focus of multiple third-party applications if more than one application is configured). For instances where multiple studies are opened simultaneously within the Workstation, a user preference is provided within the Workstation to automatically synchronize (*i.e.*, move the focus of) the application(s) to the newly opened study. If this preference is *not* selected, users can manually synchronize a third-party application to a study by clicking the Third-party Application Synchronization button for the study.

For information on using the Merge PACS Workstation to integrate with third-party dictation/report applications, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.1.14.OrthoLink

Merge PACS supports direct API-based interaction with orthopedic templating toolkit applications such as Merge OrthoCase™. Integration is initiated by the user from within the Merge PACS Workstation via the OrthoLink button, and clicking on the OrthoLink button will launch the application prepopulated with patient and study information.

For more information on using the Merge PACS Workstation to integrate with third-party orthopedic templating toolkit applications via OrthoLink, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.1.15. Merge RadStream™ Communication

If the optional Merge RadStream Communication component is enabled, a new role of Operator is created. Using a special Communication WorkPanel interface from within the Merge PACS Workstation, the Operator can see contact information for the various physicians associated with a study; view a copy of the report associated with this study, if any, with relevant portions highlighted; enter comments regarding the contact made or attempted with each Physician and view a log of previously entered comments; and set the Communication and Connection status for this study.

For more information on using the Communication WorkPanel, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.2. Configurable Settings

In conjunction with the various components described above, Merge PACS has a number of additional features that can be optionally enabled and/or configured for use at a particular site.

1.2.1. Multi-Volume Support

By default, when Merge PACS is configured to run in **Standalone** mode, study image data is stored directly on the Merge PACS Server in as many as three separate directories (or “volumes”): one for the original uncompressed DICOM images, one for the lossy images that are used for enterprise distribution of studies, and one for the lossless images that are used to perform primary interpretation. A Merge PACS Server can be also configured to store images across multiple volumes located on multiple machines.

Multi-Volume Support serves two primary functions:

- It can **reduce Server downtime** in the event that one drive becomes corrupted or otherwise unavailable
- It allows for **gradual storage capacity expansion**, since additional disk drives can be added at any time to the Server's queue of available volumes. This is especially useful given the fact that the price of storage media tends to drop over time, and with Multi-Volume Support enabled there is no need to purchase many years' worth of storage capacity up front.

Multi-volume support is typically used with **compressed** (both **lossy** and **lossless**) images to expand the number of images available to end-users via Merge's *Forever Priors* technology, and these volumes can be either local or on one or more remote non-Merge PACS Servers accessible via a network. Multi-volume support can also be used with **uncompressed** images, and these volumes would consist of different hard drives located locally on the Merge PACS Server itself and/or remotely via Network Attached Storage (NAS) or Storage Area Network (SAN).

A volume should not exceed 4 TB. If a site's annual volume is far below this number; then there is not a need to create a single volume this large. For example:

- If a site produces less than 2 TB of studies in a one year period, you should start with a volume that is twice the annual capacity.
- If a site produces more than 2 TB of studies in a one year period, you should start with a 4 TB GB volume or volumes as needed.

In all cases, new volumes should be added 2-3 months in advance of the date when they are expected to be needed.

For additional information about how Multi-Volume Support works, refer to Section 2.5 below.

1.2.2. Auto-Routing

A major feature of Merge PACS is that all image studies can be viewed by accessing one central Web server. When Merge PACS is configured to run in **Standalone** mode, however, a Merge PACS Server can be also configured to automatically forward (or Auto-Route) select studies in their original DICOM format to other Merge PACS Servers (i.e., for central archiving purposes) or to one or more third-party radiology workstations. In addition, auto-routing can be used to continually send copies of the DICOM studies from a site's Production Server to its Fail-over Server in a Direct Attached Storage (DAS) environment.

Custom auto-routing rules can be set up to determine which studies are forwarded and where. In addition, Study Validation, as described in subsection 1.2.5 below, can be configured to prevent auto-routing of invalid studies, a process known as "Controlled Delivery."

For additional information on how auto-routing is handled by Merge PACS, refer to Sections 3.6 and 3.7 below.

NOTE: If Merge PACS has been configured with autorouting rules to send studies to other DICOM devices (other than the attached DICOM archive), these rules will not be reapplied after operations are performed from the QC Editor except for copy operations. If you need a DICOM device to be kept in sync after QC operations are performed, the affected studies will need to be manually sent to that device using the Send Studies tool, as described in Section 9.9.3 below.

1.2.3. Telemedicine

As described in the previous subsection, when Merge PACS is configured to run in **Standalone** mode, Auto-Routing can be used to transfer studies in their original DICOM format from one Merge PACS Server to other Merge PACS Servers. The size of DICOM images, however, makes Auto-Routing feasible primarily for the purpose of consolidating DICOM images from multiple "spoke" Servers onto a central Merge PACS Server for archiving. For sharing studies among multiple Servers for viewing purposes when Merge PACS is configured to run in **Standalone** mode, the Merge PACS Telemedicine feature can be used to transfer only the compressed (Lossy or Lossless) copies of desired studies. This can be done using both secure and non-secure HTTP protocols to meet the needs of a particular site.

By default, Telemedicine is set up to send Lossy compressed versions of the studies. However, a Merge PACS Server can be custom configured to send Lossless compressed versions instead. If so configured, the Server can also be configured to still send Lossy images if the Lossless versions are unavailable for a particular study.

In both **Standalone** and **Integrated** mode, Telemedicine can also be configured to transmit annotations, comments, status, spine labels, study presentations, study worklist associations and/or key images.

NOTE: In a hub and spoke situation, once a study has been sent to the hub from the spoke and has been modified on the hub (via QC operations, resending the study from the spoke will not revert the QC operations.

NOTE: Contingency Servers can be used as spoke Servers, but they need to be sized to match the expected load.

Custom Telemedicine transfer rules can be set up to determine which studies are forwarded and where. In addition, the transfer rules can be configured to only operate during certain hours of the day.

1.2.4. RealTime Worklist Configuration

Once the RealTime Worklist component has been enabled, individual worklists can be created, customized, and associated with specific groups of users. Users can then view a list of studies that have been assigned to them and set a workflow status for that study that can be viewed by other users.

By default, communication between the RTWL Client and the Merge PACS Server uses HTTP protocol. If secure communication is desired, however, this can be changed to use HTTPS

By default, the information used to generate individual worklists is read directly from Server's hard drive. Depending on the study volume and other factors, this may result in an unacceptable delay in the generation of Worklist data, in which case the Server can be configured to store this information in memory (RAM).

1.2.5. Controlled Access

Merge PACS provides three methods for controlling access to patient exams (including studies, reports and orders):

a. User and Group Privileges

As described in Section 9.11 below, access to exams within Merge PACS is primarily controlled by user and group privileges. By default, all "advanced" users (Radiologists, Technologists, etc.) have the privilege to view all exams (including studies, reports and orders) on the Server via the Merge PACS Workstation.

Other users, on the other hand, do not have access to all exams by default and can only view those exams for which they or their user group have been specifically granted access. This is true whether the user runs queries from within the Viewer or from any of the Management pages. This feature is used primarily to provide referring physician with very limited access to exams.

If a site has a RIS, a Merge PACS Server can also be configured to automatically assign exams to users based on the Provider Code contained in the orders generated by the RIS.

A Merge PACS Server can also be configured to automatically assign exams to groups based on the specific modalities or other Merge PACS Servers from which the exams were forwarded.

For more information on how User and Group privileges can be used to provide Controlled Access, refer to subsection 5.8 below.

CAUTION: If a study is contained in a RealTime Worklist or RealTime Study List worklist that a user has access to (i.e., the user belongs to a group that has been assigned that worklist), the user will be able to access that study regardless of the user's access privileges.

b. Study Validation

Study Validation can be used to restrict delivery of “invalid” studies via Auto-Routing and/or Telemedicine (a process known as “Controlled Delivery”), as well as to prevent access to those studies from within the Merge PACS Workstation (one example of “Controlled Access,” as described in the previous subsection). In addition, individual worklists used with the Merge PACS Workstation can be custom configured to only include studies that have been validated.

Study Validation can be done **manually** from the Merge PACS Workstation Technologist WorkPanel or **automatically**, according to one of the following three validation methods:

- If the **Default Auto Validation** method has been enabled, the Importer marks each study as being “Valid/Unread” or “Invalid/Unverified” according to a customizable definition of validity. The three possible definitions of validity are as follows:
 - There is an order that has the same MRN and Accession Number as found in the DICOM header information of the study (a “matching order”)
 - This study has no jailed images (when Merge PACS is configured to run in **Standalone** mode) or the study does not have a QC status of “QC Exception” or “QC Exception Pending” (when Merge PACS is configured to run in **Integrated** mode).
 - There is a matching order for the study *and* the study has no jailed images
- If the **Order Matching** Validator class has been enabled, the Importer will check if there is a **completely** matching order for each study (i.e., all the DICOM header information matches the order information).
 - If there is a completely matching order, the study will be given a workflow status of “Unread”.
 - If there is no completely matching order, the following steps are performed:
 - The study will be assigned a transient workflow status of “OM” (Order Matching).
 - The Auto PDE Process will be launched in an attempt to reconcile the DICOM header information with the order information.
 - If the Auto PDE process is successful, the study will be given a permanent workflow status of “Unread” and the validation process will be run again.
 - If the Auto PDE process is unsuccessful, the study will be given a permanent workflow status of “Unverified”.

NOTE: Order Matching Validation should not be enabled when merge PACS is configured to run in Integrated Mode.

- If the **Institution/Modality Validator** class has been enabled, the Importer will mark studies as being “Valid/Unread” if the Institution Name and/or Modality in the studies’ DICOM header matches a customizable list of institutions and modalities contained in a separate XML file.

NOTE: Manual Validation to link an order with a study can also be done via the QC Editor.

All three validation classes can run in conjunction:

- If using the Default Auto Validator in conjunction with the Order Matching Validator, the Default Auto Validator would be set to only check for jailed images (as opposed to checking for orders).
- When multiple validation classes are run in conjunction, a study will be given a workflow status of “Unverified” if it fails any of the validation classes.

Manual and automatic validation can be used together, allowing a user to manually validate a study after it has been rejected by the automatic validation rule.

For information on using the Technologist WorkPanel within the Merge PACS Workstation, refer to *Merge PACS 8.2 Workstation Users Guide*.

c. DICOM Jail

When Merge PACS is configured to run in **Standalone** mode, it can also be configured to prevent Viewer access to individual study images that have been placed in the DICOM Jail, as described in subsection 1.2.22 below.

1.2.6. Automatic Granting Access to Exams

By default, all users who have access to the Merge PACS Server can view all exams (including studies, reports and orders) on the Server from the Query Page of the Merge PACS Workstation or from any of the Management Pages. It is possible, however, to give particular users and/or groups restricted access to exams by removing the default privilege that grants them access to all exams. Once that privilege has been removed, a user can only view exams from the Query Page of the Merge PACS Workstation or from any of the Management Pages that have been specifically assigned to the user or group to which the user belongs.

NOTE: The automatic granting of access to exams is separate from the automatic assignment of exams described in Section 10.4 below.

CAUTION: If a study is contained in a RealTime Worklist or RealTime Study List worklist that a user has access to (i.e., the user belongs to a group that has been assigned that worklist), the user will be able to access that study regardless of the user’s access privileges.

Merge PACS can be configured so that exams are automatically assigned to a user or group in any of the following ways:

- If the Site has a RIS, the Merge PACS Server can be configured to automatically assign exams to a user if the user’s Provider Code matches any of following fields of an order generated by the RIS:
 - **Attending Doctor**
 - **Referring Doctor**
 - **Consulting Doctor**
 - **Admitting Doctor**
 - **Order Doctor**
 - **Other Health Care Provider**

- The Merge PACS Sever can also be configured to automatically assign exams that are forwarded from specified modalities or Merge PACS Servers to specified groups, based on the AETitle of the forwarding modality or Server.

CAUTION: If auto-assignment of exams to groups based on AETitle is incorrectly configured, there may be a delay in treatment due to exams assigned to the wrong group or to nobody. It is therefore recommended that a global “Unread Exams” worklist be configured to catch unread exams if this feature is being used.

1.2.7. Automatic Setting of Study Status Based on AETitle, Date and Time

Merge PACS can be configured to automatically set the status of incoming studies based on the AETitle of the forwarding modality, the day of the week they were received and the time of day they were received. This might be done for enterprises that have both onshore and offshore reading sites, for example, and would allow studies that come in during normal business hours to be set to one initial status and studies that come in outside of normal business hours to be set to a different initial status. Worklists could then be configured for users at the different sites that would filter on those different statuses

For example, Merge PACS could be configured so that studies that arrive during normal business hours are set to “Unread” status, whereas studies that arrive outside of normal business hours are set to “Unverified.” One worklist could then be configured for use by onshore reading sites that would only display “Unread” studies and another worklist could be configured for use by offshore sites that would only display “Unverified” studies.

1.2.8. Remote Notification

Remote Notification is the process of sending out information via email regarding the performance of a Merge PACS Server. In particular, update status notifications contain such information as performance statistics that over time can be used to monitor system performance and to alert a site’s PACS administrator or other on-site personnel to a potential problem before it happens. Remote notification can also be used to send out alerts that one or more images in a study failed to import properly and have been placed in the DICOM Jail, as described in Section 2.6 below.

NOTE: In order to use the Remote Notification feature, you must have your own SMTP server configured.

NOTE: By default, DICOM Jail alerts will not have any patient information in the notifications, but this can be changed on a site-by-site basis. In addition, you can also turn DICOM Jail alerts off or on entirely if so desired.

1.2.9. VoiceClip

VoiceClip provides the ability to record, listen to, and overwrite brief audio annotations that can be saved with a particular study. These annotations are stored as DICOM Basic Audio Waveform Storage instances (a.k.a. AUs). In standalone mode, the VoiceClip annotations are stored directly in the image volumes since they are DICOM instances as opposed to needing a separate voice clip volume. In integrated mode, they are stored in EA instead.

The ability to listen, record, and overwrite is based on each user's individual login privileges so that certain users can be given the ability to listen only, while other users can be given the ability to both listen and record or listen, record, and overwrite.

Each workstation where VoiceClip will be used needs to be connected to a correctly configured set of speakers in order to listen to saved audio annotations, and workstations where users will be recording and/or overwriting annotations will need to be equipped with a properly configured microphone.

For more information on using VoiceClip, refer to the *Merge PACS 8.2 Workstation User's Guide*.

1.2.10. Maximum Length of VoiceClip Audio Annotations

By default, the maximum length of VoiceClip annotations is 30 seconds. This can be changed, however, on a site-wide basis to meet the needs of a particular site.

1.2.11. Optional 3D Tools

By default, the Merge PACS Viewer provides its own set of inherent 3D tools that consists of Multi-Planar Reconstruction (MPR) and Maximum Intensity Projection (MIP) and Color Volume Rendering (CVR). The Merge PACS Viewer can be configured, however, to work with stand-alone third-party 3D applications for sites that prefer to use those applications instead. Currently, the applications that are supported are TeraRecon's AquariusNET™ Server and Thin-Viewer as well as Vital Images' Vitrea®.

The TeraRecon software consists of a stand-alone server component that stores the 3D images and a separate client application that can be launched from within the Merge PACS Workstation if so configured. The integration procedure involves configuring the Merge PACS Server to send studies to the AquariusNET Servers and then configuring each individual Merge PACS Workstation to launch the TeraRecon AquariusNET Thin-Viewer. This is typically done by a site's TeraRecon administrator.

Vitrea, by contrast, consists solely of a client application that is installed on each Merge PACS Workstation. The integration procedure involves configuring the Merge PACS Server to send studies directly to one or more Merge PACS Workstations and then configuring each workstation to send study information to the Vitrea application. Assuming the Vitrea application is running on the workstation, it will then display the study in question.

1.2.12.Global Update

By default, if a Merge PACS Server receives an update or merge request from a HIS/RIS that has an MRN that matches a study in its database, but a different Patient Name, the request is ignored. The Server can be configured, however, to ignore the Patient Name and perform the update based solely on matching MRNs. In addition, the Server can be configured to automatically update patient demographics in its databases, for both the merge-from patient and the merge-to patient, with the patient demographics in the merge request.

For more information, refer to Section 3.9 below and the *Merge PACS 8.2 HL7 Interface Specification*.

1.2.13.Multi-Site Modality Worklist Support

A Merge PACS Server allows users at a DICOM modality to query Merge PACS for a list of scheduled procedures known as a Modality Worklist (MWL). The MWL is stored on the Merge PACS by the customer's HIS/RIS.

In cases where a single Merge PACS Server is receiving orders from multiple facilities, the Server can be configured so that a MWL request from a modality in one facility will only get the orders sent from that facility. The procedure to do this involves creating individual MWL rules in the form of SQL statements. Each rule can apply to one or more modality, based on AETitle, which allows a separate rule to be created for each facility.

1.2.14.DICOM Deletion

When Merge PACS is configured to run in **Standalone** mode, it can be configured to automatically delete uncompressed DICOM images from the Server after a specified period of time and/or when available disk space reaches a predefined “low water mark.” By default, deletion will occur after **one hour** or when the available space drops below **1750MB**, whichever comes first. The deletion scheme (time-based or watermark based) as well as the settings for each scheme can be changed to meet the needs of a particular site, however.

1.2.15.Scheduled Database Backups

By default, a Merge PACS Server is configured to run nightly backup of each of the two main Merge PACS DB2 databases at 12:15 A.M. Additionally, the Server is configured to always keep the last two backups.

If a site is planning to use the optional Watch feature, the Watch database (ANETDB) will need to be added to the list of databases to be backed up. In addition, you can change the backup schedule and the number of old backups to keep to meet the needs of a particular site.

For more information on the various Merge PACS databases, refer to Chapter 6 below.

1.2.16. Scheduled Database Checks

If site needs to check databases on a regular basis, it is possible to define scheduled tasks to be run in a desired weekly intervals (e.g., every four weeks, every eight weeks, etc.). It is not defined by default and will need to be configured by Merge Healthcare Support to meet the needs of your site.

1.2.17. Compressed Image Deletion

When Merge PACS is configured to run in **Standalone** mode, the Compressed and Lossless Free Space High and Low Water Mark settings are used to determine when JPEG2000 lossy and JPEG2000 lossless compressed image data should be deleted from one or more compressed image volumes. When the total available free space on all compressed volumes falls below the **low water mark**, the system will begin deleting studies on a first-in, first-out basis until the available free space reaches the **high water mark**.

Image deletion for compressed images can be turned off completely to prevent the loss of any of a site's online store of "Forever Prior" images. If this is the case, however, Merge PACS will stop importing new images entirely when a specified **absolute minimum free space** setting is reached, and importation will not resume until additional space is added of the absolute minimum free space setting is reduced.

If image deletion for compressed images is enabled, by default the **low water mark** is set at **675MB** and the **high water mark** is set at **1000MB** for each type of compressed image. These settings can be adjusted to meet the needs of a particular site, however.

CAUTION: If the lossy copy of a study is deleted, the lossless copy of that study will also be deleted.

1.2.18. Additional Security Settings

A Merge PACS Server can be configured to provide various optional levels of security:

Type of Security	Description
Integrated Windows Authentication	Integrated Windows Authentication can be used to authenticate users based on their Windows login without challenging them a second time when they access the various Merge PACS client and server applications.
Transport Layer Security (TLS)	TLS is used to encrypt all data that passes between a Merge PACS Server and a user via the Server's Web site. This prevents the data from being intercepted and read by third parties that have not been granted direct access to the website.
IP Address and Domain Name Restrictions	This security feature can be used to enable or prevent specific users, computers, or domains from accessing Merge PACS.
Lightweight Directory Access Protocol (LDAP)	LDAP uses a separate computer that contains a group of user accounts and associated privileges. Users attempting to establish a Web (HTTP) connection to Merge PACS can be challenged to authenticate using existing account information that is stored on the LDAP server.

Type of Security	Description
Controlled Access	Access to exams within Merge PACS is primarily controlled by user and group privileges. By default, all “advanced” users (Radiologists, Technologists, etc.) have the privilege to view all exams (including studies, reports and orders) on the Server via the Merge PACS Workstation. Other users, on the other hand, do not have access to all exams by default and can only view those exams specifically assigned to them or to their group. This is true whether the user runs queries from within the Viewer or from any of the Management pages.
Virtual Private Network (VPN)	A Virtual Private Network (VPN) uses advanced encryption and tunneling to permit organizations to establish secure, end-to-end, private network connections over third-party networks, such as the Internet or extranets

For additional information on the various security options, refer to Chapter 5 below.

1.2.19. Changing the CheckResources Settings for Image Deletion

As described in Section 3.8 below, when Merge PACS is configured to run in **Standalone** mode, the CheckResources periodic command can automatically delete DICOM and compressed (Lossy and Lossless) images from Merge PACS. For DICOM images, the deletion can be based on both the length of time each image has been on the Server and high and low free space watermarks. For compressed images, the deletion can only be based on high and low free space watermarks.

In addition, when studies are deleted due to high watermarks being reached, the order of study removal can either be based on study date by default or last modified (“import”) date. If based on study date, the time for the deletion logic can also be specified.

By default, the CheckResources command is configured to delete DICOM based on both time and watermarks (whichever occurs first), to delete Compressed based on watermarks and to delete studies based upon study date older than 720 hours. The settings for the CheckResources command can be changed, however, as needed.

1.2.20. Changing the DICOM Deletion time

As described in subsection 1.2.19 above, when Merge PACS is configured to run in **Standalone** mode, a Merge PACS Server can be configured to delete DICOM images based on a watermark scheme and/or on a time scheme. If the deletion by time scheme is enabled, the default deletion time is set to **1 hour**, but this can be changed to meet the needs of a particular site. In particular, if the optional Merge Archive component is being used to archive DICOM images, it is important that this setting be changed so that images are not deleted until they have first been copied to the Archive.

NOTE: DICOM Deletion Time will work in conjunction with any Deletion Rules configured for your site. The DICOM Deletion Time will delete all DICOM images for ALL studies after a specified time, but will not delete other DICOM objects such as Study Presentation files and CADSR files. The Deletion Rules, on the other hand, will delete ALL Dicom objects (including images) after a specified time, but only for those studies that match the specified criteria.

For example, if the DICOM Deletion Time is set to 48 hours (two days) and the Deletion Rules are set to one day, DICOM images for studies that match the Deletion Rules criteria will be deleted after one day and the rest of the DICOM images will be deleted after two days. Alternatively, if the DICOM Deletion time is set to 24 hours (one day) and the Deletion Rules are set to two days, all DICOM images will be deleted after one day and the remaining DICOM objects (such as Study Presentation files and CADSR files) will be deleted for those studies that match the Deletion Rules criteria after two days.

1.2.21. Relaxing DICOM Jail Criteria

When Merge PACS is configured to run in **Standalone** mode, every image that is sent to Merge PACS from a modality has a DICOM header. This header contains a variety of attributes that are used to uniquely identify the image. These attributes define the study the image belongs to, the series it belongs to, and various image-specific items such as the type of image, the number of the image in the series, etc. Once the first image for a particular study is received, Merge PACS checks the headers for all subsequent images in that study to make sure that attributes match those of the first image.

A DICOM header may contain a large number of attributes, but not all of them are relevant in determining whether an image should be imported or jailed. By default, Merge PACS checks a defined set of these attributes to determine whether the image should be imported or placed in the DICOM Jail -- if all the attributes match, the image is imported; otherwise, the image is jailed. Any of these necessary criteria can be turned off (or “relaxed”), however, to meet the needs of a specific site.

For additional information on the DICOM Jail, refer to Section 2.6 below.

For information on using the Problem Studies Tool to remove images from the DICOM Jail, refer to subsection 9.9.4 and Section 17.5 below.

1.2.22.Preventing Access to Jailed Images

When Merge PACS is configured to run in **Standalone** mode, if a study contains one or more images that have been sent to the DICOM jail, the rest of the study can still be accessed via the query page within the Merge PACS Workstation. Any study with jailed images will be marked with a warning flag, and users who click on that flag will be shown a Comments window that explains that one or more images for the study have been jailed as in the following example:

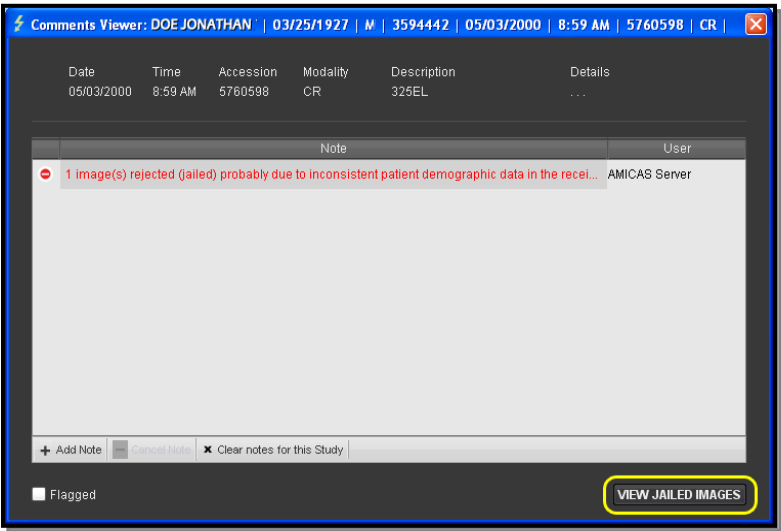


Figure 1: Option to View Jailed Images

By default, the Comments window will also display a “VIEW JAILED IMAGES” button that will allow users to view the jailed images for the study in uncompressed DICOM format within a special Viewer window. A Merge PACS Server can be configured, however, to not display that button and therefore prevent users from viewing jailed images.

Note that Merge PACS also includes a flexible *Problem Studies Tool* that allows an administrator to fix many of the problems that might have caused a study’s images to be jailed.

For information on using the Merge PACS Workstation to view jailed images, refer to *Merge PACS 8.2 Workstation Users Guide*.

For additional information on the DICOM Jail, refer to Section 2.6.

1.2.23.Configuring Viewer Warning Flags

By default, a Merge PACS Server displays a red warning flag when a user attempts to view a study with one or more jailed images from various places within the Merge PACS Workstation, as shown in the following example:



Figure 2: Warning Flag for Problem Studies

This flag can optionally be turned off to meet the needs of a particular site. In addition, the flag can optionally also be displayed for studies with one or more images where image compression failed.

1.2.24. Configuring a Remote Compressed Image Volume

As mentioned in Section 2.5 below, when Merge PACS is configured to run in **Standalone** mode, multiple compressed image volumes (*i.e.*, directories set up to store JPEG 2000 lossy and JPEG 2000 lossless images) can either be located **locally** on the Merge PACS Server itself or **remotely** on separate non-Merge PACS servers accessible via a network. In order for a user to view images stored on a remote compressed image volume, the DICOM user (which is used by the Apache service) must have access privileges to the images stored on the remote volume.

1.2.25. Configuring the Number of Trace and Error Logs Retained

As described in Sections 3.9 and 3.11 below, both ImageManager and WorklistServer generate error and trace logs whose size is limited to approximately 10 MB, and when the logs exceed this size limit, the current file is renamed and the log is rolled to a new file. By default, Merge PACS will retain a maximum of 500 old log files. This number can be changed, however, to meet the needs of a particular site.

1.2.26. Adding a Customized Logo to the Merge PACS Client Applications and Management Pages

By default, the Merge PACS Login Screen (for both the Workstation and the Management Pages) display the standard Merge PACS logo. If desired, a client-customized logo can also be displayed there, either a banner logo above the login credentials or a smaller logo to the right of the login credentials. In addition, a client-customized logo can be displayed at the top right corner of the Management Pages. These logos should be created with the following specifications:

- **Banner Logo** (for use on top of Login Screen)
 - The logo should be at least **450 pixels wide**.
 - The logo can be in **.jpg**, **.gif** or **.png** format and be given one of the following filenames as appropriate:
 - **custom-logo.jpg**
 - **custom-logo.gif**
 - **custom-logo.png**
 - To provide a transparent background, **.gif** or **.png** format should be used.
 - If **.jpg** is used, the background color of the logo should be set to **#383838** in order to match the log-in screen background.
- **Small Logo** (for use on right of Login Screen and top of Management Pages):
 - The logo should be **155 pixels wide** by **50 pixels high** in size.

- The logo can be in **.jpg**, **.gif** or **.png** format and be given one of the following filenames as appropriate:
 - **CustomLogo.jpg**
 - **CustomLogo.gif**
 - **CustomLogo.png**

1.2.27. Changing How Merge PACS Deals with Updated Orders

When Merge PACS is connected to a HIS/RIS, incoming orders can be viewed via RealTime Worklist and the order information is reflected in the DICOM headers on the images viewed from within the Merge PACS Workstation.

By default, once an order has been received by Merge PACS and had one or more studies associated with it, any updates to the order will be rejected by the system. Merge PACS can be configured, however, to accept changes to orders even after studies have been associated with those orders, in which case the updated order information will be visible via RTWL.

If Merge PACS is configured to accept changes to orders after studies have been associated with the order, by default these changes will not be reflected in the DICOM headers on the study images themselves (these changes would need to be made manually using the patient Date Editor). Merge PACS can also be configured, however, to promulgate these changes to the study images automatically without requiring somebody to manually make the changes.

NOTE: Having the study description update from the order is the recommended mode.

1.2.28. Configuring Whether to Include Procedure Codes in Study Description Fields

By default, ImageManager places an order's Procedure Code into the Requested Procedure Description and Scheduled Procedure Step Description fields (a.k.a. the Study Description fields). Merge PACS can be configured to omit the Procedure Code from these fields if so desired, however.

1.2.29. Configuring Status Changes when Images Arrive after Study Is Read

By default, if new images arrive for a study after the study has been marked "Read," the status of that study will automatically be changed to "Unread." Merge PACS can be configured, however, to instead change the status to a custom "Unverified-NewImages" status in these cases. A comment identifying the new images will also automatically be generated and associated with the study.

NOTE: If Auto Validation is enabled, as described in subsection 1.2.5 above, the Auto Validation process will be skipped for new images that arrive after a study has been read.

1.2.30. Allowing External Applications to Update Status via XML Integration

Merge PACS supports xml file-based integration with a number of third-party applications that can receive patient and study data from within the Merge PACS Workstation and RealTime Worklist. These include dictation/report applications (such as Epic and SoftMed), document management applications (such as OnBase), and any other application that supports XML file-based integration. Merge PACS integrates with these applications by writing patient and study information to an XML file that can be read by the application when that application is separately launched (note that some applications can be configured to launch automatically when the XML file is created).

In addition, a URL is provided that the Dictation/Document Software vendor may use to send a status update notifying Merge PACS that the user has finished a dictation. Once the external application has sent a call to this URL, Merge PACS will process and update the study status to DICTATED. This will save the Radiologist time by removing the need to click the appropriate status update in the Merge PACS Workstation.

1.2.31. DICOM Header Modification

When Merge PACS is configured to run in **Standalone** mode, it can be configured to modify DICOM header information for incoming studies based on information contained in a separate rule file. Rules can be set up to add one or more DICOM attributes, delete one or more DICOM attributes, add a prefix to one or more DICOM attributes, add a suffix to one or more DICOM attributes, replace one or more DICOM attribute with static values, replace one or more DICOM attribute with other DICOM attributes and/or truncate the length of one or more DICOM attributes. Furthermore, these rules can be set up to make these changes based on matching AETitle, Host Name, IP Address or DICOM attributes.

A typical use case scenario would be a hub and spoke situation where a rule could be set up to append a different prefix or a suffix to the “Patient ID” DICOM attribute based on the AETitle of the spoke server. Similarly, a rule could be set up to replace the “Institution Name” DICOM attribute with specific text or a private DICOM tag for each originating AETitle.

In general, rules will typically be set up to match based on the AETitle of the originating spoke Server. In cases where the studies are originating somewhere other than a spoke Server and there is a chance that the originating device may not have a unique AETitle, the rules can be set up to match based on Host Name (“HostName”) and IP Address (“IPAddress”)

Note that the **Institution Name**, **IPID** and **Department** fields in particular are typically handled poorly on the source modality devices. They may be missing entirely, spelled incorrectly or abbreviated inconsistently, and in many cases these can only be set by the modality field service personnel and not by the PACS Administrators. These fields are used not only by PACS, however, but also by other “downstream” systems for things such as worklist filtering (OrthoPACS, iConnect Access), patient ID matching, viewing scope control (iConnect Access), Policy Based Deletion rules (iConnect Enterprise Archive), etc. As such, the values need to be cleaned up or “enforced” via Header Modification Rules so that they can be used as filters input values. The Institution Name, for example, nearly always will need to be overridden due to differences in spelling, capitalization and abbreviations (e.g., St. vs. ST vs. St vs. Saint, Comm. vs. Community, Mem’l vs. Memorial, etc.).

NOTE: When Merge PACS is configured to run in **Integrated** mode, DICOM Header modifications are performed on iCEA via the Morphers accessible via WebAdmin.

For more information on automatic DICOM Header Modification, refer to Subsection 3.2.1 below.

1.2.32. Local Study Import Header Modification

Merge PACS can be configured to modify DICOM header information for incoming studies received from the Viewer's local import feature (usually studies on CDs from other sites). Rules can be set up to extract one or more characters from a DICOM attribute, change a DICOM attribute to be in all uppercase letters, change a DICOM attribute to be in all lowercase letters, replace some or all of the contents of a DICOM attribute, remove an entire DICOM attribute and/or set the value of a DICOM attribute to nothing if the attribute is present.

An example usage would be when a site wants to tag all studies imported from CDs with a certain prefix to the accession number to make sure there is no collision between their own system of generating accession numbers and studies from external sources.

1.2.33. Custom MRN Labels

By default, Merge PACS uses a patient's Medical Record Number (MRN) as the primary ID for that patient's records within the system. Within the various user interfaces, this number is labeled as either "MRN" or "ID". Some customers, however, use something other than the Medical Record Number to uniquely identify each patient, especially in cases where an enterprise consists of multiple sites and the MRNs from each site might overlap. In these cases, the site may instead use an Enterprise Master Patient Index/Master Person Index (EMPI/MPI) number.

If a site uses something other than MRN to uniquely identify patients, the labels that are used in the various user interfaces can be customized to indicate what is actually being used by that site. A single change to a configuration file can uniformly replace the default MRN/ID label shown in the following places:

- **RealTime Worklist data columns**
- **Query Page – search parameter**
- **Query Page – Query Results column header**
- **ER WorkPanel**
- **Technologist WorkPanel**
- **Communication WorkPanel**
- **DICOM Q/R Window**
- **Merge PACS Viewer – DICOM Overlay**

1.2.34. System-generated Orders and Accession Numbers

In situations where a site does not have a HIS/RIS (or in a hub/spoke model where one or more spoke servers do not have a HIS/RIS), Merge PACS can be configured to generate and send out study-based orders to external applications such as PowerScribe which require order information. Merge PACS can also be configured to generate unique accession numbers that can be used for delivery of an outbound order message for these study-based orders. If desired, each DICOM device that sends data to Merge PACS can be given its own unique ID code, and this ID code will then be added as a prefix to the generated accession number.

If the Outbound HL7 Interface is configured to send out system-generated orders, by default the orders are sent after the entire study has been imported. If desired, however, the Outbound HL7 Interface can be configured to send the system-generated orders as soon as the first image is received.

Note that if Merge PACS is configured to send out system-generated orders, an order will be sent whenever a study is imported into Merge PACS regardless of whether an order already exists. Therefore, to avoid duplication of orders, the Outbound HL7 Interface should only be configured to send out system-generated orders in situations where a site does not have a HIS/RIS already sending order information to the external application.

In situations where some studies may have accession numbers already assigned to them (e.g., in a hub/spoke model where some of the spoke servers have a HIS/RIS), Merge PACS can also be configured to respect any existing accession numbers instead of using the system-generated accession number in all cases.

1.2.35. Customizable Worklist Action Icons

By default, there are a set number of action icons available for use with RealTime Worklist, and those icons perform pre-defined functions. These action icons can be customized, however, to enhance the default actions (e.g., to launch the Viewer together with the Order Viewer icon) and to create brand new actions complete with custom icons. These custom actions can be constructed to call URLs or applications and pass through parameters such as Patient ID, ACC Num, User ID, etc.

Note that this configuration must be performed by Merge Healthcare Support and all applications must be first be validated by Merge Healthcare to not interfere with the proper operation of the Merge PACS system. Please contact Merge Healthcare Support to inquire about details and cost.

1.2.36. Configuring PeriodicCleanupDB to Improve Performance

By default, the PeriodicCleanupDB process, described in Section 3.8 below, runs once per day and retains database log files for 30 days (720 hours). The process can be configured, however, to run more frequently and to retain log files for a shorter number of days. This would be especially useful for large sites with extremely large forwarding logs where performance is affected by the amount of time it takes to delete the logs.

1.2.37. Allowing HL7 Change Orders to Insert Reading Physician into Study Table

By default, the ReadingPhysician column of the AMICASStudy table is populated from the DICOM header of the study images. This value is not provided via the Modality Worklist, however, and must be entered manually at the modality, which is not practical if a customer wants to use RTWL to have worklists that are specific to their Reading Radiologists.

In order to allow worklists to be created that are specific to Reading Radiologists, Merge PACS can be configured to automatically populate the ReadingPhysician column of the AMICASStudy table based on OBR.32 of the HL7 message. In addition, the following optional configurations are available:

- The status to which a study is automatically set once the ReadingPhysician column has been populated can be custom defined.
- The Reading Physician ID Number can be automatically placed into the Reading Physician Suffix field to deal with situations where there may be multiple Reading Physicians with the same name.

1.2.38. Configuring the Scheduled Procedure Start/Date Time

By default, the Scheduled Procedure Step Start Date/Time used for ORM messages is taken from OBR-7. If desired, however, Merge PACS can be configured to take it from OBR-27.4 instead.

Also, by default, if the Scheduled Procedure Step Start Date/Time is not specified, the value will be left empty in ORM messages. If desired, however, Merge PACS can be configured to set the Scheduled Procedure Step Start Date/Time to a specified offset from the current date/time when the Scheduled Procedure Step Start Date/Time is not provided.

1.2.39. Custom Accession Number Adaptor for Use with PowerScribe

If Third-party Application Synchronization is being used with PowerScribe, a customized accession number adaptor can be used to add information to accession numbers before the patient information is passed to PowerScribe. The default accession number adaptor checks to see whether the accession number already has a prefix. If it doesn't, and if the patient MRN has a prefix, the MRN prefix will be pre-pended to the accession number.

NOTE: In order to match the report to studies and/or orders, it is the customer's responsibility to strip out the prefix when the report is sent out from PowerScribe.

1.2.40. Disabling Multi-Dimensional Statusing

As described in Chapter 13 below, workflow status dimensions allow multiple statuses to be assigned to a single order simultaneously instead of sequentially. By default, these dimensions include **Diagnostic**, **Communication**, **Connection**, **Reservation**, **ImageTransfer**, **StudyLocking**, **ER**, **OrderMatching** and **UpdateOrder**. Each dimension can be customized with regard to available statuses and new dimensions can be added.

If a site wants to be able to use Merge PACS 8.2 without the need of training physicians on multi-dimensional statuses, however, multi-dimensional statusing can be disabled. Once multi-dimensional statusing is disabled, all dimensions other than Diagnostic, ImageTransfer and StudyLocking are hidden, custom statuses can only be added to the Diagnostic dimension and no new dimensions can be created.

NOTE: Multi-dimensional statusing should not be disabled if the optional RadStream component, as described in subsection 1.1.15 above, is being used.

1.2.41. Disabling Compressed Transfer Syntax

When Merge PACS is configured to run in **Standalone** mode, when studies are requested from a Merge PACS Server for the purpose of DICOM Q/R or Study Forwarding, by default if the uncompressed DICOM images are not available on the Merge PACS Server, DICOM images are reconstituted from lossless images, if available, and are sent using compressed transfer syntax if the requesting Dicom Device accepts compressed transfer syntax. If the requesting DICOM Device does not accept compressed transfer syntax, the reconstituted DICOM images are sent using uncompressed transfer syntax.

If neither DICOM nor lossless are available, DICOM images are reconstituted from lossy images and are sent using compressed transfer syntax if the requesting Dicom Device accepts compressed transfer syntax. If the requesting DICOM Device does not accept compressed transfer syntax, the reconstituted DICOM images are sent using uncompressed transfer syntax.

If desired, however, the Merge PACS Server can be configured to always send images using uncompressed transfer syntax, even if compressed transfer syntax is accepted by the requesting device.

1.2.42. Disabling Reverse DNS Lookup

By default, Merge PACS looks to resolve the IP address after each image is received from a modality. The Operating System HOSTS table must be maintained for every scanner sending studies to the PACS or a 60 second timeout occurs after each image the IP cannot be resolved. The reverse DNS lookup can be disabled, however, so that the customer no longer needs to keep the HOSTS table up to date to avoid this timeout.

1.2.43. Order Linking via Outbound HL7

Merge PACS can be configured to provide an “Order Link Notification” link on the Technologist WorkPanel within the Merge PACS Workstation. Once enabled, users can click on this link, search for an order based on various criteria, select the desired order from the list of matching results and then click on the “Notify Order Link” button. Once this is done, an Image Status Outbound HL7 message composed of the selected order information and the related DICOM study instance UID will be sent via the Outbound HL7 Interface. The value of Results Status field, OBR.25, is 2, which means the objects have been verified.

A use case for this would be a chest/abdomen/pelvis study where multiple orders are generated but the study only links to one of those orders. The user needs to send an outbound HL7 message to the dictation system for the other unlinked orders so the report is also generated for those accession numbers.

1.2.44. Enabling Automatic Conversion of HL7 XO Messages to NW Orders

By default, Merge PACS will accept a change order request (XO message) only if there is *currently* an order in the system with a matching Patient ID and Accession Number. Once an order has been deleted, any change order requests will be ignored.

If desired, however, Merge PACS can be configured so that a change order request will automatically generate a new order if the patient has *ever* had a record in the system (whether created by an earlier new order request or by a study), even if the original order has been deleted.

Note the following:

- The XO message will be NAKed even if a new order is created, and the reason will be stated in the NAK message.
- The receiving system can be configured to ignore such NAK messages.

1.2.45. Configuring the Maximum Number of Unassigned Exams

As described in subsection 9.9.9 below, the Manage Unassigned Exams page allows you to view and grant access to exams that are not currently assigned to anybody. By default, the list of matching unassigned exams can display a maximum of 20 results. This number can be configured, however.

1.2.46. Enabling and Configuring Multiple Order Association

If Merge PACS has been configured to provide Viewer XML Integration, it can be configured to associate multiple studies / orders with a primary study. Once this type of XML integration is enabled, it can also be configured in the following ways:

- By default, orders are eligible for association if they have one of the following statuses: **Ordered** (OR), **Scheduled** (SC), **Unverified** (UV) or **Unread** (UR). This can be changed, however.

- The default time tolerance for orders that will appear in the Associate Multiple Orders GUI is **1440 min** (1 day). This is the time tolerance relative to the primary study. Orders outside the time tolerance will not appear on the GUI. This can also be changed.

1.2.47. Configuring RTWL to Launch the iConnect Access™ Viewer

The Merge iConnect Access™ Web Viewer is a web-based viewer that Provide diagnostic quality image review for referring physicians from any browser-based device with a zero-download image and XDS viewer. If Merge PACS is integrated to work with Merge RIS, a custom action icon can be created by Merge Healthcare Support for use with RTWL to launch the iConnect Access Web Viewer.

NOTE: Merge PACS is validated to work with iConnect Access v4.0 and later.

1.2.48. Configuring RTWL to Launch Merge RIS Tech Entry and Patient File

If Merge PACS is integrated to work with Merge RIS, custom action icons can be created for use with RTWL to launch the RIS **Tech Entry** module or **Patient File: Demographics** page to view information about a particular patient.

NOTE: The ability to access the **Patient Files: Demographics** page from within RTWL is only supported with RIS 8.0 and higher.

NOTE: Each Merge PACS user will need a valid login account within RIS that has access to the appropriate area within RIS. For access to Tech Entry, the user should also be configured within RIS to have the Tech Entry module as the default homepage.

1.2.49. Configuring RTWL to Launch CADStream™

CADStream™ supports analysis packages for breast, liver and prostate analysis. If Merge PACS is integrated to work with CADStream, a custom action icon can be created by Merge Healthcare Support for use with RTWL to launch CADStream for a particular study.

When data is selected, CADStream automatically determines the correct analysis package. Therefore, only one action icon is required for all CADStream analysis packages.

NOTE: In order for CADStream to be launched by RTWL, the permissions for the install directory for the CADStream (*i.e.*, **c:\confirma**) application need to be updated to give the **Users** group **Full Control**.

NOTE: When selecting data from RTWL, the same data must be available on the CADStream server in order for CADStream to launch.

1.2.50. Configuring How Embedded Viewer is Launched from Internet Explorer

By default, when the Merge PACS Viewer is launched from within Internet Explorer as an embedded viewer, the original browser is left open. If desired, however, Merge PACS can be configured so that the original browser window is closed when the embedded Merge PACS Viewer is launched from within Internet Explorer.

1.2.51. Configuring Study Import Notification

As described in Section 3.7 below, the Forwarder supports the **Study Import Notification Service** that provides **DICOM Instance Availability Notification (IAN)** and **Basic Study Content Notification (BSCN)** notification messages when studies are imported into the Merge PACS Server. Notification messages are sent after a study is imported into the Server when it becomes available for reviewing (*i.e.*, 2 minutes after receiving the last image of the study) even if there is no matching order for the study. The content of the notification messages is described in *Merge PACS 8.2 DICOM Conformance Statement*. The notifications are sent by the Forwarder and follow the same association rules as for the forwarded images (AssociationIdleTimeout, MaximumJobDispatcherSleepTime, DownDestinationCheckInterval, etc.). The forwarder prioritizes IAN and BSCN messages over image forwarding.

Study Import Notification can be configured in the following ways:

- The Study Import Notification Service can be enabled or disabled completely. By default, the service is disabled.
- Destinations where IAN messages should be sent can be defined. If not defined, no IAN messages will be sent.
- Destinations where BSCN notification should be sent can be defined. If not defined, no BSCN messages will be sent.
- The directory where the server will temporarily store the IAN and BSCN messages until they are sent successfully can be defined. There is no default value and this must be defined if Study Import Notification will be used.

1.2.52. Configuring Whether PDE Requests Update Both Files and the Database

Prior to Merge PACS 8.2, it was possible to configure whether PDE requests updated both files and the database or just the database. As of Merge PACS 8.2, however, when PDE, QC events or HL7 updates are processed, the database will be updated and not the files. When studies are transferred from the system (DICOM, cd burning, teled transfer, etc.), the updated demographics will automatically be updated based on the current database value.

1.2.53. Preserve Tech EE Notes by Migrating Them into Study Comments

If desired, Merge PACS can be configured to preserve Tech EE notes by migrating them into study comments.

1.2.54. Configuring Color Maps

Merge PACS comes with a default list of color maps that can be used in MPR Viewports within the Merge PACS Workstation. This list of available color maps can be configured, however, on a site-by-site basis.

1.2.55. Enabling and Configuring EMRLink

If your site has OrthoEMR and sharing of images is required between Merge PACS and OrthoEMR, a feature called EMRLink can be enabled on the Merge PACS Server. When EMRLink is enabled, a JPEG copy of each image will be created and exported to the local file system on import to the Merge PACS database. A service (daemon) on the system will then send these images to the OrthoEMR database and they will become eligible for inclusion in an EMR report.

1.2.56. Configuring Compression Preferences

When Merge PACS is configured to run in **Standalone** mode, the preferences for each modality type that can send images to Merge PACS include a “Recompressed Lossy Ratio” that determines whether and how uncompressed JPEG lossless and JPEG 2000 lossless images that have the “Lossy Compression” flag set (*i.e.*, the “Lossy Image Compression” DICOM attribute is set to “1”) will be recompressed. By default, the value for each modality type is set to “1,” which will result in JPEG 2000 Lossless being used for recompression. Two other options are available if a site desires, however:

- Setting the Recompressed Lossy Ratio to a value greater than “1” will result in that value being used as the target compression ratio for compression/recompression.
- Setting the Recompressed Lossy Ratio to a value of “0” will result in the Lossy Compression ratio being pulled from the “Lossy Image Compression Ratio” DICOM attribute or, if that attribute is not available, from the “Derivation Description” attribute. That value will then be used as the target compression ratio for compression/recompression. If there are multiple values in the DICOM attribute, the highest value will be used.

1.2.57. Adding a Buffer for Watch Workflow Statistics Reports

If a site is using Watch, users can generate workflow statistics reports for specified time periods. By default, the reports will display all studies that have transitioned from the initial workflow event status to the ending workflow event status during the selected time period. Some studies, however, might have an intermediary transition between the defined initial workflow event status and ending workflow event status and, as a result, transition from the requested initial workflow event status before the defined time period, even though they transitioned to the ending workflow event status during the time period.

For example, if the defined period of time for a report is January 15 to January 31, the beginning workflow event is a transition from “Unverified” status, and the ending workflow event is a transition to “Final” status, the report will not normally include a study that transitioned from “Unverified” to “Unread” on January 14 and from “Unread” to “Final” on January 16. If the system is configured to add a buffer of two days to the start time for all studies that transition to the ending workflow event status during that time period, however, this study will be included since the transition from “Unverified” occurred within two days prior to the specified start date.

1.2.58. Changing Default Behavior of Keep Links to Orders and Reports Checkbox

When performing patient data edits from the Quality Control Editor, the user is presented with a **Keep links to orders and reports** checkbox that determines whether links between this patient and any orders or reports associated with the patient should be preserved. This checkbox is also displayed within the Merge PACS Workstation at the Patient and Study Demographics windows.

By default, this checkbox will be unselected when a user accesses one of the locations where it is displayed (on the Workstation side, any selection made by a user will be retained for future use). Merge PACS can be configured, however, so that the checkbox will be automatically selected by default instead.

1.2.59. Configuring Data Masking for Additional Public or Private Tags

By default, the following patient information is replaced with non-identifiable information when a user clicks the **Data Masking** icon on a Teaching Worklist within the Merge PACS Workstation:

- | | | |
|------------------------------|------------------------------|-----------------------|
| • Study Instance UID | • Referring Physician | • Series Time |
| • Series Instance UID | • Patient Name | • Study Date |
| • SOP Instance UID | • Patient ID | • Study Time |
| • Accession Number | • Patient Birth Date | • Content Date |
| • Institution Name | • Series Date | • Content Time |

In addition, the following patient information will be removed entirely:

- | | |
|--|--|
| • Institution Address | • Patient Mother Birth Name |
| • Physicians Of Record | • Responsible Person |
| • Physicians Of Record ID Sequence | • Acquisition Date |
| • Performing Physician Name | • Acquisition Time |
| • Performing Physician ID Sequence | • Admitting Date |
| • Name Of Physician Reading Study | • Admitting Time |
| • Physician Reading Study ID Sequence | • Overlay Date |
| • Issuer Of Patient ID | • Overlay Time |
| • Other Patient IDs | • Performed Procedure Step Date |
| • Other Patient Names | • Performed Procedure Step Time |
| • Other Patient IDs Sequence | • Scheduled Procedure Step Start Date |
| • Patient Birth Name | • Scheduled Procedure Step Start Time |
| • Patient Address | • Scheduled Procedure Step End Date |
| • Insurance Plan Identification | • Scheduled Procedure Step End Time |

All of items described above are considered “public tags” within DICOM. If desired, however, Merge PACS can be configured to mask or remove additional tags, both public and private

1.2.60. Limiting MWL Responses to Eliminate Content of Certain Fields

Merge PACS requires orders to populate a DICOM Modality Work List (MWL) response in the field. The required information is usually provided by an ORM feed, but some sites (typically OrthoPACS customers) do not have an ORM feed, just ADT or SIU feeds from their Practice Management systems that do not provide all the information required by Merge PACS.

Merge PACS can be configured, however, to define certain DICOM attributes whose values should be blank when sending out MWL responses. This will prevent incorrect default values from being inserted. A typical MWL response should be patient demographics, including Patient ID, Patient Name, DOB, and Patient Sex.

1.2.61. Configuring whether iCEA or Merge PACS has Precedence over Incompatible Watermarked Studies

By default, when Merge PACS is configured to run in **Standalone** mode, if images are received from iConnect® Enterprise Archive (iCEA) for a watermarked study and the normalized MRN, normalized name, or normalized IPID of the incoming data do not match that of the watermarked study record, Merge PACS will modify the data attributes to match the Merge PACS watermarked study record before the data is imported, and the updated study will be propagated back to iCEA after it is imported. In other words, it is assumed that the information in Merge PACS is correct and that information will be given precedence over the information received from iCEA.

If desired, however, Merge PACS can be configured so that the information received from iCEA will be given precedence over the information stored within Merge PACS and, as a result, these studies will be placed in the DICOM Jail instead of being modified.

NOTE: If images are received from a source other than ICEA for a watermarked study where the normalized MRN, normalized name, or normalized IPID of the incoming data do not match that of the watermarked study record, the study will be placed in the DICOM Jail regardless of how Merge PACS is configured.

1.2.62. Configuring Whether QC Operations can Be Performed on Read/Viewed Studies

By default, the operations available from the Quality Control Editor cannot be applied to studies that have a diagnostic status of Viewed or a diagnostic status equal to or greater than Read. If desired, however, Merge PACS can be configured so that QC operations can be performed on all studies, regardless of their diagnostic status.

1.2.63. Configuring Workstation to Listen for RTWL Messages on Alternate URL

By default, the Merge PACS Workstation is configured to listen for RTWL messages on port 8080. This is done via a URL that is passed from the Merge PACS server to the worklist clients. If a site has blocked access to port 8080, however, the URL must be configured to reference a port that server-side RTWL has been configured to listen on.

NOTE: In addition to changing the port referenced in the URL, you will also need to set up some proxy server to handle the RTWL messages. This means you will also need to change the IP Address referenced in the URL.

1.2.64. Configuring Patient Comparison Strategy

As described in Section 2.10.1 below, Patient Comparison Strategy is a virtual layer added to PACS Server to provide fine grain control on patient update and merges via HL7 interface or Patient Data edit via the Merge PACS Quality Control Editor. When a study is imported into Merge PACS, a Patient IDentity for the new study is created and a unique identifier is generated. Based on the unique identifier, the study is either matched to a pre-existing patient and stored as such or stored as a new patient. When patient demographics are updated or merged with another patient, the unique identifier is used to group all related studies and process the update or merge request. With Patient Comparison Strategy, Merge PACS can be configured to use other patient demographic values to identify and group studies related to a given patient.

1.2.65. Configuring Modality Types for QC Merge Operations

By default, users can only merge two series together from the QC Editor if the modalities of the two series are the same. If desired, however, Merge PACS can be configured to allow merging of specific modality types.

1.2.66. Configuring iConnect® Enterprise Archive Compatibility

As described in subsection 3.9.4 below, when Merge PACS is configured to run in **Standalone** mode and is processing A08 (Patient Update), A34 (Patient merge) and A40 (List of Patients merge) requests from a HIS/RIS, if multiple patients having the same MRN are found, the request is rejected. However, when Merge PACS Server is configured with iConnect® Enterprise Archive, you may want to process HL7 message if the patient's name (depending on Patient Comparison Strategy) matches a single patient when you have duplicate patient records for a given MRN. This feature will perform additional checks when duplicate patients are found for a given MRN by comparing the Patient's name to the list of patients matching the MRN and then processing the patient if there is a name match to only one patient.

In addition, when processing A34 (Patient merge) and A40 (List of Patients merge) requests from a HIS/RIS, you may want/need to enforce extra IPID checks. These checks will ensure that the IPIDs in Patient Demographics segment and the merge segments of a HL7 message are present, are the same (case sensitive comparison) and if default configured IPID should be used.

1.2.67. Configuring Laterality for ORM Messages

In an ORM message, the laterality value is expected to come from either the first or second subcomponent of the OBR-15.5 segment. By default, Merge PACS is configured with the assumption that the HL7 Interface will be sending the message from the entire OBR-15.5 segment or from the first subcomponent of that segment, but this can be changed if the HL7 Interface will be sending the message from the second subcomponent instead.

1.2.68. Configuring Sender Monitor Page

As described in subsection 9.6.9 below, the **Sender Monitor** page displays numerous statistics and details regarding the current activity of the **Sender** process when Merge PACS is configured to run in **Integrated** mode. The following configuration options can be set to control how much information is actually displayed on that page:

- The maximum number of failed jobs that can be displayed on the **Study Instance Send Tracker** page (available via the **Study Send Tracking link** on the main Sender Monitor page). The default value is 5000.
- The number of minutes after which successful jobs should be removed from the Study Instance Send Tracker page. The default is 120 (two hours).
- The maximum total number of jobs (both successful and failed) that can be displayed on the Study Instance Send Tracker page. The default value is 5000.
- The size of the “bucket” used when displaying information for dispatching and job queue delay times on the **Sender Job Queue Stats** page (available via the Sender Job Queue Stats link on the main Sender Monitor page). The default value is 5.
- The maximum number of entries that can be displayed for both the dispatching and job queue delay times logs on the Sender Job Queue Stats page. The default value is 1500.

1.2.69. Configuring Automatic Comparison Study Retrieval

When Merge PACS is configured to run in **Integrated** mode, studies can be retrieved from the paired iCEA in either **Full Folder** or **Half Folder** mode as follows:

Mode	Description
Full Folder	PACS has all the information of the study that is necessary to directly load the study, including all instances of the study.
Half Folder	PACS has the knowledge of the existence of the study and its location without the knowledge of the instances contained in the study. It allows the study to be listed in RTWL/RTSL, Query Page, Prior Lists, etc., without the additional overhead of using Extended Query, but requires restoration of the study before it can be used in the PACS viewer.

Typically, a site migrating from pure iCEA to Merge PACS running in Integrated mode will fully import only a small percentage of available studies (since the process can be very time intensive) and then import the remaining studies in Half Folder format. Merge PACS will then automatically retrieve a study that only exists in Half Folder form when a user attempts to load that study within the PACS Viewer. By default, however, only the primary study is retrieved, meaning that each comparison study associated with that study would need to be individually retrieved.

If desired, however, Merge PACS can be configured to automatically convert to Full Folder status any study in Half Folder mode as soon as another order or study for the same patient is received by Merge PACS. This is based on the assumption that the receipt of an order or a study will likely be soon followed by an attempt to view that study along with one or more comparison studies associated with the same patient, and converting those comparison studies to full folder status at this time will allow the user to seamlessly access all studies when needed.

NOTE: By default, the comparison study retrieval process can also be triggered manually from various locations within the Merge PACS Workstation, regardless of whether automatic retrieval is enabled or not.

NOTE: The progress of the comparison study retrieval process can be monitored from the **Image Manager Monitor** management page (as described in subsection 9.6.8 below) for retrievals triggered by orders, from the **EA Notification Processor Monitor** management page (as described in subsection 9.6.3 below) when triggered by studies while Merge PACS is configured to run in **Integrated** mode, the **Importer Monitor** management page (as described in subsection 9.6.5 below) when triggered by studies while Merge PACS is configured to run in **Standalone** mode, and the **Application Server Monitor** management page (as described in subsection 9.6.12 below) when triggered manually.

NOTE: This feature was developed specifically for use when Merge PACS is configured to run in **Integrated** mode, for restoring half folders for studies in the paired iCEA, but it can also be used for other purposes. When Merge PACS is configured to run in **Standalone** mode, for example, this feature can potentially replace the prefetching script that is currently used by Support. It can also be used in **Integrated** mode for cases where the associated study is not in the paired iCEA, but rather is in an archive behind the paired iCEA. This feature can be configured to perform the restore in this case as well, but keep in mind that this will take more time (because it will first trigger a move from the archive to the paired iCEA).

1.2.70. Configuring Study End Date/Time

As described in subsection 3.9.6 below, RTWL worklists can be configured to display Study End Date/Time instead of or in addition to Study Date/Time. By default, the value for Study End Date/Time is taken from the OBR-7.1 field of the ORM (Order Complete) message. Merge PACS can be configured, however, to pull the information from the ORC segment of the ORM message instead and can also be configured to pull the information from any specified field and sub-component of the ORM or ORC segment.

1.2.71. Configuring Integration with Patient Synopsis

Patient Synopsis™ provides meaningful cognitive insights related to ordered exams by allowing Radiologists to view patients' clinical insights while interpreting images in imaging reporting workstations, which can contribute to more precise decisions while reading imaging exams, expedite image reading, and improve the quality of diagnosis.

Merge PACS can be configured to launch studies within Patient Synopsis from the Merge PACS Workstation in any of the following ways:

- Merge PACS can be configured to automatically launch Patient Synopsis when launching a study within the Merge PACS Viewer in order to display the clinical history for the patient.
- Individual worklists can be configured to include a Patient Synopsis action icon that can be configured to launch Patient Synopsis for a study without opening the study in the Merge PACS Viewer.
- A custom macro can be configured to allow users to manually launch Patient Synopsis after launching a study in the Merge PACS viewer to see more details about the patient's clinical history.

In addition, Merge PACS will need to be configured in order to apply privilege control and encryption on the URL parameter list for Patient Synopsis.

1.2.72. Configuring Integration with OrthoView (v. 7.3 or Later)

In addition to using the OrthoLink feature to provide direct API-based interaction with orthopedic templating toolkit applications such as Merge OrthoCase™, as described in subsection 1.1.14 above, Merge PACS can also be configured to intergrate with the OrthoView™ orthopedic templating toolkit as long as it is v 7.3 or later. Once configured, studies can be launched into OrthoView from the Merge PACS Workstation by way of a custom macro. In addition, Merge PACS can be configured to require that the user have the OrthoLink privilege to launch OrthoView.

1.2.73. Configuring the Default Facility

By default, if a study or order is imported into Merge PACS with an Institution Name attribute that is not currently known to Merge PACS, Merge PACS will attempt to match it to an existing Facility, if any, that matches the exam or order's Institution Name information (case insensitive matching) or create a new Facility with this name if no matching Facility is found.

If desired, however, Merge PACS Can be configured to instead map the Institution name for the study or order to a Facility that has been defined as the “default” facility for the site.

1.2.74. Configuring Advanced Workflow Properties

The behavior of the Advanced Distribution Engine and other Advanced Workflow features can be configured as described below.

a. Configuring Automatic Retry of Auto-Assigned Exams

By default, the Advanced Distribution Engine will automatically retry auto-assignment when an unassigned exam is sent to the Advanced Distribution Engine and no assignee is found. This is useful, for example, in situations where a qualified reader may log on soon after the initial assignment attempt failed. Also by default, the system will wait five minutes before attempting to retry the assignment and will continue retrying every 20 minutes up to a maximum of 50 attempts. Also by default, exams currently in the retry queue will be given priority over newly imported exams (meaning that, when an exam is sent through the Advanced Distribution Engine, the Advanced Distribution Engine will first attempt to assign all exams in the retry queue that are closer to or further past their “due time” than that exam and then attempt to assign that exam).

These defaults are configurable on a site-by-site basis, however.

b. Configuring Trigger Options

In addition to the Status and Time triggers that can be configured from the Auto Assignment Menu, as described in subsection 10.4.4 below, a variety of trigger options can be manually configured to control when automatic assignment (or reassignment) will actually be performed.

- **Preloading of Users** – By default, the Advanced Distribution Engine will preload users as they login so that, rather than the user having to wait for the next exam to be acquired and sent into PACS and possibly be selected as the assignee for reading, the system can look at all incomplete exams in the system and check to see if the Advanced Distribution Engine would select this new user as the best candidate for assignment (and, if so, reassign the exam to the newly-logged-in user). In addition, preloading of a user will stop when a configured number of work items (5 by default) have been assigned to that user, or when the system runs out of incomplete work items to consider.
- **Rebalancing a User's Work Items** – By default, the Advanced Distribution Engine will rebalance a user's work items when the user's user profile properties change so that all of the automatically-assigned work items currently assigned to that user (and not complete) will be sent for automatic reassignment. For example, if a user has specific modalities associated with him or her and studies have been assigned to that user based on modality, changing the user's profile to remove those modalities may cause the user to no longer be the best reader for those studies.
- **Automatically Offloading Users** – By default, the Advanced Distribution Engine will offload a user when he or she manually logs out, causing all of the automatically-assigned work items currently assigned to that user (and not complete) to be sent for automatic reassignment.
- **Automatic Assignment Due to New Orders** – By default, when an order arrives and is matched to an existing study, the Advanced Distribution Engine will send the study for automatic assignment/reassignment. It is considered acceptable to reassign back to the same user.
- **Automatic Assignment Due to Order Updates** – By default, when an order update is processed, the Advanced Distribution Engine will send the associated study for automatic assignment/reassignment. It is considered acceptable to reassign back to the same user.

c. Configuring the Name of the Reader Availability Custom Tag

As described in subsection 10.5.5 below, you can define one or more custom “Reader Availability” tags that can be used when defining Auto Assignment rules. By default, this is referred to as a “Shift” tag, but the name can be custom configured on a site-by-site basis.

d. Configuring the Identification of Lone Studies

Lone studies are studies for which no HL7 order is expected due to the workflow of a particular site (e.g., the site does not use orders at all or only uses orders for some of their exams). By default, when Merge PACS is configured to integrate with an Analytics / Dashboard Server to provide Advanced Workflow statistics, studies without associated orders will be ignored by the Analytics / Dashboard Server. If desired, however, Merge PACS itself can be configured to identify which lone studies should be recognized by the Analytics / Dashboard Server.

1.2.75. Configuring Security ID Applications for Launch and Control Properties

The Launch and Control feature within Merge PACS, as described in Chapter 15 below, allows users to set up a macro that can be used to determine which available applications (including the Merge PACS Viewer and applications currently supported through XML Integration such as M*Modal Fluency, Epic, etc.) should be launched when users click the View Study worklist action icon under various conditions (e.g., the type of study, the specific workstation or facility where the study is being read, etc.).

By default, Merge PACS is configured to include a security ID for iConnect Access in the XML files dropped by PACS. If necessary, this can be custom configured to add additional applications for which the security ID should be dropped.

1.2.76. Configuring Outside Exam Properties

Outside exams (a.k.a. “alien” exams) are exams whose images are stored in 3rd-party systems and not currently available within Merge PACS, but for which Merge PACS has received an HL7 order message for the exam with a status indicating that the order is complete. By default, outside exams are ignored by Merge PACS, but Merge PACS can be configured to treat them the same as regular exams as follows:

- They will be included in lists of prior exams in worklists
- They will be included in the counts shown in the Priors column in worklists.
- They will be included in search results on the Query Search page.
- Events for the orders will be shown in the Event Log.

When this feature is enabled, there are also a number of ways in which its behavior can be configured:

- **Configuring Which HL7 Sources can Send “Outside Exams”** – Individual sites may have a workflow whereby PACS receives both the order and the images and in some cases the complete order can arrive just after the images. For this reason, no HL7 sources are configured by default as able to provide the “completed” orders that would cause an exam to be considered an “outside exam,” so this will need to be configured on a site-by-site basis.
- **Configuring Which Completed Orders Should Be Considered “Outside Exams”** – Once Merge PACS has configured which HL7 sources can provide completed orders that might be considered “outside exams”, it can then be configured to determine which orders from these sources should actually be considered outside exams. In general, an order will be considered “complete” if ORC-5 is set to “CM” for that order or if it has a numeric value that has been mapped to the display name “Complete” at the HL7 Order Status Mapping management page, as described in subsection 9.13.9 below. Since the display name is custom configurable on a site-by-site basis, however, you can change the default value to indicate a different display name that will be used to consider when an order is complete for use with outside exam determination.
- **Configuring the Diagnostic Status that Outside Exams should be Set to Automatically** – By default, outside exams that are recognized by Merge PACS will automatically transition to the Unread status in the Diagnostic dimension, but this is also configurable.
- **Configuring Whether Cancel and Change Order Messages for Outside Exams Should Be Accepted or Rejected** – By default, Merge PACS will accept cancel order messages for outside exams (causing the order to be deleted) as well as change order message, but this can be changed on a site-by-site basis.
- **Configuring Automatic Deletion of Outside Exams** – By default, the orders for outside exams will be automatically deleted from Merge PACS after 30 days, but this period is configurable on a site-by-site basis.

1.2.77. Configuring Password Requirements for Local Users

By default, any local database (as opposed to LDAP) users who attempt to log into Merge PACS, whether the Workstation or the Management Pages, will be locked out after 10 failed attempts. The number of failed attempts allowed before lockout occurs is configurable on a site-by-site basis, however.

In addition, by default the minimum password length required for any local user is set to one character, but this can also be configured on a site-by-site basis.

1.2.78. Configuring Auto Registration

The Merge PACS Viewer includes several tools that allow users to link multiple series together, even from separate studies. By default, these tools can only work if the series have a common geometric frame of reference (typically, within the same study and produced by the same scanner). If your site has a registration server that stores DICOM registration information for studies (even if they do not share a common frame of reference), Merge PACS can be configured to access this information and allow the various tools to work with those studies.

1.2.79.Site-Wide Workstation Preferences – User Configurable

Merge PACS lets administrators configure a number of preferences for the Merge PACS Workstation that will be applied site-wide to all users, including the availability of certain options, how prior studies are determined, the default status and dimension options, etc.

For information on changing any of these preferences, refer to subsection 16.1 below.

NOTE: The ability to change site-wide preferences from within the Merge PACS Workstation can be disabled on a site-by-site basis.

1.2.80.Site-Wide Workstation Preferences – Non-User Configurable

The following site-wide preferences can be set for the Merge PACS Workstation by Merge Healthcare Technical Support:

a. Enabling Auto-Logout

By default, there is no auto-logout period for users of the Merge PACS Workstation. Merge PACS can be configured by Merge Healthcare Technical Support, however, to set a system-wide auto-logout period for all users. Once this feature is enabled, individual users or groups can be given the privilege to override this setting from within the Workstation itself.

CAUTION: If system-wide Auto-Logout is enabled, user credentials will not be saved even if the site is otherwise configured to allow users to save their credentials.

b. Configuring the Behavior of the “Low Memory Warning” Message

The Merge PACS Workstation is designed to display a “Low Memory Warning” message in a status balloon in the Windows system tray and/or a pop-up dialog box under certain circumstances. The behavior of this message is defined by four properties with the following default values:

Property	Default Value	Description
Amicas.Viewer.UsageThresholdPercentage	85	When memory usage exceeds this percentage of available memory, a garbage collection will be performed to try and reclaim memory.
Amicas.Viewer.CollectionThresholdPercentage	80	If memory usage still exceeds this percentage of available memory after a garbage collection, user will be warned of low memory situation.

Property	Default Value	Description
Amicas.Viewer.MemoryWarningInterval	300	After the user has been warned of a low memory situation, they will be warned again after this number of seconds if the memory usage is still above the Collection Threshold Percentage and will continue to be warned repeatedly on this interval until memory usage falls below the Collection Threshold Percentage.
Amicas.Viewer.DisableLowMemoryStatusMessage	false	If set to true, the status balloon in the Windows system tray will not be shown (<i>i.e.</i> , only the dialog box will appear).

Merge PACS can be configured by Merge Healthcare Technical Support, however, to change one or more of these properties for a specific site.

c. Changing the Query Page Timeout

As stated in the preceding paragraph, by default users will be prompted to narrow their search criteria if their query returns more than the specified number or if the query takes more than 10 seconds to run.

It is strongly recommended that this timeout period not be changed, due to performance concerns, but if necessary Merge PACS can be configured by Merge Healthcare Technical Support to change the timeout period.

d. Hiding Action Icons on the Query Page

By default, the following Action Icons are available on the Query Page component of the Merge PACS Workstation:

- **Open Study**
- **Open Study in Secondary Viewer**
- **Change Status**
- **Study Tagging**
- **Access Control**
- **VoiceClip**
- **Burn CD**
- **Associate/Dissociate Study**
- **Route Study**
- **Show Comments**
- **Tech WorkPanel**
- **View Reports**
- **Retrieve Comparison Study**

NOTE: If the Merge Archive component is enabled for this Server, the **Retrieve from Archive** action icon will also be displayed by default.

NOTE: The Availability of each of these Action Icons to individual users will depend on their individual login privileges.

Merge PACS can be configured by Merge Healthcare Technical Support, however, to hide one or more of these Action Icons so that they are not available on the Query Page on a site-wide basis.

e. Configuring Query Page Date Range Options

By default, the Date Range pull-down menu on the Query Page component of the Merge PACS Workstation contains the following options:

- Last 3 Hours
- Last 3 Days
- Last 30 Days
- Last 12 Months
- This Month
- Last Year
- Last 12 Hours
- Last 7 Days
- Last 60 Days
- Today
- Last Month
- Custom
- Last 24 Hours
- Last 15 Days
- Last 6 Months
- Yesterday
- This Year
- No Restriction

Merge PACS can be configured by Merge Healthcare Technical Support, however, to remove one or more of these options and/or add additional numbered day ranges (e.g., Last 45 Days) on a site-wide basis.

f. Configuring Query Page Study Criteria Search Options

By default, the Query Page component of the Merge PACS Workstation has the following Study Criteria search options available for users to select:

- DOB
- MRN
- Patient Name
- Patient Code
- Patient Age
- Accession Number
- Study Description
- Physician
- Station Name
- Modality
- IPID

Merge PACS can be configured by Merge Healthcare Technical Support, however, to remove one or more of these options on a site-wide basis. In addition, a site can define up to 10 custom query fields based on the attributes in DICOM headers, as described in subsection 9.13.18 below, and choose to display one or more of them.

The list of available options determines which options are available to users on a site-wide basis. Except as noted below, options that are configured to be available will be visible (*i.e.*, displayed on the Query Page) by default for individual users, but individual users can customize their Workstations to further reduce the number of options displayed. They cannot, however, add any options that are not configured to be in the list of available options.

NOTE: The **Date of Birth** search criteria option will be **available** (added to the list of available options) and **visible** to users on the Query Page by default for new installs of Merge PACS. However, for upgrades from a previous version of Merge PACS that did not already have it configured to be available (*i.e.*, pre-8.2 versions), the option will not be available by default and must be manually added here by Merge Healthcare Technical Support.

NOTE: The **Patient Age** search criteria option will not be **visible** (displayed on the Query Page) by default for individual users with new installs of Merge PACS even though it will be configured to be **available**. This means that users will need to manually enable it from within the Workstation if they want to have that field displayed.

NOTE: In addition to the study criteria described above, individual status dimensions can be configured to be available as query filter options as described in subsection 9.13.7 below.

g. Configuring Query Page Modality Search Criteria Values

If the Modality Study Criteria search option has been enabled, as described in the preceding paragraph, it will have the following default values available to the user:

- | | | | | | |
|-----------|----------|------------|------------|------|----------|
| • AU | • BI | • CD | • CR | • CT | • DD |
| • DG | • DX | • ECG | • ES | • GM | • HC |
| • IO | • LS | • MG | • MR | • NM | • OT |
| • PR | • PT | • PX | • RF | • RG | • RTDOSE |
| • RTIMAGE | • RTPLAN | • RTRECORD | • RTSTRUCT | • SM | • ST |
| • TG | • US | • XA | • XC | | |

Merge PACS can be configured by Merge Healthcare Technical Support, however, to remove one or more of these options on a site-wide basis.

NOTE: This determines which options are available to users on a site-wide basis. Individual users can customize their Workstations to further reduce the number of options displayed, but they cannot add any that are not included here.

h. Configuring Query Page Search Criteria Case-Sensitivity

By default, the following fields are **case-sensitive** on the Query Page:

- | | |
|---|--------------------------------|
| • Patient Age (with regard to units) | • Accession Number |
| • Study Description | • Station Name |
| • Accreditation | • Notable Structure |
| • Syndrome/Abnormality | • Secondary Description |
| • Use/Conference | • Other |

Merge PACS can be configured by Merge Healthcare Technical Support, however, to turn off case-sensitivity a site-wide basis.

i. Configuring Default Teaching Worklist Action Icons

By default, newly created Teaching Worklists (including Briefcases) will have the following action icons available:

- | | |
|---------------------------------------|--------------------------------|
| • Open Study in Primary Viewer | • Data Masking |
| • Report Viewer | • Retrieve from Archive |
| • CD Burn | • View Comments |
| • Study Tagging | • QC Editor |
| • Route Study | • Activity Log |

Merge PACS can be configured by Merge Healthcare Technical Support, however, add or remove action icons from this list.

NOTE: This only applies to newly created Teaching Worklists. Existing Teaching Worklists can be manually configured on an individual basis as described in Chapter 13 below.

j. Configuring the Teaching Worklist Double-click Action

By default, double-clicking on a Teaching Worklist entry within the Workstation Browser will cause the selected study to be loaded into the Merge PACS Viewer. Merge PACS can be configured by Merge Healthcare Technical Support, however, to perform any other available action upon double-clicking.

k. Configuring Date and Time Format

By default, dates and time are displayed within the Merge PACS Workstation in the following formats:

- Date = MM/dd/yyyy
- Short Time Display = HH:mm
- Long Time Display = HH:mm:ss

If desired, however, Merge PACS can be configured by Merge Healthcare Technical Support to use different date and time formats.

l. Disabling Drag and Drop RealTime Study List Functionality

By default, the RealTime Study List includes the ability to quickly copy studies from one worklist (whether a RTWL worklist, a Teaching Worklist or a Query Worklist) to another by dragging and dropping within the RealTime Study List.

Merge PACS can be configured by Merge Healthcare Technical Support, however, to disable this feature on a site-wide basis.

m. Configuring Availability of RealTime Study List

By default, RealTime Study List is made available to users with Merge PACS Servers that are installed with the OrthoPACS profile. Users can then select via their personal Viewer preferences whether to have RealTime Study List or RealTime Worklist displayed.

Merge PACS can be configured by Merge Healthcare Technical Support, however, to either disable this feature on Servers that were installed with the OrthoPACS profile or else to enable it for Servers without the OrthoPACS profile.

n. Configuring Display Columns and Action Icons for Query Worklists

Query Worklists are Query searches that can be saved and accessed from within the RealTime Study List. By default, the following **display columns** are available on Query Worklists:

- | | | | |
|------------------------|------------------------------|---------------------------|--------------------|
| • Diagnostic | • Patient ID | • Modality | • Locked By |
| • Patient Name | • IPID | • Institution Name | • Due In |
| • Date of Birth | • Date – Time | • Priors | |
| • Age / Sex | • Referring Physician | • Images | |

NOTE: These default settings can be overridden on a site-wide basis from within the Merge PACS Workstation, as described in subsection 16.2 below and can also be overridden by individual users as part of their user-saved preferences.

By default, the following **action icons** are available on Query Worklists:

- **Open Study in Primary Viewer**
- **Open Study in Secondary Viewer**
- **CD Burn**
- **Quick Add to Default Teaching Worklist**
- **OrthoLink**
- **Open Study in iConnect Access**
- **View Comments**
- **Save Series**
- **QC Editor**

If desired, however, Merge PACS can be configured by Merge Healthcare Technical Support to add or remove columns or action icons.

o. Configuring RealTime Study List Date Range Options

By default, the Date Range drop-down menu on the Query section of the RealTime Study List contains the following options:

- **Last 3 Hours**
- **Last 3 Days**
- **Last 30 Days**
- **Last 12 Months**
- **This Month**
- **Last Year**
- **Last 12 Hours**
- **Last 7 Days**
- **Last 60 Days**
- **Today**
- **Last Month**
- **Custom**
- **Last 24 Hours**
- **Last 15 Days**
- **Last 6 Months**
- **Yesterday**
- **This Year**
- **No Restriction**

If desired, however, Merge PACS can be configured by Merge Healthcare Technical Support to remove one or more of these options and/or add additional numbered day ranges (e.g., Last 45 Days) on a site-wide basis.

p. Configuring Available RealTime Study List Search Criteria Options

By default, the following Study Criteria search options are available for use at the Quick Filters section of the RealTime Study:

- **Date Range**
- **MRN**
- **IPID**
- **Patient Name**
- **Accession Number**
- **Study Description**
- **Physician**
- **Modality**
- **Facility**
- **Location**
- **Patient Class**
- **Order Status**
- **Specialty**
- **Order Priority**
- **Region**
- **Due In**
- **Date of Birth**
- **Station Name**
- **QC Search Criteria**
- **Custom Tags 1-10**

NOTE: Up to ten custom tags (based on the attributes in DICOM headers) can be defined on a site-by-site basis, as described in subsection 9.13.18 below.

NOTE: In addition to the study criteria described above, individual status dimensions can be configured to be available as query filter options as described in subsection 9.13.7 below.

Based on this master list of available options, a site can configure which options will be visible by default, as described in paragraph q below. In addition, individual users can customize their Workstations to display or hide any of these options but cannot choose to display any that are not listed here.

The master list of available search criteria options can be configured on a site-by-site basis by Merge Healthcare Support to remove available options or to add additional available options from the list of supported criteria.

q. Configuring RealTime Study List Search Criteria Visibility

Users can configure their Workstations to specify which search criteria options should appear on the RealTime Study List “Quick Filters” section. Upon login to the Workstation, if a user has not already custom-configured which criteria should be visible, a default set of criteria will be displayed. This default set of criteria can be changed on a site-by-site basis as needed.

NOTE: The default list of visible search criteria can only contain search criteria that are available for the site as a whole, as defined in paragraph p above.

r. Configuring RealTime Study List Modality Search Criteria Values

If the Modality Study Criteria search option has been enabled, as described in the preceding paragraph, it will have the following default values available to the user:

- | | | | | | |
|-----------|----------|------------|------------|------|----------|
| • AU | • BI | • CD | • CR | • CT | • DD |
| • DG | • DX | • ECG | • ES | • GM | • HC |
| • IO | • LS | • MG | • MR | • NM | • OT |
| • PR | • PT | • PX | • RF | • RG | • RTDOSE |
| • RTIMAGE | • RTPLAN | • RTRECORD | • RTSTRUCT | • SM | • ST |
| • TG | • US | • XA | • XC | | |

If desired, however, Merge PACS can be configured by Merge Healthcare Technical Support to remove one or more of these options on a site-wide basis.

NOTE: This determines which options are available to users on a site-wide basis. Individual users can customize their Workstations to further reduce the number of options displayed, but they cannot add any that are not included here.

s. Disabling Display of Worklist Count in RealTime Study List

By default, worklist counts are displayed within RealTime Study List. This adds additional traffic between the Workstation and the Server, however, and can be disabled if performance is adversely affected.

t. Configuring Availability of the Briefcase Feature

The OrthoPACS profile includes by default the “Briefcase” feature, which creates a default Teaching Worklist for every user who logs into the Merge PACS Workstation. Merge PACS can be configured by Merge Healthcare Technical Support, however, to either disable this feature on Servers that were installed with the OrthoPACS profile or else to enable it for Servers without the OrthoPACS profile.

u. Configuring Local Study Import

The Local Study Import feature of the Merge PACS Workstation can be configured in the following ways on a site-wise basis by Merge Healthcare Support:

- Users that have the **Allow User to Import Local Studies** privilege, as described in Section 9.11 below, will have access to a special **Import Selected** button on the Local Study page of the Merge PACS Workstation. The availability of this button can be disabled for an entire site, however, so that no user can import local studies regardless of their privileges.
- A special **MRN Prefix** can be defined that, if selected by the user when importing a study, will be appended to the MRN of the local study. The availability of this prefix can also be disabled entirely if so desired.
- A special **Accession Prefix** can be defined that, if selected by the user when importing a study, will be appended to the Accession Number of the local study. The availability of this prefix can also be disabled entirely if so desired.
- By default, the Local Study Import feature displays and queries for information regarding **IPID**. If desired, all IPID information can be hidden within Local Study Import on a site-wide basis for sites that do not use IPID.
- If desired, all **Accession** information displayed and queried for at Local Study Import can be hidden on a site-wide basis. If Accession information is set to not be displayed, the “Ensure Unique Accession” option will be enabled by default and hidden from view.

v. Enabling Automatic CD Search for Local Study

By default, if the OrthoPACS profile has been installed, the Workstation will automatically search the attached CD drive, if any, when the Local Study page is first accessed. Merge PACS can be configured by Merge Healthcare Technical Support, however, to either disable this feature on Servers that were installed with the OrthoPACS profile or else to enable it for Servers without the OrthoPACS profile.

w. Configuring the Query Worklist Double-click Action

By default, if the OrthoPACS profile has been installed, double-clicking on a Query Worklist entry within the RealTime Study List will cause the selected study to be loaded into the iConnect Access Viewer. Merge PACS can be configured by Merge Healthcare Technical Support, however, to perform any other available action upon double-clicking.

x. Configuring Global Color Scheme for the Workstation Browser

For standard Merge PACS installations, the Merge PACS Workstations is displayed with a “light on dark” color scheme by default. For Merge PACS installations with the OrthoPACS profile, the Merge PACS Workstations is displayed with a “dark on light” color scheme by default. In either case, the default scheme can be changed on a site-wide basis by Merge Healthcare Technical Support.

y. Configuring Behavior of Worklist “Due In” Column

If a site’s workflow includes deadline tracking due to **Service Level Agreements** (SLAs) and uses the “Due In” column within worklists to display SLA information, the following properties can be custom configured on a site-by-site basis:

Property	Description
Due In Time	Provides a list of available options for the Due In drop-down menu that can be used on the Quick Filters panel of the RTSL as well as the Due In filter used when defining worklists.
Study Timeout	Defines the Inactivity Timeout (in seconds) that is added to the time the last image for a study is received to calculate the “Arrived” time displayed in the Due In column hover text. Default is 120 seconds (2 minutes).
Study Completion Status	Defines the minimum status that causes a study to be considered “completed” when transitioned to that status or beyond. The Due In column hover text will display the time the study transitioned to or beyond the defined Study Completion Status.

NOTE: “Due In” will only apply to studies arriving into the system following the upgrade to Merge PACS 6.5 or higher.

z. Configuring the Mammography Priors Stacking Depth

By default, if the **Automatically Stack/Scroll MG Priors** modality preference is selected within the Workstation, a maximum of four prior Mammography studies will be stacked for every primary Mammography study being viewed. The maximum number of stacked MG priors can be configured on a site-by-site basis, however, to meet the needs of the site.

NOTE: The stacking depth for MG priors should be no larger than the cache depth for priors in general.

aa. Configuring Base Status for Quick Filter Study Status Exclusion

The Quick Filters section of the RealTime Study List includes a **Study Status** field that lets users filter a worklist based on a specific status (Read|Unread|Unverified or their equivalents). In addition, the Study Status field includes the following options with regard to whether an exam has an order or study associated with it:

Option	General Description
Any	Include exams with all statuses, including those without orders or studies associated with them.
Include Orders	Only include exams with orders associated with them but no studies.
Include Studies	Only includes exams with studies associated with them, whether or not there are also orders.

These options are determined by the statuses that indicate when an order and study has arrived for a particular study. By default, the “Include Orders” option will only include exams that have a study status less than Unverified (UV) and the “Include Studies” will include exams that have a study status equal to or greater than Unverified (UV). Depending on the workflow of a particular customer site, however, the base status of Unverified (UV) can be changed to a different status as needed.

bb. Configuring How User Names are Displayed

By default, user names are displayed in the format “[FirstName] [LastName] [Prefix]” within RTWL and RTSL in such places as the “Assigned To” and “Reserved By” data columns. This can be changed on a site-by-site basis if desired, however, to display different or additional name components and display them in a different order (e.g., [LastName] [First Name] [Degree]).

cc. Configuring How Briefcase Names are Displayed within RTSL

By default, briefcases that are created automatically for users for use with the RealTime Study list are displayed in the format “[FirstName] [LastName]’s Briefcase” (e.g., “John Smith’s Briefcase”). This can be changed on a site-by-site basis if desired, however.

NOTE: Changing this server property will not affect existing briefcase names. It only affects the names of briefcases created after the property value is changed.

dd. Configuring Delay after Worklist Navigation Operations

By default, when a user clicks on the “Open Next Study” or “Open Previous Study” or “Get Next Study and Mark current as Read” button within the Merge PACS Viewer, the operation is performed immediately. Merge PACS can be configured, however, to wait a specified amount of time before loading the next study when there are no other studies currently available in the Worklist to read. Once the delay is completed, if the study has arrived in the worklist it will load; otherwise, the worklist will be displayed.

ee. Configuring Header Text for Comments Viewer Printing

By default, when users print information from within the Comments Viewer, header text reading, “Merge PACS Comments” is printed at the top of each page. If desired, however, this header text can be customized on a site-by-site basis.

ff. Configuring Optional Information in Order and Comments Viewers

By default, the Order Viewer and Comments Viewer include **Allergies** and **Diagnosis** panels, even if there is no such information available. In addition, the “General Patient Information” section of the Order Viewer includes a **Pregnancy Status** indicator even when no pregnancy status information is available. This information is received directly from third-party Electronic Medical Record (EMR)/Ordering Systems, if Merge PACS is so configured.

If a site is not configured to integrate with an EMR, this information can be disabled on a site-wide basis so that it never appears within the Order and Comments Viewers.

gg. Configuring Maximum Number of Automatic Viewer Windows for Studies Retrieved from Archive

If Merge PACS is configured with Merge Archive or iConnect® Enterprise Archive, an archive retrieval request will automatically be triggered when a user attempts to open an offline study in the Primary or Secondary Viewer. Once the retrieval process has finished, the study will automatically be loaded into the selected Viewer as long as no other study was opened during the retrieval process.

By default, retrieved studies can be loaded into a maximum of four simultaneous Viewer instances, including the Primary Viewer and three Secondary Viewers, and if a user attempts to retrieve an archived study when there are already four Viewer instances running, the fifth study will be retrieved in the background but will not open automatically.

If desired, the maximum number of simultaneous Viewer instances that can be available for automatic study opening can be changed on a site-wide basis. Automatic opening for Secondary Viewers can also be disabled entirely.

hh. Configuring Polling Preferences for Availability Status

If Merge PACS is configured with Merge Archive or iConnect® Enterprise Archive, the Workstation can be configured to show an Availability Status icon that will change to indicate the current availability of studies.

By default, the Workstation will poll the Server to check the availability of a study once per second after an archive retrieval request has been submitted by a user, but this can be changed on a site-wide basis if desired. Note that it is not recommended to set this to a smaller value, due to potential performance issues, but it may be desirable to set this to a higher value if your site has hundreds of Workstations running at the same time. Doing so will cause the Availability Status icon to be updated less frequently as a study is being retrieved from the Archive, which may lessen the performance load on the Server.

In addition, by default the Workstation will stop polling the Server after three minutes have elapsed with no change in status, and this can also be changed on a site-wide basis if desired. In particular, it may be desirable to increase the timeout period for sites where it routinely takes several minutes for the archive to process a retrieval request, but this is not recommended unless absolutely necessary.

ii. Configuring Blinded Reading Preferences

By default, all users and groups within Merge PACS can view patient, series and study-specific information for the studies that they are accessing within the Image Viewer. As described in Section 9.11 below, however, users and groups can be configured without the default **“allow access to patient/study/series information”** Image Viewer privilege. Without this privilege, the users will not be able to view patient, series and study-specific information in the following locations within the Merge PACS Workstation:

- **The Viewer Titlebar**
- **All DICOM Overlays** (including the DICOM Attributes Viewer)
- **Series Descriptions** (both within navigation thumbnails and individual Viewports).

In addition, access to the DICOM Attributes Viewer, the Order Viewer and the Report Viewer will all be disabled from within the Viewer.

The primary use case for this is to provide for “blinded reading” for use with clinical trials where a user may be required to review/read a study without seeing the identifying information for that study/patient.

By default, the following information will be hidden for users who do not have this privilege enabled:

- | | |
|-------------------------------------|---------------------------------------|
| • Patient Name | • Manufacturer |
| • Patient ID | • Series Description |
| • Issuer of Patient ID | • Frame of Reference UID |
| • Patient Birth Date | • Position Reference Indicator |
| • Patient's Size | • Patient Position |
| • Patient's Weight | • Series Number |
| • Patient Sex | • Study Instance UID |
| • Patient's Age | • Series Instance UID |
| • Referring Physician's Name | • Study Date |
| • Reading Physician Name | • Series Date |
| • Accession Number | • Study Time |
| • Station Name | • Series Time |
| • Institution Name | • Study Description |
| • Modality | |

The list of attributes to be hidden can be changed on a site-wide basis, however.

CAUTION: Refer to Appendix A.1.3 below for important notes and cautions regarding the use of Blinded Reading.

NOTE: Removing the “allow access to patient/study/series information” privilege can be done in conjunction with removing the “allow user to access other series/studies” privilege. The latter is typically disabled in situations where users access studies directly from a URL instead of via the RTWL, Query Page, etc., and you want to prevent them from accessing any other series or studies via the Patient Record, Load Comparison Study option, etc.

jj. Configuring the Minimum Slice Limit for Automatic MR Splitting

If the "Automatically Split MR Series" user preference is selected within the Workstation, the Viewer looks first to each image's echo number and then, for images with no echo number, to each image's Temporal Position Identifier (TPI) value in order to group the images into subseries.

Images will be eligible for splitting based on TPI value if, among other factors, the number of images containing the same TPI value is greater than the Minimum Slice Limit defined for Merge PACS. By default, the Minimum Slice Limit is set to 8, but this can be changed on a site-wide basis.

For example, if the Minimum Slice Limit is set to 10 and there are only 6 images within a series with a particular TPI value, those images will not be grouped together as a separate series based on TPI value.

NOTE: Images that do not have a TPI value or which do not meet the criteria for grouping based on TPI value can still be grouped into subseries based on Positional Sequence or Stack Axis/Positional Break.

kk. Configuring Full-Page Printing

When images are printed from within the Viewer by selecting **Print** from the Series Right-click Menu, by default the aspect ratio of the image will be preserved, even if that causes white space to be included on the page. If desired, however, Merge PACS can be configured on a site-wide basis so that printed images will automatically be zoomed and cropped to fill the entire page, as in the following example:



Figure 3: Normal Printing



Figure 4: Full-Page Printing

ll. Configuring Continuous Drawing Mode (Superimposed Measurements)

Continuous drawing mode allows users to superimpose one measurement on top of an existing measurement by clicking on the yellow line of the existing annotation, holding the mouse button down until the cursor changes to the selected measurement tool and then using the annotation tool as normal to draw the new measurement.

By default, users need to hold the mouse button down for half a second (500 milliseconds) in order to activate continuous drawing mode and they cannot move the cursor more than three pixels in any direction during that period. Both the time required to activate and the movement tolerance can be configured on a site-by-site basis, however.

mm. Changing the Value for Data Masked IPID

By default, when a user chooses to data mask (“anonymize”) a study on a Teaching Worklist, the value the Issuer of Patient ID (IPID) associated with that study will be changed to “ANONIPID”. If desired, however, Merge PACS can be configured so that a different value is used instead of ANONIPID.

NOTE: The configured value will apply to all data masked studies going forward and cannot be applied selectively to specific studies.

nn. Configuring Workflow Status Dimensions as Prior Search Filters

If desired, Merge PACS can be configured so that one or more Workflow Status Dimensions (and their associated Workflow Statuses) are presented as filtering options for determining prior studies at the Site Preferences menu within the Merge PACS Workstation, as described in Appendix A.3.2 below. The primary workflow for this is when Merge PACS is configured to run in **Integrated** mode and the attached iCEA Server has been set up with Study Types and Study Type Groups that have been automatically mapped to Worklist Status options and Worklist Status dimensions.

oo. Enabling Availability of a Workflow Status “None” Option for Query Page and RTSL

When Merge PACS is configured to run in **Integrated** mode and the attached iCEA Server can be set up to tag studies according to Study Type (e.g., Cardiology, Radiology, Neurology, etc.) with the study type groups automatically mapped to default workflow status dimensions, Merge PACS can also be configured to include a “None” option for these workflow dimensions on both the Query Page and the Quick Filter panel of the RealTime Study List.

pp. Changing the Default Timeout Period for Querying Priors

When users first attempt to open a study within the Merge PACS Viewer, the system will begin the process of identifying all prior/comparison studies for the selected study so that they will all be available as needed and so that the correct Hanging Protocol, if applicable, is applied. If the system is not able to identify all the priors within a configurable time period (typically three seconds), a warning message will be displayed that will give the user an option to load the primary study without further waiting for the priors to be identified. This timeout period can be custom configured on a site-by-site basis, however.

qq. Configuring Report Icon Tool-tip Text

By default, when a users hover his or her mouse over the “Launch Report Viewer” icon for an exam anywhere within the Workstation, summary information for the latest report associated with the exam will be displayed as a pop-up tool-tip as long as the report is an HTML-formatted HL7 report. Also by default, the summary information is defined as any text between the text strings “Impression” and “End Impression” in the report, and any report that has the word “Addendum” in it will be ignored. Finally, by default the searches to determine the summary information and whether or not a report is an addendum will **not** be case-sensitive.

If desired, however, this feature can be disabled for a site and it is also possible to configure how the summary information is determined, whether addenda should be ignored and whether searches should be case-sensitive.

1.3. Component Overview

Merge PACS comprises a group of interrelated components. In general, Merge PACS integrates the Merge PACS Server and Client components with additional software components from other companies. Note that Merge PACS can either be configured to run in **Integrated** mode (*i.e.*, using Merge iConnect® Enterprise Archive for image storage) or **Standalone** mode (*i.e.*, using Merge PACS for image storage). This is set during initial installation and cannot be changed afterwards.

1.3.1. Standalone Mode

a. Merge PACS Server Components

When running in **Standalone** mode, the Merge PACS Server includes the following Merge PACS components, some of which are optional:

Component	General Description
Amicas Service	Service that launches the background processes mentioned below
Amicas Web Service	Service that handles the distribution of non-image data via http and also handles Merge PACS's internal web security
Amicas Messaging Service	Service that handles communications via JMS between the various Merge PACS components. Includes the Auditor process that monitors study events (primarily the status of studies being imported into Merge PACS) and marks them as being closed [<i>i.e.</i> , successfully imported] or closed with errors
StorageServer	Process that handles communications between Merge PACS and DICOM devices
Importer	Process that compresses the DICOM images received by the StorageServer
AutoRouter	Process that uses customer-defined routing rules to determine whether an image needs to be routed
Forwarder	Process that forwards DICOM images received by the StorageServer to other DICOM devices
CommandScheduler	Process that executes periodic system maintenance commands and user-generated explicit ones. Also sends and receives images via Telemedicine and sends messages via the Outbound HL7 Interface.
ImageManager	Process that serves as the system's inbound interface with HL7 systems
Listener	Process that enables user requested query and retrieval of studies from other DICOM devices
WorklistServer	Process that keeps track of studies and orders, responds to search requests from the Worklist client and notifies clients about changes in studies and orders
DICOMPrint	Process that monitors incoming DICOM Print requests, and submits the jobs to the configured DICOM printers
ECDR Manager	Process that monitors the CD Queue for CD writing requests and then composes and sends CD writing jobs studies to the shared CD writing device

Component	General Description
AmicasWatch Service	Process running on a Watch Monitoring Station that receives data sent by Merge PACS and sends out alert e-mails when defined thresholds have been met
Merge PACS file system	The various locations on disk that store Merge PACS, related software, and both uncompressed and compressed images
Archiver	Process that bundles compressed and/or non-compressed images and copies them to an archival storage device, as well as retrieves those images
IMServer	Process that provide instant messaging functionality to clients

b. Merge PACS Client Component

The Merge PACS Workstation is a Full-featured radiology workstation that uses lossless and/or lossy image compression and is designed as a tool for primary interpretation of patient images. Consists of the following main components:

- **RealTime Worklist/RealTime Study List**
- **Patient Record**
- **Merge PACS Viewer**

c. Third-Party Components

Merge PACS also incorporates the following third-party software components:

Component	General Description
Database	An IBM <i>DB2</i> database that stores all demographic, configuration, and queue information.
HTTP Server	Apache server serving as the Internet/Intranet interface and providing enterprise-wide access to medical images over the Intranet/Internet. Note that, by default, Apache uses port 80 (443 if TLS is being used), which may conflict with other third-party components such as HL7 systems.
JRE	Oracle's <i>Java Runtime Environment</i> , providing the necessary libraries and components for Java programs.
WildFly	An open source J2EE application server that is used to relay worklist and event logging messages among the various Merge PACS components.

d. General Data Flow Mapping

The following diagram provides a detailed view of data flow mapping when merge PACS is configured to run in Standalone Mode:

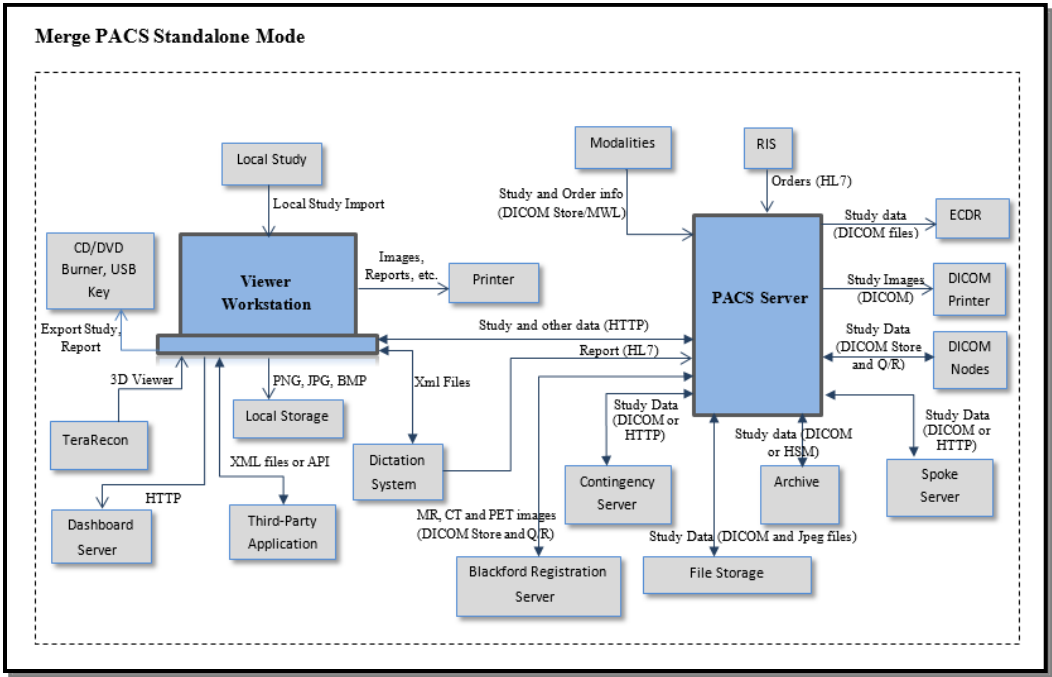


Figure 5: Merge PACS Data Flow Mapping – Standalone Mode

1.3.2. Integrated Mode

As an alternative to the architecture described in subsection 1.3.1 above, Merge PACS can operate in an Integrated mode with Merge's iConnect® Enterprise Archive (iCEA). When operating an Integrated Mode, iCEA acts as the DICOM store and removes the need for image storage (Original, Uncompressed and Compressed) on the Merge PACS server.

NOTE: In Integrated mode, if iCEA is the final archive, no watermarking is used (*i.e.*, all images will be immediately available from the archive). When used with an external Enterprise Archive or a 3rd-party vender neutral archive, watermarking of the cache can be enabled on iCEA.

a. Merge PACS Server Components

Since the DICOM instances are kept on iCEA in Integrated Mode, the following processes described in subsection 1.3.1 above are not needed and will not be run in Integrated Mode:

Component	Reason Why Not Needed
StorageServer	DICOM instances are no longer sent to Merge PACS and therefore this component is not needed. However, since Storage Server provides all of the DICOM functionality for the Merge PACS server and since Modality Worklist (MWL) can still be run from the Merge PACS server, a stripped down version of the Storage Server called the MWL Query Service is used in its place. This component provides MWL services but no other DICOM services.

Component	Reason Why Not Needed
Importer	Since there are no DICOM instances stored on the Merge PACS server, the Importer is not required.
AutoRouter	If routing is required it will be set up on iCEA.
Forwarder	If forwarding is required it will be set up on iCEA.
Archiver	If archiving is required it will be set up on iCEA.

However, to provide the integration with iCEA, the following additional components are needed on the Merge PACS server in Integrated Mode:

Component	General Description
iCEA Notification Processor	Receives notifications from iCEA about new or changed DICOM instances available in the archive as well as for quality control (QC) messages indicating various changes to the DICOM instances already available (e.g. demographic changes, splits, merges, deletes).
Sender	Forwards DICOM instances received at the Merge PACS server on to iCEA for storage. DICOM instances will still arrive at the Merge PACS server for items created by the viewer (e.g. KO, PR, SC) as well as those sent via telemedicine transfer from other Merge PACS servers. Since Merge PACS does not store DICOM instances in integrated mode it will store these instances temporarily in a cache and the Sender will send them on to iCEA, deleting them from the cache once they have been accepted.

b. Merge PACS Client Components

In Integrated Mode, the Merge PACS Viewer retrieves its images from the iCEA Server rather than from the Image Stores via the HTTP Server. It uses secured web services to support secure communication over LAN or the internet.

c. Merge PACS/iCEA Interaction

The following diagram provides a detailed view of the interactions between the Merge PACS and iCEA components in Integrated Mode:

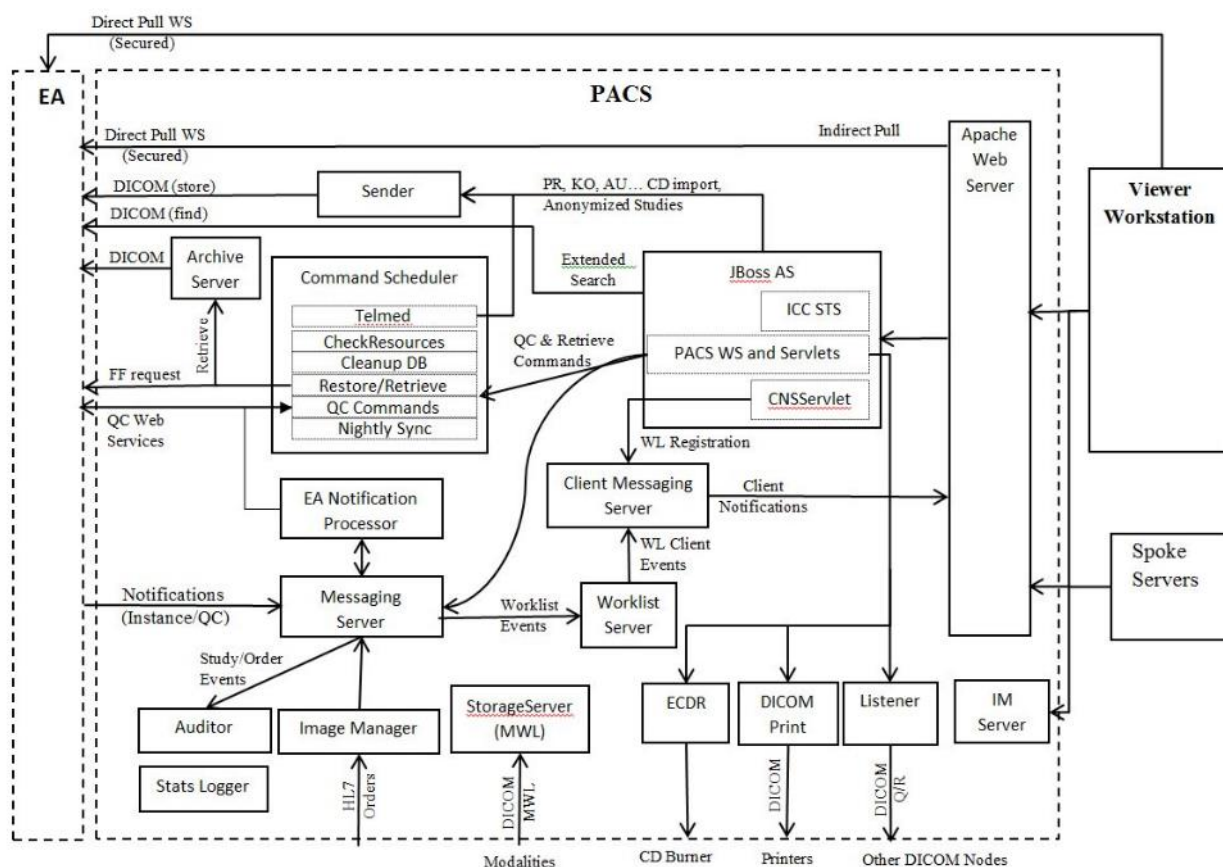


Figure 6: Merge PACS and iCEA Component Integration

d. General Data Flow Mapping

The following diagram provides a detailed view of data flow mapping when Merge PACS is configured to run in Integrated Mode:

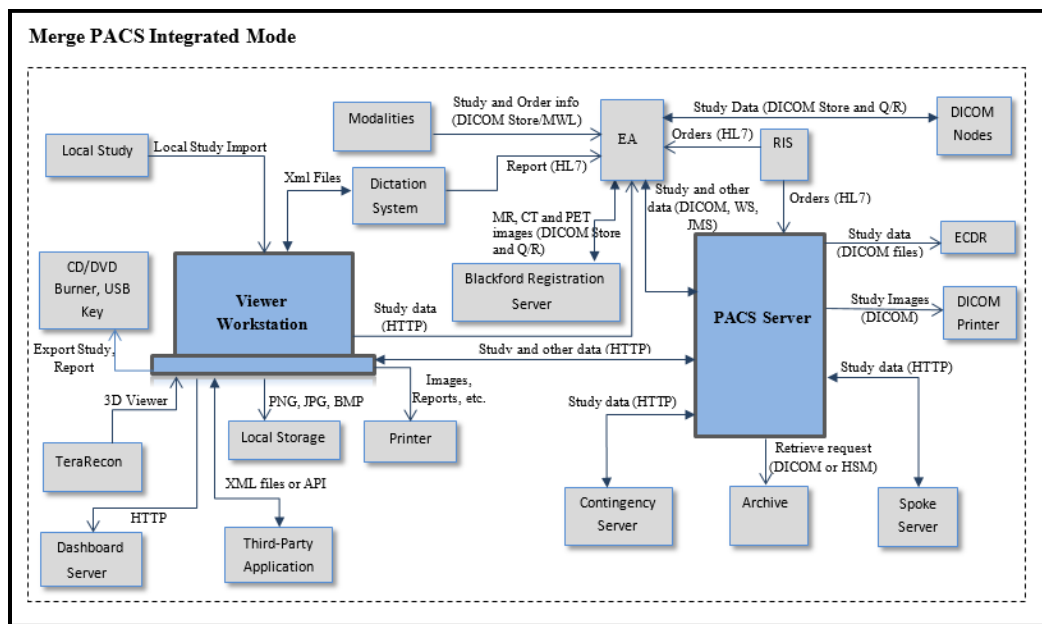


Figure 7: Merge PACS Data Flow Mapping – Integrated Mode

1.4. Contingency Server Failover Environment

In order to provide ongoing ability to access patient images in the event of a hardware failure with the Merge PACS Server, Merge Healthcare Support recommends the use of a backup or “fail-over” server that can be put into service and temporarily take the place of the main production server until the problem is resolved. This takes the form of a “contingency” server that is set up to share the applications, databases and images with the production server.

CAUTION: Failure to employ a contingency server may result in a delay in exams being read.

CAUTION: Failover environments that do not use a Contingency Server (e.g., using a SAN or DAS) are not supported with Merge PACS.

The contingency server failover environment includes the following assumptions:

- The archive configuration is either a Merge PACS Archive Server or an iConnect Enterprise Archive.
 - The limitation existing for a Merge PACS Archive Server solution is that items will not be available during a failover to Contingency.
 - An Enterprise Archive solution would provide DICOM Query/Retrieve Functionality during a failover to Contingency.
- A virtual IP switch such as an F5 Big IP or similar device. Note that these are virtual IP switches and not used as load balancers.

- All DICOM and Web communication (http / https gets) from systems other than between Contingency, Production and Watch, which will be done directly using the systems IP address, will be done through the single IP switch.
- HL7 communication will be sent directly to the two systems as well.
- DICOM and Lossless storage is sized to be able to handle an extended downtime period.

The following diagram illustrates production/contingency failover environment using the local database:

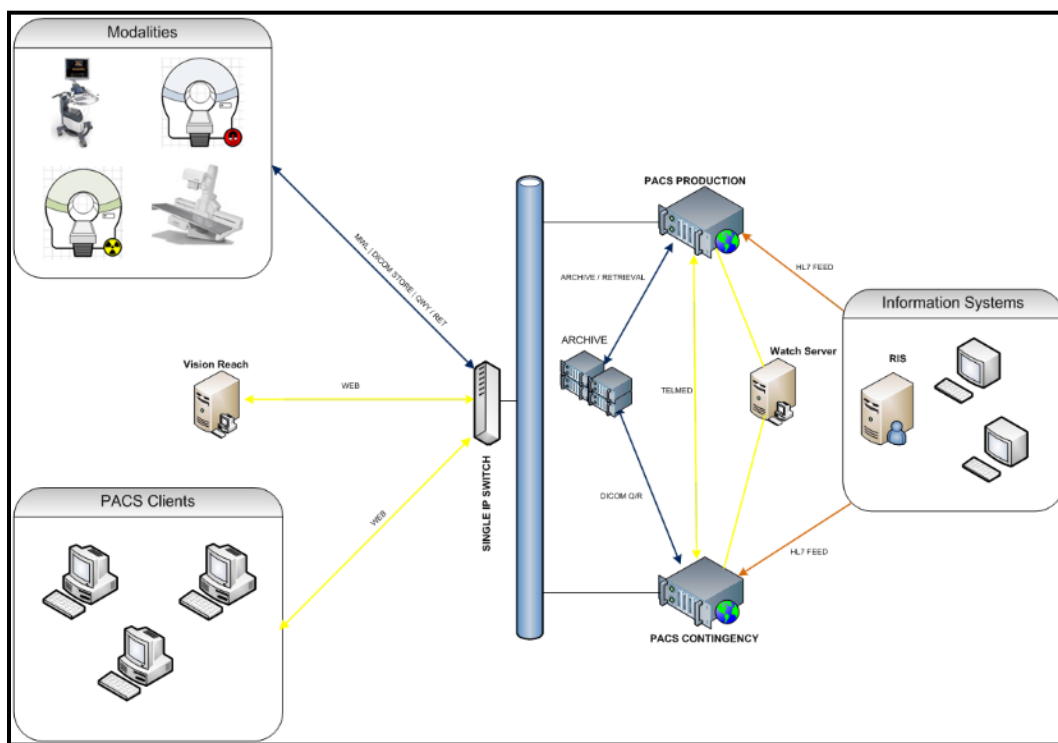


Figure 8: Merge PACS Fail-over in a Contingency Server Environment

Note that, when setting up Merge PACS in **Integrated** mode, the HL7 messages in the following table have to be routed to the appropriate system, using either iCEA's Multicaster or Mirth:

Message	Send to Merge PACS?	Send to EA?
ADT^A01 (Patient Visit)	Production and Contingency	No
ADT^A02 (Patient Transfer)	Production and Contingency	No
ADT^A03 (Patient Discharge)	Production and Contingency	No
ADT^A04 (Patient Registration)	Production and Contingency	No
ADT^A08 (Demographics Update)	Production and Contingency	Production and Contingency
ADT^A11 (Cancel Admission)	Production and Contingency	No
ADT^A12 (Cancel Transfer)	Production and Contingency	No
ADT^A13 (Cancel Discharge)	Production and Contingency	No
ADT^A34 (Merge PID)	No	Production and Contingency

Message	Send to Merge PACS?	Send to EA?
ADT^A40 (Merge)	No	Production and Contingency
ADT^A47 (Change ID)	No	No
ORM^O01 (Order)	Production and Contingency	Production and Contingency
ORU^R01 (Observation)	No	Prod
VQQ^Q01 (Query)	Production	No
ZSU (Link)	Production and Contingency	No

1.5. Backup and Restore Policies

Merge PACS is designed to be integrated into a hospital IT environment. As such, it provides an internal mechanism to make all critical files available for backup using standard third-party backup software (e.g., Veritas). Once these files are placed in their secure location on the server, the facility's backup software should then be used to write them to tape. Note that it is the responsibility of the facility to ensure that their backup software is functioning and that they have tapes properly loaded in tape drives to accommodate the receipt of these backups before Merge PACS goes live at the site.

The back up and restore procedures for Merge PACS involve the following four software components, the first two of which are considered critical:

Component	Description
Merge PACS DB2 databases	These allow access to the actual image data, and are stored in a separate location from the image data. Merge PACS provides an automated procedure that backs up the database nightly to another distinct disk location.
Merge PACS image data	All Merge PACS installations configured to run in Standalone mode require that compressed image data be written to a redundant storage device such as RAID or NAS. This is to prevent the loss of image data even in the event of a disk failure. If Merge PACS has the optional Archive component installed, there will also be a second copy of the image data in its original lossless form on tape.
System files	A backup of system files should also be created as a reference for the current configuration of a Merge PACS Server. Although not critical for a full restore of Merge PACS, having backed up copies of all system files will make the restore process smoother and faster.

There are primarily three cases of failure requiring restoration:

Failure	Description
Corruption of the database	In the case of database corruption, the restore process will allow the user to restore the database from a previously saved working copy. The user can then use the transaction logs to roll forward from the saved database copy. Using this method, with the actual image data secure on its RAID, the user should be able to restore the system, providing image access to studies that are as new as 24 hours old and perhaps newer. All Merge PACS Servers have the capacity to be restored in this manner.

Failure	Description
Server failure	In the case of a sever failure requiring the rebuilding or replacement of the server, it is the responsibility of the site to return the server to the condition it was in at the time of the original Merge PACS installation. This will typically involve restoring backed up system files and/or reinstalling necessary software such as the server's operating system. Once this is done, the server configuration files can be restored. The database can then be restored using the technique mentioned above. This will allow the server to return to normal operation, and access to studies safely stored on their RAID device. All Merge PACS servers have the capacity to be restored in this manner.
Loss of image data	If Merge PACS has the optional Archive component installed, the user will have the ability to restore both operation and access to images in the event of a complete loss of image data from the server (i.e., including the loss of redundant data stored on the RAID device). The loss of image data may or may not be accompanied by the corruption of the database or loss of the Merge PACS Server. If it does, the restore procedures would be as outlined above. In addition, the user would be able to retrieve the lossless copy of the image data that had been written to tape. Any study whose image data had been lost would be eligible to be restored.

The Merge PACS backup and restore procedures are intended to provide the end user with the minimum amount of redundancy needed to provide a quality system. Merge Healthcare Support recognizes that there will be users who require redundancy that exceeds this minimum level. It is the responsibility of the site coordinator to define what level of tolerance is applicable to their deployment. The Merge Healthcare Sales and Project Management teams can then work together to provide a properly designed system addressing the customer's concerns.

Note that, in addition to the DB2 Databases, which are backed up automatically by Merge PACS on a nightly basis, it is strongly recommended that you back up the rest of your system as well for disaster recovery purposes. There are a variety of third party software packages available, most notably Veritas, and you should consult your IT department to decide which software will best suit your needs.

The following items on your server should definitely be included in your general backup procedure:

- The system root directory and the Windows Registry
- The Merge PACS programs and configuration settings located at:
- <DriveLetter>\Amicas2
- The DB2 database backup files: IMGxxx and ANETDB.
- The DB2 Transaction logs

The following items on your server should **NOT** be included in your general backup procedure:

- The active DB2 database directory, located at <DriveLetter>:\DB2
- The active database log directory, located at <DriveLetter>:\DBLog
- The Dicom, compressed, Lossless directories, located at:
 - <DriveLetter(s)>:\PACS2DATA\DICOM
 - <DriveLetter(s)>:\PACS2DATA\COMPRESSED
 - <DriveLetter(s)>:\PACS2DATA\LOSSLESS

NOTE: For server that were upgraded from a version or Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

When backing up your system, keep in mind that it will probably be a lengthy procedure and should be done during off-peak hours whenever possible. Also, be careful that the general backup is scheduled at a time that is subsequent to the time when the DB2 backup should be complete.

1.6. General IT Recommendations

The following are general recommendations for your site's IT staff that will help keep your Merge PACS system running smoothly and minimize the risk of any downtime.

1.6.1. Settings and Conventions

The following is a list of recommended general practices that should be set up and maintained:

- **Avoid unnecessary software** – The Merge PACS Server is meant to be a dedicated server, and should not have any software on it other than the approved Merge PACS software and the various third-party applications used in conjunction with Merge PACS. This prevents problems arising due to incompatible programs and allows all available system resources to be allocated to Merge PACS.
- **Restrict Web browsing** – Use of a Web browser on the Server should be limited to managing the Server via the Merge PACS Administration pages. Using the Web browser for other purposes can consume necessary system resources and can potentially allow viruses and/or potentially incompatible programs to be automatically installed on the server.
- **Restrict access to server** – Log off or lock the server console when not in use. If possible, use a blank screen saver with a password, being sure to avoid graphically intense screen savers that can consume necessary system resources.
- **Emergency repair disk** – Use standard Windows procedures to create an emergency Windows repair disk for the server and keep it updated.
- **Renew TLS Certificates** – If your site requires TLS Communication for access to the Merge PACS Server by the end users, remember to renew TLS certificates before they expire. If you have created your own certificate authority, remember to back up the Certificate Authority's private key and certificate via the Certificate Authority's management screen.
- **Telephone access** – Have a phone installed near the server for support calls.
- **Spare hardware** – Have spare hardware for the server on-site if possible (hard drive, NIC, Monitor, Network switch, cables, etc.)
- **Service contract information** – Keep service contract and contact information in a handy place near the server.
- **Antivirus Software:**
 - Make sure your Merge PACS Server and all Workstations are protected with a full-featured (i.e., “corporate” or “enterprise”) version of an industry-standard antivirus program. Merge Healthcare Support has specifically tested **Computer Associates eTrust Antivirus** for use with Merge PACS; if you wish to use a different program, please check with your Merge PACS Project Manager to make sure it is fully compatible.

- Configure your server's antivirus software to update virus signature files automatically. Set up infected files to be quarantined and not deleted.
- When configuring your antivirus software, certain files and directories should be **excluded** from virus scanning [these files are not likely to be infected, and scanning them on a regular basis may slow down your server dramatically]. These files and directory exclusions should be enabled on both the **Real Time Protection** as well as the **Scheduled Scan**. In addition, be sure to select the "**Check for file exclusion before scanning**" option during the scan/protection setup.

- **Merge PACS Server:**

- File Types:

• .cas • .dcm • .dem • .demo • .errors • .jpg • .jp2
• .log • .nisin • .ren • .sin • .trace • .xml • .zip

- Database and DB2 Directories:

Description of Directory	Typical Location
Active Database directory	<DriveLetter>:\DB2
Active Database Logs directory	<DriveLetter>:\dblog
DB2 Application directory	<DriveLetter>:\SQLLIB
DB2 Instance Home directory	<DriveLetter>:\ProgramData\IBM\DB2COPY1
DB2 Backup directory	<DriveLetter>:\DB-BACKUPS

NOTE: Note that your site may have chosen to use custom locations for the DB2 directories described above during DB2 installation. To verify the locations, run the following command: **db2set -all**.

- Merge PACS Directories:

Description of Directory	Typical Location
Merge PACS Database directory	<DriveLetter>:\AMICAS_DATABASE
Merge PACS Data directory	<DriveLetter>:\PACS2DATA
Merge PACS Process Executables	<DriveLetter>:\amicas2
All DICOM Image Volumes	<DriveLetter>:\PACS2DATA\Dicom
All Lossless Image Volumes	<DriveLetter>:\PACS2DATA\Lossless
All Lossy Image Volumes	<DriveLetter>:\PACS2DATA\Compressed
Rhapsody directory (and Rhapsody database directory if PostgreSQL is used instead of MySQL)	<DriveLetter>:\Rhapsody <DriveLetter>:\Program Files\PostgreSQL

NOTE: For server that were upgraded from a version or Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

- **Merge PACS Workstation:**

- C:\Program Files\Merge Healthcare
- C:\Users\<loginid>\AppData\Local\Merge Healthcare\

- C:\Users\<loginid>\AppData\LocalLow\AMICAS\
- C:\Program files (x86)\Merge Healthcare (64-bit OS only)
- **Original software availability** – Keep all original software media in a safe place that is easily accessible in the event that it needs to be reinstalled. This includes Windows and Merge PACS.

1.6.2. Repeated Activities

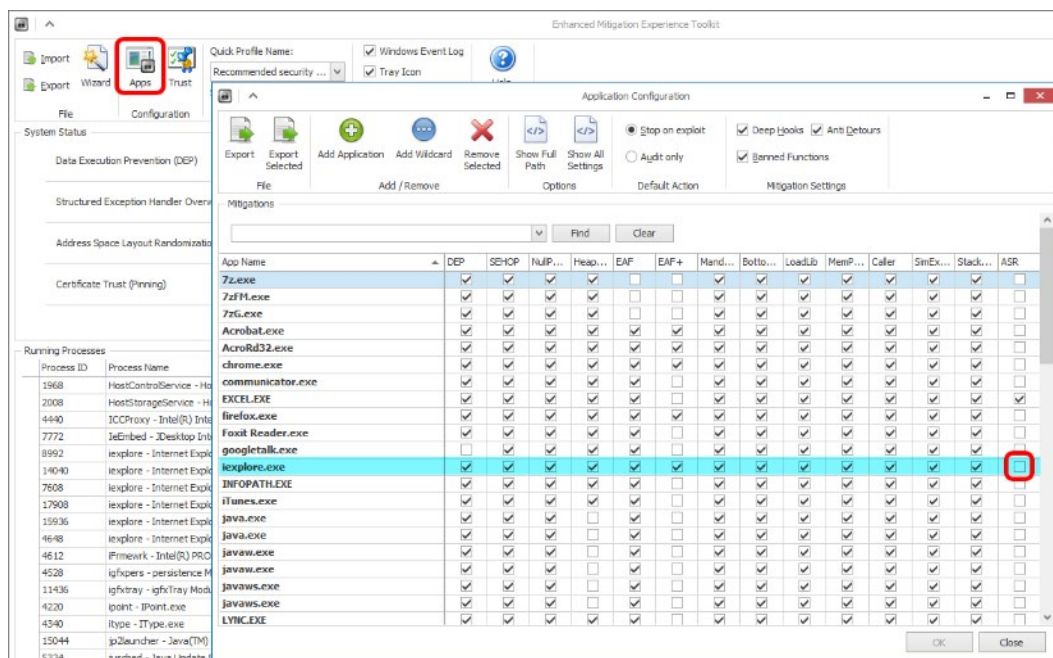
The following is a list of recommended activities that should be performed on a regular basis:

- **Review Windows event logs** – Review the Windows event logs for errors at least once a week.
- **Review Web server logs** – Check Web Server logs frequently for suspicious activity.
- **Check Merge PACS System Monitor** – Check the Merge PACS System Monitor regularly for errors, being sure to contact Merge Healthcare Support with regard to any errors that are displayed in red.
- **Check DICOM Jail** – Use the problem Studies Tool to check the Merge PACS DICOM Jail on a regular basis to monitor any studies that have not been imported correctly.
- **Check Archive tapes** – If using the Merge Archive, be sure to check the archive tapes regularly.
- **Restrict access to administrative passwords** – Review who has administrative access to your server. Do not give the administrative password to many people, and make sure not to forget it.
- **Change Administrative passwords** – Change administrative passwords on a regular basis, and especially after a staff member leaves, using complex passwords with a mix of letters, numbers, uppercase and lowercase. Note that Merge PACS specific user accounts and passwords should NOT be changed at any time, as these are used by Merge Healthcare Support.
- **Monitor free disk space** – Check free disk space at least once a week. If disk space becomes low, delete unnecessary files in the server's "temp" directory [e.g., c:\windows\temp]. If more space is needed, contact Merge Healthcare Support immediately.
- **Service pack and patch updates** – Merge Healthcare Support will send out periodic information with regard to available and compatible service packs and patches for your server's operating system, and you should update your server and workstations accordingly.
- **Document network settings** – Document as much as you can about the server and network settings, including a drawing of the network, and keep all documentation up to date.
- **Monitor UPS** – Test your server's Uninterrupted Power Supply (UPS) batteries monthly and make sure that the UPS is configured to properly shut down the server in the case of a power failure.
- **Turning server off and on** – Post the instructions described in Section 17.3 below on how to properly turn off/on the server near the server console, and update them as needed to keep make sure they're current. Remember to always shut down the server properly; the power/reset buttons should only be used as a last resort.

1.6.3. Enhanced Mitigation Experience Toolkit (EMET) Settings

If your site is using Microsoft's Enhanced Mitigation Experience Toolkit (EMET) as an enterprise security tool, it must be configured to allow Java to run from within Internet Explorer. This involves performing the following steps:

1. Open EMET.
2. Click on the Apps button in the top-left to display the Application Configuration dialog.
3. Ensure that the ASR column is not checked for iexplore.exe, as in the following example:



ASR is NOT Checked for iExplore.exe

4. Click the OK button at the bottom of the Application Configuration dialog if you made a change and then exit EMET.

Chapter 2. Merge PACS Server

2.1. Overview

This chapter describes the main components of the Merge PACS Server:

- Services
- Background processes
- Directories and files
- Multi-Volume Support
- User accounts
- System variables

In addition, this chapter describes the optional Extended Query feature that can be used in conjunction with the Merge PACS Server as well as the various ways Merge PACS can be configured to consolidate patient information..

The following concepts and terminology are discussed in detail in this chapter:

Term	General Description
Amicas service	Service that launches the Merge PACS background processes providing Merge PACS functionality
Amicas Web Service	Service that handles the distribution of non-image data via http and also handles Merge PACS's internal web security
Amicas Messaging Service	Service that handles communications via JMS between the various Merge PACS components. Includes the Auditor process that monitors study events (primarily the status of studies being imported into Merge PACS) and marks them as being closed [<i>i.e.</i> , successfully imported] or closed with errors.
Apache Service	Service that makes image data that was imported and stored on a Merge PACS Server available over the Internet/Intranet
DB2 Services	IBM DB2 services that are required for Merge PACS's database operations
AmicasWatch Service	Service that runs on a Merge PACS Server being used as an AMICAS Watch Monitoring Station, both to receive data and send out alert e-mails
Merge PACS background processes	Background processes launched by the AMICAS service and the AMICAS Messaging Service
Merge PACS application directory	Directory in which Merge PACS application is stored
Merge PACS data directory	Directory in which process logs and miscellaneous Merge PACS data are stored
DB2 application directory	Directory in which the IBM DB2 application is stored
PACS_DATABASE directory	Directory in which the actual data for the IMG database is stored
DICOM images directory	Directory in which uncompressed DICOM images are stored, when Merge PACS is configured to run in Standalone mode

Term	General Description
Lossy images directory	Directory in which JPEG 2000 Lossy compressed images are stored, when Merge PACS is configured to run in Standalone mode
Lossless images directory	Directory in which JPEG 2000 Lossless compressed images are stored, when Merge PACS is configured to run in Standalone mode
External Volume	Directory in which non-image DICOM instances are stored, when Merge PACS is configured to run in Standalone mode
DB2 log directory	Directory in which the active logs for Merge PACS databases are stored
DB2 transaction log directory	Directory in which the inactive (“archived”) database logs are stored
DB2 backup directory	Directory in which the backup of the database created during the Merge PACS installation procedure is stored
Apache directory	Directory in which Apache is stored
Local user accounts	Local user accounts set up during Merge PACS installation: Local Admin, and Database Admin user accounts
Merge PACS environment variables	Environment variables loaded and set during Merge PACS installation
Multi-Volume Support	Configuration of a Merge PACS Server to store image data across multiple drives, when Merge PACS is configured to run in Standalone mode
DICOM Jail	Directory where “problem studies” (i.e., studies that have one or more images with missing or mismatched DICOM attributes) are stored until the problems can be manually fixed, when Merge PACS is configured to run in Standalone mode
Extended Query	The ability to configure one or more DICOM devices within Merge PACS as external nodes so that studies from those external nodes can be included in specific workflows.

2.2. Services

A fully functional Merge PACS system requires the following services to be running:

- **Amicas service** (the main service for Merge PACS)
- **Amicas Web Service** (service that handles distribution of non-image data over the Web and also handles Merge PACS’s internal web security)
- **AmicasWatch Service** (if this Server is being used as a Watch Monitoring Station)
- **Amicas Messaging Service** (Service that handles communications via JMS between the various Merge PACS components)
- **Apache Service** (Service that makes image data that was imported and stored on a Merge PACS Server available over the Internet/Intranet).

- **IBM DB2 Services:**
 - **DB2 – DB2 Service**
 - **DB2DAS – DB2DAS00 Service**
 - **DB2 Governor [optional]**
 - **DB2 License Server**
 - **DB2 Remote command server**
 - **DB2 Management service**
- **All Standard Server-related Services**

The Status of all these services can be monitored and, if necessary, changed from the **Services Control Panel (Start → Settings → Administrative Tools → Services)**.

NOTE: When Merge PACS is configured to run in **Integrated** mode, iCEA needs to be running and accessible when PACS services are started, otherwise PACS needs to be restarted after iCEA comes online.

2.2.1. Amicas Service

All of the Merge PACS Server's components mentioned in the previous chapter, with the exception of the Amicas Web Service, are bundled into the AMICAS service.

The Services Control Panel must always indicate the following for the AMICAS service:

- Status = *Started*
- Startup = Automatic (Delayed Start)

2.2.2. Amicas Web Service

The AMICAS Web Service handles the distribution of non-image data via http. In addition, it handles Merge PACS's internal web security by providing a web cookie to authenticated users and then restricting access to the various Merge PACS web directories to only those users who have a valid cookie.

The Services Control Panel must always indicate the following for the Amicas Web Service:

- Status = *Started*
- Startup = *Automatic*

2.2.3. Amicas Messaging Service

The AMICAS Messaging Service handles communications via JMS between the various Merge PACS components. Includes the Auditor process that monitors study events (primarily the status of studies being imported into Merge PACS) and marks them as being closed [*i.e.*, successfully imported] or closed with errors.

The Services Control Panel must always indicate the following for the Amicas Messaging Service:

- Status = *Started*
- Startup = *Automatic*

2.2.4. Apache Service

Merge PACS uses the Apache service to make image data that was imported and stored on a Merge PACS Server available over the Internet/Intranet. The Apache is the Web Server Component. The Apache Service must always be running on the Merge PACS Server.

The Services Control Panel must always indicate the following for the Apache service:

- Status = *Started*
- Startup = *Automatic*

2.2.5. DB2 Services

IBM's DB2 manages Merge PACS's databases.

Merge PACS uses the following six services that are installed with DB2:

- DB2 – DB2 service
- DB2DAS – DB2DAS00 service
- DB2 Governor [optional]
- DB2 License Server
- DB2 Remote command server
- DB2 Management service

The Services Control Panel must always indicate the following for these three DB2 services:

- Status = *Started*
- Startup = *Automatic*

NOTE: The remaining DB2 services that are displayed in the Services Control Panel were automatically installed but should not be operational.

2.2.6. AmicasWatch Service

This service monitors the various statistics that are sent by a Merge PACS Server to a Watch Monitoring Station, as well as issues alerts when predetermined thresholds are crossed.

If this Merge PACS Server is going to be used as a Monitoring Station, whether for the server itself or remotely for another Merge PACS Server, the Services Control Panel must always indicate the following for the AmicasWatch Service:

- Status = *Started*
- Startup = *Automatic*

For detailed information about Watch, refer to the *Merge PACS 8.2 Watch User's Guide*.

2.2.7. Standard Server-related Services

In addition to the specific services mentioned above, you should make sure that all the standard services that are normally required to run on the server are, in fact, running and set to startup automatically. This includes, but is not limited to, the **Net Logon** and **Server** services.

2.3. Background Processes

Every time Merge PACS starts up, Amicas service, the Apache services, and the IBM DB2 services launch a unique set of background processes on the computer on which they are installed.

The status of the various processes can be viewed from the **Processes** tab of the Windows Task Manager, as in the example below:

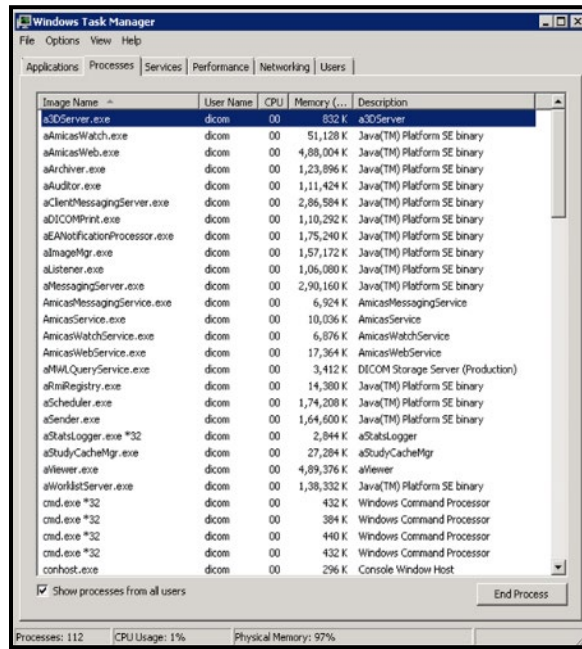


Figure 9: Viewing Processes in the Task Manager

Each of the processes shown in the figure above belongs to a particular service that may be required for Merge PACS to function properly, depending on which optional components are running. For example, if your system does not use an HL7 interface, it does not need the *aImageManager.exe* process to run, and if the Enterprise CD Writer component is not being used, the *aECDRMgr.exe* process will not be running.

The following table shows, for each service, the background processes it launches together with the tools for monitoring those processes (if any are available):

Service	Background Process	Monitoring Tools
AmicasService	aAmicasWeb AmicasService AmicasWebService aImageMgr aStorageServer aWorklstServer aAssignmentEngine	N/A
	aArchiver aAutorouter aECDRMgr aForwarder aImporter aListener aScheduler aDICOMPrint aSender (For Integrated Mode) aEANotificationProcessor(for Integrated Mode)	Merge PACS Management Web Pages
Amicas Messaging Service	aMessagingServer aAuditor	Merge PACS Management Web Pages
AmicasWatch Service	aAmicasWatch AmicasWatchServ	Amicas Watch Monitoring Station
Apache Service	httpd	N/A
DB2 – DB2	db2syscs	DB2 Control Center
DB2DAS – DB2DAS00	db2dasrrm	DB2 Control Center
DB2 Control Server	db2syscs	DB2 Control Center
DB2 Security Server	db2sec	DB2 Control Center
DB2 Remote command server	db2rcmd	DB2 Control Center
DB2 Management service	db2mgntsvc	DB2 Control Center

Table 2-1: Background Process and Monitoring Tools

2.4. Files and Directories

2.4.1. In General

In general, no matter how you configure Merge PACS Server, you should keep the structure of the files and directories close to the architecture described in this section. The file and directory architecture is established when the system is installed.

CAUTION: All these volumes should be prevented from the unauthorized user access by your site's System Administrator.

a. Standard (RAID) Installations

A typical default Merge PACS installation is spread across various separate drive letters on three physically separate drives. This presumes a Direct Attached Storage (DAS) environment using RAID technology. The following tables illustrate the standard configuration:

- **Standalone Mode:**

Drive	Description	Contents	Internal Disk
C:\	Application Drive	Operating System and various applications used by the system, including third-party programs such as DB2 and Java.	RAID GROUP 1 RAID 1
E:\	DB Log Drive	Active DB Logs	
F:\	Database Drive	Merge PACS Databases	RAID GROUP 2
G:\	DICOM Drive	Uncompressed DICOM images	RAID 5
I:\	Lossless Drive	JPEG 2000 lossless compressed images (if lossless Images are being used)	
K:\	Merge PACS Data Drive	All Merge PACS logs, DICOM Jail, database backups, and archived (inactive) DB logs.	RAID GROUP 3 RAID 5
M\	Lossy Drive	JPEG 2000 lossy compressed images	
C:\	External Drive	Non-Images	

Table 2-2: Typical Drive Configuration for a Standard Merge PACS Server in Standalone Mode

- **Integrated Mode:**

Drive	Description	Contents	Internal Disk
C:\	Application Drive	Operating System and various applications used by the system, including third-party programs such as DB2 and Java.	RAID GROUP 1 RAID 1
E:\	DB Log Drive	Active DB Logs	
F:\	Database Drive	Merge PACS Databases	RAID GROUP 2 RAID 5
K:\	Merge PACS Data Drive	All Merge PACS logs, database backups, and archived (inactive) DB logs.	RAID GROUP 3 RAID 5
C:\	Cache Directory	Location where PACS temporarily stores DICOM instances while they are processed	

Table 2-3: Typical Drive Configuration for a Standard Merge PACS Server in Integrated Mode

The following tables show the typical location of the Merge PACS directories on the various drives:

- **Standalone Mode:**

Directory	Drive with Full Pathname
Applications	
Merge PACS application directory	C:\amicas2
DB2 application directory	C:\SqlLib
Apache directories	C:\AMICAS2\Apache2.4.51
Java runtime environment directory	C:\amicas2\private_jre
License directory	C:\amicas2\licenses
DB2 data	
DB2 database directory	E:\DB2
Tablespace userspace1 for IMG	E:\PACS_DATABASE**
Merge PACS data directory	K:\PACS2DATA
DB2 backup directory	K:\DB-BACKUPS
DB2 log directory	E:\DBLog
DICOM images directory	G:\PACS2DATA\Dicom *
Lossless images directory	I:\PACS2DATA\Lossless *
Lossy images directory	M:\PACS2DATA\Compressed *
External Drive	C:\PACS2DATA\External

Table 2-4: Typical Location of Merge PACS Directories on a Standard Merge PACS Server in Standalone Mode

NOTE: * If Merge PACS is configured to run in **Standalone** mode and Multi-Volume Support, as defined below, is configured on this Server for this type of image data, the image data will be spread across multiple drives. The name of the directory for each type of data will be the same on each drive, however.

NOTE: ** If the size allocation of the IMG Database has been increased by spanning disks, there will be additional PACS_DATABASE directories on other drives as well.

NOTE: For Servers that were upgraded from a version of Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

- **Integrated Mode:**

Directory	Drive with Full Pathname
Applications	
Merge PACS application directory	C:\amicas2
DB2 application directory	C:\SqlLib
Apache directories	C:\AMICAS2\Apache2.4.51
Java runtime environment directory	C:\amicas2\private_jre
License directory	C:\amicas2\licenses
DB2 data	
DB2 database directory	E:\DB2
Tablespace userspace1 for IMG	E:\PACS_DATABASE*
Merge PACS data directory	K:\PACS2DATA
DB2 backup directory	K:\DB-BACKUPS
DB2 log directory	E:\DBLog
Cache Directory	<User>\Appdata\Cache

Table 2-5: Typical Location of Merge PACS Directories on a Standard Merge PACS Server in Integrated Mode

NOTE: * If the size allocation of the IMG Database has been increased by spanning disks, there will be additional PACS_DATABASE directories on other drives as well.

NOTE: For Servers that were upgraded from a version of Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

b. High Availability (SAN) Installations

Merge PACS Servers that handle large amounts of patient data may be optionally configured as “high availability” servers, using Storage Area Network (SAN) technology to provide failover capacity for all data and applications. A typical high availability Merge PACS installation is spread across various separate drive letters on four physically separate drives (note that the Server’s operating system is on a separate physical drive). The following tables illustrate this configuration:

- **Standalone Mode:**

Drive	Description	Contents	SAN Disk
C:\	Application Drive	Operating System and Various applications used by the system, including third-party programs such as DB2 and Java.	RAID GROUP 1 RAID 1
E:\	DB Log Drive	Active DB Logs	
F:\	Database Drive	Merge PACS Databases	RAID GROUP 2
G:\	DICOM Drive	Uncompressed DICOM images	RAID 5
I:\	Lossless Drive	JPEG 2000 Lossless compressed images (if lossless Images are being used)	
K:\	Merge PACS Data Drive	All Merge PACS logs, DICOM Jail, database backups, and archived (inactive) DB logs.	RAID GROUP 3 RAID 5
M:\	Lossy Drive	JPEG 2000 Lossy compressed images	
C:\	External Drive	Non-Images	

Table 2-6: Typical Drive Configuration for a High Availability Merge PACS Server in Standalone Mode

- **Integrated Mode:**

Drive	Description	Contents	SAN Disk
C:\	Application Drive	Operating System and Various applications used by the system, including third-party programs such as DB2 and Java.	RAID GROUP 1 RAID 1
E:\	DB Log Drive	Active DB Logs	
F:\	Database Drive	Merge PACS Databases	RAID GROUP 2 RAID 5
K:\	Merge PACS Data Drive	All Merge PACS logs, database backups, and archived (inactive) DB logs.	RAID GROUP 3 RAID 5
C:\	Cache Directory	Location where PACS temporarily stores DICOM instances while they are processed	

Table 2-7: Typical Drive Configuration for a High Availability Merge PACS Server in Integrated Mode

The following tables show the typical location of the Merge PACS directories on various drives:

- **Standalone Mode:**

Directory	Drive with Full Pathname
Applications	
Merge PACS application directory	C:\amcas2
DB2 application directory	C:\SqlLib
Apache directories	C:\AMICAS2\Apache2.4.51
Java runtime environment directory	C:\amcas2\private_jre
License directory	C:\amcas2\licenses
DB2 data	
DB2 database directory	E:\DB2
Tablespace userspace1 for IMG	E:\PACS_DATABASE**
Merge PACS data directory	K:\PACS2DATA
DB2 backup directory	K:\DB-BACKUPS
DB2 log directory	E:\DBLog
DICOM images directory	G:\PACS2DATA\Dicom *
Lossless images directory	I:\PACS2DATA\Lossless *
Lossy images directory	M:\PACS2DATA\Compressed *
External Volume	C:\PACS2DATA\External

Table 2-8: Typical Location of Merge PACS Directories on a High Availability Merge PACS Server in Standalone Mode

NOTE: * If Merge PACS is configured to run in **Standalone** mode and Multi-Volume Support, as defined below, is configured on this Server for this type of image data, the image data will be spread across multiple drives. The name of the directory for each type of data will be the same on each drive, however.

NOTE: ** If the size allocation of the IMG Database has been increased by spanning disks, there will be additional PACS_DATABASE directories on other drives as well.

NOTE: For Servers that were upgraded from a version of Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

- **Integrated Mode:**

Directory	Drive with Full Pathname
Applications	
Merge PACS application directory	C:\amicas2
DB2 application directory	C:\SqlLib
Apache directories	C:\AMICAS2\Apache2.4.51
Java runtime environment directory	C:\amicas2\private_jre
License directory	C:\amicas2\licenses
DB2 data	
DB2 database directory	E:\DB2
Tablespace userspace1 for IMG	E:\PACS_DATABASE*
Merge PACS data directory	K:\PACS2DATA
DB2 backup directory	K:\DB-BACKUPS
DB2 log directory	E:\DBLog
Cache Directory	<User>\Appdata\Cache

Table 2-9: Typical Location of Merge PACS Directories on a High Availability Merge PACS Server in Integrated Mode

NOTE: * If the size allocation of the IMG Database has been increased by spanning disks, there will be additional PACS_DATABASE directories on other drives as well.

NOTE: For Servers that were upgraded from a version of Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

2.4.2. Merge PACS Application Directory

The Merge PACS application is typically stored in the C:\amicas2 directory, as shown in the example below:

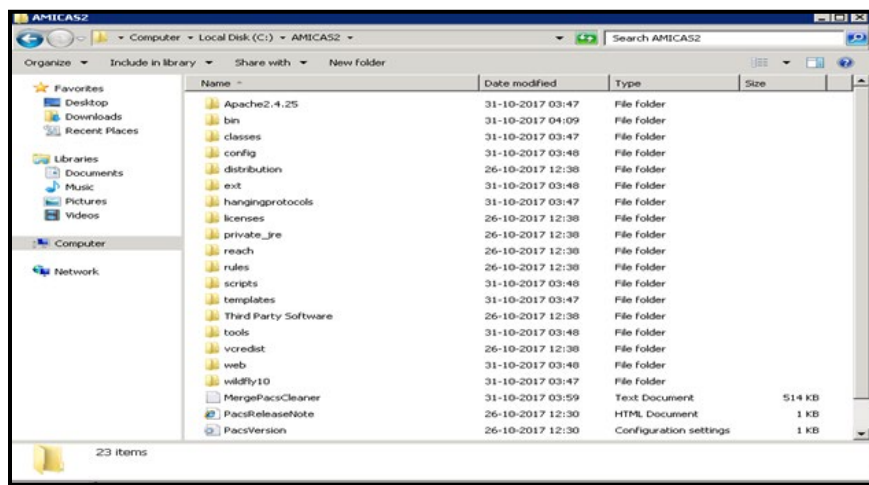


Figure 10: Merge PACS Application Directory

2.4.3. Merge PACS Data Directory

The main Merge PACS data directory typically contains all process logs, any jailed images (as described in Section 2.6 below), any temporary files, and images that are received via Telemedicine that have not yet been imported into Merge PACS, as shown in the example below:

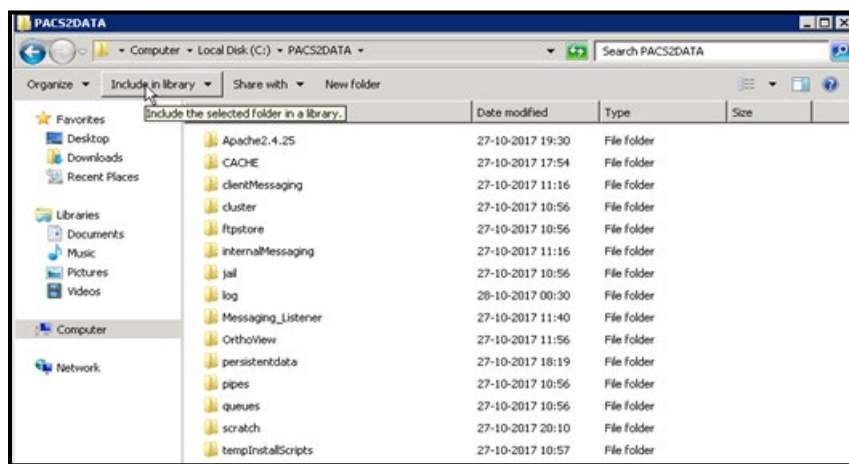


Figure 11: Merge PACS Data Directory

Typically, this directory is stored on the **K:** drive (specifically at **K:\PACS2DATA**), but it can also be on any other drive on your server.

NOTE: For server that were upgraded from a version of Merge PACS prior to 7.0, the PACS2DATA directory will be called AMICAS2DATA.

2.4.4. DB2 Data Directory

The DB2 data directory stores the ANETDB database and is also the root of the IMG database. It is typically set up on a different drive than that containing the DB2 application, and should be on a separate physical drive from the rest of the various directories described in this Section, both for performance reasons and to make sure that both the databases and the database backups are not lost in the event of a drive failure. Typically, this directory is stored on the **F:** drive (specifically at **F:\DB2**).

Note that the bulk of the IMG database is located in the PACS_DATABASE directory described below.

2.4.5. PACS_DATABASE Directory

The PACS_DATABASE directory contains the actual data (tablespace userspace1) for the IMG database that holds information about patients, studies, series, and images, both uncompressed and compressed. Typically, this directory is stored on the **F:** drive (specifically at **F:\PACS_DATABASE**). Note that if, at any time, the size allocation of the IMG Database has been increased by spanning disks, there will be additional PACS_DATABASE directories on other drives as well.

2.4.6. DICOM Images Directory

When Merge PACS is configured to run in **Standalone** mode, the DICOM images directory is where the uncompressed DICOM images are stored. Typically, this directory is stored on the **G:** drive (specifically at **G:\PACS2DATA\Dicom**), but it can also be on any other drive on your server. For performance reasons, it is recommended that this drive be a separate physical drive from the rest of the various directories described in this Section.

Note that if Multi-Volume Support is enabled for uncompressed images, as described in Section 2.5 below, there will be more than one **PACS2DATA\Dicom** directory, each on a separate drive.

2.4.7. Lossy Images Directory

When Merge PACS is configured to run in **Standalone** mode, the Lossy images directory is where the JPEG 2000 Lossy compressed images are stored. Typically, this directory is stored on the **M:** drive (specifically at **M:\PACS2DATA\Compressed**), but it can also be on any other drive on your server. For performance reasons, it is recommended that this drive be a separate physical drive from the rest of the various directories described in this Section.

Note that if Multi-Volume Support is enabled for JPEG2000 lossy and JPEG2000 lossless images, as described in Section 2.5 below, there will be more than one **PACS2DATA\Compressed** directory, each on a separate drive.

2.4.8. Lossless Images Directory

When Merge PACS is configured to run in **Standalone** mode, the Lossless images directory is where the JPEG 2000 Lossless compressed images are stored. Typically, this directory is stored on the **I:** drive (specifically at **I:\PACS2DATA\Lossless**), but it can also be on any other drive on your server.

Note that if Multi-Volume Support is enabled for lossless images, as described in Section 2.5 below, there will be more than one **PACS2DATA\Lossless** directory, each on a separate drive.

NOTE: If Merge PACS receives an image that is currently lossy compressed (*i.e.*, the transfer syntax indicates lossy), or has at any point in its past been lossy compressed (*i.e.*, the Lossy Image Compression flag is set to non-0 value), Merge PACS will compress the image as per the currently configured Recompressed Lossy Ratios (lossless or lossy) and place the output in the **Lossy** folder. A copy of this image will never be placed in the **Lossless** folder. As a result, if lossy compression has been configured as the Recompressed Lossy Ratio, there will be no lossless compressed version of the study saved.

2.4.9. External Volume

When Merge PACS is configured to run in **Standalone** mode, the External Volume is where non-image DICOM instances such as Structured Report (SR) and VoiceClip (AU) files are stored. Typically, this directory is stored on the **C:** drive, but it can also be on any other drive on your server.

Note the following:

- All non-image DICOM instances are stored in the External Volume for watermarked studies (*i.e.*, studies that have been archived and deleted from the Server). If a request is made from the Merge PACS Workstation to access a watermarked study, Merge PACS will retrieve the associated non-image DICOM instances from the External Volume instead of retrieving them from the archive.
- Instances in the External Volume are included in DICOM query/retrieve requests initiated from within the Merge PACS Workstation.

2.4.10. Cache Directory

When Merge PACS is configured to run in **Integrated** mode, the Cache Directory is the location where PACS temporarily stores DICOM instances while they are processed. The following features make use of the Cache Directory:

- Viewer-generated instances (PR, KO, AU) being sent to the Enterprise Archive by the sender
- Instances imported from a CD waiting to be sent to the Enterprise Archive by the sender
- Instances retrieved from the Enterprise Archive for data masking and the masked instances being sent back to iCEA
- Instances being migrated from PACS to the Enterprise Archive (full file to full folder migration)
- Instances received through telmed in the process of being sent to the Enterprise Archive

2.4.11. DB2 Log Directory

The DB2 log directory contains the active transaction logs of database activity. Typically, this directory is stored on the **E:** drive (specifically at **E:\DBLog**). Note that this drive is typically a logical partition on the same physical drive as the Merge PACS application drive

2.4.12. DB2 Transaction Log Directory

The path to the archived DB transaction logs is stored in the registry entry [\\HKEY_LOCAL_MACHINE\SOFTWARE\AMICAS\InstallSettings\ArchivedDBLogPath](#). By default the archive folder for the transaction logs is set to the following folder during the installation:

C:\DB-BACKUPS\TransactionLogs (the drive can be different).

NOTE: AMICAS2DATA\log\TransactionLogs is a legacy path for the archived logs, while the default root path for archived transaction logs now is C:\DB-BACKUPS (the drive can be different).

CAUTION: The DB2 Transaction Log Directory should never be on the same physical storage device as the DB2 Data Directory or the DB2 log Directory.

2.4.13.DB2 Backup Directory

The DB2 backup directory, which stores the backup of the database created during the Merge PACS installation procedure, should be on the same drive as the Merge PACS data directory described above. If this is the **K:** drive, the directory path would be **K:\DB-BACKUPS**.

CAUTION: The DB2 Backup Directory should never be on the same physical storage device as the DB2 Data Directory or the DB2 log Directory.

2.4.14.Apache Directory

Apache is installed during the installation of PACS Server. This process creates the **C:\amcas2\Apache2.4.51** directory.

2.5. Multi-Volume Support

As described above, when Merge PACS is configured to run in **Standalone** mode, both the uncompressed and compressed images on a Merge PACS Server are typically stored in separate **PACS2DATA** directories on separate disk drives. By default, there is one directory (or “volume”) for uncompressed images and one for compressed images. A Merge PACS Server can be configured, however, to store both uncompressed and compressed images across multiple volumes located locally on the Merge PACS Server.

For uncompressed images, the image volumes would consist of different hard drives located locally on the Merge PACS Server itself. For compressed images, these volumes can be either local or on one or more remote non-Merge PACS servers accessible via a network. In order for a user to view images stored on a **remote** compressed image volume, either the compressed image volume must be on a Network Attached Storage (NAS) server or else the user accounts must be in a domain to which the Merge PACS Server and the remote server belong, and there must be an NT domain controller managing the user accounts.

Compressed studies (both JPEG2000 lossy and JPEG2000 lossless) are written to the various volumes *sequentially*, meaning that when one volume becomes full the next designated volume is written to. Uncompressed studies, on the other hand, are written according to an algorithm that optimizes disk space allocation among the various volumes. Additional volumes may be added at any time for both compressed and uncompressed studies, although in typical usage scenarios only compressed volumes would need to be added in order to expand the number of images available to end-users via Forever Priors.

A volume should not exceed 4 TB. If a site’s annual volume is far below this number; then there is not a need to create a single volume this large. For example:

- If a site produces less than 2 TB of studies in a one year period, you should start with a volume that is twice the annual capacity.
- If a site produces more than 2 TB of studies in a one year period, you should start with a 4 TB GB volume or volumes as needed.

In all cases, new volumes should be added 2-3 months in advance of the date when they are expected to be needed.

2.6. DICOM Jail

When Merge PACS is configured to run in **Standalone** mode, the “DICOM Jail” is a subdirectory of the Merge PACS Data Directory, as described in subsection 2.4.3 above, typically at the following location:

J:\PACS2DATA\jail

DICOM images are routed to the DICOM Jail instead of being imported into Merge PACS when any of the following circumstances occur:

- An image is missing a required DICOM attribute, whether at the study, series or image level.
- An image has a study or series level DICOM attribute that is different from the first image that was imported into the System for that same series or study.
- The same study is sent to Merge PACS twice, but an image has an image-specific attribute that is not the same for both studies.
- A study that has been imported into Merge PACS and watermarked is later pushed to Merge PACS or restored from Enterprise Archive with the same Study Instance UID but a different patient identity (normalized MRN, normalized Patient Name, Issuer of Patient ID). Note that, with regard to Enterprise Archive in particular, studies will only be jailed if Merge PACS has been configured so that the information received from EA will be given precedence over the information stored within Merge PACS, as described in subsection 1.2.61 above.

There are several circumstances in which some of the images of a study can be “jailed,” including where the radiology technician changes accession numbers part way through a study, a user-name mismatch caused by the first and last name being reversed for some slides in the study, the patient ID being left blank for a study, etc.

Merge PACS performs two different types of checks on these DICOM headers to determine whether or not an incoming DICOM image is acceptable or whether it should be placed in the DICOM Jail:

- First, it checks to see whether an image has the headers that are defined as being required by the system. By default, every image must have either a Patient ID or a Patient Name associated with it. If there is a Patient Name but no Patient ID, the system will automatically use the Patient Name for both. A Merge PACS Server can be custom configured, however, to require any of the following combinations of Patient ID and Patient Name for reconstituted DICOM images:
 - **Patient ID or Patient Name** (default)
 - **Patient ID and Patient Name**
 - **Patient ID only** (i.e., it doesn’t matter if there is a Patient Name or not)
 - **Patient Name only** (i.e., it doesn’t matter if there is a Patient ID or not)
 - **None** (i.e., the study’s images will not be jailed even if both the Patient ID and Patient Name are missing).

- Second, Merge PACS checks to see whether a defined set of attributes for that image match those of the first image for that study (*i.e.*, the first image with the same StudyInstanceUID as the incoming image). The following is a list of the attributes that a Merge PACS checks by default:

Study Specific Attributes	Series Specific Attributes	Image Specific Attributes
• StudyID • AccessionNumber • PatientName • PatientID • IPID • PatientBirthDate • PatientSex	• SeriesNumber • Modality • StationName	• ImageType • SOPClassUID • ImageNumber • ImageDate • ImageTime • ImagePositionPt • ImageOrientationPt • SliceLocation • PixelSpacing • NRows • NColumns • LossyCompression

Images that have been placed in the DICOM Jail can be deleted or fixed by using the Problem Studies Tool, as described in subsection 9.9.4 below.

A Merge PACS Server can also be optionally configured to prevent users from viewing jailed study images within the Merge PACS Viewer.

2.7. Users

The Merge PACS installation process always creates a **Merge PACS Super User** local user account. Optionally, the Merge PACS installation process can create the following local user accounts:

- Merge PACS Administrator
- Telemedicine User
- Watch Administrator User Account

2.7.1. Merge PACS Super User Account

This user account has administration privileges. It is authorized to start and stop the Amicas service. It is also used by Merge Healthcare Technical Support when accessing the Management Web Page described in Chapter 9 below. The Super User account has access to the entire set of tools available on that Web page, including tools used to configure and maintain Merge PACS. Each Merge PACS installation will have its own unique Password for this account.

Login details:

Username	superuser
Password	<Contact Merge Healthcare Technical Support>

2.7.2. Merge PACS Administrator User Account

This user account can be used by an on-site Merge PACS Administrator for accessing the Management Web Page described in Chapter 9 below. The Merge PACS Administrator user account has access a limited set of tools available on that Web page, including tools to monitor the various processes and edit patient data, but not to configure and maintain Merge PACS. Each Merge PACS installation will have its own unique Password for this account.

Login details:

Username	admin
Password	<Contact Merge Healthcare Technical Support>

2.7.3. Telemedicine User

When the Telemedicine feature is enabled, this is a generic user account that can be used for receiving Merge PACS data from the sending Merge PACS Server. The sending site logs in to the receiving site using this user account.

Each Merge PACS installation will have its own unique Password for this account.

Login details:

Username	telmeduser
Password	<Contact Merge Healthcare Technical Support>

2.7.4. DB2 User and DB2 Admin User Account

The DB2 user account is used by various PACS applications to connect to PACS databases.

The DB2 admin user account is used during installation to create and to manage PACS databases, as well as used for routine database backups.

DB2 User and DB2 Admin User Accounts do not have privileges in PACS applications.

2.7.5. Watch Administrator User Account

This user account is set up during the standard Merge PACS Server installation process. If the optional Watch component is being used at a site, this account can be used to access the various Watch administration functions on the monitoring station. Each Merge PACS installation will have its own unique Password for this account.

Login details:

Username	watchadmin
Password	<Contact Merge Healthcare Technical Support>

NOTE: Regular users who have access to Merge PACS can also be provided with administrative access to Watch via user privileges. It is recommended that this method be used to access Watch instead of the using the default Watch administrator user account.

2.8. Environment Variables

The Merge PACS installation process sets certain environment variables. You can use the System Properties Window to examine these environment variables.

NOTE: The third-party components such as DB2 may also set environment variables. Check the vendor documentation for more information about these variables.

To view environment variables:

1. From the **Start** Menu, right-click on **My Computer** and select **Properties** from the pop-up menu.

*The **System Properties** window is displayed:*

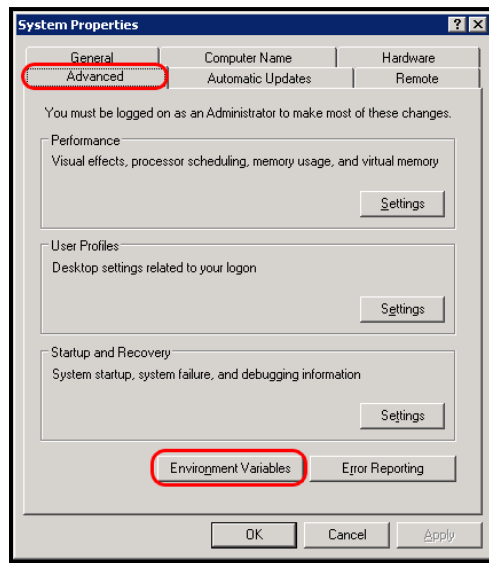


Figure 12: System Properties Window - Advanced Tab

- Click on the **Advanced** tab, and then click on the **Environment Variables** button.

The **Environment Variables** window is displayed:

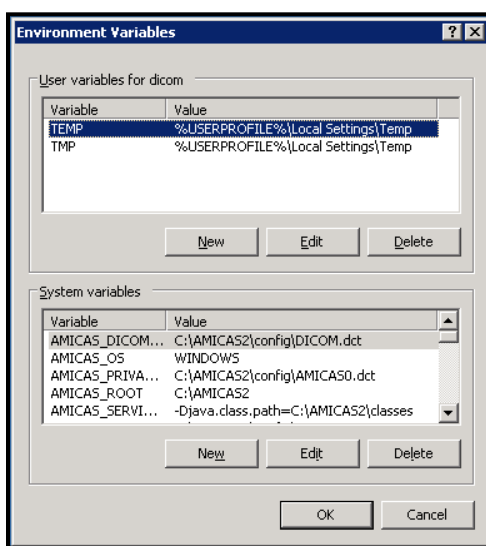


Figure 13: Viewing Environment Variables

The following table lists all of the system variables set by Merge PACS and their values.

Variable	Value
AMICAS_ARCHIVE_DB_LOG_PATH	<DriveLetter>:\DB-BACKUPS
AMICASDATA_ROOT	<DriveLetter>:\PACS2DATA
AMICAS_DICOM_DICTIONARY	<DriveLetter>:\amicas2\config\DICOM.dct
AMICAS_OS	Windows
AMICAS_PRIVATE_DICTIONARY	<DriveLetter>:\amicas2\config\AMICAS0.dct
AMICAS_ROOT	<DriveLetter>:\Amicas2
AMICAS_SERVICE_CLASSPATH	-Djava.class.path=<DriveLetter>:\Amicas2\classes
AMICAS_UID_DICTIONARY	<DriveLetter>:\amicas2\config\UID.dct
ASNPATH	<DriveLetter>:\PACS2DATA\Cluster
AUTOROUTER_PROFILE	<DriveLetter>:\Amicas2\config\StorageServer.ini
FORWARDER_PROFILE	<DriveLetter>:\Amicas2\config\StorageServer.ini
STORAGE_SERVER_PROFILE	<DriveLetter>:\Amicas2\config\StorageServer.ini

Table 2-6: Merge PACS system Variables

The installation also adds the following pathnames to the server's PATH statement:

- <DriveLetter>:\amicas2\bin
- <DriveLetter>:\amicas2\private_jre

2.9. Extended Query

Extended Query provides a quick way to let Merge PACS be aware of studies that do not currently reside on Merge PACS. In both **Standalone** mode and **Integrated** mode, Extended Query allows you to configure multiple DICOM devices in Merge PACS as external nodes and the studies from those external nodes can then be included in specific workflows.

Note that in **Integrated** mode, Extended Query would typically be used in either of the following cases:

- After upgrading an existing RadSuite Server to Merge PACS, there are still studies that have not been migrated to Merge PACS.
- Enterprise Archive has one or more DICOM devices that are exposed through its proxy and are not otherwise available to Merge PACS.

The studies from the extended node will show up in the following study lists within the Merge PACS Workstation:

- Query Search Page results
- RTSL custom searches / Filters
- **Prior lists** (worklists, Patient Record, Load Comparison Studies)

When Merge PACS is configured to run in **Standalone** mode, any DICOM device (including iConnect Enterprise Archive) can be configured as an extended query node. In **Integrated** mode, the remote devices in the paired ICEA can be configured as extended query nodes. In either case, the studies from extended node stay in offline state until they are retrieved into Merge PACS.

In general, it is not possible to query another DICOM node for search criteria that are specific to Merge PACS. For example, “Study Status” is a Merge PACS concept, so attempting to query another DICOM node for studies with “status = Unread” will return nothing, since that other node doesn't maintain the Merge PACS diagnostic status. The following query criteria are specific to Merge PACS:

- **Study Status**
- **Facilities**
- **Location**
- **Order status**
- **Specialty**
- **Patient Class**
- **Order Priority**
- **Region**
- **Due in**
- **Assigned by**
- **Reserved by**
- **Locked by**
- **QC Criteria**

2.10. Patient Consolidation and Determination of Priors

Merge PACS uses one or more of the following methods to consolidate patient information and to determine how prior studies should be identified in cases where a patient may have different identities within PACS (e.g., different last names, different MRNs, different IPIDs, etc.):

- **Patient Comparison Strategy (PCS)**
- **Master Patient Index (MPI)**
- **Selection of Priors**

2.10.1. Patient Comparison Strategy

Patient Comparison Strategy (PCS) is a virtual layer added to PACS Server to provide fine grain control on patient update and merges via HL7 interface or Patient Data edits via the Merge PACS Quality Control Editor. When a study is imported into Merge PACS, a Patient Identity for the new study is created and a unique identifier is generated. Based on the unique identifier, the Study is either matched to a pre-existing patient and stored as such or stored as a new patient. With Patient Comparison Strategy, Merge PACS can be configured to use other patient demographic values to identify and group studies related to a given patient.

NOTE: Patient Comparison Strategy does not alter how incoming data is initially stored in the database or DICOM files, but it does affect how the storage of data is changed after specific operations are performed. For example, if a patient-level edit operation is performed within the QC Editor, the system will perform the requested edit operation and will apply the modifications to all studies grouped under the modified patient based on the configured Patient Comparison Strategy. Patient Comparison Strategy is also used by other Merge PACS Server components when updating or merging patients/studies to identify related studies based on configuration.

NOTE: For orders, the Patient Comparison Strategy is always based on MRN + IPID.

NOTE: For all patient comparison strategies, IPID and MRN matching is performed based on exact values (e.g., “ABC” will be considered different from “abc” and “123” will be considered different from “1.2.3”).

a. Available Patient Comparison Strategies

There are three Patient Comparison Strategies available within Merge PACS (in Standalone mode, these are configured on the Merge PACS Server; in Integrated mode, the settings are inherited from the paired iConnect® Enterprise Archive):

Comparison Strategy	Description
StrictComparisonStrategy	The default strategy on installing Merge PACS, this strategy uses the Patient’s full name (all five components of DICOM person name), Patient MRN and Patient IPID to group related studies. This is the recommended strategy when running Merge PACS in Standalone mode.

Comparison Strategy	Description
LastNameOnlyComparisonStrategy	Uses the Patient's last name (Last Name component of DICOM person name), Patient MRN and Patient IPID to group related studies. In Integrated mode, the Enterprise Archive will typically use this strategy if Multiple Patient Identity (MPI), as described in subsection 2.10.2 below, is not enabled.
PatientIDOnlyComparisonStrategy	Uses the Patient's last name (Last Name component of DICOM person name), Patient MRN and Patient IPID to group related studies. In Integrated mode, the Enterprise Archive will use this strategy if the MPI is enabled

NOTE: The Patient Comparison Strategy can be changed at any point in time. Prior to changing Patient Comparison Strategy, however, customers should assess the impact of their workflow to ensure that no unintended updates/merges occur.

NOTE: When patient comparison strategy is applied in Merge PACS for a strategy including patient name, only the part before the first backslash is considered.

b. Consolidating Patient Information

When patient information is grouped using Patient Comparison Strategy, Merge PACS applies a “Last Wins” rule (*i.e.*, based on patient information in the most recent study) to calculate the final patient information and for all related studies.

The following pieces of information are calculated:

- **Patient Name**
- **Patient Date of Birth**
- **Patient Sex**

Under “Last Wins” rule, Merge PACS uses the last piece of data that is not null or empty. For example, look at the three studies in the following example:

Study #	Patient Name	DOB	Sex
1	Doe, Jonathan	19770101	O
2	Doe, Jonathan	19770101	
3	Doe, Jonathan Q		M

After consolidation, the patient information for all studies would be consolidated as follows:

Patient Name	DOB	Sex
Doe, Jonathan Q	19770101	M

NOTE: HL7 A08, A34 and A40 messages missing patient name, date of birth or sex will implicitly use consolidated patient information.

NOTE: HL7 A08, A34 and A40 messages will also use consolidated patient information if Image Manager is configured to not use patient demographics in HL7 message, as described in subsection 3.9.2 below.

NOTE: Modifications affecting a Patient or Study in the Quality Control Editor tool such as Copy, Move, Split, Delete, etc., will also cause a job to be posted in the Command Scheduler to consolidate patient information. The job will appear as “**PIConsolidator**” in the Command Scheduler.

2.10.2. Multiple Patient Identity (MPI)

Multiple Patient Identity (MPI) allows patients who might not otherwise be identified as the same patient (e.g., because of different names, IPIDs and/or MRNs) to be linked together by referencing some external “source of truth” such as an Enterprise Master Patient Index (EMPI) or a HIS/RIS. For example, if a patient “Jane Smith” had a procedure at one institution with an IPID of “ABC” and was assigned an MRN of “123456789,” then got married and changed her name to “Jane Brown” and then had a procedure at another institution with an IPID of “XYZ” and an MRN of “987654321,” Merge PACS would normally not be able to recognize that they were the same patient. MPI, however, would allow the system to recognize that “Jane Smith” and “Jane Brown” were the same patient under certain circumstances.

MPI is used within the Merge PACS Workstation in the following two workflows:

- To determine the list of prior/comparison studies for each primary study
- To determine the search results when performing a query on a particular patient

In order for Merge PACS to use MPI, the following must be true:

- Merge PACS must be configured to run in **Integrated** mode with iCEA v. 11.4 or higher.
- The attached iCEA must be configured to receive information from some external source of truth such as an EMPI or a HIS/RIS.
- The **Patient Comparison Strategy**, as described in subsection 2.10.1 above, must be set to **IPID + MRN**.
- Merge PACS has been configured to enable MPI.

Whatever the external source of truth is, it makes iCEA aware of the various identities and tells iCEA which identities belong to the same person via standard HL7 ADT messages. The PID segment of these messages can contain one or multiple identities; where multiple identities are included, they will be considered linked (i.e., be considered the same person). iCEA can cache this linking information and Merge PACS can then ask iCEA for alternate identities of a person and use this information to present other studies for that person.

Once MPI is enabled, Merge PACS can be customized to determine which ID or IDs should be displayed for each study (“ID” here refers to IPID and MRN). Every study has an ID associated with it, referred to as the **Local ID**. That person may also have other identities from MPI. In some cases, all identities for a person are considered “peers” (equally important). In other cases, however, one ID is considered more important or more significant than the others, which is referred to as the **Master ID** (in the UK, for example, this would be the **NHS Number**). In Merge PACS, the Master ID is configured as a **Master IPID**. If the Master IPID were set to “NHS”, for example, any MRN associated with an IPID of “NHS” will be considered the Master ID for that person.

NOTE: Master ID has no special significance in priors lists and in search results since these lists will include exams associated with any ID of the patient. Instead, the distinction between Master and Local ID only comes into play with the actual *display* of the IDs.

In addition, when MPI support is enabled, Merge PACS can be configured to display some or all of the following ID components for each study in places such as the Viewer Titlebar and Study Titlebar, as well as many other places where patient demographics are displayed such as the Report Viewer and the Change Workflow Status dialog:

ID Component	Description
Local IPID	The Issuer of Patient ID (IPID) associated directly with the study
Local MRN	The Medical Record Number (MRN) associated directly with the study
Master IPID	The Master IPID for Merge PACS, if this patient has an MRN for this IPID.
Master MRN	The MRN for this patient associated with the Master IPID. A patient may have multiple studies with different MRNs associated with them, but will only have one Master MRN.

In most cases, this setting is honored anywhere in the Merge PACS Viewer that previously showed IPID and MRN, and that has the context of one identity. So, for example, this setting is honored in the viewer titlebar and study titlebar, but is not honored in the worklist, Query page, or recently viewed studies lists, because PACS doesn't have a single identity in context there.

By default, if the system is configured to display Master IDs (or both Local and Master IDs), only the **Master MRN** is displayed. If desired, however, Merge PACS can be configured to display both the Master IPID and the Master MRN wherever appropriate.

2.10.3. Selection of Priors

When determining whether a particular Study is considered a “prior” or “comparison” Study of another Study within Merge PACS, Merge PACS first looks to the Patient Comparison Strategy configured for a site to determine a match and, if applicable, the MPI. Merge PACS then looks to the “Selection of Priors” option that is selected for the site to see if any additional studies should be included in the list of priors. Studies that do not match the Patient Comparison Strategy but do match the “Selection of Priors” option are displayed with a “+” from various places within the Workstation.

The following is a list of “Selection of Priors” options that are available for Merge PACS:

Preference	Description
MRN	Determine priors based solely on MRN
IPID and MRN	Determine priors based on a combination of Issuer of Patient ID (IPID) and MRN (<i>i.e.</i> , both must match to be considered a prior).
MRN and Name	Determine priors based on a combination of MRN and Patient Name (<i>i.e.</i> , both must match to be considered a prior).
IPID and MRN and Name	Determine priors based on a combination of IPID, MRN and Patient Name (<i>i.e.</i> , all must match to be considered a prior).
MRN or Name/DOB/Sex	Determine priors based on either MRN or a combination of Patient Name, Date of Birth and Sex.
(MRN and Name) or Name/DOB/Sex	Determine priors based on either a combination of MRN and Patient Name or a combination of Patient Name, Date of Birth and Sex.
Name/DOB/Sex	Determine priors based on a combination of Patient Name, Date of Birth and Sex.

NOTE: For all selection of priors options that involve IPID and MRN, matching is performed based on exact values (e.g., “ABC” will be considered different from “abc” and “123” will be considered different from “1.2.3”).

For information on setting the Selection of Priors option for Merge PACS, refer to Appendix A.3.2 below.

Chapter 3. Amicas Service

3.1. Overview

This chapter discusses in detail the individual processes that are launched by the Amicas service. Each of these processes is a functional module of Merge PACS.

The following concepts and terminology are discussed in this chapter:

Term	General Description	Mode
Merge PACS log files	Files that contain all communications and errors handled by Merge PACS processes.	Standalone and Integrated
Merge PACS configuration files	Files that contain parameters and settings controlling the behavior of the Merge PACS processes.	Standalone and Integrated
Archiver	Process that bundles compressed and/or non-compressed images and copies them to an archival storage device, as well as retrieves those images.	Standalone
AutoRouter	Process that uses customer-defined routing rules to determine whether an image needs to be routed.	Standalone
CommandScheduler	Process that executes periodic system maintenance commands and user-generated explicit ones.	Standalone and Integrated
EA Notification Processor	Process that receives notifications from iConnect Enterprise Archive about new or changed DICOM instances available in the archive as well as for quality control (QC) messages indicating various changes to the DICOM instances already available	Standalone and Integrated
ECDR Manager	Process that monitors the CD Queue for CD writing requests and then composes and sends CD writing jobs studies to the shared CD writing device.	Standalone
Forwarder	Process that forwards DICOM images received by the StorageServer to other DICOM devices.	Standalone and Integrated
ImageManager	Process that serves as the system's interface for receiving information from HL7 systems.	Standalone
Importer	Process that compresses the DICOM images received by the StorageServer. Also marks a study as being "valid" or "invalid" according to a customizable definition of validity.	Standalone and Integrated

Term	General Description	Mode
Listener	Process that enables communications between the Merge PACS Workstation and DICOM devices.	Standalone and Integrated
Outbound HL7 Interface	A thread running under the Command Scheduler that sends messages in HL7 format. These are either status or order messages sent to an attached HIS/RIS.	Standalone
StorageServer	Process that handles communications between Merge PACS and DICOM devices.	Standalone and Integrated
STSServer	Process that handles the DirectPullSTS and MergePacsSTS requests which come from iConnect Enterprise Archive.	Integrated
WorklistServer	Process that keeps track of studies and orders, and responds to search requests from the Worklist client.	Standalone and Integrated

3.2. StorageServer

When Merge PACS is configured to run in **Standalone** mode, the StorageServer process (aStorageServer.exe) does the following:

- Stores DICOM images that are transferred from imaging modalities.
- Deletes DICOM images from the Merge PACS Server after a specified period.
- Performs DICOM Query/Retrieve requests to third-party DICOM archives.
- Provides Modality Worklist information to DICOM modalities.
- Processes DICOM Header Modification rules

The following StorageServer parameters are critical information for DICOM devices trying to communicate with Merge PACS:

Parameter	General Description
AETitle	Application Entity Title. Unique DICOM identifier used by DICOM devices to send DICOM messages to Merge PACS.
Port #	Port on which all DICOM communications occurs (for the Merge PACS Server, this is typically Port 104).
IP address	Valid IP address for the Merge PACS Server.

3.2.1. Process activities

StorageServer handles the following types of activities:

- Image storage
- Query/Retrieve requests
- Modality Worklist delivery

- Automatic DICOM Header Modification

Each of these activities is described in further detail below.

a. Image Storage

When StorageServer receives an image, it stores the image in the `<DriveLetter(s)>:\PACS2DATA\Dicom` directory and sends information about the image to the StorageServer database tables in the Image database. The image is stored as an uncompressed .dcm (for DICOM) file that is only kept for a finite amount of time until it is deleted. StorageServer can also accept images with compressed transfer syntax.

b. Image Deletion

DICOM images are deleted from the Merge PACS Server by the CommandScheduler, as defined below. Images can be deleted according to two different criteria:

Criterion	General Description
Time	The Server can be configured to delete DICOM images after a specified period. The default period is 1 hour , but this can be manually changed to meet the needs of a specific site.
Watermarks	The Server can be configured to delete DICOM images when the amount of free space on the DICOM volume (or volumes) falls below a specified amount. This setting is determined by the DICOM free space high and low watermarks.

By default, DICOM images are deleted based on both time and watermarks.

NOTE: If the optional Merge Archive component is being used to archive DICOM images, the DICOM deletion time will be ignored until after all DICOM images for a study have been archived. However, it is still recommended that the DICOM deletion time be set longer than the Archive qualification time.

c. Query/Retrieve Requests

StorageServer supports DICOM Query/Retrieve requests. In other words, DICOM Workstations can query the StorageServer database tables in Merge PACS for all (or specific) uncompressed patient data. If the user finds the study they are looking for they can retrieve it to their workstation. Storage Server will convert images in real time based on the transfer syntax requested by the retrieval target.

d. Modality Worklist Delivery

StorageServer provides DICOM devices with Modality Worklists for scheduling studies. The Modality Worklist data had been stored in a Merge PACS database after being received through ImageManager from a HL7 system, as described in Section 3.9 below.

e. Automatic DICOM Header Modification

When Merge PACS is configured to run in Standalone mode, StorageServer can modify the DICOM header information for incoming studies based on a defined set of rules contained in a separate rules file on the Merge PACS Server. Typical use case scenarios would be to have a different prefix or suffix appended to the Patient ID attribute for each study depending on the originating spoke Server, to overwrite the existing Institution Name attribute, if any, based on the originating spoke Server, or to truncate the length of one or more attributes.

NOTE: Although not handled by StorageServer, Merge PACS can also be configured to modify DICOM header information for incoming studies received from the Viewer's local import feature (usually studies on CDs from other sites). Rules can be set up to extract one or more characters from a DICOM attribute, change a DICOM attribute to be in all uppercase letters, change a DICOM attribute to be in all lowercase letters, replace some or all of the contents of a DICOM attribute, remove an entire DICOM attribute and/or set the value of a DICOM attribute to nothing if the attribute is present.

Note that the **Institution Name**, **IPID** and **Department** fields in particular are typically handled poorly on the source modality devices. They may be missing entirely, spelled incorrectly or abbreviated inconsistently, and in many cases these can only be set by the modality field service personnel and not by the PACS Administrators. These fields are used not only by PACS, however, but also by other "downstream" systems for things such as worklist filtering (Merge PACS, iConnect Access), patient ID matching, viewing scope control (iConnect Access), Policy Based Deletion rules (iConnect Enterprise Archive), etc. As such, the values need to be cleaned up or "enforced" via Header Modification Rules so that they can be used as filters input values. The Institution Name, for example, nearly always will need to be overridden due to differences in spelling, capitalization and abbreviations (e.g., St. vs. ST vs. St vs. Saint, Comm. vs. Community, Mem'l vs. Memorial, etc.).

NOTE: For use with telemedicine, rules can only be defined to match based on AETitle.

NOTE: DICOM header modification rules can be applied to private DICOM tags as well as public DICOM tags.

CAUTION: If DICOM header values are changed on a PACS Server that already has data on it, objects imported to the system before the change will not be updated to the new values. As a result, when those objects are sent out of the system they will reflect the old values from the time they were imported.

Each rule refers to the applicable DICOM attribute by its Group and Element number (e.g., "Patient ID" is referred to as "(0x0010, 0x0020)"). The following is a list of common DICOM attributes that might be used when defining DICOM header modification rules:

Attribute	Group	Element	Data Type
Accession Number	0x0008	0x0050	SH
Issuer of Patient ID	0x0010	0x0021	LO
Institution Name	0x0008	0x0080	LO
Institutional Department Name	0x0008	0x1040	LO
Patient's ID	0x0010	0x0020	LO
Study Description	0x0008	0x1030	LO

Attribute	Group	Element	Data Type
Series Description	0x0008	0x103e	LO

In addition, DICOM header modification rules can reference any of the **custom DICOM tags** that can be defined for Merge PACS, as described in subsection 9.13.18 below. These tags will all have the private creator ID of “AMICAS0” and can be referenced as follows:

Custom Tag	Group	Element
Merge Custom Private Tag One	0x0023	0x1055
Merge Custom Private Tag Two	0x0023	0x1056
Merge Custom Private Tag Three	0x0023	0x1057
Merge Custom Private Tag Four	0x0023	0x1058
Merge Custom Private Tag Five	0x0023	0x1059
Merge Custom Private Tag Six	0x0023	0x1060
Merge Custom Private Tag Seven	0x0023	0x1061
Merge Custom Private Tag Eight	0x0023	0x1062
Merge Custom Private Tag Nine	0x0023	0x1063
Merge Custom Private Tag Ten	0x0023	0x1064

NOTE: Custom tags can only be used with the **ReplacementAlways** and **ReplacementWhenExist**, as described below, and only as a left-hand-side (*i.e.*, assign **to**; not assign **from**).

In general, DICOM attributes with the following data types are supported for use with the DICOM Header Modification rules:

Type of Rule	Supported Data Types
Replacement	AE, AS, CS, DA, DS, DT, IS, LO, LT, PN, SH, ST, TM, UI, UT
Removal	Any
Prefix/Suffix Appendage	AE, AS, CS, DA, DS, DT, IS, LO, LT, PN, SH, ST, TM, UI, UT
Truncate	AE, AS, CS, DA, DS, DT, IS, LO, LT, PN, SH, ST, TM, UI, UT
Validate	AE, AS, CS, DA, DS, DT, IS, LO, LT, PN, SH, ST, TM, UI, UT
Transform	AE, AS, CS, DA, DS, DT, IS, LO, LT, PN, SH, ST, TM, UI, UT

The following types of DICOM Header Modification rules can be created:

Rule	Description
AEHostOrIPMatcher	A DICOM instance will match the AEHostOrIPMatcher matcher if the source of the DICOM instance is the same as AE Title, Host Name or IP Address specified in the matcher.
DICOMAttributeMatcher	A DICOM instance will match the DICOMAttributeMatcher matcher if the tag specified in the matcher is present in the DICOM instance and the value specified in the matcher matches that of the attribute in the DICOM instance.

Rule	Description
Prefix	The Prefix modification will add a prefix to the given DICOM attribute. This modification is valid for string-type attributes. If the resulting value would be too long for the corresponding VR (i.e. 64, 16, 16 respectively), an error will be logged and the instance will be rejected.
Suffix	The Suffix medication will add a prefix to the given DICOM attribute. This modification is valid for string-type attributes. If the resulting value would be too long for the corresponding VR (i.e. 64, 16, 16 respectively), an error will be logged and the instance will be rejected.
ReplacementAlways	The ReplacementAlways modification will replace the value in the given DICOM attribute with a new value regardless of whether the DICOM instance already has such an attribute. This modification is valid for attributes with string or numeric types. If the value to place in the attribute is invalid for the VR (e.g. too long for a string, not a number for an integer, etc.) then an error will be logged and the value will not be changed.
ReplacementWhenExist	The ReplacementWhenExist modification will replace the value in the given DICOM attribute with a new value only if the DICOM instance already contains a value for the attribute. If the DICOM instance does not contain a value for the specified attribute, no action will be taken.
Removal	The Removal modification will remove the specified DICOM attribute.
Truncate	The Truncate modification will trim the value of the specified DICOM attribute to a certain number of characters. This modification is valid for string attributes. If the specified attribute is not present in the DICOM instance no action is taken.
Null	The Null modification does nothing and is typically only needed to support a global rule.
ValidatePresenceFix	A modification that will check that a string or numeric DICOM attribute is present and if not it will log a warning and add the attribute with the given value.
ValidatePresenceReject	The ValidatePresenceReject modification will check that a DICOM attribute of any type is present and if not it will log a warning and reject the instance.
ValidatePresence	The ValidatePresence modification will check that a DICOM attribute of any type is present and if not it will log a warning and continue to accept the DICOM instance as-is.
ValidateVRFix	The ValidateVRFix modification will check that if a string or numeric DICOM attribute is present it matches the specified VR (either type, length or both). If the DICOM attribute is not present, the modification does nothing. If the DICOM attribute is present and doesn't match the VR a warning is logged and the VR and value of the attribute are changed. When fixing, the VR type and length used for the fixed value will always be the implicit type and length for the attribute since it doesn't make sense to create a value that doesn't match the DICOM standard.
ValidateVRReject	The ValidateVRReject modification will check that if a string or numeric DICOM attribute is present it matches the specified VR. If the DICOM attribute is not present, the modification does nothing. If the DICOM attribute is present and doesn't match the VR a warning is logged and the instance is rejected.

Rule	Description
ValidateVR	A modification that will check that if a string or numeric DICOM attribute is present it matches the specified VR. If the DICOM attribute is not present, the modification does nothing. If the DICOM attribute is present and doesn't match the VR a warning is and the instance is accepted as-is.
ValidateValueFix	The ValidateValueFix modification will check that if a string or numeric DICOM attribute is present its value matches the specified regular expression(s). If the DICOM attribute is not present, the modification does nothing. If the DICOM attribute is present and its value doesn't match the regular expression a warning is logged and the value is changed.
ValidateValueReject	The ValidateValueReject modification will check that if a string or numeric DICOM attribute is present its value matches the specified regular expression. If the DICOM attribute is not present, the modification does nothing. If the DICOM attribute is present and its value doesn't match the regular expression a warning is logged and the instance is rejected.
Transform	The Transform modification can set the value of a DICOM attribute based on the value of a single other DICOM attribute (including itself) in an if/elseif/else type condition. If the DICOM attribute is not present it will be added and if it is present its value will be replaced.

3.2.2. StorageServer Configuration files

The following configuration files (located in <DriveLetter>:\amicas2\config\) control StorageServer's behavior:

Configuration File	Description
AETitletohost.map	Maps DICOM device identifiers to the Merge PACS Server and port (AETitle to IP address/Port map file). StorageServer uses this file for autorouting incoming images and for responding to Query/Retrieval requests.
storageserver.ini	Contains many of the system's specific settings, such as location and extent of error and trace logging, DICOM port used, etc.
storageserversop.lst	Lists of SOP instance UIDs. Any incoming DICOM associations will be verified against this file.
DicomHeaderModification.rul	Defines the rules that are used to perform automatic DICOM Header Modification.

3.3. Importer

When Merge PACS is configured to run in **Standalone** mode, the Importer:

- Compresses DICOM images using both JPEG 2000 lossy and lossless compression, stores them in the Merge PACS Data directory, and updates the Image database with the demographic data.
- Stores compressed images received via Telemedicine in the Merge PACS Data directory, and updates the Image database with demographic information.

- If the Default Auto Validation scheme has been enabled, marks studies as being “valid” or “invalid” according to a customizable definition of validity. Valid studies are then assigned a worklist status of “Unread” while invalid studies are assigned a status of “Unverified”.
- If the Order Matching Validation scheme has been enabled, assigns studies a worklist status of “Order Matching” if the DICOM header information those studies contain incorrect or incomplete information when compared to the associated orders. An automated Patient Data Editor process is then launched to reconcile the DICOM information with the order information. If successful, the studies are assigned a worklist status of “Unread”; otherwise, they are assigned a status of “Unverified”.
- If the AETitle Validation scheme has been enabled, assigns studies to groups based on the AETitle of the forwarding modality or Server.
- If the SourceTimeValidator Validation scheme has been enabled, automatically sets the status of incoming studies based on the AETitle of the forwarding modality, the day of the week they were received and the time of day they were received.
- Associates studies with groups based on the Provider Code in various fields of the study’s order (this is an optional feature and assumes that a RIS is being used to send orders to the Merge PACS Server).
- Optionally, can generate accession numbers to be used for delivery of outbound order messages and queues these messages for delivery by the Outbound HL7 Interface.

NOTE: If Merge PACS is configured to run in **Integrated** mode, all validation tasks are handled by the EA Notification Processor, as described in Section 3.4 below.

When Merge PACS is configured to run in **Standalone** mode, the following study level attributes will only be updated once based on the first instance of the study imported (if the initial values are empty, they will not be updated with a non-empty value):

- **Study Description**
- **Referring Physician's Name**
- **Name of Physician(s) Reading Study**
- **Study Date**
- **Study Time**
- **Patient's Weight**
- **Patient's Age**
- **Patient's Size**
- **Pregnancy Status**
- **Institution Name**

NOTE: If Merge PACS is configured to run in **Integrated** mode, the same attributes will be updated with the latest incoming value.

3.3.1. Process activities

a. Image Compression

The Importer takes a copy of the images that are saved to disk by StorageServer (in `<DriveLetter(s)>:\PACS2DATA\Dicom`) and compresses them using **JPEG 2000** compression algorithm as either **lossless** or **lossy** images.

- The target **lossy** compression ratio with JPEG 2000 lossy images is determined by the type of modality:

Modality	JPEG 2000 Compression Ratio (DICOM/Merge PACS)
CAT Scan (CT)	6/1
Magnetic Resonance Imaging (MR)	6/1
Single-Frame Ultrasound (US)	6/1
Computed Radiography (CR)	20/1
Digital Radiography (DR, DX)	20/1
Secondary Capture (SC)	20/1
Other Radiography (RG)	20/1
Other (including RF, VI, and PT)	10/1

NOTE: Some types of images, such as multi-frame US and multi-frame MR, are not compressed by the system. These files are stored as **.dcm** files. It is recommended to have compression enabled at the modality, especially for breast tomosynthesis (DBT) images.

NOTE: Non-image objects, such as DICOM SR and DICOM AU, are not compressible objects. Hence these files are not stored in Lossless volume and are instead directly stored into the compressed volumes.

- Images using JPEG 2000 lossy compression are stored as ***.jp2** files.
- Images using JPEG 2000 lossless compression are stored as ***_ll.jp2** files.
- Lossy compression is applied to images with a DICOM photometric interpretation of MONOCHROME1 or MONOCHROME2 (both 8 bit and 16 bit data).
- Files received using compressed transfer syntax are also stored as **.dcm** files.
- The following transfer syntax UIDs are used for uncompressed and compressed DBT:

Transfer Syntax UID	Transfer Syntax Name
1.2.840.10008.1.2	Implicit VR Endian: Default Transfer Syntax for DICOM
1.2.840.10008.1.2.1	Explicit VR Little Endian
1.2.840.10008.1.2.4.90	JPEG 2000 Image Compression (Lossless Only)
1.2.840.10008.1.2.4.70	JPEG Lossless, Nonhierarchical, First- Order Prediction (Processes 14 [Selection Value 1]): Default Transfer Syntax for Lossless JPEG Image Compression

- The manner in which Merge PACS treats uncompressed, JPEG 14 lossless and JPEG 2000 lossless images whose “Lossy Image Compression” DICOM attribute is set to “1” depends on the Recompressed Lossy Ratio that is set for each modality within the Merge PACS database as follows:
 - Setting the Recompressed Lossy Ratio to a value of “1” will result in JPEG 2000 Lossless being used for recompression, and “Lossless” will be displayed in the Recompressed Lossy Ratio column (this is the default value).
 - Setting the Recompressed Lossy Ratio to a value greater than “1” will result in that value being used as the target compression ratio for compression/recompression, and that value will be displayed in the Recompressed Lossy Ratio column.
 - Setting the Recompressed Lossy Ratio to a value of “0” will result in the Lossy Compression ratio being pulled from the “Lossy Image Compression Ratio” DICOM attribute or, if that attribute is not available, from the “Derivation Description” attribute. That value will then be used as the target compression ratio for compression/recompression and “Original” will be displayed in the Recompressed Lossy Ratio column. If there are multiple values in the DICOM attribute, the highest value will be used.

NOTE: If Merge PACS receives an image that is currently lossy compressed (*i.e.*, the transfer syntax indicates lossy), or has at any point in its past been lossy compressed (*i.e.*, the Lossy Image Compression flag is set to non-0 value), Merge PACS will compress the image as per the currently configured Recompressed Lossy Ratios (lossless or lossy and place the output in the **Lossy** folder. A copy of this image will never be placed in the **Lossless** folder. **As a result, if lossy compression has been configured as the Recompressed Lossy Ratio, there will be no lossless compressed version of the study saved.**

- By default, the Importer stores the JPEG2000 lossy and JPEG2000 lossless data in the following directories:
 - <DriveLetter(s)>:\PACS2DATA\Compressed
 - <DriveLetter(s)>:\PACS2DATA\Lossless
- Each directory [or directories, if Multi-Volume support is enabled] contains folders for each patient; the folders contain a subfolder for each separate patient study. The Importer also adds the patient demographics to the AmicasDemographic tables in the Image database.
- In addition, the Importer creates several other files for each study:
 - An **XML** file, stored in the patient folder and used by the Merge PACS Viewer.
 - A **DEMO** file, containing demographics for each image.

b. Image Deletion

JPEG2000 lossy and JPEG2000 lossless images are stored on the server until disk space reaches a pre-determined limit, at which point the oldest studies are deleted to make room for the newest by the CommandScheduler, as defined below. When images are deleted and how many images are deleted is determined by the **Compressed Free Space Low Water Mark** and **Compressed Free Space High Water Mark** settings for **lossy** images, and by the **Lossless Free Space Low Water Mark** and **Lossless Free Space High Water Mark** settings for **lossless** images. When the available free space on a compressed volume falls below the **low water mark**, the CommandScheduler will begin deleting studies from that volume on a first-in, first-out basis until the available free space reaches the **high water mark**.

CAUTION: If the lossy copy of a study is deleted, the lossless copy of that study will also be deleted.

Image deletion for compressed images can be turned off completely by changing the settings for the CheckResources command described in Section 3.8 below. This may be desired if a site wished to prevent the loss of any of a site's online store of "Forever Prior" images. If this is the case, however, Merge PACS will stop importing new images entirely when a specified **absolute minimum free space** setting is reached, and importation will not resume until additional space is added of the absolute minimum free space setting is reduced.

By default, the Compressed Free Space Low Water Mark is set to **675 MB** for each compressed image volume, and the Compressed Free Space High Water Mark is set to **1000 MB**.

c. Automatic Validation

The Importer processes the five types of Automatic Validation classes that are available with Merge PACS:

- If the **Default Auto Validator** class has been enabled, the Importer marks each study as being "Valid/Unread" or "Invalid/Unverified" according to a customizable definition of validity. The three possible definitions of validity are as follows:
 - There is an order that has the same MRN and Accession Number as found in the DICOM header information of the study (a "matching order")
 - This study has no jailed images
 - There is a matching order for the study *and* the study has no jailed images
- If the **Order Matching Validator** class has been enabled, the Importer will check if there is a **completely** matching order for each study (*i.e.*, all the DICOM header information matches the order information).
 - If there is a completely matching order, the study will be given a workflow status of "Unread".
 - If there is no completely matching order, the following steps are performed:
 1. The study will be assigned a transient workflow status of "OM" (Order Matching).
 2. The Auto PDE Process will be launched in an attempt to reconcile the DICOM header information with the order information.
 3. If the Auto PDE process is successful, the study will be given a permanent workflow status of "Unread" and the validation process will be run again.

4. If the Auto PDE process is unsuccessful, the study will be given a permanent workflow status of “Unverified”.
- If the **Institution/Modality Validator** class has been enabled, the Importer will mark studies as being “Valid/Unread” if the Institution Name and/or Modality in the studies’ DICOM header matches a customizable list of institutions and modalities contained in a separate XML file.

NOTE: The two validation classes described above can run in conjunction, in which case you would use only the Default Auto Validator to check for jailed images. When both validation classes run in conjunction, a study will be given a workflow status of “Unverified” if it fails either of the two validation classes.

- If the **AETitle Validator** class has been enabled, the Importer will automatically assign studies to groups based on the AETitle of the forwarding modality. Note that the AETitle Validator class does not actually change the validation status of the affected studies and can therefore be used in conjunction with either of the other two Validation classes.
- If the **SourceTimeValidation** class has been enabled, the Importer will automatically set the status of studies based on the AETitle, date and time of the studies, according to the rules that can be customized on a site-by-site basis.

Merge PACS can be optionally configured to prevent invalid studies from being auto-routed (or forwarded via Telemed, if Order Matching Validation is enabled), a process referred to as “Controlled Delivery.” It can also be configured to prevent invalid studies from appearing in query lists within the Merge PACS Workstation, an example of a process referred to as “Controlled Access.”

Automatic Validation can also be used to automatically assign studies to groups based on the Provider Code found in various fields of the orders generated by a site’s RIS.

d. Accession Number Generation

In situations where a site does not have a HIS/RIS (or in a hub/spoke model where one or more spoke servers do not have a HIS/RIS), Merge PACS can be configured to generate and send out study-based orders to external applications such as PowerScribe which require order information. In addition, Merge PACS can be configured to generate unique accession numbers that can be used for delivery of an outbound order message for these orders.

Once configured to do this, the Importer generates an accession number every time an incoming DICOM image does not contain an accession number. This number is then used for delivery of an outbound order message and sent to the HL7 Interface Queue (as described in subsection 3.8.1 below).

3.3.2. About Image Compression

Merge PACS uses two types of image compression to facilitate enterprise distribution of image files:

Compression	Description
Irreversible	Otherwise known as “ lossy ” compression, irreversible image compression removes redundant or unimportant information in the original image data with no reduction in clinical diagnostic image quality.

Reversible

Otherwise known as “**lossless**” compression, reversible image compression reduces the file size of images by encoding them more efficiently without removing any information.

The American College of Radiologists (ACR) has issued the following statement with regard to image compression:

Data compression may be performed to facilitate transmission and storage. Several methods, including both reversible and irreversible techniques, may be used under the direction of a qualified physician, with no reduction in clinical diagnostic image quality. The types and ratios of compression used for different imaging studies transmitted and stored by the system should be selected and periodically reviewed by the responsible physician to ensure appropriate clinical image quality. (ACR Standards. *ACR Standard for Digital Image Data Management*. American College of Radiology, 1998, revised 2001, p. 3.)

Where irreversible image compression is being used, it is the user's responsibility to determine the trade-off between compression ratio and diagnostic effectiveness for specific image study indications. To facilitate this determination, the Merge PACS Viewer component of the Merge PACS Workstation displays the compression ratio along with the images in the Web browser or imprints the compression ratio on images that may be viewed through standard DICOM workstations.

The following are references to peer-reviewed articles on wavelet compression in general, as well as JPEG 2000 in particular:

- Goldberg MA, Pivavarov M, Mayo-Smith WW, et al. *Application of wavelet compression to digitized radiographs*. AJR 1994; 163:463-468
- Goldberg MA, Gazelle GS, Boland GW, et al. *Focal hepatic lesions: effect of three-dimensional wavelet compression on Detection at CT*. Radiology 1997; 302:159-65
- Erickson BJ, Manduca A, Pallison P, et al. *Wavelet Compression of Medical Images*. Radiology 1998; 206:599-607
- Savcenko V, Erickson BJ, Palisson PM, et al. *Detection of Subtle Abnormalities on Chest Radiographs after Irreversible Compression*. Radiology 1998; 206:609-616
- Foos D, Muka E, Slone R, et al: *JPEG 2000 compression of medical imagery*. The International Society For Optical Engineering (SPIE), San Diego, CA 2000

3.3.3. Supported Image Types

Merge PACS supports the following image types:

Image Type	Photometric Interpretation	Store	Process		View		Overall Support
			Lossless	Lossy	Lossless	Lossy	
Single-Frame	MONOCHROME1	DICOM	JPEG2000	JPEG2000	JPEG2000	JPEG2000	Yes
	MONOCHROME2	DICOM	JPEG2000	JPEG2000	JPEG2000	JPEG2000	Yes
	PALETTE COLOR	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	RGB	DICOM	DICOM	JPEG	DICOM	JPEG	Yes
	HSV	DICOM	DICOM	DICOM	N/S	N/S	No
	ARGB	DICOM	DICOM	DICOM	N/S	N/S	No
	CMYK	DICOM	DICOM	DICOM	N/S	N/S	No

Image Type	Photometric Interpretation	Store	Process		View		Overall Support
			Lossless	Lossy	Lossless	Lossy	
	YBR_FULL	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_FULL_422	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_PARTIAL_422	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_ICT	DICOM	DICOM	DICOM	N/S	N/S	No
	YBR_RCT	DICOM	DICOM	DICOM	N/S	N/S	No
Multi-Frame	MONOCHROME1	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	MONOCHROME2	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	PALETTE COLOR	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	RGB	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	HSV	DICOM	DICOM	DICOM	N/S	N/S	No
	ARGB	DICOM	DICOM	DICOM	N/S	N/S	No
	CMYK	DICOM	DICOM	DICOM	N/S	N/S	No
	YBR_FULL	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_FULL_422	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_PARTIAL_422	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_ICT	DICOM	DICOM	DICOM	DICOM	DICOM	Yes
	YBR_RCT	DICOM	DICOM	DICOM	N/S	N/S	No
Lossy Compressed	Any	DICOM	DICOM	DICOM	DICOM	DICOM	Yes*

N/S - Not supported

DICOM - Store DICOM images as received

JPEG2000 - Use JPEG2000 compression

JPEG - Use JPEG compression

Table 3-3: Merge PACS Supported Image Types

NOTE: * Supported transfer syntaxes are: Run Length Encoding, JPEG Baseline (Lossy) (Process 1), JPEG Extended (Lossy) (Process 2 (8-bit) and Process 4 (12-bit)), JPEG Lossless Non-hierarchical (Process 14), JPEG Default Lossless Non-hierarchical, first-order prediction (Process 14 selection value 1), JPEG 2000 (Lossless only), JPEG 2000 (Lossy or Lossless).

NOTE: For DBT modalities such as Hologic Selenia Dimensions and GE SenoClaire that support BTO images (multi-frame images) it is strongly recommended to have the compression enabled on the modality device. BTO images can be very big in size (~2.4GB). Compression factor for JPEG Lossless compression is ~ 4:1 and for JPEG 2000 Lossless is ~8:1.

3.4. EA Notification Processor

When Merge PACS is configured to run in **Integrated** mode, the EA Notification Processor receives notifications from iConnect Enterprise Archive (ICEA) about new or changed DICOM instances available in the archive as well as for quality control (QC) messages indicating various changes to the DICOM instances already available (*e.g.*, demographic changes, splits, merges, deletes).

In addition, the EA Notification Processor performs the following tasks:

- If the Default Auto Validation scheme has been enabled, marks studies as being “valid” or “invalid” according to a customizable definition of validity. Valid studies are then assigned a worklist status of “Unread” while invalid studies are assigned a status of “Unverified”.
- If the AETitle Validation scheme has been enabled, assigns studies to groups based on the AETitle of the forwarding modality or Server.
- If the SourceTimeValidator Validation scheme has been enabled, automatically sets the status of incoming studies based on the AETitle of the forwarding modality, the day of the week they were received and the time of day they were received.
- Associates studies with groups based on the Provider Code in various fields of the study’s order (this is an optional feature and assumes that a RIS is being used to send orders to the Merge PACS Server).
- Optionally, can generate accession numbers to be used for delivery of outbound order messages and queues these messages for delivery by the Outbound HL7 Interface.

NOTE: If Merge PACS is configured to run in **Standalone** mode, these tasks are handled by the **Importer**, as described in Section 3.3 above.

CAUTION: In integrated mode, if ICEA fails to store, notify, or maintain objects/studies, those objects/studies may not be present/accessible in Merge PACS.

3.5. Sender

When Merge PACS is configured to run in **Integrated** mode, the Sender forwards DICOM instances received at the Merge PACS server on to ICEA for storage. DICOM instances will still arrive at the Merge PACS server for items created by the viewer (*e.g.* KO, PR, SC) as well as those sent via telemedicine transfer from other Merge PACS servers. Since Merge PACS does not store DICOM instances in integrated mode it will store these instances temporarily in a cache and the Sender will send them on to ICEA, deleting them from the cache once they have been accepted.

3.6. AutoRouter

When Merge PACS is configured to run in **Standalone** mode, AutoRouter (working with Forwarder) sends copies of a DICOM image to a DICOM workstation or archive at the same time the image is being stored for processing by Merge PACS.

This process routes files based on rules created by your organization. Each rule consists of at least two parts:

- Criteria for routing.

- Location or identifier of the DICOM target.

Rules are stored in **autorouter.rul**. The typical full pathname for this file is:

```
<DriveLetter>:\amicas2\config\autorouter.rul
```

This file can be configured to route images based any element of a DICOM header. The file identifies the target DICOM device for each set of identifying characteristics. Some of the most common criteria are:

- Time of day
- Source of data
- Study description
- AETitle of the sending device

NOTE: When a study is edited using the Problem Studies Tool (described in subsection 9.9.4 below), the AETitle of the study's "sending device" is changed to "PROBSTDTool." Therefore, if a study that was a candidate for auto-routing based on AETitle was edited using the Problem Study Tool, the study will need to be manually routed via the Send Studies tool (as described in subsection 9.9.3 below).

3.7. Forwarder

When Merge PACS is configured to run in **Standalone** mode, Forwarder transmits DICOM images based on the entries in the Forwarding Queue created by the AutoRouter. In other words, AutoRouter identifies and lists the routing jobs and then Forwarder executes these jobs. Images are converted by Forwarder in real time based on the transfer syntax requested by the receiving target.

Forwarder also supports the **Study Import Notification Service** that provides **DICOM Instance Availability Notification (IAN)** and **Basic Study Content Notification (BSCN)** notification messages when studies are imported into the Merge PACS Server. Notification messages are sent after a study is imported into the Server when it becomes available for reviewing (*i.e.*, 2 minutes after receiving the last image of the study) even if there is no matching order for the study. The content of the notification messages is described in *Merge PACS 8.2 DICOM Conformance Statement*. The notifications are sent by the Forwarder and follow the same association rules as for the forwarded images (AssociationIdleTimeout, MaximumJobDispatcherSleepTime, DownDestinationCheckInterval, etc.). The forwarder prioritizes IAN and BSCN messages over image forwarding.

Forwarder monitors the Forwarding Queue. When it finds an entry in the queue, it follows its instructions to route the appropriate DICOM image to the target DICOM device. After transferring the DICOM image, Forwarder removes the entry from the queue.

3.8. CommandScheduler

Merge PACS performs two types of maintenance activities:

Type of Command	Description
Periodic commands	Commands routinely executed on a regular schedule
Explicit commands	Commands entered explicitly by a user that are executed one time

The CommandScheduler stores a command in its queue when:

- For **periodic** commands, the command is automatically scheduled according to the timetable specified for it.
- For **explicit** commands, the command is manually submitted by an administrator or automatically submitted by one of the other Merge PACS processes.

Once it executes the command, the CommandScheduler removes the explicit commands from the CommandScheduler Queue and reschedules the periodic commands.

3.8.1. Periodic commands

Merge PACS can be configured so that CommandScheduler routinely executes any of the periodic commands listed below:

Parameter	Description
UpdateStatus	Generates an email to your technical support staff and to Merge Healthcare Technical Services about the condition of Merge PACS. This email contains performance statistics that over time can be used by Merge Healthcare support to monitor system performance and to alert Technical Services to a potential problem before it happens. You can define the intervals at which the <i>UpdateStatus</i> commands is executed.
CheckResources (Standalone Mode only)	When Merge PACS is configured to run in Standalone mode, checks disk space resources periodically (default setting is once every minute) to determine if compressed and lossless images need to be deleted to free up space on any of the image data drives. Images are deleted on a first-in, first-out basis, and the determination when to delete and how many images to delete is based on the compressed free space high and low water mark settings. CheckResources is also responsible for deleting image data from Merge PACS. The CheckResources command can be configured to delete DICOM images based on time and/or watermark settings, and to delete Lossy and Lossless images based on watermark settings. While watermarking the Compressed volume, CheckResources will also move the specific SOP classes to the External volume and remove the Study fully from Compressed volume. CheckResource is also has the additional responsibility of watermarking the External volume based on the watermark settings configured for External volume.
SystemMonitor	Triggers an email notification when the Image Import Queue exceeds a certain threshold.
PeriodicCleanupDB	Periodically removes old database entries.
AnetCommunicatorEntries	Places a command in the Merge PACS Queue database table to send data to Watch.
ImportLogExtractorInvoker	Extracts the import log to a text file so it can be imported into a spreadsheet.
AuditExtractorInvoker	Initiates extracting study, access, and performance log commands

NOTE: Two other commands, **ovRIS** and **AccessLogExtractorEntries** are only used by single sites and are specially customized for use at those sites.

3.8.2. Explicit commands

The following explicit commands can be executed by a user with the appropriate access authorizations.

Command	Description
Delete	Deletes a specified study from the Merge PACS Server
Validate	Modifies study demographics
Update	Modifies patient demographics
Merge	Merge two patients together

The following explicit commands may be initiated by other processes running within Merge PACS [note that some of the commands may also be user initiated as described above]:

Command	Description
Update	Modifies patient demographics
Validate	Modifies study demographics
CleanupDB	Cleans up database log tables
SendMail	Sends general e-mail messages
SendStatusMessage	Sends update e-mail messages
AnetCommunicator	Transfers study event log entries to Watch Monitoring Station
InportLogExtractor	Extract import logs to files
AuditLogExtractor	Extract study, access, and performance logs to files
QCFunctionMove	Moves a specified image/series/study from the Merge PACS Server
QCFunctionMerge	Merges a specified series/study from the Merge PACS Server
QCFunctionDelete	Deletes a specified image/series/study from the Merge PACS Server
INIMAGEQUEUE	Copies the specified images/series from the Merge PACS Server
StudyRetrieve	Retrieve the Study from the configured Archive. In Integrated mode, this command is used to retrieve Studies from integrated EA Server and Extended nodes, if configured.
QCWaterMarkStudy	In Integrated mode, the command intended to WaterMark the study on receiving the notification from the EA Server.
PIConsolidator	Automatically consolidate patient identity based on PCS. This command will be queued after every QC request to PACS.

3.8.3. Outbound HL7 Interface

The Merge PACS HIS/RIS Integration component described in subsection 1.1.3 above is made up of two primary components:

Component	Description
ImageManager	Allows Merge PACS to receive information from HIS/RIS systems (see the following section).
Outbound HL7 Interface	A thread running within CommandScheduler and which can be used to send various status messages about studies back to the HIS/RIS systems if so desired. In addition, in situations where a site does not have a HIS/RIS (or in a hub/spoke model where one or more spoke servers do not have a HIS/RIS), Merge PACS can be configured to generate and send out study-based orders to external applications such as PowerScribe which require order information. Finally, Merge PACS can also be configured to generate unique accession numbers that can be used for delivery of an outbound order message.

CAUTION: If the Outbound HL7 Interface is configured to send out system-generated orders, an order will be sent whenever a study is imported into Merge PACS regardless of whether an order already exists. Therefore, to avoid duplication of orders, the Outbound HL7 Interface should only be configured to send out temp orders in situations where a site does not have a HIS/RIS already generating order information.

CAUTION: If the Outbound HL7 Interface is configured to send out system-generated orders, a new system-generated order will be sent if a study's accession number is manually changed from the Quality Control Editor or the Patient and Study Demographics windows from within the Merge PACS Workstation. A new system-generated order will also be sent if additional images are received for an existing study.

When the Outbound HL7 Interface is enabled to send out status messages, each message will include the following information:

- study status
- a link to the images associated with this study (StudyInstanceUID)
- number of images
- the name of the modality technician/operator

The study status included in the message will be one of the following:

- **Study Imported**
- **Study Validated**
- **Study Deleted**

Note that entries with “status = OK” are automatically deleted from the Outbound HL7 Queue after seven days.

3.9. ImageManager

3.9.1. In General

ImageManager allows Merge PACS to receive information from systems that use an HL7 interface, such as HIS/RIS systems. The ImageManager then either performs the requested task directly or sends the information to the appropriate process to be performed.

Typical actions performed by the ImageManager include:

- Storing orders for Modality Worklists in the order table of Merge PACS database so that they can be retrieved by StorageServer for DICOM modalities.
- Updating patient and study demographics.
- Receiving and storing reports associated with particular studies.
- Reconciling orders and order requests.

ImageManager is much like StorageServer in that it listens on a specific port (port 1001 by default) for HL7 events. The ImageManager properties can be modified by editing the ImageManager properties file located at:

```
<DriveLetter>:\amicas2\config\ImageManager.props
```

NOTE: The port used by ImageManager should be something **other** than **80** wherever possible, so as to not conflict with the **WildFly** application described in subsection 1.3.1 above.

3.9.2. Global Update

Merge PACS qualifies studies by MRN when processing A08 (Patient Update) and A34 (Patient merge) requests from a HIS/RIS. By default, if an A08 or A34 request is received that has an MRN that matches a patient but has the wrong Patient Name, the request will be rejected by the Merge PACS. Merge PACS can be configured, however, to not check to see if the Patient Name stored in its database matches that sent by the HIS/RIS and will instead return all studies that have an MRN that matches that sent by the HIS/RIS (a process known as “Global Update”).

In addition, Merge PACS can be configured to automatically update patient demographics in its databases, for both the merge-from patient and the merge-to patient, with the patient demographics in the A34 request.

3.9.3. Error and Trace Logs

As described in Section 3.16 below, error and trace logs for ImageManager are stored in the `<DriveLetter>:\amicas2DATA\Log` directory on the Merge PACS Server. Each log file is limited in size to approximately 10MB, and the logs will be rolled when size of current log exceeds the 10MB threshold. The current log is named **ImageManager.errors** or **ImageManager.log**, and old logs have an integer appended to the end of the name (e.g., **ImageManager.log.1**, **ImageManager.log.2**, etc.).

By default, a maximum of 500 old logs will be retained. This number can be changed to meet the needs of a specific site, however.

3.9.4. iConnect® Enterprise Archive™ Compatibility

When processing A08 (Patient Update), A34 (Patient Merge) and A40 (List of Patients Merge) requests from a HIS/RIS, if multiple patients having the same MRN are found, the request is rejected by default. However, when Merge PACS Server is configured with iConnect® Enterprise Archive™, you may want to process an HL7 message if the patient's name (depending on Patient Comparison Strategy) matches a single patient when you have duplicate patient records for a given MRN. This feature will perform an additional check when duplicate patients are found for a given MRN by comparing the Patient's name to the list of patients matching the MRN and then processing the patient if there is a name match to only one patient.

NOTE: This feature applies when the Patient Comparison Strategy is set to **LastNameOnlyComparisonStrategy** or **StrictComparisonStrategy**. For more information on Patient Comparison Strategy, refer to subsection 2.10.1 above.

In addition, when processing A34 (Patient merge) and A40 (List of Patients merge) requests from a HIS/RIS, you may want/need to enforce extra IPID checks. These checks will ensure that the IPIDs in Patient Demographics segment and the merge segments of a HL7 message are present, are the same (case sensitive comparison) and if default configured IPID should be used.

NOTE: Although this feature is meant to be used when iConnect® Enterprise Archive is set up as an archive, you can still use this feature when deploying Merge PACS server without an archive.

3.9.5. Encapsulated PDF Reports

Merge PACS supports Encapsulated PDFs for reports. Currently neither Merge PACS nor iConnect Enterprise Archive generates reports in this format, although Merge Cardio™ does. If the encapsulated PDFs come into Merge PACS from a 3rd-party system, they will likely always display as “Preliminary” status regardless of the actual report status. If the Encapsulated PDFs come into Merge PACS from Merge Cardio, however, the report status will be set appropriately (Preliminary or Final).

NOTE: If the **Completion Flag** DICOM attribute (0040,A491) is set to COMPLETE and **Verification Flag** (0040,A493) is set to VERIFIED, the status will be considered Final; otherwise, the status will be considered Preliminary.

3.9.6. Study End Date/Time

RTWL worklists can be configured to display Study End Date/Time instead of or in addition to Study Date/Time. By default, the value for Study End Date/Time is taken from the OBR-7.1 field of the ORM (Order Complete) message. Merge PACS can be configured, however, to pull the information from the ORC segment of the ORM message instead and can also be configured to pull the information from any specified field and sub-component of the ORM or ORC segment.

Since Study End Date/Time can only be pulled from completed orders, Merge PACS should be configured to only look for orders that have the letter or numeric value corresponding to “Complete” status for this particular site.

NOTE: Since Merge PACS pulls the Study End Date/Time value from orders, this information will disappear if and when the orders are purged (similar to other order information). When this happens, the End Date/Time column will show as empty for the exam.

3.10. Listener

The Merge PACS Viewers can retrieve DICOM images from external DICOM archives using DICOM Query/Retrieve requests. In the event a user cannot locate a study in Merge PACS, they send a request from the Merge PACS Workstation to Listener (assuming they have been given the appropriate privileges). Listener performs a DICOM Query/Retrieve of the DICOM archive.

The archive responds back to the viewer client via Listener with a list of all studies meeting the request criteria. The user can then select which studies are to be retrieved to Merge PACS. StorageServer receives all the retrieved studies, processing them in the same way it does new studies. The user can then request the retrieved studies from Merge PACS.

The following figure illustrates this process:

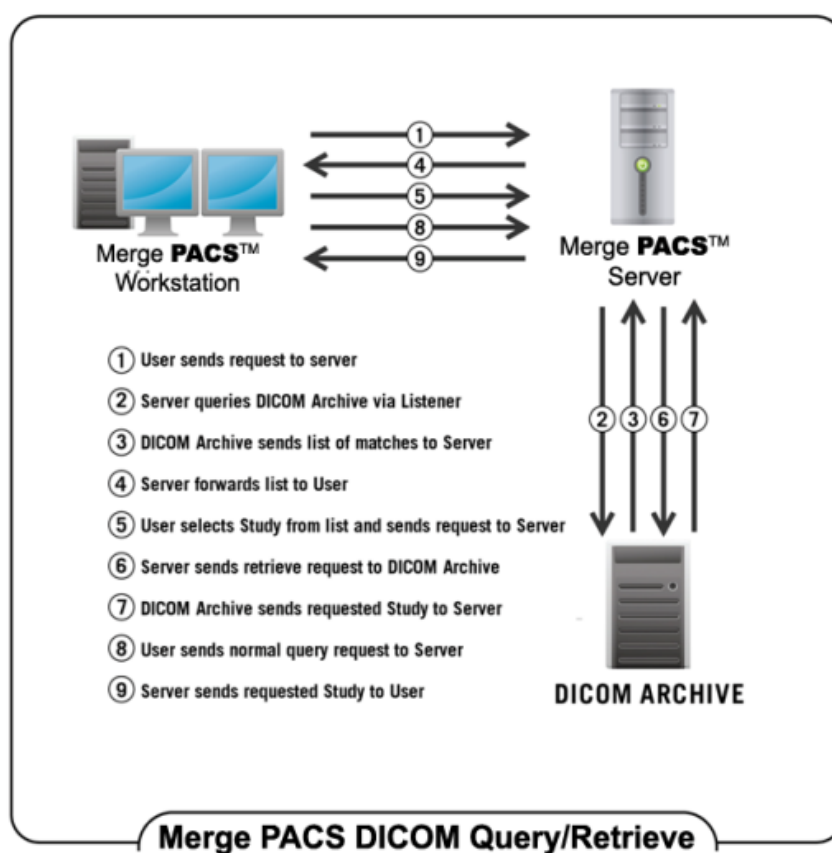


Figure 14: DICOM Query/Retrieve via Listener

When Merge PACS is configured to run in **Integrated** mode and Multiple Patient Identity (MPI) support is enabled, as described in subsection 2.10.2 above, Listener is also responsible for sending and receiving the HL7 messages used for MPI to and from iCEA.

3.11. WorklistServer

3.11.1. In General

The WorklistServer process keeps track of all orders and studies and responds to search requests from the Worklist client. In addition, it populates individual worklists with data when a user first subscribes to a worklist.

Once a newly subscribed worklist is populated by the WorklistServer, further updates to the worklist will be done from other Amicas Services by notifying the WorklistServer with the changes via JMS Message for the topic to which the WorklistServer is subscribed.

3.11.2. Error and Trace Logs

As described in Section 3.16 below, error and trace logs for WorklistManager are stored in the `<DriveLetter>:\amicas2DATA\Log` directory on the Merge PACS Server. Each log file is limited in size to approximately 10MB, and the logs will be rolled when size of current log exceeds the 10MB threshold. The current log is named `Worklistserver_1.errors` or `Worklistserver_1.log`, and old logs have an integer appended to the front of the name (e.g., `WorklistServer_1.log.1`, `WorklistServer.log.2`, etc.).

By default, a maximum of 500 old logs will be retained. This number can be changed to meet the needs of a specific site, however.

3.12. DICOMPrint

The DICOMPrint process monitors incoming DICOM Print requests, and submits the jobs to the configured DICOM printers.

3.13. ECDR Manager

The ECDR Manager process monitors the CD Queue for CD/DVD writing requests and then composes and sends CD/DVD writing jobs studies to the shared Enterprise CD Writer.

CD/DVD writing requests can be entered into the CD Queue in one of two ways:

- A request can be manually entered from the Enterprise CD Writer section of the Merge PACS Management Web Page.
- A request can be entered by the Importer in response to an Auto Writing rule.

NOTE: Enterprise CD Writer is not supported for use with Windows 2012 R2.

For detailed information on manually creating CD writing requests and viewing the CD Queue, refer to Section 9.10 below.

3.14. Archiver

When Merge PACS is configured to run in **Standalone** mode, Archiver is a background process that runs on the Merge Server. It bundles DICOM images and copies them to an archival storage device, as well as retrieves those images.

For information on using the Merge Archive component, refer to *Merge PACS 8.2 Archive User's Guide*.

3.15. STSServer

When Merge PACS is configured to run in **Integrated** mode, STSServer is a background process that runs on the Merge Server and that handles the DirectPullSTS and MergePacsSTS requests which come from iConnect Enterprise Archive.

3.16. Merge PACS Logs

There are three different types of log files generated by the processes mentioned above: ERROR files, TRACE files and LOG4J log files. For example, when the Amicas service is first started, the StorageServer process is launched and a **StorageServer.error** file and a **StorageServer.trace** are created. These logs and the information they contain are vital for troubleshooting problems that can arise with Merge PACS.

- The **error file** contains information relating to any errors that may have occurred (if no errors have occurred this file should remain empty).
- The **trace file** contains information relating to any tasks that the process has performed or any communication it has received from other sources. How verbose the trace file is depends on how the **StorageServer.ini** file is configured.
- The **log4j** log file is only generated by java processes and contains informational, warning and error messages

Merge PACS logs (including ERROR and TRACE files for the **Archiver** process) are stored in the **<DriveLetter>:\PACS2DATA\Log** directory and can be configured via configuration files located in the **<DriveLetter>:\amicas2\config** directory. Note that if **Multi-Volume Support** is configured on the Server, only the main PACS2DATA directory (typically on the **J:** drive) will contain the system logs.

Chapter 4. Amicas Messaging Service

This chapter discusses in detail the individual processes that are launched by the Amicas Messaging Service. Each of these processes is a functional module of Merge PACS.

The following concepts and terminology are discussed in this chapter:

Term	Description
Auditor	Process that monitors study events (primarily the status of studies being imported into Merge PACS) and marks them as being closed [<i>i.e.</i> , successfully imported] or closed with errors.
MessagingServer	A process which provides interprocess messaging other services.

4.1. Auditor

The Auditor process's primary function is to monitor studies as they are being imported into Merge PACS and indicate when the importation has terminated (whether successfully or unsuccessfully) by marking the study "closed."

A study is considered "active" while it is in the process of being imported and until it is marked "closed" by the Auditor process. The Auditor process will mark a study "closed" under a variety of different circumstances, including the following:

- The study has been successfully imported and validated, either manually or as the result of an auto-validation rule.
- The study has been successfully imported and a set amount of time has elapsed without it being validated.
- The study was not successfully imported and a set amount of time has elapsed.

Note that every study, whether successfully imported or not, will eventually be marked as "closed" and will appear on the Auditor Monitor page, as described in Section 9.6.2 below.

If a study has not been validated, the length of time that Auditor waits before closing a study is determined by the type of **Study Validation** scheme that is being used by the system:

Validation Scheme	Length of Time before Closing Study
No Validation	Two minutes
Auto Validation	Five minutes
Manual Validation	Thirty minutes

NOTE: By default, Auditor uses port 11100 in Merge PACS Server for web monitoring, which may conflict with other third-party components such as HL7 systems.

4.2. MessagingServer

MessagingServer is the server process which provides interprocess communication for other services running on the MERGE PACS Server. This has an embedded HornetQ messaging server which takes care of the server's internal communication. In **Integrated** mode, MessagingServer has the additional responsibility of dispatching notifications received from the EA Server.

NOTE: By default, MessagingServer uses port 11104 in Merge PACS Server for web monitoring, which may conflict with other third-party components such as HL7 systems.

Chapter 5. Security Overview

NOTE: Administration and management privileges (including DB2 administration privileges) are protected by default during the installation of Merge PACS.

There are a number of options available for applying security to the images stored on Merge PACS:

- **Merge PACS Authentication** (default security)
- **Windows-based Authentication** (a.k.a. NT Authentication)
- **Transport Layer Security** (TLS)
- **Merge PACS Web Security**
- **Integrated Windows Authentication**
- **Controlled Access**
- **IP Address and Domain Name Restrictions**
- **Lightweight Directory Access Protocol** (LDAP)
- **Virtual Private Network** (VPN)

NOTE: Transport Layer Security (TLS) is the new industry standard that has replaced the now deprecated Secure Sockets Layer (SSL), although the functionality is the same.

5.1. Merge PACS Authentication

5.1.1. In General

All Merge PACS user interfaces, including the Management Pages and the Merge PACS Workstation have a front-end page that challenges users for a valid username and password.

Merge PACS Authentication authenticates users in the following way:

- Merge PACS will first look to its own internal database to see if the user is authorized.
- If the user cannot be found in the Merge PACS internal database, Merge PACS will then perform Windows-based authentication (as described in Section 5.2 below) to see if the user is authorized to access the system.

If the user has a valid Windows account but does not have an account in the Merge PACS database, he or she will be logged in as a special “guest” account within Merge PACS that, by default, will have access to all exams via the Merge PACS Workstation but can be denied such access if desired.

5.1.2. Merge PACS Authentication and TLS

Merge PACS Authentication is typically used in conjunction with **Transport Layer Security (TLS)**, as defined above. Without TLS in place, usernames, passwords, and all other information sent between the Server and the client are sent in “clear” (non-encrypted) text, meaning they could theoretically be intercepted and used by unauthorized users.

- If Merge PACS is being used over the **Internet**, Merge PACS Authentication **must** be used in conjunction with **TLS** in order to provide reliable security.
- If Merge PACS is being used solely over an **Intranet**, TLS is still recommended to provide end-to-end encryption (“Zero-Trust”).

5.2. Windows-based Authentication

5.2.1. In General

Windows-based Authentication is used to prevent unauthorized users from establishing a Web (HTTP) connection to a Merge PACS Server. The authentication process involves determining whether a user has a valid Windows user account.

When used as part of Merge PACS, Windows authentication provides security in the following two ways:

- It prevents unauthorized access to patient images by requiring a valid user name and login.
- It allows Merge PACS to keep a log of who has accessed the system.

Windows-based Authentication is enabled by default on all Merge PACS 8.2 Merge PACS Servers for existing domain and local users. As described in Section 5.1 above, the default Merge PACS Authentication security scheme will look to the local machine or Windows domain if a user fails to be authenticated against the Merge PACS database.

Note that, by default, any authorized user within the domain will be able to access the Merge PACS Server when Windows-based Authentication is enabled. They will not, however, have access to all exams unless they have been specifically given that privilege (or placed within a group that has that privilege), as described in Section 9.11 below.

By default, domain users must specify their domain name as part of their username in order to be successful authentication. They may do so using any of the following formats:

- domain\username
- domain/username
- username@domain

The system may be configured with a default domain name which will allow domain users to login without specifying their domain.

5.2.2. Windows-based Authentication and TLS

Windows-based Authentication is typically used in conjunction with **Transport Layer Security (TLS)**, as defined above. Without TLS in place, user names, passwords, and all other information sent between the Server and the client are sent in “clear” (non-encrypted) text, meaning they could theoretically be intercepted and used by unauthorized users.

- If Merge PACS is being used over the **Internet**, Basic Authentication **must** be used in conjunction with **TLS** in order to provide reliable security.
- If Merge PACS is being used solely over an **Intranet**, TLS is still recommended to provide end-to-end encryption (“Zero-Trust”).

5.3. Transport Layer Security (TLS)

NOTE: Transport Layer Security (TLS) is the new industry standard that has replaced the now deprecated Secure Sockets Layer (SSL), although the functionality is the same.

Transport Layer Security (TLS) is an industry standard method for providing security for Web-based (HTTP) transmissions over the Internet. With TLS in place, users will have access to patient images and data via a “secure” website, meaning that all data that passes between the user and your Merge PACS Server via that website will be encrypted. In addition, patient images that are transferred from one Merge PACS Server to another via Telemedicine will also be encrypted. This prevents the data being intercepted and read by third parties that have not been granted direct access to the website.

TLS uses the public-and-private key encryption system from RSA, which includes the use of a digital “certificate” to guarantee the server's authenticity to the user [this certificate will need to be obtained from a certifying authority; see below for a list of supported TLS vendors]. A “key” is a long string of characters that the encryption software uses to encrypt and decrypt data. A server that has TLS enabled will have both a “private” key that is not given out to anybody and a “public” key (based on the private key) that is made available to anyone who asks. Anybody who receives a copy of a server's public key can use it to encrypt data, but that encrypted data can then only be decrypted using the server's private key.

When a user first logs onto the Merge PACS Server's secure website, the Server provides a copy of its digital certificate to the user's web browser. This certificate both authenticates the Server's identity to the browser and gives the browser a copy of the Server's “public” key. The browser then creates a private “session” key [*i.e.*, one that will expire once the connection between the browser and the Server is broken], which it encrypts using the Server's public key and sends back to the Server. Once the browser and the Server both have a copy of this private session key, they can use it to encrypt and decrypt data back and forth between themselves. This same process is followed when data is sent between two Merge PACS servers via Telemedicine.

TLS can be used to provide secure transmission of data:

- Within a hospital, if the internal private network is not considered a secure network (*e.g.*, a University network or wireless network)
- Outside a hospital (*e.g.*, to remote users accessing the Server from home, especially those without access to a secure Virtual Private Network, as described below, or to another Merge PACS Server via Telemedicine).

Merge PACS is compatible with any certificate that is supported by Java, including certificates provided by the following third-party vendors:

Verisign	RSA Data Security	USERTRUST Network / DATACorp SGC
Thawte	Equifax Secure	Starfield Technologies
Baltimore Cybertrust	GeoTrust Global	AAA Certificate Services
Entrust	AddTrust AB	Sonera
GTE	ValiCert	Go Daddy Group

NOTE: Each client site is responsible for acquiring the TLS certificate mentioned above. It can take up to 48 hours for this to be generated, and cannot be requested until the Merge PACS server has been installed at the site. Please factor this potential time delay into your planning if appropriate. Also, if a customer plans to have a **backup server**, a second certificate will be required for that server.

CAUTION: If a site needs to replace or add an TLS certificate, please keep in mind that the certificate needs to be added to both IIS and Tomcat and therefore requires the assistance of a Merge Healthcare support representative.

5.4. Merge PACS Web Security

Merge PACS Web Security provides an additional layer of security on top of whatever other security options are in place. After a user is authenticated (whether by Merge PACS Authentication, Windows-based Authentication, Integrated Windows Authentication, or LDAPS), Merge PACS Web Security determines what access privileges the user has with relation to the various web resources (e.g., Java Server Pages, servlets, and image data). This is done by issuing the authenticated user a web cookie and then only granting access to the various web resources to those users who have a valid cookie.

Merge PACS Web Security is enabled for both the all Merge PACS Web Pages (management and non-management) and can be configured but not disabled (*i.e.*, you can configure it to work with Basic Authentication or Integrated Windows Authentication, but can not turn it off entirely).

5.5. Integrated Windows Authentication

5.5.1. In General

As with Basic Authentication (described above), Integrated Windows Authentication is used to prevent unauthorized users from establishing a Web (HTTP) connection to your Merge PACS Server. Integrated Windows Authentication offers two benefits over Basic Authentication, however:

- If a user has already logged on to Windows, Integrated Windows Authentication can use this logon information to authenticate the user if the Client machine has been properly configured. This means the user will not be prompted for a username and password when accessing the Merge PACS Web Server.

- Username and password information is not sent in clear text; instead, the user's web browser proves its knowledge of the password through a cryptographic exchange with the Merge PACS Web server. **Note that this does NOT encrypt the actual patient information and is therefore not a substitute for TLS.**

Although Integrated Windows authentication is more secure and convenient than Basic Authentication, it does have two limitations:

- Integrated Windows Authentication is only supported with Internet Explorer version 5.0 or later on the client side.
- In order for a user to be able to make use of Integrated Windows Authentication's "single logon" feature, his or her web browser must be individually configured to do so.

For information on configuring a Merge PACS Workstation to work with Integrated Windows Authentication's single logon feature, refer to Section 16.3 below.

5.5.2. Integrated Windows Authentication and TLS

Integrated Windows Authentication is typically used in conjunction with **Transport Layer Security (TLS)**, as defined above. Although usernames and passwords are encrypted, all other information sent between the Server and the client is still sent in "clear" (non-encrypted) text without TLS in place, meaning it could theoretically be intercepted and used by unauthorized users.

- If Merge PACS is being used over the **Internet**, Integrated Windows Authentication **must** be used in conjunction with **TLS** in order to provide reliable security.
- If Merge PACS is being used solely over an **Intranet**, TLS is still recommended to provide end-to-end encryption ("Zero-Trust").

5.6. IP Address and Domain Name Restrictions

This security feature can be used to enable or prevent specific users, computers, or domains from accessing Merge PACS. For example, you can:

- Exclude a particular individual or group by denying access to your server from a particular IP address
- Prevent entire networks from accessing your server, while allowing all others to have access.

IP address security is probably most useful on the Internet, to exclude everyone except known users.

For more information, consult your Web server documentation.

5.7. Lightweight Directory Access Protocol (LDAP)

LDAP is a specification for a client-server protocol to retrieve and manage directory information. It is intended as a means for clients on PCs to access a directory system that follows the X.500 data models.

In other words, and insofar as it relates to Merge PACS, LDAP is a separate computer that contains a group of user accounts and associated privileges. Users attempting to establish a Web (HTTP) connection to Merge PACS can be challenged to authenticate using existing account information that is stored on the LDAP server.

Merge PACS can be configured to just authenticate against LDAP, to authenticate against its own database (“Merge PACS Web Security”) or to authenticate against both (“Mixed Mode”).

Merge PACS also supports authentication against **Active Directory (AD)** Servers. AD is a specific type of LDAP implementation created by Microsoft for use with Windows.

NOTE: Secure LDAP (*i.e.*, LDAP using TLS) is referred to as LDAPS and is the recommended way of implementing LDAP.

5.8. Controlled Access

As described in Section 9.11 below, access to exams within Merge PACS is primarily controlled by user and group privileges. By default, all “advanced” users (Radiologists, Technologists, etc.) have the privilege to view all exams (including studies, reports and orders) on the Server via the Merge PACS Workstation.

Other users, on the other hand, do not have access to all exams by default and can only view those exams from the Workstation Query Page or from any of the Management Pages for which they or their user group have been specifically granted access.

CAUTION: If a study is contained in a RealTime Worklist or RealTime Study List worklist that a user has access to (*i.e.*, the user belongs to a group that has been assigned that worklist), the user will be able to access that study regardless of the user’s access privileges.

Exams can be assigned to a user or group in any of the following ways:

- If the Site has a RIS, Merge PACS can be configured to automatically assign exams to a user if the user’s Provider Code matches any of following fields of an order generated by the RIS:
 - Attending Doctor
 - Referring Doctor
 - Consulting Doctor
 - Admitting Doctor
 - Order Doctor
 - Other Health Care Provider
- The Merge PACS Sever can also be configured to automatically assign exams that are forwarded from specified modalities or Merge PACS Servers to specified groups, based on the AETitle of the forwarding modality or Server.
- Authorized users can manually assign exams to users or groups from the Access Control Management Page, as described in subsection 9.9.8 below.
- Authorized users can manually assign exams to users or groups from within the Merge PACS Workstation.

For information on configuring a Merge PACS Server to automatically assign exams based on Provider Codes or AETitles, refer to Section 17.4 below.

5.9. Virtual Private Network (VPN)

A Virtual Private Network (VPN) uses advanced encryption and tunneling to permit organizations to establish secure, end-to-end, private network connections over third-party networks, such as the Internet or extranets. In other words, a VPN allows you to connect multiple computers together over the public Internet, while maintaining the security of a local (or “private”) network connection. A VPN can also be used to connect both a single computer to a network as well as two separate networks together.

If your hospital has a VPN set up, doctors working at home can use it to connect with your Merge PACS Server with the same level of security as if they were at the office. In addition, your Merge PACS Server can communicate with another Merge PACS Server over the Internet as if they were both connected to the same local network.

A VPN may be used to provide:

- Secure encrypted transmission of data to and from other Merge PACS Servers, via the optional Telemedicine feature
- Secure transmission of data to and from remote users (e.g., doctors accessing your Merge PACS Server from home)
- Secure access to Merge Healthcare Support

5.10. Firewalls

A firewall is a system designed to block unauthorized access to or from a private network. Firewalls may be used both within an enterprise and for restricting access to resources from the Internet, and internal firewalls are generally configured differently than external (Internet) firewalls. Firewalls are frequently used to prevent unauthorized Internet users from accessing private networks connected to the Internet, especially intranets. All messages entering or leaving the intranet pass through the firewall, which examines each message and blocks those that do not meet the specified security criteria.

CAUTION: All internet connections must be via encrypted means (whether TLS or VPN).

If you are using a firewall to restrict access to your network, the following TCP/IP ports must be opened through the firewall for **internal** and **VPN** users:

TCP/IP Port	Purpose	Users
80, 8080	HTTP Traffic	Remote Merge PACS Workstation; Merge Healthcare Technical Support.
443, 444, 8443	HTTPS Traffic (if TLS is enabled on Server)	Remote Merge PACS Workstation users (if any); Merge Healthcare Technical Support
3389	Remote Desktop	Merge Healthcare Technical Support
5222	Merge PACS Workstation Instant Messenger	Remote Merge PACS Workstation users (if any)

NOTE: Increasingly customers, especially large IDNs, are deploying SSL/TLS certificates even for internal users.

CAUTION: When a server is internet facing and using a SSL/TLS certificate, port 8080 should be blocked as secured traffic will go over port 443/444/ 8443 as described below.

For **Internet** users without VPN, the following TCP/IP ports must be opened through the firewall:

TCP/IP Port	Purpose	Users
443, 444, 8443	HTTPS Traffic	Remote Merge PACS Workstation users (if any); Merge Healthcare Technical Support
5222	Merge PACS Workstation Instant Messenger	Remote Merge PACS Workstation users (if any)

CAUTION: The “bypass authentication” for Merge embedded viewer must **NEVER** be used on a system that is directly exposed to the internet. That authentication mechanism should only be used for internal users.

Chapter 6. System Databases

6.1. Overview

This chapter discusses Merge PACS's DB2 databases. It describes their contents, their functions within Merge PACS, and the tools and practices for maintaining them.

The following concepts and terminology are discussed in this chapter:

Term	General Description
ANETDB database	Database on a Merge PACS Server that has been set up as a Watch Monitoring Station that receives and stores data from the Merge PACS Server or Servers being monitored.
Image database	Database that stores information about patients, studies, series, and images.
online, unattended backups	Regularly scheduled backups automatically performed by DB2.
offline, attended backups	Unscheduled backup initiated by the System Database Administrator.

NOTE: DB2 10.x does not come with a GUI-based tool for administering databases. Customers are encouraged to install **IBM Data Studio** or **IBM Data Studio Web Console** to monitor all DB2 servers.

6.2. The Databases

Merge PACS has the following DB2 databases:

- **Image** database
- **ANETDB** database (if the Server is acting as a Watch Monitoring Station)

6.2.1. Image database

The Image database contains information about patients, studies, series, and images, together with the various queue and log files. Primarily, this database has the following sets of tables:

Table Set	Description
Queue Tables	Contain records of the periodic and specific commands executed by the CommandScheduler
StorageServer Tables	Contain information regarding uncompressed images that are stored on the server
Log Tables	Contain records of user access and Importer processing
AmicasDemographic Tables	Contain information regarding the compressed images that are stored on the server
ImageManager Tables	Contain information received through the HL7 interface

The Image database is named **IMG<StationID>**, with **<StationID>** being Merge PACS's StationID (an Merge PACS-specific identifier) – for example, **IMG2002**.

6.2.2. ANETDB database

The ANETDB database is found only Merge PACS Servers that have been set up as a Watch Monitoring Station. The database receives and stores data from the Merge PACS Server or Servers being monitored by the Monitoring Station.

The ANETDB database is named **ANETDB**.

Chapter 7. Supported Character Sets

When performing searches with accented characters throughout the system (both on the Server and within the client applications), Merge PACS uses an approach that can be described as "accent-insensitive searching" (in analogy with "case-insensitive searching"). This means that attributes such as Patient Names are stored in the database in "normalized" form with any accents removed and with each Latin1 character converted to its nearest uppercase ASCII equivalent. When a search string is entered, it is also normalized before the search is run. Thus, for example, A, À, Á, Â, Ã, Ä, and Å will all be mapped to A, so they will all match each other.

The following table shows the complete set of mappings that are used:

Character	Decimal Code	ASCII Mapping
À	192	A
Á	193	A
Â	194	A
Ã	195	A
Ä	196	A
Å	197	A
Æ	198	AE
Ç	199	C
È	200	E
É	201	E
Ê	202	E
Ë	203	E
Ì	204	I
Í	205	I
Î	206	I
Ï	207	I
Ð	208	TH
Ñ	209	N
Ò	210	O
Ó	211	O
Ô	212	O
Õ	213	O
Ö	214	O
×	215	null
Ø	216	O
Ù	217	U
Ú	218	U
Û	219	U
Ü	220	U

Character	Decimal Code	ASCII Mapping
Ý	221	Y
Þ	222	TH
ß	223	SS
à	224	A
á	225	A
â	226	A
ã	227	A
ä	228	A
å	229	A
æ	230	AE
ç	231	C
è	232	E
é	233	E
ê	234	E
ë	235	E
ì	236	I
í	237	I
î	238	I
ï	239	I
ð	240	TH
ñ	241	N
ò	242	O
ó	243	O
ô	244	O
õ	245	O
ö	246	O
÷	247	null
ø	248	O
ù	249	U
ú	250	U
û	251	U
ü	252	U
ý	253	Y
þ	254	TH
ÿ	255	Y

Chapter 8. Merge PACS Client Applications

8.1. Overview

The Merge PACS Workstation is a full-featured radiology workstation that is designed as a tool for primary interpretation of patient images. It consists of the following main components:

Component	Description
RealTime Worklist™ (RTWL)	Optional component that lets users view and manage one or more lists of patient studies (“worklists”) that have been assigned to them.
Teaching Worklists	Optional component that lets users view and manage one or more lists of patient studies that users have tagged for teaching and/or conferencing purposes.
Query Search	Lets users search for a particular patient based on such criteria as patient name, physician’s name, modality type, etc.
RealTime Study List (RTSL)	Optional combined interface that lets users view and manage RTWL worklists and Teaching Worklists on the same page, as well as create and save special Query Worklists on the fly. Note that users can either have RealTime Study List or RealTime Worklist enabled, but not both at the same time.
Local Study	Lets users open studies that have been previously saved to their local workstation or on a network accessible drive.
Recently Viewed Studies	A combined interface that lets users view RTWL worklists and Teaching Worklists on the same page, along with special Query Worklists that can be created and saved on the fly.
Patient Record	Once a study has been opened, the Patient Record displays Series Navigation thumbnails for that study as well as any prior studies for the same patient. These Series Navigation thumbnails can be used to open the study into the Merge PACS Viewer or into a separate pop-up Series Viewport window.
Merge PACS Viewer	The primary application for viewing and manipulating patient images. It includes basic features as well as numerous optional advanced features that can be made available to different classes of users based on login privileges.
Quality Control Editor	Provides authorized users the ability to perform a wide variety of Quality Control (QC) operations. It can be launched by authorized users from within various components of the Merge PACS Workstation.

NOTE: All the components described above except for the Merge PACS Viewer and the Quality Control Editor are displayed within the **Workstation Browser**. The Merge PACS Viewer and the Quality Control Editor are displayed in separate windows.

For complete information about using the Merge PACS Workstation, refer to *Merge PACS 8.2 Workstation User’s Guide*.

8.2. The Merge PACS Workstation

The Merge PACS Workstation is a full-featured radiology workstation that is designed as a tool for primary interpretation of patient images. It consists of the Workstation Browser, which provides a number of different ways to locate patient images, and the Merge PACS Viewer, which is the primary application for viewing and manipulating patient images.

The Merge PACS Workstation can be run as a stand-alone application on a client machine or as an Embedded Viewer within a third-party application.

The Merge PACS Workstation can be launched in any of the following ways:

- From the Merge PACS Workstation **Login Page**
- Via a **desktop icon**
- From a **command line**
- **Automatically** (when distributed via an MSI file)

NOTE: When launched as an Embedded Viewer, from the Login Page or Automatically, if the Merge PACS Workstation is currently running and the user attempts to launch the Workstation a second time from the Embedded Viewer or the Login Page, the first instance of the Workstation will be reused instead of launching a second instance. When launched via a desktop icon or a command line, however, any attempt to launch a second instance of the Workstation will, by default, cause a second instance of the Workstation to be launched that will run concurrently with the first instance. This default behavior can be changed as described in subsection 16.4 below.

8.2.1. Merge PACS Viewer

The *Merge PACS Viewer* provides full-fidelity viewing and a complete set of viewing tools, including:

- Image manipulation (Window/Level, Zoom, Invert, Rotate, Sharpen, Color Maps, etc.)
- Annotation (text and audio)
- Measurements
- DICOM overlays
- Comparisons
- Cross-reference lines
- Multiple series display
- Compare studies mode
- Intelligent linking of series based on 3D position of table
- 3D cursor to allow the rapid review of multiple series
- Ability to create and edit Hanging Protocols and Study Presentations
- Inherent 3D support, including Multi-Planar Reconstruction (MPR), Maximum Intensity Projection (MIP), Color Volume Rendering (CVR) and Image Fusion (e.g., PET/CT)
- Spine labeling

- Mammography support
- Lossless image support
- Ability to create and save Key Images
- Integration with third-party applications, such as dictation systems and orthopedic templating toolsets.
- Ability to copy patient studies to a CD, together with a stand-alone copy of the Viewer (requires an attached CD writer)
- Ability to save images as JPEGs

Note that many of the features and components of the Merge PACS Viewer, such as inherent 3D support, the ability to create hanging protocols, spine labeling, the use of lossless vs lossy image compression, etc., are optional and made available to the users based on their login privileges, as described in Chapter 11 below. Users with “advanced” privileges (typically radiologists) will be able to make use of the full range of features, whereas other users (e.g., referring physicians) will only have access to a subset of those features.

NOTE: If Merge PACS receives DICOM images from external systems (another PACS, for example), The Merge PACS Viewer will display any associated Grey Scale Presentation State (GSPS) object. If multiple GSPS objects are sent, the user can select the desired presentation state to apply. In addition, a user preference exists that will determine whether or not to automatically apply the latest GSPS when loading a study.

8.2.2. Workstation Browser

The Merge PACS Workstation Browser provides a front end to the Merge PACS Workstation and provides access to a number of different optional methods for locating studies, including:

- **RealTime Worklist**
- **Patient Record**
- **Query Search**
- **RealTime Study List**
- **Local Study Open**
- **Recently Viewed Studies**
- **Teaching Worklists**

In addition, the Workstation Browser provides quick access to other components of Merge PACS, such as the Management Pages.

Access to the various components within the Workstation Browser, as well as specific functionality, is made available to the users based on their login privileges, as described in Chapter 11 below. Users with “advanced” privileges (typically radiologists) will be able to make use of the full range of features, whereas other users (e.g., referring physicians) will only have access to a subset of those features.

Note that privileges that are set for the Workstation Browser as a whole apply to the **Patient Record, Query Search** and the **Merge PACS Viewer**. They do not, however, apply to RealTime Worklist, since Worklist privileges are defined on a per worklist basis, as described in Chapter 13 below. They also do not apply to the **Local Study Open** feature, since that feature has a very limited set of tools available by design.

8.2.3. RealTime Worklist

RealTime Worklist™ is an optional component that provides a powerful “front end” to the Merge PACS Workstation that allows users to view and manage a list of patient studies that have been assigned to them. Because the Worklist information is stored on a centralized Merge PACS Server and not on individual machines, users can view their individual study lists regardless of which Workstation they log into. RealTime Worklist can also be configured to work with the optional Merge RadStream™ Acuity component. With Merge RadStream Acuity enabled, studies can be assigned an “acuity score” that can be used to determine priority within individual worklists. Acuity can be based, in part, on the answers to questions inputted by a Technologist, but the final acuity score is determined automatically by the system according to a complex algorithm.

a. Merge Radstream

If the Optional Merge RadStream™ component is enabled, the following special features will be available for use with RealTime Worklist:

- **Prioritization Based on Acuity** – Individual worklists can be automatically sorted based on acuity scores, so that exams with higher acuity scores appear higher up on the worklist. Acuity scores are determined based on a variety of factors, including input by Technologists at the Technologist WorkPanel accessible from the Worklist.
- **Merge RadStream Communication** - A new role of Operator is created. Using the Communication WorkPanel described above, an Operator can see contact information for the various physicians associated with a study; view a copy of the report associated with this study, if any, with relevant portions highlighted; enter comments regarding the contact made or attempted with each Physician and view a log of previously entered comments; and set the Communication and Connection status for this study.

b. Remote Users of RealTime Worklist

Remote users (*i.e.*, users from outside a site’s Intranet) can access RealTime Worklist via **http** by default. The Merge PACS Server can also be configured to provide access via **https** by Merge Healthcare Technical Support, if desired

NOTE: In order for remote users to access RealTime Worklist via HTTPS, the URL of the Merge PACS Server must be entered into the browser exactly as it appears on the Server’s TLS Certificate. In most cases, this is the Fully Qualified Domain Name.

8.2.4. RealTime Study List

RealTime Study list is a combined interface that lets users view and manage RTWL worklists and Teaching Worklists on the same page, as well as create and save special Quick Filter Worklists on the fly. Note that a Workstation can either have RealTime Study List or RealTime Worklist enabled, but not both at the same time.

8.2.5. Quality Control Editor

The Quality Control Editor (QC Editor) can be accessed by authorized users from within various components of the Merge PACS Workstation and lets those users perform the following functions for a selected study:

- Copy one or more series into the same study.
- Copy images from one series into an existing or new series within the study.
- Move one or more studies to another existing patient or a new patient.
- Link a study to an existing order.
- Move one or more series to a new study in a new patient, to a new or existing study for the current patient, to a new or existing study for a different patient or to an existing order (thereby creating a new study linked to that order)
- Move images from one series to another existing series within the study.
- Merge a patient with another patient in the system.
- Merge a study with another study for the same patient or for another existing patient.
- Merge a series with another existing series for the same patient or for another existing patient.
- Split a series into multiple series within the same study.
- Delete one or more studies from the patient.
- Delete one or more series from a study.
- Delete one or more images from a series.
- Edit DICOM attributes for individual series, studies and patients.
- Perform patient and study level demographic edits

While performing these functions, users can view the affected images and, if desired, change the presentation state of those images and the series associated with those images (magnification, Window/Level, annotations, etc.).

For more information on the QC Editor, refer to the *Merge PACS 8.2 Quality Control Editor Users Guide*.

8.3. Report Integration

As described above, the Merge PACS Workstation can be configured to allow users to view reports that have been created for a particular study and that have been made available to Merge PACS. “Report integration” refers to the method by which those reports are made available to Merge PACS.

There are currently two methods of integration for report viewing by the Merge PACS viewer:

- The default Merge PACS report servlet
- HL7 integration

Each of these methods uses a specific “report servlet” to communicate data to Merge PACS. The report servlet for each integration method is described below.

8.3.1. Default Merge PACS Report Servlet

“ReportPanel” is the default report servlet that is shipped with Merge PACS. It is designed primarily as a proof of concept tool, and is not intended for clinical use. This servlet does not utilize a database, nor does it provide for automated mechanism for receiving and storing studies. In a tightly monitored environment that does not have the benefit of a HIS/RIS system, but does assign Accession numbers for all its studies, it is possible to use this servlet to view reports with studies.

8.3.2. HL7 Report Integration

HL7 report integration uses Merge PACS’s ability to receive and store HL7 report information and is clinically viable. This technique requires that the HIS/RIS system be configured to provide HL7 report information to the Merge PACS server. Once these setup items have been completed on the HIS/RIS and on the Merge PACS Server, the mechanism is in place for the receipt and storage of report information. There may still be a need for two other configuration changes on the Merge PACS Server. First, it is possible based on different HIS/RIS vendors and their individual use of HL7 fields that there will need to be some configuration of field mapping within the HL7 listening of the Merge PACS Server. Next, depending on the use of HL7 information as it relates to the formatting of the report text, there may be a need to edit the XML style sheet used to display the report information. The sharing of HL7 information prior to installation will reduce the amount of work needed at installation time, however, it should be expected that there will be some time needed to insure the best integration of the two systems.

Chapter 9. Merge Administration & Management Tools

9.1. Overview

The Merge PACS Management Web Pages are the primary tool for monitoring, reconfiguring, and troubleshooting Merge PACS. The availability of the various pages, as well as what tools are on each page, is determined by login access rights. The following table lists the complete set of Management pages available by default to members of the Administrators group:

Page:	Function:
Home	View the most recent activity on the Server, including studies currently being imported and those that were imported within the last five minutes, with or without errors.
Monitoring Pages	
Archiver	View a current log of the Archiver process
Auditor	View a current log of the study auditing process.
EA Notification Processor	View a current log of the EA Notification Processor process, when Merge PACS is configured to run in Integrated Mode.
ECDR Manager	View a current log of the ECDR manager process
Importer	View a current log of the study importing process.
Listener	View a current log of the DICOM query/retrieve listening process.
Scheduler	View a current log of the command scheduler process.
Image Manager	View a current log of the ImageManager process.
Sender	View a current log of the Sender process, when Merge PACS is configured to run in Integrated Mode.
Worklist Servers	View a current log of the Servers(s) on which RealTime Worklist is running
Client Notification Server	View a current log of the process responsible for sending notification messages from the server to clients (Workstations)
Application Server	View a current log of web server responsible for processing requests from clients (Workstations), other Merge PACS servers (via telmed) and third-party applications
Messaging Server	View a current log of the server that is responsible for the communication between server processes and, in Integrated Mode, dispatching notifications received from iCEA
Queue Pages	
Archive Queue	Display and manage the list of studies scheduled to be archived or retrieved from the Archive. If the optional Archive feature has not been enabled, this option will not appear on the Queues menu.
Autorouting Queue	Display and manage the list of studies scheduled to be autorouted.
Command Queue	Display and manage the list of studies scheduled for execution by the CommandScheduler.

Page:	Function:
Forwarding Queue	Display the studies that are scheduled to be forwarded to a DICOM workstation or archive.
Image Import Queue	Display and manage the list of DICOM images currently scheduled for the Importer.
Insert File Queue	Display the images that have been sent from a Merge PACS Workstation via HTTP.
Interface Queue	Displays the list of HL7 messages that have been sent or are waiting to be sent via the Outbound HL7 Interface
Telemed Import Queue	Display and manage the list of studies currently scheduled to be imported from another Merge PACS Server.
Telemed Export Queue	Display and manage the list of studies currently scheduled to be exported to another Merge PACS Server.
DICOM Print Queue	Display any studies that are currently waiting to be sent to a DICOM printer, as well as successfully-submitted jobs, failed jobs, and cancelled jobs that have not yet been cleared (manually by the user) or automatically cleaned up (by the cleanup DB script).
Sender Job Queue	Display information about instances being sent to EA (Integrated Mode only)
DataStorage Cache	Display information about instances that are temporarily stored in the Server and are waiting to be processed in one way or another.
Logs	
Study Log	Query and display study logs.
Access Log	Query and display access logs.
Event Log	Query and display event logs.
Performance Log	Query and display performance logs.
User Login / Logout	Query and display user login / logout logs.
User Modifications	Query and display user modifications Logs
Tools	
Patient Data Editor	No longer available. Refer to the <i>Merge PACS 8.2 Quality Control Editor Users Guide</i> for information on this functionality.
Restore Deleted Files	Restores files that were deleted via the Quality Control Editor
Send Studies	Manually send studies to the Telemed Export Queue or a local DICOM device.
Problem Studies	View the list of DICOM studies that the Importer could not process successfully and fix the problems.
Patient Information	View detailed patient and study information for a selected patient.
Archive Tool	Request studies that have been sent to the Merge Archive Server and deleted from the main Merge PACS Server to be reimported.
Archive Info	Get detailed study archive information.
Access Control	Manually assign exams to select users and/or groups to provide restricted access to these exams.
Manage Unassigned Exams	View and grant access to exams that are not currently assigned to anybody.
Enterprise CD Writer	
Create CDs	Manually create CDs for patients.

Page:	Function:
CD Queue	Manage job queues for CD writers.
CD Writers	Manage CD writer devices.
Auto Writing Rules	Manage automatic CD writing rules.
Users, Groups & Site	
Users	Manage account information for individual users, specify which groups they belong to, manage their Credentialing and Licensure settings, if applicable.
Groups	Enter or edit the general description of each group, manage the members of each group, manage LDAP groups (if applicable), manage the worklists associated with each group (if any), configure Viewer XML integrations used by this each group (if any), specify the parent of each group for use with Templated User Management, configure special privileges for each group (such as Merge Instant Messenger and Study Assignment privileges) and configure the relative priorities between all existing groups for use with Templated User Management.
Site	Enter or edit the general description of this site for use with Templated User Management,
Services	View the status, stop, and restart the various Merge PACS services and processes.
Configuration	
LDAP Servers	Define LDAP servers this Merge PACS server connects to.
DICOM Devices	Edit the DICOM targets with which Merge PACS is set up to communicate.
Modalities	Create a master list of modalities, some or all of which can be used to create filters for individual worklists
Procedures	Create a master list of procedures, some or all of which can be used to create filters for individual worklists
Provider Types	Create a master list of provider types, some or all of which can be used to create filters for individual worklists
Workflow Priorities	Edit existing workflow priority items that can be used to sort individual worklists.
Workflow Status Dimensions	Edit existing or create custom workflow status dimensions
Workflow Status	Edit the master list of workflow status options, some or all of which can be used to create filters for individual worklists
HL7 Order Status Mapping	Define rules for mapping Order Status values obtained from HL7 messages to Viewer display names for use with RealTime Worklist and RealTime Study List.
Credentialing and Licensure	Define Licensure Regions.
Locations	Create a master list of locations, some or all of which can be used to create filters for individual worklists
Patient Classes	Edit the master list of patient classes, some or all of which can be used to create filters for individual worklists
Teaching Tags	Manage the teaching tags used with the Teaching Files feature.

Page:	Function:
IPID Configuration	Configure the default IPID (Issuer of Patient ID) value as well and define the list of available IPID values that can be used as filters for use with RealTime Worklist and RealTime Study List.
Workstation Phone Link	Add, modify or delete phone links for individual Merge PACS Workstations that can be selected from the Merge PACS RadStream Communication Workpanel.
Standard Comments	Add, modify or delete standard comments that can be selected from various places within the Merge PACS Workstation.
Refresh Rate	Specify how often the data on the various pages should be refreshed.
Facility Configuration	Define Facilities that can be used as filters for use with RealTime Worklist and RealTime Study List and map one or more existing Institution Names to each Facility.
CustomTag Configuration	Map specific DICOM tags to Display Names that can be used as query fields, quick filters and worklist column headers.
Upload Action Button Icons	Upload custom icons that will be added to the list of available icons when editing toolbar action buttons.

In addition, the following configuration pages are available from the **Settings** dialog within the Merge PACS Workstation, as described in Section 10.6 below:

- **Practices**
- **Specialties** (formerly known as “Reading Pools”)
- **Procedures**
- **Orderers**
- **Order/Practice Associations**
- **Orderer/Reader Associations**

9.2. Wildcard Searches

Many of the individual Merge PACS Management Web pages allow you to search for information regarding a specific patient, referring physician, and/or physician. Such searches may include various “wildcard” searches as follows:

- Enter an asterisk (*) to find all entries.
- An “implicit wildcard” is automatically added to the end of each query, meaning that “**BL**” would find “Black, John” as well as “Blake, James” and “**B**” would find all people whose last name begins with the letter “B.”
- You can also manually insert one or more asterisks [*] as wildcard characters to fill in for letters you are unsure of at the beginning or middle of a name. For example:
 - “**B*K**” would find “Black, John” as well as “Beckman, Timothy” and “Brown, Kelly.”
 - “***SMITH**” would find both “Goldsmith, David” and “DrSmith, Robert J.”
 - “***B*G**” would find both “Goldberg, Benjamin” and “Cable, George.”

- Names are stored in the Merge PACS database in whatever format they arrive from the DICOM modality and/or the HIS/RIS. Typically, this format is **last name, first name, middle name (or initial), prefix, suffix**, meaning that “Dr. John Q. Smith” would be stored as “SmithJohnQDr.” This practice can vary from site to site, however, and may affect how wildcard searches work for you. Some sites, for example, may append the **prefix** at the beginning of the name (e.g., “DrSmithJohnQ”).

9.3. Opening the Management Web Pages

To open the Management Web pages:

- Launch your Web browser and enter the URL for the Merge PACS Administration pages, usually <http://<your-server>/merge-management>.
- Log in using a username and password with appropriate authorization. Note that the **Home** page is displayed by default as in the example below:

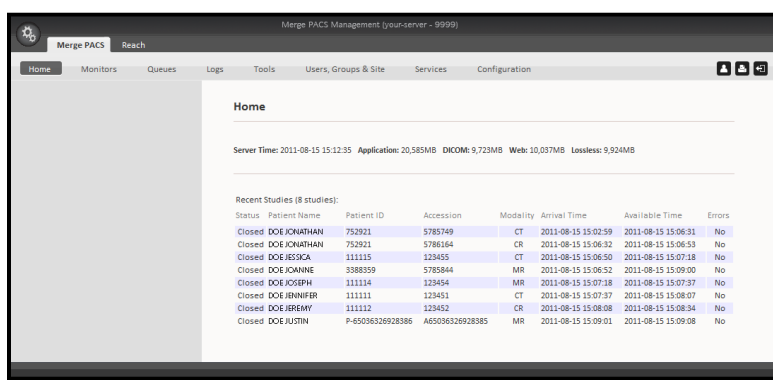


Figure 15: Main Management Page

- To view a particular management page, click the appropriate link on the top category menu to display a submenu of options for that category.

NOTE: If using Internet Explorer to access the Management Pages, **active scripting** must be set to “**Enable**” under **Internet Options** → **Security** → **Internet** → **Custom Level** → **Scripting** → **Active Scripting**, as in the following example:

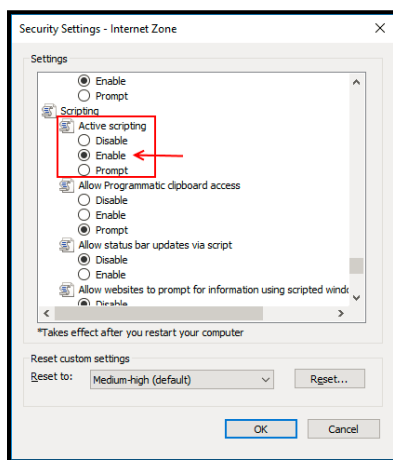


Figure 16: Enabling Active Scripting

9.4. General Management Page Icons

The top of every Management Page displays the same three icons, as in the following example:

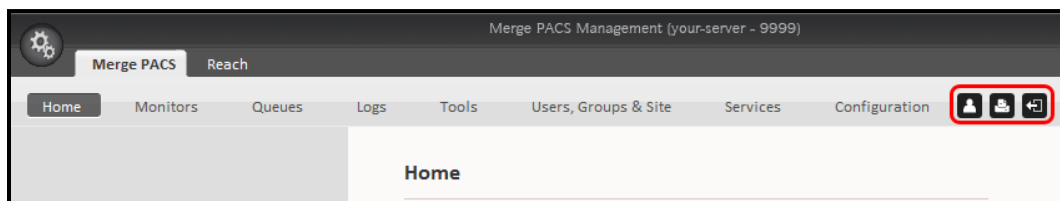


Figure 17: Management Page General Icons

These icons perform the following functions:

Icon	Description
	Hovering your mouse cursor over this icon will display your login name as a ToolTip pop-up.
	Clicking this icon will print the page currently being viewed.
	Clicking this icon will log you out of the Management Pages.



Hovering your mouse cursor over this icon will display your login name as a ToolTip pop-up.



Clicking this icon will print the page currently being viewed.



Clicking this icon will log you out of the Management Pages.

9.5. Home

The Home page displays the general status of the Merge PACS Server, as well as the status of studies that are either in the process of being imported or that have recently finished being imported, as shown in the example below:

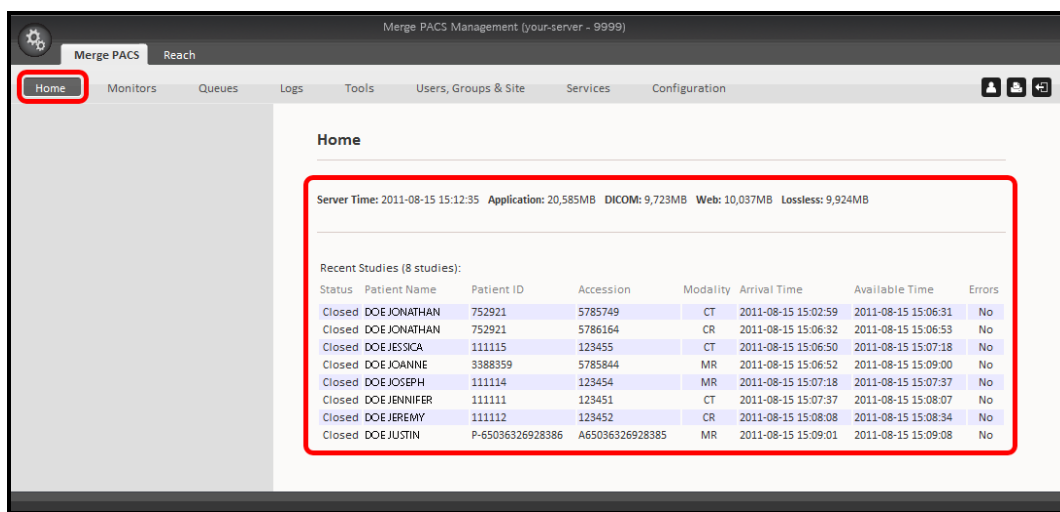


Figure 18: The Management Home Page

9.5.1. Server Status

The top section of the Home page displays the current Server time, as well as the amount of space available on the drive where the application data is stored, the drive where the DICOM images are stored, the drive where the JPEG2000 lossy images (“Web”) are stored, and the drive where the JPEG2000 lossless images are stored.

9.5.2. Recent Studies

The bottom section of the Home page displays information regarding studies that are either in the process of being imported or that have been marked as “closed” in the last five minutes.

Studies listed on the Home page may have one of the following two status types:

Status	Description
Active	The study is currently in the process of being imported into Merge PACS from the modality.
Closed	The study is no longer being imported into Merge PACS from the modality and has been marked “closed” by the Auditor process [see subsection 9.6.2, below]. Note that a closed study may or may not have been successfully imported, and any errors will be referenced in the “Errors” column of the Recent Studies table.

9.6. Monitor Pages

The various monitor pages display the current activity of the Merge PACS Server. Clicking on the “Monitors” link on the top category menu will display a submenu of the available individual monitoring pages as shown below:

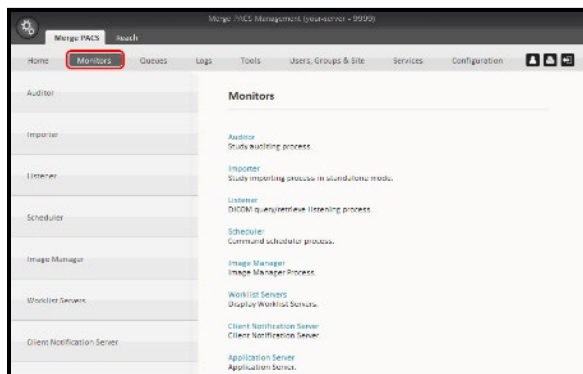


Figure 19: The Monitor Pages

To select a particular monitoring page, click on the appropriate link in the Monitors submenu, or the link displayed on the main Monitors page.

Note that some of the individual monitor pages are for optional components of Merge PACS and will only appear if those components have been enabled.

NOTE: You can set how often the data on the screen is updated for each monitoring page by setting the **refresh** rate as described in subsection 9.13.17 below.

9.6.1. Archiver

If Merge PACS is running in **Standalone** mode and the optional Merge Archive component has been enabled, as described in the *Merge PACS 8.2 Archive Installation and Configuration Guide*, the Archiver monitoring page displays the current activity of the **Archiver** process (defined in Section 3.14 above), as seen in the following example:



Figure 20: The Archiver Monitor Page

The Archiver Monitor displays the progress of each job that has been sent to the Archiver, including both requests for a study to be archived and requests to retrieve a study from the Archive, as well as any error messages that are generated by the system.

9.6.2. Auditor

The Auditor Monitor displays the current activity of the **Study Auditor** process, as defined in Section 4.1 above, listing studies as they are marked “closed” as shown in the following example:

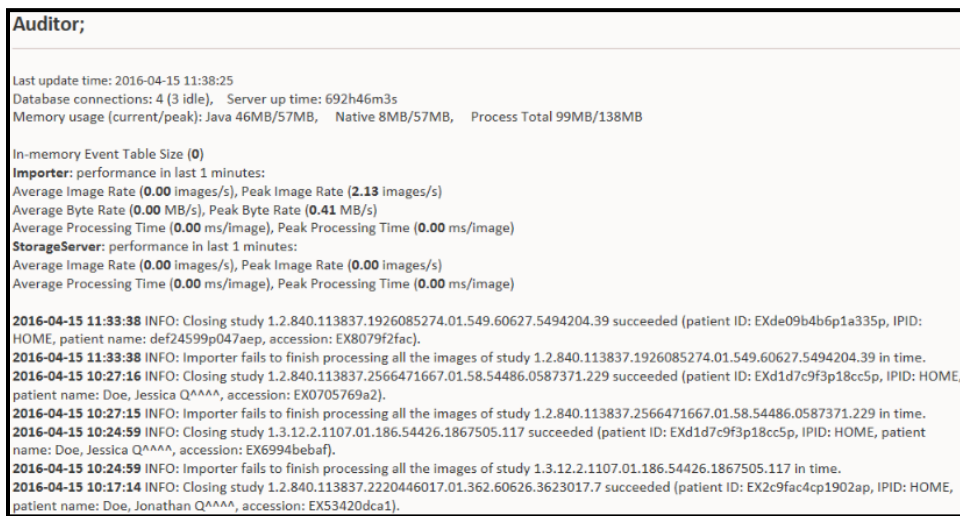


Figure 21: The Auditor Monitor

As mentioned in Section 4.1 above, a study is considered “active” while it is in the process of being imported and until it is marked “closed” by the Study Auditor process. The Study Auditor process will mark a study “closed” under a variety of different circumstances, including the following:

- The study has been successfully imported and validated, either manually or as the result of an auto-validation rule.

- The study has been successfully imported and a set amount of time (default is two minutes) has elapsed without it being validated.
- The study was not successfully imported and a set amount of time (default is two minutes) has elapsed.

Every study, whether successfully imported or not, will eventually be marked as “closed” and will appear on the Auditor Monitor page.

9.6.3. EA Notification Processor

When Merge PACS is configured to run in **Integrated** mode, the EA Notification Processor Monitor displays the current activity of the **EA Notification Processor** process (as defined in Section 3.4 above) as seen in the following example:

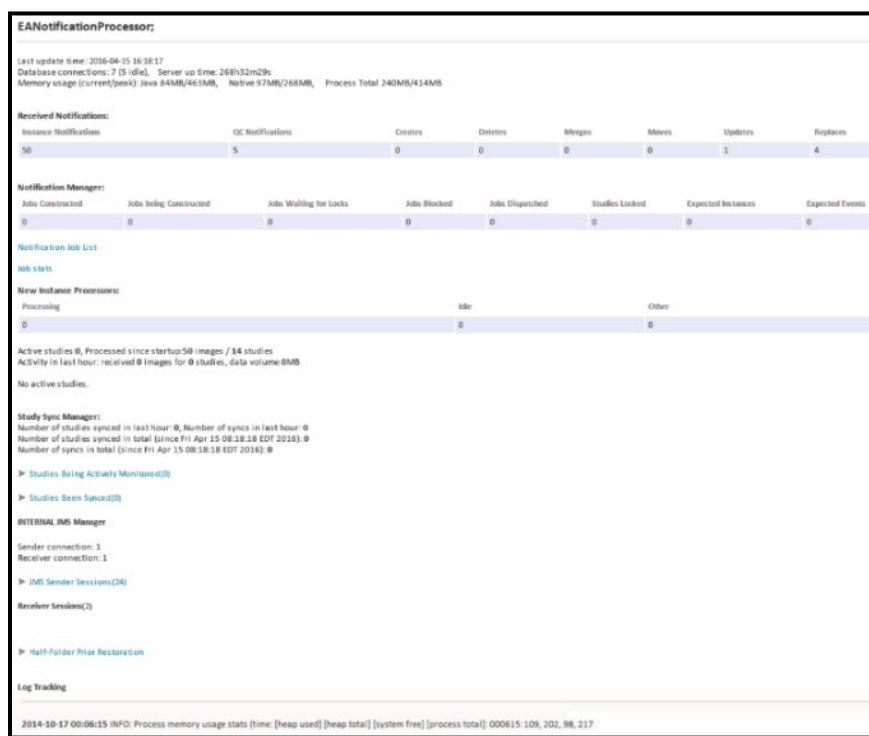


Figure 22: EA Notification Processor Monitor

If automatic comparison study retrieval has been enabled for this site, as described in subsection 1.2.69 above, you can also monitor the progress of the current retrieval process triggered by the receipt of studies by clicking on the **Half Folder Prior Restoration** link at the bottom of the page, as in the following example:

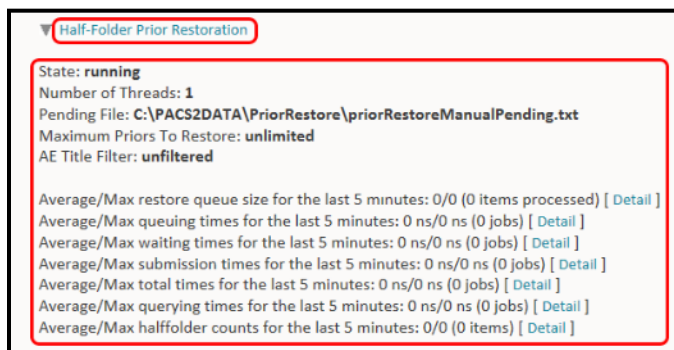


Figure 23: Half Folder Prior Restoration Progress

9.6.4. ECDR Manager

If the optional Enterprise CD Writer component of Merge PACS has been enabled, the ECDR Manager monitoring page displays the current activity of the **ECDR Manager** process (defined in Section 3.12 above), as seen in the following example:

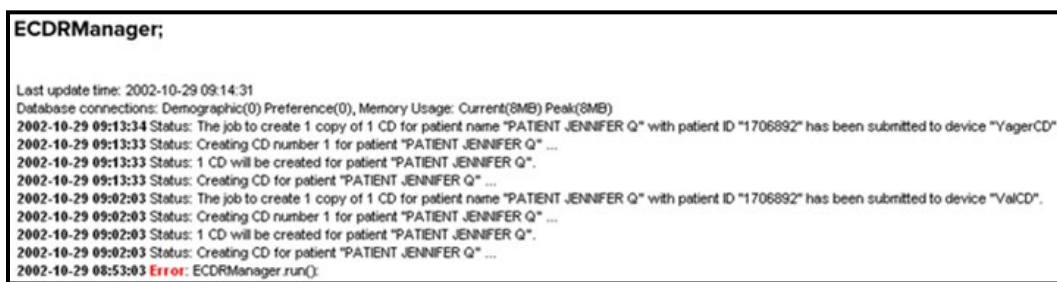


Figure 24: The ECDR Manager

The ECDR Manager displays the progress of each job that has been sent to the Enterprise CD Writer.

9.6.5. Importer

When Merge PACS is configured to run in **Standalone** mode, the Importer Monitor displays the current activity of the **Importer** process (as defined in Section 3.3 above) as seen in the following example:

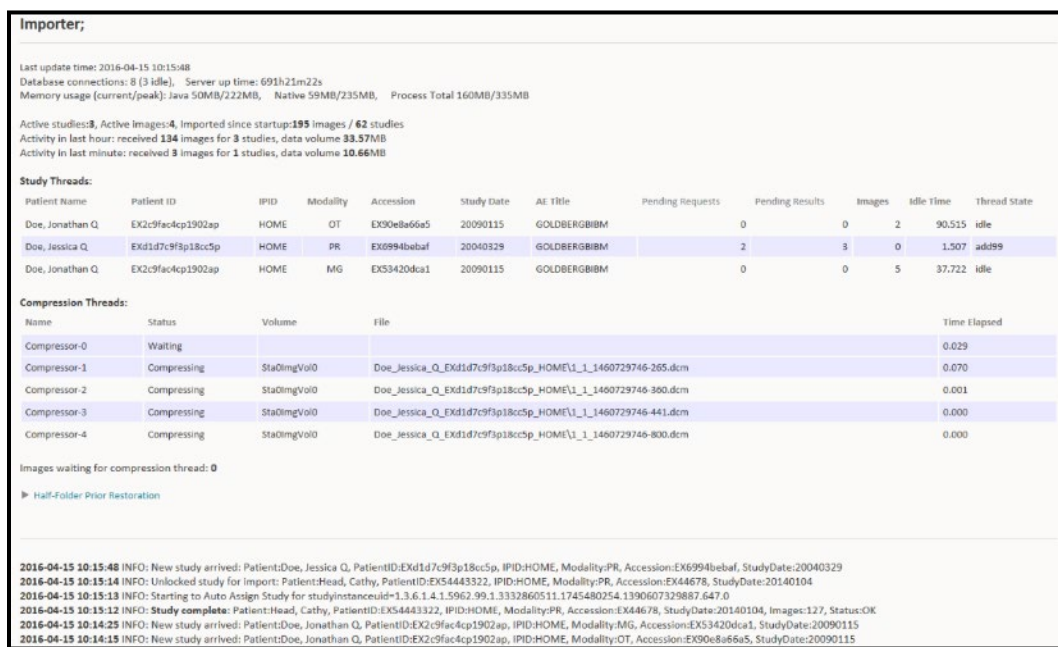


Figure 25: The Importer Monitor

- The top part of the window displays statistical information about the Server, including the software version, the last time the monitor display was updated, the number of active studies and images, the total number of images imported since the Importer has been running, and the length of time the Importer has been in continuous operation.
- If there are any studies currently being imported into the System, information about these studies will be displayed in a table listing the Patient Name, Patient ID, Modality, Accession Number and other information for each study.
- The middle section of the window displays status information for the various compression threads running within Importer.

- If Merge PACS has been configured to run in Standalone mode and automatic comparison study retrieval has been enabled for this site, as described in subsection 1.2.69 above, you can also monitor the progress of the current retrieval process triggered by the receipt of studies by clicking on the **Half Folder Prior Restoration** link, as in the following example:

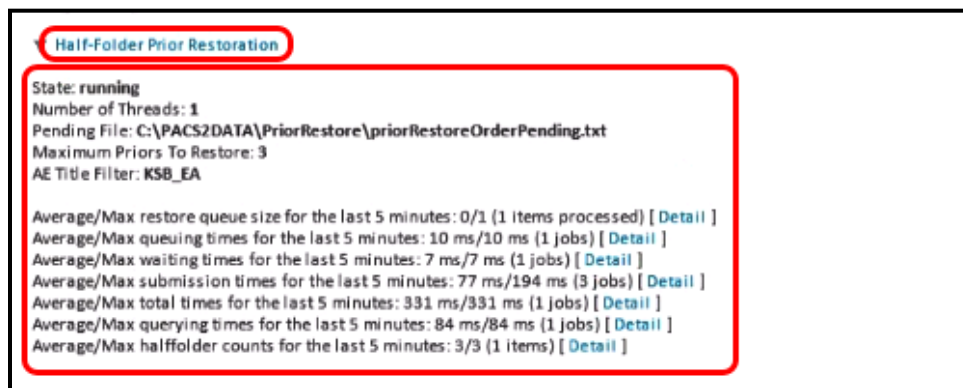


Figure 26: Half Folder Prior Restoration Progress

- The rest of the window displays any messages generated by the Importer, whether status updates or error messages.

9.6.6. Listener

As shown in the example below, the Listener Monitor displays the current activity of the **Listener** process as defined in Section 3.10 above:

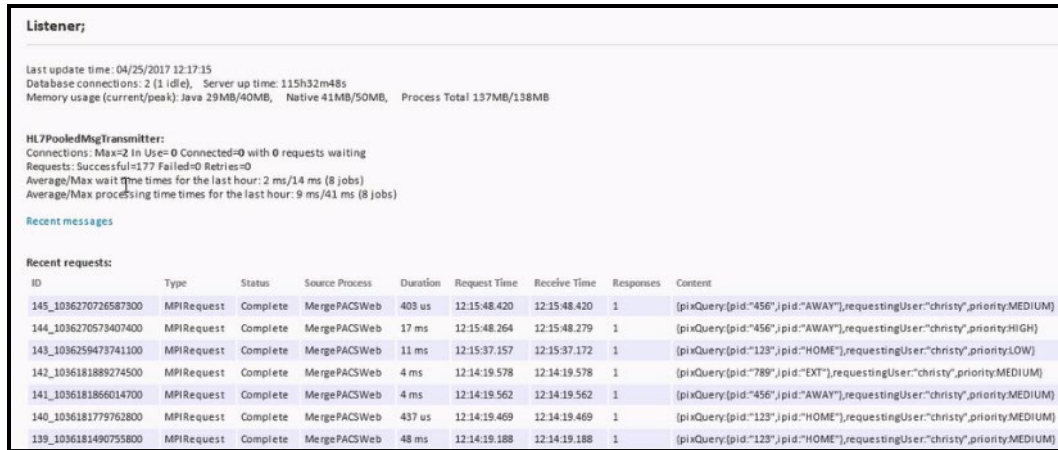


Figure 27: The Listener Monitor

Note that the Listener process will be active only if the Merge PACS Server has been configured to query and retrieve from an external DICOM archive or if Merge PACS is configured to run in **Integrated** mode and Multiple Patient Identity (MPI) support is enabled.

When MPI support is enabled, as described in subsection 2.10.2 above, the Listener Monitor will capture the PIX HL7 messages sent from Merge PACS to the attached iCEA to get identity information. For each request for information, the Listener Monitor will display certain information, including the following:

- The identity being passed
- The Merge PACS User being used
- The amount of time (“Duration”) it took iCEA to respond
- The priority of the message. For MPI requests, the priority is set based on the type of request as follows:

Priority	When Used
High	Used when requesting priors during a study open operation.
Medium	Used for “show priors” operations.
Low	Used for worklist precaching priors requests

Clicking the **Recent messages** link at the top of the Listener Monitor will let you view the raw MPI messages being exchanged, as in the following example:

[illegible]

Figure 28: The Listener Monitor – Raw MPI Messages

The parsed details from the MPI messages are also displayed at the bottom of the Listener Monitor in a more user-friendly format, as in the following example:

Queue list:			
Name	Type	Message Count	Consumer Count
WorklistServer_Resp_MPI	Queue	0	1
com.merge.pacs.mpi.MPIRequest	Queue	0	1
MergePACSWeb_Resp_MPI	Queue	0	1


```

2017-04-25 12:15:48 INFO: PIX Query from christy / MergePACSWeb Identity: (AWAY,456) Response: (from cache)
2017-04-25 12:15:48 INFO: PIX Query from christy / MergePACSWeb Identity: (AWAY,456) Response:
2017-04-25 12:15:57 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 12:14:19 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 101648: 38, 44, 134, 100 101649: 38, 44, 158, 100 101650: 38, 44, 174, 82 104403: 38, 44, 153, 86
104404: 38, 44, 107, 122 124149: 30, 50, 98, 184
2017-04-25 12:14:19 INFO: PIX Query from christy / MergePACSWeb Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890]
2017-04-25 12:14:19 INFO: PIX Query from christy / MergePACSWeb Identity: (AWAY,456) Response:
2017-04-25 12:14:19 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789] (from cache)
2017-04-25 12:14:19 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 12:14:19 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 12:12:33 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 12:12:27 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 10:43:11 INFO: PIX Query from lmgrv / MergePACSWeb Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890]
2017-04-25 10:43:22 INFO: PIX Query from christy / WorklistServer Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890]
2017-04-25 10:43:04 INFO: PIX Query from lmgrv / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 10:40:20 INFO: PIX Query from christy / WorklistServer Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890]
2017-04-25 10:39:19 INFO: PIX Query from christy / WorklistServer Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890]
2017-04-25 10:38:17 INFO: PIX Query from christy / MergePACSWeb Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890] (from cache)
2017-04-25 10:38:17 INFO: PIX Query from christy / WorklistServer Identity: (EXT,789) Response: (HOME,123)[NHS,123 456 7890]
2017-04-25 10:22:56 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 10:19:23 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 10:19:02 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]
2017-04-25 10:17:17 INFO: PIX Query from christy / MergePACSWeb Identity: (HOME,123) Response: (NHS,123 456 7890)[EXT,789]

```

Figure 29: The Listener Monitor – Parsed Details of MPI Messages

9.6.7. Scheduler

As shown in the example below, the Scheduler Monitor displays the current activity of the **Command Scheduler** process, as defined in Section 3.8 above:

```
Scheduler;

Last update time: 2014-09-26 20:57:29
Database connections: Demographic(1) Preference(0), Server up time: 505h32m32s
Memory usage (current/peak): Java 46MB/83MB, Native 49MB/99MB, Process Total 149MB/181MB

2014-09-26 20:55:08 INFO: DICOM Deletion Qualifier query:select AmicasStudy.StudyInstanceUID from AmicasStudy where AmicasState in ('IMPORTED',
'FAILED'), 'PARTIALR') and AmicasStudy.ModifiedDate < ? and not exists (select StudyInstanceUID from InImageQueue where
InImageQueue.StudyInstanceUID = AmicasStudy.StudyInstanceUID) and not exists (select StudyInstanceUID from ForwardingQueue where
ForwardingQueue.StudyInstanceUID = AmicasStudy.StudyInstanceUID) and not exists (select StudyInstanceUID from StudyLock where
StudyLock.StudyInstanceUID = AmicasStudy.StudyInstanceUID) order by ModifiedDate asc fetch first 200 rows only for read only optimize for 200 rows
NOTE That param=:2014-09-26 19:55:08.832'
2014-09-26 20:55:08 INFO: Running CheckResources Command [MaxDeleteMisc: 200, DeleteCompressedBasedOnWatermarks: false, MaxDeleteExternal:
200, DeleteLosslessBasedOnWatermarks: true, DeleteDicomBasedOnWatermarks: true, MaxDeleteCompressed: 200, DeleteExternalBasedOnWaterMarks:
false, MaxDeleteLossless: 200, DeleteLosslessCompressedOlderThanHours: 720, DeleteExternalBasedOnTime: false,
DeleteCmprBasedOnWatermarksOrderByStudydateandImportdate: false, DeleteDicomBasedOnTime: true, MaxDeleteDICOM: 200,
DeleteCacheEmptyFolderOlderThanMinutes: 5, DeleteLLBasedOnWatermarksOrderByStudydateandImportdate: false, DeleteExternalOlderThanDays: 360]
```

Figure 30: The Scheduler Monitor

9.6.8. Image Manager

As shown in the example below, the Image Manager Monitor displays the current activity of the **ImageManager** process, as defined in Section 3.9 above:

```
ImageManager;

Last update time: 2016-04-13 15:36:29
Database connections: 3 (1 idle), Server up time: 648h42m5s
Memory usage (current/peak): Java 121MB/357MB, Native 266MB/365MB, Process Total 433MB/512MB

Image Manager status
▶ Half-Folder Prior Restoration

2016-03-25 00:30:59 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 100055: 96, 406, 200, 383 161402: 105, 406, 197, 386 214927: 115, 406, 171, 389 003059: 119, 406, 99, 389
2016-03-24 09:29:02 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 092902: 95, 406, 56, 415
2016-03-24 03:33:10 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 033310: 85, 406, 57, 414
2016-03-24 00:15:13 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 080520: 334, 385, 144, 282 141018: 344, 385, 229, 289 181158: 349, 385, 199, 290 181203: 65, 406, 163, 404 210734: 75, 406, 119, 410 001513: 79, 406, 99, 414
2016-03-23 07:42:34 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 074234: 334, 385, 86, 313
2016-03-23 02:03:07 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 020307: 325, 385, 98, 303
2016-03-23 00:15:16 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 100050: 300, 385, 223, 274 155550: 309, 385, 178, 283 195409: 315, 385, 143, 293 001516: 323, 385, 98, 300
2016-03-22 08:52:51 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 085251: 297, 385, 70, 304
2016-03-22 04:09:33 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 040933: 287, 385, 100, 302
2016-03-22 02:00:02 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 020002: 284, 385, 100, 301
2016-03-21 16:26:23 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 150515: 266, 385, 164, 298 162623: 269, 385, 98, 298
2016-03-21 15:05:11 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 181216: 118, 385, 129, 409 000452: 128, 385, 396, 414 064633: 138, 385, 369, 416 114132: 145, 385, 268, 426 114133: 145, 385, 281, 412 172643: 154, 385, 252, 413 002536: 164, 385, 226, 413 055336: 173, 385, 221, 413 092657: 183, 385, 233, 415 100245: 184, 385, 227, 415 100246: 184, 385, 378, 383 163330: 193, 385, 480, 386 230203: 203, 385, 457, 387 050008: 212, 385, 442, 387 102010: 222, 385, 447, 387 171535: 232, 385, 448, 387 224656: 241, 385, 441, 387 040653: 251, 385, 433, 389 111333: 261, 385, 425, 389 150208: 266, 385, 192, 389 150209: 266, 385, 272, 357 150210: 266, 385, 298, 357 150211: 266, 385, 330, 325 150510: 266, 385, 90, 325
2016-03-17 15:13:48 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 145613: 113, 385, 185, 434 151348: 116, 385, 100, 437
2016-03-17 14:55:50 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 145429: 112, 383, 191, 455 145431: 261, 383, 154, 479 145432: 312, 383, 144, 482 145433: 323, 383, 113, 482 145434: 333, 383, 135, 458 145437: 333, 383, 172, 457 145438: 37, 385, 139, 463 145450: 106, 385, 128, 466 145550: 113, 385, 98, 466
2016-03-17 14:54:40 INFO: HL7 Listener started
```

Figure 31: The Image Manager Monitor

If automatic comparison study retrieval has been enabled for this site, as described in subsection 1.2.69 above, you can also monitor the progress of the current retrieval process triggered by the receipt of orders by clicking on the **Half Folder Prior Restoration** link, as in the following example:

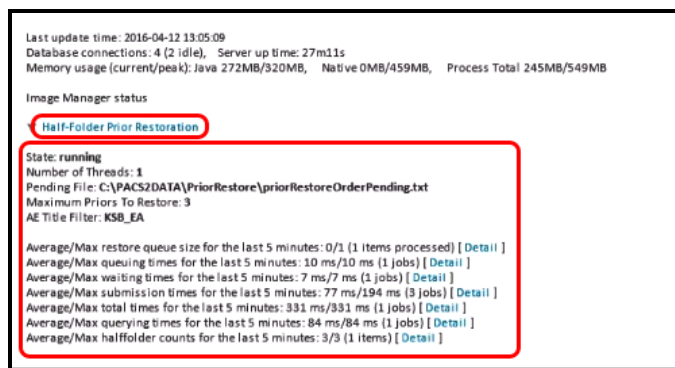


Figure 32: Half Folder Prior Restoration Progress

9.6.9. Sender

When Merge PACS is configured to run in **Integrated** mode, the Sender Monitor displays the current activity of the **Sender** process (as defined in Section 3.5 above) as seen in the following example:



Figure 33: Sender Monitor

- The top section of the Sender Monitor displays **general statistics** for the Sender process, as in the following example:

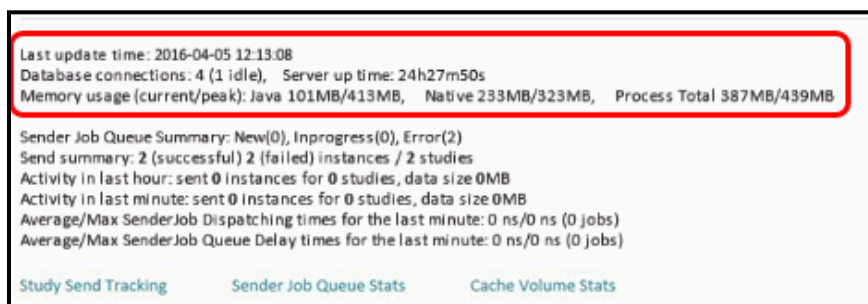


Figure 34: General Sender Process Statistics

- The next section of the Sender Monitor displays **specific statistics** for the current run of the Sender process, as in the following example:

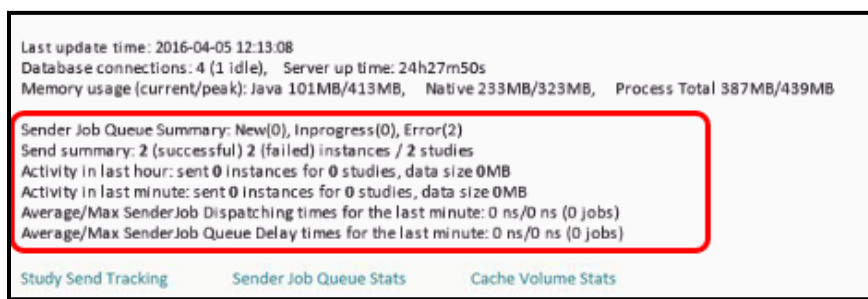


Figure 35: General Sender Process Statistics

The following information is displayed:

- Number of jobs in each status (New, Inprogress, and Error).
- Number of instances that were successfully and/or failed to send since the Sender was last started, along with the number of studies these represent.
- Details about activity in the last hour and last minute, including size.
- Statistics for the average and maximum dispatching and delay times for the last minute.

NOTE: Dispatch times are calculated as the time from when Merge PACS is ready to send to iCEA until it receives the permanent response from iCEA.

NOTE: Delay times are calculated as the time from when the job enters the queue (or re-enters the queue in the case of a retry) until the Sender actually sends the instance (regardless if the send succeeds or fails).

- The **Study Send Tracking** link, as shown in the following example, allows you to track individual studies that have been sent to ICEA (whether successfully or not):

Last update time: 2016-04-05 12:13:08
 Database connections: 4 (1 idle), Server up time: 24h27m50s
 Memory usage (current/peak): Java 101MB/413MB, Native 233MB/323MB, Process Total 387MB/439MB
 Sender Job Queue Summary: New(0), Inprogress(0), Error(2)
 Send summary: 2 (successful) 2 (failed) instances / 2 studies
 Activity in last hour: sent 0 instances for 0 studies, data size 0MB
 Activity in last minute: sent 0 instances for 0 studies, data size 0MB
 Average/Max SenderJob Dispatching times for the last minute: 0 ns/0 ns (0 jobs)
 Average/Max SenderJob Queue Delay times for the last minute: 0 ns/0 ns (0 jobs)

[Study Send Tracking](#) [Sender Job Queue Stats](#) [Cache Volume Stats](#)

Figure 36: Study Send Tracking Link

When you click on the **Study Send Tracking** link, the **Study Instance Send Tracker** page will be displayed, as in the following example:

Last update time: 2016-04-05 12:17:06
 Send summary: 4 (successful) 2 (failed) instances / 3 studies
 Activity in last hour: sent 2 instances for 2 studies, data size 9MB
 Activity in last minute: sent 0 instances for 0 studies, data size 0MB

Study Instances Send Tracker:

StudyInstanceUID	PatientName	MRN	IPID	Accession	First Instance Processed	Last Instance Processed	Instances Pending	Instances Failed
1.2.840.113837.3434743105.2843.1.7326	YOUNGDOE*EUNOR***	40041222		AL01444	2016-04-05 12:14:16	2016-04-05 12:14:16	0	0
1.2.840.113837.3421260597.51923.13.6851	HOWARDOE*CLINTON	184964402		ID09663	2016-04-04 12:19:40	2016-04-04 12:19:40	0	2

Figure 37: Study Instance Send Tracker Page

- The top of the Study Instance Send Tracker page displays some of the same statistics that were displayed on the main Sender Monitor page.
- Below the statistics is a list of studies that have been processed by the Sender. This may include studies processed in this Sender session (*i.e.*, since the last startup of the Sender), as well as studies whose instances failed to send (regardless of whether they were processed in this Sender session or a previous Sender session).

NOTE: By default, up to 5000 failed studies will be displayed on this page (with the oldest touched studies removed when the limit is reached) along with any studies processed within the last 2 hours (or since the Sender last started, whichever is shorter). This is all configurable, however.

- For each failed job listed, you can click on the **Restart** link to retry all the failed instances for that job or the **Delete** link to delete all failed instances for that job.
- For each failed job, you can also click on its **StudyInstanceUID** link to display detailed information for that job, as in the following example:

Failed Instances:

Job ID	Source	InstanceUID	ImagePath	Fail Reason	Restart	Delete	View
5	christy-apr06	1.2.840.113837.1421260597.11959.15.1540	christy-apr06\1.2.840.113837.1421260597.51923.13.6851\5bd36383-0ba8-40ac-b0ed-a7fc1756f9f5	Store Exception - ORA-12899: value too large for column "CLARCD80"."STUDY"."REASONFORSTUDY" (actual: 136, maximum: 64)	Restart	Delete	View
4	christy-apr06	1.2.840.113837.1421260597.13539.16.9545	christy-apr06\1.2.840.113837.1421260597.51923.13.6851\0850ba75-7a09-408e-a360-7c68f00c5394	Store Exception - ORA-12899: value too large for column "CLARCD80"."STUDY"."REASONFORSTUDY" (actual: 136, maximum: 64)	Restart	Delete	View

Figure 38: Details of Failed Job

NOTE: When viewing details of a failed job, you can choose to **Delete**, **Restart** or **View DICOM** attributes for any specific instance.

- The **Sender Job Queue Stats** link, as shown in the following example, allows you to view a variety of detailed statistics about the Sender Job Queue (as described in subsection 9.7.11 below):

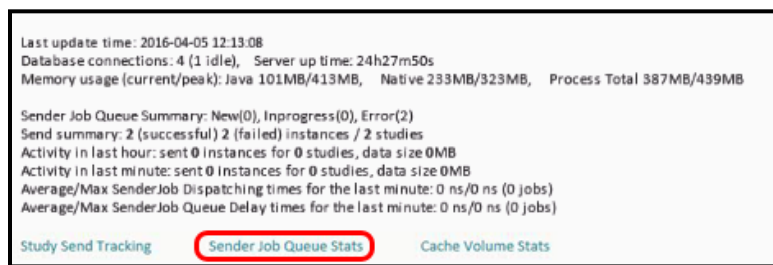


Figure 39: Sender Job Queue Stats Link

When you click on the **Sender Job Queue Stats** link, the **Sender Job Queue Stats** page will be displayed, as in the following example:

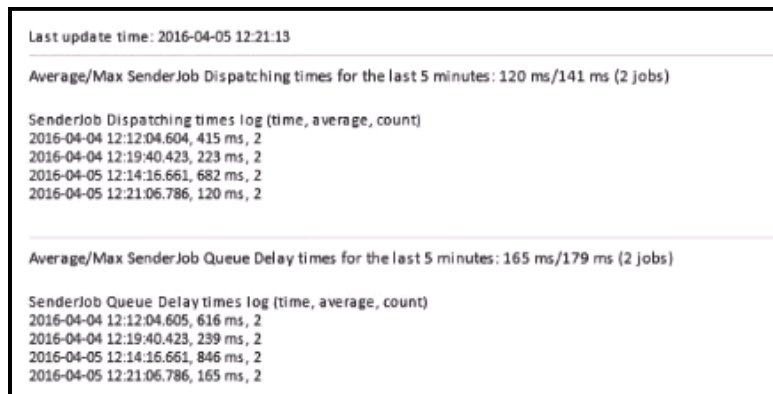


Figure 40: Sender Job Queue Stats Page

The following statistics are displayed:

Statistic	Description								
Average/Max Sender Job Dispatching times for the last 5 minutes	Displays the average and maximum “dispatching time” within the last five minutes. Only jobs that have been started (<i>i.e.</i> , loaded from the database) within the last five minutes will be included in this calculation).								
Sender Job Dispatching times log (time, average, count)	Displays the dispatching time information for several time periods since the most recent startup of the Sender. Each line displays the following information: <table border="1"> <thead> <tr> <th>Statistic</th><th>Description</th></tr> </thead> <tbody> <tr> <td>Time</td><td>Dispatching date and time of the first instance that falls into this bucket</td></tr> <tr> <td>Average</td><td>Average dispatching time for all instances in this bucket</td></tr> <tr> <td>Count</td><td>Number of instances that were sent in that time period</td></tr> </tbody> </table>	Statistic	Description	Time	Dispatching date and time of the first instance that falls into this bucket	Average	Average dispatching time for all instances in this bucket	Count	Number of instances that were sent in that time period
Statistic	Description								
Time	Dispatching date and time of the first instance that falls into this bucket								
Average	Average dispatching time for all instances in this bucket								
Count	Number of instances that were sent in that time period								

Statistic	Description								
	NOTE: By default, each "bucket" is five minutes, but this is configurable on a site-by-site basis. NOTE: By default a maximum of 1500 entries or time periods will be displayed on this page, but this is configurable on a site-by-site basis.								
Average/Max Sender Job Queue delay times for the last 5 minutes	Displays the average and maximum job queue delay times within the last five minutes. Only jobs that have been sent within the last five minutes will be included in this average and maximum calculation.								
Sender Job Queue delay times log (time, average, count)	Displays the delay time information for several time periods since the most recent startup of the Sender. Each line displays the following information: <table> <tr> <th>Statistic</th><th>Description</th></tr> <tr> <td>Time</td><td>Date and time of the first instance that falls into this bucket</td></tr> <tr> <td>Average</td><td>Average queue delay time for all instances in this bucket</td></tr> <tr> <td>Count</td><td>Number of instances that were sent in that time period</td></tr> </table> NOTE: By default, each "bucket" is five minutes, but this is configurable on a site-by-site basis. NOTE: By default a maximum of 1500 entries or time periods will be displayed on this page, but this is configurable on a site-by-site basis.	Statistic	Description	Time	Date and time of the first instance that falls into this bucket	Average	Average queue delay time for all instances in this bucket	Count	Number of instances that were sent in that time period
Statistic	Description								
Time	Date and time of the first instance that falls into this bucket								
Average	Average queue delay time for all instances in this bucket								
Count	Number of instances that were sent in that time period								

- The **Cache Volume Stats** link, as shown in the following example, allows you to view details about the cache volume on disk, as in the following example:

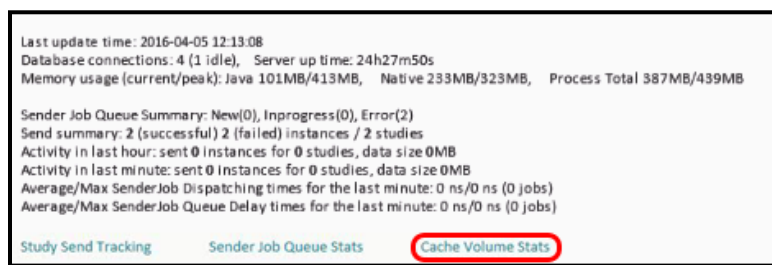


Figure 41: Cache Volume Stats Link

When you click on the **Cache Volume Stats** link, the **Cache Volume** page will be displayed, as in the following example:

Last update time: 09/28/2018 13:45:24			
Cache Volume:			
Cache Volume Data size	Total Instances	Total Folders	Total Empty Folders
42496	20	2	0

Figure 42: Sender Job Queue Stats Page

The Cache Volume page displays the size (*i.e.*, how much space these failed items are taking up on disk) and the current number of instances, folders, and empty folders.

NOTE: Folders are automatically deleted when empty, but not immediately.

- The **Sender Process** section of the Sender Monitor page displays the instances that sent successfully and failed to send since the Sender last started, as well as the In-memory Queue size (*i.e.*, the size of the requested jobs that are currently in memory), as in the following example:

Study Send Tracking		Sender Job Queue Stats		Cache Volume Stats	
Sender Process:					
Instances Sent		Instances Failed		In-memory Queue Size	
2		2		0	
DICOM Sender:					
DICOM Sender Threads		Active Associations		Idle Associations	
2		0		0	
				Presentation Contexts	
				3	
INTERNAL JMS Manager					
Sender connection: 0					
Receiver connection: 1					

Figure 43: Sender Process Information

NOTE: Not all requested jobs are loaded from the database at all times. If this list is long, there will be requested jobs in the database that are not yet loaded into the In-memory Queue.

- The **DICOM Sender** section of the Sender Monitor page displays details about the DICOM Sender, providing a view into the DICOM association and sending level, as in the following example:

Study Send Tracking	Sender Job Queue Stats	Cache Volume Stats	
Sender Process:			
Instances Sent	Instances Failed	In-memory Queue Size	
2	2	0	
DICOM Sender:			
DICOM Sender Threads	Active Associations	Idle Associations	Presentation Contexts
2	0	0	3
INTERNAL JMS Manager			
Sender connection: 0			
Receiver connection: 1			

Figure 44: DICOM Sender Information

- The **Internal JMS Manager** section of the Sender Monitor page displays details about the current connection state of JMS Manager used by Sender, as in the following example:

Study Send Tracking		Sender Job Queue Stats		Cache Volume Stats	
Sender Process:					
Instances Sent		Instances Failed		In-memory Queue Size	
2		2		0	
DICOM Sender:					
DICOM Sender Threads		Active Associations		Idle Associations	
2		0		0	
				Presentation Contexts	
				3	
INTERNAL JMS Manager					
Sender connection: 0					
Receiver connection: 1					
Receiver Sessions: 1					

Figure 45: Internal JMS Manager Information

- The bottom section of the Sender Monitor displays the contents of the Sender.log file, as in the following example:

```

INTERNAL IMS Manager
Sender connection: 0
Receiver connection: 1

Receiver Session(s)

2016-04-05 13:01:45 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 113145: 91, 551, 17, 415
2016-04-05 10:00:57 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 103057: 82, 551, 41, 418
2016-04-05 09:48:28 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 094828: 77, 551, 55, 418
2016-04-05 08:48:58 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 084858: 70, 551, 41, 417
2016-04-05 07:49:58 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 074958: 60, 551, 70, 416
2016-04-05 07:07:24 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 065056: 45, 551, 127, 411 070724: 54, 551, 90, 412
2016-04-05 06:50:28 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 051961: 128, 568, 118, 368 030254: 137, 568, 152, 373 060455: 144, 568, 152, 378 065026: 150, 568, 111, 383 065028: 14, 551, 85, 411
2016-04-05 02:59:53 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 025953: 128, 568, 67, 398
2016-04-05 02:00:05 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 011920: 104, 568, 125, 391 015821: 114, 568, 105, 395 020005: 121, 568, 98, 396
2016-04-05 00:15:25 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 214640: 73, 568, 186, 376 223347: 82, 568, 177, 379 235517: 88, 568, 157, 385 001525: 95, 568, 97, 388
2016-04-04 23:42:49 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 148828: 850, 565, 201, 264 140042: 855, 565, 855, 270 144237: 184, 565, 240, 278 152642: 188, 565, 236, 285 161342: 177, 565, 196, 295 172733: 386, 565, 162, 304 183735: 395, 565, 135, 314 193413: 400, 565, 140, 320 203214: 409, 565, 133, 327 205022: 413, 565, 130, 330 205026: 51, 548, 237, 461 205028: 57, 548, 201, 402 205046: 61, 568, 201, 402 205529: 70, 568, 174, 403 214249: 73, 568, 100, 407
2016-04-04 13:48:18 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 121023: 279, 565, 213, 294 121033: 284, 565, 232, 302 121044: 300, 565, 241, 313 121049: 305, 565, 156, 312 121041: 318, 565, 156, 315 121648: 327, 565, 118, 317 121649: 327, 565, 278, 285 124837: 336, 565, 285 132219: 345, 565, 192, 290 134830: 350, 565, 99, 294
2016-04-04 12:03:48 ERROR: Failed to send instance 1.2.840.113837.3421260597.13598.16.9545: response status 272 - StoreException - GRA-12899: value too large for column "CLARCD00" "STUDY" "REASONFORSTUDY" (actual: 135, maximum: 64)
2016-04-04 12:03:48 ERROR: Failed to send instance 1.2.840.113837.3421260597.13598.15.1540: response status 272 - StoreException - GRA-12899: value too large for column "CLARCD00" "STUDY" "REASONFORSTUDY" (actual: 135, maximum: 64)
2016-04-04 12:07:21 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 120533: 279, 565, 117, 340 120534: 279, 565, 142, 326 120721: 279, 565, 81, 326
2016-04-04 12:05:13 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 120531: 279, 565, 64, 366
2016-04-04 12:05:29 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 120529: 279, 565, 18, 396
2016-04-04 12:05:13 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 120513: 279, 565, 39, 408
2016-04-04 12:03:48 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 120949: 279, 565, 74, 416
2016-04-04 12:03:04 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 120904: 279, 565, 96, 422
2016-04-04 13:47:45 INFO: Process memory usage stats (Time: [heap used] [heap total] [system free] [process total]) 114690: 116, 565, 158, 421 114691: 129, 565, 155, 425 114692: 143, 565, 134, 425 114693: 166, 565, 157, 425 114694: 185, 565, 141, 434 114695: 252, 565, 102, 439 114696: 261, 565, 154, 439 114735: 270, 565, 109, 435 114745: 274, 565, 94, 435
2016-04-04 11:46:35 INFO: Trigger handler registered
2016-04-04 11:46:35 INFO: EA Proxy node AET is ECM_PRC_70
2016-04-04 11:46:35 INFO: EA DICOM node AET is ECM_PRC_70
2016-04-04 11:46:35 INFO: EA DICOM WS URI is /dicom/3
2016-04-04 11:46:35 INFO: EA DICOM WS UniqueName is DICOMWebService3
2016-04-04 11:46:35 INFO: EA PRD is: disjoint is false
2016-04-04 11:46:35 INFO: EA PCS is: IsDataOnlyCompliant is true
2016-04-04 11:46:35 INFO: Initializing INTERNAL IMS Manager
2016-04-04 11:46:35 INFO: Registering trigger handler
2016-04-04 11:46:35 INFO: Sender thread-0 started
2016-04-04 11:46:35 INFO: Sender thread-1 started
2016-04-04 11:46:34 INFO: Sender jobs are changed from in-progress to new status
2016-04-04 11:46:34 INFO: Background Thread started
2016-04-04 11:46:34 INFO: Study Send Track Cleaner Started
2016-04-04 11:46:34 INFO: Creating Service http://www.merge.com/services/ConfigurationWebService from WSDL: http://10.4.19.70:12310/config?wsdl
2016-04-04 11:46:34 INFO: Created DICOM sender: StorageDicu(DestinationID: ECM_PRC_70)DestinationHost: 10.4.19.70DestinationPort: 12000LocalAE: HPHESTAE
2016-04-04 11:46:34 INFO: Initializing service
2016-04-04 11:46:32 INFO: Creating Service http://www.merge.com/services/ConfigurationWebService from WSDL: http://10.4.19.70:12310/config?wsdl
2016-04-04 11:46:30 INFO: Sender process started (control port 11118) Version 7.0.0.88888
  
```

Figure 46: Sender Log

9.6.10. Worklist Servers

The Worklist Servers Monitor displays the current activity and general statistics of the Servers(s) on which RealTime Worklist is running, as in the following example:

WorklistServer;						
Last update time: 2016-04-15 11:01:18						
WorklistServer:						
ID	IP Address	Status	DB Connections	Memory Usage Used/Allocated/Peak	Uptime	Stats
1	127.0.0.1	Running	3	106MB / 187MB / 644MB	692h6m25s	Clients:2 Exams:63 Worklists:9
						<button>Restart</button>

Figure 47: The Worklist Servers Monitor

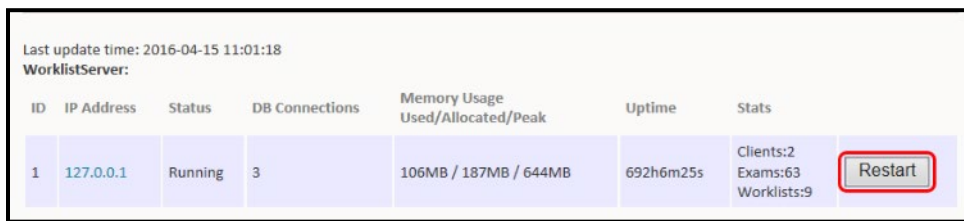
NOTE: Currently, Merge PACS supports the use of only one Worklist Server at a time.

At the Worklist Servers Monitor you can do the following:

- Restart a Worklist Server
- View detailed information about a Worklist Server

a. Restarting a Worklist Server

To restart a Worklist Server, click on the **Restart** button for that Server, as in the following example:



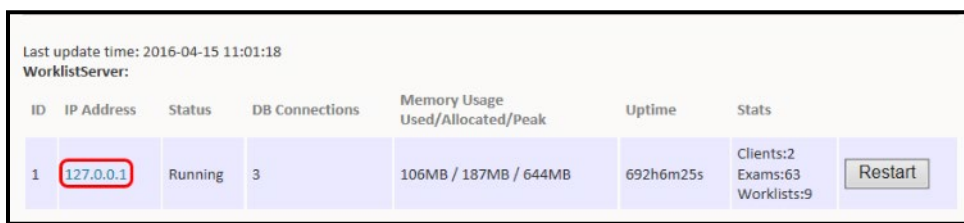
The screenshot shows a web interface for managing Worklist Servers. At the top, it says 'Last update time: 2016-04-15 11:01:18' and 'WorklistServer:'. Below this is a table with the following columns: ID, IP Address, Status, DB Connections, Memory Usage (Used/Allocated/Peak), Uptime, and Stats. There is one row with ID 1, IP Address 127.0.0.1, Status Running, DB Connections 3, Memory Usage 106MB / 187MB / 644MB, Uptime 692h6m25s, and Stats Clients:2, Exams:63, Worklists:9. To the right of the Stats column is a 'Restart' button, which is highlighted with a red rectangle.

ID	IP Address	Status	DB Connections	Memory Usage Used/Allocated/Peak	Uptime	Stats	
1	127.0.0.1	Running	3	106MB / 187MB / 644MB	692h6m25s	Clients:2 Exams:63 Worklists:9	Restart

Figure 48: Restarting a Worklist Server

b. Viewing Detailed Information about a Worklist Server

To view detailed information about a Worklist Server, click on the **IP Address** for the desired Server, as in the following example:



The screenshot shows the same web interface as Figure 48. In this view, the 'IP Address' '127.0.0.1' in the first row is highlighted with a red rectangle. The 'Restart' button is still visible but not highlighted.

ID	IP Address	Status	DB Connections	Memory Usage Used/Allocated/Peak	Uptime	Stats	
1	127.0.0.1	Running	3	106MB / 187MB / 644MB	692h6m25s	Clients:2 Exams:63 Worklists:9	Restart

Figure 49: Worklist Server IP Address

The **Worklist Server Details** page is displayed, as in the following example:



Figure 50: Worklist Server Details Page

- The top section of the Worklist Server Details page displays general statistics about the Worklist Server, as in the following example:

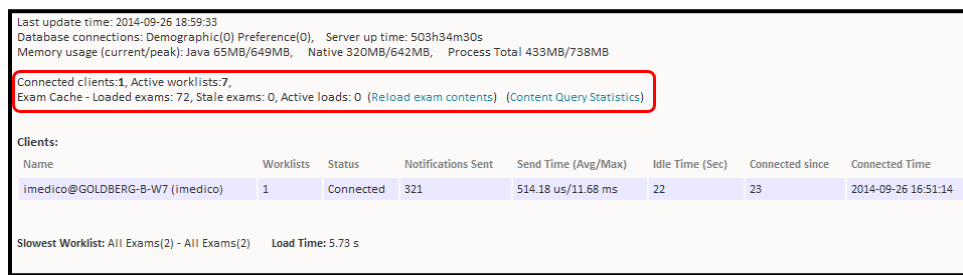


Figure 51: General Statistics about Worklist Server

- The following information is displayed:

Statistic	Description
Connected clients	Number of client workstations currently connected to this Worklist Server
Active worklists	Number of currently active worklists on this Worklist Server
Exam Cache:	
Loaded exams	Studies and orders that are currently cached in memory (these do not need to be retrieved from the database)
Stale exams	Studies and orders that are not currently in use but are still in memory (the system waits a short time before clearing these)
Exam loads	Studies and orders that are currently being loaded into memory (this number is typically very small since the process is very quick).

- Click on the **Reload exam contents** link to reload the database entries to Worklist Server, thereby and synching the worklists with the latest information from the database. This should be done in any of the following cases where changes to the database are not synched with the worklist servers by default:
 - Any changes have been made to a particular user's **Credentials & Licensing** information, as described in subsection.9.11.1 below.
 - Any changes have been made at the **Credentialing and Licensure** management page, as described in subsection 9.13.10 below.
 - Any changes have been made at the **Facility Configuration** management page, as described in subsection 10.5.2 below.
- Click on the **Content Query Statistics** link to view detailed information about the content queries run against this Worklist Server, as in the following example:

Content Queries:						
Name	DB Thread	Runs	Fails	First Run	Duration (Min/Avg/Max)	Query
CustomTag	5	205	0	2.59 ms	561.60 us / 12.66 ms / 978.85 ms	SELECT CustomTag_1,CustomTag_2,CustomTag_3,CustomTag_4,CustomTag_5,CustomTag_6,CustomTag_7,CustomTag_8,CustomTag_9,CustomTag_10 FROM CustomTag WHERE StudyInstanceUID = ? FOR READ ONLY
InstitutionGroup	4	27	0	8.05 ms	557.70 us / 50.19 ms / 731.53 ms	SELECT INSTGROUP.REGION, INST.INSTITUTIONGROUP FROM INSTITUTIOND INST, INSTITUTIONGROUP INSTGROUP WHERE INST.INSTITUTIONNAME = ? AND INST.INSTITUTIONGROUP = INSTGROUP.GROUPNAME FOR READ ONLY
StudyAcuityAnswers	5	205	0	5.70 ms	507.70 us / 13.42 ms / 653.03 ms	SELECT c.Question_text, c.Factor, a.Weight FROM Answer as a, Patient_Answer as b, Question as c WHERE b.StudyInstanceUID = ? and b.Question_id = c.Id and b.Answer_id = a.Id FOR READ ONLY
StudyAcuityAnswers	3	255	0	5.03 ms	531.30 us / 12.77 ms / 648.51 ms	SELECT c.Question_text, c.Factor, a.Weight FROM Answer as a, Patient_Answer as b, Question as c WHERE b.StudyInstanceUID = ? and b.Question_id = c.Id and b.Answer_id = a.Id FOR READ ONLY
DeadlineTracking	5	205	0	4.24 ms	672.70 us / 14.28 ms / 637.66 ms	SELECT ARRIVETIME, DUETIME, COMPLETEDTIME, HL7Priority FROM STUDYDEADLINETRACKING WHERE STUDYINSTANCEUID=? FOR READ ONLY
InstitutionGroup	2	36	0	34.00 ms	595.60 us / 43.45 ms / 631.50 ms	SELECT INSTGROUP.REGION, INST.INSTITUTIONGROUP FROM INSTITUTIOND INST, INSTITUTIONGROUP INSTGROUP WHERE INST.INSTITUTIONNAME = ? AND INST.INSTITUTIONGROUP = INSTGROUP.GROUPNAME FOR READ ONLY
InstitutionGroup	3	23	0	404.28 ms	570.80 us / 31.10 ms / 605.78 ms	SELECT INSTGROUP.REGION, INST.INSTITUTIONGROUP FROM INSTITUTIOND INST, INSTITUTIONGROUP INSTGROUP WHERE INST.INSTITUTIONNAME = ? AND INST.INSTITUTIONGROUP = INSTGROUP.GROUPNAME FOR READ ONLY

Figure 52: Content Query Statistics

- The next section of the Worklist Server Details page displays details about each of the **current client Workstations** retrieving data from this Worklist Server, including usernames, number of worklists, etc., as in the following example:

Last update time: 2014-09-26 18:59:33
 Database connections: Demographic(0) Preference(0), Server up time: 503h34m30s
 Memory usage (current/peak): Java 65MB/649MB, Native 320MB/642MB, Process Total 433MB/738MB

Connected clients:1, Active worklists:7,
 Exam Cache - Loaded exams: 72, Stale exams: 0, Active loads: 0 ([Reload exam contents](#)) ([Content Query Statistics](#))

Clients:							
Name	Worklists	Status	Notifications Sent	Send Time (Avg/Max)	Idle Time (Sec)	Connected since	Connected Time
imedico@GOLDBERG-B-W7 (imedico)	1	Connected	321	514.18 us/11.68 ms	22	23	2014-09-26 16:51:14

Slowest Worklist: All Exams(2) - All Exams(2) Load Time: 5.73 s

Figure 53: Current Client Information

- The next section of the Worklist Server Details page displays details about the **slowest worklist**, including the worklist name, worklist ID, worklist block name and worklist block ID, as in the following example:

Last update time: 2014-09-26 18:59:33
 Database connections: Demographic(0) Preference(0), Server up time: 503h34m30s
 Memory usage (current/peak): Java 65MB/649MB, Native 320MB/642MB, Process Total 433MB/738MB

Connected clients:1, Active worklists:7,
 Exam Cache - Loaded exams: 72, Stale exams: 0, Active loads: 0 ([Reload exam contents](#)) ([Content Query Statistics](#))

Clients:							
Name	Worklists	Status	Notifications Sent	Send Time (Avg/Max)	Idle Time (Sec)	Connected since	Connected Time
imedico@GOLDBERG-B-W7 (imedico)	1	Connected	321	514.18 us/11.68 ms	22	23	2014-09-26 16:51:14

Slowest Worklist: All Exams(2) - All Exams(2) Load Time: 5.73 s

Figure 54: Slowest Worklist Information

- The next section of the Worklist Server Details page displays details about all active or recently active worklists, including custom searches performed from the RealTime Study List, as in the following example:

Worklists:

Name	Clients Content/Counts	Exams total/loading/pending	Pending Client Events	Status	Idle/Load time	Actions
▶ Unread Exams(1) - Unread Exams(1)	1/0	5/0/0	0	EXAM_CONTENT_LOADED	0/97.94 ms	Content Reload Filters
▶ All Tech Worklist(7) - Problem Studies(7)	1/0	0/0/0	0	EXAM_CONTENT_LOADED	0/1.04 s	Content Reload Filters
▶ Test(12) - All Exams(2)	1/0	72/0/0	0	EXAM_CONTENT_LOADED	0/5.60 s	Content Reload Filters
▶ CTest(5) - All Exams(2)	1/0	72/0/0	0	EXAM_CONTENT_LOADED	0/1.15 s	Content Reload Filters
▶ All Tech Worklist(7) - All Exams(2)	1/0	72/0/0	0	EXAM_CONTENT_LOADED	0/1.30 s	Content Reload Filters

Worklist Timing Trace is OFF

Worklist PreLoading is ON

Drop unused worklists

Figure 55: Slowest Worklist Information

- For each worklist displayed, you can do the following:
 - Click on the name of the worklist to toggle the display of timing-related information for the worklist, as in the following example:

Worklists:

Name	Clients Content/Counts	Exams total/loading/pending	Pending Client Events	Status	Idle/Load time	Actions																																
Unread Exams(1) - Unread Exams(1)	1/0	5/0/0	0	EXAM_CONTENT_LOADED	0/97.94 ms	Content Reload Filters																																
<table><tr><th>Name</th><th>Value</th></tr><tr><td>1 Creation Time</td><td>2014-09-26 15:41:23</td></tr><tr><td>2 Exam List Queries</td><td>97.49 ms</td></tr><tr><td>3 Load Duration</td><td>97.94 ms</td></tr><tr><td>4 Load Time</td><td>2014-09-26 16:51:14</td></tr><tr><td>5 Query ExplicitLinkedOrdersQuery-General</td><td>97.49 ms</td></tr><tr><td>6 Query ExplicitLinkedProceduresQuery-General</td><td>96.89 ms</td></tr><tr><td>7 Query ExplicitLinkedStudiesQuery-General</td><td>18.90 ms</td></tr><tr><td>8 Query OrderOnlyQuery-General</td><td>67.54 ms</td></tr><tr><td>9 Query StudyOnlyQuery-General</td><td>53.40 ms</td></tr><tr><td>10 Query StudyWithOrderQuery-General</td><td>82.45 ms</td></tr><tr><td>11 Status EXAM_CONTENT_LOADING</td><td>32.90 us</td></tr><tr><td>12 Status EXAM_LIST_LOADED</td><td>98.40 us</td></tr><tr><td>13 Status EXAM_LIST_LOADING</td><td>97.90 ms</td></tr><tr><td>14 Status INITIALIZING</td><td>1.35 s</td></tr><tr><td>15 Status WAITING_TO_LOAD</td><td>1h9m49s</td></tr></table>							Name	Value	1 Creation Time	2014-09-26 15:41:23	2 Exam List Queries	97.49 ms	3 Load Duration	97.94 ms	4 Load Time	2014-09-26 16:51:14	5 Query ExplicitLinkedOrdersQuery-General	97.49 ms	6 Query ExplicitLinkedProceduresQuery-General	96.89 ms	7 Query ExplicitLinkedStudiesQuery-General	18.90 ms	8 Query OrderOnlyQuery-General	67.54 ms	9 Query StudyOnlyQuery-General	53.40 ms	10 Query StudyWithOrderQuery-General	82.45 ms	11 Status EXAM_CONTENT_LOADING	32.90 us	12 Status EXAM_LIST_LOADED	98.40 us	13 Status EXAM_LIST_LOADING	97.90 ms	14 Status INITIALIZING	1.35 s	15 Status WAITING_TO_LOAD	1h9m49s
Name	Value																																					
1 Creation Time	2014-09-26 15:41:23																																					
2 Exam List Queries	97.49 ms																																					
3 Load Duration	97.94 ms																																					
4 Load Time	2014-09-26 16:51:14																																					
5 Query ExplicitLinkedOrdersQuery-General	97.49 ms																																					
6 Query ExplicitLinkedProceduresQuery-General	96.89 ms																																					
7 Query ExplicitLinkedStudiesQuery-General	18.90 ms																																					
8 Query OrderOnlyQuery-General	67.54 ms																																					
9 Query StudyOnlyQuery-General	53.40 ms																																					
10 Query StudyWithOrderQuery-General	82.45 ms																																					
11 Status EXAM_CONTENT_LOADING	32.90 us																																					
12 Status EXAM_LIST_LOADED	98.40 us																																					
13 Status EXAM_LIST_LOADING	97.90 ms																																					
14 Status INITIALIZING	1.35 s																																					
15 Status WAITING_TO_LOAD	1h9m49s																																					

Figure 56: Slowest Worklist Information

- Click on the **Content** link in the Action column to view all exams that are included in the worklist.
- Click on the **Reload** link in the Action column to unload and then reload the worklist, similar to if you had changed the worklist definition.
- Click on the **Filters** link in the Action column to show the SQL statements that are run when that worklist is loaded.
- Below the list of active worklists are the following links:

Link	Description
Worklist Timing Trace	Toggle on and off whether this information should be automatically dumped to a CSV file in PACS2DATA\log every time a worklist is loaded
Worklist Preloading	Toggle on and off whether worklist preloading should be enabled for this session of the Worklist Server (<i>i.e.</i> , until the Worklist Server is next restarted). If worklist preloading is turned on, worklists that have been loaded during the last configurable number of hours that took longer than a configurable number of seconds to load will be preloaded into memory, even if no client is requesting them. This will make loading faster if and when a client does request those studies
Drop unused worklists	Forces worklists without clients to be unloaded. Note that if worklist preloading is turned on, the worklists that qualify for preloading will be preloaded again shortly thereafter

- The next section of the Worklist Server Details page displays details about the current state and general statistics of all active database connections as in the following example:

Thread	DB	Jobs (Pending/Done)	Current job	Job Duration (Crt/Avg/Max)	Active Queries	Worklists	Slowest Job
1	Disconnected	0/49		43.51 s / 58.59 ms / 314.60 ms	0	0	ExplicitLinkUpdateJob
2	Disconnected	0/33		43.44 s / 116.03 ms / 1.96 s	0	0	StudyChangeJob
3	Disconnected	0/29		43.51 s / 51.94 ms / 209.83 ms	0	0	ExamsSelectionJob
4	Disconnected	0/23		43.53 s / 268.57 ms / 3.76 s	0	0	StudyChangeJob
5	Disconnected	0/23		43.50 s / 153.19 ms / 1.34 s	0	0	StudyChangeJob
6	Disconnected	0/21		43.44 s / 276.52 ms / 4.06 s	0	0	StudyChangeJob
7	Disconnected	0/15		49.91 s / 148.93 ms / 1.19 s	0	0	StudyChangeJob
8c	Connected	0/14		6.85 s / 115.99 ms / 938.61 ms	0	0	ExamLoadJob
9c	Connected	0/2		6.80 s / 42.71 ms / 60.42 ms	0	0	ExamLoadJob
10c	Disconnected	0/0		1.22 s / 0 / 0 ns	0	0	
11c	Disconnected	0/0		2.78 s / 0 / 0 ns	0	0	
12c	Disconnected	0/0		1.22 s / 0 / 0 ns	0	0	

Total pending content/list jobs: 0/0
Average wait times for the last 5 minutes: 8.29 ms (3 jobs)
Average processing times for the last 5 minutes: 39.55 ms (3 jobs)

Figure 57: Database Threads Information

Below the list of Database threads are the following links:

Link	Description
Pending jobs	List jobs waiting for a database thread.
Job stats	Shows the statistics of the slowest jobs.

- The next section of the Worklist Server Details page displays details about the current connection state of **CLIENT JMS Manager** as in the following example:

Sender connection: 1 Receiver connection: 0		
JMS Sender Sessions(2)		
Sender Sessions:		
Thread ID	Thread Name	Session Name
25	Worklist-Most recent 2000 studies(1) - All Studies(2)	b88a0c16-5488-11e4-988a-00505696f54
57	Thread-5 (HornetQ-client-global-threads-595243056)	20112f5d-5488-11e4-988a-00505696f54

Figure 58: Client JMS Manager Information

Click on **JMS Sender Sessions** link to toggle the display of details of current active JMS sender sessions.

- The next section of the Worklist Server Details page displays details about the current state of the worklist server as in the following example:

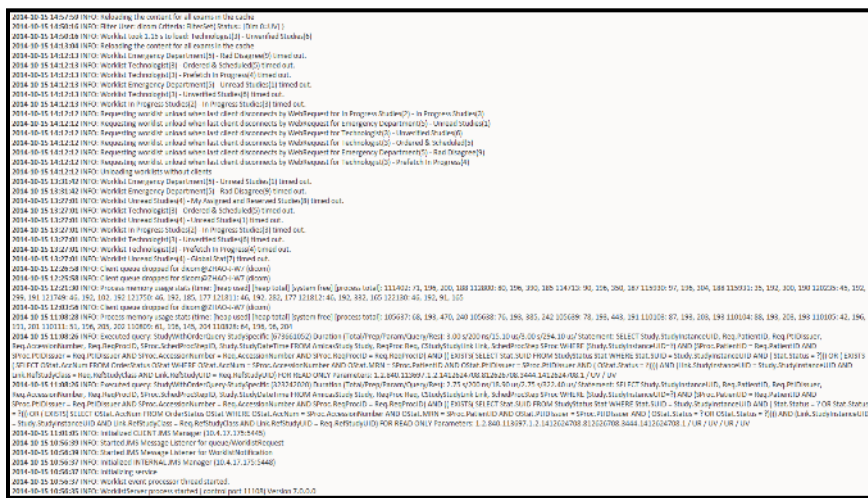


Figure 59: Worklist Server Information

9.6.11. Client Notification Server

The **Client Notification Server Monitor** page displays information about the process that is responsible for sending notification messages from the server to clients (Workstations) as in the following example:

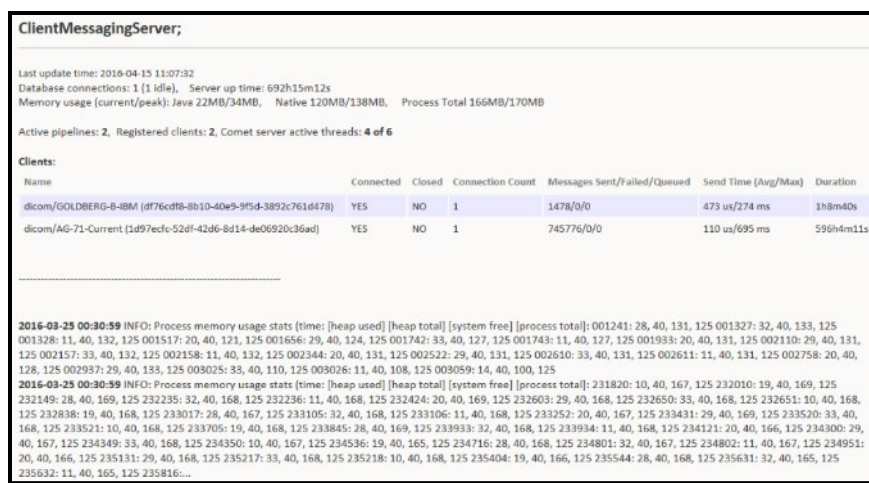


Figure 60: Client Server Notification Monitor

This process is mostly used for sending RTWL notifications to the Workstation when an order or a study changes. As with other monitor pages, the Client Notification Server Monitor page displays the memory status of the process, the number of database connections, up time and also some specific information on the message flow to each client currently connected.

9.6.12. Application Server

The **Application Server Monitor** page displays information about the web server (WildFly) responsible for processing requests from clients (Workstations), other Merge PACS servers (via telmed) and third-party applications, as in the following example:

MergePACSWeb;

Last update time: 04/25/2017 12:32:49
 Database connections: 5 (5 idle). Server up time: 96h27m50s
 Memory usage (current/peak): Java 251MB/380MB, Native 227MB/486MB, Process Total 753MB/976MB

Searches (0 running, 0 priority waiting, 0 waiting):

ID	Status	Type	Priority	User	Results	Duration	Start Time	Wait Time	Filters
64	Complete	PriorSearch	HIGH	chrissy	3	67 ms	04/25/2017 12:15:48	0 ns	PatientID=456,JPID=AWAY,StudyInstanceUID=1.2.840.113837.2566471667.01.58.54486.0587371.229,UVas Unread,Extended
63	Complete	PriorSearch	LOW	chrissy	5	122 ms	04/25/2017 12:15:37	0 ns	PatientID=123,JPID=HOME,DOB=19670512,Sex=F,PatientName=IONES*KATE***,StudyInstanceUID=1.3.12.2.1107.01.186.54426.1867505.117,VoiceClipsInc
62	Complete	PriorSearch	MEDIUM	chrissy	5	151 ms	04/25/2017 12:14:19	16 ms	PatientID=123,JPID=HOME,StudyInstanceUID=1.3.12.2.1107.01.186.54426.1867505.117,UVas Unread
61	Complete	PriorSearch	HIGH	chrissy	5	158 ms	04/25/2017 12:12:27	16 ms	PatientID=123,JPID=HOME,StudyInstanceUID=1.3.12.2.1107.01.186.54426.1867505.117,UVas Unread,Extended
60	Complete	StudySearch	MEDIUM	chrissy	1	63 ms	04/25/2017 10:38:17	0 ns	(patientID:789)(originalPatientID:789)(JPID:EXT)(userid:21)(CannedDate:StartDate:20160425 EndDate:20170425)
59	Complete	StudySearch	MEDIUM	chrissy	4	106 ms	04/25/2017 10:22:56	0 ns	(patientID:123)(originalPatientID:123)(JPID:HOME)(userid:21)
58	Complete	PriorSearch	LOW	chrissy	5	96 ms	04/25/2017 10:19:23	16 ms	PatientID=123,JPID=HOME,DOB=19670512,Sex=F,PatientName=IONES*KATE***,StudyInstanceUID=1.3.12.2.1107.01.186.54426.1867505.117,VoiceClipsInc
57	Complete	PriorSearch	LOW	chrissy	5	97 ms	04/25/2017 10:19:02	0 ns	PatientID=123,JPID=HOME,DOB=19670512,Sex=F,PatientName=IONES*KATE***,StudyInstanceUID=1.3.12.2.1107.01.186.54426.1867505.117,VoiceClipsInc
56	Complete	PriorSearch	HIGH	chrissy	5	226 ms	04/25/2017 10:16:47	0 ns	PatientID=123,JPID=HOME,StudyInstanceUID=1.3.12.2.1107.01.186.54426.1867505.117,UVas Unread,Extended

Figure 61: Application Server Monitor

Currently, the only process specific information on this monitor page is related the following:

- DICOM associations for processing extended search requests
- MPI operations, if Merge PACS is configured to run in **Integrated** mode and MPI support is enabled.

If the retrieval of any comparison studies have been triggered manually, as described in Section 1.2.69 above, you can also monitor the progress of the current retrieval process triggered by the receipt of studies by clicking on the **Half Folder Prior Restoration** link, as in the following example:

Half-Folder Prior Restoration

State: **running**
 Number of Threads: **1**
 Pending File: C:\PACS2DATA\PriorRestore\priorRestoreManualPending.txt
 Maximum Priors To Restore: **unlimited**
 AE Title Filter: **unfiltered**

Average/Max restore queue size for the last 5 minutes: 0/0 (0 items processed) [[Detail](#)]
 Average/Max queuing times for the last 5 minutes: 0 ns/0 ns (0 jobs) [[Detail](#)]
 Average/Max waiting times for the last 5 minutes: 0 ns/0 ns (0 jobs) [[Detail](#)]
 Average/Max submission times for the last 5 minutes: 0 ns/0 ns (0 jobs) [[Detail](#)]
 Average/Max total times for the last 5 minutes: 0 ns/0 ns (0 jobs) [[Detail](#)]
 Average/Max querying times for the last 5 minutes: 0 ns/0 ns (0 jobs) [[Detail](#)]
 Average/Max halffolder counts for the last 5 minutes: 0/0 (0 items) [[Detail](#)]

Figure 62: Half Folder Prior Restoration Progress

For any MPI operations, click on the triangle to the left of the operation to view detailed information about that operation, as in the following example:

Searches (0 running, 0 priority waiting, 0 waiting) :

ID	Status	Type	Priority	User	Results	Duration	Start Time	Wait Time	Filters
64	Complete	PriorSearch	HIGH	christy	3	67 ms	04/25/2017 12:15:48	0 ms	PatientID=456,IPID=AWAY,StudyInstanceUID=1.2.840.113837.2566471667.01.58.0
5	Complete	PriorSearch	LOW	christy	5	122 ms	04/25/2017 12:15:37	0 ms	PatientID=123,IPID=HOME,DOB=19870512,Sex=F,PatientName=JONES*KATE***

Name	Status	Start Time	Duration	Results (found/accepted)	Message	Criteria	Timing
1 ExtSOP1-63	Complete	04/25/2017 12:15:37	66 ms	2/1	Success	(patientID:123) (patientName:JONES*)	Exec=3 us Completion=3 us Query=65 ms CreateQuery=622 us TimeToStart=8 ms
2 ExtSOP2-63	Complete	04/25/2017 12:15:37	92 ms	3/2	Success	(patientName:JONES*) (patientDOB:19870512)	Exec=2 us Completion=3 us Query=91 ms CreateQuery=405 us TimeToStart=13 ms
3 ExtPCS-IPID+MRN-63	Complete	04/25/2017 12:15:37	79 ms	2/2	Success	(patientID:123) (IPID:HOME)	Exec=3 us Completion=800 ms Query=79 ms CreateQuery=485 us TimeToStart=12 ms

Figure 63: MPI Operation Details

9.6.13. Messaging Server

The **Messaging Server Monitor** page displays information about the server responsible for the communication between server processes and, in integrated mode, is also responsible for dispatching notifications received from EA, as in the following example:

MessagingServer;

Last update time: 2016-04-15 11:10:59
 Database connections: 1 (1 idle), Server up time: 692h18m38s
 Memory usage (current/peak): Java 26MB/131MB, Native 140MB/262MB, Process Total 189MB/299MB

JMS server:

Connection Count: 10
 Version : 2.2.13.Final (HQ_2_2_13_FINAL_A57, 122)

Queue list:

Name	Type	Message Count	Consumer Count
Event.StudyRetrieve	Topic	0	0
Event.StudyXMLInvalidator	Topic	0	0
EANotification	Queue	0	0
zdfdf7f1-d7c5-4b66-8766-8c2af8918cfa	Temp	0	1
•			
•			
•			
Event.StorageServer	Topic	0	0

2016-03-25 00:31:43 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 003143: 57, 153, 97, 200
2016-03-25 00:31:00 INFO: Process memory usage stats (time: [heap used] [heap total] [system free] [process total]): 234307: 20, 153, 167, 200 234515: 29, 153, 166, 200 234720: 38, 153, 168, 200 234921: 47, 153, 166, 200 235125: 56, 153, 168, 200 235331: 65, 153, 169, 200 235536: 74, 153, 168, 200 235741: 83, 153, 165, 200 235946: 92, 153, 167, 200 000151: 101, 153, 167, 200 000316: 107, 153, 168, 200 000517: 20, 153, 168, 200 000521: 29, 153, 164, 200 000727: 38, 153, 167, 200 000932: 47, 153, 129, 200 001136: 56, 153, 132, 200 001341: 65, 153, 132, 200 001547: 74, 153, 105, 200 001751: 83, 153, 127, 200 001957: 92, 153, 129, 200 002201: 101, 153, 132, 200 002322: 107, 153, 131, 200 002323: 20, 153, 131, 200 002532: 29, 153, 132, 200 002737: 38, 153, 129, 200

Figure 64: Messaging Server Monitor

The monitor page lists the status of all message queues and topics currently active in the server.

9.7. Queues

The various Queue pages display events that are currently waiting to be processed by Merge PACS. Clicking on the “Queues” link on the top category menu will display a submenu of the individual available queue pages as shown below:

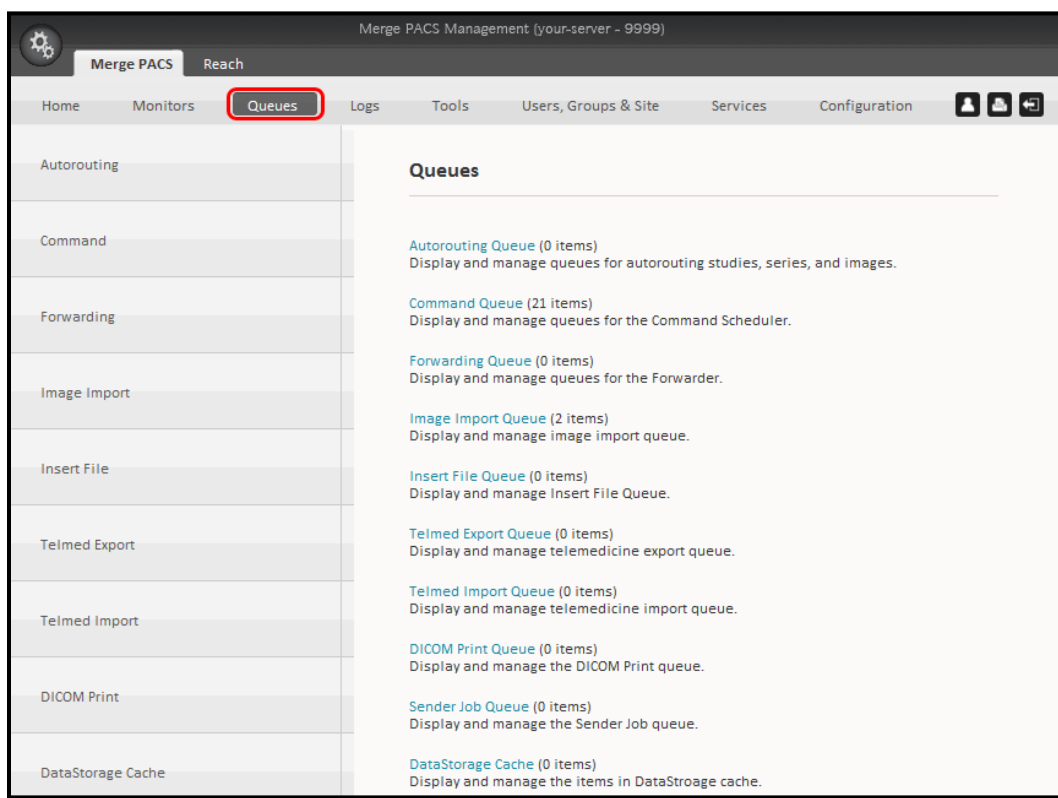


Figure 65: The Queues Pages

To select a particular queue page, click on the appropriate link in the Queues submenu, or the link displayed on the main Queues page. Some of the individual queue pages are for optional components of Merge PACS and will only appear if those components have been enabled.

NOTE: On each queue page, you can sort the data by clicking on a column heading and then clicking on the little blue triangle next to the column heading to select whether to sort in ascending or descending order.

NOTE: You can set how often the data on the screen is updated for each queue page by setting the **refresh** rate as described in subsection 9.13.17 below.

9.7.1. Archive/Retrieve Queue

The Archive Queue/Retrieve page lists the studies that are currently being transferred to or retrieved from the Archive, are waiting to be transferred or retrieved, have successfully been transferred or retrieved, or are remaining after archiving or retrieval failed, as shown in the example below:

The screenshot shows the 'Archive/Retrieve Queue' interface. At the top, there are buttons for 'Delete Failed Jobs' and 'Restart Failed Jobs'. Below these are checkboxes for 'Include completed jobs' and 'Include jobs with start time in the future', and radio buttons for 'Archive jobs' (selected) and 'Retrieve jobs'. A search bar with a dropdown menu set to 'Patient Name' and a 'Search' button is present, along with an 'Archive Selected Studies' button. Below the search bar is a table with columns: Patient Name, Patient ID, IPID, DOB, Sex, Accession, Study Date, and Images. The table shows two jobs in the queue. The first job is an 'ARCHIVE' job for 'GYN MR' with a status of 'Requested' and a completion ratio of '0/0'. The second job is an 'ARCHIVE' job for 'CMT5 CT' with a status of 'Requested' and a completion ratio of '0/0'. Both jobs have a 'Delete' button next to them.

Figure 66: The Archive/Retrieve Queue

The Archive Queue/Retrieve page will be available either if Merge PACS is configured to run in **Standalone** mode and the optional Merge Archive feature has been enabled or if Merge PACS is configured to run in **Integrated** mode, with the following differences:

- In **Standalone** mode, there are separate radio buttons for **Archive jobs** and **Retrieve jobs**.
- In **Integrated** mode only **Retrieve jobs** are applicable, and the Archive jobs radio button is hidden.

a. Archive Jobs

By default, the Archive/Retrieve Queue will display information about archive jobs. If you are currently viewing retrieve jobs and want to view archive jobs, click on the **Archive jobs** radio button at the top of the page, as in the following example:

This screenshot is similar to Figure 66, but the 'Archive jobs' radio button is highlighted with a red circle, indicating it is the selected option for viewing archive jobs.

Figure 67: The Archive Jobs Button

NOTE: All studies sent to the Merge PACS Server will be qualified for archiving after the number of minutes, hours, days, etc., that have been configured for Merge PACS. Each qualified study is inserted into the Archive Queue for archiving.

At the Archive Jobs Queue, you can do the following:

- Click the **Delete Failed Jobs** button to remove all failed jobs from the Archive Jobs Queue.

- Click the **Restart Failed Jobs** button to restart all failed jobs in the Archive Jobs Queue.
- Select the **Include Completed Jobs** button to display archive jobs that have already been completed.
- Select the **Include jobs with start time in the future** option to display archive jobs that have been scheduled to start in the future.
- Increase or decrease the priority of one or more pending jobs in the queue by clicking on the **Increase Priority** or **Decrease Priority** button for individual jobs. Note that “1” is the highest priority available and “5” is the lowest.
- Manually request archiving of one or more studies that are not currently qualified for archiving or in the process of being archived as follows:
 - Use the search fields at the top of the window to locate the studies you want to archive, as in the following example:

The screenshot shows the 'Archive / Retrieve Queue' window. At the top, there are buttons for 'Delete Failed Jobs' and 'Restart Failed Jobs'. Below these are checkboxes for 'Include completed jobs' (unchecked), 'Include jobs with start time in the future' (checked), and radio buttons for 'Archive jobs' (selected) and 'Retrieve jobs' (unselected). A search bar is highlighted with a red box, containing 'Search By: Patient Name' and a dropdown menu set to 'Doe, Jill'. To the right of the search bar is a 'Search' button and an 'Archive Selected Studies' button. Below the search bar is a table with columns: Patient Name, Patient ID, IPID, DOB, Sex, Accession, Study Date, and Images. Two rows are visible, both for 'DOE JILL' with Patient ID 'AM-0107' and IPID 'Home'. The first row has DOB '19800405', Sex 'F', Accession 'SE0000184', Study Date '20031020', and Images '1'. The second row has DOB '19800405', Sex 'F', Accession 'SE0000181', Study Date '20031019', and Images '1'. Both rows have a checkbox in the first column, which is highlighted with a red box.

Figure 68: Searching for Studies Not in the Archive Queue

- Select the studies you want to be archived by clicking on the checkbox next to those studies (or click on the checkbox to the left of the Patient Name column header to select all studies listed).
- Click on the Archive Selected Studies button.

b. Retrieve Jobs

To view information about retrieve jobs, click on the **Retrieve jobs** radio button at the top of the page, as in the following example:

The screenshot shows the 'Archive / Retrieve Queue' window. At the top, there are buttons for 'Delete Failed Jobs' and 'Restart Failed Jobs'. Below these are checkboxes for 'Include completed jobs' (unchecked), 'Include jobs with start time in the future' (checked), and radio buttons for 'Archive jobs' (unselected) and 'Retrieve jobs' (selected). The 'Retrieve jobs' radio button is highlighted with a red box.

Figure 69: The Retrieve Jobs Button

The Retrieve Jobs queue will be displayed, showing the studies that are currently being retrieved from the Archive, are waiting to be retrieved, have successfully been retrieved, or are remaining after retrieval failed, as shown in the example below:

Figure 70: The Retrieve Jobs Queue

At the Retrieve Jobs Queue, you can do the following:

- Click the **Delete Failed Jobs** button to remove all failed jobs from the Retrieve Jobs Queue.
- Click the **Restart Failed Jobs** button to restart all failed jobs in the Retrieve Jobs Queue.
- Select the **Include Completed Jobs** button to display retrieve jobs that have already been completed.
- Select the **Include jobs with start time in the future** option to display retrieve jobs that have been scheduled to start in the future.
- Select the **Delete All Jobs** button to remove all jobs from the Retrieve Jobs queue.
- Increase or decrease the priority of one or more pending jobs in the queue by clicking on the **Increase Priority** or **Decrease Priority** button for individual jobs. Note that “1” is the highest priority available and “5” is the lowest.

9.7.2. Autorouting Queue

When Merge PACS is configured to run in **Standalone** mode, the Autorouting Queue page lists the images that are scheduled to be qualified by the AutoRouter for forwarding to a DICOM device, as shown in the following example:

Time Queued	Start Time	End Time	Submitter	Originator	Status	User Name	Delete
2016-04-18 12:49:05	2016-04-18 12:49:05	2016-04-18 12:49:05	PACS-W2KB-04	storecu	Requested	Unknown	Delete
2016-04-18 12:49:07	2016-04-18 12:49:07	2016-04-18 12:49:07	PACS-W2KB-04	storecu	Requested	Unknown	Delete
2016-04-18 12:49:08	2016-04-18 12:49:08	2016-04-18 12:49:08	PACS-W2KB-04	storecu	Requested	Unknown	Delete

Figure 71: The Autorouting Queue

Once an image has been qualified by the AutoRouter, information regarding that image and where it is supposed to be forwarded is sent to the Forwarder Queue for execution.

At the Autorouting Queue page, you can **delete** any pending job by clicking on the “Delete” button for the job in question.

9.7.3. Command Queue

This page lists the commands currently scheduled for execution by the CommandScheduler, including jobs performed with Quality Control Editor or the Patient and Study Demographics windows from within the Merge PACS Workstation, and update commands received via the **HIS/RIS Interface** as described in Section 3.8 above. An example of the Command Queue page is shown below:

Time Queued	Command	Type	Description / Error Message	State	Delete	Restart
2016-03-17 14:54:33	CheckResources	Periodic	Monitors disk space and deletes compressed images when necessary	Queued	Delete	
2016-03-17 14:54:34	PeriodicCleanupDB	Periodic	Submits command to delete old database log entries	Queued	Delete	
2016-03-17 14:54:36	SystemMonitor	Periodic	Notifies PACS support of image importing queue backlog	Queued	Delete	
2016-03-17 14:54:36	AnetCommunicatorEntries	Periodic	Submits command to transfers event data PACS Watch	Queued	Delete	
2016-03-17 14:54:36	ImportLogExtractorInvoker	Periodic	Submits command to extract import log to file	Queued	Delete	

Figure 72: The Command Queue

If you are logged into the system as a member of the “Super User” group, you can perform the following tasks at the Command Queue page:

- **Delete** individual commands
- **Restart** individual failed commands
- **Delete** all failed commands.

Note that **Periodic** commands are only visible to members of the “Super User” group. All other users will only be able to see **Explicit** commands.

9.7.4. Forwarding Queue

When Merge PACS is configured to run in **Standalone** mode, the Forwarding Queue page lists the studies that are scheduled to be forwarded to a DICOM workstation or archive, as shown in the following example:

Min Submission Time	Job Type	Patient Name	Patient ID	IPID	Accession	Modality	Destination	Status	Images	User Name	Delete
2016-04-18 12:49:26	FWD	DEV258 CT	DEV258_2001	ANONIPID	DEV258_MISC100	CT	PACS-WK28-05	Requested	10	Unknown	Delete
2016-04-18 12:49:26	FWD	CHR123 US	CHR123_123	ANONIPID	CHR123_MISC100	MR	PACS-WK28-05	Started	6	Unknown	Delete
2016-04-18 12:49:26	FWD	CAN500_US	CAN500_100	ANONIPID	CAN500_MISC100	US	PACS-WK28-05	Requested	19	Unknown	Delete

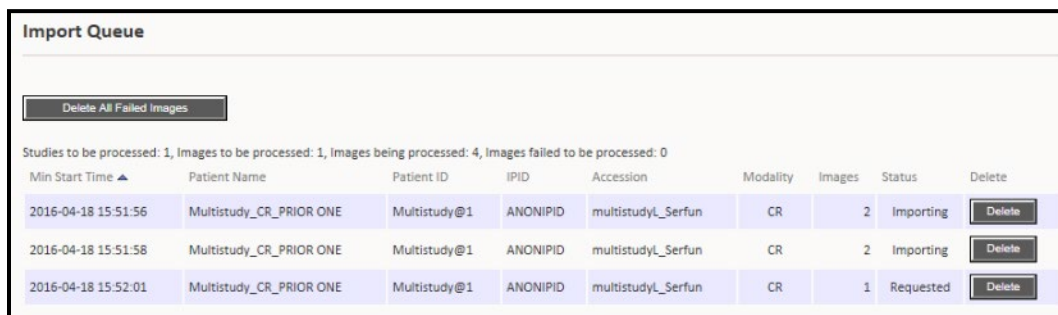
Figure 73: The Forwarding Queue

At the Forwarding Queue page you can:

- **Delete** any pending job by clicking on the “Delete” button for the job in question.
- **Delete all jobs** being routed to a particular destination by clicking on the **Delete All For Destination** button for the desired destination at the top of the page.

9.7.5. Image Import Queue

When Merge PACS is configured to run in **Standalone** mode, this page lists the DICOM images that have been transferred to Merge PACS and are currently being processed, waiting to be compressed, or remaining after processing failed, as shown in the following example:



Min Start Time ▲	Patient Name	Patient ID	IPID	Accession	Modality	Images	Status	Delete
2016-04-18 15:51:56	Multistudy_CR_PRIOR ONE	Multistudy@1	ANONIPID	multistudyL_Serfun	CR	2	Importing	<button>Delete</button>
2016-04-18 15:51:58	Multistudy_CR_PRIOR ONE	Multistudy@1	ANONIPID	multistudyL_Serfun	CR	2	Importing	<button>Delete</button>
2016-04-18 15:52:01	Multistudy_CR_PRIOR ONE	Multistudy@1	ANONIPID	multistudyL_Serfun	CR	1	Requested	<button>Delete</button>

Figure 74: The Import Queue

When Merge PACS successfully compresses an image, it removes that image from this queue.

At the Image Import Queue page, you can:

- **Delete** any pending job by clicking on the “Delete” button for the job in question.
- **Delete All Failed Images** by clicking on the “Delete All Failed Images” button at the top of the window. This will remove from the queue any jobs with a “Failed” status and also permanently delete any images that the system was unable to import.

9.7.6. Insert File Queue

This page lists the images that have been sent from a Merge PACS Workstation via HTTP (as would be the case when a user is appending a new series to a study using the 3D Image Capture or OrthoLink features), as shown in the following example:

Insert File Queue

Show New Jobs Show Completed Jobs Show Failed Jobs

Completed Jobs: 1222 [Delete All Completed Jobs](#)

Submission Time	Patient ID	IPID	Patient Name	Accession Number	Modality	Storing AE Title	Status	Delete	Restart
04/15/2016 10:12:28	54443322		Head Cathy	44678	PR	GOLDBERG@BM	Completed	Delete	
File: C:\PACS2DATA\FTPstore\GOLDBERG-8-BM\1.2.840.113837.37845701202.1460729549.1_fixed.dcm (Not Present)									
04/15/2016 10:12:30	54443322		Head Cathy	44678	PR	GOLDBERG@BM	Completed	Delete	
File: C:\PACS2DATA\FTPstore\GOLDBERG-8-BM\1.2.840.113837.3536050575.1460729549.0_fixed.dcm (Not Present)									
04/15/2016 10:12:30	54443322		Head Cathy	44678	PR	GOLDBERG@BM	Completed	Delete	
File: C:\PACS2DATA\FTPstore\GOLDBERG-8-BM\1.2.840.113837.3845701202.1460729550.0_fixed.dcm (Not Present)									
04/15/2016 10:12:30	54443322		Head Cathy	44678	PR	GOLDBERG@BM	Completed	Delete	
File: C:\PACS2DATA\FTPstore\GOLDBERG-8-BM\1.2.840.113837.371151534.1460729550.1_fixed.dcm (Not Present)									

Figure 75: The Insert File Queue

By default, the Insert File Queue only displays new jobs, but you can also display **Completed** and **Failed** jobs by clicking on the relevant links at the top of the page.

At the Image Insert File Queue page, you can:

- **Delete** any pending job by clicking on the “Delete” button for the job in question.
- **Restart** any failed job by clicking on the “Restart” button for the job in question.

9.7.7. Outbound HL7 Interface Queue

If the optional Outbound HL7 Interface feature, as described in subsection 3.8.1 above, has been enabled, this page displays the status of any HL7 messages created by the Merge PACS Server, as shown in the example below:

HL7 Outbound Interface Queue

Total Number of jobs in queue: 0
Number of jobs in this search: 2

Search entries from days ago to days ago and/or for destination [Go](#)

Destination	Time Queued	Time Processed	Queue Status	Patient Name	Patient ID	Patient ID Accession	Study Status	Details	Delete
TEMPORDER	2016-04-18 18:50:10	2016-04-18 18:50:10	NE	CST	CST150	HOME CTEST	Imported	Details	
TEMPORDER	2016-04-18 17:49:07	2016-04-18 17:50:07	NE	BEL	BEL1@100 HOME	BEL1_MISC100	Imported	Details	

Figure 76: The Outbound HL7 Interface Queue

- The queue page will display a maximum of 1000 jobs, including jobs that are pending and those that have been completed.
- You can use the query fields at the top of the queue page to perform the following tasks:
 - **Search** for jobs that were entered into the queue during a specified period of time and/or that were sent to a specified destination

- **Delete** all jobs that were entered into the queue during a specified period of time and/or that were sent to a specified destination
- **Resend** all jobs that were entered into the queue during a specified period of time and/or that were sent to a specified destination
- Jobs can have one of the following **Queue Status** options:

Status	Description
NE	Newly received job, still pending delivery
OK	Job successfully completed
ER	Error – job could not be completed.

- Jobs that were unsuccessful (“**ER**”) can be resent by clicking the **Resend** button in **Queue Status** column.
- Jobs that have been completed (“**OK**”) or that were unsuccessful (“**ER**”) can be **removed** from the queue by clicking on the **Delete** button for that job.
- Click on the **Details** button for a job to view detailed information about that job, as shown in the following example:

HL7 Outbound Interface Queue Details

Patient/Order Information:		
Patient ID: JPEG2000-RG2	Name: JPEG 2000*Medical Hip Joint X-Ray	Date Of Birth: null
Account:	Sex: M	
Order Placer Number:		
Study Information:		
Accession: null	Study Description: CR/HIP JOINT	Study Date/Time: 19970314/091300
Study Status: Imported	Image Count: 1	
Study Instance UID: 1.2.840.113619.6.48.1.2.0.889519491		
HL7 Message Information:		
Destination: GENERIC	Host: 127.0.0.1	Port: 1234
HL7 Message: <pre> MSH ^~\& GENERIC 20061002115611623 ORM^O01 S041 P 2.3 AL NE PID JPEG2000-RG2 JPEG 2000*Medical Hip Joint X-Ray M ORC NW OBR 1 CR*HIP JOINT 199703140913000000 1.2.840.113619.6.48.1.2.0.889519491 \EX1 OBX 1 RP StudyInstanceUID=1.2.840.113619.6.48.1.2.0.889519491 F </pre>		
Error Information:		
Retry count: 10		
Last Error Message: Can't connect to 127.0.0.1:1234		

Figure 77: Outbound HL7 Interface Queue Details

9.7.8. Telemed Export Queue

If the Server has been configured as a **Sending** server for Telemedicine, the Telemedicine Export Queue Monitoring Page displays any studies that are currently waiting to be exported to another Merge PACS Server, as shown in the following example:

Telemed Export Queue

Display Telemed Export jobs where Job Type is and Current Status is

Number of jobs in queue: 5

Job Type	Time Queued	Time Started	Time Finished	Requester	Patient Name	Patient ID	IPID	Accession	StudyInstanceUID	Images	Target	Status	Priority	Delete	Restart
Status	04-18 14:55:26	04-18 14:55:33	04-18 14:55:33	WkPwstVgr (-UnSet-)(4)					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.152.0	0/0	PACS-WK28-05	Completed	5	Delete	
Status	04-18 14:55:27	04-18 14:55:33	04-18 14:55:33	WkPwstVgr (-UnSet-)(4)					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.152.0	0/0	PACS-WK28-05	Completed	5	Delete	
Status	04-18 14:55:27	04-18 14:55:33	04-18 14:55:33	WkPwstVgr (-UnSet-)(4)					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.152.0	0/0	PACS-WK28-05	Completed	5	Delete	
Status	04-18 14:55:28	04-18 14:55:33	04-18 14:55:33	WkPwstVgr (-UnSet-)(4)					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.152.0	0/0	PACS-WK28-05	Completed	5	Delete	
Study	04-18 14:55:28	04-18 14:55:33	04-18 14:55:33	Auto Validation	DCI11111 OT	DCI11111_11111	ANONIPID	DCI11111_MISC100	1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.152.0	10/10	PACS-WK28-05	Completed	5	Delete	

Figure 78: The Telemed Export Queue

At the Telemed Export Queue page you can:

- **Filter** the list of jobs by Job Type and Current Status
- **Delete** a particular job
- **Restart** a job if it is in “Error” or “In-Progress” state
- **Change** the priority values by clicking one of the up or down arrows in the **Priority** column

9.7.9. Telemed Import Queue

If the Server has been configured as a **Receiving** server for Telemedicine, the Telemedicine Import Queue Monitoring Page displays any studies that are currently waiting to be imported from another Merge PACS Server, as shown in the following example:

Telemed Import Queue

Display Telemed Import jobs where Job Type is and Current Status is

Number of jobs in queue: 5

Job Type	Time Queued	Time Started	Time Finished	Patient Name	Patient ID	IPID	Accession	StudyInstanceUID	Images	Source	Status	Priority	Delete	Restart
Status	04-18 14:47:44	04-18 14:47:46	04-18 14:47:46					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.142.0	0/0	RAVHAL0008	Completed	5	Delete	
Status	04-18 14:47:59	04-18 14:48:05	04-18 14:48:05					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.142.0	0/0	RAVHAL0008	Completed	5	Delete	
Status	04-18 14:47:59	04-18 14:48:02	04-18 14:48:05					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.142.0	0/0	RAVHAL0008	Completed	5	Delete	
Status	04-18 14:47:59	04-18 14:48:05	04-18 14:48:05					1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.142.0	0/0	RAVHAL0008	Completed	5	Delete	
Study	04-18 14:48:17	04-18 14:48:32	04-18 14:48:41	DEV2000 US	DEV2000_2001	ANONIPID	DEV2000_MISC100	1.3.6.1.4.1.5962.99.1.3500277488.87886405.1459494223600.142.0	87/87	RAVHAL0008	Completed	5	Delete	

Figure 79: The Telemed Import Queue

At the Telemed Import Queue page you can:

- **Filter** the list of jobs by Job Type and Current Status

- **Delete** a particular job
- **Restart** a job if it is in “Error” or “In-Progress” state
- **Change** the priority values by clicking one of the up or down arrows in the **Priority** column

NOTE: When studies are sent from the Sending Server, the telemed job status will be set to “On Hold” on the Receiving Server until all images have arrived. If the images still haven’t been received after 10 minutes, however, the telemed job will fail.

9.7.10.DICOM Print Queue

If one or more DICOM printers have been configured for Merge PACS, the DICOM Print Queue Monitoring Page displays any studies that are currently waiting to be sent to a DICOM printer, as well as successfully-submitted jobs, failed jobs, and cancelled jobs that have not yet been cleared (manually by the user) or automatically cleaned up (by the cleanup DB script), as shown in the following example:

Creation Date	Creator	Priority	Status	# of Retries	Patient Name	Accession Number	Cancel	Promote
2013-01-21 09:03:55.942	bgoldberg	(UNSPECIFIED)	READYTOSUBMIT	1	Doe^Jonathan^Q	123456789	Cancel	Promote
2013-01-21 09:23:15.357	bgoldberg	(UNSPECIFIED)	SUBMITTED	1	Doe^Joanne^P	999999999		Promote

Figure 80: The DICOM Print Queue

At the DICOM Print Queue page you can:

- **Select** the DICOM Printer whose queue you wish to view
- **Cancel** a print job that has not yet been submitted to the printer (READYTOSUBMIT status).
- **Promote** a particular job that has not yet been submitted to the printer to the top of the queue.
- **Clear All Stale Jobs** to delete all jobs that have failed or been canceled from the queue.

NOTE: By default, failed and cancelled jobs remain in the queue for troubleshooting purposes for a configurable amount of time (set by default to 48 hours).

9.7.11. Sender Job Queue

When Merge PACS is configured to run in **Integrated** mode, the **Sender Job Queue** page provides information on instances being sent to iCEA. These are usually instances generated by the Viewer (presentation states, key objects or CD upload) or instances received through telmed from other Merge PACS Servers, as shown in the following example:

Time Queued	Patient Name	Patient ID	IPID	Accession	StudyInstanceUID	Images	Source	Status	Delete	Restart
2016-02-19 10:18:44.517	DEANN*SNOODGRASS	101651807	HOME4	CF05573	1.2.840.113837.1455892097.50208.1.3069	6	TELMEDSPOKE	Failed	Delete	Restart
Error Message: Click here to see error details										
2016-02-19 10:18:44.571	DEANN*SNOODGRASS	101651807	HOME4	DH05574	1.2.840.113837.1455892102.22923.9.1931	1	TELMEDSPOKE	Failed	Delete	Restart
Error Message: Click here to see error details										
2016-02-19 10:18:44.655	DEANN*SNOODGRASS	101651807	HOME4	AH05575	1.2.840.113837.1455892107.21892.16.207	5	TELMEDSPOKE	Failed	Delete	Restart
Error Message: Click here to see error details										
2016-02-19 10:18:44.664	DEANN*SNOODGRASS	101651807	HOME4	BP05576	1.2.840.113837.1455892112.22744.23.29	4	TELMEDSPOKE	Failed	Delete	Restart
Error Message: Click here to see error details										
2016-02-19 10:18:44.742	DEANN*SNOODGRASS	101651807	HOME4	EE05577	1.2.840.113837.1455892118.8760.29.1595	5	TELMEDSPOKE	Failed	Delete	Restart
Error Message: Click here to see error details										
2016-02-19 14:37:16.171	EARLE*LILLEY	954791119		DO06046	1.2.840.113837.1455910623.68955.8.9155	3	TELMEDSPOKE	Requested	Delete	
2016-02-19 14:37:16.201	EARLE*LILLEY	954791119		PP06047	1.2.840.113837.1455910628.48805.13.1998	4	TELMEDSPOKE	Requested	Delete	
2016-02-19 14:37:31.128	EARLE*LILLEY	954791119		NK06048	1.2.840.113837.1455910633.6912.19.9567	5	TELMEDSPOKE	Requested	Delete	
2016-02-19 14:37:31.235	EARLE*LILLEY	954791119		OC06049	1.2.840.113837.1455910638.33773.20.4215	6	TELMEDSPOKE	Requested	Delete	
2016-02-19 15:19:01.546	SHIRLENE*BILLINGS	275541282		MQ07026	1.2.840.113837.1455913129.90007.1.6470	5	TELMEDSPOKE	Requested	Delete	

Figure 81: The Sender Job Queue

The Sender Job Queue page will display one row per study per source per status, with the rows listed in reverse chronological order. The default maximum number of rows that can be displayed on the Sender Job Queue page is 1000, but this can be changed on a per-site basis.

NOTE: When things are running smoothly, there should be very little (or nothing) displayed on this page.

At the Sender Job Queue page, you can do the following:

- Choose the status of jobs you want to see by selecting the desired status from the drop-down menu at the top of the page and clicking the **Search** button. The following status options are available:
 - All** (this is the default)
 - Requested**
 - In Progress**
 - Failed**

NOTE: There is no option to see successfully completed jobs because the jobs drop out of the queue when they complete successfully.

- Click on the **Delete** button for a particular row to delete all instances (“images”) captured by that row and remove that job from the queue.
- Click on the **Restart** button for a particular row to retry sending all instances captured by that row.

- View the **Error Message** associated with any failed job. If the error message is too long to fit on the screen, click on the “**Click here to see error details**” link to view the message on a new page, as in the following example:

Sender Job Queue Error Details for Study: 1.2.840.113837.1421260597.51923.13.6931

Time Queued	Time Modified	Patient Name	Patient ID	IPID	Accession	SourceName	Status	JobID	StorageID	View
2016-04-04 12:19:40.206	2016-04-04 12:19:40.438	HOWARDG001*CLINTON	184964002		ID09663	chirstyap04	Failed	4	6	View
Image Path:		chirsty-ap04\1.2.840.113837.1421260597.51923.13.6931\0056ba76-7a09-4696-a360-7c46f0d032e4								
Error Message:		Store Exception - ORA-12859: value too large for column "CLARCD00"."STUDY"."REASONFORSTUDY" (actual: 136, maximum: 64)								
2016-04-04 12:19:40.162	2016-04-04 12:19:40.438	HOWARDG001*CLINTON	184964002		ID09663	chirstyap04	Failed	3	4	View
Image Path:		chirsty-ap04\1.2.840.113837.1421260597.51923.13.6931\5ed16383-d0a8-46a-cbde-d-affc1756f5f3								
Error Message:		Store Exception - ORA-12859: value too large for column "CLARCD00"."STUDY"."REASONFORSTUDY" (actual: 136, maximum: 64)								

[Back](#)

Figure 82: Sender Job Queue Error Details

- The error details page displays the full text of the error message for each underlying instance as well as the location on disk where the .dcm file for that instance has been placed (within the PACS2DATA directory).
- Click on the **View** button for a particular instance to view the DICOM attributes for that instance.
- Click on the **Back** link at the bottom of the Error Details page to return to the main Sender Job Queue page.

9.7.12.DataStorage Cache

The **DataStorage Cache Queue** page provides information on instances that are temporarily stored in the Server and are waiting to be processed in one way or another. These can be instances received through telmed, instances being anonymized, instances received from the viewer, legacy PACS instances being migrated to iCEA, etc., as in the following example:

DataStorage Cache

Display DataStorage Items with Status:

Total Number of Items: 2; Items with Status ERROR: 0; Items with Status DELETE: 0; Total Size: 4 (Bytes)

ID	Source	Group	Status	Data Length	Data Path	Created On
24	StorageSCU_main	G1	0	2	D:\AMICAS2DATA\CACHE\StorageSCU_main\G1\d1bf3650-75bf-44fa-8b2e-7fe732915409	2016-04-18 17:49:07.19
25	StorageSCU_main	G1	0	2	D:\AMICAS2DATA\CACHE\TEST\StorageSCU_main\G1\9b221ee2-abb2-4cb9-a587-545701488d97	2016-04-18 18:04:00.2

Figure 83: The DataStorage Cache

By default, all items with a status of **OK** will be displayed, but you can change this by selecting a different option from the drop-down **Display DataStorage items with Status** menu.

9.8. Logs

The various Log pages display events that have previously been processed by Merge PACS. Clicking on the “Logs” link on the top category menu will display a submenu of the individual available log pages as shown below:

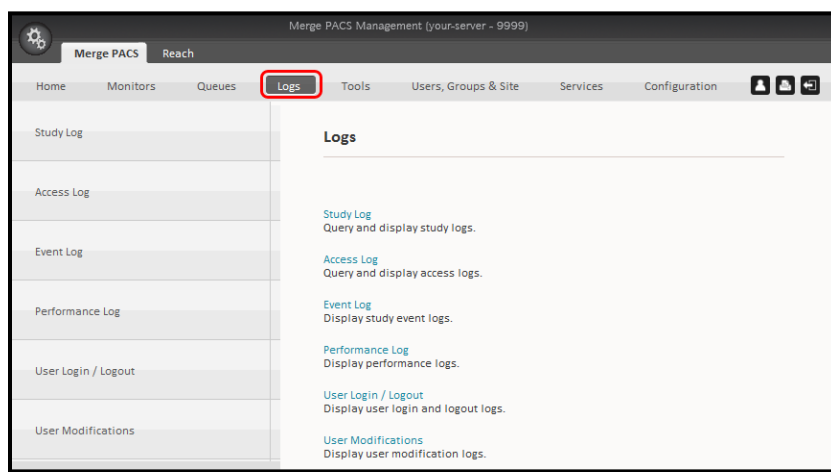


Figure 84: The Log Pages

To select a particular log page, click on the appropriate link in the Logs submenu, or the link displayed on the main Log page.

NOTE: On the **Study Log**, **Access Log**, and **Event Log** pages, you can sort the data by clicking on a column heading and then clicking on the little blue triangle next to the column heading to select whether to sort in ascending or descending order.

9.8.1. Study Log

At the Study Log page, you can view information for studies that have been imported into Merge PACS (and which have been marked as “closed” by the Auditor process), as well as those studies that have been deleted from Merge PACS.

When you first click on the Study Log page, you will see a query interface as shown in the following example:

A screenshot of the 'Study Log' query interface. It features a 'Search by:' dropdown menu currently set to 'Patient Name', followed by a text input field. Below this is a 'Date:' dropdown menu set to 'Today'. A 'Search Study Log' button is positioned below the date field. At the bottom of the interface, a message reads: 'Please select a date and click Submit to view the Study Log.'

Figure 85: The Study Log

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:

The screenshot shows a search interface. On the left, there are two labels: "Search by:" and "Date:". To the right of "Search by:" is a pull-down menu currently showing "Patient Name". To the right of "Date:" is a pull-down menu currently showing "Patient Name". To the right of the "Search by:" menu is a text input field containing "DOE". Below the "Date:" menu is a button labeled "Search Study Log".

Figure 86: Selecting a Search Criterion

- If searching by **Patient Name**, you can enter the name you are searching for in the format **“last name, first name”** or simply enter the **last name** to find all people with that last name.
 - You can also use **wildcard searches** for Patient Name, as described in Section 9.2 above.
- By default, the Study Log will display studies for the current date (“Today”). If you want to set the search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

The screenshot shows the same search interface as Figure 86, but the "Date:" pull-down menu is open, displaying a list of options: "No Restriction", "Today", "Yesterday", "This month", "Last month", "This year", "Last year", and "Custom". A mouse cursor is pointing at "Last month". Below the menu, there is a text prompt "Please select a date range to view the Study Log."

Figure 87: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

The screenshot shows the search interface with the "Date:" pull-down menu set to "Custom". Below the menu, there is a red-bordered box containing the text "From: 10/31/2017" followed by a calendar icon, and "To: 10/31/2017" followed by a calendar icon. Below this box is a button labeled "Search Study Log".

Figure 88: Specifying a Custom Date Range

- When you have entered the desired search criteria, click on the **Search Study Log** to display the study log, as shown in the following example:

The screenshot shows the 'Study Log' interface. At the top, there are search filters: 'Search by:' with a dropdown menu set to 'Patient Name' and a text input field containing 'doe'. Below this is a 'Date:' dropdown menu set to 'No Restriction'. A red box highlights the 'Search Study Log' button. Below the filters is a table with the following columns: Log Time, Arrival, Available, Last Image, Patient ID, IPID, Patient Name, Accession, Modality, Images, and Log Type. The table contains three rows of data.

Log Time	Arrival	Available	Last Image	Patient ID	IPID	Patient Name	Accession	Modality	Images	Log Type
2014-05-12 15:49:11.406	2014-05-12 15:46:09.283	2014-05-12 15:47:06.915	2010-02-19 19:43:13.0	123456789	HOME	Doe, Jonathan	12345abc	CT	243	DICOM
2014-05-12 15:49:06.384	2014-05-12 15:46:54.447	2014-05-12 15:46:59.888	2013-08-20 13:38:39.0	999999999	HOME	Doe, Joanne	99999xyz	CR	1	DICOM
2014-05-12 15:49:01.364	2014-05-12 15:46:48.143	2014-05-12 15:46:54.184	2013-08-19 10:00:41.0	1234512345	HOME	Doe, Jessica	987654cba	SR	6	DICOM

Figure 89: Generated Study Log

- The **Arrival** column indicates when Merge PACS began to receive the first image of a study.
- The **Available** column indicates when the last image of the study is available on the Merge PACS web server.
- The following **Log Types** may be used to describe the study event:

Log Type	Description
DICOM	This log entry represents a DICOM import.
AMICAS	This log entry represents a Merge PACS Telemedicine transfer.
DELETE	This log entry represents a delete event.

9.8.2. Access Log

At the Access Log page, you can view summary of the Event Log (described in subsection 9.8.3, below) that reflects the end users' access to Merge PACS. In essence, it shows activity associated with all or selected studies, including when a study was viewed, deleted, routed, etc.

When you first click on the Access Log page, you will see a query interface as shown in the following example:

The screenshot shows the 'Access Log' query interface. It has a 'Search by:' dropdown menu set to 'Patient Name' and a text input field. Below this is a 'Date:' dropdown menu set to 'Today'. A 'Search Access Log' button is located below the date dropdown. At the bottom of the interface, there is a message: 'Please enter desired search entry and click Submit to view Access Log.'

Figure 90: The Access Log

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:

Search by: Patient MRN

Date: Patient MRN

Patient Name

User ID

HP Name

Please enter desired search entry and click Submit to view Access Log.

Figure 91: Selecting a Search Criterion

- If searching by **Patient Name**, you can enter the name you are searching for in the format “**last name, first name**” or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for Patient Name, as described in Section 9.2 above.
- If searching by **HP Name**, you can enter the full name of the Hanging Protocol or use wildcard searches.
- By default, the Access Log will display access for the current date (“Today”). If you want to set the search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

Search by: Patient Name

Date: Today

No Restriction

Today

Yesterday

This month

Last month

This year

Last year

Custom

Please enter desired search entry and click Submit to view Access Log.

Figure 92: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

Search by: Patient Name

Date: Custom

From: 10/31/2017 To: 10/31/2017

Search Access Log

Figure 93: Specifying a Custom Date Range

- When you have entered the desired search criteria, click on the **Search Access Log** to display the access log, as shown in the following example:

Time	Event Type	Patient ID	IPID	Patient Name	Accession	Hostname	User ID	Event Source	Process Time	Target	HP Name
07/03/2020 15:08:54	View	000054321	Home	JPG_LOSSY_SF_MONO, PATIENT	OS000034	192.168.140.39	superuser	Viewer	0	192.168.140.39(1)	
07/03/2020 15:08:11	StartingViewer	000054321	Home	JPG_LOSSY_SF_MONO, PATIENT	OS000034	192.168.140.39	superuser	Viewer	0		

Figure 94: Generated Access Log

- The Access Log will display information for the following types of events:

Event	Description
AssignOrder	A user assigned an order from a worklist to another
RTWLCache	Caching Study Finished Event
StatusChange	One of the workflow statuses of this study is changed.
CDBurning	This event will be logged when a CD is burned for a study
DeleteStudy	A user submitted a request to delete a study
DICOMPrintJob	A user successfully sent a print job to a DICOM printer
Download	A user downloaded or updated the Merge PACS Workstation.
Comments	A user entered a comment or warning for a study
SettingKeyImage	A user set a key image within the Merge PACS Workstation
ModifyPatient	A user requested to modify demographic information for a patient.
StudyModify	A change to the demographics or accession number of a study, as requested by a user or via HL7, has finished.
ModifyStudy	A change to the demographics or accession number of a study has been requested by a user or via HL7.
RemoveOrder	A user removed an order from a worklist
View	A user accessed a study within the Merge PACS Viewer.
Login	A user logged into a LDAP server via the Merge PACS Workstation
Search	A user searched for a list of matching studies via the Merge PACS Workstation Query Search Page.
StartingViewer	A user launched the Merge PACS Workstation
DICOMQ/R	A user searched for a list of matching studies via DICOM Q/R.
HP Edit	A user saved a Hanging Protocol for a particular user.
HP Delete	A user deleted a particular Hanging Protocol.
HP Subscribe	A user subscribed a Hanging Protocol for a particular user.
HP Unsubscribe	A user unsubscribed a Hanging Protocol for a particular user.

- Because the Access Log is a summary of the **Events Log**, you can view details for each access activity by looking for it in the Event Log, as described in the next subsection.

9.8.3. Event log

At the Event Log page, you can view all study-related events for an entire day or for a particular patient.

When you first click on the Event Log page, you will see a query interface as shown in the following example:

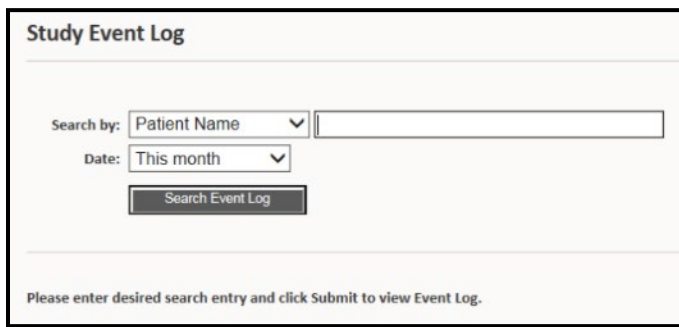


Figure 95: The Study Event Log

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:

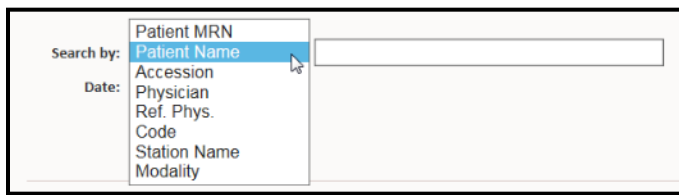


Figure 96: Selecting a Search Criterion

- If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, **Referring Physician** and **Physician**, as described in Section 9.2 above.
- By default, the Event Log will display events for the current date ("Today"). If you want to set the search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

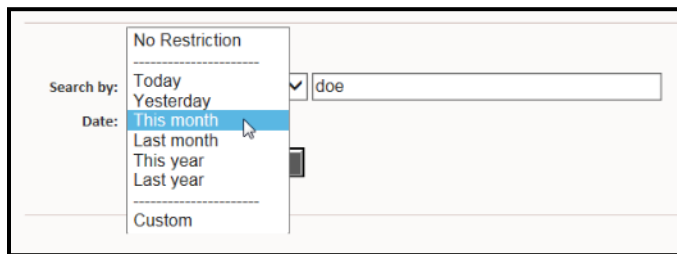


Figure 97: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

Search by: Patient Name

Date: Custom

From: 10/31/2017 To: 10/31/2017

Figure 98: Specifying a Custom Date Range

- When you have entered the desired search criteria, click on the **Search Event Log** to display the event log, as shown in the following example:

Study Event Log

Search by: Patient Name

Date: Custom

From: 10/31/2017 To: 10/31/2017

Min Event Time	Patient Name	Patient ID	IPID	Accession	Study Date	Study Description	Details
2016-04-15 10:15:46	Doe, Jessica Q.	EXd1d7c9f3p18cc5p	HOME	EX6994bebf	2004-03-29		Details
2016-04-15 10:14:24	Doe, Jonathan Q.	EX2c9fac4cp1902ap	HOME	EX53420dca1	2009-01-15		Details
2016-04-15 10:14:12	Doe, Jonathan Q.	EX2c9fac4cp1902ap	HOME	EX90e8a6a5	2009-01-15		Details
2016-04-15 10:22:56	Doe, Jessica Q.	EXd1d7c9f3p18cc5p	HOME	EX0705769a2	2004-03-29		Details

Figure 99: Generated Study Event Log

- The main Event Log page provides a description of which studies were affected and when. Clicking on the **Details** button for a particular study will display information regarding the actual events that affected that study, as shown in the following example:

Study Event Log

Patient Name: Doe, Jessica Q. Patient ID: EXd1d7c9f3p18cc5p, IPID: HOME, Accession Number: EX6994bebf, Study Date: 2004-03-29

Study Description:

Event Time	Event Name	Event Source	Source	Destination	User	Error Message
2016-04-15 10:15:46	started queuing images for import	StorageServer	GCS.DRE.HUBB4			
2016-04-15 10:15:48	Non-Image DICOM file import started	Importer				GrayScaleSoftcopyPresentationState Non-Image import started
2016-04-15 10:15:48	workflow status changed to Image Transferring	StatusManager			Importer	Importer
2016-04-15 10:15:49	importing images started	Importer				
2016-04-15 10:15:50	Non-Image file import completed	Importer				GrayScaleSoftcopyPresentationState finished import
2016-04-15 10:17:52	Unset ImageTransfer status	StatusManager				SetStatus CMD - null(UNSET)
2016-04-15 10:22:55	Finished queuing images for import	StorageServer				
2016-04-15 10:22:55	import completed	Importer				
2016-04-15 10:24:56	Unset ImageTransfer status	StatusManager			Importer	Importer(UNSET)
2016-04-15 10:24:56	could not find an existing order that matched the new study	Importer				
2016-04-15 10:24:59	study closed	Auditor				
2016-04-15 10:25:32	study cached on client	RTWL-Cacher	AG-71-Current	AG-71-Current	dicom	
2016-04-15 11:52:45	online DICOM images deleted	CommandScheduler				
2016-04-15 13:05:28	study cached on client	RTWL-Cacher	AG-71-Current	AG-71-Current	dicom	

Figure 100: Study Event Details

Note that if a particular study event was manually initiated by a user, as opposed to being automatically performed by the System, the user's name will be displayed in the appropriate column.

9.8.4. Performance Log

At the Performance Log page, you can view study and image related performance statistics.

When you first click on the Performance Log page, you will see a query interface as shown in the following example:

Figure 101: The Performance Log

Select the date or date range for the statistics you wish to view and then click on the appropriate **Submit** button to generate the performance log, as in the following example:

Performance Log

Display Performance Log on: 07/01/2017

Display Performance Log from: 07/01/2017 00:00:00

to: 10/31/2017 13:25:56

Sampling Time Period: From: 07/01/2017 00:00:00 To: 10/31/2017 13:25:56

Elapsed Time: 176485.94 minutes

Average Image Work Load: 0.00 images/hour (0.00 images/minute)

Average Volume Work Load: 0.00 MB/hour (0.00 MB/minute)

Processing Performance by Modality:

Modality	Total Studies (count)	Total Images (count)	Total Volume (MB)	Average Latency (s)	95th Percentile Latency (s)
CT	3	3	0.39	2.77	3.78
DX	1	1	1.23	3.06	3.06
MR	4	4	0.28	7.71	16.44
PT	1	1	0.02	4.94	4.94
Total	9	9	1.92	-	-

Viewer Access Performance by Modality:

Modality	Total Accesses (count)	Average Latency to First Image (s)	95th Percentile Latency to First Image (s)
CT	2	0.0	0.0
MG	2	0.0	0.0
MR	3	0.0	0.0
null	5	0.0	N/A
Total	12	-	-

Viewer Search Access Performance:

Average response time for search requests: 0.00s

Hourly Throughput:

Hour	Image Throughput (images/minute)	Byte Throughput (MB/minute)
07/01/2017 00-01	0.00	0.00
07/01/2017 01-02	0.00	0.00
07/01/2017 02-03	0.00	0.00

Figure 102: The Performance Log

The following statistics are monitored:

- Processing Performance by Modality

Statistic	Description
Total Studies	Total number of studies that were imported from this modality
Total Images	Total number of images that were imported from this modality
Total Volume	Size of all images that were imported from this modality
Average Latency	Average delay, in seconds, for importing studies from this modality, measured from the time the first image for each study arrived from the modality to the time the last image for that study was imported into Merge PACS.
95th Percentile Latency	95% of studies from this modality took less time than this, in seconds, to be imported. Only 5% of studies from this modality took longer than this to be imported.

- Viewer Access Performance by Modality

Statistic	Description
Total Accesses	Total number of studies from this modality that were accessed by a user
Average Latency to First Image	Average delay, in seconds, for each study accessed, measured from the time the user clicked on the study from the Search Page to the time the first image was displayed in the Viewer
95th Percentile Latency to First Image	95% of the studies from this modality that were accessed by users were displayed within this amount of time, in seconds. Only 5% of studies from this modality took longer than this for the first image to be displayed.

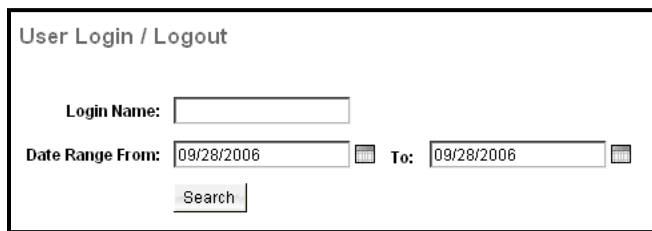
- Viewer Search Access Performance** – Average delay, in seconds, for search requests to be completed
- Hourly Throughput** – For each hour, the total number of images per minute and megabytes per minute that were imported into Merge PACS.

You can select the previous or next day's log, if available, by clicking on the dated links above and below the log display. You can also select a log for a specific date or for a particular time span by entering the information in the appropriate fields and clicking the **Submit** button.

9.8.5. User Login / Logout

At the User Login / Logout page, you can view a log of when individual users logged into and out of the Merge PACS system.

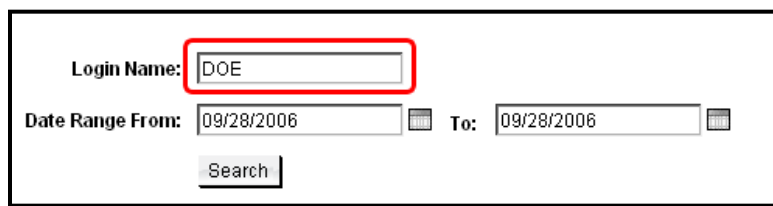
When you first click on the User Login / Logout page, you will see a query interface, as shown in the following example:



The screenshot shows a web form titled "User Login / Logout". It contains three input fields: "Login Name:" followed by an empty text box, "Date Range From:" followed by a date field containing "09/28/2006", and "To:" followed by a date field containing "09/28/2006". Below these fields is a "Search" button.

Figure 103: The User Login / Logout Page

- Enter the login name of the user you wish to track in the **Login Name** field, as shown in the following example:

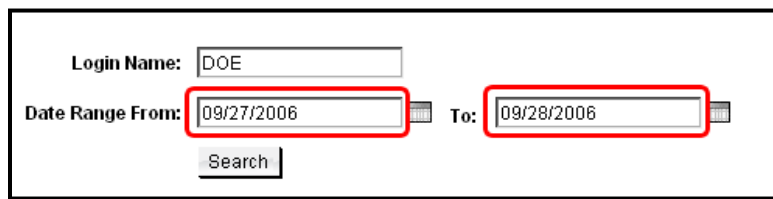


The screenshot shows the same form as Figure 103, but the "Login Name:" field now contains the text "DOE". The "Date Range From:" and "To:" fields remain "09/28/2006". The "Search" button is still present.

Figure 104: Entering the Login Name

If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.

- Enter the desired date range in the fields provided, as shown in the following example:



The screenshot shows the same form as Figure 104, but the "Date Range From:" field now contains "09/27/2006" and the "To:" field contains "09/28/2006". The "Login Name:" field still contains "DOE". The "Search" button is still present.

Figure 105: Entering the Date Range

NOTE: You can also select dates by clicking on the little calendar icon next to each date field. This will cause a small calendar to be displayed in a separate pop-up window and you can click the < and > icons surrounding the month and year at the top of the calendar to navigate forward and backwards through the calendar, if necessary, and then click the desired day

- When you have entered the login name and date range, click the **Search** button to generate the log for that user, as shown in the following example:

Time	Login Name	User Name	Login/Logout	Hostname/IP Address
2016/06/08 06:37:48	doej	Doe, Jonathan	Login	10.10.7.145
2016/06/08 06:37:48	doej	Doe, Jonathan	Login	10.10.7.145
2016/06/08 13:34:10	doej	Doe, Jonathan	Login	10.10.7.145
2016/06/08 13:35:43	doej	Doe, Jonathan	Logout	10.10.7.145
2016/06/08 13:35:49	doej		Login Failed	10.10.7.145
2016/06/08 13:35:54	doej	Doe, Jonathan	Login	10.10.7.145

Figure 106: User Login / Logout Search Results

9.8.6. User Modifications

At the User Modifications page, you can view a log of when the login accounts of individual users have been created or modified. Modifications include such things as changing a user's personal information and enabling or disabling a user.

When you first click on the User Modifications page, you will see a query interface, as shown in the following example:

Figure 107: The User Modifications Page

- Enter the login name of the user you wish to track in the **Login Name** field, as shown in the following example:

Figure 108: Entering the Login Name

If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.

- Enter the desired date range in the fields provided, as shown in the following example:

Figure 109: Entering the Date Range

NOTE: You can also select dates by clicking on the little calendar icon next to each date field. This will cause a small calendar to be displayed in a separate pop-up window and you can click the < and > icons surrounding the month and year at the top of the calendar to navigate forward and backwards through the calendar, if necessary, and then click the desired day

- When you have entered the login name and date range, click the **Search** button to generate the log for that user, as shown in the following example:

Time	Operation	User Modified	Who Modified
2016/06/07 09:14:30	User Created	Doe, Jonathan (doej)	Admin, Joseph (admin)
2016/06/08 13:53:36	User Modified	Doe, Jonathan (doej)	Admin, Joseph (admin)

Figure 110: User Modifications Search Results

9.9. Tools

The Tools page provides you with various study-related tools, including the ability to modify study information, route copies of studies to other locations, and manage studies that are located in the DICOM jail. Clicking on the “Tools” link on the top category menu will display a submenu of the individual available log pages as in the following example:

Figure 111: The Tools Pages

To select a particular tools page, click on the appropriate link in the Tools submenu, or the link displayed on the main Tools page.

NOTE: The **Patient Data Editor** is no longer available from the Tools Management Page. All PDE functions can now be performed from the Quality Control Editor. Refer to the *Merge PACS 8.2 Quality Control Editor Users Guide* for more information.

9.9.1. Report Deletion

The Report Deletion page allows you to delete a report associated with a particular study stored on the Merge PACS Server. Note the following:

- This feature is only available for upgrades from previous versions of Merge PACS and will not be available for new installations of Merge PACS 8.2.
- The Report Deletion tool only shows reports stored on the Merge PACS Server in proprietary XML format.
 - Text reports copied manually to the Server will not be displayed.
 - Structured Reports can be deleted from the QC Editor, as described in the *Merge PACS 8.2 Quality Control Users Guide*.
- The Report Deletion tool is slated for complete removal some time within the next two releases of Merge PACS.

When you click on the Delete Report page, you will first need to search for the report you wish to delete from the Report Deletion Query Page, as shown below:

112: Report Deletion Query Page

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:

Figure 113: Selecting a Search Criterion

- If searching by **Patient Name**, you can enter the name you are searching for in the format “last name, first name” or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, as described in Section 9.2 above.

- When you have entered the desired search criteria, click on the **Search** button to display a list of matching reports, as shown in the following example:

Search by: Patient Name = DOE Search

Patient Name	Received Date/Time	Patient ID	Accession	Status	Delete Report	View Report
DOE/JONATHAN	2008-05-14 14:42:26.963	111113	123453	Final	Delete Report	View Report
DOE/JONATHAN	2008-05-14 14:42:20.716	111113	123453	Final	Delete Report	View Report
DOE/JOANNE	2008-05-13 16:03:45.57	123456	654321	Final	Delete Report	View Report

Reason for Deletion:

Change status to: Read

Figure 114: List of Matching Reports

- If desired, you can click on the **View Report** button for a particular report to see that report in a separate pop-up Report Viewer window.
- Enter the reason for deletion in the space provided.
- Click on the **Delete Report** button for the report you want to delete.

NOTE: By default, after the report is deleted from a study, the study/order status will be kept at “Final” if the study still has Final report and “Preliminary” if the study has Preliminary report. If a study has no more reports associated with it, the status will be set to whatever you select from the pull-down **Change status to** menu.

9.9.2. Restore Deleted Files

When Merge PACS is configured to run in **Standalone** mode, the Restore Deleted Files page allows you to manually restore deleted files, as in the following example:

Restore Deleted Files

Search for Deleted Files

Figure 115: Restore Deleted Files Page

NOTE: Other QC operations, such as deleting all or part of a study, splitting a study into multiple series, merging series, etc., can now be performed from within the Quality Control Editor. For more information, refer to the *Merge PACS 8.2 Quality Control Editor Users Guide*.

CAUTION: Restored items may not have the latest demographics; therefore, it is important to confirm that the demographics are correct after using this feature.

- Eligible studies, series and images are stored in a “trash bin” for seven days before being permanently deleted. If you need to undelete images or series within that time frame, **Search for Deleted Files** button, as in the following example:

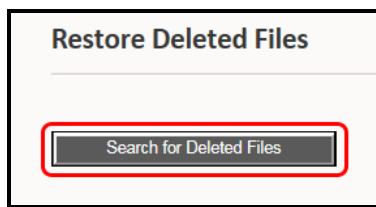


Figure 116: Search for Deleted Files

A list of all deleted images or series that are eligible for restoration is displayed, as in the following example:



Figure 117: List of Deleted Files

NOTE: All images and series that were deleted together are stored in a single directory and only that entire directory can be restored. You cannot restore individual images or series.

- Click on the **Restore** button to the right of the directory of images and/or series you wish to undelete.
- If desired, click on your web browser’s **Refresh** button to verify that the restoration process has completed. Once a directory has been restored, it will be removed from the list.
- Click on the **Close** button to exit this window.

The eligibility of a file to be restored once deleted is determined as follows:

- Studies will be eligible for restoration if they are not entirely watermarked from PACS.
- If lossless images are watermarked from PACS only the lossy version will be backed up in the trash bin.

9.9.3. Send Studies

When Merge PACS is configured to run in **Standalone** mode, the Send Studies page allows you to send studies manually to another Merge PACS Server (via the Telemet Export Queue) or to a local DICOM device (via the Command Queue).

If the DICOM images for the selected study are currently available on the Merge PACS Server (*i.e.*, not solely on the Merge Archive), the DICOM images will be sent. If the DICOM images are not currently available, the Lossy images for this study will be decompressed and sent with compression tags burned on the images.

NOTE: Merge PACS can be configured to prevent DICOM studies with certain SOP Classes from being sent to specified DICOM devices, as described in subsection 9.13.2 below.

NOTE: If Merge PACS has been configured with autorouting rules to send studies to other DICOM devices (other than the attached DICOM archive), these rules will not be reapplied after operations are performed from the QC Editor except for copy operations. If you need a DICOM device to be kept in sync after QC operations are performed, the affected studies will need to be manually sent to that device using the Send Studies tool.

When you click on the Send Studies page, you will first need to locate a study to send from the Send Studies Query Page, as shown below:

Figure 118: The Send Studies Query Page

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:

Figure 119: Selecting a Search Criterion

- If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, **Referring Physician** and **Physician**, as described in Section 9.2 above.

- If you want to restrict your search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

The screenshot shows a search form with 'Search by:' set to 'Accession' and a value of '123456789'. The 'Date:' dropdown menu is open, showing options: 'No Restriction', 'Today', 'Yesterday', 'This month' (highlighted with a mouse cursor), 'Last month', 'This year', 'Last year', and 'Custom'.

Figure 120: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

The screenshot shows the search form with 'Search by:' set to 'Patient Name'. The 'Date:' dropdown is set to 'Custom'. Below it, the 'From:' date is '10/31/2017' and the 'To:' date is '10/31/2017', both highlighted with a red rectangle. A 'Search Studies' button is visible below the date range.

Figure 121: Specifying a Custom Date Range

- When you have entered the desired search criteria, click on the **Search Studies** to display a list of matching studies, as shown in the following example:

The screenshot shows the 'Send Studies' page. The search criteria are 'Patient Name' and 'doe'. The 'Date:' dropdown is set to 'No Restriction'. The 'Search Studies' button is highlighted with a red rectangle. Below the search criteria, there is a section for 'Send checked studies below to' with a dropdown set to 'No target selected' and a 'Send' button. At the bottom, there is a table of matching studies.

	Patient Name	Patient ID	IPID	DOB	Sex	Accession	Study Date	Study Description	Images
☒	Doe, Jonathan	123456789	HOME	20140512	O	12345abc	20030609		89
☒	Doe, Joanne	99999999	HOME	20140512	O	99999xyz	20030609		96

Figure 122: List of Matching Studies

- Select the desired target from the pull-down **Target** menu, click on the checkbox for each of the studies you want to send, and then click on the **Send** button.

NOTE: In order for a Merge PACS Server or a local DICOM device to appear on the pull-down Target menu, it must first be entered on the appropriate **configuration** page as described in Section 9.13 below.

- Once you have sent a study, the appropriate **Queue** page for the type of target will appear in a separate pop-up window. If sending to another Merge PACS Server, for example, the **Telemed Export Queue** described in subsection 9.7.8, above, will be displayed. This new window can be closed at any time.

9.9.4. Problem Studies Tool

When Merge PACS is configured to run in **Standalone** mode, the Problem Studies Tool page allows users to fix various problems that have caused one or more of a study's images to be placed in the DICOM Jail (as described in Section 2.6). These problems typically involve one of three situations:

- One or more of the images in a study are missing one or more DICOM attributes that are required by Merge PACS.
- One or more images in a study have one or more DICOM attributes that do not match the attributes for the first image that was received by Merge PACS.
- The same study has been sent to Merge PACS twice, but one or more images have an image-specific attribute that is not the same for both studies.
- A study that has been imported into Merge PACS and watermarked is later pushed to Merge PACS or restored from Enterprise Archive with the same Study Instance UID but a different patient identity (normalized MRN, normalized Patient Name, Issuer of Patient ID). Note that, with regard to Enterprise Archive in particular, studies will only be jailed if Merge PACS has been configured so that the information received from EA will be given precedence over the information stored within Merge PACS, as described in subsection 1.2.61 above.

NOTE: When a study is edited using the Problem Studies Tool, the AETitle of the study's "sending device" is changed to "PROBSTDTool." Therefore, if a study that was a candidate for auto-routing based on AETitle was edited using the Problem Study Tool, the study will need to be manually routed via the Send Studies tool (as described in subsection 9.9.3 above).

NOTE: If Auto Validation has been enabled, new validation will be triggered after the Problem Studies Tool is used to change any attributes of the jailed images. However, new validation will not be triggered if the Problem Studies Tool is used solely to delete duplicate images.

CAUTION: Great care should be exercised when changing attributes via PDE operations. Mistakes could result in problems such as delayed reading of studies, studies read out of order, etc.

By default, the Problem Studies Tool page displays all problem studies for the current day, regardless of the originating source (the "Station"), Patient ID or Patient Name, as in the following example:

Problem Studies Tool

Station Name: All Stations

Patient ID:

Patient Name:

Date Search By: 11/01/2017

Problem Studies Found:

Study Date	Patient Name	Patient ID	IPID	DOB	Sex	Station Name	Problems	Action
02/18/2016	CINIII MARION J r1	HOME	05/28/1948	F	BAY10W0	DICOM study attribute mismatch	Fix	

Figure 123: The Problem Studies Tool

You can limit or expand the list of studies shown by selecting a particular **Station Name** from the pull-down menu provided or entering a specific **Patient ID** or **Patient Name**, and then clicking on the appropriate **Search** button.

You can also specify either a specific date or a range of dates by selecting “Date” or “Range” from the pull-down **Date Search By** menu and entering the desired study date or date range, and then clicking on the appropriate **Search** button. Alternatively, you can click on the **Today** and **Yesterday** buttons to see all problem studies for those days.

Each study listed may have one or more problems associated with it. To address the problems of a particular study, click on the **Fix** button next to that study to display a “Fix a Problem Study” page in a separate pop-up window for that study, as shown in the following example:

Fix a Problem Study

Study Information

Study Date	Patient Name	Patient ID	IPID	DOB	Sex	Accession #	Modality
2004-10-04 11:52:09	d1881463p0d95dp	6ea96c11p18df7p	HOME	2012-12-20	O	0a2d73aa4	MR

Study Description

DICOM study attribute mismatch

Attribute #	PACS Name	Images	Value	Incoming Values	Actions
1	Accession Number	1	0a2d73aa4	3772a2aa5	Delete Jailed: ☐ New Value: 0a2d73aa4 0a2d73aa4

Apply All Changes **Delete All Jailed Images**

Figure 124: Fixing a Problem Study

At the top of the page will be information about the study, followed by a list of the various problems with that study. In the example above, the study has two mismatched DICOM attributes.

Depending on the type of problem, you will have the option to enter the correct value and/or delete the affected images. Whenever possible, the system will offer a default suggested value as well as a pull-down list with other available values. You will have the option in many cases to enter a new value from scratch. You can delete the images that have a specific attribute problem by clicking on the “Delete Jailed” button next to that attribute and you can also delete all jailed images for this study by clicking on the **Delete All Jailed Images** button at the bottom of the screen.

When you have made all the desired changes, click on the **Apply All Changes** button at the bottom of the screen. When the changes have been made, a status message will be displayed as in the following example:

Fixing Problem Study ... Checking data ... Preparing fixes ... Updating Database ... Applying fixes ... Fix complete

Successfully applied all changes to the study.

Summary of changes:
 2 Jailed Images edited.
 118 Amicas Images edited.

Figure 125: Problem Study Fixed

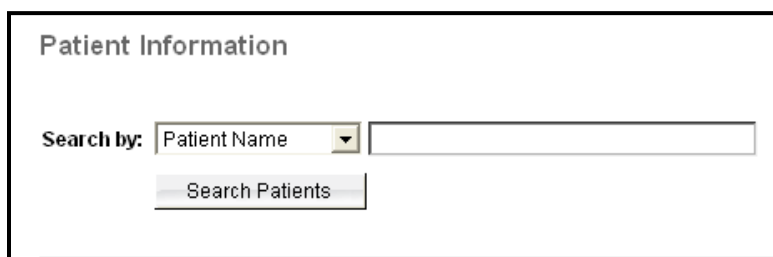
NOTE: If a particular image has been jailed for more than one reason (*i.e.*, it has multiple attributes that are different from what is already listed in the Merge PACS database), you will have to use the Problem Studies Tool multiple times to address each problem.

For detailed instructions on using the Problem Studies Tool to fix a wide variety of problems, refer to Section 17.5.

9.9.5. Patient Information

The Patient Information page allows you to view detailed patient and study demographic information.

When you click on the Patient Information page, you will first need to search for the patient whose information you want to view. Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:



Patient Information

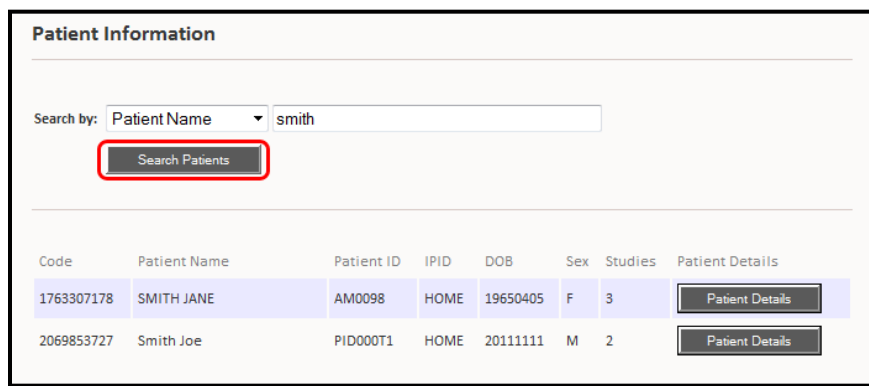
Search by: Patient Name

Search Patients

Figure 126: Selecting the Search Criterion

- If searching by **Patient Name**, you can enter the name you are searching for in the format “**last name, first name**” or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for Patient Name, as described in Section 9.2 above.

Once you have entered the desired search criterion, click on the **Search Patients** button search button to produce a list of matching patients as shown in the example below:



Patient Information

Search by: Patient Name

Search Patients

Code	Patient Name	Patient ID	IPID	DOB	Sex	Studies	Patient Details
1763307178	SMITH JANE	AM0098	HOME	19650405	F	3	<button>Patient Details</button>
2069853727	Smith Joe	PID000T1	HOME	20111111	M	2	<button>Patient Details</button>

Figure 127: List of Matching Patients

Click on the **Patient Details** button for the patient whose information you wish to view. This will cause the general patient and study information for this patient to be displayed as in the following example:

Patient Information

Patient Information:

Code	Normalized Patient Name	Display Patient Name	Normalized Patient ID	IPID	Creation Date	Modified Date	Accessed Date	Patient File Root
1763307178	null	SMITH JANE	AM0098	HOME	2013-04-03 16:21:21.829	2014-02-06 19:12:06.318	2013-04-03 16:21:21.829	SMI1763307178

Studies: 3

Study Date	Study Time	Last Modified	Accession	Modality	Refer. Phys.	Station Name	State	Validation Error	Series	Image	Study Details
20030609	133134.328000	2014-02-06 19:12:06.318	SE0000166	MR	JONES DOCTOR	mrc11076	DELETED_U	U	0	10	91
Study Description: CSPINE^_C SYN_											<button>Study Details</button>
20030609	140949.093000	2014-02-06 19:12:06.318	SE0000167	MR	JONES DOCTOR	mrc11076	DELETED_U	U	0	10	96
Study Description: LSPINE^_C SYN_											<button>Study Details</button>
20031130	135428.921000	2014-02-06 19:12:06.318	SE0000168	MR	JONES DOCTOR	mrc11076	DELETED_U	U	0	10	82
Study Description: TSPINE^_C SYN_											<button>Study Details</button>

[Return to Search](#)

Figure 128: General Patient and Study Information

- The top section of the screen displays general patient level information for this patient.
- The bottom section of the screen displays the general study information for each study.

For each study listed, you can click on the **Study Details** button to display detailed information for that study as shown in the following example:

[illegible]

Figure 129: Detailed Study Information

- At the top of the screen is the same general patient information that was shown on the previous page.

- Below the patient information is detailed demographic information for the study, as shown in the following example:

Study Instance UID: 1.2.134.1.1553.132.183.36.32.20000506.112933.323369C
 Patient Name (R Value): DOW, JENNIFER (DOW, JENNIFER)*5
 Patient ID: 8076542
 Patient DOB: 12/28/1976
 Patient Sex: M
 Accession Number: 5256154
 Modality: CR
 Study ID: 5256154
 Study Date: 20080506
 Study Time: 14:07:58.000000
 Study Description: 80LULJL0
 Station Name: Ellison 2 ADC 2
 Referring Physician (R Value): MODJERA, RAUL O (MODJERA, RAUL O)
 Reading Physician (R Value): ADULT (ADULT)
 Patient Age:
 Patient Weight:
 Storing AE Title: LOCALAE
 Modified Date: 2008-09-25 13:19:57.401
 Access Time: 11/29/2013 9:39
 AMICAS State: DELETED_U
 Archive State 1:
 Archive Location 1:
 Archive State 2:
 Archive Location 2:
 Compressed Image Flag: 0
 Reconstructed Image Flag: 0
 Telemedicine Image Flag: 0
 Archive Restricted Flag: 0
 Blurred Image Flag: 0
 Error Severity: 0
 Validation Flag: U
 Study XML File: [Show Study XML File](#)
 Patient Demo Changes: [Show Patient Demographics Changes](#)

Figure 130: Detailed Study Demographics

- NOTE:** You can view the XML file for this study, if available, within your browser by clicking on the **Show Study XML File** button provided at the bottom of the study demographic information. You can also view any changes that have been made to the patient's demographics by clicking on the **Show Patient Demographics Changes** button.

- Below the study demographic information is a list of any comments associated with this study, as in the following example:

Comments:			
Date & Time	User	Type	Error Flag
2014-02-03 16:16:17.356	amicas		E
Comments: 1 image(s) rejected, most probably due to inconsistent patient demographic or order data			

Figure 131: Study Comments

- Below the list of study comments is detailed information for each image and each series for this study as shown in the following example:

Series and Images:		
Series Number	Series Description	Series Instance UID
1	shoulder AP	1.3.51.5145.15015.20000506.1130417
Series Date		
Series Time		
Institution Name		
Station Name		
Operator Name		
Storing AE Title		
Patient Position		
Position Reference		
Frame of Reference UID		
Image Number	SOP Instance UID	Orig. SOP Inst. UID
1	1.2.840.113837.3794671460.20060925121733.2	1.3.51.5145.5015.20000506.1130417.1.0.1
Lossless SOP Inst. UID		
Compressed File		
Compressed File		
Lossless File		
Lossless File		
DICOM File		
Return to Patient		

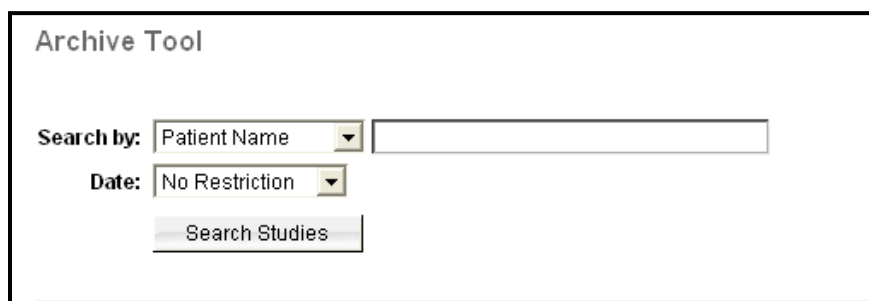
Figure 132: Series and Image Information

- When you have finished viewing this patient's information, click the **Return to Patient** button at the bottom of the screen to return to the previous screen.

9.9.6. Archive Tool

If Merge PACS is configured to run in **Standalone** mode and the optional Merge Archive component has been enabled, the Archive Tool page allows you to retrieve the DICOM images for studies that have been archived and whose DICOM images have since been deleted from the main Merge PACS Server. Note that this page will only appear on the main Tools menu if the Archive component has been enabled.

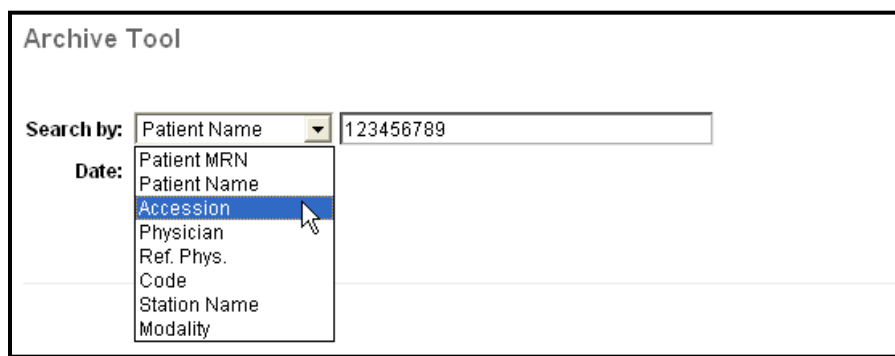
When you click on the Archive Tool page, you will first need to search for the studies whose DICOM images you want to retrieve, as shown in the following example:



The screenshot shows the 'Archive Tool' interface. It has a 'Search by:' dropdown menu currently showing 'Patient Name' and an adjacent text input field. Below this is a 'Date:' dropdown menu showing 'No Restriction'. At the bottom of the search section is a button labeled 'Search Studies'.

Figure 133: Searching for Studies

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:



This screenshot shows the 'Archive Tool' interface with the 'Search by:' dropdown menu open. The dropdown list contains the following options: Patient MRN, Patient Name, Accession (which is highlighted with a blue background and a mouse cursor), Physician, Ref. Phys., Code, Station Name, and Modality. The text input field to the right of the dropdown menu contains the value '123456789'.

Figure 134: Selecting a Search Criterion

- If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, **Referring Physician** and **Physician**, as described in Section 9.2 above.

- If you want to restrict your search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

The screenshot shows a search form with 'Search by:' set to 'Accession' and the value '123456789'. The 'Date:' dropdown menu is open, showing options: 'No Restriction', 'Today', 'Yesterday', 'This month' (highlighted), 'Last month', 'This year', 'Last year', and 'Custom'.

Figure 135: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

The screenshot shows the 'Date:' dropdown set to 'Custom'. Below it, a red box highlights the 'From:' and 'To:' date fields, both set to '10/31/2017'. A 'Search Studies' button is visible at the bottom.

Figure 136: Specifying a Custom Date Range

- When you have entered the desired search criteria, click on the **Search Studies** to display a list of matching studies, as shown in the following example:

The screenshot shows the 'Archive Tool' interface. The search criteria are 'Patient Name' and 'doe'. The 'Date:' dropdown is set to 'No Restriction'. Below the search fields is a 'Search Studies' button. The results are displayed in a table with columns: Patient Name, Patient ID, IPID, DOB, Sex, Accession, Study Date, AMI/CAS State, Archive State, Images, Request, and Action.

Patient Name	Patient ID	IPID	DOB	Sex	Accession	Study Date	AMI/CAS State	Archive State	Images	Request	Action
DOE, JESSICA	63981642	HOME	19851210	F	AR05130	20120104	IMPORTED	Not Archived	2		Request to Archive

Figure 137: List of Matching Studies

- Studies that are eligible for retrieval will have a button labeled **Request to Retrieve** next to their listing. Click this button for the study you want to retrieve.

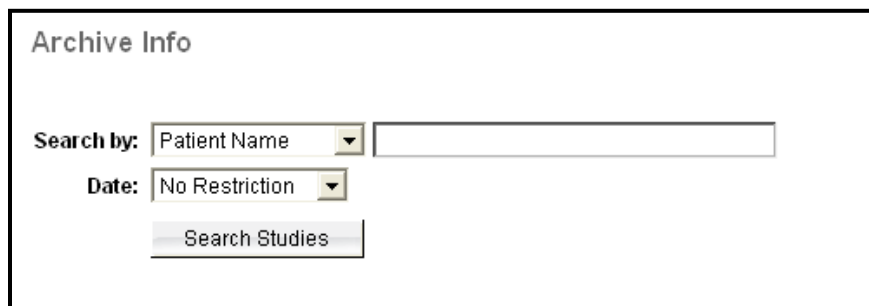
NOTE: A study is eligible for retrieval if its DICOM images are either **NEARLINE** (*i.e.*, on the Merge Archive Server) or **ONLINE** (*i.e.*, still on the main Merge PACS Server). If the **Archive State** of a study is listed as either **SKIPPED** or is left blank, the study’s DICOM images are not available for retrieval.

- If desired, the status of studies actively being retrieved can be viewed using the **Merge Archive Queue Monitor** page described in subsection 9.6.1, above.

9.9.7. Archive Info

If Merge PACS is configured to run in **Standalone** mode and the optional Merge Archive component has been enabled, the Archive Info page allows you to view detailed information about studies that have been sent to the Archive. In addition, if the study is still nearline (*i.e.*, on a tape inside the Archive tape device), the Archive info page also allows you to verify whether specified studies can be retrieved from the Archive. Note that this page will only appear on the main Tools menu if the Archive component has been enabled.

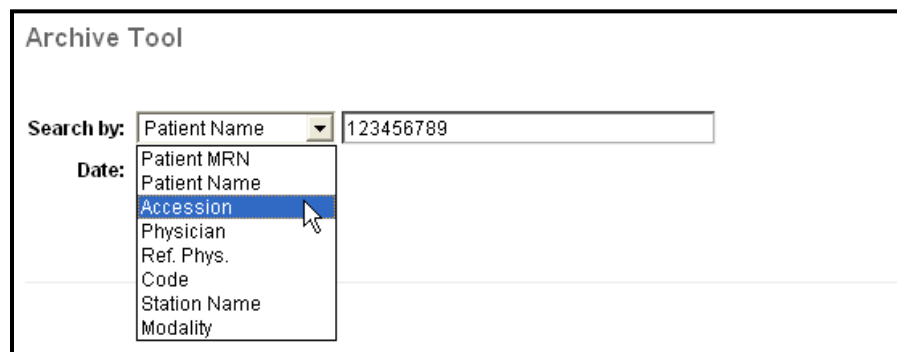
When you click on the Archive Info page, you will first need to search for the study, as shown in the following example:



The screenshot shows a web form titled "Archive Info". It contains two search criteria: "Search by:" with a dropdown menu set to "Patient Name" and an adjacent text input field; and "Date:" with a dropdown menu set to "No Restriction". Below these fields is a "Search Studies" button.

Figure 138: Searching for Studies

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:



The screenshot shows a web form titled "Archive Tool". The "Search by:" dropdown menu is open, displaying a list of search criteria: "Patient MRN", "Patient Name", "Accession" (highlighted with a mouse cursor), "Physician", "Ref. Phys.", "Code", "Station Name", and "Modality". The text input field to the right of the dropdown contains the value "123456789".

Figure 139: Selecting a Search Criterion

- If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, **Referring Physician** and **Physician**, as described in Section 9.2 above.

- If you want to restrict your search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

The screenshot shows a search form with 'Search by:' set to 'Accession' and the value '123456789'. The 'Date:' dropdown menu is open, showing options: 'No Restriction', 'Today', 'Yesterday', 'This month' (highlighted), 'Last month', 'This year', 'Last year', and 'Custom'.

Figure 140: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

The screenshot shows the search form with 'Search by:' set to 'Accession' and the value '123456789'. The 'Date:' dropdown menu is set to 'Custom'. Below it, a red box highlights the 'From:' and 'To:' date fields, both set to '10/31/2017'. A 'Search Studies' button is visible at the bottom.

Figure 141: Specifying a Custom Date Range

- When you have entered the desired search criteria, click on the **Search Studies** to display a list of matching studies, as shown in the following example:

The screenshot shows the search results page. The search criteria are 'Patient Name' and 'DOE'. The 'Search Studies' button is highlighted with a red box. Below the search criteria, there is a table of matching studies.

	Patient ID	Patient Name	Sex	DOB	Study Date	Accession	Study Description
Study	123456789	DOE JONATHAN Q	M	19460325	20031130	SE000089	Abdomen
Archive Details	Archive Library: TSM, Archive Type: DICOM, Validate: true, Last Validation Time: 2006-10-02 15:55:58.075, Next Validation Time: 2006-11-01 14:55:58.075, Archive Location: 2003111301.3.12.2.1107.5.4.5.25089.4.0.2809759022451446.zip 1: Device: UD01DISK, Volume: TSMDB_BACKUP1, Size: 2864916 bytes, Time Inserted: 2006-10-02 15:54:45, Time Expires: never 2: Device: UD01DISK, Volume: TSMDB_BACKUP2, Size: 2864916 bytes, Time Inserted: 2006-10-02 15:54:45, Time Expires: never						
	Test						
Study	987654321	DOE JUSTIN	O	19060118	19420103	A65036329287681	
Archive Details	Archive Library: TSM, Archive Type: DICOM, Validate: true, Last Validation Time: 2006-10-02 15:55:57.546, Next Validation Time: 2006-11-01 14:55:57.546, Archive Location: 1942010311.2.840.113837.20010612155350.27.3839.zip 1: Device: UD01DISK, Volume: 1, Size: 139701 bytes, Time Inserted: 2006-10-02 15:54:45, Time Expires: never 2: Device: UD01DISK, Volume: TSMDB_BACKUP1, Size: 139701 bytes, Time Inserted: 2006-10-02 15:54:45, Time Expires: never						
	Test						

Figure 142: List of Matching Studies

- The Archive Details section provides the following information about the study:

Information	Description
Archive Library	Type of storage management software being used (HSM = Microsoft Remote Storage Management, DICOM = ECM Archive)
Archive Type	Format of the archived study (DICOM or Lossless)
Validate	Whether the archived study is in a queue to be validated
Last Validation Time	Time of last validation (“null” indicates that no validation has yet occurred)
Next Validation Time	Proposed time of next validation. Note that the validation process runs only when there is no other activity in the Archive and this may affect the actual time when validation next occurs.
Archive Location	For tape libraries, the ID string that uniquely identifies the file containing the archived study. For ECM, the Application Entity Title of the ECM Archive.
Device	The type of archive media to which the study has been written (UDO/LTO, etc.). If the study has not yet been written to tape/disk (or has not been validated since being written to tape/disk), the value will be “null”.
Volume	The volume name of the media to which the study has been written. If the study has not yet been written to tape/disk (or has not been validated since being written to tape/disk), the value will be “null”.
Size	The size of the archived study that was written to tape. If the study has not yet been written to tape/disk (or has not been validated since being written to tape/disk), the value will be “0 bytes”.
Time Inserted	Time when the study was first sent to the Archive Server from the main Merge PACS Server. In the case of studies that have been rearchived, this reflects the last time the study was sent to the Archive.
Time Expires	This should always have a value of “never”.

- Studies that have been archived will have a **Test** button below the general archive details for the study, as in the following example:

	Patient ID	Patient Name ▲	Sex	DOB	Study
Study	123456789	DOE JONATHAN Q	M	19460325	2003
Archive Details	Archive Library: TSM, Archive Type: DICOM, Validate: true, Last Vali 14:55:58.075, Archive Location: 20031113011.3.1.2.1107.5.4.5.25089.4.0.28097 1: Device: UDO1DISK, Volume: TSMDB_BACKUP1, Size: 2864916 2: Device: UDO1DISK, Volume: TSMDB_BACKUP2, Size: 2864916				
	Test				

Figure 143: The Archive Info Test Button

- If the study is still nearline (*i.e.*, on a tape inside the Archive tape device), when you click on the Test button a separate pop-up window will appear that displays the results of the test in real time, as in the following example:

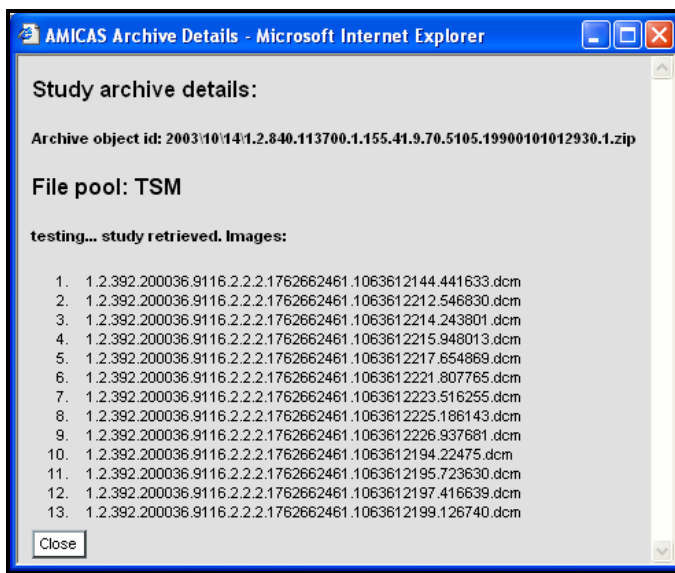


Figure 144: Study Archive Details

- If the study is offline, when you click the Test button an error message will be displayed stating that the study is not currently available, as in the following example:

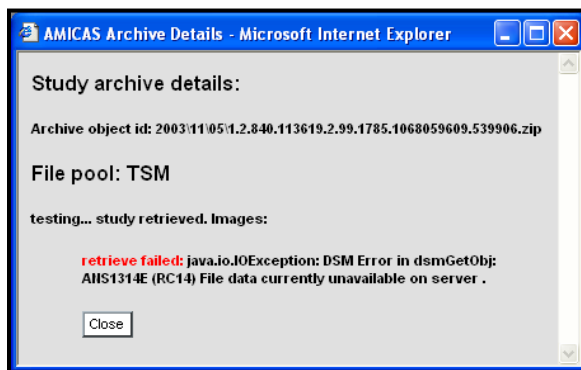


Figure 145: Study Not Available

9.9.8. Access Control

As described in Section 9.11 below, it is possible to give specific users or groups unrestricted access to all exams within the system. An “exam” can be a study, an order and/or a report. Users and groups who do not have such unrestricted access will only have access to exams that have been specifically assigned to them, either automatically (based on the Referring Physician field of the orders produced by a site’s RIS) or manually (via the Access Control page).

CAUTION: If a study is contained in a RealTime Worklist or RealTime Study List worklist that a user has access to (*i.e.*, the user belongs to a group that has been assigned that worklist), the user will be able to access that study regardless of the user’s access privileges.

a. Searching for an Exam

When you click on the Archive Info page, you will first need to search for the exam, as shown in the following example:

Figure 146: Searching for Exams

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:

Figure 147: Selecting a Search Criterion

- If searching by **Patient Name**, **Referring Physician** ["Ref. Phys."] or **Physician**, you can enter the name you are searching for in the format "**last name, first name**" or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, **Referring Physician** and **Physician**, as described in Section 9.2 above.
- If you are searching by anything other than Patient MRN or Patient Name and want to restrict your search to a specific date or date range, select the desired date range from the pull-down **Date** menu, as shown below:

Figure 148: Selecting a Date Restriction

If you select **custom** from the pull-down Date menu, you will then be able to specify an exact date range either by manually entering the “from” and “to” dates or using the calendar icons to select them, as in the following example:

The screenshot shows a search form with the following elements:

- Search by:** Patient Name (dropdown menu) with the value "smith" entered in the adjacent text field.
- Date:** Custom (dropdown menu).
- From:** 11/01/2017 (text field) with a calendar icon to its right.
- To:** 11/01/2017 (text field) with a calendar icon to its right.
- Search** button.

A red rectangle highlights the "From" and "To" date fields and their associated calendar icons.

Figure 149: Specifying a Custom Date Range

- When you have entered the desired search criteria, click the **Search** button to display a list of matching patients or exams, as shown in the following examples:

The screenshot shows the search results for "Patient Name" with the value "doe". The "Date" is set to "No Restriction". The "Search" button is highlighted with a red rectangle.

Below the search criteria, the text "Select Patient:" is displayed. To the right of this text, the MRN "987654cba" is listed. Below this, a table lists three patients:

MRN	IPID	Patient Name	Date Of Birth	Sex	Latest Update Date	
123456789	HOME	Doe, Jonathan	05/12/2014	O	2003-11-30 13:52:48.921	Select Patient
99999999	HOME	Doe, Joanne	12/20/2012	O	2000-05-05 22:16:01.0	Select Patient
1234512345	HOME	Doe, Jessica	12/20/2012	O	2004-04-03 08:15:47.0	Select Patient

Figure 150: List of Matching Patients

The screenshot shows the search results for "Modality" with the value "CT". The "Date" is set to "No Restriction". The "Search" button is highlighted with a red rectangle.

Below the search criteria, a table lists three exams:

Patient Name	MRN	IPID	Accession	Study Date	Study Time	Study Description	Station Name	Modality	Status	Online	
Doe, Jonathan	123456789	HOME	12345abc	20000505	17.27.57			CT	DELETED_U	Online	Select Study
Doe, Joanne	99999999	HOME	99999xyz	20100219	19.27.16			CT	DELETED_U	Online	Select Study
Doe, Jessica	1234512345	HOME	987654cba	20111031	14.01.00			CT	DELETED_U	Online	Select Study

Figure 151: List of Matching Exams

NOTE: Searching by **Patient MRN** or **Patient Name** will produce a list of matching **patients**. Searching by any other criteria will produce a list of matching **exams**.

- If you searched by Patient MRN or Patient Name, click the **Select Patient** button next to the desired patient's entry, as in the following example:

Search by: Patient Name | doe

Date: No Restriction

Search

12345abc
99999xyz
987654cba

Select Patient:

MRN	IPID	Patient Name	Date Of Birth	Sex	Latest Update Date	
123456789	HOME	Doe, Jonathan	05/12/2014	O	2003-11-30 13:52:48.921	Select Patient
99999999	HOME	Doe, Joanne	12/20/2012	O	2000-05-05 22:16:01.0	Select Patient
1234512345	HOME	Doe, Jessica	12/20/2012	O	2004-04-03 08:15:47.0	Select Patient

Figure 152: Selecting a Patient

- If that patient has more than one exam associated with him or her, a list of those exams will then be displayed, as in the following example:

Access Control (Find Exam for)

Selected Patient

Patient ID: AM-0098
Patient Name: SMITH, JANE
Date of Birth: 04/05/1965
Sex: F

Select an Exam

Date	Accession	Modality	Description	
2004-11-24 13:31:34.328	SE0000166	MR	CSPINE	Modify Access Control
2003-11-30 13:52:48.921	SE0000168	MR	TSPINE^_C SYN_	Modify Access Control
2003-06-09 14:09:49.093	SE0000167	MR	LSPINE	Modify Access Control

Figure 153: Selecting an Exam for this Patient

Click the **Modify Access Control** button next to the desired exam's listing to display the Modify Access Control page for this exam, as described in paragraph b below.

- If the patient has only one exam, the Modify Access Control page for this exam will be displayed, as described in subsection paragraph b below.
- If you searched by anything other than Patient MRN or Patient Name, click the **Select Study** button next to the desired study's listing, as in the following example:

Search by: Modality | CT

Date: No Restriction

Search

Patient Name	MRN	IPID	Accession	Study Date	Study Time	Study Description	Station Name	Modality	Status	Online	
Doe, Jonathan	123456789	HOME	12345abc	20000505	17:27:57			CT	DELETED_U	Online	Select Study
Doe, Joanne	99999999	HOME	99999xyz	20100219	19:27:16			CT	DELETED_U	Online	Select Study
Doe, Jessica	1234512345	HOME	987654cba	20111031	14:01:00			CT	DELETED_U	Online	Select Study

Figure 154: Selecting a Study

- If that study has more than one exam associated with it, a list of those exams will then be displayed, as in the following example:

Access Control (Find Exam for)

Selected Patient

Patient ID: AM-0098
 Patient Name: SMITH, JANE
 Date of Birth: 04/05/1965
 Sex: F

Select an Exam

Date	Accession	Modality	Description	
2004-11-24 13:31:34.328	SE0000166	MR	CSPINE	Modify Access Control
2003-11-30 13:52:48.921	SE0000168	MR	TSPINE^_C SYN_	Modify Access Control
2003-06-09 14:09:49.093	SE0000167	MR	LSPINE	Modify Access Control

Figure 155: Selecting an Exam for this Patient

Click the **Modify Access Control** button next to the desired exam's listing to display the Modify Access Control page for this exam, as described in subsection paragraph b below.

- If the study has only one exam associated with it, the Modify Access Control page for this exam will be displayed, as described in subsection paragraph b below.

b. Managing Access Control for the Selected Exam

Once you have selected an exam, as described in the preceding subsection, the Modify Access Control page for that exam will be displayed, as in the following example:

Access Control (Assign / Unassign Users & Groups)

Selected Exam:

Patient Name	MRN	Date and Time	Status	Modality	Accession Number	Description
SMITH, JANE	AM-0098	11/24/2004 13:31:34		MR	SE0000166	CSPINE

Users & Groups with access to this Exam:

Existing Users:

Login	Name	Code	Phone	Email	Address
☒	xcarten	Carten, Xavier C	10003 617-555-1212	xcarten@hospital.com	3 Microsoft Way, Redmond, WA 98073

Message to the users being notified:

Existing Groups:

	Name	Description
☐	East	East
☒	West	West

Search for Physicians to Grant Access:

Provider Name

Figure 156: The Modify Access Control Page

At the Modify Access Control page, you can revoke access to this exam from users and/or groups that currently have access, grant access to one or more physicians as well as the groups to which they belong, and send a notification e-mail to all users who currently have access to this exam.

- The top section of the Modify Access Control page displays separate lists of all users and all groups that currently have access to this exam, as shown in the following example:

Users & Groups with access to this Exam:

Existing Users:

	Login	Name	Code	Phone	Email	Address
☒	xcarter	Carter, Xavier C	10003	617-555-1212	xcarter@hospital.com	3 Microsoft Way, Redmond, WA 34567

Unassign User Access

Message to the users being notified:

Notify all Existing Users

Existing Groups:

	Name	Description
☐	East	East
☒	West	West

Unassign Group Access

Figure 157: Users and Groups with Access to this Exam

NOTE: Clicking the **Users & Groups with access to this Exam** link at the top of the section will toggle the display of this section in its entirety.

- To revoke access from a one or more users included in the list of Existing Users, click the checkboxes next to their Login and then click the **Unassign User Access** button, as in the following example:

Existing Users:

	Login	Name	Code	Phone	Email	Address
☒	xcarter	Carter, Xavier C	10003	617-555-1212	xcarter@hospital.com	3 Microsoft Way, Redmond, WA 34567

Unassign User Access

Figure 158: Revoking User Access to this Exam

- To revoke access from a one or more groups included in the list of Existing Groups, click the checkboxes next to their Login and then click the **Unassign Group Access** button, as in the following example:

Existing Groups:

	Name	Description
☐	East	East
☒	West	West

Unassign Group Access

Figure 159: Revoking Group Access to this Exam

- The top section of the Modify Access Control page also allows you to send a notification e-mail to all users who currently have access to this study, as shown in the following example:

Existing Users:

Login	Name	Code	Phone	Email	Address
☒ xcarter	Carter, Xavier C	10003	617-555-1212	xcarter@hospital.com	3 Microsoft Way, Redmond, WA 34567

Unassign User Access

Message to the users being notified:

Notify all Existing Users

160: Sending a Notification E-mail

To send a notification e-mail, enter the desired text in the field provided and then click the **Notify all Existing Users** button.

- The bottom section of the Modify Access Control page allows you to search for physicians to whom you want to grant access to this exam, as shown in the following example:

Search for Physician(s) to Grant Access:

Provider Name = Find Physician

Figure 161: Search for Physicians to Grant Access

- Select either Provider Name or Provider Code from the pull-down search options menu, enter the desired text in the field provided, and then click the **Find Physician** button to display a list of matching physicians, as in the following example:

Search for Physician(s) to Grant Access:

Provider Name = Find Physician

Provider Name	Code	Phone	Email	Address
☐ Imedico, Ima	10001	617-555-1111	imedico@hospital.com	59 Waverly St., Somerville, MA 01760
☐ Imedico, yura	10002	617-555-2222	ymedico@hospital.com	14 Wedgewood Rd., Malden, MA 01244

☐ Assign Group Access Also

☐ Send Notification

Assign User Access

Figure 162: List of Matching Physicians

- Click on the checkboxes next to the Login of the physicians to whom you want to grant access to this exam.
- If you want to grant access to all groups this physician is a member of, click the checkbox next to the **Assign Group Access Also** option.
- If you want a notification e-mail sent to this physician saying that he or she has been granted access to this exam, click the checkbox next to the **Send Notification** button.
- Click the **Assign User Access** button to perform the selected actions.

9.9.9. Manage Unassigned Exams

The Manage Unassigned Exams page allows you to view and grant access to exams that are not currently assigned to anybody.

a. Searching for Unassigned Exams

At the Manage Unassigned Exams page, you will first need to search for the specific unassigned exam to which you want to assign access, as shown in the following example:

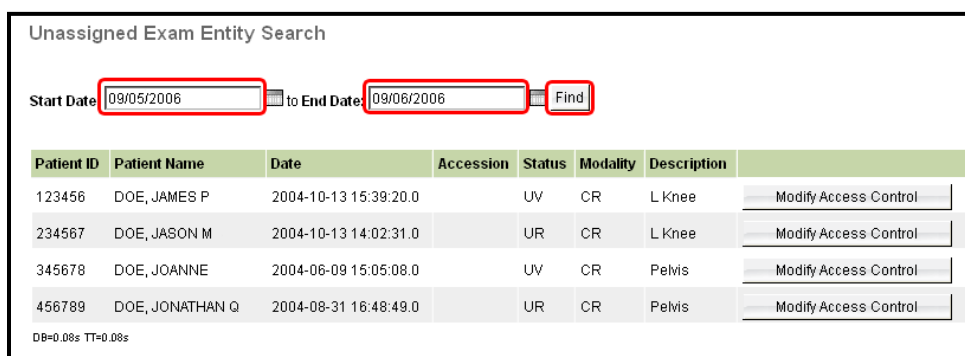


Unassigned Exam Entity Search

Start Date: 10/03/2006 to End Date: 10/03/2006 Find

Figure 163: Searching for an Exam

- Enter a **Start Date** and an **End Date** in the fields provided and then click the **Find** button to display a list of all unassigned exams that have an exam date during the specified date range, as in the following example:



Unassigned Exam Entity Search

Start Date: 09/05/2006 to End Date: 09/06/2006 Find

Patient ID	Patient Name	Date	Accession	Status	Modality	Description	
123456	DOE, JAMES P	2004-10-13 15:39:20.0		UV	CR	L Knee	Modify Access Control
234567	DOE, JASON M	2004-10-13 14:02:31.0		UR	CR	L Knee	Modify Access Control
345678	DOE, JOANNE	2004-06-09 15:05:08.0		UV	CR	Pelvis	Modify Access Control
456789	DOE, JONATHAN Q	2004-08-31 16:48:49.0		UR	CR	Pelvis	Modify Access Control

DB=0.06s TT=0.06s

Figure 164: List of Matching Unassigned Exams

NOTE: You can also select dates by clicking on the little calendar icon next to each date field. This will cause a small calendar to be displayed in a separate pop-up window and you can click the < and > icons surrounding the month and year at the top of the calendar to navigate forward and backwards through the calendar, if necessary, and then click the desired day

NOTE: By default, the list of matching unassigned exams can display a maximum of 20 results. This number can be configured, however, by Merge Healthcare Technical Support.

b. Managing Unassigned Exams

Once you have generated a list of unassigned exams, click the **Modify Access Control** button next to the exam to which you want to grant access, as shown in the following example:

Unassigned Exam Entity Search

Start Date: 09/05/2006 to End Date: 09/06/2006 Find

Patient ID	Patient Name	Date	Accession	Status	Modality	Description	
123456	DOE, JAMES P	2004-10-13 15:39:20.0		UV	CR	L Knee	Modify Access Control
234567	DOE, JASON M	2004-10-13 14:02:31.0		UR	CR	L Knee	Modify Access Control
345678	DOE, JOANNE	2004-06-09 15:05:08.0		UV	CR	Pelvis	Modify Access Control
456789	DOE, JONATHAN Q	2004-08-31 16:48:49.0		UR	CR	Pelvis	Modify Access Control

DB=0.08s TT=0.08s

Figure 165: Selecting an Unassigned Exam

When you click the Modify Access Control button for an exam, the **Modify Access Control** page for that exam will be displayed, as in the following example:

Access Control (Assign / Unassign Users & Groups)

Selected Exam:

Patient Name	MRN	Date and Time	Status	Modality	Accession Number	Description
SMITH, JANE	AM-0098	11/24/2004 13:31:34	MR	SE0000166		CSPINE

▼ Users & Groups with access to this Exam:

Existing Users:

Existing Groups:

Search for Physician(s) to Grant Access:

Provider Name = Find Physician

Figure 166: The Modify Access Control Page

The bottom section of the Modify Access Control page allows you to search for physicians to whom you want to grant access to this exam, as shown in the following example:

Search for Physician(s) to Grant Access:

Provider Name = Find Physician

Figure 167: Search for Physicians to Grant Access

- Select either Provider Name or Provider Code from the pull-down search options menu, enter the desired text in the field provided, and then click the **Find Physician** button to display a list of matching physicians, as in the following example:

Search for Physician(s) to Grant Access:

Provider Name = Imedico Find Physician

Provider Name	Provider Code	Name	Code	Phone	Email	Address
☐	imedico	Medico, Ima	10001	617-555-1111	imedico@hospital.com	59 Waverly St., Somerville, MA 01760
☐	ymedico	Medico, yura	10002	617-555-2222	ymedico@hospital.com	14 Wedgewood Rd., Malden, MA 01244

☐ Assign Group Access Also

☐ Send Notification

Assign User Access

Figure 168: List of Matching Physicians

- Click on the checkboxes next to the Login of the physicians to whom you want to grant access to this exam.
- If you want to grant access to all groups this physician is a member of, click the checkbox next to the **Assign Group Access Also** option.
- If you want a notification e-mail sent to this physician saying that he or she has been granted access to this exam, click the checkbox next to the **Send Notification** button.
- Click the **Assign User Access** button to perform the selected actions.

9.10. Enterprise CD Writer

As described in subsection 1.1.12 above, the *Enterprise CD Writer* is an optional feature that adds the ability to send patient studies to a shared CD or DVD writing device, both manually and automatically according to custom-defined “auto-writing” rules. The device itself can be configured to write just the compressed or uncompressed images or both the compressed and uncompressed images.

Once the Enterprise CD Writer feature has been fully enabled, the Enterprise CD Writer page will provide you with the ability to:

- Manually send patient studies to a shared CD or DVD writing device
- Monitor the CD Writing Queue

NOTE: Enterprise CD Writer is supported for use with **Windows 2012 R2** as long as Sorna **Vertex** software is used. As an alternative (regardless of the operating system), if Merge PACS is configured to run in **Standalone** mode, a shared CD/DVD writing device (such as Sorna Vertex) can be configured as a DICOM Device, as described in subsection 9.13.2 below, and studies can be sent to it manually either via the Send Studies page, as described in subsection 9.9.3 above, or routed to it from the Merge PACS Workstation. If Merge PACS is configured to run in **Integrated** mode, a shared CD/DVD writing device can be configured as a DICOM Device on the paired iCEA Server and studies can be routed to it from the Merge PACS Workstation.

Clicking on the “Enterprise CD Writer” link on the top category menu will display a submenu of the individual available CD writing pages as shown below:

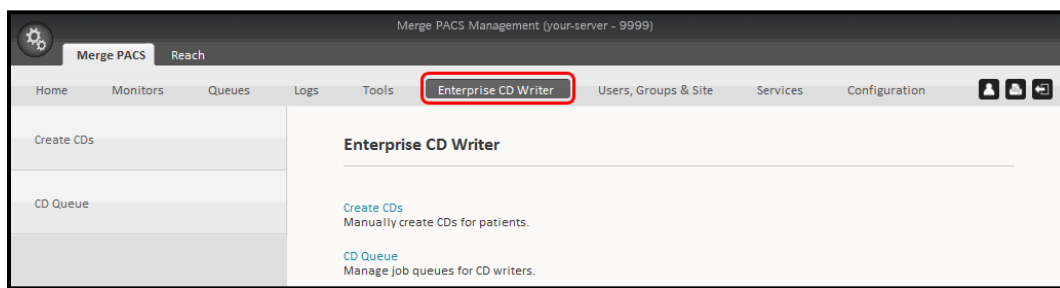


Figure 169: The Enterprise CD Writer Main Page

9.10.1. Manually Sending Studies to a Shared CD/DVD Writing Device

Studies can be manually sent to the configured CD/DVD writing device from the **Create CDs** section of the Enterprise CD Writer Management pages, which will bring up the “Create CDs” window as shown in the example below:



Figure 170: Manually Creating a CD

- Select a search criterion from the pull-down **Search by** menu and enter the desired value in the field to the right of the menu, as shown in the following example:



Figure 171: Selecting a Search Criterion

- If searching by **Patient Name**, you can enter the name you are searching for in the format “**last name, first name**” or simply enter the **last name** to find all people with that last name.
- You can also use **wildcard searches** for **Patient Name**, as described in Section 9.2 above.

- When you have entered the desired search criteria, click on the **Search Patients** to display a list of matching patients, as shown in the following example:

Code	Patient Name	Patient ID	DOB	Sex	Studies	Select Studies
☐ 123456789	DOE JESSICA	9876543	19320714	F	1	
☐ 234567890	DOE JONATHAN Q	8765432	19260110	M	3	Select studies
☐ 345678901	DOE JUSTIN	7654321	19350527	M	2	Select studies

Figure 172: Selecting a Patient

- If necessary, select a CD/DVD writing device from the pull-down “Create CD on CD Writer” menu.
- Enter the number of copies of this CD or DVD you wish to create.
- If more than one patient matched your search criteria, select the desired patient by clicking on the check box to the left of the patient’s name (if there is only one matching patient, the box will be checked automatically).
- If a patient has more than one study and you want to specify which studies should be included on the CD or DVD, click on the **Select Studies** button for that patient. You will then be able to select the studies you do want to appear on the CD or DVD, as in the following example:

Create CDs

Create CD on CD writer: FilmX1270 Number of Copies: 1

Create Study CD

Patient Information:

Code	Patient Name	Patient ID	DOB	Sex
234567890	DOE JONATHAN Q	8765432	19260110	M

Accession	Modality	Station Name	Study Date	Study Time	Study Desc.
☐ 12345678	CT	CT04_OC0	20061003	190728.00000	679/087/NE/3D
☐ 23456789	MR	MRS30C0	20061003	215056.00000	580/NE
☐ 34567890	CR	Ellison 2 ADC	20061004	140746.00000	603EL/BO

Figure 173: Specifying Which Studies to Be Included on CD

By default, all the studies will initially be selected, so you can actually **deselect** those studies you don’t want to be included on the CD or DVD by clicking on the appropriate boxes to remove the check marks.

NOTE: You **cannot** use the browser’s **Back** button to return to the list of patients after selecting studies.

- When you have selected the desired patient and/or studies, click on the **Create CD** button.

The creation request will be entered into the CD Creation Queue, which can be viewed and managed as described in the next Section.

9.10.2. Monitoring and Managing the CD Creation Queue

All CD creation requests, whether entered manually or automatically, can be monitored and managed from the CD Creation Queue Manager. This is done from the **CD Queue** section of the Enterprise CD Writer Management pages, which will bring up the “Manage CD Writing Queue” window, as shown in the example below:

Number of jobs in queue: 3												
Time Queued ▲	Time Started	Time Completed	Job Type	Patient Name	Patient ID	Accession	CD Writer	Copies	Priority	Status	Delete	Restart
2006-10-03 11:27:09	2006-10-03 11:27:16	2006-10-03 11:27:16	patient	DOE JESSICA	9876543		FilmX1270	1	3	Completed	Delete	
2006-10-03 11:27:09	2006-10-03 11:27:11	2006-10-03 11:27:15	patient	DOE JONATHAN Q	8765432		FilmX1270	1	3	Completed	Delete	
2006-10-03 11:27:09	2006-10-03 11:27:15	2006-10-03 11:27:16	patient	DOE JUSTIN	7654321		FilmX1270	1	3	Completed	Delete	

Figure 174: The Enterprise CD Writer's Queue Manager

Each creation request (or “job”) will display one of the following status abbreviations:

Status	Description
Requested	The job is in the queue, but the studies have not yet begun to be sent to the CD/DVD writing device.
In Progress	The studies are currently being sent to the CD/DVD writing device.
Completed	The studies for this patient have all been sent to the CD/DVD writing device.
Failed	The studies for this patient could not be sent to the CD/DVD writing device.

If the Enterprise CD Writer was unable to send one or more studies to the CD/DVD writing device (e.g., because the images had not finished being imported from the main Merge PACS Server when the CD creation request was entered), an error message will be displayed as in the following example:

Number of jobs in queue: 3												
Time Queued ▲	Time Started	Time Completed	Job Type	Patient Name	Patient ID	Accession	CD Writer	Copies	Priority	Status	Delete	Restart
2006-10-03 11:27:09	2006-10-03 11:27:16	2006-10-03 11:27:16	patient	DOE JESSICA	9876543		FilmX1270	1	3	Completed	Delete	
2006-10-03 11:27:09	2006-10-03 11:27:11	2006-10-03 11:27:15	patient	DOE JONATHAN Q	8765432		FilmX1270	1	3	Completed	Delete	
2006-10-03 11:27:09	2006-10-03 11:27:15	2006-10-03 11:27:16	patient	DOE JUSTIN	7654321		FilmX1270	1	3	Failed	Delete	Restart
Error Message: At least one study for the patient is being imported. When all studies for the patient are imported, try again by restarting this job.												

Figure 175: Error Message

At the Manage CD Writing Queue page, you can:

- **Sort** the queue by clicking on any column heading. Once a column heading has been selected, you can also choose whether to sort in ascending or descending order by clicking on the blue triangle to the right of the column heading
- **Restart** a failed job
- **Change the Priority** of a job (1 is highest, 5 is lowest) by clicking one of the up or down arrows in the Priority column
- **Delete** a completed or failed job from the queue (note that you cannot delete a job that is requested or in progress).

9.11. Users, Groups & Site

The Users, Group & Site management pages allows you to manage the following information:

Page	Description
Users	Manage account information for individual users, specify which groups they belong to, manage their Credentialing and Licensure settings, if applicable.
Groups	Enter or edit the general description of each group, manage the members of each group, manage LDAP groups (if applicable), manage the worklists associated with each group (if any), configure Viewer XML integrations used by this each group (if any), specify the parent of each group for use with Templated User Management (as described in Chapter 11 below), configure special privileges for each group (such as Merge Instant Messenger and Study Assignment privileges) and configure the relative priorities between all existing groups for use with Templated User Management.
Site	Enter or edit the general description of this site for use with Templated User Management, as described in Chapter 11 below.

NOTE: The Users, Groups & Site management page allows you to define the various entities within Merge PACS. Once these entities are defined, you can assign privileges and preferences to them via Templated User Management, as described in Chapter 11 below.

NOTE: Users inherit their privileges and default set of preferences from their “parents”, which can include one or more groups as well as the site as a whole.

NOTE: The use of LDAP / Active Directory is the recommended means of user and credential management.

Clicking on the “**Users, Groups & Site**” link on the top category menu will display a submenu of the individual available Users, Groups & Site pages as shown below:

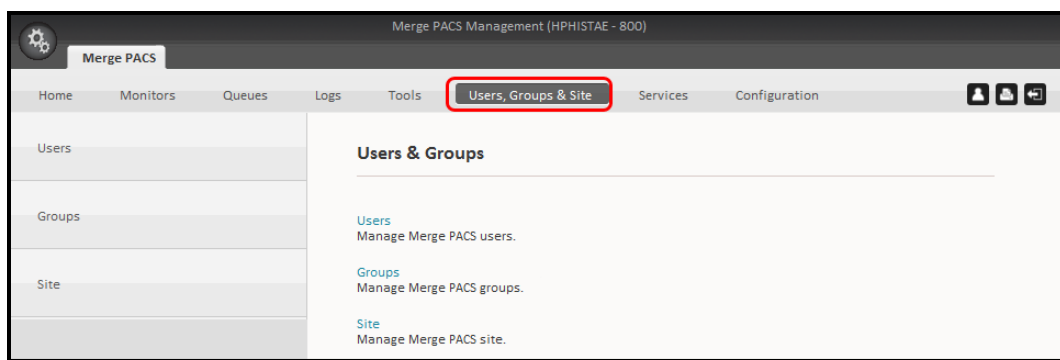


Figure 176: The Users, Groups & Site Pages

To select either the Users, the Groups or the Site page, click on the appropriate link in the Users, Groups & Site submenu, or the link displayed on the main Users, Groups & Site page.

Note that Merge PACS can be configured to authenticate users against its own database (“DB mode”), against one or more LDAP servers (“LDAP mode”), or both (“Mixed mode”). The availability and functionality of the various Users and Groups Management Pages will depend on the specific authentication mode being used:

Mode	Description
DB	The Merge PACS database will be responsible for the authentication and privileges (“Merge PACS Web Security”). All User and Group Management Pages function as described. This is the default, non-LDAP setting.
LDAP	LDAP will be responsible for the authentication and privileges will be assigned based on the LDAP and PACS Group mapping. Note the following: • The PACS Admin will not have the privilege to create a new user. • The PACS Admin can only change the account information for an existing LDAP user that is not provided by the LDAP server. • The PACS Admin will not have the privilege to modify the LDAP user's group information.
MIXED	The domain drop-down menu on the client login pages will have the option “Merge PACS” as well as the configured LDAP domains. The privileges are maintained only in the Merge PACS database. Note the following: • DB users will need to select “Merge PACS” from the domain drop-down in order to log in and will get their privileges from the database depending on the Group level privileges assigned. • LDAP users will need to select the appropriate LDAP domain from the domain drop-down in order to log in. If the LDAP User has not logged in before and the database does not have any reference for this LDAP user, the system will automatically create this user and assign the privileges based on the LDAP Group Mapping. • The PACS Admin will not have the privilege to create new LDAP users, although they can create local users. • The PACS Admin will not have the privilege to modify the LDAP user's group information. NOTE: Prior to Merge PACS 8.2, changes in group mapping on the LDAP server did not affect Merge PACS. This behavior changed in Merge PACS 8.2 and group mapping within Merge PACS will automatically be updated to reflect changes within LDAP. • The PACS Admin can only change the account information for an existing LDAP user that is not provided by the LDAP server.

9.11.1. Manage Users Page

The Manage Users page allows you to view and manage all Merge PACS user accounts and is accessed by clicking on the **Users** link as shown in the following example:

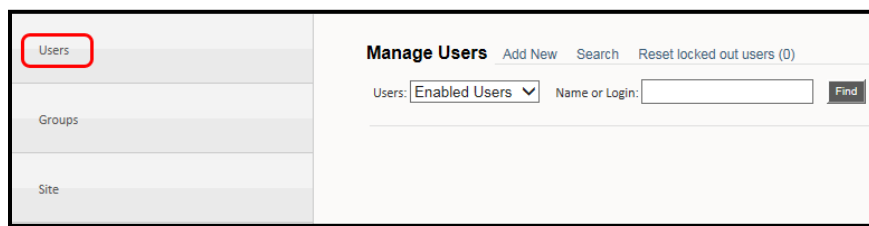


Figure 177: The Manage Users Page

When you first access the Manage Users page, you will need to choose to either add a new user or locate an existing user. By default, the Manage Users page will present you with the **Search** option, but you can access the other options at any time from the top of the page, as shown in the following example:

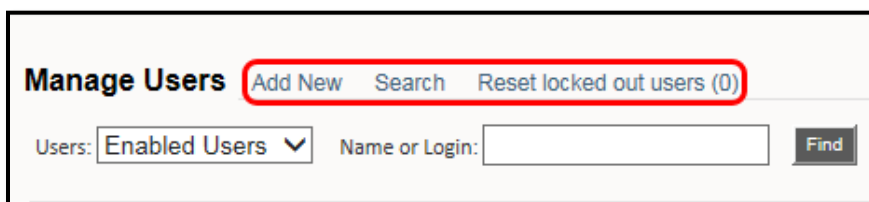


Figure 178: Manage Users Options

a. Adding a New User

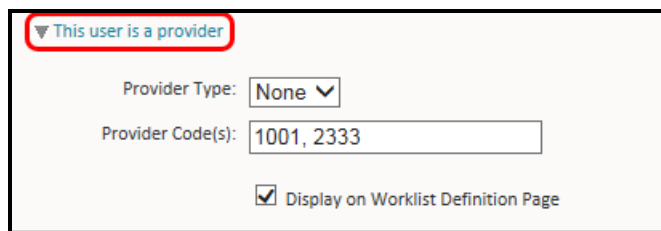
To add a new user account from scratch, click on the **Add New** link at the top of the Manage Users page to bring up the **Add New User** page, as shown in the following example:

 A screenshot of the 'Add New User' form. The form is titled 'Manage Users' and includes links for 'Add New', 'Search', and 'Reset locked out users (0)'. It is divided into several sections: 'Login Information' with fields for 'Login Name *', 'Password *', and 'Confirm Password *'; 'User Information' with fields for 'Prefix', 'First', 'Middle', 'Last', 'Suffix', and 'Degree'; 'Address' with fields for 'Address', 'Address 2', 'City', 'State' (a dropdown menu), and 'ZIP'; 'Phone' and 'Fax' fields; and 'Email' with fields for 'Email', 'Email 2', and 'Email 3'. At the bottom, there is a checkbox labeled 'This user is a provider' and an 'Enabled' checkbox. The form also includes 'Add User' and 'Cancel' buttons.

Figure 179: Adding a New User

NOTE: User accounts can also be imported in batch from an external Comma Separated Value (CSV) file, as described in Section 17.4 below.

- Enter the user's information in the provided fields (fields marked with a "*" are required).
- If this user is a physician, click on the **This user is a provider** link at the bottom of the page to enter additional information, as in the following example:



▼ This user is a provider

Provider Type: None ▼

Provider Code(s): 1001, 2333

☒ Display on Worklist Definition Page

Figure 180: Additional Information for Physicians

- If desired, select the user's **Provider Type** from the drop-down menu.
- Enter the user's **Provider Code** in the field provided, if desired. If the user has more than one provider code, you can enter them all in this field separated by commas (e.g., 10001, 10002). You don't need to put a provider code for the user if the user is not a physician and will not order exams.

NOTE: The provider code is the unique identifier the RIS uses to identify physicians that order exams in the RIS. The provider codes for the referring physician, ordering physician, attending physicians, admitting physicians, consulting physicians, and other healthcare providers that are related to this exam are embedded in the order or report messages that are sent to the Merge PACS Server. The Merge PACS Server uses the provider codes to link the exam to the Merge PACS users so that those users can have access to exams.

CAUTION: In order for the link to function correctly, the provider codes must be entered precisely and corresponds to the provider codes that are sent from the RIS.

- If this user should appear in the "providers" drop down lists for in the PACS Worklist Definition Page, select the **Display on Worklist Definition Page**. Refer to Chapter 13 below for additional information on this option.
- Click on the **Enabled** option to give this user access to the system as a whole. Note that the user's group privileges, as described in Chapter 11 below, will determine his or her access to specific areas and features.

- When you have entered all the necessary information, click on the **Add User** button to save the information and create an account for this user. The Manage Users page will now display a listing for this user and provide additional options for editing his or her account information, as shown in the following example:

The screenshot shows the 'Manage Users' interface. At the top, there are links for 'Add New' and 'Search'. Below this is a table with columns: ID, Login, Name, and Status. A single row is displayed with the following data: ID 45, Login imedico, Name Dr Ima Real Medico, MD, and Status Enabled. Below the table, there are two tabs: 'User Information' (selected) and 'Group Membership'. Under the 'User Information' tab, the following details are shown: Login Name: imedico (with a 'Change Password' button), Name: Dr Ima Real Medico, MD, Address: 100 Main St., Melrose, MA 02176, Phone: 781-555-1234, Fax: (empty), Email: imedico@hospital.com, Email 2: (empty), and Email 3: (empty). A 'Change Account Information' button is located at the bottom right of the form.

Figure 181: User Has Been Successfully Added

At the Manage Users page for this specific user, you can view and/or modify the user's account information, group membership, Credentials and Licensing information, and/or covering physicians, as described below.

b. Searching for an Existing User

To search for an existing user's account, click on the **Search** link at the top of the Manage Users page to bring up the Search for Users page, as shown in the following example:

The screenshot shows the 'Search for Users' interface. At the top, there are links for 'Add New', 'Search' (highlighted with a red box), and 'Reset locked out users (0)'. Below this, there is a dropdown menu labeled 'Users:' with 'Enabled Users' selected. To the right of the dropdown is a text input field labeled 'Name or Login:' and a 'Find' button.

Figure 182: Searching for an Existing User

NOTE: By default, the Search for Users page will be displayed when you first access the Manage Users page.

- From the pull-down **Users** menu, select the type of user you wish to search for, as in the following example:

Figure 183: Selecting the Type of User

- The following types of users can be searched:

User	Description
All Local Users	All users currently in the Merge PACS database
Enabled Local Users	Users in the Merge PACS database who have access to the system because the “Enabled” option is selected on their individual user account page
Disabled Local Users	Users in the Merge PACS database who do not currently have access to the system because the “Enabled” option is not selected on their individual user account page
All Mapped LDAP Users	Users on LDAP servers who are currently in an LDAP group that is currently mapped to a Merge PACS group.
All Unmapped LDAP Users	Users on LDAP servers who are not currently in an LDAP group that is currently mapped to a Merge PACS group.

- If LDAP authentication has been enabled, the following additional types of users can be searched:

User	Description
All Mapped LDAP Users	Users on LDAP servers who are currently in an LDAP group that is currently mapped to a Merge PACS group.
All Unmapped LDAP Users	Users on LDAP servers who are not currently in an LDAP group that is currently mapped to a Merge PACS group.

- If LDAP authentication has been enabled and you are searching for **Mapped** or **Unmapped LDAP Users**, select the LDAP domain you want to search from the drop-down **Domain** menu.
- Once you have selected a type of user, enter all or part of the user’s name or login name and click the **Find** button to generate a list of matching users, as in the following example:

Manage Users [Add New](#) [Search](#) [Reset locked out users \(0\)](#)

Users: Name or Login:

#	ID	Login	Name	User Status
1	45	imedico	Dr Ima Real Medico, MD	Enabled
2	46	ymedico	Dr Yura Great Medico, MD	Enabled

Figure 184: List of Matching Users

NOTE: Enter the first letter of the user's name or login to display all users starting with that letter.

NOTE: You can also leave the **Name or login** field blank to display a list of all users of the selected type.

- Click the **Login** for the desired user to bring up that user's individual account page, as in the following example:

Manage Users [Add New](#) [Search](#)

ID	Login	Name	Status
27	imedico	Dr. Ima Medico, MD	Enabled

User Information **Group Membership**

Login Name: imedico

Name: Dr. Ima Medico, MD
 Address: 100 Main St,
 Melrose, MA 02176
 Phone: 7815551234
 Fax:
 Email:
 Email 2:
 Email 3:

Figure 185: Manage This User's Account

At the Manage Users page for this specific user, you can view and/or modify the user's account information, group membership, Credentials and Licensing information, and/or covering physicians, as described below.

c. Editing a User’s Account Information

As described in the preceding subsections, once you have added or searched for a particular user, the Manage Users page for that user’s account will be displayed as in the following example:

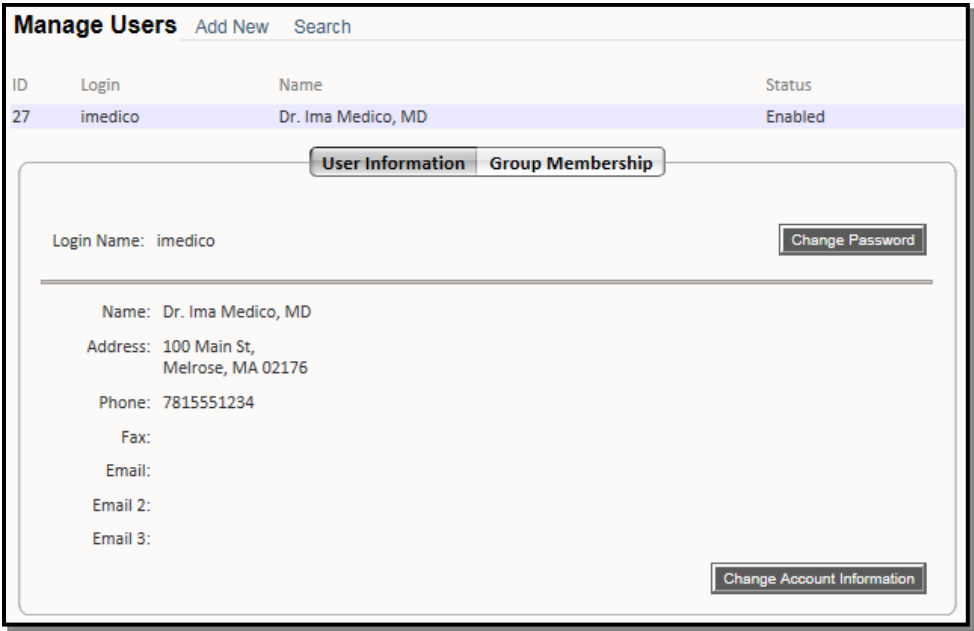


Figure 186: Manage This User's Account

The top of the Manage Users page contains various tabs to access different sections, as shown in the following example:

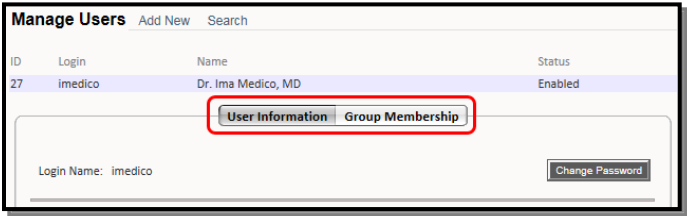


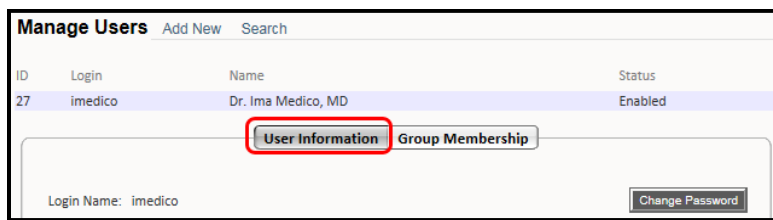
Figure 187: Manage Users Tabs

The following tabs are available, each of which is described in further detail below:

Tab	Description
User Information	View and/or edit the general account information for this user, including name, e-mail address, phone number, etc., as well as change the user’s password.
Group Membership	View and edit the list of groups to which this user belongs.

d. User Information

Click on the **User Information** tab to access the User Information section, as shown in the following example:



The screenshot shows the 'Manage Users' interface. At the top, there are links for 'Add New' and 'Search'. Below is a table with columns: ID, Login, Name, and Status. A single row is visible with ID 27, Login 'imedico', Name 'Dr. Ima Medico, MD', and Status 'Enabled'. Below the table, there are two tabs: 'User Information' (which is highlighted with a red box) and 'Group Membership'. At the bottom left, it says 'Login Name: imedico'. At the bottom right, there is a 'Change Password' button.

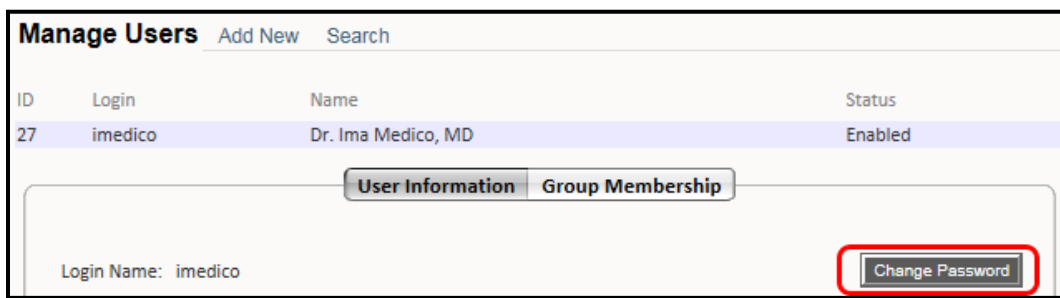
Figure 188: Managing User Information

NOTE: The User Information section will be displayed by default whenever you first access the Manage Users page.

The User Information section allows you to view and/or edit the general account information for this user, including name, e-mail address, phone number, etc., as well as change the user's password.

NOTE: Additional user information for use with Advanced Workflow and the Launch & Control feature, such as Credentials & Licensing, availability for auto-assignment of exams, practices, etc., can be defined as part of the user's Advanced Workflow Information, as described in Section 12.4 below,

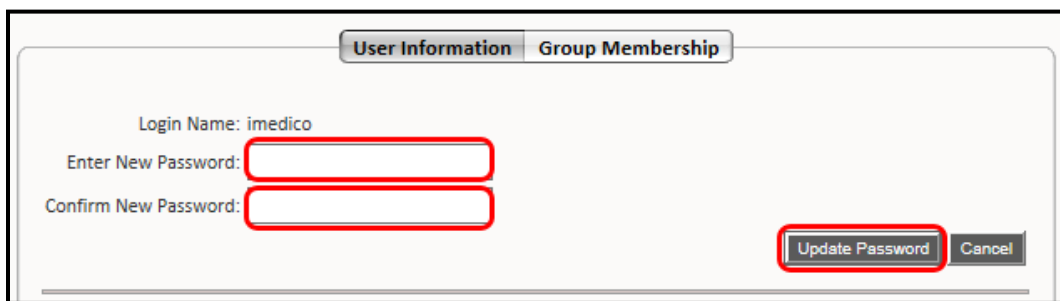
- To change this user's password, click the **Change Password** button, as shown in the following example:



This screenshot is similar to Figure 188, showing the 'Manage Users' page with the 'User Information' tab selected. However, the 'Change Password' button at the bottom right is highlighted with a red box.

Figure 189: The Change Password Button

The **Change Password** section will be displayed, as in the following example:



The screenshot shows the 'Change Password' section. At the top, there are two tabs: 'User Information' and 'Group Membership'. Below the tabs, the 'Login Name: imedico' is displayed. There are two input fields: 'Enter New Password:' and 'Confirm New Password:'. Both input fields are highlighted with red boxes. At the bottom right, there are two buttons: 'Update Password' (highlighted with a red box) and 'Cancel'.

Figure 190: Changing a User's Password

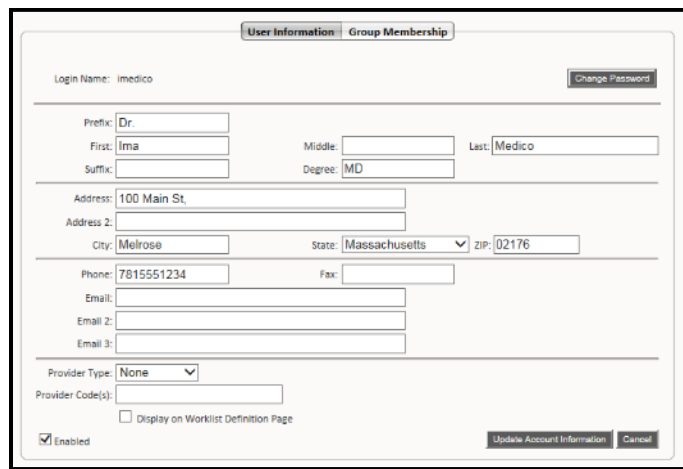
- Enter and confirm the new password in the fields provided (note that passwords can be any length but are case-sensitive), and click the **Update Password** button to save the change.
- Alternatively, you can click the **Cancel** button (next to the Update Password button) to return to the previous screen without making any changes.
- To change general account information for this user, click the **Change Account Information** button, as shown in the following example:



The screenshot shows a user profile for 'Dr. Ima Medico, MD'. The fields displayed are: Name, Address (100 Main Street, Anytown, MA 02176), Phone (617-555-5555), Fax, Email (imedico@anytownclinic.com), Email 2, and Email 3. A button labeled 'Change Account Information' is highlighted with a red rectangle in the bottom right corner.

Figure 191: The Change Account Information Button

The **Change Account Information** section will be displayed, as in the following example:



The screenshot shows the 'Change Account Information' form. It has tabs for 'User Information' and 'Group Membership'. The 'User Information' tab is active. The form contains fields for: Login Name (imedico), Prefix (Dr.), First (Ima), Middle, Last (Medico), Suffix, Degree (MD), Address (100 Main St), Address 2, City (Melrose), State (Massachusetts), Zip (02176), Phone (7815551234), Fax, Email, Email 2, Email 3, Provider Type (None), Provider Code(s), and a checkbox for 'Display on Worklist Definition Page'. There are buttons for 'Update Account Information' and 'Cancel' at the bottom right.

Figure 192: Changing a User's Account Information

- Make any changes in the appropriate fields and click the **Update Account Information** button to save the changes. Note that if the user has more than one provider code, you can enter them all in the Provider Code field separated by commas (e.g., 10001, 10002).
- Alternatively, you can click the **Cancel** button (next to the Update Account Information button) to return to the previous screen without making any changes.

NOTE: For LDAP Users, when Merge PACS is configured to run in “LDAP” or “MIXED” mode, only information that is not provided by the LDAP server will be editable here, including **Suffix**, **Degree**, **Email 2**, **Email 3**, and **Provider Codes**. All other fields will be disabled and the information will need to be edited directly within LDAP.

e. Group Membership

Click on the **Group Membership** tab to access the Group Membership section, as shown in the following examples:

Figure 193: Managing Group Membership in “DB” Mode

Figure 194: Viewing Group Membership in “LDAP” or “MIXED” Mode

When Merge PACS is configured to run in “DB” mode (as opposed to “LDAP” or “MIXED” mode), the Group Membership section allows you to view the groups to which this user currently belongs and edit that group membership. Members of a group can all be given the same general access privileges and default preferences, can cover for other members of the group, and can share access to exams with other members of the group.

- The bottom section of the Group Membership tab displays a list of the groups to which this member currently belongs, as shown in the following example:

Figure 195: This User's Current Group Membership

Click the **Remove** button to the right of a particular group’s description to remove the user from this group (the Remove button will not be visible for LDAP users).

- To add a user to a new group, enter all or part of the group’s name in the “Find Groups” field and then click the **Find Groups** button, as in the following example:

Figure 196: Searching for Groups

NOTE: You can also leave the **Find Groups** field blank to display a list of all available groups.

Click the check box next to the name of the desired group or groups, and then click the **Add User to Selected Groups** button.

- To create a new group for a user (to which other users can be added later), click the **Add New Group** link to display the Add New Group section as in the following example:

Figure 197: Creating a New Group

- Enter a brief name for the group as well a description in the appropriate fields.
- If for any reason you want to create this group without adding the current user to it, deselect the “Add user... to this group” checkbox.
- When you are ready, click the **Create New Group** button to create the group.

NOTE: You can edit this group’s properties, including and members of the group, from the Manage Groups page, as described in subsection f below.

When Merge PACS is configured to run in “LDAP” or “MIXED” mode, the Group Membership section will simply display the group(s) to which this user currently belongs.

f. Resetting Locked Out Users

By default, any local database (as opposed to LDAP) users who attempt to log into Merge PACS, whether the Workstation or the Management Pages, will be locked out after 10 failed attempts, although the lockout feature and the number of failed login attempts allowed before lockout occurs is configurable on a site-by-site basis. If any users are currently locked out, the number of such users will be displayed at the top of the main Manage Users page in parentheses next to the “Reset locked out users” link, as in the following example:

Figure 198: One Locked Out User

Clicking on the “Reset locked out users” link will cause all locked out users to become unlocked. Note the following:

- Clicking the link will not cause a separate page to be displayed.

- Clicking the link will cause **all** locked users to become unlocked. There is no way to see which users are currently locked or select individual users to become unlocked.

9.11.2. Manage Groups Page

The Manage Groups page allows you to create a new group, edit the general description of existing groups, manage the members of each group, manage LDAP groups mapped to each group (if applicable), manage the worklists associated with each group (if any), configure Viewer XML integrations used by this each group (if any), specify the parent of each group for use with Templated user Management (as described in Chapter 11 below) and configure special privileges for each group (such as Merge Instant Messenger and Study Assignment privileges). You can also configure the relative priorities between all existing groups for use with Templated User Management.

The manage Groups page is accessed by clicking on the **Groups** link as shown in the following example:

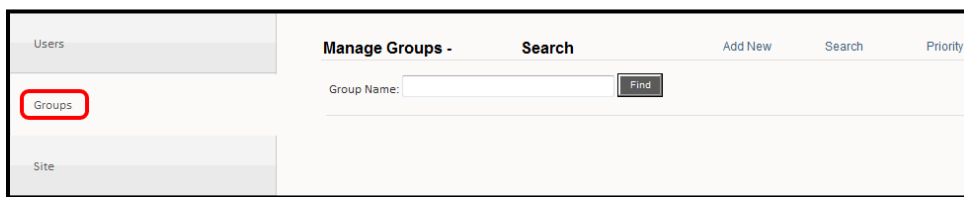


Figure 199: The Manage Groups Page

When you first access the Manage Groups page, you will need to choose to add a new user group, locate an existing group or set the priority for all groups. By default, the Manage Groups page will present you with the **Search** option, but you can access the other options at any time from the top of the page, as shown in the following example:

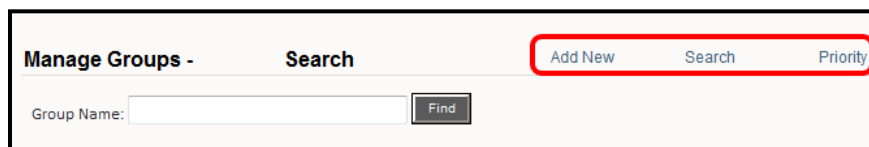


Figure 200: Manage Groups Options

For new installations of Merge PACS, the following pre-defined groups are provided by default, each with their parent set to “site” and with their priority set as shown below:

Group	Priority
Administrators	2
ER Physicians	4
Guests	10
Merge Support	1
Radiologists	3
Referring Clinicians	7
Surgeons	5
Technologists	6

Group	Priority
Telmed Service	8
Watch Admin	12

NOTE: Prior to 7.0, the “Merge Support” group was referred to as Super Users.

a. Adding a New Group

To add a new user group from scratch, click on the **Add New** link at the top of the Manage Groups page to bring up the **Add New Groups** page, as shown in the following example:

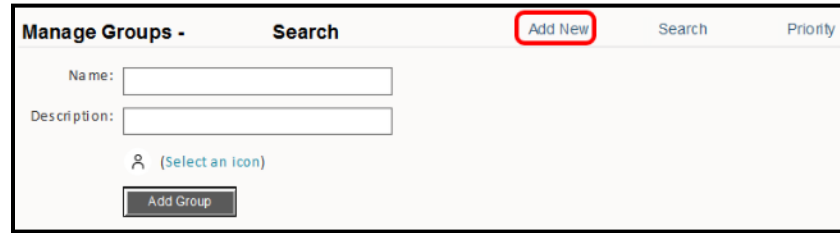


Figure 201: Adding a New Group

- Enter a brief **name** and **description** for this group in the fields provided.
- If you want to change the default icon associated with the group (for display on the ER Workpanel within the Workstation), click the **Select an icon** link.

The **Define Group Icons** pop-up screen is displayed, as in the following example:

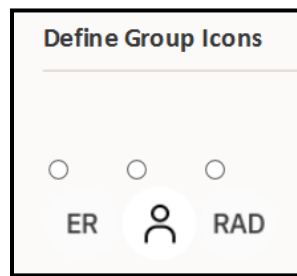


Figure 202: Selecting a Group Icon

Click the radio button for the desired icon.

NOTE: If desired, a site can provide other icon options by storing additional icons (in .png or .jpg format) in the following directory on the Server: **C:\amicas2\wildfly25\standalone\deployments\amicas.war\images\rtwl\groupicons**

- When finished, click the **Add Group** button, as in the following example:

The screenshot shows the 'Manage Groups' form. At the top, there are tabs for 'Add New', 'Search', and 'Priority'. Below these, there are input fields for 'Name' (containing 'City Clinic') and 'Description' (containing 'City Clinic Referring Physicians'). There is also a 'Group Icon' selection area with a person icon and the text '(Select an icon)'. At the bottom, the 'Add Group' button is highlighted with a red rectangular box.

Figure 203: Adding a New Group

- The Manage Groups page will now display a listing for this group and provide additional options for editing its information, as shown in the following example:

The screenshot shows the 'Manage Groups' page. At the top, there are tabs for 'Add New', 'Search', and 'Priority'. Below these, there is a table with the following data:

ID	Group Name	Description	Group Icon
21	City Clinic	City Clinic Referring Physicians	(Select an icon)

Below the table, there is a detailed view of the group information. It includes tabs for 'Group Information', 'Members', 'LDAP Groups', 'Worklist', 'Viewer XML Integration', 'Parent', and 'Special Privileges'. The 'Group Information' tab is selected, showing the 'Name' (City Clinic), 'Description' (City Clinic Referring Physicians), and 'Group Icon' (Select an icon) fields. A 'Submit Changes' button is at the bottom.

Figure 204: Newly Added Group

At the Manage Groups page for this specific user group, you can view and/or modify the group's account information, group membership, LDAP group mapping, XML integration, parent information and special privileges, as described below.

b. Searching for an Existing Group

To search for an existing user group, click on the **Search** link at the top of the Manage Groups page to bring up the Search for Groups page, as shown in the following example:

The screenshot shows the 'Search' page for groups. At the top, there are tabs for 'Add New', 'Search', and 'Priority'. The 'Search' tab is selected and highlighted with a red rectangular box. Below the tabs, there is a 'Group Name' input field and a 'Find' button.

Figure 205: Searching for an Existing User Group

NOTE: By default, the Search for Groups page will be displayed when you first access the Manage Groups page.

- Enter all or part of the group's name and click the **Find** button to generate a list of matching groups, as in the following example:

ID	Group Name	Description	Group Icon	Priority
21	City Clinic	City Clinic Referring Physicians		10
8	Referring Clinicians	Referring Clinicians		7

Figure 206: List of Matching User Groups

NOTE: Enter the first letter of the group's name to display all groups starting with that letter.

NOTE: You can also leave the **Group Name** field blank to display a list of all available groups.

- Click the **Group Name** for the desired user group to bring up that group's account page, as shown in the following example:

ID	Group Name	Description	Group Icon
21	City Clinic	City Clinic Referring Physicians	

Group Information | Members | LDAP Groups | Worklist | Viewer XML Integration | Parent | Special Privileges

Name:

Description:

Group Icon: (Select an icon)

Figure 207: Manage This Group's Account Information

At the Manage Groups page for this specific user group, you can view and/or modify the group's account information, group membership, LDAP group mapping, XML integration, parent information and special privileges, as described below.

c. Editing a Group’s Account Information

As described in the preceding subsections, once you have added or searched for a particular user group, the Manage Groups page for that group will be displayed as in the following example:

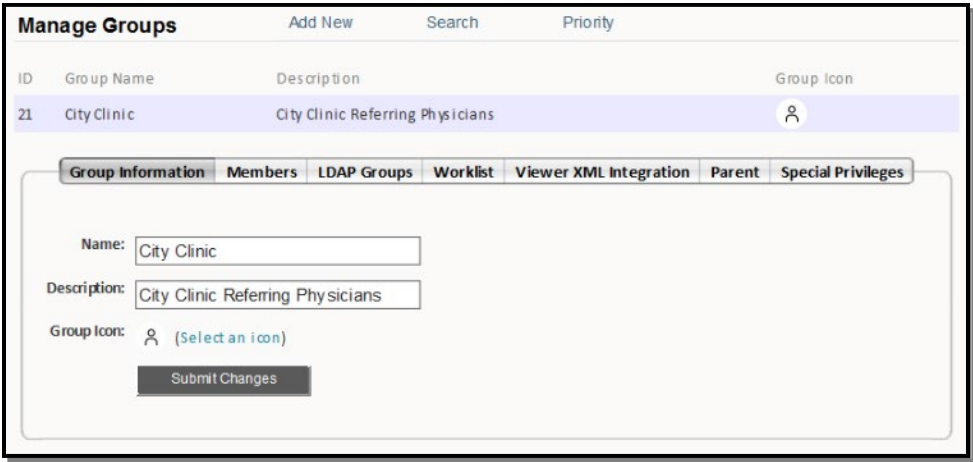


Figure 208: Manage This Group’s Information and Privileges

The top of the Manage Groups page contains various tabs to access different sections, as shown in the following example:

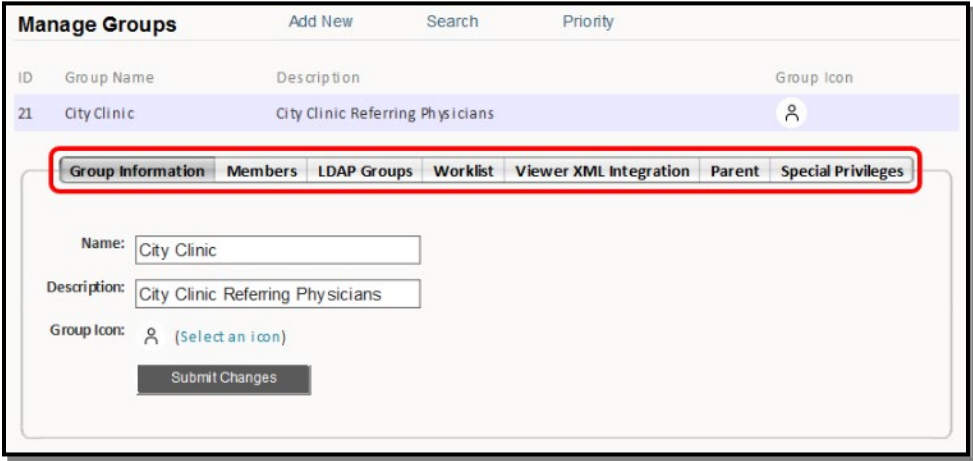


Figure 209: Manage Groups Tabs

The following tabs are available, each of which is described in further detail below:

Tab	Description
Group Information	View and change the name and/or description for this group, as well as whether members of this group as a whole should be enabled or not.
Members	View and/or edit the list of users who belong to this group.
LDAP Groups	Map this group to an existing group on an external LDAP server against which Merge PACS is authenticating, if any.
Worklist	View and/or edit the list of worklists currently associated with this group.

Tab	Description
Viewer XML Integration	View and/or edit the list of third-party applications that this group currently has access to via XML file-based integration.
Parent	Specify the parent of this group for use with Templated user Management, as described in Chapter 11 below.
Special Privileges	Provide members of this group with special privileges, such as access to the Instant Messenger feature within the Merge PACS Workstation and the ability to be assigned studies by other users.

d. Group Information

Click on the **Group Information** tab to access the Group Information section, as shown in the following example:

The screenshot shows the 'Manage Groups' interface. At the top, there are buttons for 'Add New', 'Search', and 'Priority'. Below this is a table with columns 'ID', 'Group Name', 'Description', and 'Group Icon'. A row is visible with ID '21', Group Name 'City Clinic', and Description 'City Clinic Referring Physicians'. Below the table is a tabbed interface with tabs: 'Group Information' (highlighted with a red box), 'Members', 'LDAP Groups', 'Worklist', 'Viewer XML Integration', 'Parent', and 'Special Privileges'. The 'Group Information' tab is active, showing fields for 'Name' (City Clinic), 'Description' (City Clinic Referring Physicians), and 'Group Icon' (a person icon with a link to 'Select an icon'). A 'Submit Changes' button is at the bottom.

Figure 210: Managing Group Information

NOTE: The Group Information section will be displayed by default whenever you first access the Manage Groups page.

The Group Information section allows you to view and/or edit the name and/or description for this group.

- To change the name or description of this user group, enter the desired information in the appropriate fields.
- Click the **Submit Changes** button to save the changes.

e. Members

Click on the **Members** tab to access the Group Membership section, as shown in the following example:



Figure 211: Managing Group Membership

The Members section displays a list of current members of this group, if any, and allows you to add and/or remove members.

NOTE: You cannot add or remove LDAP users from a group when Merge PACS is configured to run in “LDAP” or “MIXED” mode.

NOTE: When Merge PACS is configured to run in “LDAP” or “MIXED” mode, only LDAP users who have logged into Merge PACS at least once (thereby automatically creating a Merge PACS user account) will be listed. In addition, the list of LDAP users will only include those LDAP users with Merge PACS accounts that are still a member of an LDAP group mapped to this Merge PACS group.

- To add a new member to this group, click on the **Add Members** link to display the Add Members section, as in the following example:

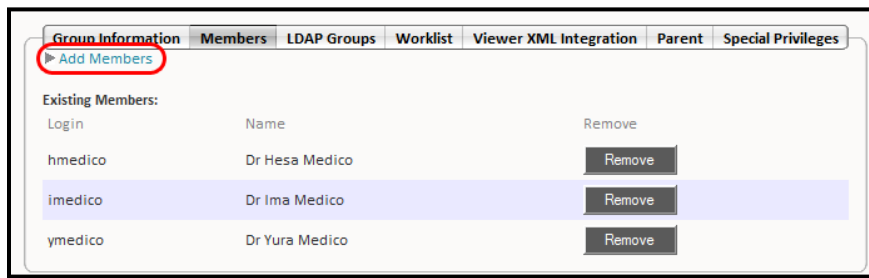


Figure 212: Adding Members to this Group

- Select a type of user from the pull-down Users menu (**All Local Users** or **Pending Local Users**), enter all or part of the user's Name or Login, and then click the **Find** button to generate a list of matching user as in the following example:

The screenshot shows the 'Members' tab of the Merge PACS 8.2 Administration interface. The 'Users' dropdown menu is set to 'All Local Users'. The 'Name' field contains the text 'medico'. The 'Find' button is highlighted with a red box. Below the search fields, the 'Existing Members' section displays a table with three rows of user information. Each row has a 'Remove' button next to it.

#	Login	Name	Remove
1	imedico	Ima Medico	Remove
2	smedico	Shesa Medico	Remove
3	ymedico	Yura Medico	Remove

Figure 213: Searching for Users to Add

A list of matching members will be displayed.

- Click the checkboxes next to the users you wish to add to this group to select them, and then click the **Add Selected Users to Group** button at the bottom of the list to add these users to the group.
- If this group currently has members, you can remove one or more members by clicking the **Remove** button next to each member's name, as shown in the following example:

The screenshot shows the 'Members' tab of the Merge PACS 8.2 Administration interface. The 'Users' dropdown menu is set to 'All Users'. The 'Name' field is empty. The 'Find' button is highlighted with a red box. Below the search fields, the 'Existing Members' section displays a table with three rows of user information. Each row has a 'Remove' button next to it, which is also highlighted with a red box.

Login	Name	Remove
hmedico	Dr Hesa Medico	Remove
imedico	Dr Ima Medico	Remove
ymedico	Dr Yura Medico	Remove

Figure 214: Removing Users from a Group

f. LDAP Groups

If Merge PACS has been configured to authenticate against one or more external LDAP servers, click on the **LDAP Groups** to access the LDAP Groups section, as shown in the following example:

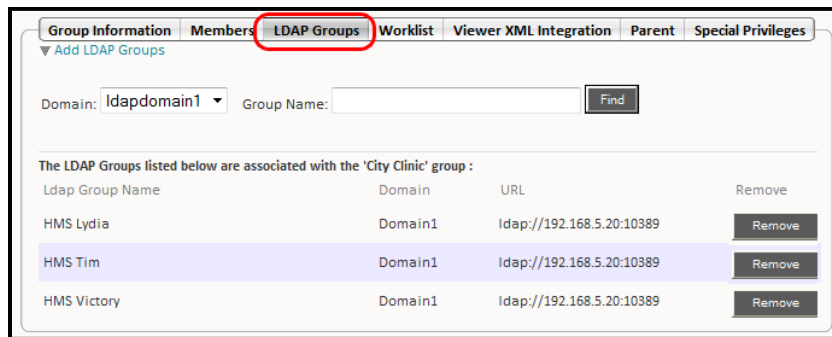


Figure 215: Managing LDAP Groups

The LDAP Groups section allows you to associate this group with one or more LDAP Groups that exist on one or more of the external LDAP Servers.

NOTE: Before you can associate a Merge PACS group with an LDAP group, you must first follow the procedures described in subsection 9.13.1 below to add one or more LDAP servers to the list of LDAP servers against which Merge PACS will be authenticating.

- To associate this Merge PACS group with a new LDAP group, click on the **Add LDAP Groups** link at the top of the section to display the LDAP Group query fields, as in the following example:

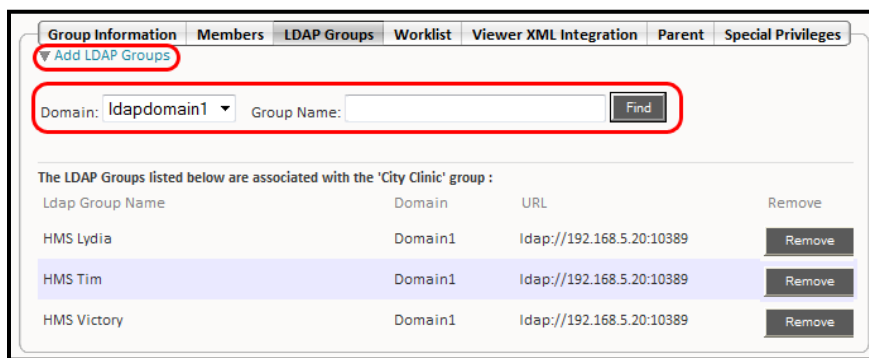


Figure 216: Adding LDAP Groups

NOTE: If the LDAP Group query fields are displayed by default, clicking the link will cause the fields to be hidden.

- Select the desired LDAP Domain from the drop-down **Domain** menu.
- Enter all or part of the desired group's name in the **Group Name** field.

- Click on the **Find** button to display a list of matching LDAP groups, as in the following example:

Group Information Members **LDAP Groups** Worklist Viewer XML Integration Parent Special Privileges

▼ Add LDAP Groups

Domain: Idapdomain1 Group Name: Radiologists **Find**

	Ldap Group Name	Domain	URL
☐	Radiologists1	Domain1	Idap://192.168.5.20:10389
☐	Radiologists2	Domain1	Idap://192.168.5.20:10389
☐	Radiologists3	Domain1	Idap://192.168.5.20:10389

Associate to pacs1 Group

Figure 217: Matching LDAP Groups

- Select one or more LDAP groups by clicking on the checkboxes and then click on the **Associate to Group** button.
- To disassociate this Merge PACS group with one or more LDAP groups to which it is currently associated, click on the **Remove** button for the LDAP group in question, as shown in the following example:

Group Information Members **LDAP Groups** Worklist Viewer XML Integration Parent Special Privileges

▼ Add LDAP Groups

Domain: Idapdomain1 Group Name: **Find**

The LDAP Groups listed below are associated with the 'City Clinic' group :

Ldap Group Name	Domain	URL	Remove
HMS Lydia	Domain1	Idap://192.168.5.20:10389	Remove
HMS Tim	Domain1	Idap://192.168.5.20:10389	Remove
HMS Victory	Domain1	Idap://192.168.5.20:10389	Remove

Figure 218: Removing LDAP Groups

g. Worklist

Click on the **Worklist** tab to access the Worklist section, as shown in the following example:

Manage Groups Add New Search Priority

Group Name Description

City Clinic City Clinic Referring Physicians

Group Information Members **Worklist** Viewer XML Integration Parent Special Privileges

► Add Worklists

Worklists: **Find**

Figure 219: Managing Group Worklists

The Worklist section displays a list of worklists currently associated with this group, if any, and allows you to associate or dissociate worklists.

NOTE: A worklist must first be defined, as described in Chapter 13 below, before it can be associated with a group.

- To associate a new worklist with this group, click on the **Add Worklists** link to display the Add Worklists section, as in the following example:

Figure 220: Adding Worklists to this Group

NOTE: The **Add Worklist** link will be selected by default when you click on the **Worklist** tab.

- Enter all or part of a worklist name in the field provided and then click the **Find** button to generate a list of matching worklists as in the following example:

Figure 221: Searching for Worklists to Add

NOTE: You can also leave the **Worklist** field blank to display a list of all available worklists.

- Click the checkbox next to the worklists you wish to associate with this group to select them, and then click the **Associate Selected Worklists to Group** button at the bottom of the screen to associate these worklists with this group.
- If this group currently has worklists associated with it, you can remove one or more of them by clicking the **Remove** button next to each worklist's name, as shown in the following example:

Figure 222: Removing Worklists from a Group

h. Viewer XML Integration

Click on the **Viewer XML Integration** tab to access the Viewer XML Integration section, as shown in the following example:

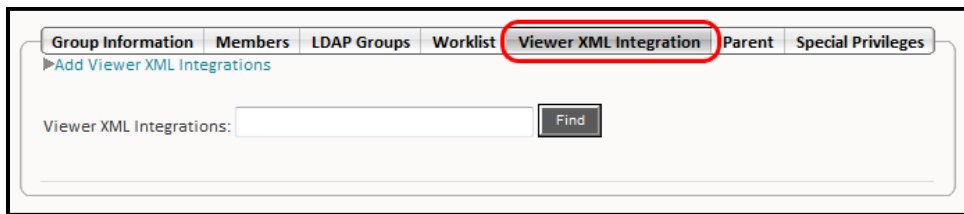


Figure 223: Managing Group Viewer XML Integration Options

The Viewer XML Integration section displays a list of third-party applications that this group currently has access to via XML file-based integration, if any, and allows you to add additional applications for this group.

NOTE: Merge PACS must be configured to provide XML file-based integration with a third-party application before XML integration for that application can be associated with a group.

NOTE: If your site is configured to just launch 3D images into iConnect Access from within the Viewer, there is no need to add iConnect Access to the list of Viewer XML Integrations for any group.

- To associate a new xml integration with this group, click on the **Add Viewer XML Integrations** link to display the Add Worklists section, as in the following example:

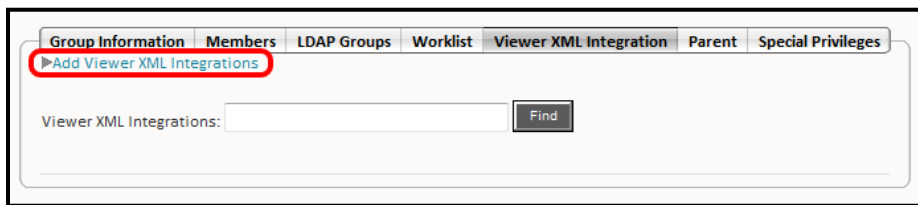


Figure 224: Adding XML Integrations to this Group

NOTE: The **Add Viewer XML Integration** link will be selected by default when you click on the **Viewer XML Integration** tab.

- Enter all or part of a XML integration's name in the field provided and then click the **Find** button, as in the following example:

Group Information Members LDAP Groups Worklist **Viewer XML Integration** Parent Special Privileges

►Add Viewer XML Integrations

Viewer XML Integrations: Epic **Find**

Application Name

☐	Epic
--------------------------	------

Associate Selected Viewer XML Integrations to Group

Figure 225: Associating XML Integrations with this Group

- Click the checkbox next to the XML Integrations you wish to add to this group to select them.
- Click the **Associate Selected Viewer XML Integrations to Group** button at the bottom of the screen to associate these XML integrations with the group.
- If this group currently has XML Integrations associated with it, you can remove one or more of them by clicking the appropriate **Remove** button, as shown in the following example:

Group Information Members LDAP Groups Worklist **Viewer XML Integration** Parent Special Privileges

►Add Viewer XML Integrations

Associated Viewer XML Integrations:

Application Name	Remove
Epic	Remove
OnBase	Remove

Figure 226: Removing XML Integrations from a Group

i. Parent

When setting up Templated User Management, as described in Chapter 11 below, each group inherits its privileges and default preferences from its “parent” (*i.e.*, the level above it in the hierarchy). Each group must have a parent defined, and this is done by clicking on the **Parent** tab, as in the following example:

Name	Description
☒ Home	Home Site
☐ Administrators	Administrators
☐ ChristyTest	ChristyTest
☐ ER Physicians	ER Physicians
☐ Guests	Guests
☐ Merge Support	Merge Support
☐ Radiologists	Radiologists
☐ Reach Users	Reach Users
☐ Referring Clinicians	Referring Clinicians
☐ Surgeons	Surgeons
☐ Technologists	Technologists

Figure 227: Defining the Parent of a Group

- By default, each group is automatically assigned the main site as its parent.
- If desired, you can filter the list of available groups by entering all or part of a group’s name in the **Find Groups** field and clicking on the **Search** button.
- To select a parent for this group, click on the radio button for the desired parent and then click the **Save** button

j. Special Privileges

Click on the **Special Privileges** tab to access the Special Privileges section, as shown in the following example:

☐ Enable Instant Messenger
☐ Allow Group to be added for Study Assignment
 Save Changes

Figure 228: Managing Special Group Privileges

- To provide all members of this group with the ability to use the Instant Messenger feature within the Merge PACS Workstation, click on the **Enable Instant Messenger** checkbox.

- To allow all members of this group to be assigned studies by other users, click on the **Allow Group to be added for Study Assignment**
- When finished, click on the **Save Changes** button to save your choices.

NOTE: Both of these privileges are **Group** level privileges only. In order for an individual user to have either privilege, that user must belong to a group with this privilege. Similarly, if a user is in a group that has one or both of these privileges, the only way to restrict that user's access would be to remove the user from the group.

NOTE: Users in this group will need to log out and log back into the Merge PACS Workstation for any changes to their group's special privileges to take effect.

k. Priority

As described in Chapter 11 below, users inherit their default preferences from the groups to which they belong. Since users can belong to multiple groups, it is possible for there to be a conflict between the preferences of one group and those of another group. To resolve these conflicts, each group is assigned a priority and the preference from the group with the highest priority will prevail.

To configure the priorities of all existing groups, click on the **Priority** link at the top of the Manage Groups page to bring up the **Priority** page, as shown in the following example:

ID	Group Name	Description	Group Icon
7	Radiologists	Radiologists	RAD
1	Merge Support	Merge Support	person icon
2	Administrators	Administrators	person icon
6	ER Physicians	ER Physicians	ER
9	Surgeons	Surgeons	person icon
3	Technologists	Technologists	person icon
8	Referring Clinicians	Referring Clinicians	person icon
4	Read Users	Read Users	person icon
5	Guests	Guests	person icon
21	City Clinic	City Clinic Referring Physicians	person icon

Figure 229: Configuring Group Priorities

Groups are listed according to their current priority, with the highest priority group at the top of the page and the lowest priority group at the bottom of the page.

To change the priority of a group relative to the other groups, click on the group once to highlight it and then use the buttons on the right to move the group up or down the list of groups, as in the following example:

ID	Group Name	Description	Group Icon
7	Radiologists	Radiologists	RAD
1	Merge Support	Merge Support	
2	Administrators	Administrators	
6	ER Physicians	ER Physicians	ER
9	Surgeons	Surgeons	
3	Technologists	Technologists	
8	Referring Clinicians	Referring Clinicians	

Figure 230: Changing a Group’s Priority

9.11.3.Manage Site Page

The Manage Site page allows you to configure the name and general description of your site and is accessed by clicking on the **Site** link as shown in the following example:

Users

Groups

Site

Manage Site

Name:

Home

Description:

Home Site

Submit Changes

Figure 231: The Manage Site Page

NOTE: This information is not currently used elsewhere in the System and is reserved for future use.

9.12. Services

The **Services** page allows you to stop or restart any of the main services used by Merge PACS, as shown below:

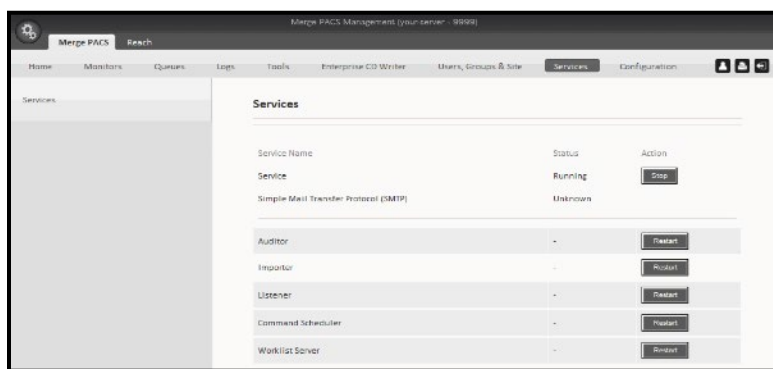


Figure 232: The Services Page

NOTE: The main Amicas service is listed here simply as “Service.”

9.13. Configuration

The Configuration pages allow you to change the configuration of your Merge PACS server. You might do this to add or remove functionality, to change how a function is implemented, or to improve the efficiency of a Merge PACS.

Clicking on the “Configuration” link on the top category menu will display a submenu of the individual available configuration pages as shown in the example below:

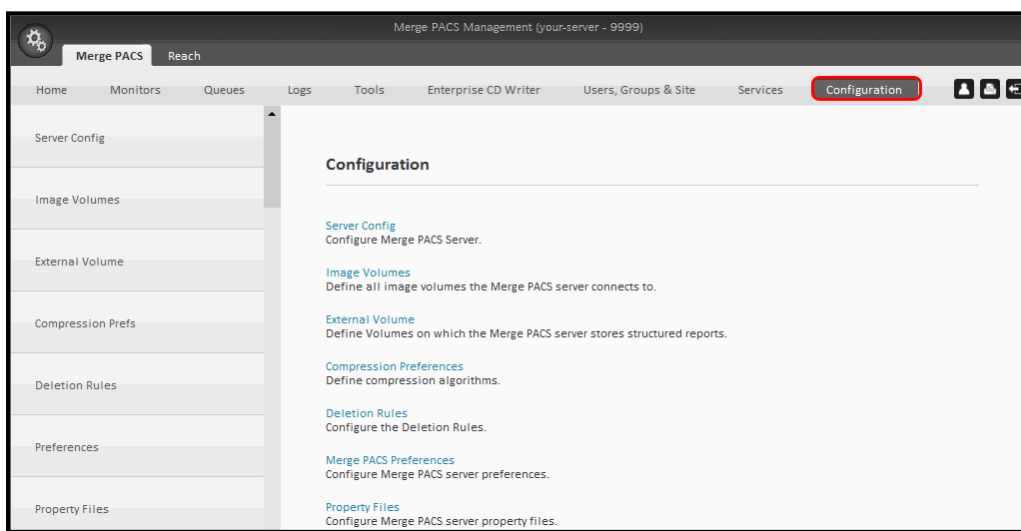


Figure 233: The Configuration Pages

The Configuration Submenu contains links to the following Configuration pages, some of which will only appear if optional features have been enabled:

Configuration Page	Description
LDAP Servers	Define LDAP servers this Merge PACS server connects to.
DICOM Devices	Edit the DICOM targets with which Merge PACS is set up to communicate.
Modalities	Create a master list of modalities, some or all of which can be used to create filters for individual worklists
Procedures	Create a master list of procedures, some or all of which can be used to create filters for individual worklists
Workflow Priorities	Edit existing workflow priority items that can be used to sort individual worklists.
Workflow Status Dimensions	Edit existing or create custom workflow status dimensions
Workflow Status	Edit the master list of workflow status options, some or all of which can be used to create filters for individual worklists
HL7 Order Status Mapping	Define rules for mapping Order Status values obtained from HL7 messages to Viewer display names for use with RealTime Worklist and RealTime Study List.
Credentialing and Licensure	Define Licensure Regions as well as associate Facilities to a specific Licensure Region for use with RealTime Worklist and RealTime Study List.
Locations	Create a master list of locations, some or all of which can be used to create filters for individual worklists
Patient Classes	Edit the master list of patient classes, some or all of which can be used to create filters for individual worklists
Teaching Tags	Manage the teaching tags used with the Teaching Files feature.
IPID Configuration	Configure the default IPID (Issuer of Patient ID) value as well and define the list of available IPID values that can be used as filters for use with RealTime Worklist and RealTime Study List.
Workstation Phone Link	Add, modify or delete phone links for individual Merge PACS Workstations that can be selected from the Merge PACS RadStream Communication Workpanel.
Standard Comments	Add, modify or delete standard comments that can be selected from various places within the Merge PACS Workstation.
Refresh	Specify how often the data on the various pages should be refreshed.
CustomTag Configuration	Map specific DICOM tags to Display Names that can be used as query fields, quick filters and worklist column headers.
Upload Action Button Icons	Upload custom icons that will be added to the list of available icons when editing toolbar action buttons.

In addition, the following configuration pages are available from the **Settings** dialog within the Merge PACS Workstation, as described in Section 10.6 below:

- **Practices**
- **Specialties** (formerly known as “Reading Pools”)
- **Procedures**

- **Orderers**
- **Order/Practice Associations**
- **Orderer/Reader Associations**

To select one of the individual configuration pages, click on the appropriate link in the Configuration submenu, or the link displayed on the main Configuration page.

9.13.1.LDAP Servers

The **LDAP Servers** page allows you to define the LDAP servers the Merge PACS Server connects to and authenticates against, as shown in the example below:

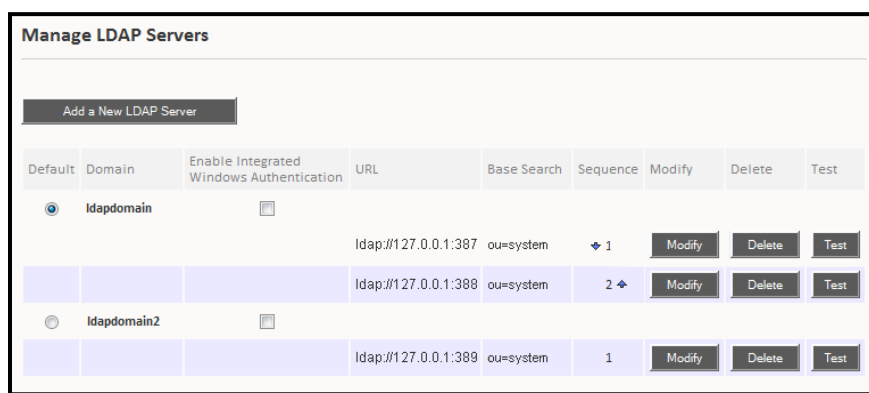


Figure 234: The LDAP ServersPage

- You can configure multiple LDAP domains and multiple LDAP Servers for each domain.
 - LDAP users that have been assigned to multiple domains will be able to select the appropriate domain when logging into the Merge PACS client applications.
 - If a domain has multiple LDAP servers associated with it, Merge PACS will attempt to authenticate the user against the first server listed under the domain on the LDAP Servers page.
 - If the first server is unavailable or the user's account is not found on that server, Merge PACS will attempt to authenticate the user against the next server.
 - If the user authentication fails against the first server (*i.e.*, the user has an account but the password is entered incorrectly), Merge PACS will not attempt to authenticate against additional servers.
- The Manage LDAP Servers page also allows you to configure authentication against **Active Directory (AD)** Servers. AD is a specific type of LDAP implementation created by Microsoft for use with Windows.
- Merge PACS can be configured to just authenticate against LDAP, to authenticate against its own database ("Merge PACS Web Security") or to authenticate against both ("Mixed Mode").

a. Adding a New LDAP Server

To add a new LDAP Server to the LDAP Servers table, click the **Add New LDAP Server** button to display the Add a New LDAP Server screen shown below:

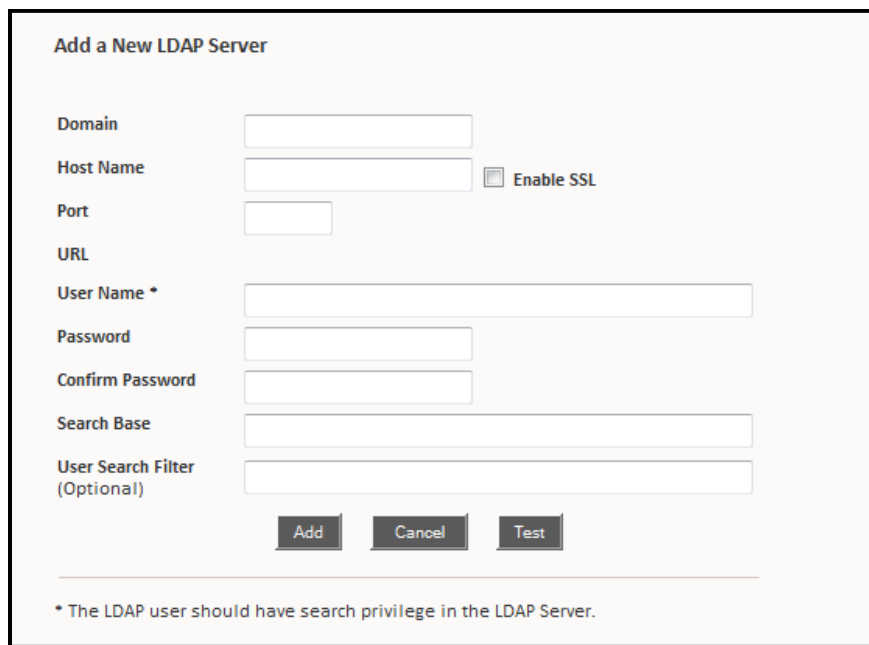


Figure 235: Adding a New LDAP Server

The following information must be correctly filled in:

Field	Description
Domain	The name of the domain to which this LDAP server belongs. Note that multiple LDAP Servers can be listed under the same domain.
Host Name	The IP Address/Fully Qualified Domain Name of the LDAP server.
Enable SSL	Check this box if the LDAP server is configured to support TLS.
Port	The valid port used by the LDAP server
URL	Filled in automatically
User Name	The entire Distinguished Name (DN) of a user on the LDAP server who has search privileges within the defined search base on the LDAP server. This user should be present in the first level of the Base Search tree node so that the user shall be used for initial binding. The User Name is typically entered as a concatenated string that will vary in form depending on whether standard LDAP or AD is being used, as shown in the following examples (do not include the quotation marks): LDAP: "uid=admin,ou=system" AD: "CN=Administrator,CN=Users,DC=addev,DC=internal" Contact your IT department to determine the actual information needed here.
Password/Confirm Password	The password for the User Name described above.

Field	Description
Search Base	The entire DN of the directory node which holds the users and groups that will be used in Merge PACS. The Search Base is typically entered as a concatenated string that will vary in form depending on whether standard LDAP or AD is being used, as shown in the following examples (do not include the quotation marks): • LDAP: "ou=system" • AD: "dc=addev,dc=internal" Contact your IT department to determine the actual information needed here.
User Search Filter	If this field is left blank, the user search filter will search on the following, depending on whether standard LDAP or AD is being used: • LDAP: uid • AD: samAccountName If users are configured on the LDAP/AD server without uid or samAccountName, enter the \$username which will be replaced with the current user name and provide the appropriate result. For example, if cn is used instead of uid/samAccountName, you would enter the following: (& (objectclass=person) (cn=\$username))

Once you have entered all the necessary information, click on the **Test** button to make sure that everything is configured correctly. A message (including the time required for testing) will be displayed as in the following example:

The screenshot shows a web-based configuration interface for LDAP. It has two input fields: "Search Base" with the value "dc=maxcsrc,dc=com" and "User Search Filter (Optional)" which is empty. Below these fields are three buttons: "Save", "Cancel", and "Test". The "Test" button is highlighted with a red border. To the right of the buttons, a red-bordered box contains the message: "*The LDAP Server URL 'ldap://192.168.20.190:389' is connected Successfully (16ms)." At the bottom of the interface, a note states: "* The LDAP user should have search privilege in the LDAP Server."

Figure 236: Testing a New LDAP Server

NOTE: If the time required for testing is greater than 20ms, the Open LDAP/AD Server might be running slow due to the existence of a large number of users/groups. Contact Merge Healthcare Support to have them implement LDAP performance tuning.

If the test is successful, click on the **Add** button to finish the process and return to the main Manage LDAP Servers page.

b. Modifying an Existing LDAP Server

To edit the information for a LDAP server already listed, click the **Modify** button for that server to display the Modify LDAP Server screen shown below:

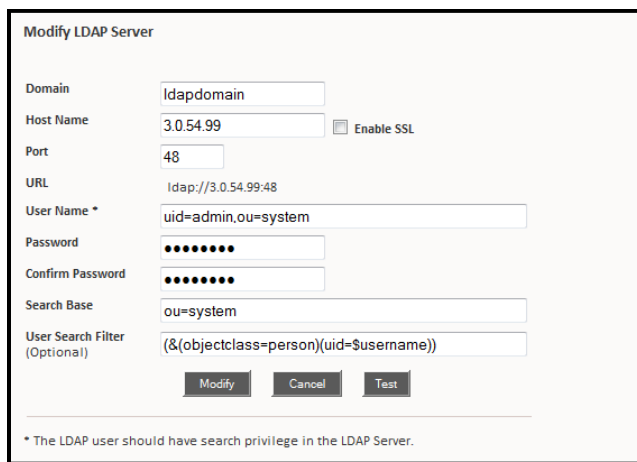


Figure 237: Modifying an Existing LDAP Server

Edit the necessary information for the server and click on the **Modify** button.

c. Deleting an Existing LDAP Server

To delete an LDAP server from the LDAP Servers table, click on the **Delete** button for that Server.

d. Testing an Existing LDAP Server

To validate the configuration of an existing LDAP server, click on the **Test** button for that server. A message (including the time required for testing) will be displayed as in the following example:



Default	Domain	Enable Integrated Windows Authentication	URL	Base Search	Sequence	Modify	Delete	Test
☒	Open LDAP	☐	ldap://192.168.20.18:389	d=maxcdnrc,d=com	1	Modify	Delete	Test
☐	LDAP Domain	☐	ldap://192.168.20.190:389	d=maxcdnrc,d=com	1	Modify	Delete	Test

Figure 238: Testing an Existing LDAP Server

NOTE: If the time required for testing is greater than 20ms, the Open LDAP/AD Server might be running slow due to the existence of a large number of users/groups. Contact Merge Healthcare Support to have them implement LDAP performance tuning.

NOTE: You can also test the configuration of an existing LDAP server by clicking on the **Modify** button and then on the **Test** button on the **Modify LDAP Server** page.

e. Changing the Sequence of Multiple LDAP Servers Under One Domain

As mentioned above, if a domain has multiple LDAP servers associated with it, Merge PACS will attempt to authenticate the user against the first server listed under the domain on the LDAP Servers page. By default, LDAP servers are listed in the order they are added, but you can change the sequence by clicking on the up or down arrows next to the LDAP servers on the main Manage LDAP Servers page, as in the following example:

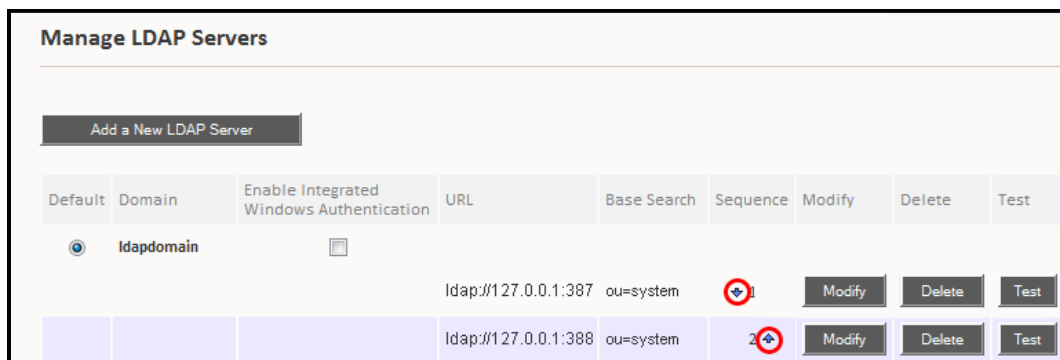


Figure 239: Changing the Sequence of LDAP Servers

f. Enabling Integrated Windows Authentication for an LDAP Server

If you are planning to use LDAP Authentication in conjunction with Integrated Windows Authentication, select the **Enable Integrated Windows Authentication** option for the applicable LDAP domain, as in the following example:

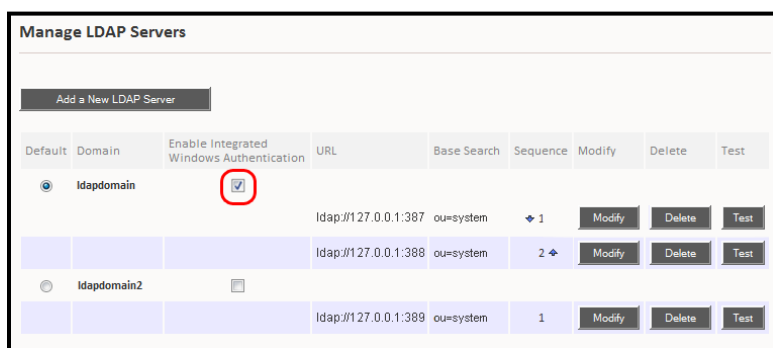


Figure 240: Enabling Integrated Windows Authentication

NOTE: Integrated Windows Authentication can only be enabled for one LDAP domain at a time.

9.13.2.DICOM Devices

When Merge PACS is configured to run in **Standalone** mode, the DICOM Devices page allows you to add or delete DICOM devices that are potential recipients (Targets) of Merge PACS image data, as shown in the example below:

AE Title	Hostname	AE Title Prefix	Port	Station Name	Response Charset	Max Number of Simultaneous Associations	Limit SOP Classes Sent to This Device?	Do Not Send These SOP Classes	Allow Custom Tag Queries	Allow Optional Tag Queries	Allow Special Tag Queries	Allow Extended Search	Modify	Delete	Ping
MRI	MRI.yourhospital.com	XYZ	4006		ASCII	4	NO		NO	NO	NO	YES	Modify	Delete	Ping

Figure 241: The DICOM Devices Page

When Merge PACS is configured to run in **Integrated** mode, the **DICOM Devices** page will simply display the DICOM devices defined within the Enterprise Archive and you will only be able to edit extended node information for a DICOM device. Other DICOM Device fields will be disabled.

Images can be sent to listed DICOM Targets either manually via the **Send Studies** page, described in subsection 9.9.3 above, or as part of a DICOM Query/Retrieve via the **Listener**, described in Section 3.10 above.

DICOM Devices listed on this page can also be used as filters when creating individual worklists, as described in Chapter 13 below.

CAUTION: Merge Healthcare's image display and archive products depend on the DICOM metadata provided by imaging equipment manufacturers to properly mark laterality, orientation, and measures on medical images. Whenever adding a new imaging device, please perform careful validation testing to ensure that such marks are accurate. On rare occasion, such testing might reveal a DICOM metadata problem or practice that can result in mislabeled images. Careful proactive validation can mitigate patient risk.

a. Adding a New DICOM Device

To add a new DICOM device to the list of available DICOM Targets, click on the **Add a New DICOM Device** button. Enter the required information, as shown in the following example, and then click on the **Add** button to save the information:

DICOM Devices

Add a New DICOM Device

DICOM Device wildcard definitions are allowed, (fields: AE Title), simply use a * to indicate wildcard. [\[example\]](#)

AE Title	
Hostname	
AE Title Prefix	
Port	
Station Name	All Stations ▼
Response Charset	ASCII ▼
Max Number of Simultaneous Associations	4

Please use a "," to separate SOP Classes in the list. Send Filter Options

Limit SOP Classes Sent to This Device? NO ▼

Do Not Send These SOP Classes

☒ Allow Extended Search on this node

☐ Allow Extended Search to access PACS custom tags
☐ Allow Extended Search to access DICOM Optional Tags
☐ Allow Extended Search to access DICOM Special Tags

C-MOVE Destination

Associated IPIDs

☐ VIRTUSA

Associated MRN Prefixes [Please use "," to separate MRN prefixes]

Figure 242: Adding a New DICOM Device

NOTE: Once you have entered the required information, you can also click on the **Ping** button to verify the connection to the device.

- If this device will be used as a **DICOM target**, enter the following information:

Field	Description								
AE Title	Enter the AE Title of the DICOM device.								
Hostname	Enter the fully qualified DNS name (e.g., yourserver.hospital.com), or the IP address of the Server if it does not have a fully qualified DNS name.								
AETitlePrefix	Enter a prefix if the site is using system-generated accession numbers and wants each accession number to include a unique prefix for each DICOM device. CAUTION: The AETitlePrefix can be a maximum of 6 characters long. If you choose to add a prefix, you must restart the AMICAS service, as described in Section 9.12 above, in order for the change to take effect.								
Port	Enter the port number of the DICOM device.								
Response Charset	Specify the character-set attribute that Merge PACS will send in response to C-FIND requests from that device. The following options are available: <table border="1"> <thead> <tr> <th>Option</th><th>Description</th></tr> </thead> <tbody> <tr> <td>ASCII</td><td>PACS will not send the character set attribute (0008,0005) in C-FIND responses.</td></tr> <tr> <td>LATIN1</td><td>PACS will set the character set attribute (0008,0005) to "ISO_IR 100" in C-FIND responses.</td></tr> <tr> <td>AUTO</td><td>PACS will set the character set attribute (0008,0005) to whatever value is in the C-FIND request.</td></tr> </tbody> </table>	Option	Description	ASCII	PACS will not send the character set attribute (0008,0005) in C-FIND responses.	LATIN1	PACS will set the character set attribute (0008,0005) to "ISO_IR 100" in C-FIND responses.	AUTO	PACS will set the character set attribute (0008,0005) to whatever value is in the C-FIND request.
Option	Description								
ASCII	PACS will not send the character set attribute (0008,0005) in C-FIND responses.								
LATIN1	PACS will set the character set attribute (0008,0005) to "ISO_IR 100" in C-FIND responses.								
AUTO	PACS will set the character set attribute (0008,0005) to whatever value is in the C-FIND request.								
Max Number of Simultaneous Associations	Specify the maximum number of simultaneous associations this DICOM device can support (default is 4). Each DICOM transaction is set up over an “association” between the sending system and the destination, which is how the transaction is negotiated, and Merge PACS can send multiple associations in parallel to increase performance. Some older PACS and Vender Neutral Archives (VNAs), however, cannot handle multiple concurrent associations for a single study transfer, however, and specifying the maximum number of simultaneous associations a particular DICOM device can handle therefore allows Merge PACS to throttle back the number of associations just to that device without compromising the performance for all destination devices. Note that the ability to have multiple concurrent associations is important to maximize performance on high latency LAN connections, since the acknowledgement cycle of DICOM gets held up by the round trip latency limiting transfer rates if the transfers are not done in parallel.								

Field	Description
Send Filter Options	If you want to prevent DICOM studies with certain SOP Classes from being sent to this DICOM target from the Send Studies tool, as described in subsection 9.9.3 above, do the following: • Select Yes from the drop-down Limit SOP Classes Sent to This Device? menu. • Enter one or more SOP Class UIDs, separated by commas, in the Do Not Send These SOP Classes text box.

- If this device will be used as a **filter** for one or more worklists, enter the following:

Field	Description
AE Title	Enter the AE Title of the DICOM device.
Station Name	Enter the Station Name for this DICOM device or select a station name already contained in the system from the drop-down menu.

- If this device will be used as an extended node, enter the following:

Field	Description
Allow Extended search on this node	If this option is selected, respective DICOM device will be used for extended search.
Allow Extended Search to access PACS custom tags	If this option is selected, Merge PACS will extend the search to this node even if the search criteria includes one or more custom private DICOM attributes (as described in subsection 9.13.36 below, and those private attributes will be included in the C-FIND request. If this option is not selected, and if the search criteria include one or more custom private DICOM attributes, Merge PACS will not extend the searches to this node. NOTE: Only iCEA10 supports this option.
Allow Extended Search to access DICOM Optional Tags	The DICOM Standard specifies which C-FIND keys are required to be supported by the SCP, as opposed to being optional. Examples of optional keys are Study Description and Referring Physician's Name. If this option is selected, Merge PACS will extend searches to this node, and these optional keys will be included as C-FIND criteria (if specified). If this option is not selected, Merge PACS will still extend searches to this node, but the optional tags will not be included in the C-FIND request. Instead, Merge PACS will apply a post-processing filter, (i.e., it will filter out the results that don't match any specified optional keys).

CAUTION: There is a configurable limit for the number of results Merge PACS can receive from an extended query node. In this case, it's possible that there are results on the extended query node that match the desired criteria (including optional keys) but, if the results are truncated and the initial set returned does not happen to include the studies that match the optional key criteria, it will appear as though the external query node had no matches.

Field	Description
Allow Extended Search to access DICOM Special Tags	If this option is selected, Merge PACS will use tags such as Modalities in Study, Number of Instances in Study, and Number of Series in Study (passed back to Merge PACS from this DICOM device) instead of issuing series-level C-FIND requests. These are DICOM attributes that let Merge PACS learn about the instances and series of a study by issuing only study-level C-FIND requests. If this option is not selected, Merge PACS will issue a series-level C-FIND for each returned study to get the information need (e.g., to fill in the Modality column in the worklist).
C-MOVE Destination	Used to specify the destination to which the extended study needs to be retrieved. The default value for Standalone mode is the AE title configured in Archive.remoteAE. The default value for Integrated mode is paired ICEA AE Title. The AE Titles of configured DICOM devices will be listed in the drop-down menu. NOTE: You can also specify a custom destination.
Associated IPIDs	One or more IPIDs can be associated to this DICOM Device. The IPID list will contain the IPIDs configured in the IPID Configuration Management Page, as described in subsection Error! Reference source not found. If the required IPID is not available, you can also add a custom IPID. NOTE: Custom IPIDs added here will not be listed on the IPID Configuration Management page. NOTE: When linking an IPID to an extended node on the DICOM Devices page, you should ensure that there is something in place to assign that IPID to studies being retrieved from that device (for example, passing through another device, or morpher/DICOM header modification rules to be applied at ingestion time).
Associated MRN Prefixes	One or more MRN prefixes can be associated to this DICOM Device. When searching against a particular extended node, if that node has any MRN prefixes configured and the MRN of the patient has one of those configured prefixes, that prefix will be excluded (stripped) from the MRN when the extended node search is performed. These MRN Prefixes will be appended to MRN while displaying results from extended node if configured with an MRN Prefix.

NOTE: When you add a new DICOM device, information for this device is automatically added to the AETitleToHost.map file.

b. Modifying Information for an Existing DICOM Device

To edit the information for a DICOM device already listed, click the **Modify** button next to the device's listing. Edit the necessary information for the DICOM device, as shown in the following example, and press the **Save** button to save your changes:

DICOM Devices

Modify DICOM Device

DICOM Device wildcard definitions are allowed, (fields: AE Title), simply use a * to indicate wildcard. [\[example\]](#)

AE Title:

Hostname:

AE Title Prefix:

Port:

Station Name:

Response Charset:

Max Number of Simultaneous Associations:

Please use a "," to separate SOP Classes in the list. Send Filter Options

Limit SOP Classes Sent to This Device?

Do Not Send These SOP Classes:

☒ Allow Extended Search on this node

☐ Allow Extended Search to access PACS custom tags

☐ Allow Extended Search to access DICOM Optional Tags

☐ Allow Extended Search to access DICOM Special Tags

C-MOVE Destination:

Associated IPIDs

☒ VIRTUSA

Associated MRN Prefixes [Please use "," to separate MRN prefixes]

Figure 243: Modifying Information for a DICOM Device

CAUTION: The AETitlePrefix can be a maximum of 6 characters long. If you choose to add a prefix, you must restart the Amicas service, as described in Section 9.11 above, in order for the change to take effect.

NOTE: When linking an IPID to an extended node on the DICOM Devices page, you should ensure that there is something in place to assign that IPID to studies being retrieved from that device (for example, passing through another device, or morpher/DICOM header modification rules to be applied at ingestion time).

c. Deleting a DICOM Device

To delete a DICOM device from the list of DICOM Targets, click on the **Delete** button next to that device's listing on the main DICOM Devices page.

9.13.3.DICOM Printers

The **DICOM Printers** page allows you to add or delete DICOM printers to which images can be sent from the Merge PACS Workstation, as shown in the example below:

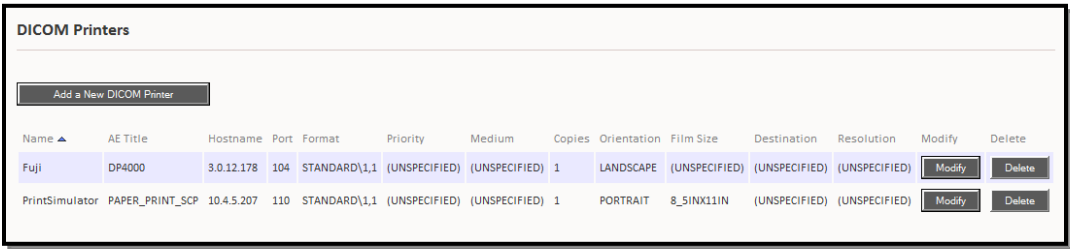


Figure 244: The DICOM Printers Page

NOTE: The calipers displayed on images on film printed with the “True Size” scaling option selected are supposed to represent 50mm in image space. Therefore, it is important to verify that this is correct with each DICOM printer if users may be selecting this option.

a. Adding a New DICOM Printer

To create a new listing for a DICOM printer that can receive images from the Merge PACS Workstation, click on the **Add a New DICOM Printer** button to bring up the Add a New DICOM Printer window shown below:

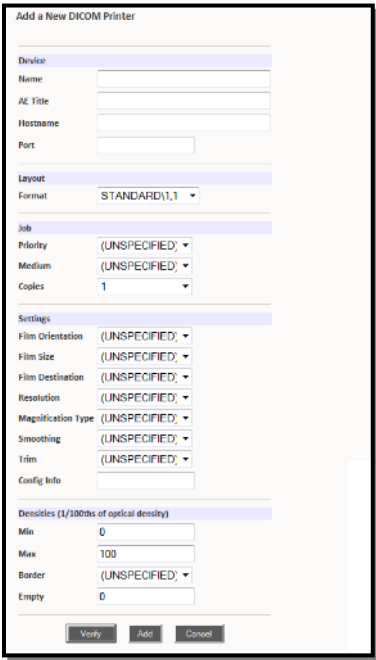


Figure 245: Adding a New DICOM Printer

- Enter the following mandatory information in the fields provided:

Field	Description
Name	Enter a user-friendly display name for this printer.

Field	Description
AE Title	Enter the AE Title for this printer
Hostname	Enter the fully qualified DNS name (e.g., yourserver.hospital.com), or the IP address of the printer if it does not have a fully qualified DNS name
Port	Enter the Port used by this printer.

- If desired, use any of the additional fields to select or enter default settings to be presented to the end-user when choosing this printer. If any field is set to “UNSPECIFIED”, the printer’s own default settings will be used. The following fields are available:

Field	DICOM Attribute	Description
Format	(2010,0010)	The standard format subdivides a film into image boxes of equal size. Therefore, the film layout is fully symmetrical, <i>i.e.</i>, the arrangement of image boxes on film is left-right and top-bottom symmetric. The following layouts are available: • STANDARD\1,1 • STANDARD\1,2 • STANDARD\2,2 • STANDARD\2,3 • STANDARD\2,4 • STANDARD\3,3 • STANDARD\3,4 • STANDARD\3,5 • STANDARD\4,4 • STANDARD\4,5 • STANDARD\4,6 • STANDARD\5,6 • STANDARD\5,7
Priority		Specify the desired job priority for this printer. The following priorities are available: • LOW • MED • HIGH
Medium		Specify the desired default output medium for this printer. The following options are available • BLUE FILM • CLEAR FILM • PAPER
Copies		Specify the number of copies (1-10) for this printing job.
Film Orientation		Specify the desired orientation for printing. The following options are available: • PORTRAIT [vertical film position] • LANDSCAPE [horizontal film position]

Field	DICOM Attribute	Description
Film Size	(2010,0050)	Specify the desired film size identification. The following options are available: • 14 inch x 17 inch • 14 inch x 14 inch • 11 inch x 17 inch • 11 inch x 14 inch • 10 inch x 14 inch [corresponds with 25.7 cm x 36.4 cm] • 10 inch x 12 inch • 8.5 inch x 11 inch • 8 inch x 10 inch • 24 cm x 24 cm • 24 cm x 30 cm • A3 [corresponds with 297 mm x 420 mm] • A4 [corresponds with 210 mm x 297 mm]
Film Destination	(2000,0040)	Specify the default film destination, which can be overridden for individual print jobs. The following options are available: • BIN_1 • BIN_2 • BIN_3 • BIN_4 • BIN_5 • BIN_6 • BIN_7 • BIN_8 • BIN_9 • PROCESSOR If no sorter bin is configured, all print jobs will be routed to the default output bin (BIN_1). The values of MAGAZINE and PROCESSOR will be mapped to BIN_1. If no value is provided, it will use the default. If no sorter bin is configured, default output bin (BIN_1) is used. BIN_1 Default can be changed through configuration.
Resolution		Specify the desired print resolution. The resolution of an image is largely dependent upon the total number of pixels contained in the image, which can be derived by multiplying the number of horizontal and vertical pixels within the image. The following options are available: • 1200 DPI • 600 DPI • 300 DPI • 200 DPI • 150 DPI • 100 DPI • 75 DPI

Field	DICOM Attribute	Description	
Magnification Type	(2010,0060)	Specify the interpolation type by which the printer magnifies or decimates the image in order to fit the image in the image box on film. The following options are available:	
		Option	Description
		NONE	No scaling. If the image box is larger than the image, the image will not completely fill the box. If the image is too large, it will be cropped (<i>i.e.</i> , borders cut off).
		REPLICATE	Scale image to fit image box (as good as the image aspect ratio allows) by replicating (duplicating) pixels.
		BILINEAR	Scale image using bilinear interpolation.
CUBIC	Scale image using cubic spline interpolation.		
	Smoothing	(2010,0080)	If you have selected CUBIC as the Magnification type, you can further specify the type of the interpolation function. The following options are available: • MEDIUM • SHARP • SMOOTH
Trim	(2010,0140)	Specify (YES/NO) whether a trim box shall be printed surrounding each image on the film.	
Config Info	(2010,0150)	Allows you to specify a character string that consists either of the ID of the printer configuration table that contains a set of values for implementation specific print parameters (<i>e.g.</i> , perception LUT related parameters) or one or more configuration data values, encoded as characters. If there are multiple configuration data values encoded in the string, they must be separated by backslashes. The definition of values must be contained in the SCP's Conformance Statement. Defined Terms: CS000-CS999: Implementation specific curve type. NOTE: It is recommended that for SCPs, CS000 represent the lowest contrast and CS999 the highest contrast levels available.	
Min	(2010,0120)	Specify the minimum density of the images on the film, expressed in hundredths of OD (Optical Density). If Min Density is lower than minimum printer density, Min Density will be set to minimum printer density.	
Max	(2010,0130)	Specify the maximum density of the images on the film, expressed in hundredths of OD. If Max Density is higher than maximum printer density, Max Density will be set to maximum printer density.	

Field	DICOM Attribute	Description
Border	(2010,0100)	Specify the density of the film areas surrounding and between images on the film. The following options are available: • BLACK • WHITE
Empty	(2010,0110)	Enter an integer specifying the desired density (in hundredths of OD) of image box areas on the film that contains no image. For example, 150 would correspond to 1.5 OD.

- Click on the **Verify** button at the bottom of the page to check the connection to the printer. A pop-up message will be displayed to indicate whether the verification was successful or not.
- When finished, click on the **Add** button at the bottom of the page.

b. Modifying Information for an Existing DICOM Printer

To edit the information for a DICOM device already listed, click the **Modify** button next to the device's listing to display the Modify DICOM Printer page for that printer. Edit the necessary information for the DICOM device and press the **Save** button at the bottom of the page to save your changes.

c. Deleting a DICOM Printer

To delete a DICOM Printer from the list of DICOM Printers, click on the **Delete** button next to that printer's listing on the main DICOM Printers page.

9.13.4. Modalities

The **Define Modalities** page allows you to populate a dictionary of modalities (MR, CR, etc.) that can be used as filters when setting up individual RTWL and Teaching worklists. They can also be used as query fields on the Query Page and in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation.

Click on the **Modalities** option to bring up the **Define Modalities** page, as seen in the following example:

Figure 246: Defining Modality Types

Note that the following modality types are included by default:

CODE	MODALITY NAME	CODE	MODALITY NAME
CR	Computed Radiography	RF	Radio Fluoroscopy
CT	Computed Tomography	RTIMAGE	Radiotherapy Image
MR	Magnetic Resonance	RTDOSE	Radiotherapy Dose
NM	Nuclear Medicine	RTSTRUCT	Radiotherapy Structure Set
US	Ultrasound	RTPLAN	Radiotherapy Plan
OT	Other	RTRECORD	RT Treatment Record
BI	Biomagnetic imaging	HC	Hard Copy
CD	Color flow Doppler	DX	Digital Radiography
DD	Duplex Doppler	MG	Mammography
DG	Diaphanography	IO	Intra-oral Radiography
ES	Endoscopy	PX	Panoramic X-Ray
LS	Laser surface scan	GM	General Microscopy
PT	Positron emission tomography (PET)	SM	Slide Microscopy
RG	Radiographic imaging (conventional film/screen)	XC	External-camera Photography
ST	Single-photon emission computed tomography (SPECT)	PR	Presentation State
TG	Thermography	AU	Audio
XA	X-Ray Angiography	ECG	Electrocardiography

a. Adding a New Modality

To add a new modality, click on the **Add a New Modality** radio button, enter the code for the modality (MR, CT, US, etc.) and a brief description, and then click on the **Add Modality** button, as shown in the following example:

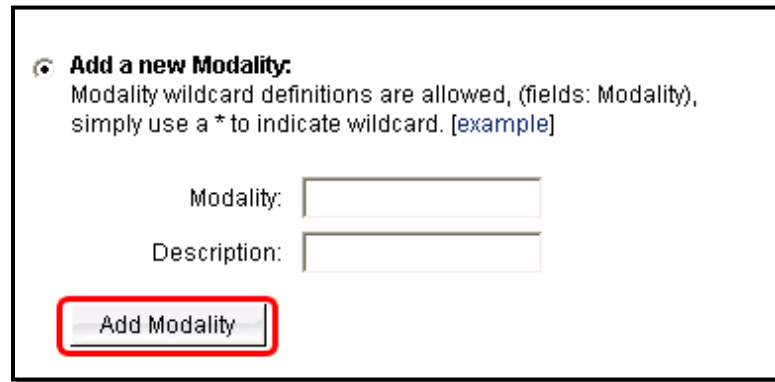


Figure 247: Adding a New Modality

Note that you can use an asterisk as a wildcard when entering the name of the Modality. To see how wild cards work, click on the [\[example\]](#) link on the Define Modalities page.

NOTE: The modality name may be up to 16 ASCII characters. Non-ASCII characters are not supported.

b. Modifying an Existing Modality

To modify information about a previously defined modality, click on the **Edit an Existing Modality** radio button, select the desired modality from the drop-down menu, modify the information as desired, and then click on the **Update Modality** button, as shown in the following example:

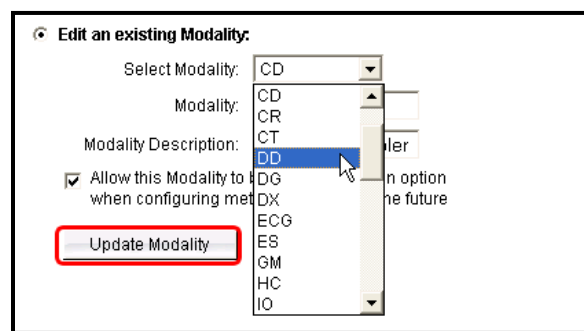
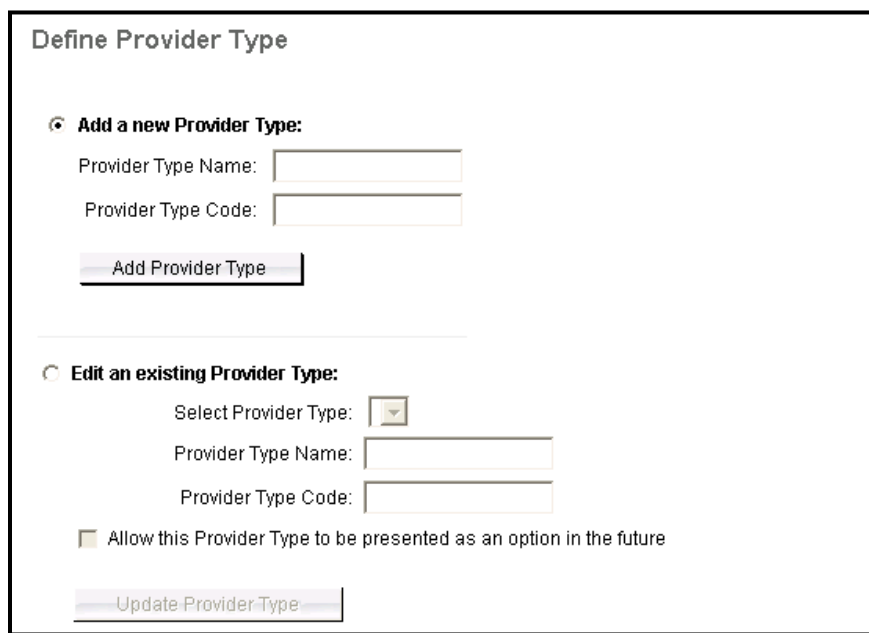


Figure 248: Modifying a Modality

NOTE: Once created, a modality cannot be deleted. To prevent a modality from showing up as an option in the various drop-down menus, **deselect** the checkbox labeled “**Allow this Modality to be presented as an option in the future.**”

9.13.5.Provider Types

The **Define Provider Types** page allows you to create a dictionary of provider types that can be used as filters when setting up individual RTWL and Teaching worklists. Click on the **Provider Types** option to bring up the **Define Provider Types** page, as seen in the following example:

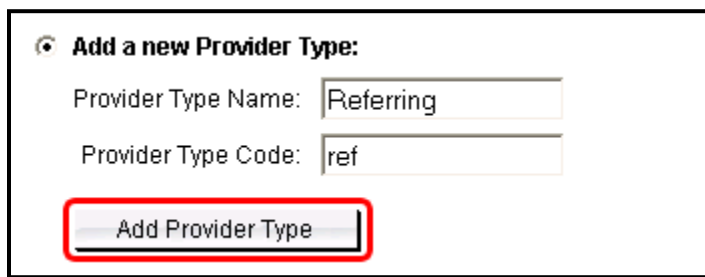


The screenshot shows the 'Define Provider Type' page. It has two main sections. The first section, 'Add a new Provider Type', is selected with a radio button. It contains two text input fields: 'Provider Type Name' and 'Provider Type Code', followed by an 'Add Provider Type' button. The second section, 'Edit an existing Provider Type', is unselected. It contains a 'Select Provider Type' dropdown menu, two text input fields for 'Provider Type Name' and 'Provider Type Code', a checkbox labeled 'Allow this Provider Type to be presented as an option in the future', and an 'Update Provider Type' button.

Figure 249: Defining Provider Types for RealTime Worklist

a. Adding a New Provider Type

To add a new provider type, click on the **Add a New Provider Type** radio button, enter the Code and Name for the type of provider, and then click on the **Add Provider Type** button, as shown in the following example:

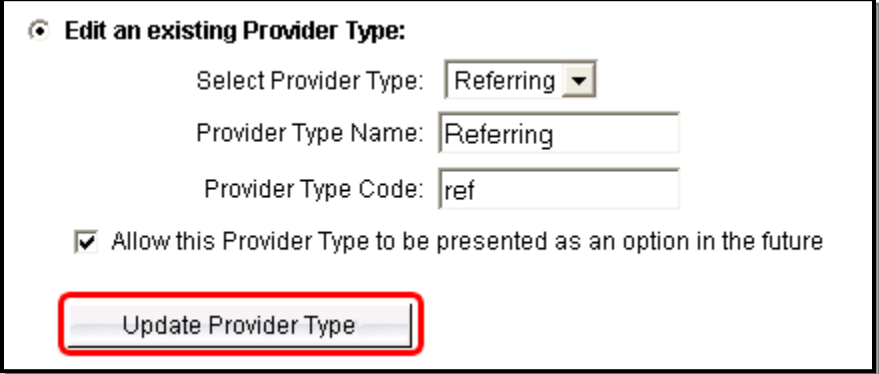


This screenshot shows the 'Define Provider Type' page with the 'Add a new Provider Type' section filled out. The 'Provider Type Name' field contains the text 'Referring' and the 'Provider Type Code' field contains the text 'ref'. The 'Add Provider Type' button is highlighted with a red rectangular box.

Figure 250: Adding a New Provider Type for RealTime Worklist

b. Modifying an Existing Provider Type

To modify information about a previously defined provider type, click on the **Edit an Existing Provider Type** radio button, select the desired provider type from the drop-down menu, modify the information as desired, and then click on the **Update Provider Type** button, as shown in the following example:



☒ **Edit an existing Provider Type:**

Select Provider Type: Referring ▼

Provider Type Name: Referring

Provider Type Code: ref

☒ Allow this Provider Type to be presented as an option in the future

Update Provider Type

Figure 251: Modifying a Provider Type for RealTime Worklist

NOTE: Once created, a provider type cannot be deleted. To prevent a provider type from showing up as an option in the various drop-down menus, **deselect** the checkbox labeled “**Allow this Provider Type to be presented as an option in the future.**”

9.13.6. Workflow Priorities

Priority items in RTWL and Teaching worklists are mapped directly from the incoming HL7 priority in the DICOM header for each order. By default, for individual worklists that are sorted by priority, orders with STAT priority will be shown first, followed by N/A (no priority), followed by the remaining status values in alphabetical order.

Worklists can be custom configured, however, to display status values in any order. In addition, each individual status value can be assigned its own display color, if so desired, so that the status is highlighted within individual worklists.

NOTE: Changing the order of priorities and/or their display colors will apply to all individual worklists. However, each individual worklist will still need to be manually configured to display and sort by priority if so desired.

NOTE: You cannot change the description of a priority item, but you can change the display name and/or the display color.

Clicking on the **Workflow Priorities** option to bring up the **Define Workflow Priority Items** page, as seen in the following example:

Define Workflow Priority Items

Sort Workflow Priorities Add new Workflow Priority

The following table lists the Workflow Priority items in the descending order they will be displayed. System priorities cannot have their name and descriptions modified.

Priority ID	DICOM Name	Display Name	Display Color	Description	SLA	Modify	Delete
S	STAT	STAT		HL7 STAT	30	Modify	
A	HIGH	ASAP		HL7 ASAP	60	Modify	
C	HIGH	Callback		HL7 Callback	90	Modify	
P	HIGH	Pre-op		HL7 Pre-op	120	Modify	
T	MEDIUM	Timing	Transparent	HL7 Timing	240	Modify	
R	ROUTINE	Routine	Transparent	HL7 Routine	480	Modify	
N/A	Unknown	Unknown	Transparent	No priority code was issued	480	Modify	

Figure 252: Define Workflow Priority Items Page

a. Sorting Workflow Priorities

To sort the existing list of workflow priorities, click on the **Sort Workflow Priorities** button. This will bring up the **Sort Priorities** page, as seen in the following example:

Sort Priorities

STAT (STAT - HL7 STAT)
ASAP (HIGH - HL7 ASAP)
Callback (HIGH - HL7 Callback)
Pre-op (HIGH - HL7 Pre-op)
Timing (MEDIUM - HL7 Timing)
Routine (ROUTINE - HL7 Routine)

↑
↓

Submit Cancel

Figure 253: Sort Workflow Priorities Page

- Priority values will be sorted in the order they appear in the **Sort Priorities** box, from top to bottom.
- To reorder a priority value, select it by clicking on it once with your mouse and then click on the up or down arrow to the right of the box to move it to the desired position.
- Repeat as necessary with other priority values
- When finished, click on the **Submit** button to save your changes and return to the previous page.

b. Adding a New Workflow Priority Item

To add a new workflow status dimension, click on the **Add New Workflow Priority Dimension** button. This will bring up the **Add a Priority** page, as seen in the following example:

Figure 254: Adding a New Workflow Status Dimension

- Enter a short (one or two letter) ID for this priority in the **Priority ID** field.
- Enter the DICOM name for this priority in the **DICOM Name** field.
- Enter a display name for this priority in the **Display Name** field.
- Choose the display color for this priority status item by clicking on the **Choose Color** link. This will bring up the Select Color window:

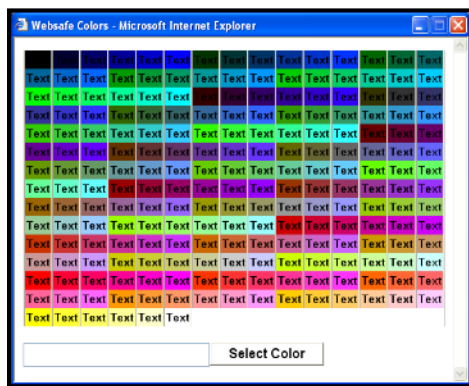


Figure 255: The Select Color Window

- Select the desired display color by clicking on it once with your mouse.
- If the **Select Color** field is left blank, the priority status will be displayed with a transparent color (*i.e.*, it will not appear highlighted). This is different from selecting white as the display color.
- Make sure that the chosen display color will not conflict with the colors chosen for individual workflow status items, as described in the following subsection.
- When finished, click on the **Select Color** button to save your selection and exit the Select Color window. If you wish to exit the window without making any changes, click on the small **"X"** in the top right corner of the window instead.

- If the site's workflow includes deadline tracking due to **Service Level Agreements (SLAs)**, you enter the following information:
 - In the **SLA** field, enter the amount of time (in minutes) required for studies to be read with this particular priority.
 - If desired, you can define up to four separate threshold limits (in minutes), each with its own display color. For example, if the SLA is set to 30 minutes, you might define the first threshold to display **RED** when 1 minute is remaining, **YELLOW** when 5 minutes are remaining and **GREEN** when 10 minutes are remaining.
 - Select a color for the **Above Display Color** to indicate what color should be displayed if the time remaining is greater than the last defined threshold.

Note the following with regard to SLA values:

- SLA values will be displayed in the **Due In** column for worklists that are configured to display that column.
- In addition to the configured color and the amount of time remaining until due (or time past due), the Due In column will display the following information as "hover" text when the mouse cursor is placed over it:

Value	Description
Arrived	The time the last image was received for the study plus a configurable Inactivity Timeout period.
Due By	The "Arrived" time plus the SLA value defined for the study's Priority.
Completed	The time the study was transitioned to (or past) the diagnostic status configured for your site to indicate completion.

- If there is no order associated with a study, the study will be considered "Routine" priority and the SLA will be whatever is currently configured for the Routine priority.
- If a new or updated order is associated with a study and that order contains a priority, the study Due In time will be updated
- If a new image is received for a study that is not yet complete, the "Arrived" and "Due By" times will be updated.
- If a new Presentation State (PR) or Key Object (KO) is received for a study that is not yet complete, the "Arrived" and "Due By" times will be not be updated
- If a new image arrives for a study that is complete, the behavior depends on the value of the configurable OverWriteFutureStatus property.
- If a new PR or KO arrives for a study that is complete, the "Arrived" and "Due By" times will be not updated and the Study Complete value will not be cleared.
- When you have finished changing the display name and/or display color, click on the **Add Workflow Priority** button to save your changes and return to the main Define Workflow Priority Items page.

c. Modifying Priority Status Items

To change the display name and/or display color of a priority status item, click on the **Modify** link to the right of the desired status item, as in the following example:

Define Workflow Priority Items

Sort Workflow Priorities Add new Workflow Priority

The following table lists the Workflow Priority Items in the descending order they will be displayed. System priorities cannot have their name and descriptions modified.

Priority ID	DICOM Name	Display Name	Display Color	Description	SLA	Modify	Delete
S	STAT	STAT		HL7 STAT	30	Modify	
A	HIGH	ASAP		HL7 ASAP	60	Modify	
C	HIGH	Callback		HL7 Callback	90	Modify	
P	HIGH	Pre-op		HL7 Pre-op	120	Modify	
T	MEDIUM	Timing	Transparent	HL7 Timing	240	Modify	
R	ROUTINE	Routine	Transparent	HL7 Routine	480	Modify	
N/A	Unknown	Unknown	Transparent	No priority code was issued	480	Modify	

Figure 256: Choosing a Priority Status Item to Modify

This will bring up the **Modify Priority** window for that status item, as in the following example:

Modify Priority STAT

Priority ID: S

DICOM Name: STAT

Display Name:

Display Color:  (Choose Color)

Description: HL7 STAT

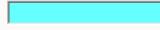
SLA: (in minutes)

Threshold 1: (in minutes) Display Color:  (Choose Color)

Threshold 2: (in minutes) Display Color: (Choose Color)

Threshold 3: (in minutes) Display Color: (Choose Color)

Threshold 4: (in minutes) Display Color: (Choose Color)

Above Display Color:  (Choose Color)

Modify Workflow Priority Cancel

Figure 257: The Modify Status Priority Window

Edit the information as desired and click on the **Modify Workflow Priority** button to save your changes and return to the main Define Workflow Priority Items page.

d. Deleting a Custom Workflow Priority

Any workflow priority that has been manually created can be deleted by clicking on the **Delete** link to the right of the desired status item, as in the following example:

N/A	Unknown	Unknown	Transparent	No priority code was issued	480	Modify	
247	Test	TestVIN		TestVIN	25	Modify	Delete

Figure 258: Deleting a Priority Status Item

9.13.7. Workflow Status Dimensions

Workflow status dimensions allow multiple statuses to be assigned to a single order simultaneously instead of sequentially. For example, instead of just a single diagnosis-based status that moves sequentially from “Ordered” to “Scheduled” to “Unverified” to “Unread” to “Read” to “Preliminary” to “Final”, a particular order could have a diagnostic status of “Unread”, a communication status of “Convey Results”, and an ER status of “ER Read”, all at the same time.

Each “dimension” corresponds to a different type of status that might need to be set for an order, whether it be a status set by a type of user (such as a radiologist or an ER doctor) or a status set automatically by the Merge PACS system (such as whether the images have been transferred). Each status dimension can be assigned a unique icon that will be displayed at the top of a separate column within individual worklists, and each dimension can also be customized to determine how it can be used within a site’s overall workflow.

By default, Merge PACS comes with a number of preset status dimensions that cannot be deleted. These default status dimensions can be modified, however, and new status dimensions can also be added from scratch.

When Merge PACS is configured to run in **Integrated** mode, the attached iCEA Server can be set up to tag studies according to Study Type (e.g., Cardiology, Radiology, Neurology, etc.). In addition, multiple Study Types can be placed together into a **Study Type Group** (e.g., an “OLOGY” group that would include the aforementioned Cardiology, Radiology and Neurology study types). Any such study type groups will automatically be mapped to a default workflow status dimension (e.g., the “OLOGY” study type group will be mapped to a default “OLOGY” workflow status dimension).

NOTE: Although an “OLOGY” group is the most common study type group that would typically be used with Merge PACS, any number of other study type groups can be configured, such as a “RETENTION” group that would describe how the data for each study type is retained.

When Merge PACS is configured to integrate with one or more **AI applications** (e.g., Watson Imaging Care Advisor for liver or Watson Imaging Care Advisor for prostate), dimensions created to display status and results for those applications can be aggregated under two pre-defined “AI” status dimensions (“AI Status” and “AI Results”). This will allow a user to see a summary of the status and results of AI processing on a particular study based on the evaluation by all associated AI applications, without needing to display separate status dimensions for each AI application. When configuring aggregation for a particular dimension, you can also choose whether the parent dimension (e.g., “AI Status”) should use the highest or lowest values from the associated “children” dimensions (e.g., “LA Status” and “PA Status”). Note the following:

- When you aggregate other dimensions, the parent dimension statuses will be automatically replicated to the child dimension(s).
- Once a “child” dimension has been aggregated to a “parent” dimension, no dimensions can be aggregated under that child dimension.
- Only dimensions with no defined statuses will be eligible to be aggregated under the “parent” dimension. So, for example, if you were to create an “LA Status” dimension for use with Watson Imaging Care Advisor for liver and wanted to have it aggregated under “AI Status,” you would not add any workflow statuses to that “LA Status” dimension as described in subsection 9.13.8 below.

Clicking on the **Workflow Status Dimensions** option to bring up the **Define Workflow Status Dimensions** page, as seen in the following example:

Add New Workflow Status Dimension

The following table lists the Workflow Status Dimensions in the order they will be displayed.

Dimension Name	Icon	RTWL Filter	Trigger Auto-cache	Enable Action Buttons	Enable Status colors	Enable User Search	Action
Diagnostic	+	☒	☒	☒		☒	Modify
Communication	☎	☒		☒			Modify
Connection	⚡	☒		☒			Modify
Reservation	👤	☒	☒	☒		☒	Modify
ImageTransfer	🔄						Modify
StudyLocking	🔒					☒	Modify
ER	👤	☒					Modify
OrderMatch	🔍			☒			Modify
UpdateOrder	🔄	☒		☒			Modify
Assignment	👤	☒	☒	☒	☒	☒	Modify
Prefetch	👤	☒	☒	☒			Modify
QC	👤	☒					Modify
AutomaticAssignment	🔄	☒					Modify
AI Status	👤	☒	☒		☒	☒	Modify
AI Results	👤	☒	☒		☒	☒	Modify

Figure 259: Define Workflow Status Dimensions Page

NOTE: Once the various status dimensions have been defined at the Workflow Status Dimensions page, the individual status options for each dimension can be defined at the Workflow Status page, as described in subsection 9.13.8 below.

a. Adding a New Workflow Status Dimension

To add a new workflow status dimension, click on the **Add New Workflow Status Dimension** button:

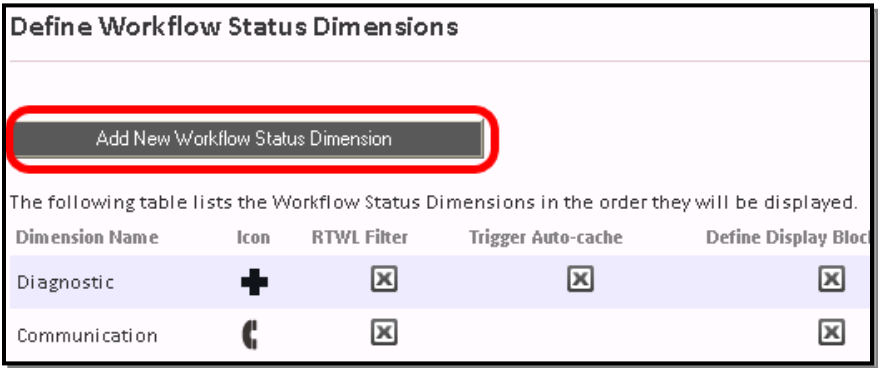


Figure 260: The Add a New Workflow Status Dimension Button

This will bring up the **Add a new Workflow Status Dimension** page, as seen in the following example:

Add a new Workflow Status Dimension:

Dimension Name:

Dimension Icon: (Select an icon)

☐ Allow Dimension to be filtered by RTWL

☐ Allow Dimension to trigger Auto-Cache

☐ Allow Dimension to enable action buttons

☐ Allow to use this Dimension's status colors

☐ Allow Dimension as a picklists option for user study search

Figure 261: Adding a New Workflow Status Dimension

At the Add a new Workflow Status Dimension page, do the following:

- Enter a **Dimension Name** for this status dimension in the field provided.
- Select a **Dimension Icon** for this status dimension by clicking on the **Select an icon** link:

Add a new Workflow Status Dimension:

Dimension Name:

Dimension Icon: (Select an icon)

Figure 262: The Select an Icon Link

- When you click on the Select and Icon link, an icon selection pop-up window similar to the following will appear:

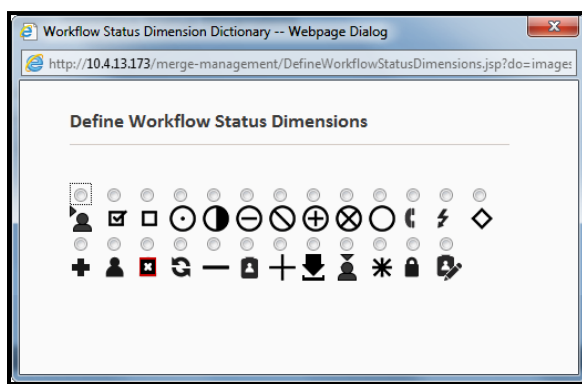


Figure 263: Selecting an Icon for this Status Dimension

- Click on the radio button above the desired icon to select that icon and close the pop-up window.
- Select the **Allow Dimension to be filtered by RTWL** option if you this dimension to be filtered by RTWL.

- Select the **Allow Dimension to trigger Auto-Cache** option if you want this dimension to be able to trigger auto-caching.
- Select the **Allow Dimension to enable action buttons** option if you want this dimension to be able to enable action buttons.
- Select the **Allow to use this Dimension's status colors** option if you want the system to use this dimension's status defined status colors.
- Select the **Allow Dimension as a picklists option for user study search** option if you want this status dimension to be available as a query filter option on both the Query Page and the Quick Filter section of the RTSL.
- Select the **Aggregate other dimensions** option to configure AI Aggregation options for this dimension, if applicable. Selecting this option will cause additional sub-options to be displayed, as in the following example:

Figure 264: Aggregate Other Dimensions

- All dimensions with no currently defined workflow statuses (as described in subsection 9.13.8 below) will be displayed as separate check box options (e.g., “LA Status” and “PA Status” in the example above). Select all dimensions you want to be aggregated to this dimension.
- Use the **Aggregate Lowest** and **Aggregate Highest** radio buttons to indicate whether this dimension should use the highest or lowest values from the associated “children” dimensions.
- When finished, click on the **Add Workflow Status Dimension** button to save the changes and return to the previous screen.

b. Modifying an Existing Workflow Status Dimension

To modify information about a previously defined workflow status dimension, click on the **Modify** link to the right of the desired status dimension, as in the following example:

Dimension Name	Icon	RTWL Filter	Trigger Auto-cache	Enable Action Buttons	Enable Status colors	Enable User Search	Action
Diagnostic	+	☒	☒	☒		☒	Modify
Communication	☎	☒		☒			Modify
Connection	⚡	☒		☒			Modify

Figure 265: The Modify an Existing Workflow Status Dimension Link

This will bring up the **Edit an existing Workflow Status Dimension** page, as seen in the following example:

Edit an existing Workflow Status Dimension:

Dimension Name:

Dimension Icon: (Select an icon)

- ☒ Allow Dimension to be filtered by RTWL
- ☒ Allow Dimension to trigger Auto-Cache
- ☐ Allow Dimension to enable action buttons
- ☐ Allow to use this Dimension's status colors
- ☒ Allow Dimension as a picklists option for study search
- ☐ Aggregate other Dimensions

Update Workflow Status Dimension

Figure 266: Editing an Existing Workflow Status Dimension

At the Edit an existing Workflow Status Dimension page, edit the information as desired.

If you are editing one of the default “AI” dimensions note the following:

- If you are editing this dimension for the first time, “Aggregate other Dimensions” will be selected by default, the “Aggregate Lowest” option will be selected by default and the workflow statuses will be displayed in ascending order, as in the following example:

Edit an existing Workflow Status Dimension:

Dimension Name:

Dimension Icon: (Select an icon)

- ☒ Allow Dimension to be filtered by RTWL
- ☒ Allow Dimension to trigger Auto-Cache
- ☐ Allow Dimension to enable action buttons
- ☐ Allow to use this Dimension's status colors
- ☒ Allow Dimension as a picklists option for study search
- ☒ Aggregate other Dimensions

- ☒ LA Status ☐ PA Status ☐ LA Result
 - ☒ PA Result
 - ☒ Aggregate Lowest ☐ Aggregate Highest
 - Status Names:**
 - Queued
 - In Progress
 - Completed
 - Failed
 - Rejected

Update Workflow Status Dimension

Figure 267: Default “AI” Dimension Options

- If you choose “Aggregate Highest,” the workflow statuses will be displayed in descending order based on the workflow status display order.

- If this dimension has already been aggregated under another dimension, you will not be able to aggregate any other dimensions and the option to aggregate will be grayed out as in the following example:

☒ Aggregate other Dimensions (Note: This Dimension being Aggregated with AI Status)

Status Names:

- Queued
- In Progress
- Completed
- Failed
- Rejected

Figure 268: Child Dimensions Cannot Be Aggregated

When finished, click on the **Update Workflow Status Dimension** button to save your changes.

CAUTION: If you change the name of a Dimension that is currently included as a display column in an existing worklist, as described in subsection 13.4.8 below, you will need to edit that worklist to remove the old column and replace it with the renamed Dimension column.

9.13.8. Workflow Status

A Merge PACS Server comes with a default set of workflow status options that can be used with the optional RealTime Worklist component. These options cannot be deleted, but their names and display colors can be modified. You can also add new status options.

Click on the **Workflow Status** option to bring up the **Define Workflow Status** page, as seen in the following example:

Define Workflow Status Items

[Add New Workflow Status](#)

The following table lists the Workflow Status items, by Dimension, in the order they will be displayed.

Dimension	Abbreviation	Status Code	Status Name	Status Color	Status Icon	Parameters	Force Comment	Action
Diagnostic	OR		Ordered				N	Modify
Diagnostic	SC		Scheduled				N	Modify
Diagnostic	UV		Unverified				N	Modify
Diagnostic	CPS		Problem Study				Y	Modify
Diagnostic	CAP		Alert Problem Studies				N	Modify
Diagnostic	UR		Unread				N	Modify
Diagnostic	CZZ		CTest				Y	Modify
Diagnostic	DI		Read				N	Modify
Diagnostic	TR		Transcribed				N	Modify
Diagnostic	PR	70	Preliminary				N	Modify
Diagnostic	FI	99	Final				N	Modify
Communication	CC		Convey Result			ME	N	Modify
Communication	CV		Conveyed				N	Modify
Connection	CT		Talk to MD			ME	N	Modify

Figure 269: Defining Workflow Status for RealTime Worklist

The following workflow statuses are provided by default:

Dimension	ID	Display Name	Default Parameters	How Set
Diagnostic	OR	Ordered		Set by ImageManager when it receives an order without any scheduled date/time.
Diagnostic	SC	Scheduled		Set by ImageManager when it receives an order with at least one scheduled date/time.
Diagnostic	UV	Unverified		Set by the user via the validation Tab, or the auto-validation process.
Diagnostic	UR	Unread		Set by Importer after it closes a study if the validation is not on. Otherwise, by the validation process.
Diagnostic	DI	Read		Set by Viewer when it closes a study.
Diagnostic	TR	Transcribed		Set by ImageManager when it receives a report with OBR-25 (Results Status) set to "TR" (or 60, by default).
Diagnostic	PR	Preliminary		Set by ImageManager when it receives a report with Final status (or 70, by default).
Diagnostic	FI	Final		Set by Viewer when it closes a study or ImageManager when it receives a report with Final status (or 99, by default).
Communication	CC	Convey Result	ME	Set by RTWL status button, Query Page status button, Viewer or Tech WorkPanel.
Communication	CV	Conveyed		Set by Communications WorkPanel.
Connection	CT	Talk to MD	ME	Set by RTWL status button, Query Page status button, Viewer or Tech WorkPanel.
Connection	CN	MD Connected		Set by Communications WorkPanel.
Reservation	RR	Reserved	ME, None, Anyone	Set by RTWL status button, Query Page status button, or Viewer
ImageTransfer	IT	Image Transferring		Set by Importer for each N (configurable) new image it processes. Reset by the Importer after it closes a study.

Dimension	ID	Display Name	Default Parameters	How Set
StudyLocking	LO	In Use	ME, None, Anyone	Set by Viewer or ADW when it opens a study. Unset by Viewer or ADW when it closes a study.
ER	MEP	ER Positive		Set by RTWL status button, Query Page status button, or Viewer
ER	MEN	ER Negative		Set by RTWL status button, Query Page status button, or Viewer
ER	EA	Rad Agree		Set by RTWL status button, Query Page status button, or Viewer
ER	ED	Rad Disagree		Set by RTWL status button, Query Page status button, or Viewer
ER	EC	ER Acknowledge		Set by RTWL status button, Query Page status button, or Viewer
ER	ER	ER Resolved		Set by RTWL status button, Query Page status button, or Viewer
OrderMatch	OM	OrderMatching		Set by Importer when Auto Validation is enabled and the Order Matching Validator class is selected.
UpdateOrder	ZA	Arrived		Set by ImageManager when it receives a change order with this value in ORC.5.
UpdateOrder	IP	Checked In		Set by ImageManager when it receives a change order with this value in ORC.5.
UpdateOrder	ZS	Exam Started		Set by ImageManager when it receives a change order with this value in ORC.5.
UpdateOrder	CM	Exam Ended		Set by ImageManager when it receives a change order with this value in ORC.5.
Prefetch	MPN	Needed		Set by the Pre-fetcher component of VNA when a Study is awaiting pre-fetch of priors.
Prefetch	MPI	In Progress		Set by the Pre-fetcher component of VNA when the pre-fetch of priors is in progress.
Prefetch	MPP	Partial Fail		Set by the Pre-fetcher component of VNA when some of the priors could not be pre-fetched.
Prefetch	MPF	Fail		Set by the Pre-fetcher component of VNA when no priors could be pre-fetched.

Dimension	ID	Display Name	Default Parameters	How Set
Prefetch	MPC	Complete		Set by the Pre-fetcher component of VNA when pre-fetch of all priors based upon pre-fetch rules is complete.
QC	MQE	QC Exception		Set by EA Notification Processor when iCEA notifies Merge PACS that it has two studies with the same Study Instance UID but different patient demographics.
QC	MQP	QC Exception Pending		Set by QC Editor after a user has manually made changes to a study with “QC Exception” status until the changes have been fully processed.
Assignment	AS	Assigned	ME, None, Anyone	Set by RTWL “Assign the Study” action icon. Can also be set by the Study Status Dialog if the "Available Status Dimensions" site preference has been updated to include the Assignment dimension
AutomaticAssignment	MAA	Assigned		Set by the Advanced Distribution Engine process when the exam has been assigned to a reader.
AutomaticAssignment	MAR	Retry		Set by the Advanced Distribution Engine process when the exam was triggered for auto assignment and failed to find a suitable assignee and the maximum configured number of retry attempts has not yet been reached.
AutomaticAssignment	MARF	Retry failed		Set by the Advanced Distribution Engine process when the exam was triggered for auto assignment and failed to find a suitable assignee after <i>n</i> attempts of retry based on Advanced Distribution Engine configuration.
AutomaticAssignment	MAC	Completed		Set by the Advanced Distribution Engine process when the exam is completed with respect to automatic Advanced Distribution Engine. The exam will now be ignored by the Advanced Distribution Engine
AI Status	AIQ	Queued		Set by RTWL status button, Query Page status button, or workflow status API.

Dimension	ID	Display Name	Default Parameters	How Set
AI Status	AII	In Progress		Set by RTWL status button, Query Page status button, or workflow status API.
AI Status	AIC	Completed		Set by RTWL status button, Query Page status button, or workflow status API.
AI Status	AIE	Error		Set by RTWL status button, Query Page status button, or workflow status API.
AI Status	AIN	No Content		Set by RTWL status button, Query Page status button, or workflow status API.
AI Status	AIF	Failed		Set by RTWL status button, Query Page status button, or workflow status API.
AI Results	AIL	Normal		Set by RTWL status button, Query Page status button, or workflow status API.
AI Results	AIM	Non-actionable		Set by RTWL status button, Query Page status button, or workflow status API.
AI Results	AIH	Non-critical Actionable - Category 3		Set by RTWL status button, Query Page status button, or workflow status API.
AI Results	AIV	Urgent Actionable - Category 2		Set by RTWL status button, Query Page status button, or workflow status API.
AI Results	AIU	Emergent Actionable - Category 1		Set by RTWL status button, Query Page status button, or workflow status API.

NOTE: Starting with Merge PACS 7.0, new system workflow statuses are prefixed by an "M" to distinguish them from custom prefixes that may be custom created by a site.

NOTE: If Merge PACS is configured to run in **Integrated** mode, the attached iCEA Server can be set up to tag studies according to Study Type (e.g., Cardiology, Radiology, Neurology, etc.). In addition, multiple Study Types can be placed together into a **Study Type Group** (e.g., an "OLOGY" group that would include the aforementioned Cardiology, Radiology and Neurology study types). Any such study type groups will automatically be mapped to a default workflow status dimension (e.g., the "OLOGY" study type group will be mapped to a default "OLOGY" workflow status dimension) and the associated study types within those groups will automatically be mapped to workflow status items.

NOTE: Merge PACS includes various default workflow statuses for the “AI Status” and “AI Results” Workflow Status Dimension for use with one or more **AI applications**. If you have created separate dimensions for any of those applications (e.g., an “LA Status” and an “LA Results” dimension for use with Watson Imaging Care Advisor for liver) and want them to be aggregated under one of the default AI dimensions, you should not define any workflow statuses for them here.

a. Adding a New Workflow Status

To add a new workflow status, click on the **Add a New Workflow Status** button to bring up the **Add a New Workflow Status** page, as shown in the following example:

The screenshot shows the 'Define Workflow Status Items' form. It contains several input fields and sections:

- Add Workflow Status:** A section with a dropdown for 'Workflow Dimension' (currently 'Diagnostic'), a text field for 'Workflow Abbreviation' (containing 'C'), a text field for 'Workflow Status Code', a text field for 'Workflow Name', a text field for 'Workflow Color' with a '(Choose Color)' link, and a button for 'Workflow Icon' labeled '(Select an Icon)'.
- Display After:** A dropdown menu currently set to '- display first -'.
- Parameters:** A text field.
- Timed Set Status:** A section with an 'Add Timed Set Status' button and a label 'Dimension: Status: Parameter: Timer (sec):'.
- Timed Clear Status:** A section with an 'Add Timed Clear Status Dimension' button and a label 'Dimension: Timer (sec):'.
- Associated Default Actions:** A grid of checkboxes for various actions such as 'View study', 'View comments', 'Burn CD', 'Add To Default Teaching Worklist', 'Launch Voice Clip', 'Patient Data Editor', 'Data Masking', 'Save Series', 'View study in the secondary viewer', 'Tech Panel', 'Route study', 'Remove From Current Worklist', 'View Access Control Page', 'Change Status MD Connected', 'Reserve the Study', 'Open study in QC Editor', 'View Order', 'ER Panel', '3rd Party Application Synchronization', 'Remove study from the associated worklist', 'OrthoLink', 'Change Status Talk to MD', 'Assign the Study', 'View Report', 'Communication Panel', 'Study Tagging', 'Add study to another worklist', 'Change Status', 'View study in the Web Viewer', and 'Retrieve'.
- Enable this workflow status and allow this to be presented as an option in picklists** and **Enable force comment to make the user add comments when updating to this status.** (both with checkboxes).
- Add Workflow Status** and **Cancel** buttons at the bottom.

Figure 270: Adding a New Workflow Status

The following information should be entered:

Field	Description
Workflow Dimension	Select the status dimension to which this status applies. Note that this list will only include dimensions that have currently been defined on the Workflow Status Dimensions page, as described in subsection 9.13.7 above. NOTE: Aggregated dimensions will not be shown in the drop-down list of available workflow dimensions.
Workflow Abbreviation	Enter a two-letter abbreviation for this status.

Field	Description
	NOTE: Starting with Merge PACS 8.2, custom workflow statuses will automatically be prefixed with a “C” to distinguish them from system workflow statuses. A new status given the abbreviation “AB”, for example, will be displayed as “CAB”. This will not apply retroactively, however, to custom statuses created prior to 7.0 in the case of upgraded systems.
Workflow Status Code	If desired, enter a numeric value that, when detected in OBR-25 field of an incoming order, will be equivalent to the Workflow Abbreviation. CAUTION: The Workflow Status Code should be entered with care after carefully reviewing the mappings. Incorrect mappings could result in studies being set to the wrong status. CAUTION: The use of Latin characters is not supported.
Workflow Name	Enter a descriptive name for this status.
Workflow Color	Click on the Choose Color link to select a color for this status from a pop-up table of available colors:

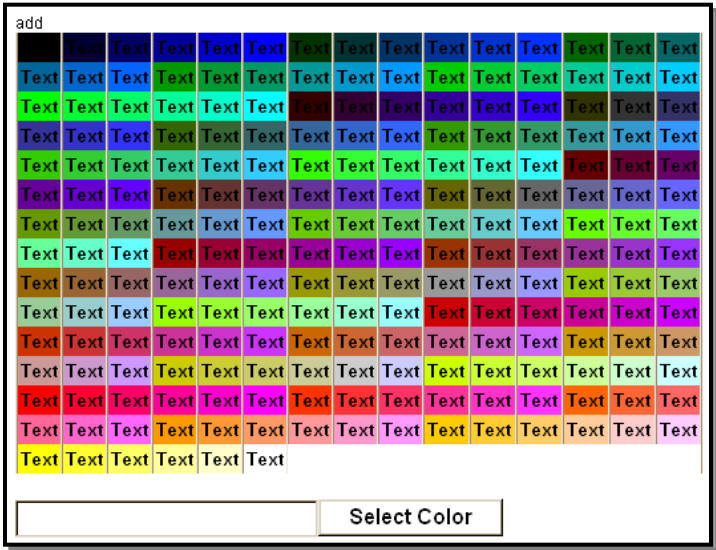


Figure 271: Selecting a Color for this Status

Select a color by clicking on the desired color and then clicking on the Select Color button. Note that the table displays how text will appear with each color.

Field	Description
Workflow Icon	Click on the Select an Icon link to choose an icon for this status option from a pop-up table of available icons: Define Workflow Status Items 
Display After	Select an icon by clicking on the radio button directly above the desired icon. Use the drop-down menu to select where this status should appear in the various workflow pick lists, in relation to the other defined status options.
Parameters	If desired, you can enter one or more parameters for this status option, separated by commas, in the field provided. When a user sets the status of an order from within a worklist, the defined parameters will then be presented as sub-status options for this status. Any text can be entered as a parameter, although “ME” is reserved as a predefined parameter. If “ME” is included as a parameter and the user selects it, it will be replaced with that user’s login name when the status is displayed within RTWL. For example, if one or more physicians were added as parameters to the “Reservation” status option, a user could then select the physician the study should be reserved for when changing the status to “Reserved.” Then, if that physician uses a worklist that is defined to show studies with a “Reserved” status and a parameter of “ME”, the worklist will show all studies that are reserved for physician.

Figure 272: Selecting an Icon for this Status

Field	Description
Timed Set Status Actions	If desired, you can have the system automatically set the status of one or more other dimensions for this order after a specified period of time after this status has been set. For each status you want to have automatically set, click on the Add Timed Set Status button:

Timed Set Status Actions: Add Timed Set Status

Dimension:

Status:

Parameter:

Timer (sec.):

Diagnostic

Dictated

- none -

Delete

Figure 273: Defining Timed Set Status Actions

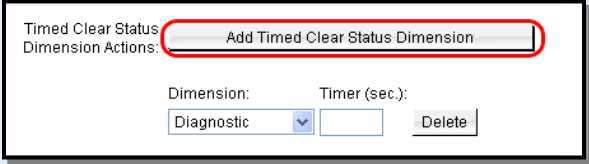
The following information can then be entered:

Dimension	Select the dimension whose status you want to automatically set.
Status	Select the status you want the dimension to be set to.
Parameter	If applicable, select the parameter for the status.
Timer	If desired, enter a delay (in seconds) for this action to take effect. If left blank, the specified dimensional status will be set as soon as this status is set.

When finished, you can click on the **Add Timed Set Status** button again to set an additional timed set status action.

You can also click on the **Delete** button to the right of a defined timed set status action to remove it.

NOTE: The Timed Set Status needs to be displayed on the Worklist in order for the user to witness the change.

Field	Description								
Timed Clear Status Dimension Actions	If desired, you can have the system automatically clear the status of one or more other dimensions for this order after a specified period of time after this status has been set. For each status you want to have automatically cleared, click on the Add Timed Clear Status Dimension button:  Figure 274: Defining Timed Clear Status Dimension Actions The following information can then be entered: <table><tr><td>Dimension</td><td>Select the dimension whose status you want to automatically set.</td></tr><tr><td>Status</td><td>Select the status you want the dimension to be set to.</td></tr><tr><td>Parameter</td><td>If applicable, select the parameter for the status.</td></tr><tr><td>Timer (sec)</td><td>If desired, enter a delay (in seconds) for this action to take effect. If left blank, the specified dimensional status will be set as soon as this status is set.</td></tr></table> When finished, you can click on the Add Timed Clear Status Dimension button again to set an additional timed set status action. You can also click on the Delete button to the right of a defined timed clear status dimension action to remove it. CAUTION: If you are using the Reservation, Assigned, or Study Locking status dimensions, it is strongly recommended that Clear Status Dimension timers be configured for each of those dimensions to ensure a study does not wait indefinitely on a user. CAUTION: If you are using the Prefetch status dimension, you should either configure a Clear Status Dimension timer for that dimension or else configure a global “Unread Exams” RealTime Worklist to ensure studies are not missed.	Dimension	Select the dimension whose status you want to automatically set.	Status	Select the status you want the dimension to be set to.	Parameter	If applicable, select the parameter for the status.	Timer (sec)	If desired, enter a delay (in seconds) for this action to take effect. If left blank, the specified dimensional status will be set as soon as this status is set.
Dimension	Select the dimension whose status you want to automatically set.								
Status	Select the status you want the dimension to be set to.								
Parameter	If applicable, select the parameter for the status.								
Timer (sec)	If desired, enter a delay (in seconds) for this action to take effect. If left blank, the specified dimensional status will be set as soon as this status is set.								
Associated Default Actions	Select the Action Icons that will be enabled for studies that have this status. NOTE: When the Retrieve from Archive action icon is included in a worklist, it will be enabled unless there is no archive configured or the study is already online, regardless of whether the option for that action icon is selected here. NOTE: The Communication Panel Action Item will only be available if the optional RadStream component has been enabled.								

If you want this status to appear as an option in the various Status pick lists, select the **Allow this Workflow Status to be presented as an option in picklists** checkbox.

If you want you require that a comment be entered before a study can be manually set to this status from the Workstation, select the **Enable force comment to make the user add comments when updating to this status** checkbox.

When finished, click on the **Add Workflow Status** button to save your changes.

NOTE: Amicas services must be restarted before the newly added status will be available from within RTWL.

b. Modifying an Existing Status

To modify an existing status option, click on the **Modify** link for that status on the main **Define Workflow Status** page. This will bring up the **Modify Workflow Status** page, as shown in the following example:

Figure 275: Modifying an Existing Status Option

- Edit the workflow status as desired and then click on the **Modify Workflow Status** button at the bottom of the screen to save your changes.
- If you want to exit this screen without making any changes, click on the **Cancel** button at the bottom of the screen.
- If this is a custom defined status (as opposed to one of the pre-defined default statuses), you can also delete it from the system by clicking on the **Delete Workflow Status** button at the bottom of the screen.

NOTE: Although you cannot delete any of the pre-defined default statuses, you can prevent them from appearing as an option in any Status pick lists by deselecting the **Allow this Workflow Status to be presented as an option in picklists** checkbox toward the bottom of the screen.

NOTE: Although you cannot modify Aggregated Dimension Statuses if the parent dimension status changed anything, that same change will be reflected in the child dimension.

9.13.9.HL7 Order Status Mapping

Order Status values can be pulled from ORC-5 of the incoming HL7 messages and can be used as filters when setting up individual RTWL and Teaching worklists. They can also be used as query fields on the Query Page and in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation. Because HL7 Order Status values are numeric, Merge PACS can be configured to map a display name for each possible value (or each range of values), and this display name will be used as a value on worklists and as an option in various pick lists.

Click on the **HL7 Order Status Mapping** option to bring up the **HL7 Order Status Mapping** page, as seen in the following example:

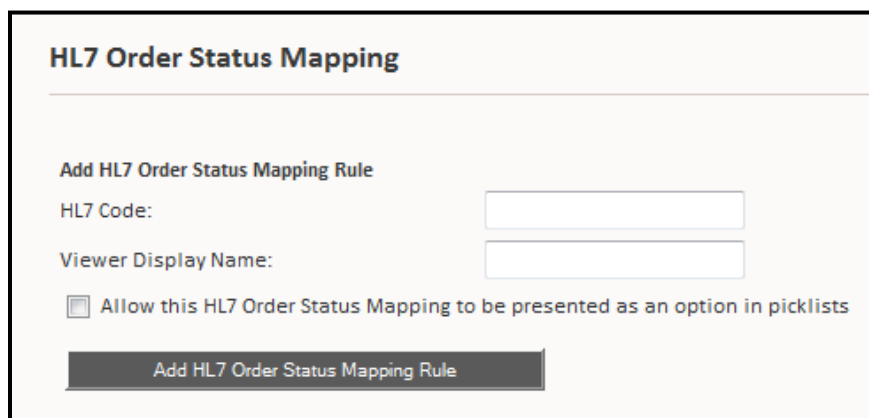
HL7 Code	Viewer Display Name	Display in List	Action
0-29	In Progress	Y	Modify Delete
30-99	Complete	Y	Modify Delete

* The HL7 Code can be configured to map a range of values. The range should be separated using delimiter "-".
Sample Range can be 1-50 for In Progress.

Figure 276: HL7 Order Status Mapping Page

a. Adding a New HL7 Order Status Mapping Rule

To add a new HL7 order status mapping rule, click on the **Add New HL7 Order Status Mapping Rule** button. This will bring up the **Add HL7 Order Status Mapping Rule** page, as seen in the following example:



The screenshot shows a form titled "HL7 Order Status Mapping". Below the title is a section "Add HL7 Order Status Mapping Rule". It contains two input fields: "HL7 Code:" and "Viewer Display Name:". Below these fields is a checkbox labeled "Allow this HL7 Order Status Mapping to be presented as an option in picklists". At the bottom of the form is a button labeled "Add HL7 Order Status Mapping Rule".

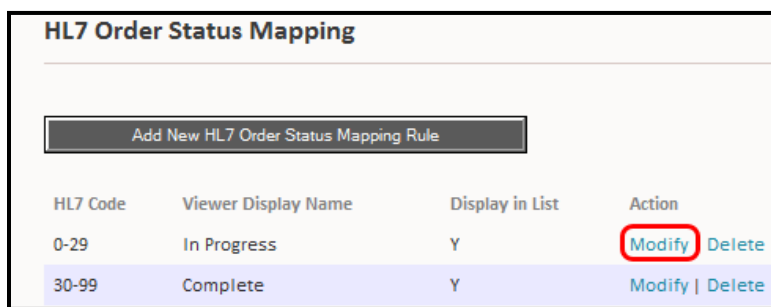
Figure 277: Adding a New HL7 Order Status Mapping Rule

At the Add HL7 Order Status Mapping Rule page, do the following:

- Enter a HL7 Code (or a range of codes, separated by a "-") in the **HL7 Code** field.
- Enter the name for this code (or range of codes) that you want to be displayed in the **Viewer Display Name** field.
- Make sure that the Allow this HL7 Order Status Mapping to be presented as an option in picklists checkbox is selected.
- Click the **Add HL7 Order Status Mapping Rule** button to save the information for this new rule and return to the main HL7 Order Status Mapping page.

b. Modifying an Existing HL7 Order Status Mapping Rule

To modify an existing HL7 order status mapping rule, click on the **Modify** link to the right of that rule's entry on the main HL7 Order Status Mapping page, as in the following example:



The screenshot shows a table titled "HL7 Order Status Mapping". Above the table is a button labeled "Add New HL7 Order Status Mapping Rule". The table has four columns: "HL7 Code", "Viewer Display Name", "Display in List", and "Action". There are two rows of data. In the "Action" column of the first row, the "Modify" link is highlighted with a red circle.

HL7 Code	Viewer Display Name	Display in List	Action
0-29	In Progress	Y	Modify Delete
30-99	Complete	Y	Modify Delete

Figure 278: Modifying an Existing HL7 Order Status Mapping Rule

This will bring up the **Edit an Existing HL7 Order Status Mapping** page, as shown in the following example:

HL7 Order Status Mapping

Edit an existing HL7 Order Status Mapping:

HL7 Code

0-29

Viewer Display Name:

In Progress

☒ Allow this HL7 Order Status Mapping to be presented as an option in picklists

Update HL7 Order Status

Figure 279: Edit an Existing HL7 Order Status Mapping Rule Page

Change the information as desired and then click on the **Update HL7 Order Status** button to record your changes.

NOTE: You can delete HL7 Order Status Mapping rules, as described in paragraph c below. If you want to simply disable this rule without permanently deleting it, however, you can select the **Allow this HL7 Order Status Mapping to be presented as an option in picklists** checkbox on this page.

c. Deleting an Existing HL7 Order Status Mapping Rule

To delete an existing HL7 order status mapping rule, click on the **Delete** button next to that rule’s listing on the main HL7 Order Status Mapping page, as shown in the example below:

HL7 Order Status Mapping

Add New HL7 Order Status Mapping Rule

HL7 Code	Viewer Display Name	Display in List	Action
0-29	In Progress	Y	Modify Delete
30-99	Complete	Y	Modify Delete

Figure 280: Deleting an Existing HL7 Order Status Mapping Rule

9.13.10. Credentialing and Licensure

In some cases, physicians are licensed to read studies only from certain geographic regions, a practice known as “Licensure.” The **Credentialing and Licensing** page allows you to define Licensure Regions as well as associate Facilities to a specific Licensure Region. Once Facilities are associated with Licensure Regions, you can associate one or more Licensure Regions to specific users, enable a filter on RTWL and Teaching worklists to only show exams that match a user’s Licensure Region(s) and enable a Region query field in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation that will let users opt to search among either all exams or only exams that match their specific Licensure Region(s).

NOTE: Changes made to Credentialing and Licensure information will not automatically be synched to the worklist server. You will need to click the **Reload exam contents** link for the worklist server at the Worklist Server Monitor page, as described in subsection 9.6.10 above.

Click on the **Credentialing and Licensing** option to bring up the **Credentialing and Licensing** page, as seen in the following example:

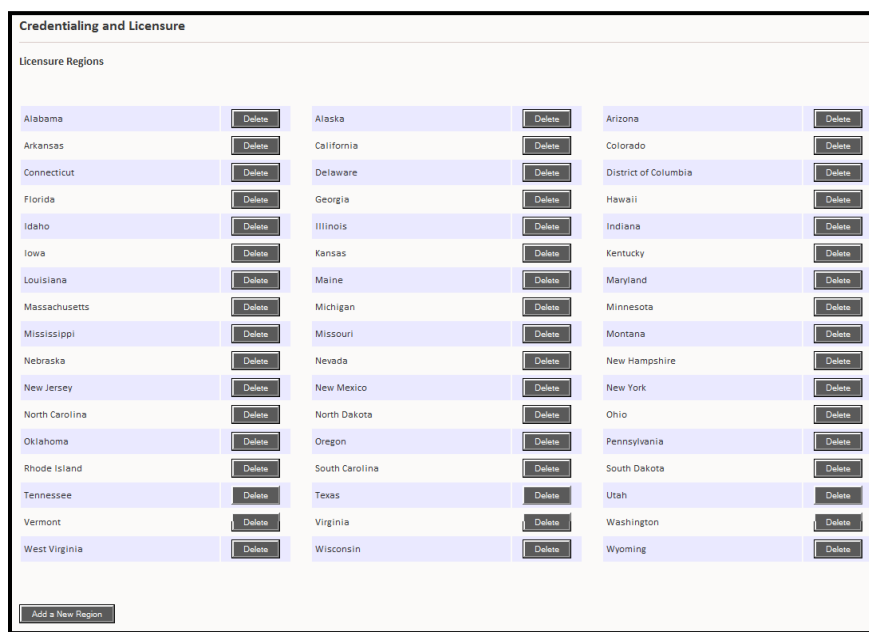
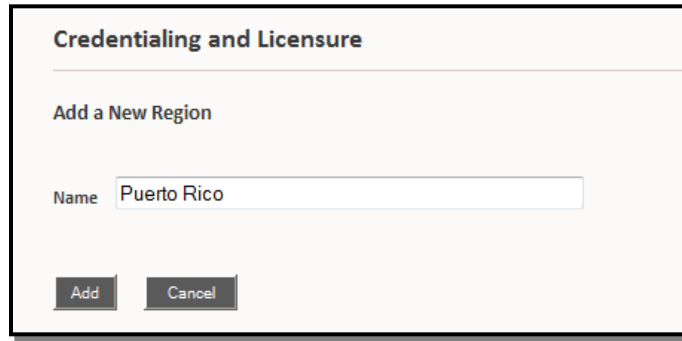


Figure 281: Licensure Regions

NOTE: By default, Merge PACS comes pre-installed with all 50 U.S. States and the District of Columbia defined as Licensure Regions.

- If you need to add a region to the list, click on the **Add a New Region** button at the bottom of the page. This will cause the **Add a New Region** page to be displayed, as in the following example:



Credentiaing and Licensure

Add a New Region

Name

Figure 282: Adding a New Licensure Region

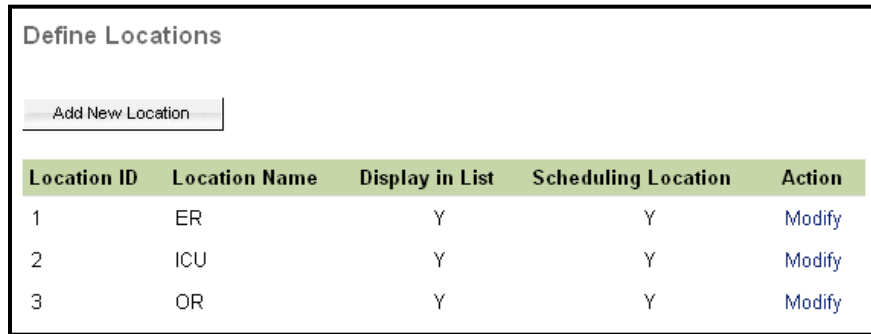
Enter the name of the region in the **Name** field and then click the **Add** button.

- If you need to remove a Licensure Region, click on the **Delete** button next to that region's entry

9.13.11. Locations

The **Define Locations** page allows you to define various locations to be used as filters when setting up individual worklists. You must also define locations if you want them to be displayed in the "View Order" pop-up window from within a worklist.

Click on the **Locations** option to bring up the **Define Locations** page, as seen in the following example:



Define Locations

Location ID	Location Name	Display in List	Scheduling Location	Action
1	ER	Y	Y	Modify
2	ICU	Y	Y	Modify
3	OR	Y	Y	Modify

Figure 283: Defining Locations for RealTime Worklist

a. Adding a New Location

To add a new location, click on the **Add a New Location** button. This will bring up the **Add a New Location** page, as shown in the following example:

Add a new Location:
 Location wildcard definitions are allowed, (fields: Location ID), simply use a * to indicate wildcard. [\[example\]](#)

Location ID:

Location Name:

Figure 284: Adding a New Location

At the **Add a New Location** page, you should enter the following information:

Field	Description
Location ID	This should be the code used by the HL7 system that is received by the HL7 Interface (contained in the PVI.3 field). Note that you can use an asterisk as a wildcard when entering the Location ID (to see how wild cards work, click on the [example] link on the Add a New Location page).
Location Name	Enter a descriptive name for this location.

When finished, click on the **Add Location** button.

b. Modifying an Existing Location

To modify information about a previously defined location, click on the **Modify** button for that location on the main **Define Locations** page. This will bring up the **Edit an Existing Location** page, as shown in the following example:

Edit an existing Location:

Location ID: 2

Location Name:

☒ Allow this Location to be presented as an option in picklists

☒ Allow requests for orders to be submitted for scheduling to this location.

Figure 285: Modifying an Existing Location

At the **Edit an Existing Location** page, you can do the following:

- Change the **Location Name** for this location
- Change whether or not this location will appear as an option in the location pick list when customizing individual worklists
- Change whether or not requests for orders may be submitted for scheduling to this location.

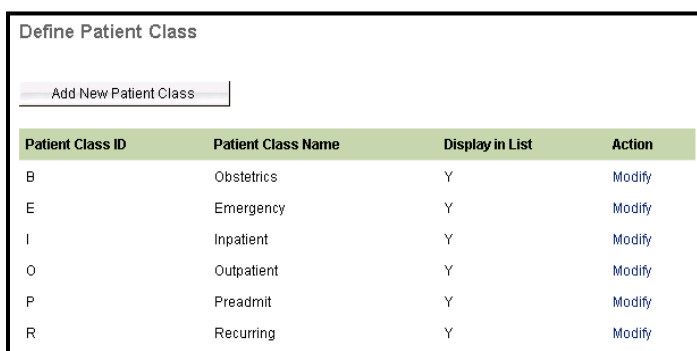
When finished, click on the **Update Location** button.

9.13.12. Patient Classes

A Merge PACS Server comes with a default set of patient class options that can be used with the optional RealTime Worklist component. These options cannot be deleted, but you can add new options, change the names of existing options, and configure whether a particular option will appear on worklist-related picklists.

CAUTION: The list of available patient classes should match those used by the client's HIS/RIS.

Click on the **Patient Classes** option to bring up the **Define Patient Class** page, as seen in the following example:

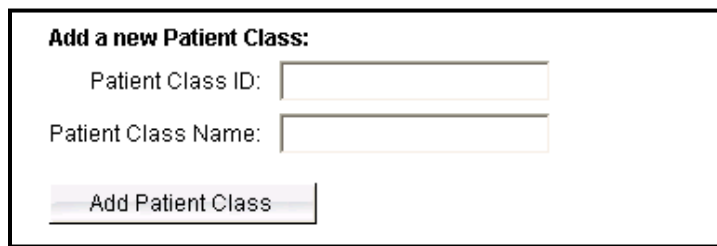


Define Patient Class			
Add New Patient Class			
Patient Class ID	Patient Class Name	Display in List	Action
B	Obstetrics	Y	Modify
E	Emergency	Y	Modify
I	Inpatient	Y	Modify
O	Outpatient	Y	Modify
P	Preadmit	Y	Modify
R	Recurring	Y	Modify

Figure 286: Defining Patient Classes for RealTime Worklist

a. Adding a New Patient Class

To add a new patient class, click on the **Add a New Patient Class** button to bring up the **Add a New Patient Class** page, as shown in the following example:



Add a new Patient Class:

Patient Class ID:

Patient Class Name:

Figure 287: Adding a New Patient Class

Enter the **Patient Class ID** and the **Patient Class Name** for this class and click on the **Add Patient Class** button.

b. Modifying an Existing Patient Class

To modify an existing patient class, click on the **Modify** link to the right of that class's entry on the main Define Patient Class page. This will bring up the **Edit an Existing Patient Class** page, as shown in the following example:

Edit an existing Patient Class:

Patient Class ID: B

Patient Class Name:

☒ Allow this Patient Class to be presented as an option in picklists

Figure 288: Modifying an Existing Patient Class

Change the information as desired and then click on the **Update PatientClass** button to record your changes.

NOTE: Although you cannot delete a patient class once it has been created, you can prevent it from appearing as an option in any Status pick lists by deselecting the **Allow this Patient Class to be presented as an option in picklists** checkbox toward the bottom of the screen.

9.13.13. Manage Teaching Tags

The Teaching Tags page allows you to manage the tags that can be applied to studies for teaching and/or conferencing purposes via the Study Tagging window within the Merge PACS Workstation, as shown in the following example:

Manage Teaching Tags

1	☒	string tag	Accreditation	Edit List
2	☒	string tag	Body Part	
3	☒	numeric tag	Quality Rank	
4	☒	string tag	Secondary Description	Edit List
5	☒	string tag	Syndrome/Abnormality	Edit List
6	☒	string tag	Use/Conference	Edit List
7	☒	date tag	Conf Date	
8	☒	numeric tag	Value	
9	☒	string tag	Notable structure	Edit List
10	☒	string tag	Other	
11	☒	string tag	Units	

Figure 289: Managing Teaching Tags

NOTE: In order to view the Manage Teaching Tags page, you must have the following user or group privilege: **Merge PACS Server (V6 Viewer and RTWL) Portal → Teaching Files Privileges → Allow user to Manage Tags**. For information on setting user and group privileges, refer to Section 9.11 above.

At the Manage Teaching Tags page, you can do the following:

- Select which tags should be available for the users to assign to studies
- Edit the name of the available tags
- Edit the picklist values for each tag

a. Configuring Availability of Tags

To configure which tags will be available to users at the Study Tagging window, select or deselect the checkbox next to the desired tags, as shown below:

Manage Teaching Tags				
1	☒	string tag	Accreditation	Edit List
2	☒	string tag	Body Part	
3	☒	numeric tag	Quality Rank	
4	☒	string tag	Secondary Description	Edit List
5	☒	string tag	Syndrome/Abnormality	Edit List
6	☒	string tag	Use/Conference	Edit List
7	☒	date tag	Conf Date	
8	☒	numeric tag	Value	
9	☒	string tag	Notable structure	Edit List
10	☒	string tag	Other	
11	☒	string tag	Units	

Figure 290: Selecting Tags to Display

b. Editing the Names of Tags

To edit the name of any existing tag (except for “Body Part” and “Units”), enter the desired name directly into the field associated with the tag, as in the following example:

Manage Teaching Tags				
1	☒	string tag	Geographic Region	Edit List
2	☒	string tag	Body Part	
3	☒	numeric tag	Quality Rank	

Figure 291: Editing a Tag Name

c. Editing Picklist Values of Tags

To edit the values that will display in the picklists associated with this tag at the Study Tagging window, click on the **Edit List** button next to the tag whose values you want to change, as shown below:

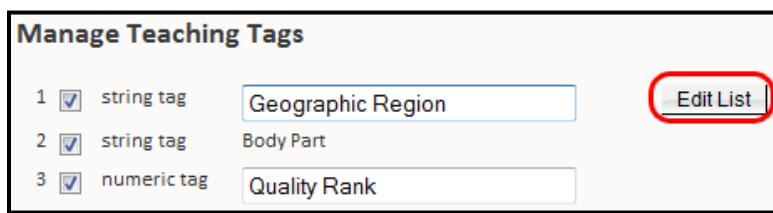


Figure 292: The Edit List Button

The Edit List interface for the selected tag will display on the right, as in the following example:

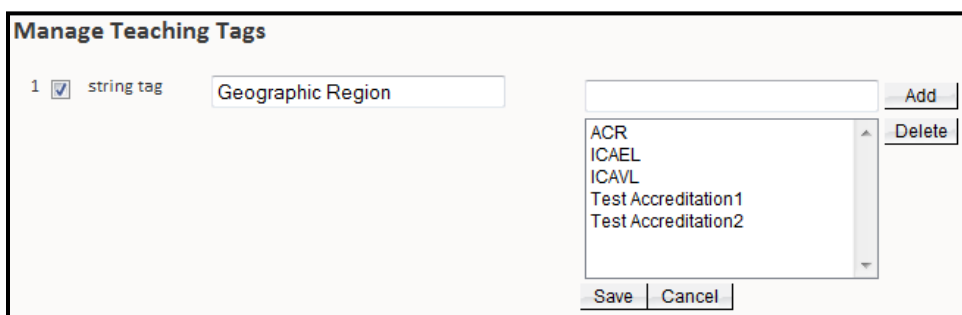


Figure 293: Editing a Teaching Tag

- To **add** a new value, enter the value in the field at the top of the Edit List interface and click on the **Add** button.
- To **delete** a value, select it from the list of existing value and click on the **Delete** button.
- When finished editing values for this tag, click on the **Save** button below the Edit List interface.

NOTE: You cannot directly rename or replace an existing tag value; instead, you must delete the old value and add a new one.

d. Accessing the Manage Teaching Tags Page with Internet Explorer 11

If you used Internet Explorer 11 to access the Manage Teaching Tags Management Page prior to Merge PACS being upgraded from v6.x to v7.1.1, any changes you made to this page may not be retained post-upgrade if you still use IE 11 to access the page from the same workstation. To avoid this issue, you can either use a lower version of IE to access the page (such as IE 10, 9 or 8) or else clear the browser cache once for each workstation after the upgrade, as described below.

To clear the browser cache for IE 11:

1. With the browser running, press **Ctrl+Shift+Del** to open the “Delete Browsing History” dialog.
2. Select only the following items:
 - **Preserve Favorites website data**
 - **Temporary Internet files and website files**

- **Cookies and website data**
- **History**

3. Click the **Delete** button.

9.13.14. IPID Configuration

Issuer of Patient ID (IPID) can be used in conjunction with MRN to identify patients, especially in cases where multiple institutions are sending data to Merge PACS and might use the same MRN for different patients. The IPID value for a particular study will be displayed within the Viewer wherever MRN is displayed. It can also be used as a filter when setting up individual RTWL and Teaching worklists and as a query field on the Query Page and in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation.

IPID values are created automatically by Merge PACS based on the Issuer of Patient ID DICOM attribute (0010, 0021) of studies that are imported into the system and PID-3.4 of incoming HL7 messages. If no IPID is found in either the DICOM header or the HL7 message, a default IPID value will be assigned to that study.

The **IPID Configuration** page allows you to configure the default IPID value as well as define the list of available IPID values that can be used as filters within the Query Page and the various types of worklists. Click on the **IPID Configuration** option to bring up the **IPID Configuration** page, as seen in the following example:

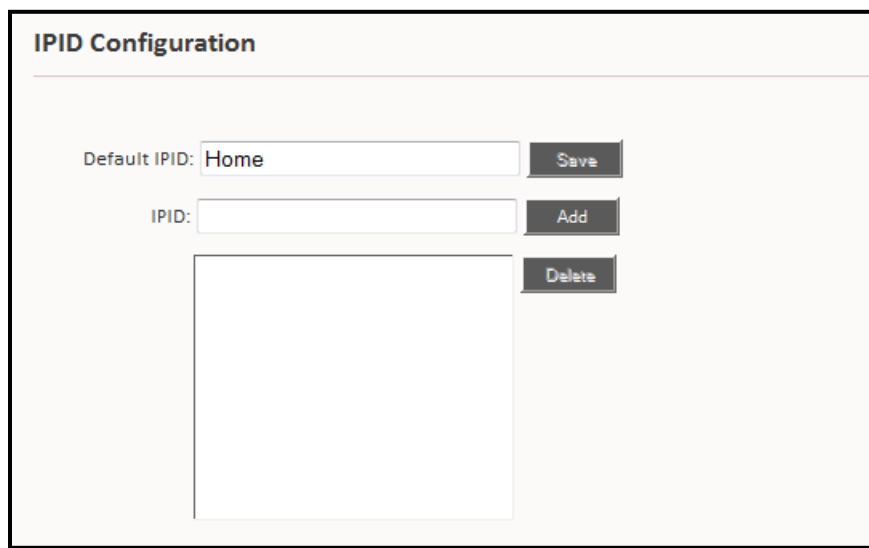


Figure 294: IPID Configuration Page

NOTE: IPID Values for existing studies, orders and reports can be changed on a system-wide basis by using the ConvertIPID tool. Contact Merge Healthcare Support for more information.

CAUTION: IPID values are case sensitive. IPID Value 'Home' and 'HOME' are treated as two separate values for patient identification.

CAUTION: For every unique patient ID numbering scheme, there must be in place a unique IPID or a unique patient ID prefix; otherwise, identity collisions/patient merges may occur.

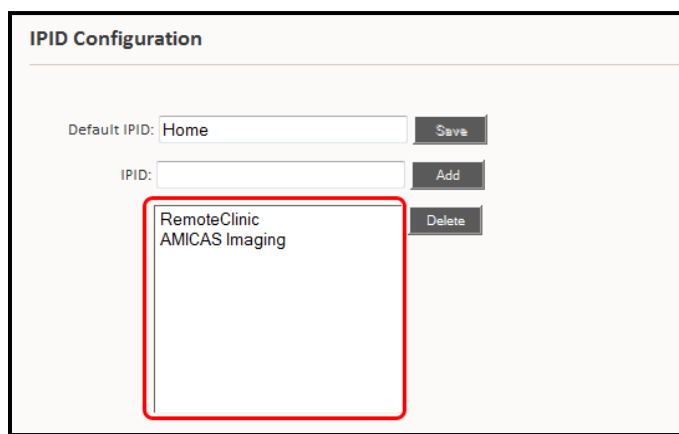
a. Changing the Default IPID Value

To change the default IPID value that will be used when no IPID is found in either the DICOM header or the HL7 message, enter the desired value in the **Default IPID** field and click on the **Save** button.

NOTE: The default IPID value will only apply to future studies that arrive with no IPID information.

b. Adding New IPID Values as Picklist Menus

To add an IPID value for use as a filter with the Query Page and various worklists, enter the desired value in the **IPID** field and click on the **Add** button. The new value will be added to the list of available IPID values for those filters, as in the following example:



The screenshot displays the 'IPID Configuration' window. It features a 'Default IPID' field with the value 'Home' and a 'Save' button. Below this is an 'IPID' field with an 'Add' button. A list of available IPID values is shown, with 'RemoteClinic' and 'AMICAS Imaging' highlighted. A 'Delete' button is also present.

Figure 295: Available IPID Values

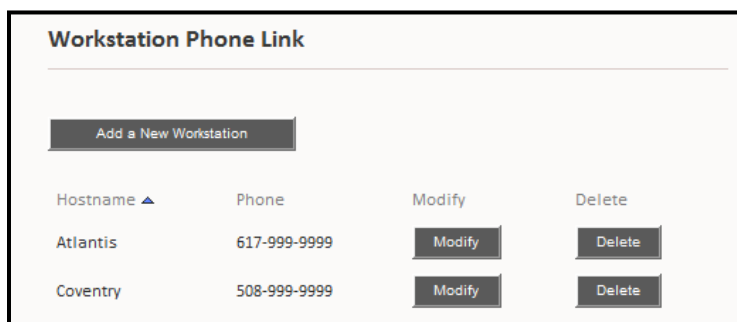
NOTE: Any IPID value entered here must match exactly the actual IPID value from the DICOM header or HL7 message.

c. Removing IPID Values from Picklist Menus

To remove an IPID value from the list of available IPID values, click on the IPID value once to highlight it and then click on the **Delete** button.

9.13.15. Workstation Phone Link

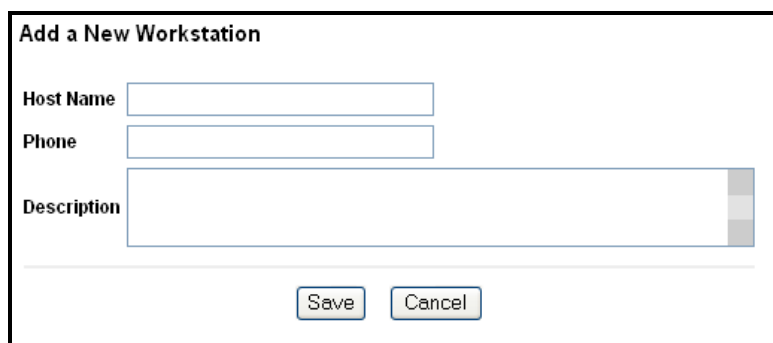
If the optional Merge RadStream component has been enabled, the **Workstation Phone Link** page allows you to add, modify or delete phone links for individual Merge PACS Workstations that can be selected from the RadStream Communication Workpanel, as shown in the example below:



Hostname ▲	Phone	Modify	Delete
Atlantis	617-999-9999	<button>Modify</button>	<button>Delete</button>
Coventry	508-999-9999	<button>Modify</button>	<button>Delete</button>

Figure 296: The Workstation Phone Link Page

- To add a new workstation to the list, click on the **Add a New Workstation** button to bring up the **Add a New Workstation** window, as in the following example:



Add a New Workstation

Host Name

Phone

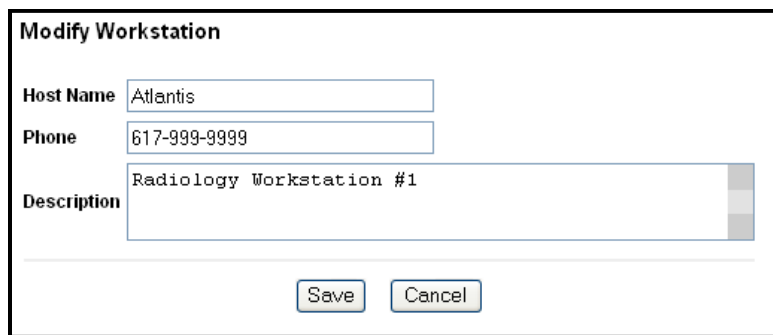
Description

Save Cancel

Figure 297: The Add a New Workstation Window

Enter the **Host Name**, **Phone Number** and **Description** in the fields provided and click on the **Save** button to record your changes.

- To modify information about a workstation already on the list, click on the **Modify** button for that workstation to bring up the **Modify Workstation** window, as in the following example:



Modify Workstation

Host Name

Phone

Description

Save Cancel

Figure 298: The Modify Workstation Window

Modify the information as necessary and click on the **Save** button to record your changes.

- To delete a workstation from the list, click on the **Delete** button for that workstation.

9.13.16. Standard Comments

The **Standard Comments** page allows you to add, modify or delete standard comments that can be selected from various places within the Merge PACS Workstation (including the **Order Viewer**, the **Communication WorkPanel**, the **ER WorkPanel** and the **VoiceClip Player**), as shown in the example below:

Comment	Type	Move Up	Move Down	Modify	Delete
Paged Doctor	OP		Down	Modify	Delete
Doctor did not return page, paged doctor again.	OP	Up		Modify	Delete

Figure 299: The Standard Comments Page

- To add a new standard comment to the list, click on the **Add a New Comment** button to bring up the **Add a New Comment** window, as in the following example:

Figure 300: The Add a New Comment Window

- Enter the text of the comment in the **Comment** field.
- If the optional Merge RadStream component has been enabled, select the type of comment (**OP**, **ER** or **TC**) from the drop-down **Type** menu (if RadStream is not enabled, only the **ER** and **TC** options will be available).
- Click on the **Save** button to record your changes.

- To modify an existing comment, click on the **Modify** button for that comment to bring up the **Modify Comment** window, as in the following example:

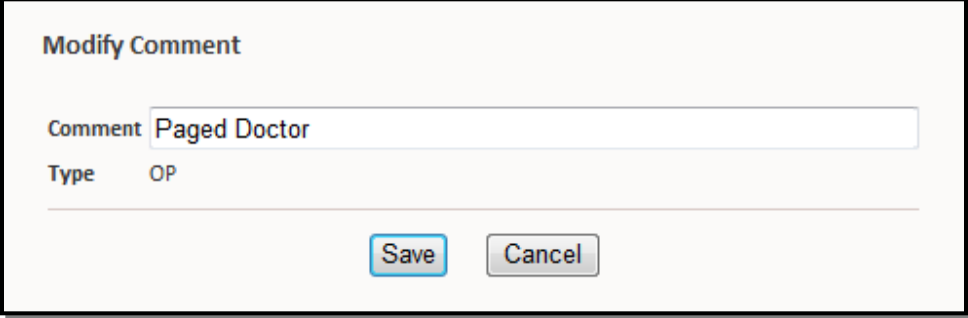


Figure 301: The Modify Comment Window

- Modify the information as necessary
 - Click on the **Save** button to record your changes.
- If the optional Merge RadStream component has been enabled, you can filter the list of comments on the main Standard Comments page by selecting the desired type of comment from the drop-down **Filter by Comment Type** menu (if RadStream is not enabled, only the **ER** option will be available).
- To change the order in which the defined comments will appear at the Merge RadStream Communication Workpanel, click on the **Up** and **Down** buttons for the various comments until they are ordered the way you would like.
- To delete a comment from the list, click on the **Delete** button for that comment.

9.13.17. Refresh Rate

The **Refresh Rate** page allows you to specify how often the data on the various management pages should be refreshed, as shown in the example below:

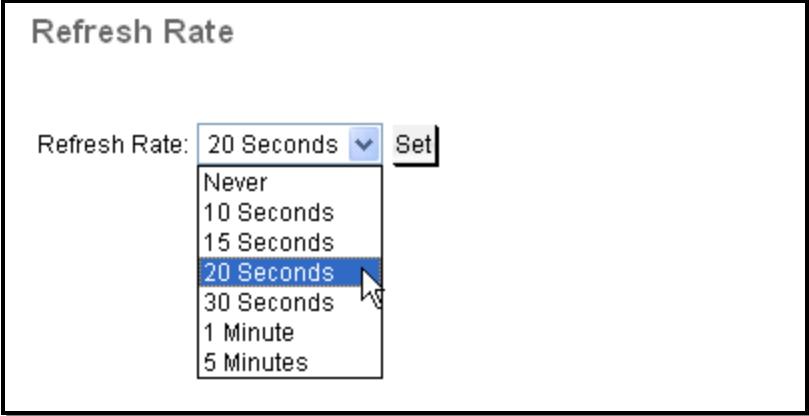


Figure 302: Setting the Refresh Rate

To set the refresh rate, select an option from the pull-down menu and click on the **Set** button.

9.13.18. CustomTag Configuration

The **Custom Tag Configuration** page allows you to map specific DICOM tags that are not typically used to values that can be used as query fields on the Query Page and in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation. They can also be used as column headers for RTWL and Teaching worklists and can be referenced in certain DICOM Header Modification Rules when Merge PACS is configured to run in **Standalone** mode, as described in subsection 1.2.31 above.

a. Configuring Custom Tags

Click on the **Custom Tag Configuration** option to bring up the **Custom Tag Configuration** page, as seen in the following example:

Custom Tag	Attributes	Display Name
1	0x0023,0x1055	Custom1
2	0x0023,0x1056	Custom2
3	0x0023,0x1057	Custom3
4	0x0023,0x1058	Custom4
5	0x0023,0x1059	Custom5
6	0x0023,0x1060	Custom6
7	0x0023,0x1061	Custom7
8	0x0023,0x1062	Custom8
9	0x0023,0x1063	Custom9
10	0x0023,0x1064	Custom10

* Empty Fields get filled by the application itself

Figure 303: Configuring Custom Tags

- Enter the name of this set of custom tags in the **Custom Tag ID** field.
- Enter the desired display name for one or more custom tags in the appropriate **Display Name** fields.
- Click on the **Save** button to record your changes,

NOTE: In addition to customizing the Display Name for each of the custom tags, you can ask Merge Healthcare Support to configure which custom tags you want to appear as options on the Query Page and in the Quick Filters panel. By default, custom tags #1-10 will be displayed as options, so you should remove any that you have not given display names on this page.

b. Accessing the Custom Tag Configuration Page with Internet Explorer 11

If you used Internet Explorer 11 to access the Custom Tag Configuration Management Page prior to Merge PACS being upgraded from v6.x to v7.1.1, any changes you made to this page may not be retained post-upgrade if you still use IE 11 to access the page from the same workstation. To avoid this issue, you can either use a lower version of IE to access the page (such as IE 10, 9 or 8) or else clear the browser cache once for each workstation after the upgrade, as described below.

To clear the browser cache for IE 11:

1. With the browser running, press **Ctrl+Shift+Del** to open the “Delete Browsing History” dialog.
2. Select only the following items:
 - **Preserve Favorites website data**
 - **Temporary Internet files and website files**
 - **Cookies and website data**
3. Click the **Delete** button.

9.13.19. Upload Action Button Icons

The **Upload Action Button Icons** page allows you to upload custom icons that will be added to the list of available icons when editing toolbar action buttons, as described in Appendix A.2.19 below, as in the following example:

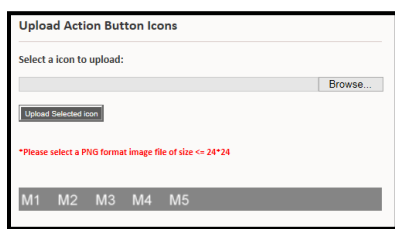


Figure 304: The Workstation Phone Link Page

Click on the **Browse** button to select the desired icon from the local Server or an available network drive or attached storage device. Then, click the **Upload Selected Icon** button to add the icon to the list of custom icons on this page.

Note the following:

- The icon must be in PNG format.
- The icon must be no larger than 24 pixels by 24 pixels
- The file name of the icon must not contain spaces
- Once uploaded, custom icons cannot be deleted from the Upload Action Button Icons management page.

NOTE: If you need a custom icon to be deleted, please contact Merge Healthcare Support.

Chapter 10. Advanced Workflow Management

10.1. Overview

The Advanced Workflow Management Pages provides you with a number of additional management tools specifically designed for advanced workflow, including the following:

- A special Administrator Dashboard that lets you quickly view information regarding the reading physicians and their current assignments.
- The ability to manage auto-assignment rules, status triggers and time triggers to provide for the auto-assignment of studies to individual users.
- The ability to configure Institution Names, Facilities, non-reading work, reader locations and create reader availability custom tags.

NOTE: The Administrator Dashboard will only have data if Merge PACS is configured to integrate with an Analytics / Dashboard Server.

NOTE: You must have the **Allow access to Advanced Workflow Configuration** privilege to access the Advanced Workflow Management pages.

In addition, the following advanced workflow items can be configured from the **Settings** dialog within the Merge PACS Workstation:

- **Practices**
- **Specialties**
- **Orderers**
- **Order/Practice Associations**
- **Orderer/Reader Associations**

NOTE: **Procedures** can also be configured from the Settings dialog within the Workstation, even though the use of Procedures is not specific to Advanced Workflow.

Once all relevant information has been configured, both from the Management Pages and the Workstation Settings dialog, you can also configure the advanced workflow preferences for individual users from the Workstation Settings dialog, as described in Section 12.4 below.

10.2. Accessing the Advanced Workflow Management Pages

The Advanced Workflow Management Pages are accessed from the Merge PACS Management Pages by clicking on the **Advanced Workflow** tab, as in the following example:

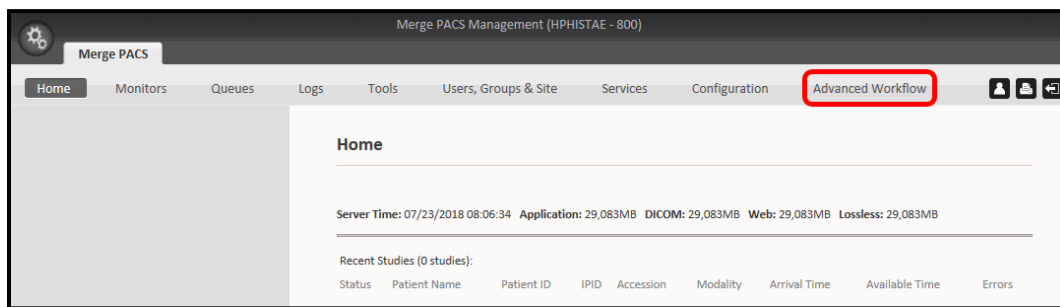


Figure 305: Accessing the Advanced Workflow Management Pages

When you click on the Advanced Workflow tab, the Advanced Workflow Management Pages will be displayed with the Home Page (the Dashboard) showing by default, as in the following example:

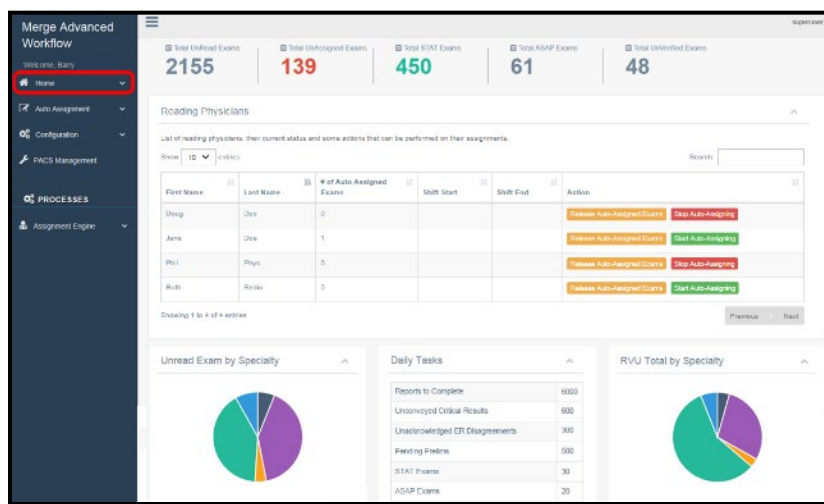


Figure 306: The Advanced Workflow Management Pages - Home

To return to the main Merge PACS Management Pages, click on the **PACS Management** link in the left-hand navigation panel, as in the following example:

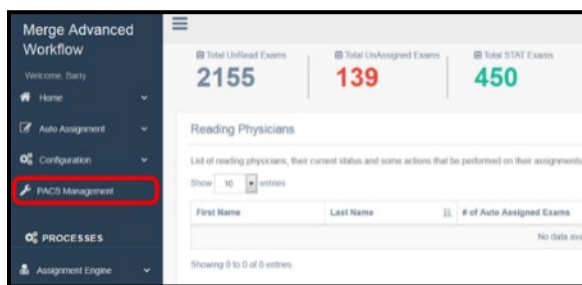


Figure 307: Link to the Merge PACS Management Pages

10.3. The Advanced Workflow Administrator Dashboard

The Advanced Workflow Administrator Dashboard (“the Dashboard”) allows you to view and analyze data from Merge PACS, including the ability to track service level agreements, productivity, turnaround time, and patient wait time. Statistical information provided on the Administrator Dashboard will only be available if Merge PACS is configured to integrate with an Analytics / Dashboard Server and, depending on how Merge PACS is configured, the Analytics / Dashboard Server will consider one or more of the following as “exams” when determining these statistics:

- Exams that have both a study and an HL7 order
- Studies that are not expected to have an HL7 order (“lone studies”).
- Completed HL7 orders that are not expected to receive a study (“outside exams”).

10.3.1. Accessing the Dashboard

The Dashboard is displayed automatically by default when you first access the Advanced Workflow Management Pages, but can be accessed at any time from within the Advanced Workflow Management Pages by clicking on the **Dashboard** link under **Home** in the left-hand navigation panel, as in the following example:

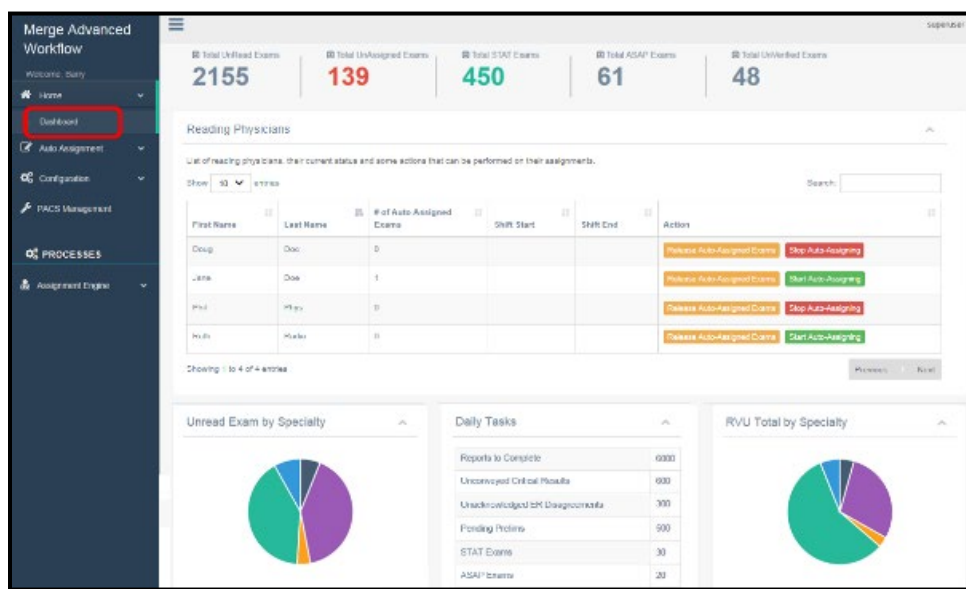


Figure 308: The Advanced Workflow Administrator Dashboard

10.3.2. Viewing Exam Status

If Merge PACS is configured to integrate with an Analytics / Dashboard Server, the top section of the Dashboard allows you to view the following statistics that provide an option for the administrator to see how many studies are read by the reader and how many studies remain to be read:

Status	Description
Total UnRead Exams	Study must be linked with order and the study status should not set to READ, TRANSCRIBED, PRELIM or FINAL. Under Diagnostic workflow dimension, a site can add custom workflow statuses based on their workflow. In that case, all the statuses that come after READ should be excluded for unread exams. For example, if a site has custom status like READREVIEW after READ, then the following statuses should be excluded for unread exams (READ, READREVIEW, TRANSCRIBED, PRELIM or FINAL). The display order of workflow status (custom and predefined) can be changed in Workflow Status page in PACS Management pages.
Total UnAssigned Exams	Total count of Unread exams which are not Assigned .
Total STAT Exams	Total count of UnRead Exams and order priority is set to STAT .
Total ASAP Exams	Total count of UnRead Exams and order priority is set to ASAP .
Total Unverified Exams	Total count of exams whose diagnostic status is Unverified . The unverified studies with or without order will be included in the count.
Server Status	The Server status indicator is used to show the health status of Merge PACS integration with Dashboard. This indicator will show proper status only if the property Analytics.Server. InstanceNames in AnalyticServerConfig.properties is configured properly.

10.3.3. Viewing Reading Physician Status

In the **Reading Physicians** area, you can view a list of reading physicians, the number of auto-assigned exams, when their shift starts and ends, and any actions that can be performed on their assignments, as in the following example:

Total UnRead Exams	2155	Total UnAssigned Exams	139	Total STAT Exams	450	Total ASAP Exams	61	Total Unverified Exams	48
--------------------	------	------------------------	-----	------------------	-----	------------------	----	------------------------	----

Reading Physicians					
List of reading physicians, their current status and some actions that can be performed on their assignments.					
Show 10 entries		Search:			
First Name	Last Name	# of Auto Assigned Exams	Shift Start	Shift End	Action
Doug	Doe	0			Release Auto-Assigned Exams Stop Auto-Assigning
Jane	Doe	1			Release Auto-Assigned Exams Start Auto-Assigning
Phil	Phys	0			Release Auto-Assigned Exams Stop Auto-Assigning
Ruth	Radio	0			Release Auto-Assigned Exams Start Auto-Assigning

Figure 309: The Advanced Workflow Administrator Dashboard – Reading Physician Status

The following information is displayed:

Label	Description
First Name	First name of the reading physician.
Last Name	Last name of the reading physician.
# of Auto Assigned Exams	Number of exams that are automatically assigned to the reading physician.
Shift Start	The reading physician shift start date and time.
Shift End	The reading physician shift end date and time.

In addition, the **Action** column includes the following buttons that you can click to perform specific actions:

Button	Description
Release Auto-Assigned Exam	Release all the currently assigned exams from that particular reading physician, which will allow you to reassign all current user exams to others.
Stop Start Auto-Assigning	Toggle on and off whether auto-assignment is enabled for this reading physician.

10.3.4.Viewing Unread Exams by Specialty

If Merge PACS is configured to integrate with an Analytics / Dashboard Server, the UnRead Exam by Specialty section provides information about the total count of the unread exams with respect to the exams assigned to the specialty, as in the following example:

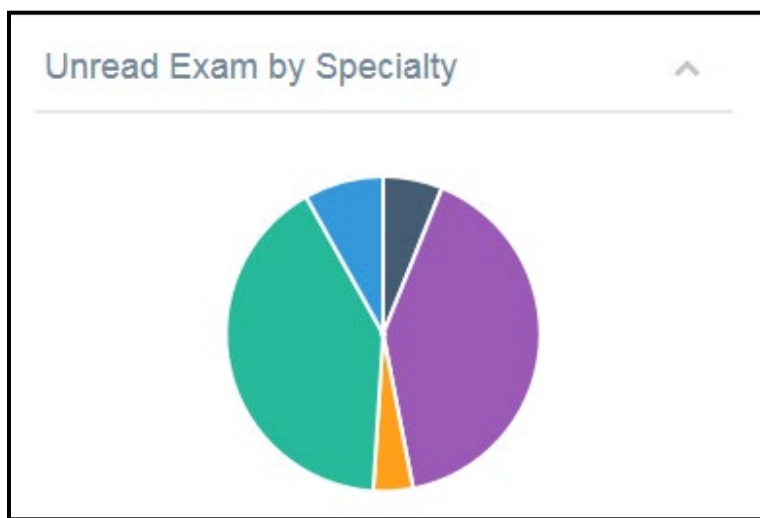


Figure 310: Unread Exam by Specialty

Each exam will be associated to one or more procedures in the system (with the help of procedure code) and every procedure in the system will be mapped to one Specialty.

Hovering your mouse over each section of the chart will display a tooltip containing the exam count for the specialty represented by that section, as in the following example:

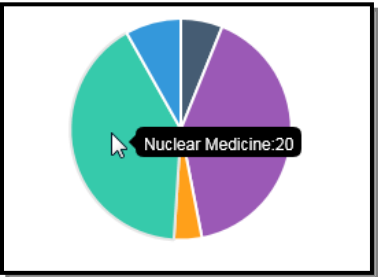


Figure 311: Unread Exam by Specialty Hover Text

10.3.5.Viewing Daily Tasks

If Merge PACS is configured to integrate with an Analytics / Dashboard Server, the Daily Tasks section shows the statistics about the current system's status, as in the following example:

Daily Tasks	
Reports to Complete	60
Unconveyed Critical Results	250
Unacknowledged ER Disagreements	23
Pending Prelims	212
STAT Exams	75
ASAP Exams	57

Figure 312: Administrator Daily Tasks

The following daily tasks are available:

Task	Description
Reports to Complete	Count of the reports that are assigned to all readers and left to approve. This count include exams that has workflow status >= READ and <=FINAL.
Unconveyed Critical Results	Count of exams in the system that has critical results that are yet to be conveyed. This count includes all the exams in the system with status 'Convey result (CC)' under 'Communication' dimension (Irrespective of study with order or without order).
Unacknowledged ER Disagreements	Count of exams in the system that has Rad disagreements yet to be acknowledged. This count includes all the exams in the system with status 'Rad Disagree (ED)' under 'ER' dimension (irrespective of study with order or without order).

Task	Description
Pending Prelims	Includes the number of preliminary exams waiting for signature. This count includes exams with Status PRELIM >= and <= FINAL. This count includes exams with custom status that comes between PRELIM and FINAL.
STAT Exams	Count of Unread exams with workflow priority 'STAT', SUPER-STAT and STROKE. Note: This count includes only unread exams.
ASAP Exams	Count of Unread exams with workflow priority 'ASAP'. Note: This count includes only unread exams.

10.3.6. Viewing RVU Total by Specialty

If Merge PACS is configured to integrate with an Analytics / Dashboard Server, the RVU Total by Specialty section provides information about the sum of Relative Value Units (RVU) totals of the unread exams read by specialty, as in the following example:

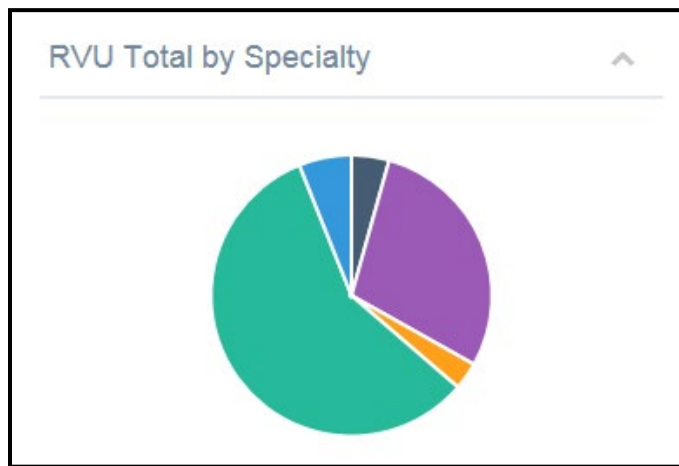


Figure 313: RVU Total by Specialty

Hovering your mouse over each section of the chart will display a tooltip containing additional information for the specialty represented by that section, as in the following example:

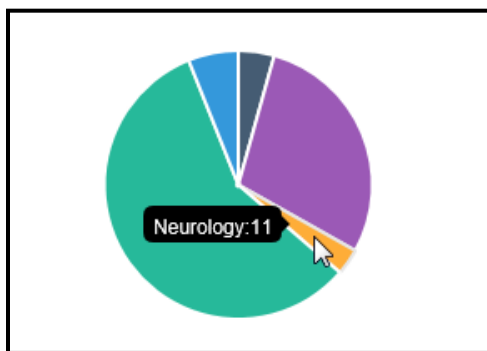


Figure 314: RVU Total by Specialty Hover Text

10.4. Auto Assignment

In addition to manually assigning exams to users (*i.e.*, using the Assign Study action in a worklist to assign each exam to the appropriate radiologist), you can also define assignment rules and triggers points to optimize workflow depending on the business or reading model. Auto Assignment allows you to generate these worklists dynamically for different professionals and to customize your work environment.

When an exam is automatically assigned to a specific user, a flag will be set indicating that the exam is so assigned. A worklist can then be configured to display all exams assigned to that user.

The exams can be automatically assigned to the best reader based on the following:

- Location
- Specialties
- Experience
- Credentials and Licenses
- Urgency

10.4.1. Overview

The Auto Assignment workflow is as follows:

- Identify the type of exam with a set of rules based on the matching criteria.
- Select the set of candidates who can be assigned this exam.
- Sort the potential candidates based on criteria from the rule so that you can determine the best candidate.

The goal for Auto Assignment is to provide the following benefits:

Benefit	Description
Improved productivity	Automates routine administrative activities and targets the work to the correct reader.
Improved satisfaction	Limits extra information for the reading physician, balances workflow, and removes certain tasks from the administrator.
Improved patient care	Decreases turnaround time and matches studies to the best reader.

By default, the Advanced Distribution Engine will automatically retry auto-assignment when an unassigned exam is sent to the Advanced Distribution Engine and no assignee is found. This is useful, for example, in situations where a qualified reader may log on soon after the initial assignment attempt failed. Also by default, the system will wait five minutes before attempting to retry the assignment and will continue retrying every 20 minutes up to a maximum of 50 attempts. Also by default, exams currently in the retry queue will be given priority over newly imported exams (meaning that, when an exam is sent through the Advanced Distribution Engine, the Advanced Distribution Engine will first attempt to assign all exams in the retry queue that are closer to or further past their “due time” than that exam and then attempt to assign that exam). These defaults are configurable on a site-by-site basis, however.

The Auto Assignment functionality is accessed by clicking the Auto Assignment link in the left-hand navigation panel to display the Auto Assignment menu, as in the following example:

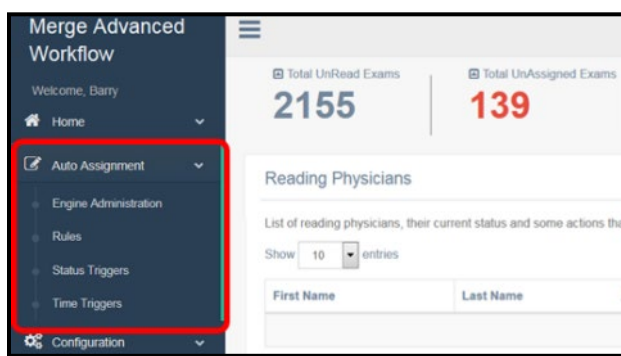


Figure 315: Auto Assignment Menu

10.4.2. Rebalancing All Assignments

From the **Engine Administration** page, you can rebalance all automatically assigned exams for all users. Rebalancing will cause the Advanced Distribution Engine to re-evaluate all exams against all available users, essentially reassigning all exams that are not completed. During reassignment, the current assignee is included.

Note the following:

- Rebalancing of assignments applies to exams in the system that are auto assigned and not completed. Auto-assigned exams are those with MAA status and completed exams are those with MAC status.
- Exams tagged for retry with status MAR or retry failed with status MARF are not included in the rebalance.
- Exams that are not tagged with an auto assignment status DO NOT qualify for this operation.

To reassign all exams:

1. Select **Engine Administration** from the Auto Assignment menu.
2. Click **Reassign all exams**.
A confirmation dialog box appears.
3. Click **OK**.

10.4.3. Configuring Rules

The Rules workflow is as follows:

- Match which conditions should apply to the rule.
- Select who should be considered for the work to be assigned.
- Sort who is best at the time to be assigned the work.

In addition to these three conditions, there is an additional mandatory condition, **Rules evaluation**, which needs to be included in each rule. This condition is used to decide if you need to proceed rule evaluation further with the next rule in the hierarchy **on no assignment done for the current rule**.

When you click on the **Rules** link from the Auto Assignment menu, a list of existing rules (if any) will be displayed in the left-hand navigation panel, as in the following example:

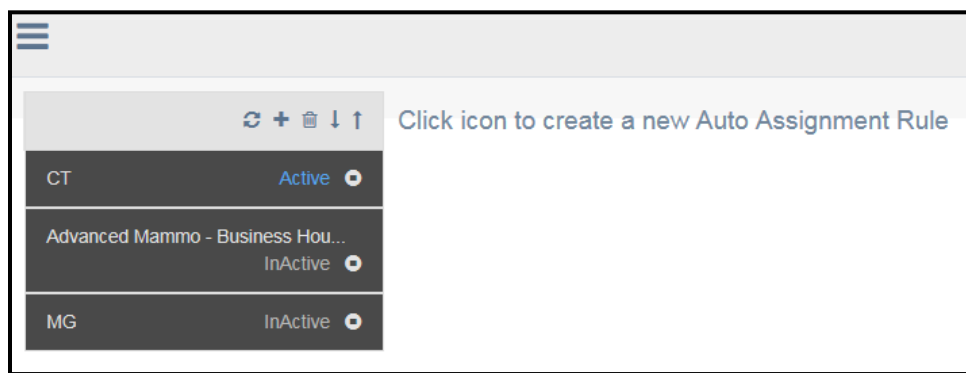


Figure 316: Existing Auto Assignment Rules

a. Creating a New Rule

When creating a new rule, you can define one or more study conditions and reader types, as well as one or more condition categories for each study condition and/or reader type and one or more specific conditions using various logical operators for each category. In addition, you can define the assignment priority for the rule and general rule evaluation properties.

NOTE: The steps below give an example of creating a specific rule, but the general concepts should be applied to create rules of any complexity desired.

To create a new rule:

1. Click the **+** sign at the top of the list of rules to create a new Auto Assignment Rule, as in the following example:



Figure 317: Plus Sign to Create a New Auto Assignment Rule

The **Create New Rule** page is displayed, as in the following example:

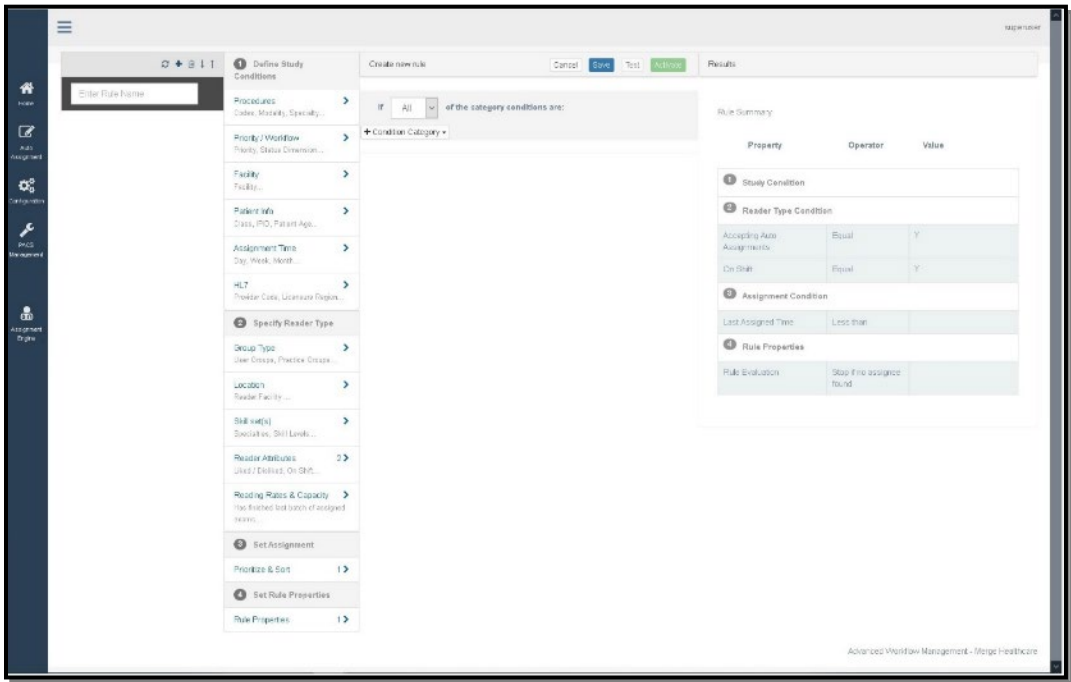


Figure 318: Create New Auto Assignment Rule Page

- 2. In the **Enter Rule Name** box, type a name for the new rule.
- 3. Set one or more study conditions from the **Define Study Conditions** menu, as in the following example:

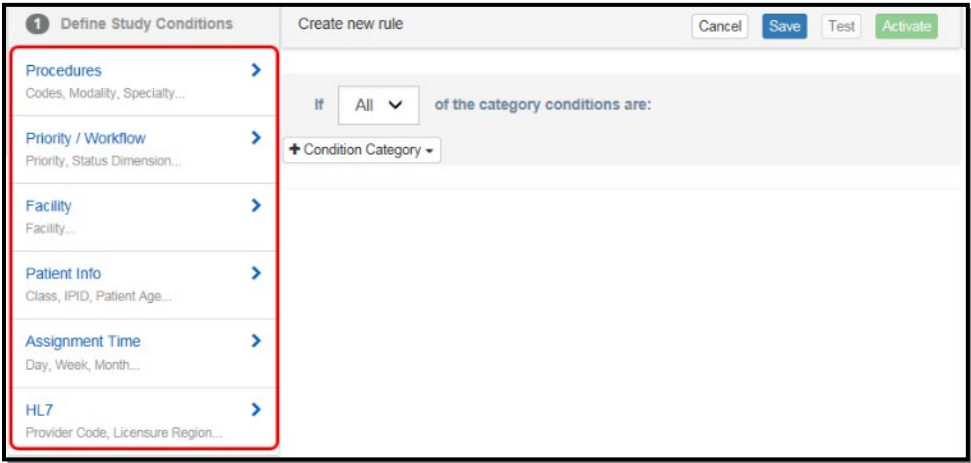


Figure 319: Define Study Conditions Menu

Study conditions determine **which studies** this rule applies to (as well as the general time, if so configured).The following study conditions are available:

Study Condition	Available Categories
Procedures	Procedure Code Description, Modality, Study Description DICOM Tag, Specialty, Measure of Difficulty / Complexity
Priority / Workflow	Priority, Value, Parameter, Note

Study Condition	Available Categories
Facility	Facility
Patient Info	Patient Class, Issuer of Patient ID (IPID), Patient Age in Days, Patient Age in Weeks, Patient Age in Months, Patient Age in years
Assignment Time	Current Time, Current Day of Week, Current Week of Month
HL7	RIS Order Status, Orderer's Provider Code, Licensure Region

- a. Once you have selected a type of study condition, click the **+ Condition Category** button and select the first desired **category** for the the selected study condition from the drop-down menu, as in the following example:

Figure 320: Selecting “Modality” as a Condition Category for the “Procedures” Study Condition

The details section for the selected category is displayed, as in the following example:

Figure 321: Details for First Modality Condition

- b. To define the first **condition** for this category, select the desired logical operator from the drop-down menu and then click on the field to the right to enter (or select from the options available for this condition, if any), as in the following example:

 A screenshot of a web interface for defining study conditions. At the top, there's a section 'If All of the category conditions are:' with a dropdown set to 'All'. Below this is a section 'Modality conditions are:' with a dropdown set to 'All' and a button 'X Delete Category'. The main area shows a logical operator dropdown set to 'Equal' and a text input field. A dropdown menu is open, showing a list of modalities: AUJ, BI, CD, CR, CT, DD, DG, DX, and ECG. The 'CT' option is highlighted with a mouse cursor.

Figure 322: Setting a Condition of Modality Is Equal to “CT”

- c. If desired, click the **+Condition** link to add additional conditions for this category, as in the following example:

 A screenshot of the same web interface as Figure 322, but now showing two conditions. The first condition has the operator 'Equal' and the value 'CT'. The second condition also has the operator 'Equal' and the value 'MG'. Below the conditions is a red-bordered button labeled '+Condition'.

Figure 323: Adding an Additional Condition

- If you include more than one condition for this category, select whether you want **Any** or **All** of the defined conditions to apply, as in the following example:

 A screenshot of the web interface showing the 'Modality conditions are:' section. The dropdown menu is now set to 'Any' and is highlighted with a red box. The two conditions from the previous figure are still present: 'Equal' with value 'CT' and 'Equal' with value 'MR'.

Figure 324: Selecting “Any” or “All”

[In the example above, the rule will apply if modality equals “CT” **or** modality = “MR” since the “Any” option has been selected for this condition category.]

- Click the **X** button to the right of a particular condition to remove that condition.
 - Click the **Delete Category** button to remove this entire condition category from this study condition.
- d. If desired, add additional condition categories to this study condition by selecting the desired category from the drop-down **+ Condition Category** button and then following the instructions in Steps b-c above to define conditions for each additional category.
- e. If desired, repeat steps a-d above to add additional study conditions to this rule.

4. Once you have defined all the desired study conditions for this rule, set one or more reader types from the **Specify Reader Type** menu, as in the following example:

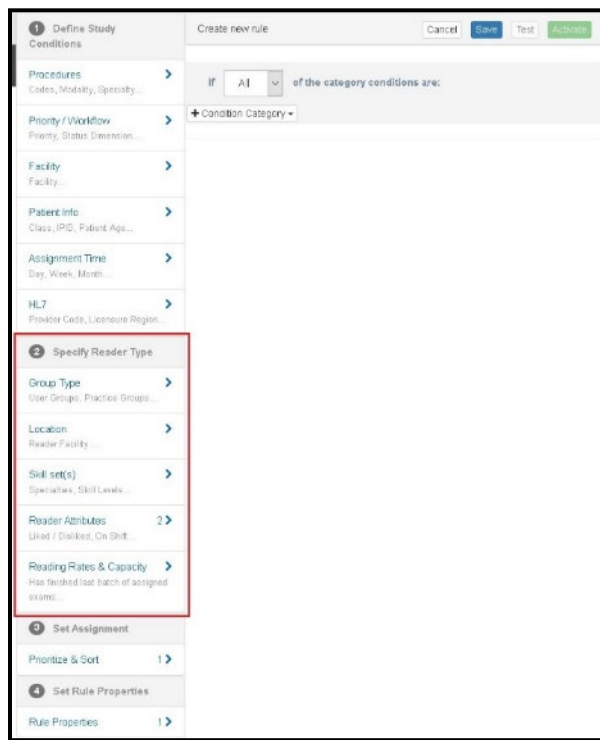


Figure 325: Specify Reader Type Menu

Reader Type determines the pool of potential readers who are eligible to have this study assigned to them. The following reader types are available:

Reader Type	Available Categories
Group Type	User Group, Practice Group, Provider Type, Facility, Modalities
Location	Reader's Facility, Reader's Workstation Location
Skill Set(s)	Specialties, Allowed Specialties per Facility, Allowable Specialties per Facility, Skill Levels per Facility
Reader Attributes	Liked / Disliked by Orderer, Practice Liked / Disliked by Orderer, Accepting Auto Assignments, On Shift, Reader Availability (Shift) Custom Tag
Reading Rates & Capacity	Hourly Reading Rates per Specialty, Default Hourly Reading Rates, Remaining Capacity, Has finished last batch of assigned exams

NOTE: If the "Has finished last batch of assigned exams" condition is enabled, once a user has flipped the hopper status from EMPTYING to FILLING the Advanced Distribution Engine will start the retry queue without waiting for the defined RETRY due time. Hence, the retry exam will be assigned to the user up to the assigned high water mark level.

For each desired Reader Type, define one or more categories and conditions as described in step 3 above.

- Once you have defined all the desired reader types for this rule, you can change the default assignment priority from the “Set Assignment” menu, as in the following example:

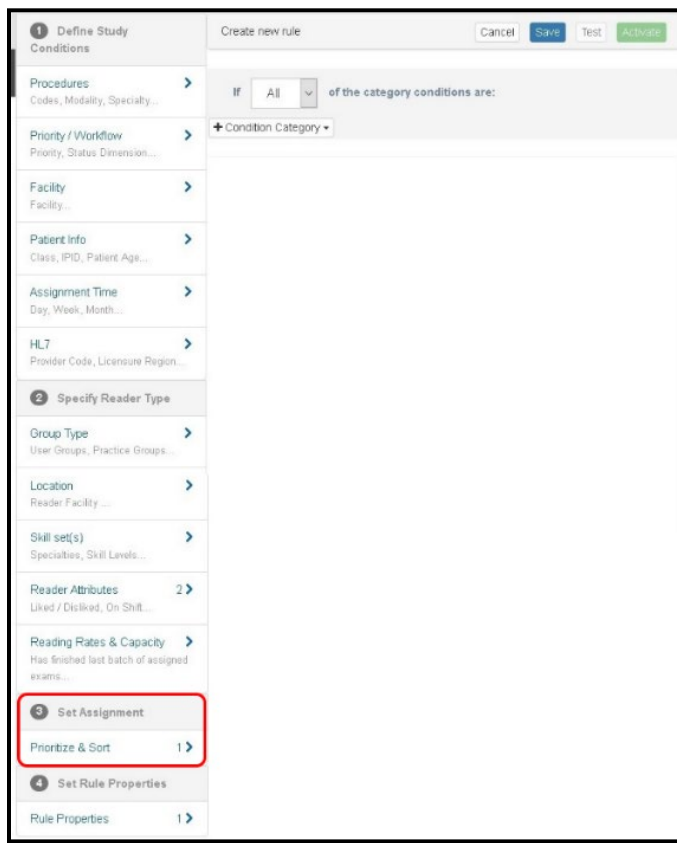


Figure 326: Set Assignment

Since a study can only be assigned to one person at a time, Set Assignment determines specifically **who** the study will be assigned to when multiple readers qualify based on the defined reader types. The only Condition Category available for Set Assignment is **Prioritize & Sort**, which has the following conditions available:

- **Current Load (study count & difficulty scale)**
- **Current Load (study count only)**
- **Last Assigned Time**
- **Remaining Capacity**
- **Skill Level for Capacity**
- **Hourly Reading Rate for Study**

Click on **Prioritize & Sort** and then select and define the desired condition.

NOTE: Only one condition can be set for Prioritize & Sort at a time.

NOTE: “Last Assigned Time” is selected by default. If you want to choose a different condition, you will need to first delete this condition by clicking the **X** to the right of it.

- Once you have defined all the desired reader types for this rule, you can change the default assignment priority from the “Set Assignment” menu, as in the following example

7. Define the general rule evaluation properties for this rule as follows:
 - a. Click the **Rule Properties** option under “Set Rule Properties”, as in the following example:

The screenshot shows the 'Create new rule' dialog box. On the left, a sidebar lists four steps: 1. Define Study Conditions, 2. Specify Reader Type, 3. Set Assignment, and 4. Set Rule Properties. Step 4 is currently selected and highlighted with a red box. Under step 4, the 'Rule Properties' option is also highlighted with a red box. The main area of the dialog shows the 'If All of the category conditions are:' section, which is currently empty. There is a '+ Condition Category' button to add conditions. At the top right, there are buttons for 'Cancel', 'Save', 'Test', and 'Activate'.

Figure 327: Rule Properties

- b. Select the desired Rule Evaluation option, as in the following example:

The screenshot shows the 'Rule Evaluation' dialog box. It contains two radio button options: 'Continue to next rule if no assignee found' and 'Stop if no assignee found'. The 'Stop if no assignee found' option is selected and highlighted with a red box. There is a close button (X) in the top right corner. At the bottom, there is a '+ Condition Category' button. At the top right, there are buttons for 'Cancel', 'Save', 'Test', and 'Activate'.

Figure 328: Rule Evaluation

NOTE: No additional condition categories can be added.

8. Click the **Save** button at the top of the Create New Rule page.
9. If desired, click on the **Test** button at the top of the Create New Rule page.

The results, including the relative size and the list of reader candidates, will be displayed at the bottom right of the page (you may need to scroll down to view the information).

10. Click the **Activate** button at the top of the Create New Rule page.

b. Editing a Rule

To edit an existing rule:

1. On the left-hand navigation bar, click on the existing rule you wish to edit.
The information for this selected rule is displayed.
2. Edit the necessary conditions.
3. If necessary, do any of the following actions:
 - If you want to undo any changes you made, click the **Cancel** button at the top of the page.
 - If you want to simulate the rule evaluation, click the **Test** button at the top of the page.
 - If you want to deactivate a rule, click the **De-activate** button at the top of the page (if the rule is already deactivated, the **Activate** button appears instead).
4. When finished, click the **Save** button at the top of the page.

c. Deleting a Rule

To delete an existing rule:

1. On the left-hand navigation bar, click on the existing rule you wish to delete.
2. Click the trashcan icon at the top of the navigation bar, as in the following example:



Figure 329: Deleting an Auto Assignment Rule

d. Reordering a Rule

When attempting to assign a work item, the Advanced Distribution Engine will select the first rule (from the ordered set of configured rules) whose study conditions are met. So if you have multiple rules, it is important to have them listed in the desired priority. Once a rule is created, it will be added to the bottom of the list by default, but you can reorder the list of rules at any time.

To reorder an existing rule:

1. On the left-hand navigation bar, click on the existing rule you wish to move to a different place in the list.

- Click the up or down arrow icons at the top of the navigation bar to move the rule to the desired location, as in the following example:



Figure 330: Reordering an Auto Assignment Rule

10.4.4. Initiating Trigger Actions

The following triggers can be configured from the Auto Assignment menu to initiate the auto-assignment process:

Type of Trigger	Description
Status Triggers	Triggers based on the workflow of the site such as exam ordered, exam scheduled, exam read, etc. For example, if you have a service level agreement and the agreement is about to expire, you might want to reassign the action.
Time Triggers	Triggers based on the passage of time in the system. For example, if a study has been assigned to a person and the person has not read it, you might want to reassign the study to someone else.

In addition, a variety of trigger options can be manually configured to control when automatic assignment (or reassignment) will actually be performed.

Note the following with regard to status and time trigger:

- One default status trigger is included for the Diagnostic dimension and UNREAD status with the default assignment action set to “Assign if unassigned.” Other Status and Time triggers must be explicitly created.
- New dimensions and statuses do NOT have status triggers automatically generated/created.
- New statuses for existing dimensions do NOT have status triggers automatically generated/created.
- The following dimensions (together with their associated statuses) are considered System/Internal cannot be selected for status triggers:
 - Assignment**
 - AutomaticAssignment**
 - ExamType**
 - StudyLocking**
 - ImageTransfer**
 - Reservation**
 - OrderMatch**
- Manually assigned exams are ignored by Advanced Distribution Engine for status and time trigger activities.

a. Configuring Status Triggers

Status triggers are actions to be performed on an exam status change notification received by the Advanced Distribution Engine. You can define status triggers to tell the Advanced Distribution Engine to attempt automatic assignment or reassignment of an exam when it has a particular status set (e.g., the Unread status in the Diagnostic dimension). These actions are performed at an exam level and apply to one exam only.

You can manage and configure status triggers from the **Manage Status Triggers** page, accessible by clicking on the **Status Triggers** link on the Auto Assignment menu, as in the following example:

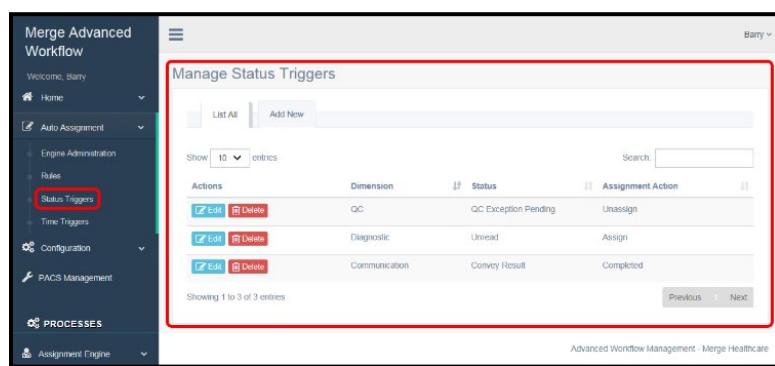


Figure 331: Auto Assignment – Manage Status Triggers

At the Manage Status Triggers page, you can do the following:

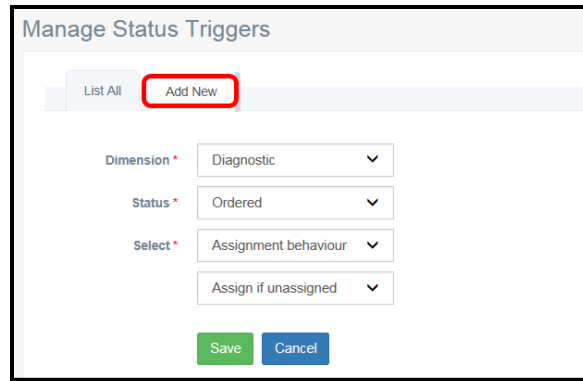
- **View** existing status triggers
- **Create** new status triggers
- **Edit** existing status triggers
- **Delete** existing status triggers

To view existing status triggers:

1. Click the **List All** tab at the top of the Manage Status Triggers page (this tab is selected by default when first accessing the page).
By default, the first 10 existing status triggers will be displayed in an unfiltered list.
2. If desired, change the maximum number of status triggers to be displayed on the page by selecting from the drop-down **Show Entries** menu,
3. To filter the list based on the defined **Dimension**, **Status** and **Assignment Action** of each trigger, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

To create a new status trigger:

1. Click the **Add New** tab at the top of the Manage Status Triggers page to display the **Add New Status Trigger** page, as in the following example:



The screenshot shows the 'Manage Status Triggers' interface. At the top, there are two tabs: 'List All' and 'Add New'. The 'Add New' tab is highlighted with a red rectangle. Below the tabs, there are four dropdown menus, each with a red asterisk indicating a required field. The first dropdown is labeled 'Dimension' and has 'Diagnostic' selected. The second is labeled 'Status' and has 'Ordered' selected. The third is labeled 'Select' and has 'Assignment behaviour' selected. The fourth is labeled 'Assign if unassigned' and has 'Assign if unassigned' selected. At the bottom of the form are two buttons: a green 'Save' button and a blue 'Cancel' button.

Figure 332: Auto Assignment – Add New Status Trigger

2. From the **Dimension** drop-down menu, select from the list of configured dimensions in the system.

NOTE: This menu will not include reserved dimensions (internal dimensions used by various subsystems for coordination such as ImageImporting).

3. From the **Status** drop-down list, select from the list of configured statuses associated with that dimension in the system.

NOTE: This menu will not include statuses explicitly barred by the configuration parameter **Pacs.AssignmentEngine.Status.Restricted** in **AmicasServer.properties**.

4. From the **Select** drop-down list, select one of the three assignment actions:

Action	Description								
Assignment Behavior	When selected, an additional drop-down menu will appear that will let you select one of the following assignment actions:								
	<table> <tr> <th>Action</th><th>Description</th></tr> <tr> <td>Assign if unassigned</td><td>Process the exam and attempt to assign it only if has not already been assigned (whether automatically or manually).</td></tr> <tr> <td>Assign if already assigned</td><td> Process the exam and attempt to assign it only if the exam can be assigned to another user. If this option is selected, an additional drop-down menu will appear that will let you select from the following: • Any User – exam can be reassigned to any user including current assignee. • New User – exam can be reassigned to any user apart from the current assignee. </td></tr> <tr> <td>Either</td><td> Process the exam and attempt to assign it if has not already been assigned or if it can be assigned to another user. If this option is selected, an additional drop-down menu will appear that will let you select from the following: • Any User – exam can be reassigned to any user including current assignee. • New User – exam can be reassigned to any user apart from the current assignee. </td></tr> </table>	Action	Description	Assign if unassigned	Process the exam and attempt to assign it only if has not already been assigned (whether automatically or manually).	Assign if already assigned	Process the exam and attempt to assign it only if the exam can be assigned to another user. If this option is selected, an additional drop-down menu will appear that will let you select from the following: • Any User – exam can be reassigned to any user including current assignee. • New User – exam can be reassigned to any user apart from the current assignee.	Either	Process the exam and attempt to assign it if has not already been assigned or if it can be assigned to another user. If this option is selected, an additional drop-down menu will appear that will let you select from the following: • Any User – exam can be reassigned to any user including current assignee. • New User – exam can be reassigned to any user apart from the current assignee.
Action	Description								
Assign if unassigned	Process the exam and attempt to assign it only if has not already been assigned (whether automatically or manually).								
Assign if already assigned	Process the exam and attempt to assign it only if the exam can be assigned to another user. If this option is selected, an additional drop-down menu will appear that will let you select from the following: • Any User – exam can be reassigned to any user including current assignee. • New User – exam can be reassigned to any user apart from the current assignee.								
Either	Process the exam and attempt to assign it if has not already been assigned or if it can be assigned to another user. If this option is selected, an additional drop-down menu will appear that will let you select from the following: • Any User – exam can be reassigned to any user including current assignee. • New User – exam can be reassigned to any user apart from the current assignee.								
Completed	The exam will be marked as completed when it is automatically assigned.								
Unassign	The exam will be marked as unassigned when it is automatically assigned (as opposed to manually assigned).								

5. Click **Save**.

To edit a status trigger:

1. Click **Edit** next to the status trigger you want to modify to display the **Modify Status Trigger** page, as in the following example:

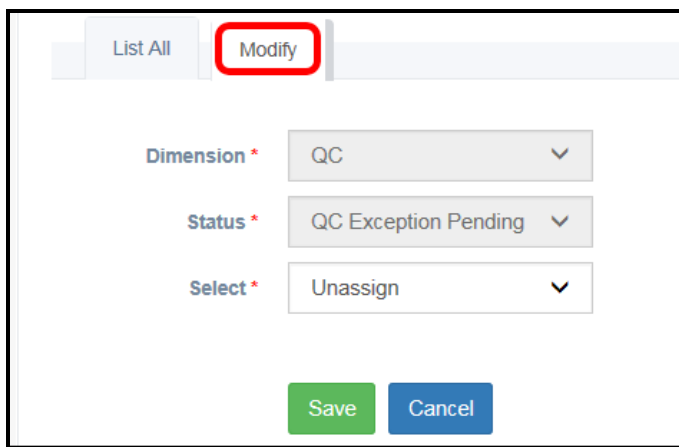


Figure 333: Auto Assignment – Modify Status Trigger

2. Edit any of the parameters for this status trigger as desired.
3. Click **Save**.

To delete a status trigger:

1. Click **Delete** next to the status trigger you want to delete.

A confirmation message appears.

2. Click **OK**.

b. Configuring Time Triggers

Time triggers are actions that are performed on an exam based on time intervals. These actions are performed at the exam level. The time intervals are either:

- **Exam age** in the system (measured by “**Due-in Time**”)
- Assignment age in the system (measured by “Last Assigned Time”)

A time trigger will invoke the Advanced Distribution Engine when the Due-in or Last Assigned time of the exam matches the time defined in the trigger and will update the assigned user based on the time specified in the trigger and the type of the trigger selected. Time triggers can Unassign, Reassign to New User, or Reassign to Any User depending upon how the specific trigger is defined.

NOTE: All time is specified in minutes.

You can manage and configure time triggers from the **Manage Time Triggers** page, accessible by clicking on the **Time Triggers** link on the Auto Assignment menu, as in the following example:

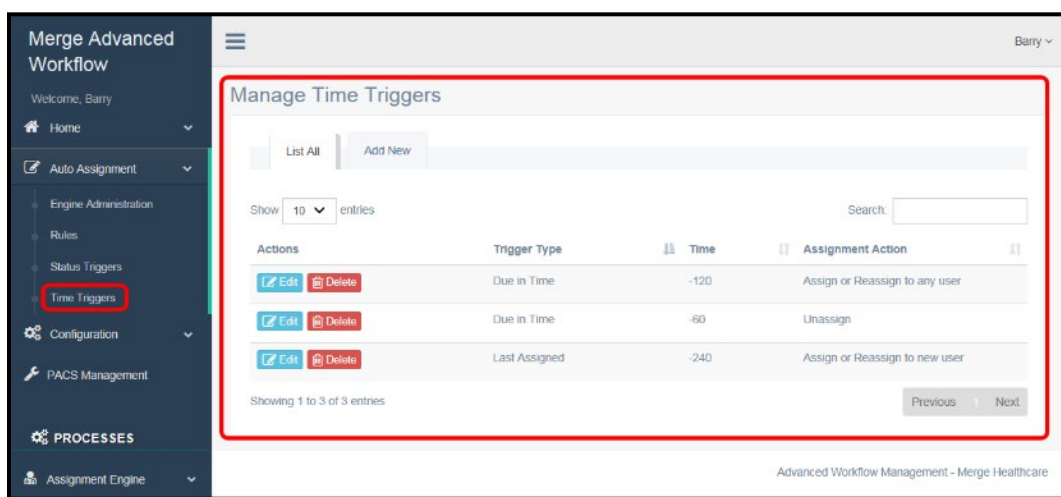


Figure 334: Auto Assignment – Manage Time Triggers

At the Manage Time Triggers page, you can do the following:

- **View** existing time triggers
- **Create** new time triggers
- **Edit** existing time triggers
- **Delete** existing time triggers

To view existing time triggers:

1. Click the **List All** tab at the top of the Manage Time Triggers page (this tab is selected by default when first accessing the page).

By default, the first 10 existing time triggers will be displayed in an unfiltered list.

2. If desired, change the maximum number of time triggers to be displayed on the page by selecting from the drop-down **Show Entries** menu,
3. To filter the list based on the defined **Trigger Type**, **Time** and **Assignment Action** of each trigger, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

To create a new time trigger:

1. Click the **Add New** tab at the top of the Manage Time Triggers page to display the **Add New Time Trigger** page, as in the following example:

Figure 335: Auto Assignment – Add New Time Trigger

2. From the **Trigger Type** drop-down menu, select one of the two types of trigger names:

Trigger	Description
Due in Time	Time relative to the due-in time of the work item. Negative numbers indicate the action to be performed before the due-in time. Positive numbers indicate the action to be performed post due-in time.
Last Assigned	Time relative to when the exam was last assigned by the Advanced Distribution Engine. Only positive numbers make sense here.

3. In the **Time** field, enter the desired time in minutes. If the trigger type is “Due in Time,” you can enter negative numbers (e.g., -120) to indicate that the trigger should be initiated a specific period of time prior to the due-in time.
4. In the **Select** drop-down list, select one of the following:

Action	Description
Unassign	Cause this exam to be unassigned. This maps to the UNASSIGN Assignment Action.
Reassign to Any User	Cause this exam to be auto-assigned again, including reassignment back to the same user if they are selected again. This maps to the REASSIGN_ANY Assignment Action.
Reassign to New User	Cause this exam to be auto-assigned again, but NOT back to the same user. This maps to the REASSIGN_NEW Assignment Action.

5. Click **Save**.

To edit a time trigger:

1. Click **Edit** next to the time trigger you want to modify to display the **Modify Time Trigger** page, as in the following example:

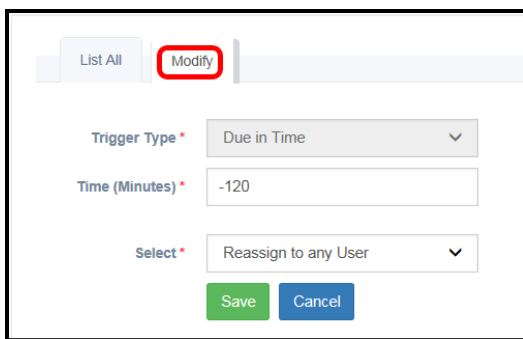


Figure 336: Auto Assignment – Modify Time Trigger

2. Edit any of the parameters for this time trigger as desired.
3. Click **Save**.

To delete a time trigger:

1. Click **Delete** next to the time trigger you want to delete.

A confirmation message appears.

2. Click **OK**.

c. Additional Trigger Configurations

In addition to the Status and Time triggers that can be configured from the Auto Assignment Menu, a variety of trigger options can be custom configured on a site-by-site basis to control when automatic assignment (or reassignment) will actually be performed.

The following trigger options can be configured:

i. Preloading of Users

By default, the Advanced Distribution Engine will preload users as they login so that, rather than the user having to wait for the next exam to be acquired and sent into PACS and possibly be selected as the assignee for reading, the system can look at all incomplete exams in the system and check to see if the Advanced Distribution Engine would select this new user as the best candidate for assignment (and, if so, reassign the exam to the newly-logged-in user).

Note the following:

- Incomplete work items are considered in order of Due In time (that is, closest Due In time first).
- Preloading of a user will stop when the configured number of work items have been assigned to that user, or when the system runs out of incomplete work items to consider. By default this is set to 5, but is configurable.
- Manually-assigned exams are not considered for this reassignment.

ii. Rebalancing a User's Work Items

By default, the Advanced Distribution Engine will rebalance a user's work items when the user's user profile properties change so that all of the automatically-assigned work items currently assigned to that user (and not complete) will be sent for automatic reassignment. For example, if a user has specific modalities associated with him or her and studies have been assigned to that user based on modality, changing the user's profile to remove those modalities may cause the user to no longer be the best reader for those studies.

Note the following:

- It is considered acceptable to reassign some or all of the work items back to the current user, since the change to the profile may not be relevant to the user's qualification to read the study.
- Manually-assigned exams are not considered for this reassignment.

iii. Automatically Offloading Users

By default, the Advanced Distribution Engine will offload a user when he or she manually logs out, causing all of the automatically-assigned work items currently assigned to that user (and not complete) to be sent for automatic reassignment.

Note the following:

- It is not considered acceptable to reassign work items back to the current user since the user is logged out.
- If no new assignee is found and if the retry feature is enabled, the work item's assignment will be retried.
- Manually-assigned exams are not considered for this reassignment.
- This functionality will not be applied to auto-logouts (exams will not be automatically reassigned in this case).

iv. Automatic Assignment Due to New Orders

By default, when an order arrives and is matched to an existing study, the Advanced Distribution Engine will send the study for automatic assignment/reassignment. It is considered acceptable to reassign back to the same user.

v. Automatic Assignment Due to Order Updates

By default, when an order update is processed, the Advanced Distribution Engine will send the associated study for automatic assignment/reassignment. It is considered acceptable to reassign back to the same user.

10.5. General Configuration

Clicking the **Configuration** link in the left-hand navigation panel will display the Configuration menu, as in the following example:

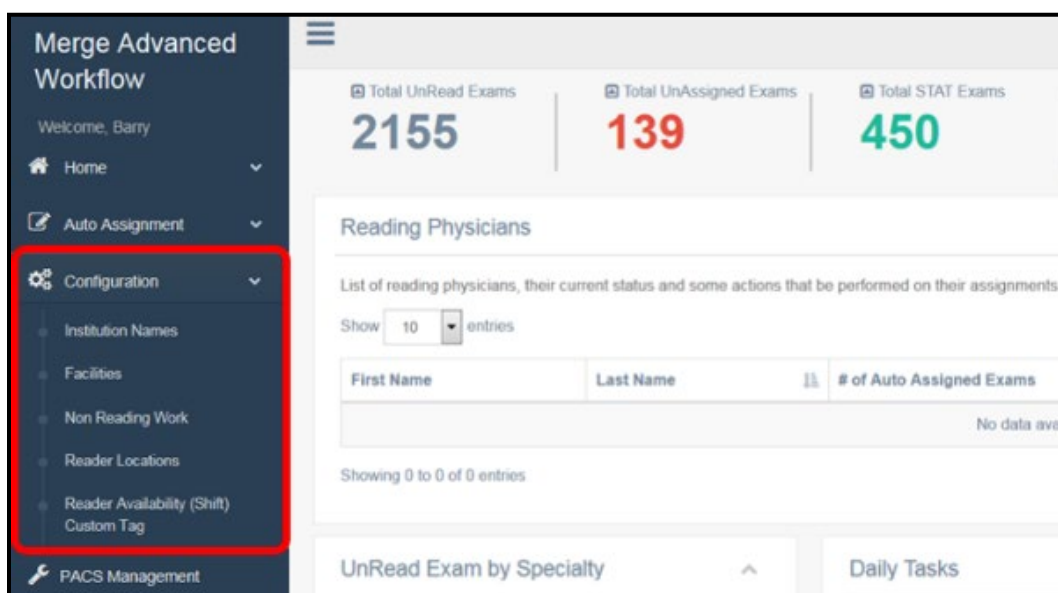


Figure 337: Configuration Menu

The configuration pages allow you to do the following:

- **Configure institution names**
- **Configure facilities**
- **Configure non-reading work**
- **Configure reader locations**
- **Configure reader availability (shift) custom tags**

Once all relevant information has been configured, both from the Management Pages and the Workstation Settings dialog (as described in Section 10.6 below), you can then configure the advanced workflow preferences for individual users from the Workstation Settings dialog, as described in Section 12.4 below.

10.5.1. Configuring Institution Names

Institution Names are automatically derived from the Institution Name DICOM attribute (0008,0080) of studies that have been imported into the system or, for orders, the applicable HL7 attribute and, as described in 10.5.2 below, are automatically mapped to a Facility that can then be used as a filter when setting up individual RTWL and Teaching worklists as well as a query field in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation.

Institution Names can also be created manually, and if a study is imported with an Institution Name DICOM attribute that matches a manually defined Institution Name, it will be mapped to the facility associated with that Institution Name.

The Manage Institution Names configuration page is accessed by clicking the **Institution Names** link under **Configuration** in the left-hand navigation panel, as in the following example:

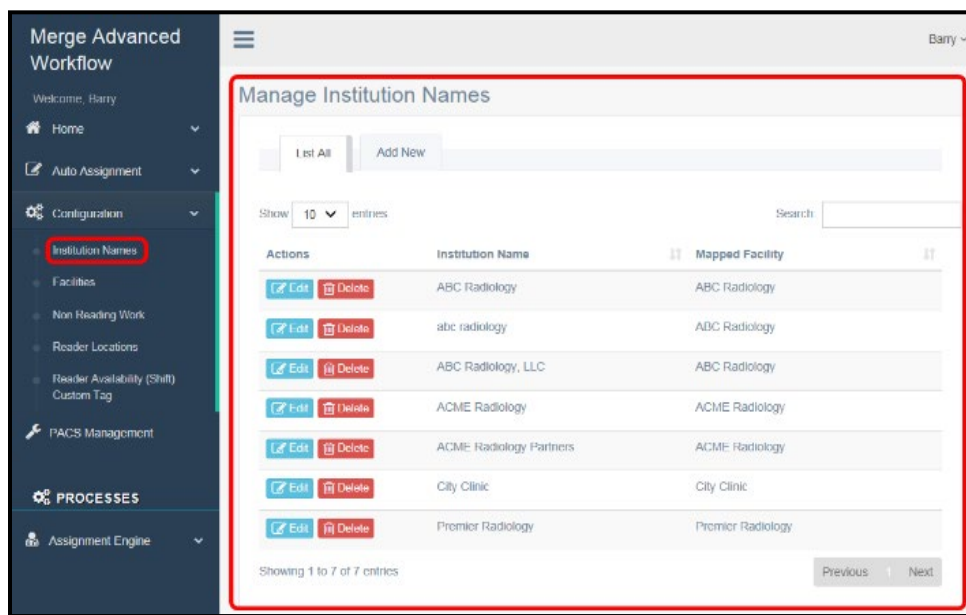


Figure 338: Manage Institution Names Page

NOTE: The list of existing Institution Names will be pre-populated with any institution names that have been automatically derived from exams or orders, as described above.

At the Manage Institution Names page, you can do the following:

- **View** existing Institution Names
- **Create** new Institution Names
- **Edit** existing Institution Names
- **Delete** existing Institution Names

a. Viewing Existing Institution Names

To view existing Institution Names:

1. Click the **List All** tab at the top of the Manage Institution Names page (this tab is selected by default when first accessing the page).
By default, the first 10 existing Institution Names will be displayed in an unfiltered list.
2. If desired, change the maximum number of Institution Names to be displayed on the page by selecting from the drop-down **Show Entries** menu.
3. To filter the list based on the defined **Institution Name** and **Mapped Facility** of each Institution Names, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

b. Creating New Institution Names

To create a new Institution Name:

1. Click the **Add New** tab at the top of the Manage Institution Names page to display the **Add New Institution Name** page, as in the following example:

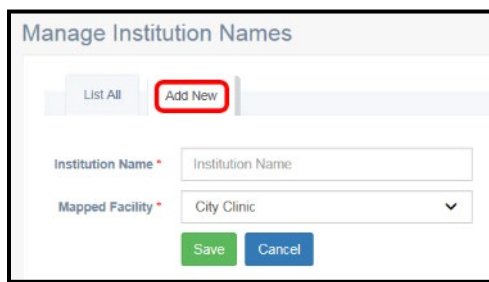
The screenshot shows the 'Manage Institution Names' page. At the top, there are two tabs: 'List All' and 'Add New'. The 'Add New' tab is selected and highlighted with a red rectangle. Below the tabs, there are two input fields: 'Institution Name *' with a text box containing 'Institution Name', and 'Mapped Facility *' with a dropdown menu showing 'City Clinic'. At the bottom, there are two buttons: 'Save' (green) and 'Cancel' (blue).

Figure 339: Add New Institution Name

2. In the **Institution Name** field, enter the desired name with the exact capitalization, spelling and punctuation.

NOTE: If there are multiple variations of this institution's name, they should be entered as separate Institution Names.

3. From the **Mapped Facility** drop-down list, select from the list of existing Facilities.

NOTE: Facilities can either be automatically created by the System or manually created, as described in 10.5.2 below.

4. Click **Save**.

c. Editing Existing Institution Names

To edit an Institution Name:

1. Click **Edit** next to the Institution Name you want to modify to display the **Modify Institution Name** page, as in the following example:

The screenshot shows the 'Manage Institution Names' page. At the top, there are two tabs: 'List All' and 'Modify'. The 'Modify' tab is selected and highlighted with a red rectangle. Below the tabs, there are two input fields: 'Institution Name *' with a text box containing 'ABC Radiology, LLC', and 'Mapped Facility *' with a dropdown menu showing 'ABC Radiology'. At the bottom, there are two buttons: 'Save' (green) and 'Cancel' (blue).

Figure 340: Modify Institution Name

2. Edit any of the information for this Institution Name as desired.
3. Click **Save**.

d. Deleting Institution Names

To delete an Institution Name:

1. Click **Delete** next to the Institution Name you want to delete.

A confirmation message appears.

2. Click **OK**.

10.5.2. Configuring Facilities

As described in subsection 10.5.1 above, Institution Names can either be derived automatically based on the information associated with exams and orders that are imported into Merge PACS or else manually defined. Since the same institution may be referred to by different Institution Names (especially due to variances in spelling, capitalization, punctuation, etc.), Merge PACS uses the concept of Facility to indicate the “actual” name of an Institution that can then be used as a filter when configuring auto-assignment rules, as described in subsection 10.4.3 above, as a filter when setting up individual RTWL and Teaching worklists and as well as a query field in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation.

Facilities are created by mapping one or more Institution Names to a Facility. In most cases, there will be a one-to-one mapping of Institution Name to Facility, but a Facility might be mapped to multiple Institution Names if there are variations in spelling, punctuation, capitalization, etc. (e.g., “St. Mary’s Hospital,” “Saint Mary’s Hospital” and “st marys hospital” could all be mapped to a single “St. Mary’s Hospital” Facility).

If a study or order is imported with an Institution Name attribute that is not currently known to Merge PACS, Merge PACS will automatically do one of the following depending on how it is configured:

- Map it to a Facility configured as the “default” facility.
- Match it to the existing Facility, if any, that matches the exam or order’s Institution Name information (case insensitive matching) or create a new Facility with this name if no matching Facility is found.

Institution Names can also be created manually, as described 0 above, and if a study or order is imported with Institution Name information that matches that Institution Name, it will be mapped to the facility associated with that Institution Name.

NOTE: Facilities will also be created automatically based on the logic described above when the system is upgraded to Merge PACS 8.2 from a previous version.

NOTE: Changes made to Facilities configuration information will not automatically be synched to the worklist server. You will need to click the **Reload exam contents** link for the worklist server at the Worklist Server Monitor page, as described in subsection 9.6.10 above.

The Manage Facilities configuration page is accessed by clicking the **Facilities** link under **Configuration** in the left-hand navigation panel, as in the following example:

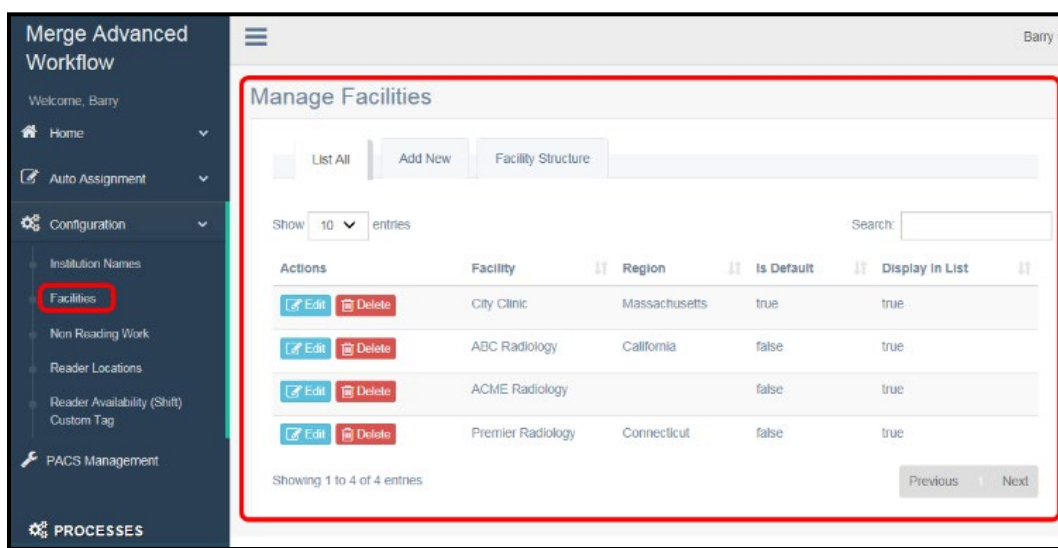


Figure 341: Manage Facilities Page

NOTE: The list of existing Facilities will be pre-populated with any Facilities that have been automatically derived, as described above.

At the Manage Institution Names page, you can do the following:

- **View** existing Facilities
- View the current **Facility Structure** (*i.e.*, which Institution names are mapped to which Facilities)
- **Create** new Facilities
- **Edit** existing Facilities
- **Delete** existing Facilities

a. Viewing Existing Facilities

To view existing Facilities:

1. Click the **List All** tab at the top of the Manage Facilities page (this tab is selected by default when first accessing the page).
By default, the first 10 existing Facilities will be displayed in an unfiltered list.
2. If desired, change the maximum number of Facilities to be displayed on the page by selecting from the drop-down **Show Entries** menu.
3. To filter the list based on the defined **Facility Name**, **Region**, **Is Default** status and **Display in List** status of each Facility, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

b. Viewing the Existing Facility Structure

To View the existing Facility structure:

1. Click the **Facility Structure** tab at the top of the Manage Facilities page to display the **Facility Structure** page, as in the following example:

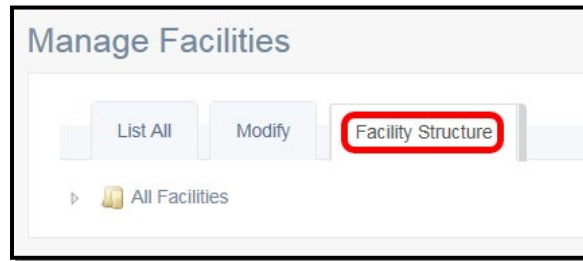


Figure 342: Facility Structure Page

2. Click on the triangle next to the **All Facilities** folder and then click on the triangles beside any listed Facility to view the Institution Names associated with those Facilities, as in the following example:

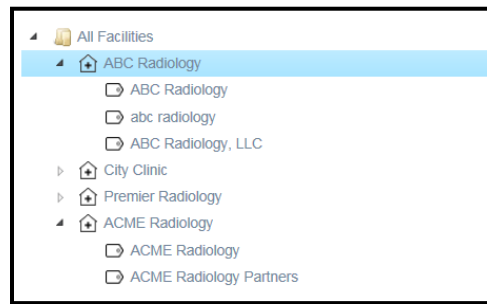


Figure 343: Facility Structure

c. Creating New Facilities

To create a new Facility:

1. Click the **Add New** tab at the top of the Manage Facilities page to display the **Add New Facility** page, as in the following example:

A screenshot of the 'Add New Facility' page. At the top, there are three tabs: 'List All', 'Add New', and 'Facility Structure'. The 'Add New' tab is highlighted with a red rectangular box. Below the tabs, there is a form with the following fields: 'Name *' (text input), 'Region' (dropdown menu), 'Is Default' (checkbox), and 'Display in List' (checkbox). At the bottom of the form, there are two buttons: 'Save' (green) and 'Cancel' (blue).

Figure 344: Add New Facility

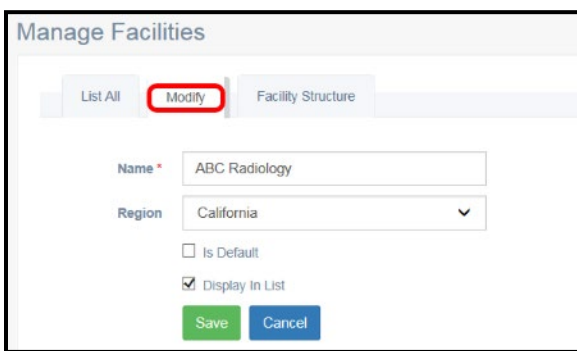
2. In the **Name** field, enter the desired name of the Facility.

3. If this site is using the Credentialing and Licensure feature, as described in subsection 9.13.10 above, select the region associated with this facility from the drop-down **Region** menu.
4. If you want this Facility to be the default facility for use with reader locations (as discussed in subsection 10.5.4 below), select the **Is Default** checkbox.
5. If you want this Facility to be listed as an option in most selectable lists throughout the system (e.g., in the RTSL Quick Filters), select the **Display in List** checkbox.
6. Click **Save**.

d. Editing Information for Existing Facilities

To edit a Facility:

1. Click **Edit** next to the Facility you want to modify to display the **Modify Facility** page, as in the following example:



The screenshot shows a web interface titled "Manage Facilities". At the top, there are three tabs: "List All", "Modify" (which is highlighted with a red rectangle), and "Facility Structure". Below the tabs, there is a form with the following elements: a text input field for "Name" containing "ABC Radiology", a dropdown menu for "Region" set to "California", an unchecked checkbox for "Is Default", and a checked checkbox for "Display in List". At the bottom of the form are two buttons: a green "Save" button and a blue "Cancel" button.

Figure 345: Modify Institution Name

2. Edit any of the information for this Facility as desired.
3. Click **Save**.

e. Deleting Existing Facilities

To delete a Facility:

1. Click **Delete** next to the Facility you want to delete.
A confirmation message appears.
2. Click **OK**.

10.5.3. Configuring Non Reading Work Categories

Auto Assignment rules can be configured to take into account the current capacity and availability of individual radiologists. If desired, a site can define various Non-reading categories of work such as tumor board or teaching that can be taken into account when determining a radiologist's capacity and availability. Once defined, individual users can log time against these tasks from within the Workstation.

The Manage Non-Reading Work configuration page is accessed by clicking the **Non Reading Work** link under **Configuration** in the left-hand navigation panel, as in the following example:

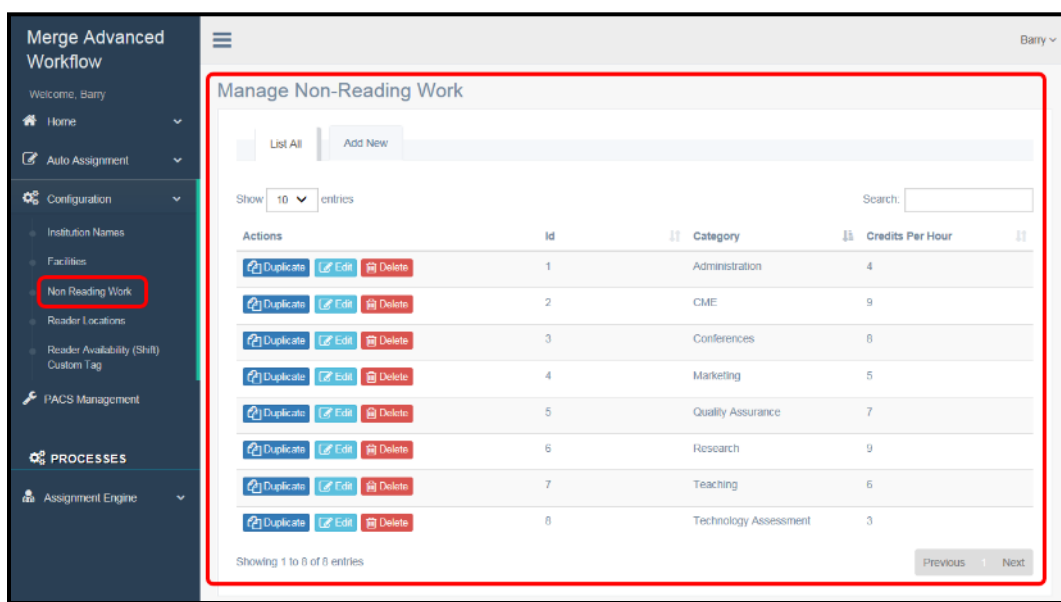


Figure 346: Manage Non-Reading Work Page

At the Manage Non-Reading Work page, you can do the following:

- **View** existing non-reading work categories
- **Create** new non-reading work categories from scratch
- **Duplicate** existing non-reading work categories
- **Edit** existing non-reading work categories
- **Delete** existing non-reading work categories

a. Viewing Existing Non-Reading Work Categories

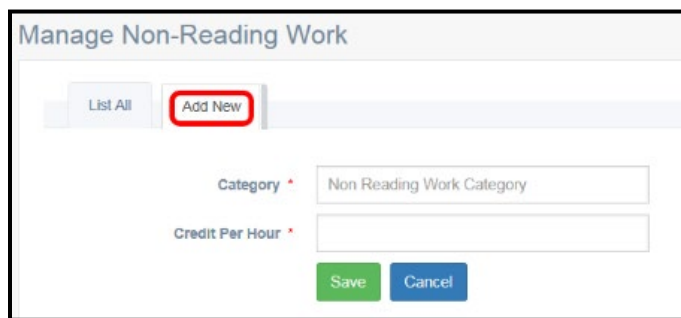
To view existing non-reading work categories:

1. Click the **List All** tab at the top of the Manage Non-Reading Work page (this tab is selected by default when first accessing the page).
By default, the first 10 existing non-reading work categories will be displayed in an unfiltered list.
2. If desired, change the maximum number of non-reading work categories to be displayed on the page by selecting from the drop-down **Show Entries** menu.
3. To filter the list based on the defined **ID**, **Category** and **Credits per Hour** of each non-reading work category, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

b. Creating New Non-Reading Work Categories

To create a new non-reading work category:

1. Click the **Add New** tab at the top of the Manage Facilities page to display the **Add New Non-Reading Work** page, as in the following example:



The screenshot shows a web interface titled "Manage Non-Reading Work". At the top, there are two tabs: "List All" and "Add New". The "Add New" tab is selected and highlighted with a red rectangle. Below the tabs, there are two input fields. The first is labeled "Category *" and contains the text "Non Reading Work Category". The second is labeled "Credit Per Hour *" and is empty. At the bottom of the form, there are two buttons: a green "Save" button and a blue "Cancel" button.

Figure 347: Add New Non-Reading Work

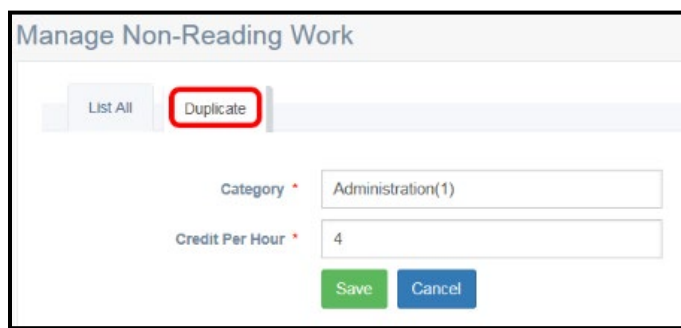
2. In the **Category** field, enter the desired name of the category.
3. In the **Credits per Hour** field, enter the desired integer value.
4. Click **Save**.

c. Duplicating Existing Non-Reading Work Categories

Instead of creating a new non-reading work category from scratch, you can duplicate an existing non-reading work category and modify as needed.

To duplicate an existing non-reading work category:

1. Click **Duplicate** next to the non-reading work category you want to copy to display the **Duplicate Non-Reading Work** page, as in the following example:



The screenshot shows a web interface titled "Manage Non-Reading Work". At the top, there are two tabs: "List All" and "Duplicate". The "Duplicate" tab is selected and highlighted with a red rectangle. Below the tabs, there are two input fields. The first is labeled "Category *" and contains the text "Administration(1)". The second is labeled "Credit Per Hour *" and contains the number "4". At the bottom of the form, there are two buttons: a green "Save" button and a blue "Cancel" button.

Figure 348: Duplicate Non-Reading Work

2. Modify any of the information for this non-reading work category as desired.
3. Click **Save**.

d. Editing Existing Non-Reading Work Categories

To edit an existing non-reading work category:

1. Click **Edit** next to the non-reading work category you want to modify to display the **Modify Non-Reading Work** page is displayed, as in the following example:

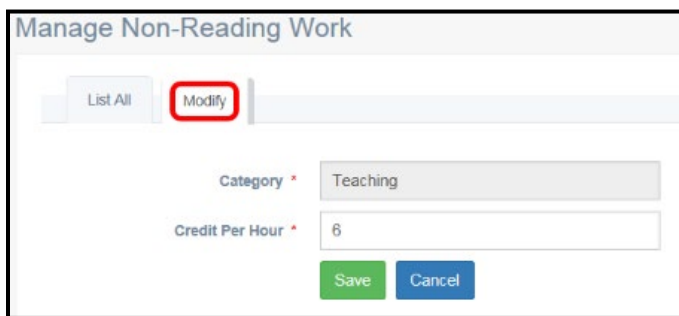


Figure 349: Modify Non-Reading Work

2. Edit any of the information for this non-reading work category as desired.
3. Click **Save**.

e. Deleting Non-Reading Work Categories

To delete a non-reading work category:

1. Click **Delete** next to the non-reading work category you want to delete.
A confirmation message appears.
2. Click **OK**.

10.5.4. Configuring Reader Locations

The Reader Location Configuration page lets you define specific Workstations in use at this site (e.g., a 3D Specialty Workstation or a Mammography Workstation) and provide specific information about those Workstations including their hostname or IP address and physical address. Once so defined, the reader location of a particular user can be set when the user logs into Merge PACS or at other points in the workflow.

The reader location of the logged-in user can be used by Merge PACS in the following ways:

- The reader location can be taken into account by a configured Auto Assignment rule (e.g., a complex mammography study can be automatically assigned to whichever user is currently at the Mammography Workstation).
- The reader location can be taken into account by the Launch & Control feature (e.g., for the user sitting at the 3D Specialty Workstation, clicking the Primary Viewer action in the worklist can be configured to launch the 3D Specialty Viewer instead of the standard PACS Viewer).
- The reader location can be used for tracking place of service, or where a study is read. PACS will log details when a study is set to a status greater than Unread (in the Diagnostic dimension) that can include the physical address of where the study was read if there is an associated reader location.

The Manage Reader Location configuration page is accessed by clicking the **Reading Locations** link under **Configuration** in the left-hand navigation panel, as in the following example:

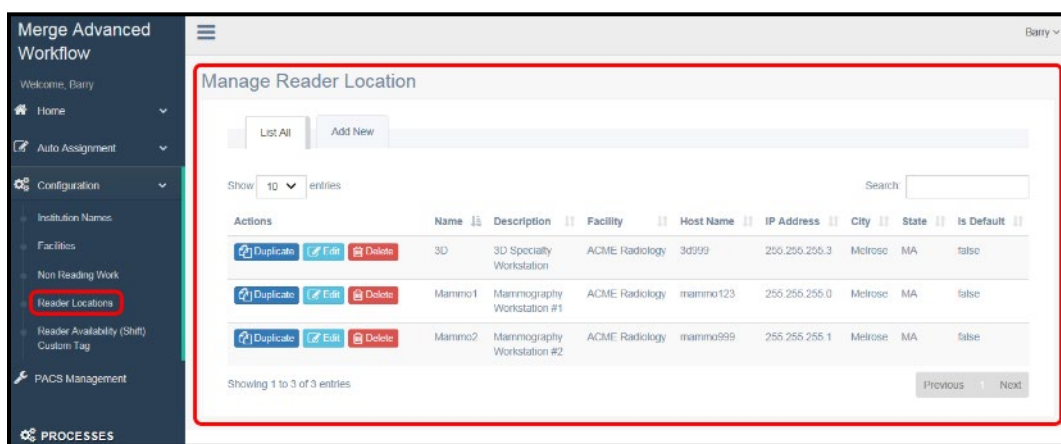


Figure 350: Manage Reader Location Page

At the Manage Reader Location page, you can do the following:

- **View** previously defined reader locations
- **Create** new reader locations from scratch
- **Duplicate** existing reader locations
- **Edit** existing reader locations
- **Delete** existing reader locations

a. Viewing Previously Defined Reader Locations

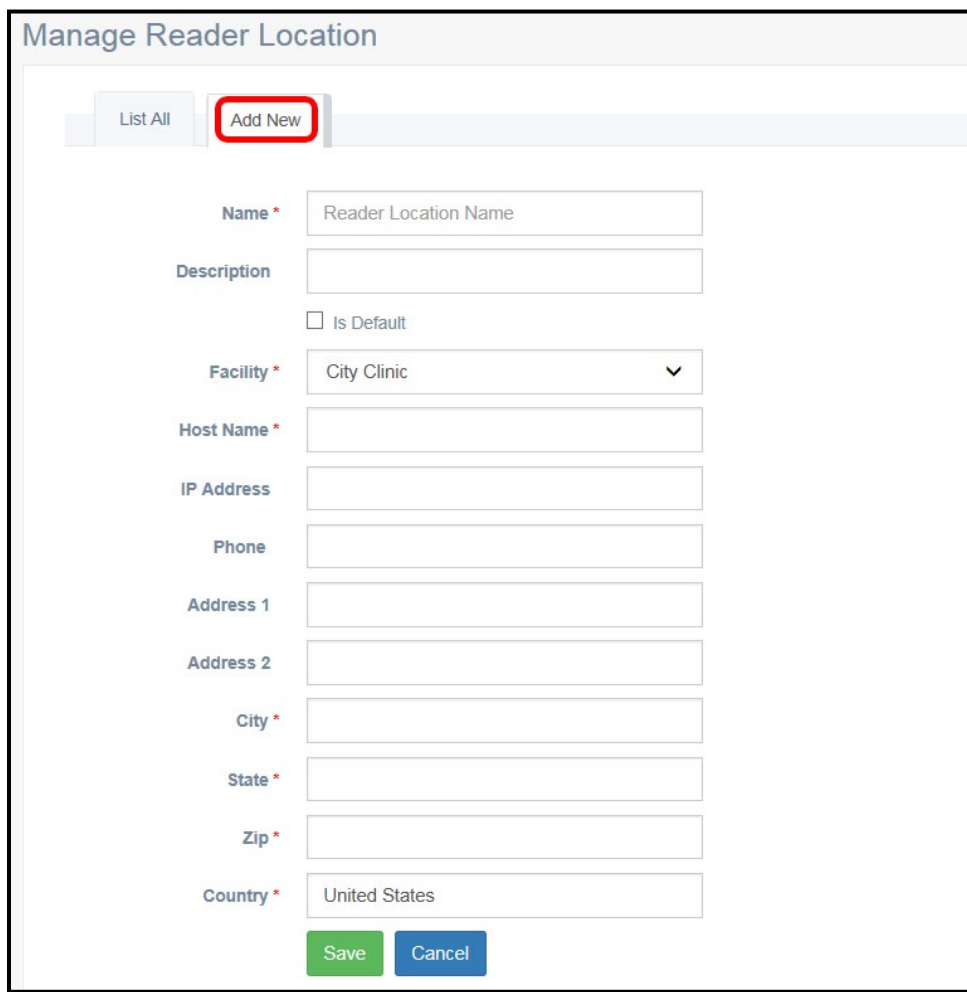
To view previously defined reader locations:

1. Click the **List All** tab at the top of the Manage Reader Location page (this tab is selected by default when first accessing the page).
By default, the first 10 existing reader locations will be displayed in an unfiltered list.
2. If desired, change the maximum number of reading locations to be displayed on the page by selecting from the drop-down **Show Entries** menu.
3. To filter the list based on the defined **Name**, **Description**, **Facility**, **Host Name**, **IP Address**, **City**, **State** and **Is Default** status of each reader location, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

b. Creating New Reader Locations

To create a new reader location:

1. Click the **Add New** tab at the top of the Manage Reader Location page to display the **Add New Reader Location** page, as in the following example:



The screenshot shows the 'Manage Reader Location' page. At the top, there are two tabs: 'List All' and 'Add New'. The 'Add New' tab is selected and highlighted with a red box. Below the tabs, the form contains the following fields:

- Name ***: Text input field with placeholder text 'Reader Location Name'.
- Description**: Text input field.
- ☐ **Is Default**: Checkbox.
- Facility ***: Dropdown menu with 'City Clinic' selected.
- Host Name ***: Text input field.
- IP Address**: Text input field.
- Phone**: Text input field.
- Address 1**: Text input field.
- Address 2**: Text input field.
- City ***: Text input field.
- State ***: Text input field.
- Zip ***: Text input field.
- Country ***: Text input field with 'United States' selected.

At the bottom of the form, there are two buttons: 'Save' (green) and 'Cancel' (blue).

Figure 351: Add New Reader Location

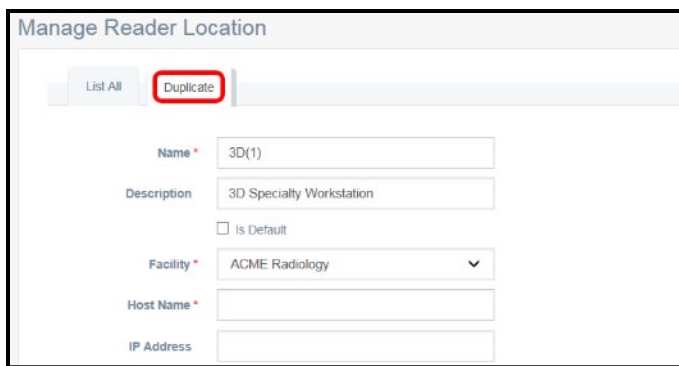
2. Enter the information for this reader location in the available fields, noting the following:
 - Fields marked with a red asterisk are required.
 - Select the **Is Default** checkbox to make this location the default.
 - The phone number, if entered, cannot contain any special characters such as parentheses or dashes (e.g., enter 6175551234 instead of (617) 555-1234).
3. Click **Save**.

c. Duplicating Existing Reader Locations

Instead of creating a new reader location from scratch, you can duplicate an existing reader location and modify it as needed.

To duplicate an existing reader location:

1. Click **Duplicate** next to the reader location you want to copy to display the **Duplicate Reader Location** page, as in the following example:



The screenshot shows the 'Manage Reader Location' form. At the top, there are two buttons: 'List All' and 'Duplicate'. The 'Duplicate' button is highlighted with a red rectangle. Below the buttons, the form contains several fields: 'Name *' with the value '3D(1)', 'Description' with the value '3D Specialty Workstation', an unchecked 'Is Default' checkbox, 'Facility *' with a dropdown menu showing 'ACME Radiology', 'Host Name *' with an empty field, and 'IP Address' with an empty field.

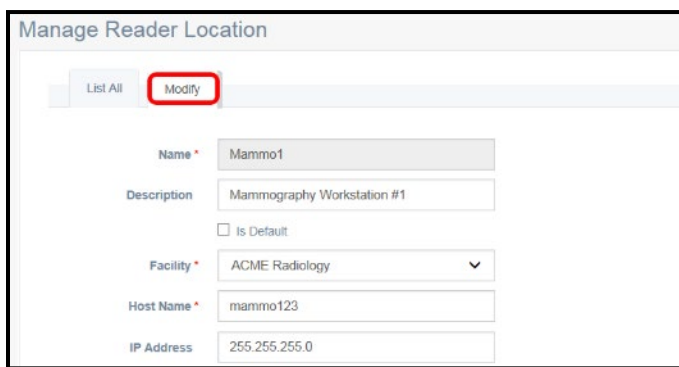
Figure 352: Duplicate Reader Location

2. Modify any of the information for this non-reading work category as desired.
3. Click **Save**.

d. Editing Existing Reader Locations

To edit an existing reader location:

1. Click **Edit** next to the reader location you want to modify to display the **Modify Reader Location** page, as in the following example:



The screenshot shows the 'Manage Reader Location' form. At the top, there are two buttons: 'List All' and 'Modify'. The 'Modify' button is highlighted with a red rectangle. Below the buttons, the form contains several fields: 'Name *' with the value 'Mammo1', 'Description' with the value 'Mammography Workstation #1', an unchecked 'Is Default' checkbox, 'Facility *' with a dropdown menu showing 'ACME Radiology', 'Host Name *' with the value 'mammo123', and 'IP Address' with the value '255.255.255.0'.

Figure 353: Modify Reader Location

2. Edit any of the information for this reader location as desired.
3. Click **Save**.

e. Deleting Reader Locations

To delete a reader location:

1. Click **Delete** next to the reader location you want to delete.
A confirmation message appears.
2. Click **OK**.

10.5.5. Configuring Reader Availability Custom Tags

The Reader Availability Custom Tag Configuration page lets you define one or more custom “Reader Availability” tags that can be used when defining Auto Assignment rules. By default, this is referred to as a “Shift” tag, but the name can be custom configured on a site-by-site basis.

For example, a Radiologist can log into the Workstation and select the “Neuro Rotation” shift tag, and if an Auto Assignment Rule is configured to assign studies to users who are on a Neuro Rotation shift then that user will be considered eligible to have a study assigned to him under that rule.

The **Manage Reader Availability Custom Tag** configuration page is accessed by clicking the **Reader Availability Custom Tag** link under **Configuration** in the left-hand navigation panel, as in the following example:

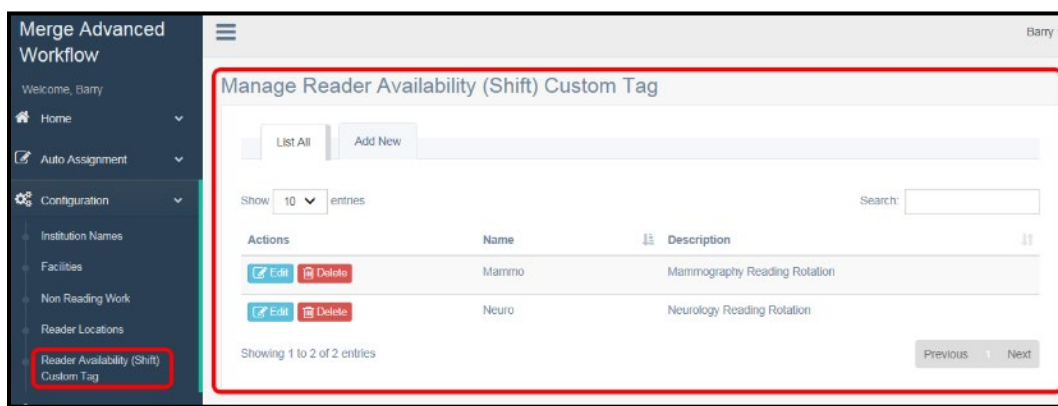


Figure 354: Manage Reader Availability Custom Tag Page

At the Manage Reader Availability Custom Tag page, you can do the following:

- **View** existing reader availability custom tags
- **Create** new reader availability custom tags from scratch
- **Edit** existing reader availability custom tags
- **Delete** existing reader availability custom tags

a. Viewing Existing Availability Custom Tags

To view existing reader availability custom tags:

1. Click the **List All** tab at the top of the Manage Reader Availability Custom Tag page (this tab is selected by default when first accessing the page).

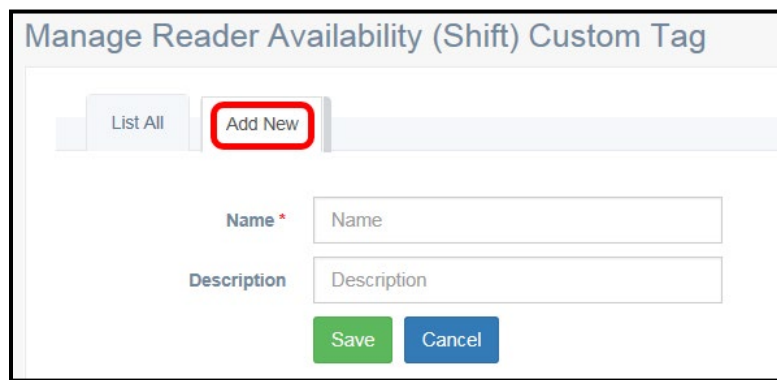
By default, the first 10 existing reader availability custom tags will be displayed in an unfiltered list.

2. If desired, change the maximum number of reader availability custom tags to be displayed on the page by selecting from the drop-down **Show Entries** menu.
3. To filter the list based on the defined **Name** and **Description** of each reader availability custom tag, enter text in the **Search** box.
4. Click on any column heading to sort the list based on that column, clicking again to toggle between alphabetical and reverse alphabetical sorting.

b. Creating New Reader Availability Custom Tags

To create a new reader availability custom tag:

1. Click the **Add New** tab at the top of the Manage Reader Availability Custom Tag page to display the **Add New Reader Availability Custom Tag** page, as in the following example:



The screenshot shows a web interface titled "Manage Reader Availability (Shift) Custom Tag". At the top, there are two tabs: "List All" and "Add New". The "Add New" tab is selected and highlighted with a red rectangle. Below the tabs, there are two input fields: "Name *" and "Description". The "Name *" field contains the text "Name" and the "Description" field contains the text "Description". At the bottom of the form, there are two buttons: "Save" (green) and "Cancel" (blue).

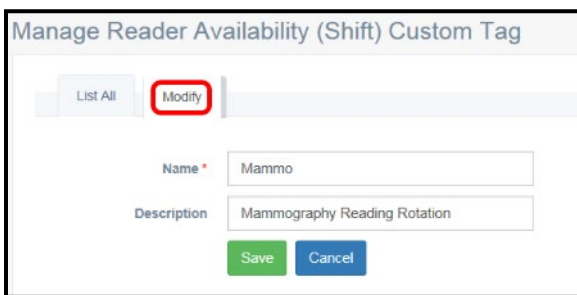
Figure 355: Add New Reader Availability Custom Tag

2. In the **name** field, enter the desired name for this availability tag.
3. If desired, enter a description of this tag in the **Description** field.
4. Click **Save**.

c. Editing Existing Reader Availability Custom Tags

To edit an existing reader availability custom tag:

1. Click **Edit** next to the reader availability custom tag you want to modify to display the **Modify Reader Availability Custom Tag** page, as in the following example:



The screenshot shows a web interface titled "Manage Reader Availability (Shift) Custom Tag". At the top, there are two tabs: "List All" and "Modify". The "Modify" tab is selected and highlighted with a red rectangle. Below the tabs, there are two input fields: "Name *" and "Description". The "Name *" field contains the text "Mammo" and the "Description" field contains the text "Mammography Reading Rotation". At the bottom of the form, there are two buttons: "Save" (green) and "Cancel" (blue).

Figure 356: Modify Reader Availability Custom Tag

2. Edit any of the information for this reader location as desired.
3. Click **Save**.

d. Deleting Reader Availability Custom Tags

To delete a reader availability custom tag:

1. Click **Delete** next to the reader availability custom tag you want to delete.
A confirmation message appears.
2. Click **OK**.

10.6. Client Side Configuration

In addition to the Configuration done via the Advanced Workflow Management Pages, the following configuration options can be performed from the **Settings** dialog within the Merge PACS Workstation:

- **Practices**
- **Specialties**
- **Orderers**
- **Order/Practice Associations**
- **Orderer/Reader Associations**

NOTE: **Procedures** can also be configured from the Settings dialog within the Workstation, even though the use of Procedures is not specific to Advanced Workflow.

Once all relevant information has been configured, both from the Management Pages and the Workstation Settings dialog, you can then configure the advanced workflow preferences for individual users from the Workstation Settings dialog, as described in Section 12.4 below.

10.6.1. Accessing the Workstation Settings Dialog

To access the Workstation Settings Dialog:

1. Log into the Merge PACS Workstation as a Merge Support or Admin user.
2. Click on the **Settings** icon in the upper right corner of the Workstation Browser, as in the following example:

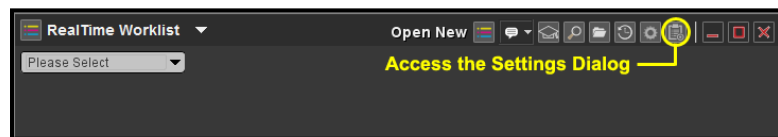


Figure 357: The Settings Icon

The **Settings** dialog is displayed with links to the various Advanced Workflow Configuration pages included on the left-hand side, as in the following example:

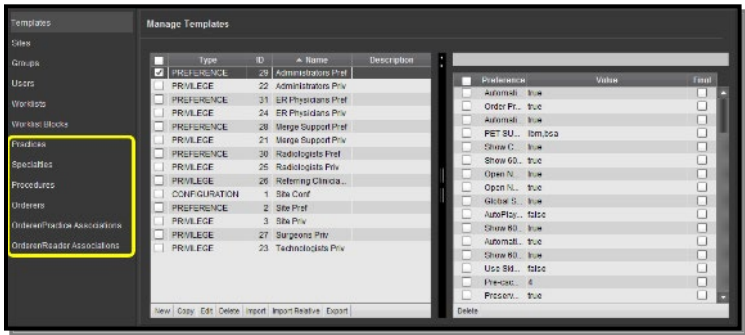


Figure 358: Advanced Workflow Configuration Pages on the Settings Dialog

10.6.2.Managing Practices

Click on the **Practices** link to display the **Manage Practices** page, as in the following example:

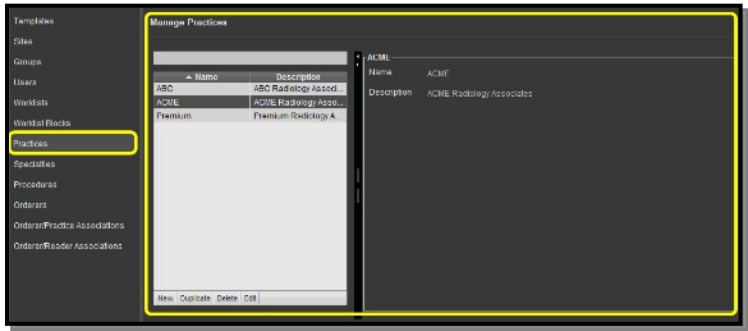


Figure 359: Manage Practices Page

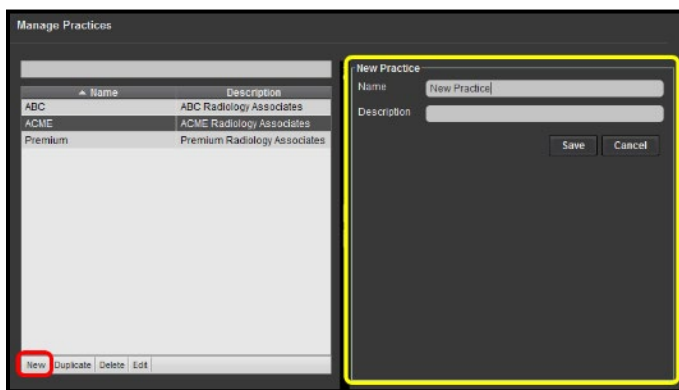
By default, all existing practices will be displayed in the left-hand panel of the Manage Practices page and you can click on any practice to see the details for that practice in the right-hand panel.

If the list of existing practices is long, you can filter the list by entering text in the field at the top of the left-hand panel to filter based on matching **Name**.

a. Defining New Practices

To define a new practice:

1. Click the **New** button at the bottom of the left-hand panel to display the **New Practice** page in the right-hand panel, as in the following example:



The screenshot shows the 'Manage Practices' window. On the left, a table lists existing practices: ABC (ABC Radiology Associates), ACME (ACME Radiology Associates), and Premium (Premium Radiology Associates). At the bottom of this panel, the 'New' button is highlighted with a red box. On the right, the 'New Practice' form is displayed, featuring input fields for 'Name' (containing 'New Practice') and 'Description', and 'Save' and 'Cancel' buttons. The entire right-hand panel is outlined with a yellow border.

Figure 360: New Practice Page

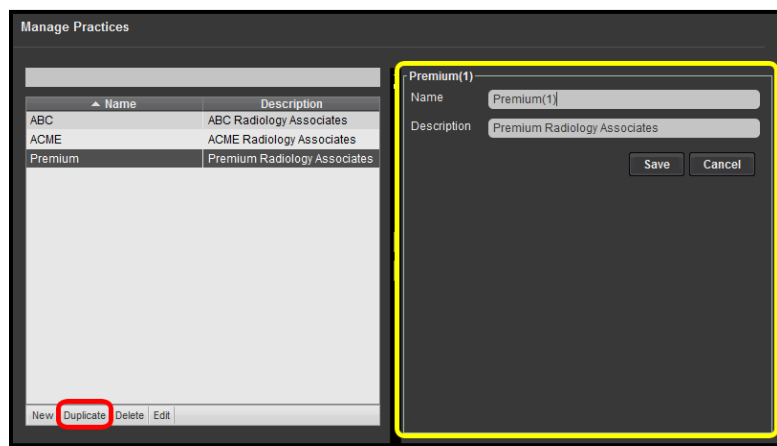
2. At the **New Practice** page, enter the name and a description for the new practice.
3. Click **Save**.

b. Duplicating Existing Practices

Instead of creating a new practice from scratch, you can duplicate an existing practice and modify it as needed.

To duplicate an existing practice:

1. Click on the practice you wish to duplicate in the left-hand panel and click on the **Duplicate** button to display the **Duplicate Practice** page, as in the following example:



The screenshot shows the 'Manage Practices' window. On the left, the same table of practices is shown. At the bottom of this panel, the 'Duplicate' button is highlighted with a red box. On the right, the 'Duplicate Practice' form is displayed, titled 'Premium(1)'. It contains input fields for 'Name' (containing 'Premium(1)') and 'Description' (containing 'Premium Radiology Associates'), and 'Save' and 'Cancel' buttons. The entire right-hand panel is outlined with a yellow border.

Figure 361: Duplicate Practice Page

2. Modify any of the information for this practice as desired.
3. Click **Save**.

c. Editing Existing Practices

To edit an existing practice:

1. Click on the practice you wish to edit in the left-hand panel and click on the **Edit** button (or double-click on the practice) to display the **Edit Practice** page, as in the following example:

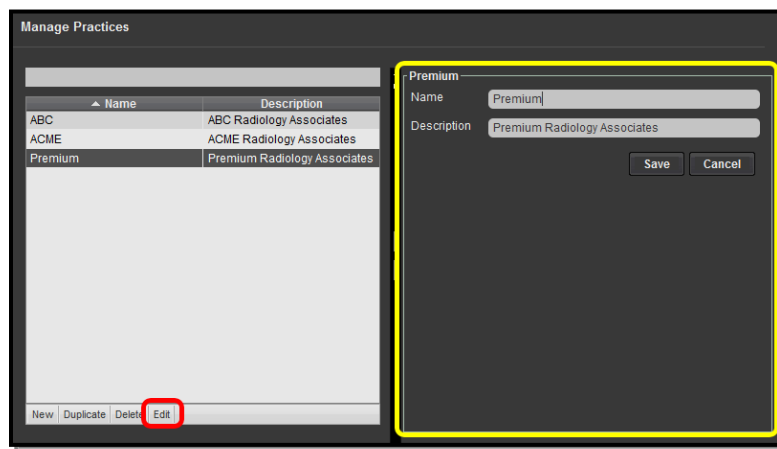


Figure 362: Edit Practice Page

2. Modify any of the information for this practice as desired.
3. Click **Save**.

d. Deleting Existing Practices

To delete an existing practice:

1. Click on the practice you wish to edit in the left-hand panel and click on the **Delete** button at the bottom of the left-hand panel.

A confirmation message appears.

2. Click **OK**.

10.6.3. Managing Specialties

Specialties (known as “Reading Pools” in previous versions of Merge PACS) can be pulled automatically from OBR-18 of the incoming HL7 messages or entered manually. Once defined in the system, specialties can be used in the following ways:

- as filters when setting up individual RTWL and Teaching worklists
- as query fields on the Query Page and in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation.
- to determine whether specific users are reading candidates for defined Auto Assignment rules.

Since Specialty values are numeric within HL7, you can specify a display name for each possible value, and this display name will be used as a value on worklists and as an option in various pick lists.

Click on the **Specialties** link to display the **Manage Specialties** page, as in the following example:

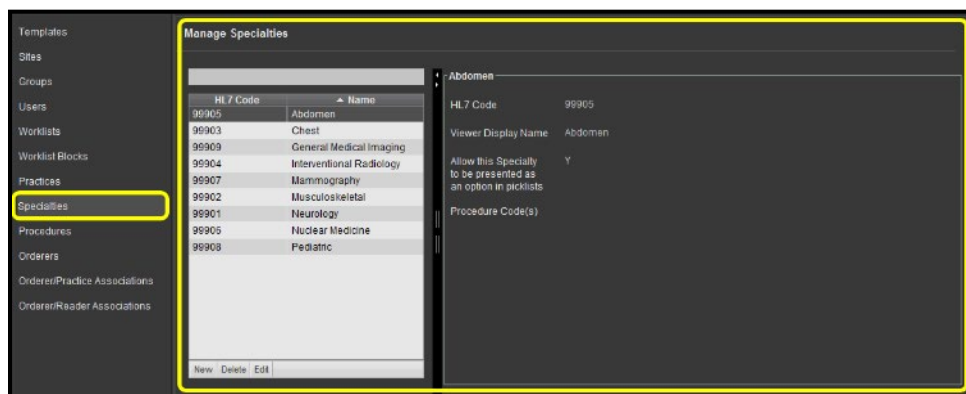


Figure 363: Manage Specialties Page

By default, all existing specialties will be displayed in the left-hand panel of the Manage Specialties page and you can click on any specialty to see the details for that specialty in the right-hand panel.

If the list of existing specialties is long, you can filter the list by entering text in the field at the top of the left-hand panel to filter based on matching **Name**.

a. Defining New Specialties

To define a new specialty:

1. Click the **New** button at the bottom of the left-hand panel to display the **New Specialty** page in the right-hand panel, as in the following example:

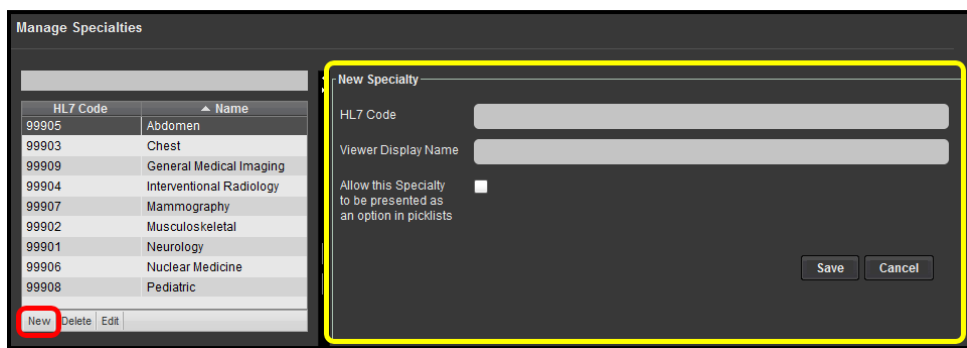


Figure 364: New Specialty Page

2. At the **New Specialty** page, enter the HL7 Code and the desired **Viewer Display Name** for the new specialty.
3. Make sure that the **Allow this Specialty to be presented as an option in picklists** checkbox is selected.
4. Click **Save**.

b. Editing Existing Specialties

To edit an existing specialty:

1. Click on the specialty you wish to edit in the left-hand panel and click on the **Edit** button (or double-click on the specialty) to display the **Edit Specialty** page, as in the following example:

The screenshot shows the 'Manage Specialties' window. On the left, a table lists specialties by HL7 Code and Name. 'Mammography' (99907) is selected. Below the table are 'New', 'Delete', and 'Edit' buttons; the 'Edit' button is circled in red. On the right, the 'Edit Specialty' form for 'Mammography' is displayed, showing the HL7 Code as 99907 and the Viewer Display Name as Mammography. There is a checkbox labeled 'Allow this Specialty to be presented as an option in picklists' which is checked. At the bottom right of the form are 'Save' and 'Cancel' buttons.

Figure 365: Edit Specialty Page

2. Modify any of the information for this specialty as desired.

NOTE: You cannot change the HL7 Code for an existing specialty.

3. Click **Save**.

c. Deleting Existing Specialties

To delete an existing specialty:

1. Click on the specialty you wish to edit in the left-hand panel and click on the **Delete** button at the bottom of the left-hand panel.

A confirmation message appears.

2. Click **OK**.

10.6.4. Managing Procedures

The **Manage Procedures** page allows you to create a dictionary of procedures that can be used as filters when setting up individual RTWL and Teaching worklists as well as to determine whether specific users are reading candidates for defined Auto Assignment rules.

Click on the **Procedures** link to display the **Manage Procedures** page, as in the following example:

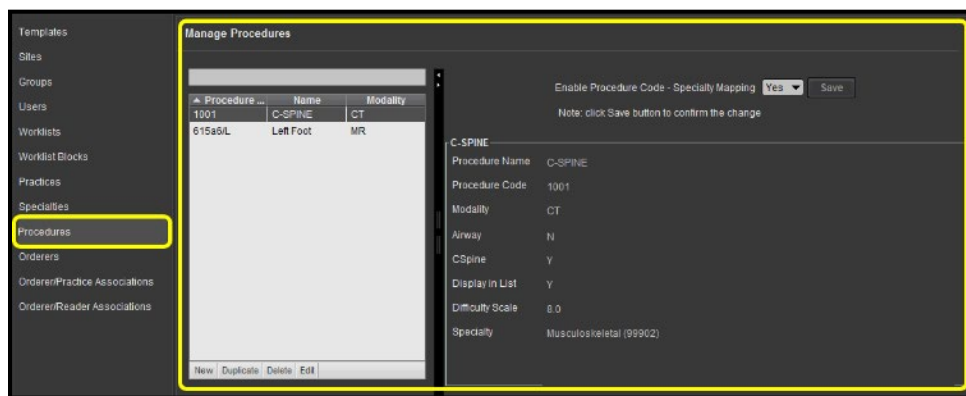


Figure 366: Manage Procedures Page

By default, all existing procedures will be displayed in the left-hand panel of the Manage Procedures page and you can click on any procedure to see the details for that procedure in the right-hand panel.

If the list of existing procedures is long, you can filter the list by entering text in the field at the top of the left-hand panel to filter based on matching **Name**.

a. Enabling Procedure Code – Specialty Mapping

As described in subsection 10.6.3 above, if a site is receiving HL7 messages with specialty values defined in the OBR-18 field, those values can be used as filters when setting up individual RTWL and Teaching worklists and as query fields on the Query Page and in the Quick Filters panel of the RealTime Study List within the Merge PACS Workstation. If the HL7 messages do not have values entered for OBR-18, however, and the site still wants query search on Specialty, Merge PACS can be configured to map incoming exams to a specialty based on the procedure codes specified in the HL7 message at OBR-4.1.

To enable Procedure Code – Specialty Mapping, select **Yes** from the drop-down menu and click on the **Save** button, as in the following example:

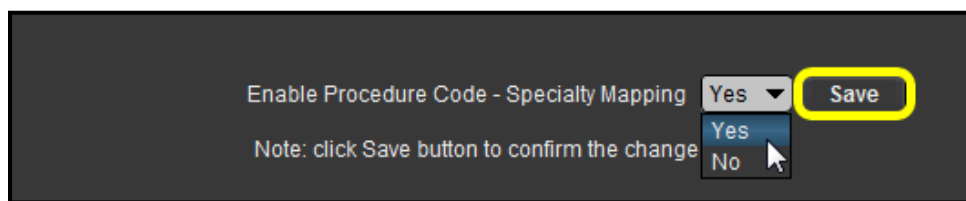


Figure 367: Procedures Configuration Page

Once Procedure Code – Specialty Mapping has been enabled, you can map individual procedures to existing specialties when defining the procedures, as described below.

NOTE: If you decide to disable Procedure Code – Specialty Mapping after you have mapped one or more procedures to a specialty, the mappings will be retained in case you later decide to re-enable Procedure Code – Specialty Mapping; they simply will not be used while Procedure Code – Specialty Mapping is disabled.

b. Defining New Procedures

To define a new procedure:

1. Click the **New** button at the bottom of the left-hand panel to display the **New Procedure** page in the right-hand panel, as in the following example:

The screenshot shows the 'Manage Procedures' window. On the left, a table lists existing procedures:

Procedure Code	Name	Modality
700	ProcedureMR	MR
701	ProcedureCT	CT
702	ProcedureOT	OT
703	ProcedureCR	CR
704	ProcedureDD	DD
705	ProcedureST	ST

At the bottom of this table is a 'New' button. The right panel shows the 'New Procedure' form with the following fields:

- Procedure Name:
- Procedure Code:
- Modality:
- Airway: ☐
- CSpine: ☐
- Display in List: ☒
- Difficulty Scale:
- Specialty:

Buttons at the bottom right of the form are 'Save' and 'Cancel'.

Figure 368: New Procedure Page

2. In the **Procedure Name** field, enter a descriptive name for this procedure.

NOTE: The Procedure Name should not contain the < or > characters.

NOTE: You can use an asterisk as a wildcard when entering the Procedure Name. To see how wild cards work, click on the [\[example\]](#) link at the top of the New Procedure section.

3. In the **Procedure Code** field, enter the code for this procedure.
4. From the **Modality** drop-down menu, select the modality type for this procedure.

NOTE: You can only select modality types that have previously been defined, as described in subsection 9.13.4 above.

5. If you are using the optional **Merge RadStream** component, you can also designate whether this procedure is related to **Airway** or **CSpine** in order to have it given top priority within the worklist.
6. Make sure that the **Display in List** checkbox is selected to have this procedure included as an option in various picklists.
7. In the **Difficulty Scale** field, enter the number of difficulty units (e.g., RVUs), if applicable.
8. If **Procedure Code – Specialty Mapping** has been enabled, you can also select a specialty from the drop-down **Specialty** menu to which this procedure will be mapped.
9. Click **Save** at the bottom of the **New Procedure** page.

c. Duplicating Existing Procedures

Instead of creating a new procedure from scratch, you can duplicate an existing procedure and modify it as needed.

To duplicate an existing procedure:

1. Click on the procedure you wish to duplicate in the left-hand panel and click on the **Duplicate** button to display the **Duplicate Procedure** page, as in the following example:

Manage Procedures

Procedure Code	Name	Modality
1001	C-SPINE	CT
515a6L	Left Foot	MR

Enable Procedure Code - Specialty Mapping

Note: click Save button to confirm the change

Left Foot(1)

Procedure Name:

Procedure Code:

Modality:

Anway: ☐

CSpine: ☐

Display in List: ☒

Difficulty Scale:

Specialty:

New Duplicate Delete Edit

Figure 369: Duplicate Procedure Page

2. Modify any of the information for this procedure as desired.
3. Click **Save**.

d. Editing Existing Procedures**To edit an existing procedure:**

1. Click on the procedure you wish to edit in the left-hand panel and click on the **Edit** button (or double-click on the procedure) to display the **Edit Procedure** page, as in the following example:

Manage Procedures

Procedure Code	Name	Modality
1001	C-SPINE	CT
515a6L	Left Foot	MR

Enable Procedure Code - Specialty Mapping

Note: click Save button to confirm the change

Left Foot

Procedure Name:

Procedure Code:

Modality:

Anway: ☐

CSpine: ☐

Display in List: ☒

Difficulty Scale:

Specialty:

New Duplicate Delete Edit

Figure 370: Edit Procedure Page

2. Modify any of the information for this procedure as desired.
3. Click **Save**.

e. Deleting Existing Procedures

To delete an existing procedure:

1. Click on the procedure you wish to edit in the left-hand panel and click on the **Delete** button at the bottom of the left-hand panel.

A confirmation message appears.

2. Click **OK**.

10.6.5. Managing Orderers

In addition to the providers who actually read the exams within Merge PACS and their practices, you can also define any number of orderers (e.g. a referring physician) so that you can later define relationships between specific providers and specific orderers and take these relationships into account when creating Auto Assignment rules. For example, Ima Medico is a referring physician who prefers that all her exams be read by radiologist Randolph Radiologist. Yura Medico, on the other hand, is another orderer who specifically prefers that Randolph Radiology does not read his exams and also does not want anybody from the practice ACME Radiology Partners to read his exams.

NOTE: By default, any Merge PACS Users that have a defined provider code (as described in subsection 9.11.1 above) will automatically be considered an orderer, and you can also add additional orderers to the system manually.

Once you have defined the set of orderers for Merge PACS, you can configure the relationships between orderers and providers/practices in the following subsections.

Click on the **Orderers** link to display the **Manage Orderers** page, as in the following example:

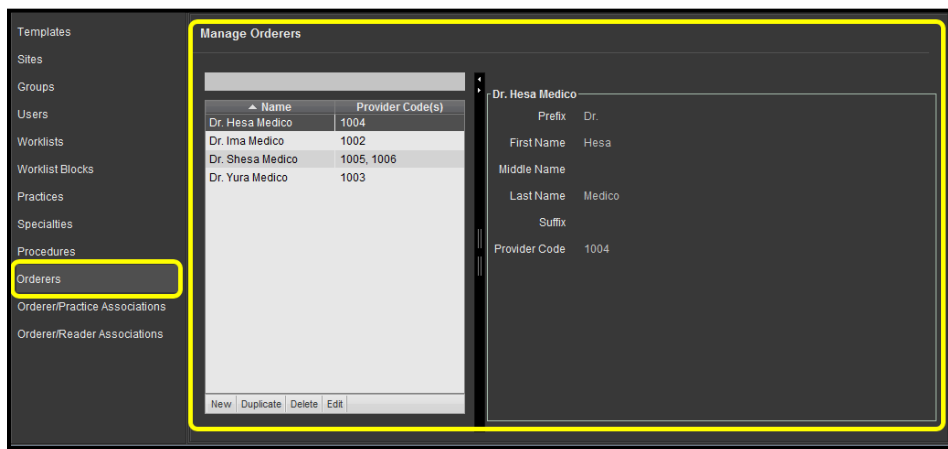


Figure 371: Manage Orderers Page

By default, all existing orderers will be displayed in the left-hand panel of the Manage Orderers page and you can click on any orderer to see the details for that orderer in the right-hand panel.

If the list of existing orderers is long, you can filter the list by entering text in the field at the top of the left-hand panel to filter based on matching **Name**.

a. Defining New Orderers

To define a new orderer:

1. Click the **New** button at the bottom of the left-hand panel to display the **New Orderer** page in the right-hand panel, as in the following example:

The screenshot shows the 'Manage Orderers' interface. On the left, there is a table with the following data:

Name	Provider Code(s)
Dr. Hesa Medico	1004
Dr. Ima Medico	1002
Dr. Shesa Medico	1005, 1006
Dr. Yura Medico	1003

At the bottom of the left panel, there are buttons: 'New' (highlighted with a red circle), 'Duplicate', 'Delete', and 'Edit'. On the right, the 'New Orderer' form is displayed, outlined in yellow. It contains the following fields:

- Prefix
- First Name (pre-filled with 'New Orderer')
- Middle Name
- Last Name
- Suffix
- Provider Code

At the bottom right of the form are 'Save' and 'Cancel' buttons.

Figure 372: New Orderer Page

2. Enter the desired information for this orderer in the fields provided.

NOTE: If the user has more than one provider code, you can enter them all in this field separated by commas (e.g., 10001, 10002).

3. Click **Save**.

b. Duplicating Existing Orderers

Instead of creating a new orderer from scratch, you can duplicate an existing orderer and modify it as needed.

To duplicate an existing orderer:

1. Click on the orderer you wish to duplicate in the left-hand panel and click on the **Duplicate** button to display the **Duplicate Orderer** page, as in the following example:

The screenshot shows the 'Manage Orderers' interface. On the left, a table lists orderers: Dr. Hesa Medico (1004), Dr. Ima Medico (1002), Dr. Shesa Medico (1005, 1006), and Dr. Yura Medico (1003). The 'Duplicate' button at the bottom is highlighted with a red box. On the right, the 'Duplicate Orderer' form for 'Dr. Ima(1) Medico' is highlighted with a yellow box. The form contains fields for Prefix (Dr.), First Name (Ima(1)), Middle Name, Last Name (Medico), Suffix, and Provider Code (1002(1)). 'Save' and 'Cancel' buttons are at the bottom right.

Name	Provider Code(s)
Dr. Hesa Medico	1004
Dr. Ima Medico	1002
Dr. Shesa Medico	1005, 1006
Dr. Yura Medico	1003

Buttons: New Duplicate Delete Edit

Form Fields:
Prefix: Dr.
First Name: Ima(1)
Middle Name:
Last Name: Medico
Suffix:
Provider Code: 1002(1)
Buttons: Save Cancel

Figure 373: Duplicate Orderer Page

2. Modify any of the information for this orderer as desired.
3. Click **Save**.

c. Editing Existing Orderers**To edit an existing orderer:**

1. Click on the orderer you wish to edit in the left-hand panel and click on the **Edit** button (or double-click on the orderer) to display the **Edit Orderer** page, as in the following example:

The screenshot shows the 'Manage Orderers' interface. On the left, a table lists orderers: Dr. Hesa Medico (1004), Dr. Ima Medico (1002), Dr. Shesa Medico (1005, 1006), and Dr. Yura Medico (1003). The 'Edit' button at the bottom is highlighted with a red box. On the right, the 'Edit Orderer' form for 'Dr. Shesa Medico' is highlighted with a yellow box. The form contains fields for Prefix (Dr.), First Name (Shesa), Middle Name, Last Name (Medico), Suffix, and Provider Code (1005, 1006). 'Save' and 'Cancel' buttons are at the bottom right.

Name	Provider Code(s)
Dr. Hesa Medico	1004
Dr. Ima Medico	1002
Dr. Shesa Medico	1005, 1006
Dr. Yura Medico	1003

Buttons: New Duplicate Delete Edit

Form Fields:
Prefix: Dr.
First Name: Shesa
Middle Name:
Last Name: Medico
Suffix:
Provider Code: 1005, 1006
Buttons: Save Cancel

Figure 374: Edit Orderer Page

2. Modify any of the information for this orderer as desired.
3. Click **Save**.

d. Deleting Existing Orderers

To delete an existing orderer:

1. Click on the orderer you wish to edit in the left-hand panel and click on the **Delete** button at the bottom of the left-hand panel.

A confirmation message appears.

2. Click **OK**.

10.6.6. Managing Orderer/Practice Associations

Once you have defined practices (as described in subsection 10.6.2 above) and orderers (as described in subsection 10.6.5 above), you can configure relationships between specific orders and specific practices to indicate the practice or practices the orderer would prefer to read his or her studies or those that the orderer would prefer to **not** read his or her studies. These preferences can then be taken into account when creating Auto Assignment rules.

Click on the **Orderer/Practice Associations** link to display the **Manage Orderer / Practice** page, as in the following example:

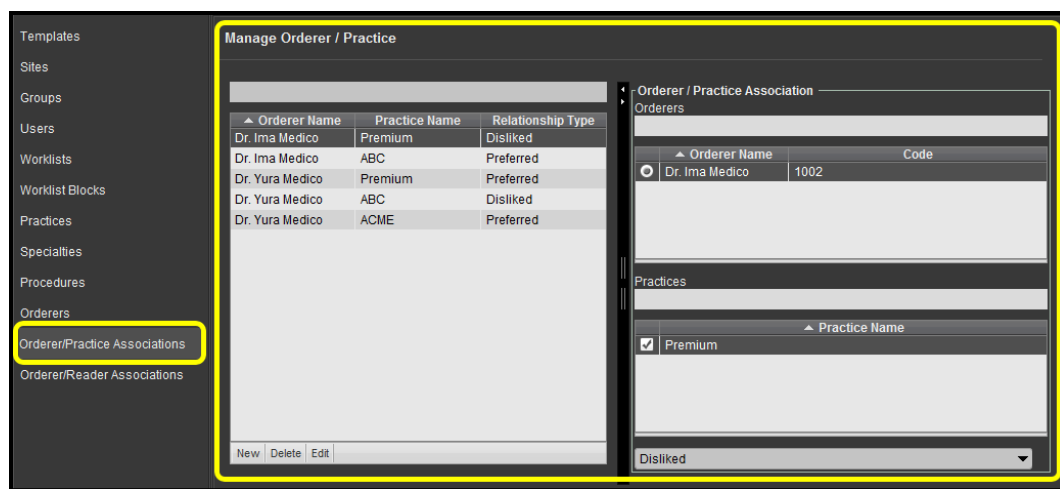


Figure 375: Manage Orderer / Practice Page

By default, all existing orderer/practice associations will be displayed in the left-hand panel of the Manage Orderer / Practice page and you can click on any association to see the details of that association in the right-hand panel.

NOTE: Although you can configure associations with multiple practices for a single orderer, the association with each practice for each orderer will be listed separately in the left-hand panel.

If the list of existing associations is long, you can filter the list by entering text in the field at the top of the left-hand panel to filter based on matching **Orderer Name**, **Practice Name** or **Relationship Type**.

a. Defining New Orderer/Practice Associations

To define a new orderer/practice association:

1. Click the **New** button at the bottom of the left-hand panel to display the **New Orderer / Practice Association** page in the right-hand panel, as in the following example:

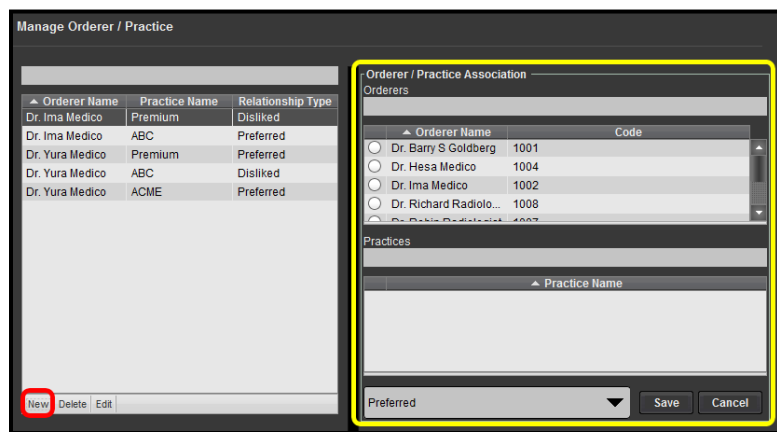


Figure 376: New Orderer / Practice Association Page

2. At the top section of the right-hand panel, select the desired orderer from the list of available orderers by clicking on the radio button next to that orderer's name.

NOTE: If the list of orderers is long, you can filter the list by entering text in the field at the top of the list to filter based on matching **Orderer Name** or **Code**.

Once you have selected an orderer, a list of available practices will be displayed in the bottom section, as in the following example:

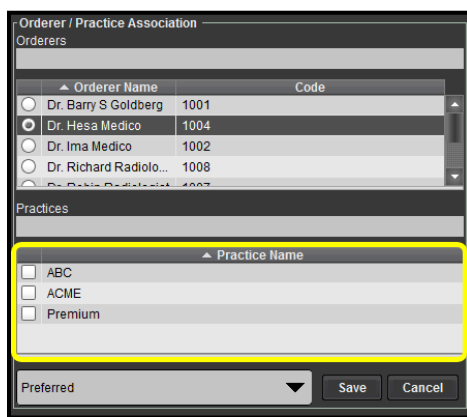


Figure 377: List of Available Practices

NOTE: If the list of practices is long, you can filter the list by entering text in the field at the top of the list to filter based on matching **Practice Name**.

NOTE: The list will not include any practices for which an association has already been defined for the selected orderer.

3. Select one or more practices by clicking the checkbox for each practice.

- From the drop-down **Relationship Type** menu, select whether the selected practice or practices are **Preferred** by this orderer or **Disliked**.
- Click **Save**.

NOTE: If you have selected multiple practices for this association, a separate association for each practice will be displayed in the main list of existing associations in the left-hand panel.

NOTE: You cannot define both Preferred and Disliked relationships for the same orderer at once. Instead, you can create two associations for that orderer (one defining the practices the orderer prefers and the other defining the practices the orderer dislikes).

b. Editing Existing Orderer/Practice Associations

To edit an existing orderer/practice association:

- Click on the orderer/practice association you wish to edit in the left-hand panel and click on the **Edit** button (or double-click on the association) to display the **Edit Orderer / Practice Association** page, as in the following example:

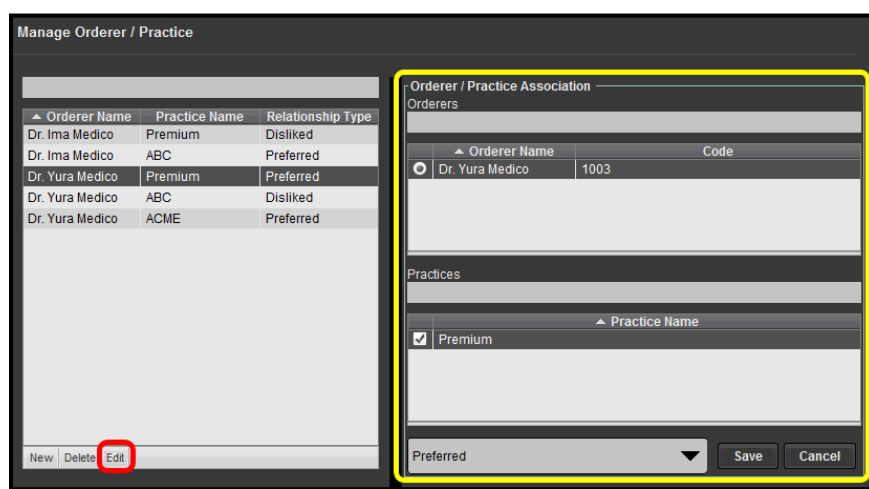


Figure 378: Edit Orderer / Practice Association Page

NOTE: Although you can create associations between one orderer and multiple practices at the same time, the association between each practice and that orderer is listed separately and will need to be edited separately.

- Modify any of the information for this association as desired.
- Click **Save**.

c. Deleting Existing Orderer/Practice Associations

To delete an existing orderer/practice association:

- Click on the association you wish to edit in the left-hand panel and click on the **Delete** button at the bottom of the left-hand panel.

A confirmation message appears.

- Click **OK**.

10.6.7. Managing Orderer/Reader Associations

Once you have defined orderers (as described in subsection 10.6.5 above), you can configure relationships between specific orders and specific readers (e.g., PACS Users) to indicate the reader or readers the orderer would prefer to read his or her studies or those that the orderer would prefer to **not** read his or her studies. These preferences can then be taken into account when creating Auto Assignment rules.

Click on the **Orderer/Reader Associations** link to display the **Manage Orderer / Reader** page, as in the following example:

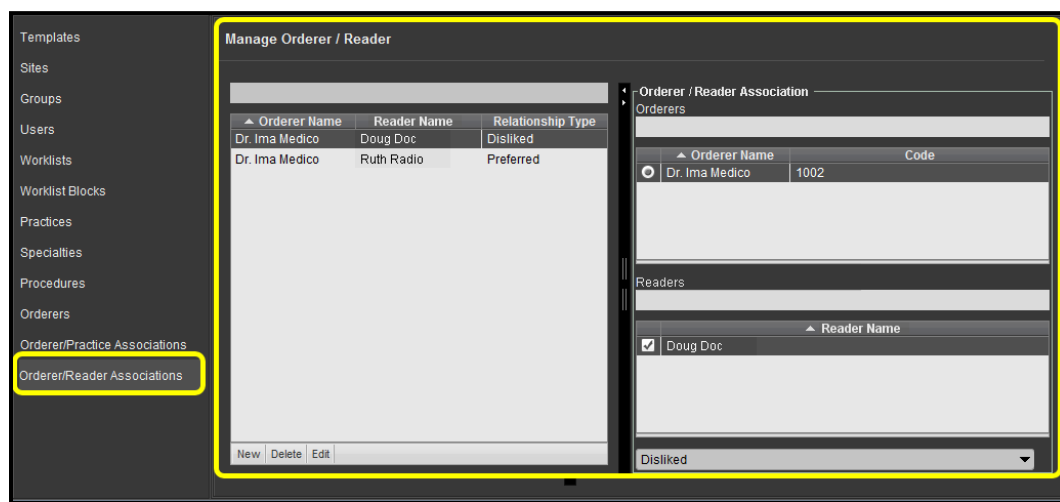


Figure 379: Manage Orderer / Reader Page

By default, all existing orderer/reader associations will be displayed in the left-hand panel of the Manage Orderer / Reader page and you can click on any association to see the details of that association in the right-hand panel.

NOTE: Although you can configure associations with multiple readers for a single orderer, the association with each reader for each orderer will be listed separately in the left-hand panel.

If the list of existing associations is long, you can filter the list by entering text in the field at the top of the left-hand panel to filter based on matching **Orderer Name**, **Reader Name** or **Relationship Type**.

a. Defining New Orderer/Reader Associations

To define a new orderer/reader association:

1. Click the **New** button at the bottom of the left-hand panel to display the **New Orderer / Reader Association** page in the right-hand panel, as in the following example:

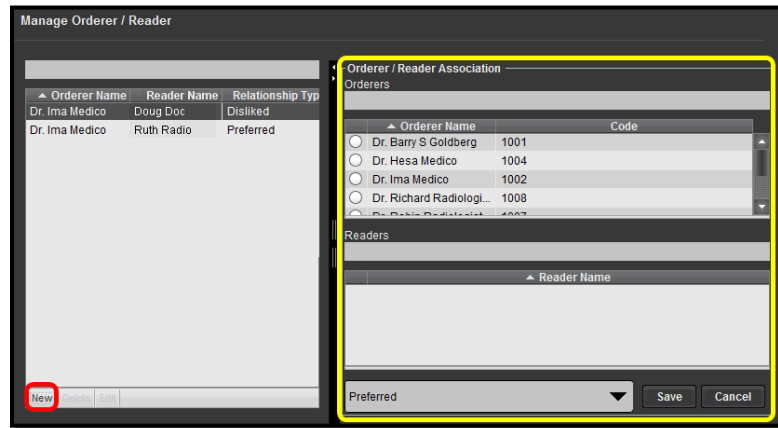


Figure 380: New Orderer / Reader Association Page

2. At the top section of the right-hand panel, select the desired orderer from the list of available orderers by clicking on the radio button next to that orderer's name.

NOTE: If the list of orderers is long, you can filter the list by entering text in the field at the top of the list to filter based on matching **Orderer Name** or **Code**.

Once you have selected an orderer, a list of available readers will be displayed in the bottom section, as in the following example:

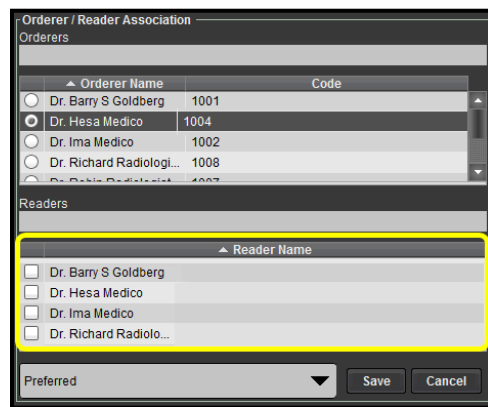


Figure 381: List of Available Readers

NOTE: If the list of readers is long, you can filter the list by entering text in the field at the top of the list to filter based on matching **Reader Name**.

NOTE: The list will not include any readers for which an association has already been defined for the selected orderer.

3. Select one or more readers by clicking the checkbox for each reader.

- From the drop-down **Relationship Type** menu, select whether the selected reader or readers are **Preferred** by this orderer or **Disliked**.
- Click **Save**.

NOTE: If you have selected multiple readers for this association, a separate association for each reader will be displayed in the main list of existing associations in the left-hand panel.

NOTE: You cannot define both Preferred and Disliked relationships for the same orderer at once. Instead, you can create two associations for that orderer (one defining the readers the orderer prefers and the other defining the readers the orderer dislikes).

b. Editing Existing Orderer/Practice Associations

To edit an existing orderer/reader association:

- Click on the orderer/reader association you wish to edit in the left-hand panel and click on the **Edit** button (or double-click on the association) to display the **Edit Orderer / Reader Association** page, as in the following example:

Figure 382: Edit Orderer / Reader Association Page

NOTE: Although you can create associations between one orderer and multiple readers at the same time, the association between each reader and that orderer is listed separately and will need to be edited separately.

- Modify any of the information for this association as desired.
- Click **Save**.

c. Deleting Existing Orderer/Reader Associations

To delete an existing orderer/reader association:

- Click on the association you wish to edit in the left-hand panel and click on the **Delete** button at the bottom of the left-hand panel.

A confirmation message appears.

- Click **OK**.

10.7. Advanced Distribution Engine

Clicking the **Assignment Engine** link in the left-hand navigation panel will display the Advanced Distribution Engine Processes menu, as in the following example:

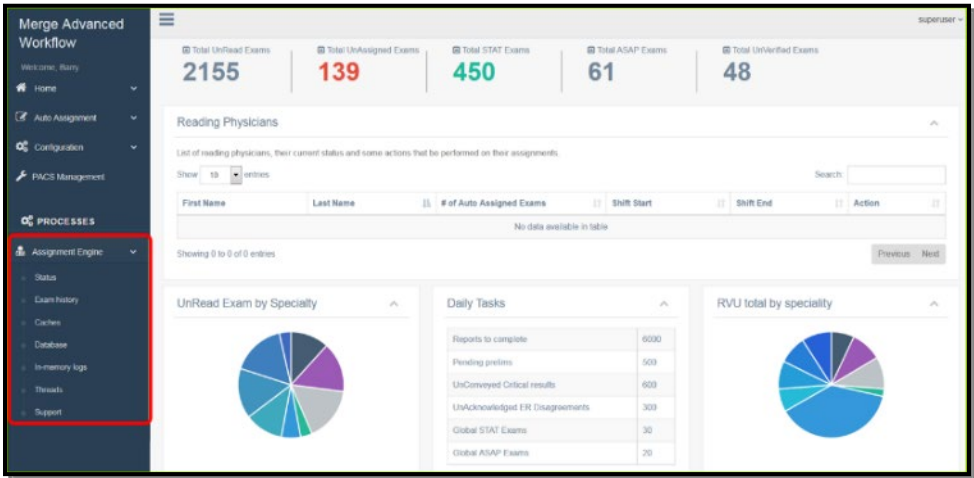


Figure 383: Advanced Distribution Engine Processes Menu

The Advanced Distribution Engine Processes menu has links to the following pages designed to help Administrators, Support, and Developers troubleshoot the Advanced Distribution Engine:

Page	Description
Status	Provides snapshots of state and data in the Advanced Distribution Engine.
Exam History	Provides query capability to search for exam events and Advanced Distribution Engine logs.
Caches	Provides state information of internal information in the Advanced Distribution Engine.
Database	Provides state information of database connections and SQL statement execution status in the Advanced Distribution Engine.
In-memory Logs	Provides snapshots of state and data in the Advanced Distribution Engine.
Threads	Provides state information of threads, monitors and locks in the Advanced Distribution Engine.
Support	Provides specific tools for collecting information needed for troubleshooting, as well as the ability to perform certain intervention actions such as restarting/killing processes.

10.7.1. Advanced Distribution Engine Status

The Advanced Distribution Engine Status page provides a quick glance at the overall state of the Advanced Distribution Engine process and is accessed by clicking the **Status** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:

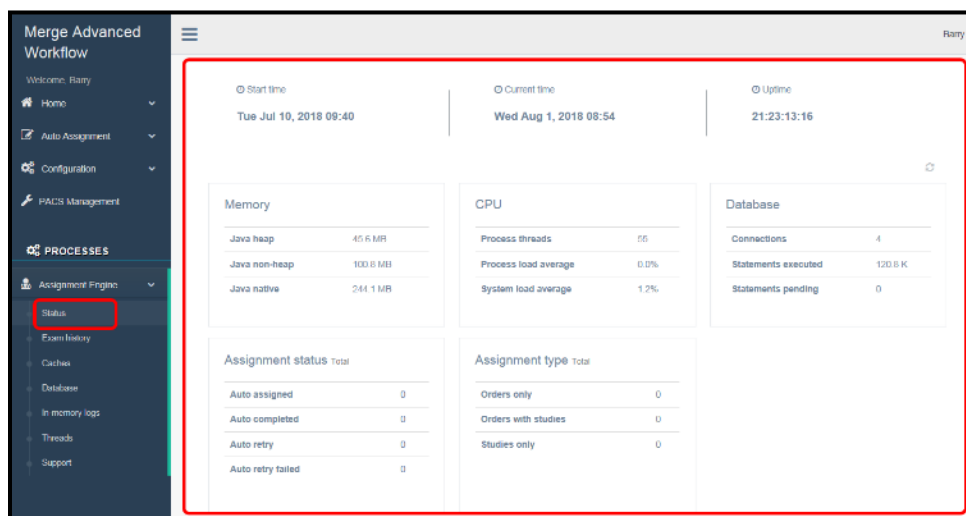


Figure 384: Advanced Distribution Engine Status

a. Refreshing Data

To refresh just the data content of the Advanced Distribution Engine Status, click the **Refresh** icon on the page itself (as opposed to using the refresh feature of your Web browser, which will refresh all the Advanced Workflow Management Pages), as in the following example:

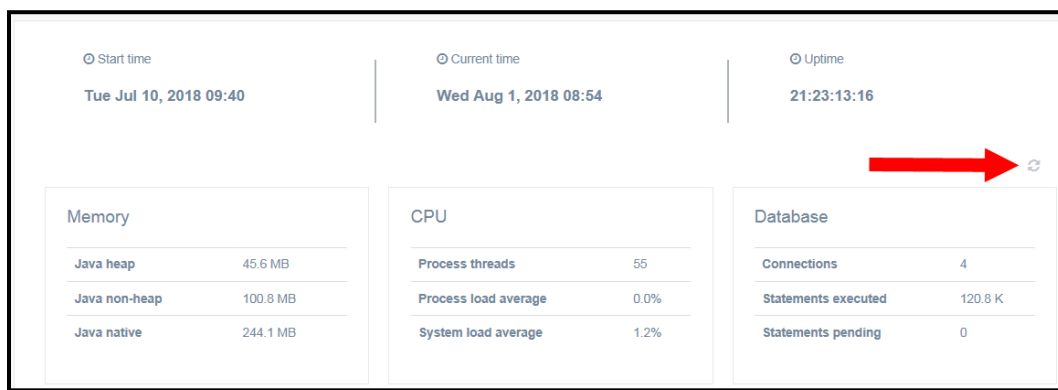


Figure 385: Advanced Distribution Engine Status Refresh

NOTE: It is strongly recommended to use the Refresh icon on the page to get current status information.

b. Time Metrics

The top section of the Advanced Distribution Engine Status page displays various time metrics, as in the following example:

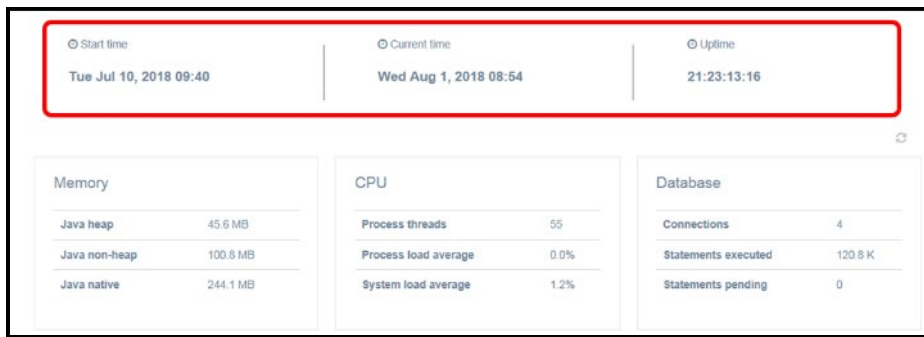


Figure 386: Advanced Distribution Engine Status -- Time Metrics

The following metrics are available:

Metric	Description
Start time	Time the process was launched. This information does not change while the process is running.
Current time	Configured and current server time when the page was last loaded. This changes on every load or refresh of the page.
Uptime	Computed time between Start and Current time for reference. This changes on every load or refresh of the page.

c. Memory CPU and Database Metrics

The middle section of the Advanced Distribution Engine Status page displays metrics related to memory, CPU and database usage by the Advanced Distribution Engine process, as in the following example:

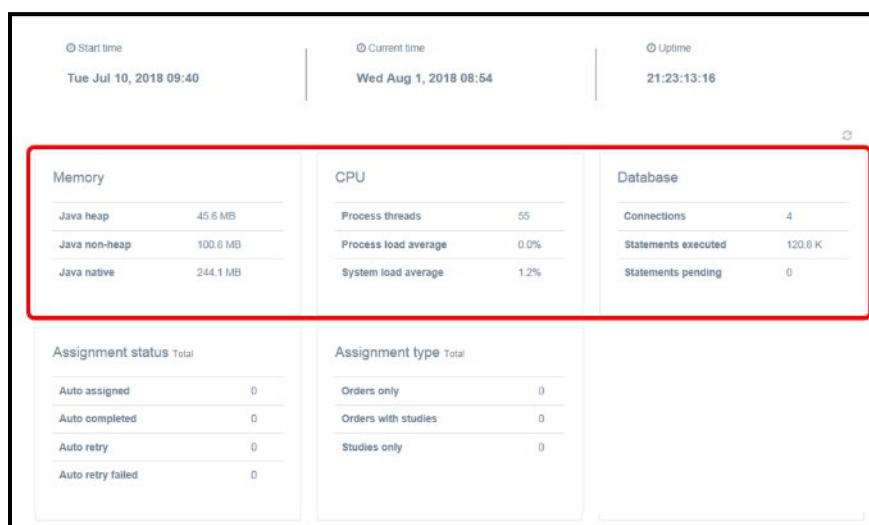


Figure 387: Advanced Distribution Engine Status -- Memory, CPU and Database Usage

- The **Memory** section displays the following metrics related to memory allocations for the Advanced Distribution Engine process:

Metric	Description
Java heap	Current Java heap committed to physical memory
Java non-heap	Non-heap allocations made by the java virtual machine. This are usually JVM specific internal memory allocations.
Java native	Allocations made by the JVM to interact with Windows-specific (native) resources and processes

- The **CPU** section displays the following aggregated and normalized metrics across physical CPU's in the system:

Metric	Description
Process threads	The number of threads currently allocated in the Advanced Distribution Engine process
Process load average	The percentage of CPU cycles used across all CPU's in the system by the Advanced Distribution Engine when calculations were made. This measurement is made by the Oracle JVM.
System load average	The percentage of CPU cycles used across all CPU's in the system by all system processes when calculations were made. This measurement is made by the Oracle JVM.

- The **Database** section of the Advanced Distribution Engine Status page displays the following metrics around database usage by the process:

Metric	Description
Connections	The number of open connections from the Advanced Distribution Engine to the database.
Statements executed	The number of statements executed by the Advanced Distribution Engine.
Statements pending	The number of statements waiting execution. The Advanced Distribution Engine manages database connection pools and statements are always executed in the context of their respective pool. This should always be zero. If it's not zero and this is observed over a period of time, it may indicate the database is busy and requires database administrator assistance.

NOTE: These metrics are not metrics about the database itself but instead how the process is using the database.

d. Assignment Status and Assignment Type Metrics

The bottom section provides quick glance information for Advanced Distribution Engine notifications, assignment status, and assignment type, as in the following example:

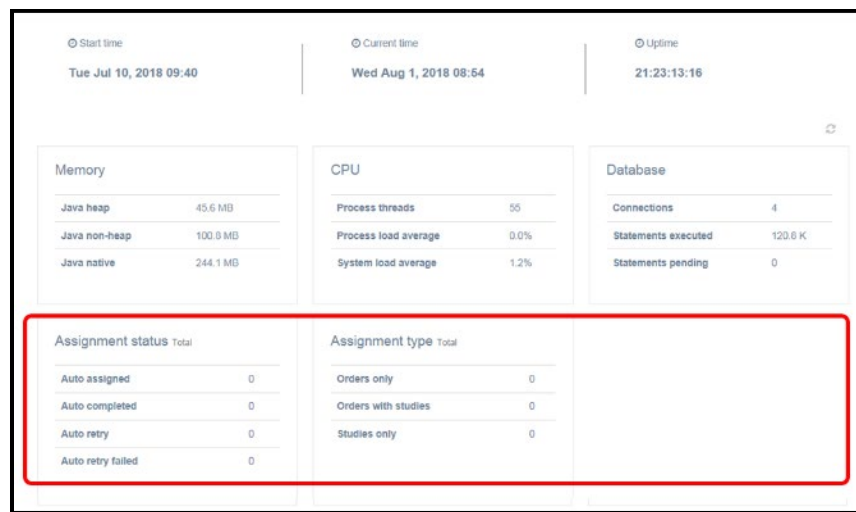


Figure 388: Advanced Distribution Engine Status -- Notifications, Assignment Status and Assignment Type

NOTE: The information in this section is temporal, not persisted, and is always calculated from the point of engine start.

NOTE: The information in this section consists of running totals. It does not indicate the state of exams and no calculations are made from this information. If you want to understand the state of exams in the system, turnaround times, or average times, it is strongly recommended that you run the reports in the Analytics / Dashboard Server.

- The **Assignment Status** section of the Advanced Distribution Engine Status page displays the following totals around exam statuses set by the Advanced Distribution Engine.

Metric	Description
Auto Assigned	Number of times an exam(s) status was set to MAA.
Auto Completed	Number of times an exam(s) status was set to MAC.
Auto Retry	Number of times an exam(s) status was set to MAR.
Auto Retry Failed	Number of times an exam(s) status was set to MARF.

NOTE: These totals are independent of the number of exams. For example, if a single exam has been reassigned 10 times and the assignment was successful all 10 times, you will see "Auto assigned" with a count of 10 and not 1.

- The **Assignment Type** section of the Advanced Distribution Engine Status page displays the following totals around type of exams that have been processed:

Metric	Description
Orders Only	Number of times an exam which comprises of an Order only was processed.
Orders with Studies	Number of times an exam which comprises of an Order with a linked study was processed.
Study Shells	Number of times an exam which comprises of a Study Shell (Order + study shell) was processed.
Studies Only	Number of times an exam which comprises of a study only was processed.

NOTE: These totals are independent of the exam(s). For example, if an exam (Study Only) has been reassigned 10 times, regardless of the outcome, you will see "Studies Only" with a count of 10 and not 1.

10.7.2. Advanced Distribution Engine Exam History

The Advanced Distribution Engine Exam History page allows you to perform exam and log searches to view the history of exams and is accessed by clicking the **Exam History** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:

The screenshot displays the 'Merge Advanced Workflow' interface. On the left, a navigation menu lists various options, with 'Exam history' highlighted under the 'Assignment Engine' section. The main area is divided into two search panels: 'Exam search' and 'Log search'. The 'Exam search' panel contains input fields for MRN, IPID, and Accession, along with a date filter set to 'Aug 1, 2018 - Aug 1, 2019'. The 'Log search' panel includes fields for Study UID, Logger UID, and a dropdown for Operation. Both panels feature a 'Max results' slider currently set to 100. At the bottom of each panel are 'Search' and 'Clear' buttons. A red rectangle highlights the entire search area.

Figure 389: Advanced Distribution Engine Exam History

NOTE: From the Exam History page you can *either* perform an exam search *or* a log search at one time, not both. When search data is entered in the fields for one section, the fields in the other section will be disabled.

a. Exam Search

Exam search is a union of events from various Advanced Workflow processes and the Advanced Distribution Engine. The results are ordered by date and time and displayed as a time line.

NOTE: The search operation does not validate the existence of the exam in the system, as the results can be for historical exam.

To perform an exam search:

1. Enter any combination of the following information in the Exam Search fields:

- **MRN + IPID + Accession**
- **MRN + IPID only**
- **Accession only**

NOTE: MRN only and IPID only are not supported.

2. Click the **Date filter** to limit the exam search to a specific date range, as in the following example:

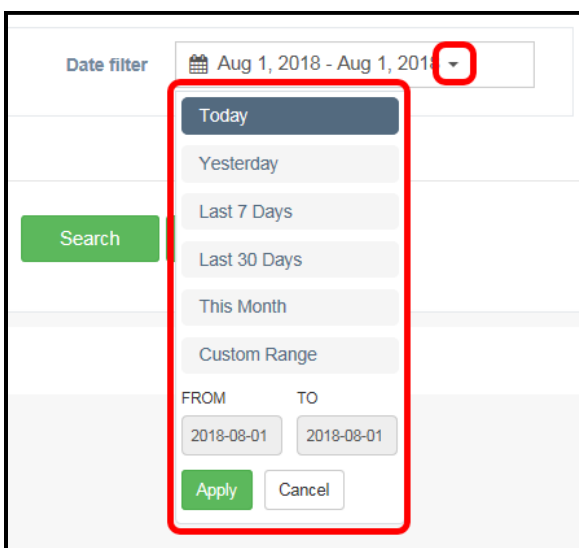


Figure 390: Advanced Distribution Engine Exam History -- Date Filter

- You can select from five preset filters and a custom date range.
 - By default, the Date filter is set to the current day and the max results are set to 50, but you can manually select a custom start date.
 - The start date cannot be before 3 months prior to today.
 - The end date is set to the current date and cannot be changed.
3. Click **Apply** to apply the date filter changes.
4. Click the **Search** button.

The search produces a list of all exams qualifying for the search data you entered, as in the following example:



Figure 391: Advanced Distribution Engine Exam History -- Exam Search Results

- 5. Click on the + symbol next to any exam listed to display the assignment exam history for that exam, as in the following example:

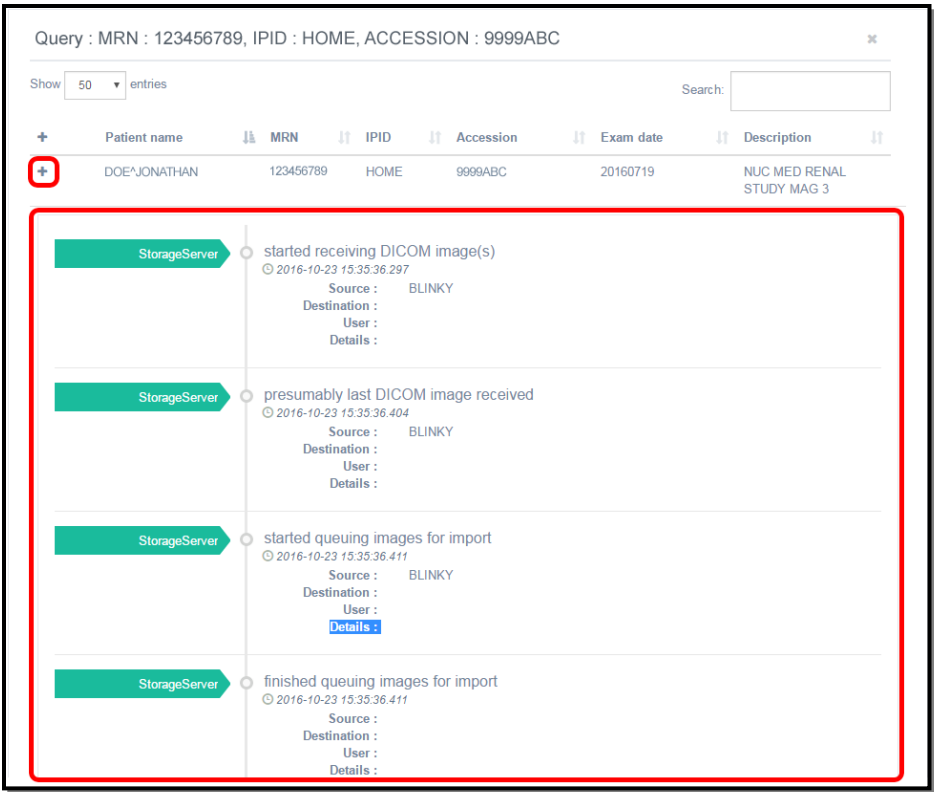


Figure 392: Advanced Distribution Engine Exam History -- Exam History Details

- NOTE:** The + symbol will actually change to – when the details are displayed and you can click on the – symbol at any time to collapse the information.
- When the result is expanded, all events in the system for the event are listed in data time order.

- The sub-system/process is identified on the left side of the time line and the data for each event is displayed on the right side of the time line.
6. If too many results are listed, you can select from the drop-down **Show entries** menu to limit the number of results you want to display.
 7. You can also enter text in the **Search** field to filter the number of results displayed to just those that match the entered text.

b. Log Search

A log search produces a list of all Advanced Distribution Engine logger events qualifying for the search data. A logger event is a collection of all events from the inception of an operation to the end. This can be 1 to n events. The results are always listed with the first event of a logger event.

To perform a log search:

1. Search for log events by entering/selecting any **one** of the following types of search criteria:

Search Criteria	Description
Study UID	Enter the Study instance UID of the exam.
Logger UID	Enter the logger UID in the Advanced Distribution Engine log.
Operation	Select the operation in the Advanced Distribution Engine.

NOTE: You cannot search based on multiple criteria at once.

2. Use the Max Results slider to limit the results to the specified maximum number of unique events.

NOTE: The minimum is pre-set to 50 results and maximum is limited to 1000. These limits cannot be changed.

3. Click the **Search** button.

The search produces a list of all exams qualifying for the search data you entered, as in the following example:

Query : 1.2.840.113837.1468955897.55546.157.8766			
Show	50	entries	Search:
+	Logger UID	Event time	Source
+	30	2016-10-23 15:42:23.77	EV_StudyStatus
+	32	2016-10-23 15:44:04.956	EV_StudyStatus
+	37	2016-10-23 15:46:41.215	EV_StudyStatus
+	39	2016-10-23 15:47:47.114	EV_StudyStatus
+	60	2016-10-27 13:46:28.369	EV_CBRReassign
Showing 1 to 5 of 5 entries			
		Previous	Next

Figure 393: Advanced Distribution Engine Exam History -- Log Search Results

- Click on the **+** symbol next to any log entry listed to display detailed information for that entry, as in the following example:

Query : 1.2.840.113837.1468955897.55546.157.8766

Show **50** entries Search:

+	Logger UID	Event time	Source	Initiator
+	30	2016-10-23 15:42:23.77	EV_StudyStatus	superuser

MRN 123456789 **IPID** HOME **Accession** 9999ABC **Exam date** 20160719

Patient name : DOE JONATHAN

Procedure name : NUC MED RENAL STUDY MAG 3

Event time	Operation	Status	Source	Target
2016-10-23 15:42:23.77	EV_StudyStatus	FILTER	superuser	[AA TBD]
Details : DISPATCHED (ExamId : S:1.2.840.113837.1468955897.55546.157.8766, dim = 0, stat = UR, initiator = superuser, parameter = null, note = , time = 1477251743753)				
2016-10-23 15:42:23.833	OP_StatusAction	PROCESSING	superuser	S:1.2.840.113837.1468955897.55546.157.8766
Details : StatusAction [actions={ASSIGN=true}]				
2016-10-23 15:42:23.835	OP_Assign	PROCESSING	[NA]	S:1.2.840.113837.1468955897.55546.157.8766
Details : Workitem : { "procedurecode": [], "specialty": null, "priority": null, "orderstatus": null, "patientclass": null, "ordererprovidercode": null, "institutiongroup": null, "institution": "ACME", "status": { "0": { "value": "UR", "parameter": null, "note": "" } }, "age": { "years": 32, "months": 387, "weeks": 1686, "days": 11805, "difficulty": null, "ipid": "HOME", "patientid": "3940927662", "patientname": "ADALBERTO BERRY", "birthdate": "1984-03-24", "sex": "M", "studydescription": "NUC MED RENAL STUDY MAG 3", "modality": ["NM"], "licenseregion": null, "procedurecodemeaning": [] } }				

Figure 394: Advanced Distribution Engine Exam History -- Log Details

NOTE: The **+** symbol will actually change to **–** when the details are displayed and you can click on the **–** symbol at any time to collapse the information.

- The top section of the expanded details provides more information about the patient or exams depending on what search criterion was used:
 - When searching by Study Instance UID, the top section contains details of the patient.
 - When searching by Logger UID or Operation the top section lists all the exams in the operation.
 - The second section contains all the details of each log event within the logger operation, which mirrors the data in the AssignmentEngine.log file. Log events only related to Advanced Distribution Engine operations on the exam are listed. Other log events are excluded.
- If too many results are listed, you can select from the drop-down **Show entries** menu to limit the number of results you want to display.
 - You can also enter text in the **Search** field to filter the number of results displayed to just those that match the entered text.

10.7.3. Advanced Distribution Engine Caches

The Advanced Distribution Engine Caches page allows you to view a variety of different types of data cached in the Advanced Distribution Engine which can be used to track data and operate in memory for performance and is accessed by clicking the **Caches** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:

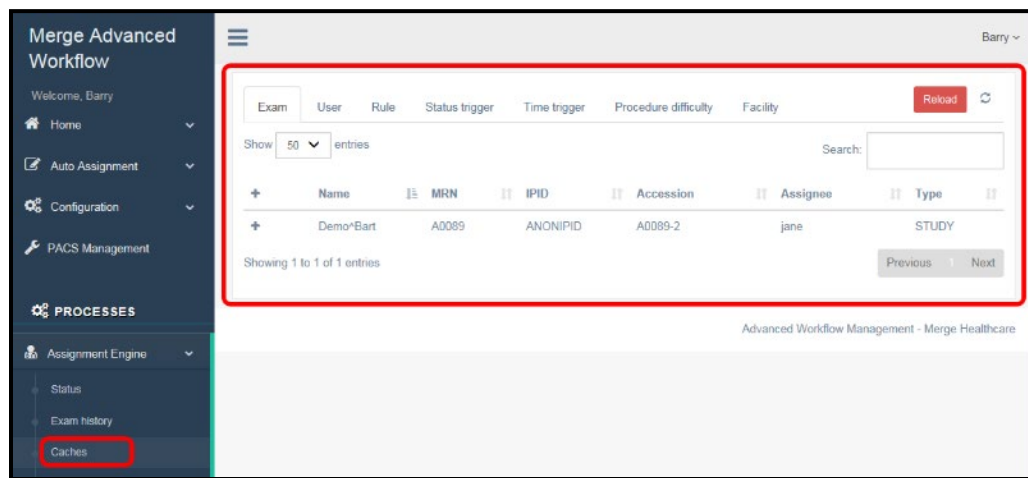


Figure 395: Advanced Distribution Engine Caches

The Caches page provides access to the following caches, each displayed as a separate tab on the page:

- **Exam**
- **User**
- **Rule**
- **Status Trigger**
- **Time Trigger**
- **Procedure Difficulty**
- **Facility**

Regardless of which cache you are currently viewing, you can perform the following actions:

- Click the red **Reload** button to cause the current cache (the selected tab) to reload the cache in the Advanced Distribution Engine. If this is the **Exam** or **User** tab, the cache data will be re-synchronized. This is an asynchronous event and the web page will not automatically refresh.
- Click the **Refresh** icon to reload the page content for the selected tab.
- Although there is no filter for limiting the number of entries in a cache, you can limit the number of cache entries to be displayed in “Show entries” drop down list for each cache. There is also a “Search” field that can be used to limit the results of the displayed cache entries.

a. Exam Cache

The **Exam** tab lists all the exams currently in the Advanced Distribution Engine (*i.e.*, exams that have been automatically assigned and yet to be completed), as in the following example:

+	Name	MRN	IPID	Accession	Assignee	Type
+	Demo^Bart	A0089	ANONIPID	A0089-2	jane	STUDY

Showing 1 to 1 of 1 entries

Previous 1 Next

Figure 396: Advanced Distribution Engine Exam Cache

- Each entry can be expanded to see additional details of the exam.
- The details are variable depending on the type of exam:
 - If the exam has order details, then they are listed.
 - If an exam has study details, then they are listed.
- Only the patient and status sections for additional details is fixed.

b. User Cache

The **User** tab lists all the users currently in the Advanced Distribution Engine (*i.e.*, users who are enabled and have selected the "Consider for auto assignment" user preference), as in the following example:

+	Login	Name	Load	Difficulty
+	doug	Doug Doc	0	0
+	jane	Jane Doe	1	0
+	phil	Phil Phys	0	0
+	ruth	Ruth Radio	0	0

Showing 1 to 4 of 4 entries

Previous 1 Next

Figure 397: Advanced Distribution Engine User Cache

- Each entry can be expanded to see additional details of the user.
- If for any reason, there are any unqualified users with automatically assigned studies, then such users will also be included until all exams assigned to user are completed or reassigned from to other users. Once that occurs, the user will be removed from the Advanced Distribution Engine and will not be listed.

c. Rule Cache

The **Rule** tab lists all the enabled Auto Assignment rules that have been created for use in evaluating assignments and reassignments, as in the following example:

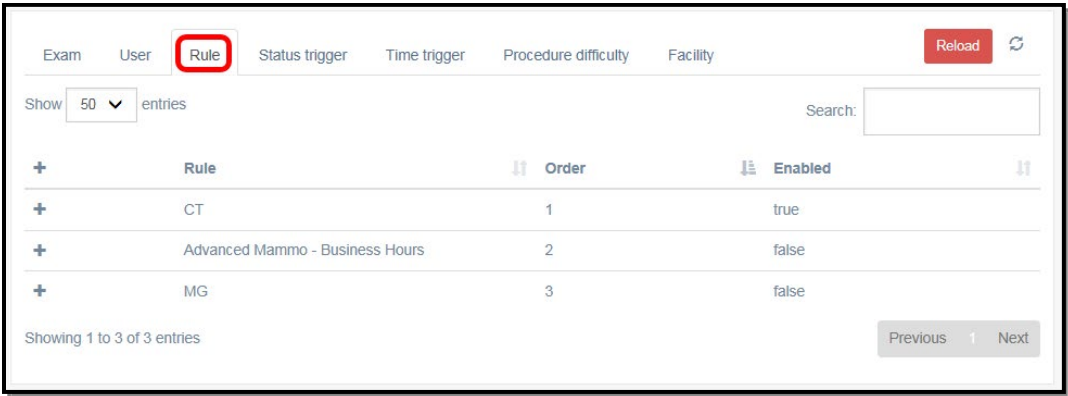


Figure 398: Advanced Distribution Engine Rule Cache

Each rule has 3 distinct sections to it as follows:

Section	Description
Match	Deals with matching a rule to an exam(s). Is always identified with ".match" as the end of the name.
Select	Deals with matching a rule to qualifying user(s). Is always identified with ".select" as the end of the name.
Sort	Deals with prioritizing qualified user(s). Is always identified with ".sort" as the end of the name.

When a rule is expanded, each of these sections is displayed as a separate tree listing on the page, as in the following examples:

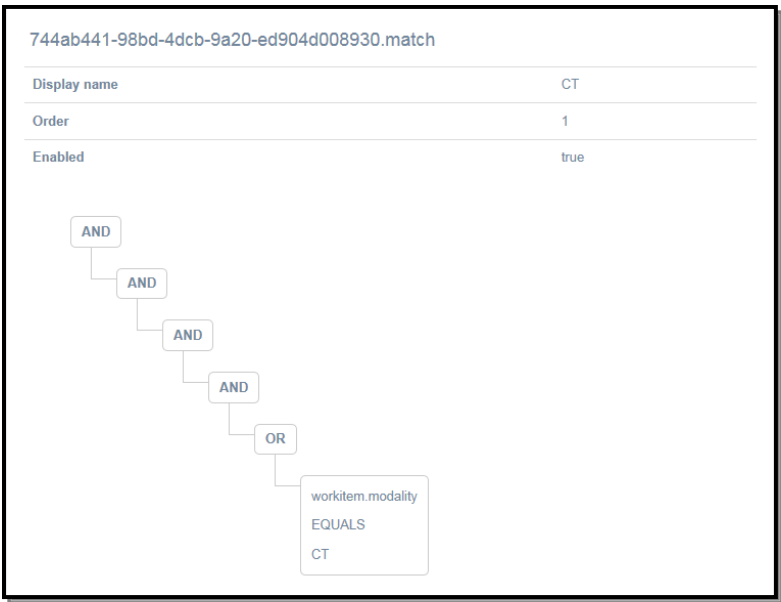


Figure 399: Advanced Distribution Engine Rule Cache – Match Details

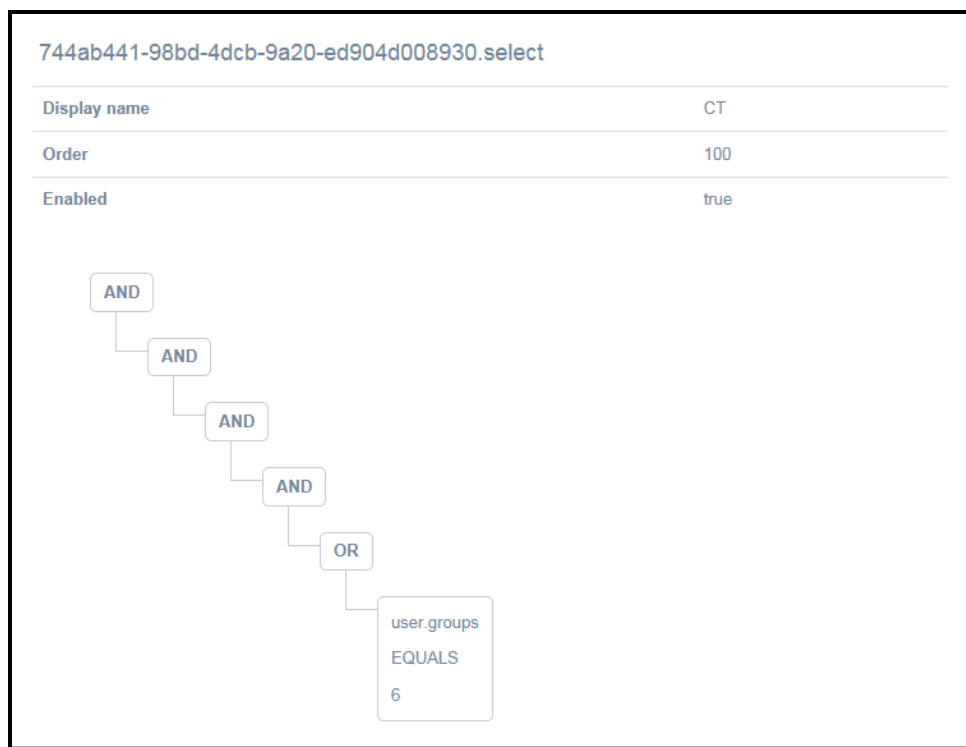


Figure 400: Advanced Distribution Engine Rule Cache – Select Details

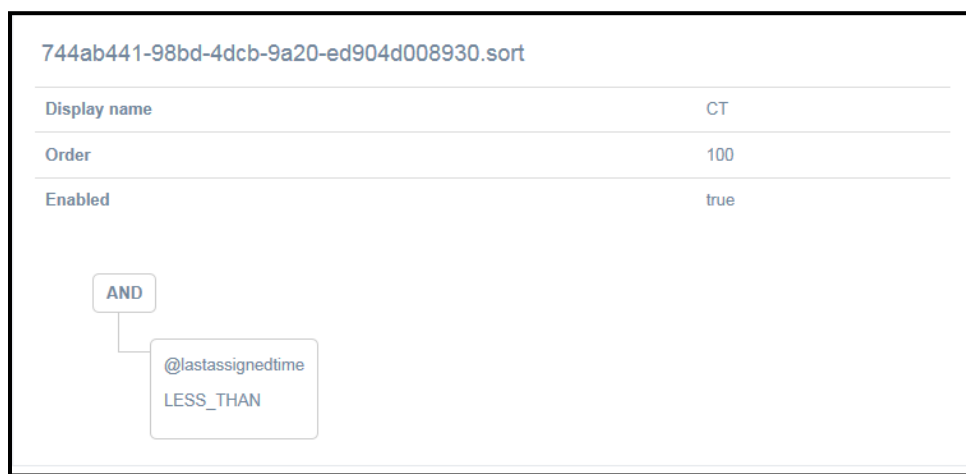
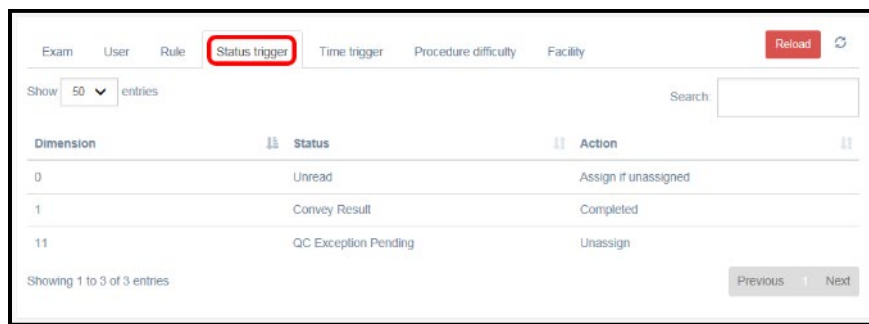


Figure 401: Advanced Distribution Engine Rule Cache – Sort Details

NOTE: A rule can only have one Match to one Select to one Sort.

d. Status Trigger Cache

The **Status Trigger** tab lists all the status triggers that have been created for use with Auto Assignment rules, as in the following example:

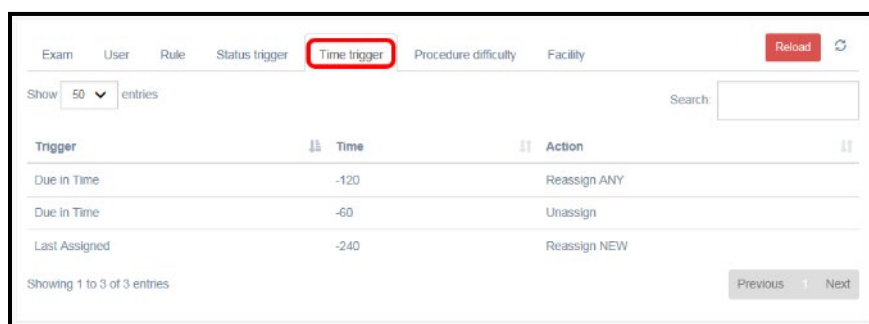


Exam	User	Rule	Status trigger	Time trigger	Procedure difficulty	Facility
Show 50 entries						
Dimension	Status	Action				
0	Unread	Assign if unassigned				
1	Convey Result	Completed				
11	QC Exception Pending	Unassign				
Showing 1 to 3 of 3 entries						
						Previous 1 Next

Figure 402: Advanced Distribution Engine Status Trigger Cache

e. Time Trigger Cache

The **Time Trigger** tab lists all the time triggers that have been created for use with Auto Assignment rules, as in the following example:



Exam	User	Rule	Status trigger	Time trigger	Procedure difficulty	Facility
Show 50 entries						
Trigger	Time	Action				
Due in Time	-120	Reassign ANY				
Due in Time	-60	Unassign				
Last Assigned	-240	Reassign NEW				
Showing 1 to 3 of 3 entries						
						Previous 1 Next

Figure 403: Advanced Distribution Engine Time Trigger Cache

f. Procedure Difficulty Cache

The **Procedure Difficulty** tab lists all the procedures and their configured difficulty that have been successfully loaded and live in the Advanced Distribution Engine, as in the following example:



Exam	User	Rule	Status trigger	Time trigger	Procedure difficulty	Facility
Show 50 entries						
Procedure	Difficulty					
C-SPINE	8.0					
Left Foot	5.0					
Showing 1 to 2 of 2 entries						
						Previous 1 Next

Figure 404: Advanced Distribution Engine Procedure Difficulty Cache

g. Facility Cache

The **Facility** tab lists all the configured facilities along with their specified regions that have been successfully loaded and live in the Advanced Distribution Engine, as in the following example:

Id	Name	Region
1	City Clinic	Massachusetts
2	ABC Radiology	California
2	ABC Radiology	California
2	ABC Radiology	California
3	ACME Radiology	Tennessee
3	ACME Radiology	Tennessee
4	Premier Radiology	Connecticut

Figure 405: Advanced Distribution Engine Facility Cache

10.7.4. Advanced Distribution Engine Database

The database page provides the state of how the Advanced Distribution Engine is using the database as well what and how many times statements from the Advanced Distribution Engine have been executed and is accessed by clicking the **Database** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:

Connections	Statements executed	Statements pending
3	50114	0

Executor	Statements	Cached statements	Duration	Current statement
Executor-1	9	0	28h33m39s	Idle
Executor-1	14410	1	46 s	Idle
Executor-2	10	0	29h4m37s	Idle
Executor-2	11662	1	21 s	Idle
Executor-3	7	0	10m33s	Idle
Executor-3	24061	2	21 s	Idle

Figure 406: Advanced Distribution Engine Database

The Advanced Distribution Engine has six pools for managing work with the database. Each pool is listed as a tab in the database page as follows:

Pool	Description
ExamPool	Used to execute queries related to exams. This involves loading of orders, studies, studysells, order status and study status. This pool is limited to select queries only.
UserPool	Used to execute queries related to users, preferences and permissions of users. This pool is limited to select queries only.
CachePool	Used to execute queries related to loading cache entries. This pool is limited to select queries only.
LoggingPool	Used to update logging and audit data of the Advanced Distribution Engine.
SearchPool	Used to perform exam and log searches from the exam history page. This involves searching Advanced Distribution Engine log tables and other study and event log tables used in the Watch process. This pool is limited to select queries only.
DefaultPool	Used to perform ad hoc queries. This pool is not used by the Advanced Distribution Engine.

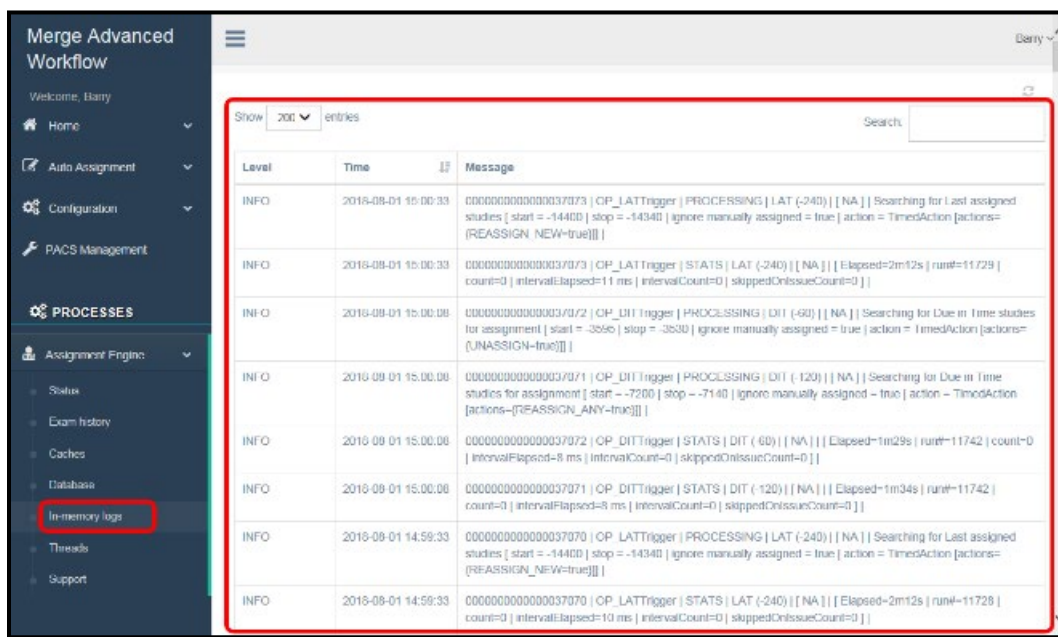
For each pool, the Summary section lists all executors for that pool and all SQL statements executed in that pool.

The data listed for each pool is from the point of start of the Advanced Distribution Engine and is temporal.

A database manager is used per pool to manage the pool. Executors are worker threads that manage parallel access to the database. They also serve as a rate limiter to manage access to the database. The database manager also manages connections to the database. The manager provides executors a connection on demand. Idle connections are automatically closed by the manager. By default, all prepared statements are cached by the database manager and removed when connections are closed.

10.7.5. Advanced Distribution Engine In-memory Logs

The In-memory Logs page provides up to a maximum of 500 most recent log messages that have been written to the AssignmentEngine.log file and is accessed by clicking the **In-memory Logs** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:



Level	Time	Message
INFO	2016-08-01 15:00:33	000000000000037073 CP_LATTrigger PROCESSING LAT (-240) [NA] Searching for Last assigned studies start = -14400 stop = -14340 ignore manually assigned = true action = TimedAction [actions= (REASSIGN_NEW=true)]
INFO	2016-08-01 15:00:33	000000000000037073 CP_LATTrigger STATS LAT (-240) [NA] Elapsed=2m12s run#=11729 count=0 intervalElapsed=11 ms intervalCount=0 skippedOnIssueCount=0
INFO	2016-08-01 15:00:08	000000000000037072 CP_DITTrigger PROCESSING DIT (-60) [NA] Searching for Due in Time studies for assignment start = -3580 stop = -3530 ignore manually assigned = true action = TimedAction [actions= (UNASSIGN=true)]
INFO	2016-08-01 15:00:08	000000000000037071 CP_DITTrigger PROCESSING DIT (-120) [NA] Searching for Due in Time studies for assignment start = -7200 stop = -7140 ignore manually assigned = true action = TimedAction [actions= (REASSIGN_ANY=true)]
INFO	2016-08-01 15:00:08	000000000000037072 CP_DITTrigger STATS DIT (-60) [NA] Elapsed=1m29s run#=11742 count=0 intervalElapsed=8 ms intervalCount=0 skippedOnIssueCount=0
INFO	2016-08-01 15:00:08	000000000000037071 CP_DITTrigger STATS DIT (-120) [NA] Elapsed=1m34s run#=11742 count=0 intervalElapsed=8 ms intervalCount=0 skippedOnIssueCount=0
INFO	2016-08-01 14:59:33	000000000000037070 CP_LATTrigger PROCESSING LAT (-240) [NA] Searching for Last assigned studies start = -14400 stop = -14340 ignore manually assigned = true action = TimedAction [actions= (REASSIGN_NEW=true)]
INFO	2016-08-01 14:59:33	000000000000037070 CP_LATTrigger STATS LAT (-240) [NA] Elapsed=2m12s run#=11728 count=0 intervalElapsed=10 ms intervalCount=0 skippedOnIssueCount=0

Figure 407: Advanced Distribution Engine In-Memory Logs

For each log entry, the following details are displayed:

- **Logging level of message**
- **Time of log event**
- **Message of log event**

You can use the “**Show Entries**” drop-down list to control the number of log entries being displayed.

You can also use the **Search** field to filter all of the log messages that are currently loaded for display. The search is across all data in all columns in the log table listed.

10.7.6. Advanced Distribution Engine Threads

The Threads page provides details of all threads and their state within the Advanced Distribution Engine process and is accessed by clicking the **Threads** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:

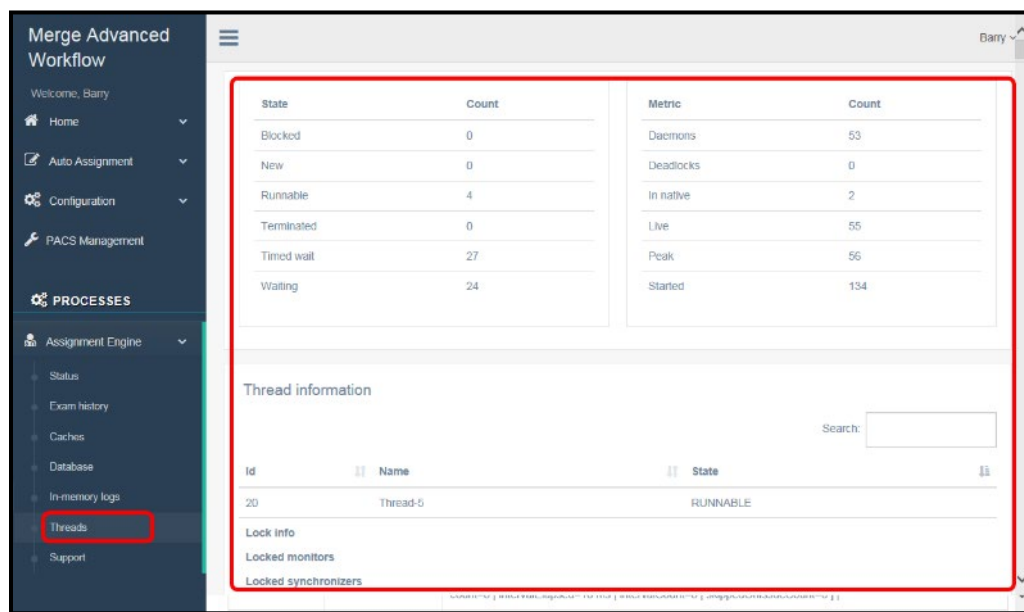


Figure 408: Advanced Distribution Engine Threads

The top section provides a summary of the state of threads required for investigation.

If a deadlock is detected in the system, the **Deadlock Analysis** section (not shown above) will be displayed and will list all the deadlocks in the process. Each deadlock is listed with grouping of information related to the deadlock that contains the following information:

- Deadlock chain of threads, with the chain of the deadlock separated by the ">" symbol and each thread identified by its thread name.
- The current list of all objects involved in the deadlock.

The **Thread Information** section lists all the threads with the following relevant information regarding each thread:

Detail	Description
Lock Info	Lists all the objects that are currently in flight. These could be locked or waiting to lock.
Locked Monitors	Lists all the locked monitors owned by the thread.
Locked Synchronizers	Lists all the locked synchronizer monitors owned by the thread.
Stack	Lists the current stack trace of the thread.

10.7.7.Advanced Distribution Engine Support

The Support page provides the actions, operating mode change, and facilities to collect troubleshooting information and is accessed by clicking the **Support** link under Advanced Distribution Engine in the left-hand navigation panel, as in the following example:

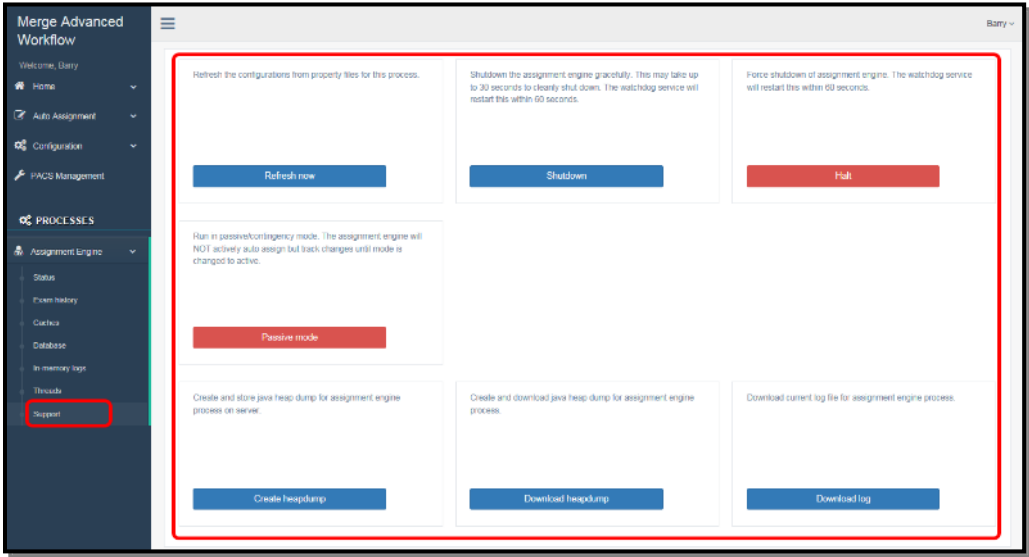


Figure 409: Advanced Distribution Engine Support

The following buttons are available for use on the Support page:

Button	Description						
Refresh now	Refreshes properties information from AmicasServer.properties and AssignmentEngine.properties . It also causes various systems to reset themselves within the engine. Not all properties can be refreshed.						
Shutdown	Shuts down the Advanced Distribution Engine. The shutdown period is capped to 30 seconds at which point the engine will forcefully exit after 30 seconds.						
Halt	Forces shutdown of Advanced Distribution Engine. While the Advanced Distribution Engine will exit and the operating system will clean up after the process, there is the potential for leaks in terms of resources in the database or file system due to abnormal termination. This is recommended ONLY as a last option to recover the process.						
Active Passive Mode	Toggles between Active Mode and Passive Mode as follows: <table><tr><th>Mode</th><th>Description</th></tr><tr><td>Active</td><td>The Advanced Distribution Engine is fully operational and is processing exams and performing assignments.</td></tr><tr><td>Passive</td><td>The Advanced Distribution Engine will NOT actively auto assign exams but will still track changes until the mode is changed to active. This mode is used when Merge PACS is deployed in passive/primary contingency mode.</td></tr></table>	Mode	Description	Active	The Advanced Distribution Engine is fully operational and is processing exams and performing assignments.	Passive	The Advanced Distribution Engine will NOT actively auto assign exams but will still track changes until the mode is changed to active. This mode is used when Merge PACS is deployed in passive/primary contingency mode.
Mode	Description						
Active	The Advanced Distribution Engine is fully operational and is processing exams and performing assignments.						
Passive	The Advanced Distribution Engine will NOT actively auto assign exams but will still track changes until the mode is changed to active. This mode is used when Merge PACS is deployed in passive/primary contingency mode.						
Create heapdump	Creates and stores java heap dump for Advanced Distribution Engine process on server. This will cause the engine to generate and place a heap dump in the PACS2DATA\log directory.						

Button	Description
NOTE: Heapdump generation is on a “best-effort” basis. If the process is low on memory, then the operation will fail. If the process is starved of CPU cycles, then the operation will take a long time and eventually time out.	
Download heapdump	Creates and downloads a java heap dump for the Advanced Distribution Engine process, which will cause the engine to generate a heapdump in memory and stream it back as a download. You will always get a compressed zip file with the heapdump.
Download log	Downloads the current log file for the Advanced Distribution Engine process, which will cause the engine to stream the contents of the current log file for the Advanced Distribution Engine process it as a download. You will always get a compressed zip file.

a. Process Restarts

Merge PACS has a service launcher with a mechanism that tracks process and restarts the process if it not found running. The watchdog works on 60-second timed intervals.

Process restarts are calculated as ranges because it depends on which of the timer interval as process exits. When the engine process is halted, it can start immediately or in up to a maximum of 60 seconds. When the engine process is gracefully shutdown, it can start immediately or in up to a maximum of 90 seconds.

b. Operating Mode for Production and Contingency Servers

As described above, the Advanced Distribution Engine can be toggled between running active mode or passive mode when Merge PACS is deployed in Primary-Contingency mode. In Primary-Contingency mode, the Primary Server is required to run the Advanced Distribution Engine in active mode, whereas the Contingency server will run the Advanced Distribution Engine in passive mode.

CAUTION: Setting the Advanced Distribution Engine in Primary and Contingency to both be active is not supported and will result in the Servers being out-of-sync with automatic assignments and cause “server stomping” (i.e., assignments by one server being stomped by the other).

Chapter 11. Templated User Management

11.1. General Concepts

11.1.1.Introduction

The operation of Merge PACS can be custom configured to provide different features and functionality to different users through the use of **privileges** and **preferences**. Privileges determine what features a user has access to (including access to specific management pages and specific features within the Merge PACS Workstation), while preferences set default values for various optional configurations within the Merge PACS Workstation.

Privileges and preferences are not assigned on a user by user basis. Instead, they are assigned on a site and group basis (which can include one or more subgroups) and these privileges and preferences flow from the topmost level down to the group (and subgroup) level. As illustrated below, individual users can belong to one or more groups (or subgroups) as well as to the entire site, and will inherit their privileges and preferences from the group(s) and site that are their “parents”:

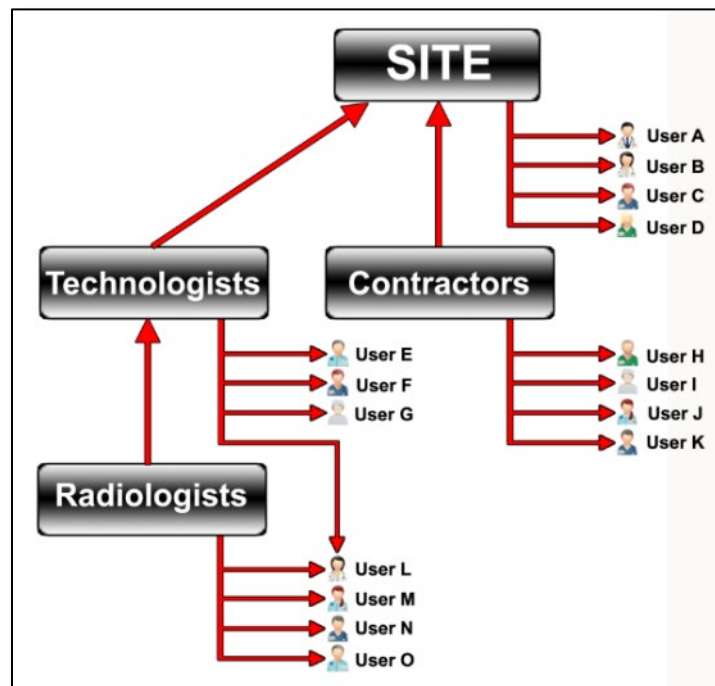


Figure 410: Example of Templated User Management

In the example above, users A, B, C and D do not belong to a group and inherit their privileges and default preferences from the site. Users E, F and G inherit their privileges and default preferences both from the Technologists group to which they belong and the site, while users L, M, N and O inherit their privileges and default preferences from the Radiologist subgroup to which they belong as well as the Technologist group and the Site (note that each group can only have one parent, but a user can belong to multiple groups as is the case with user L).

The way the actual inheritance process works differs between privileges and preferences, as described in detail below.

NOTE: The creation and management of users and groups, including defining the members and setting the parent of each group, is done from the **Users, Groups and Site** management pages, as described in Section 9.11 above.

Note that, instead of assigning privileges and preferences directly to the site and to groups, you actually create **templates** that store a set of privileges or preferences and then link templates to a site or to a group. Also, at the site level you can also create and link special site preference templates called **configuration** templates (in addition to privilege and preference templates) that cannot be overridden at the group or user level.

NOTE: Each template can contain a set of privileges, a set of preferences **or** (at the site level) a set of configurations.

11.1.2.Privileges

Privilege templates can be created to store a set of privileges that provide access to specific functions and management pages. The creation and inheritance of privileges operates according to the following principals:

- The site can only have a maximum of one privilege template linked to it (or it can have no privilege template linked to it).
- Each group can have one or more privilege templates linked to it (or none).
- Privileges are **cumulative** as you down up the hierarchy and a user will inherit all the privileges in the group(s) he or she belongs to, all the privileges in the groups those groups belong to, and so on until you get to the site's privileges (if any). It doesn't matter if one level in the hierarchy has a privilege and another doesn't – the user will get the privilege if any level in that user's direct line has that privilege.
- There are no privileges enabled by default within Merge PACS and you will typically want to create at least one privilege template that can be linked to the site or to one or more groups.

- For new installations of Merge PACS 8.2, a number of privilege templates are provided by default and linked to the site and default groups within the system. In the following table, X represents a template that is directly linked to a group or site:

Site/Group	Privilege Template						
	Administrators Priv	ER Physicians Priv	Radiologists Priv	Referring Clinicians Priv	Merge Support Priv	Surgeon Priv	Technologists Priv
Site							
Administrators	X						
ER Physicians		X					
Radiologists			X				
Referring Clinicians				X			
Merge Support					X		
Surgeons						X	
Technologists							X

NOTE: If a user's privileges are changed (e.g., by changing the privileges template of a group the user belongs to) while that user is currently logged into Merge PACS (either the Workstation or the Management Pages), he or she will need to log out and log back in to see the change.

11.1.3. Preferences

Preference templates can be created to specify default values related to the operation of the Merge PACS Workstation, such as how comparison studies should be loaded, what type of PET SUV measurements should be used, the shape of the Magnify tool, modality-specific Window/Level preferences, etc. The creation and inheritance of privileges operates according to the following principals:

- Merge PACS ships with a default set of preference values contained within the Merge PACS database. As a result, there will always be a set of preference values that exist for each user, even if there are no templates linked at any point in the hierarchy and no user overrides from the Workstation Preferences menu.
- As with privileges, preferences flow down the hierarchy from the site level to the individual group level and a user inherits his or her preferences from his or her group and all the levels above that group.
- Unlike privileges, however, preferences are **not** cumulative (since they can have different values instead of just being turned on or off) and can (except for "final" privileges as described below) be overridden as you move down the hierarchy. For example, if the site is linked to a template that specifies a preference that the Magnify tool should be square, but a user belongs to a group that is linked to a template with the preference that the Magnify tool should be round, the user will get a round magnify tool by default since the lower level (the group) takes precedence of the higher level (the site).
- Although preferences are not assigned in advance at the user level, users can (except for "final" privileges as described below) override the default preferences they inherit by changing their preferences within the Workstation. In general, inherited preferences are defaults that can be overridden by individual users from the Preferences menu of the Workstation.

- At any level in the hierarchy, a preference can be set as “final”, which will prevent it from being overridden at a lower level (even if it is also set to “final” with a different value at a lower level). In the example above, if the site preference setting the Magnify tool were set to “final”, the user would have a square Magnify tool even if he or she were in a group linked to a template that set the preference to “round,” and would be unable to change this preference from the Preferences menu within the Workstation.

NOTE: In general, if a preference is marked “final” in one or more levels in the hierarchy above the user, the highest level “final” preference value will prevail. If a preference is not marked “final” in any of the levels above the user, however, the lowest level preference value will prevail.

- If multiple templates are linked to a single group, each template must be assigned a priority. Then, if there is a conflict between multiple templates regarding a particular non-final preference, the setting from the template with the highest priority will be applied to members of the group.
- If a user belongs to multiple groups, preference value conflicts between these groups are resolved as follows:
 - If a preference is marked as “final” in one or more parents of this user, the preference value shall be taken from the parent with the highest group priority (as described in subsection 9.11.2.k above) and shall be considered final.
 - If a preference is set but **not** marked as “final” in any of the parents of this user:
 - If the user has selected a value for the preference from the Preferences menu within the Workstation, the value selected by the user will be applied.
 - If the user has **not** selected a value within the Workstation, the preference value shall be taken from the parent with the highest group priority.
 - If a preference is not set in any of the parents of the user (either because the group has no template linked to it or because the template(s) linked to that group do not set a value for that preference), the preference value shall be taken from the parent with the highest group priority which will, in turn inherit the value from its parent (e.g., the site) or from the default preferences in the database if not explicitly set anywhere higher up in the hierarchy.

- For new installations of Merge PACS 8.2, a number of preference templates are provided by default and linked to the site and default groups within the system. In the following table, X represents a template that is directly linked to a group or site:

Site/Group	Privilege Template						
	Administrators Priv	ER Physicians Priv	Radiologists Priv	Referring Clinicians Priv	Merge Support Priv	Surgeon Priv	Technologists Priv
Site							
Administrators	X						
ER Physicians		X					
Radiologists			X				
Referring Clinicians				X			
Merge Support					X		
Surgeons						X	
Technologists							X

11.1.4. Configurations

Configurations are site-level preferences that apply to all users and cannot be overridden at the group or user level. As with regular preferences, new installations of Merge PACS 8.2 include a default set of configuration values contained within the database. Although these configurations cannot be overridden at the group or user level, they can be overridden at the site level from the Preferences menu within the Workstation. In addition, you can create a configuration template and link it to the site.

Configuration values for a site are determined as follows:

- If a configuration is set via the Preferences menu within the Workstation (*i.e.*, is part of the site's "implicit" configuration template), this configuration value will be used.
- If a configuration is not part of the site's implicit template but is contained in the configuration template linked to the site (the site's "explicit" configuration template), this configuration will be used.
- If a configuration is neither set in the site's implicit configuration template nor the explicit configuration template, the default configuration from the database will be used.

11.2. Configuring Templates

Privilege, preference and configuration templates can be configured from the **Settings** dialog within the Merge PACS Workstation. At the Settings dialog, you can do the following:

- Add, edit, copy, delete, export and import (complete or relative) templates that can be used at various levels.
- Configure the privilege, preference and/or configuration template linked to the whole site (note that a site can only have one of each type linked to it at a time or none).

- Configure the privilege and/or preference templates linked to individual groups (note that each group can have multiple privilege and/or preference templates linked to it or none).
- View and/or edit the preferences and view the privileges currently configured for individual users (note that privileges cannot be edited at the user level).

11.2.1. Accessing the Template Management Pages

The Templates Management Pages are accessed via the Merge PACS Workstation.

To access the Template Management Pages:

1. Log into the Merge PACS Workstation as an Admin user.
2. Click on the **Settings** icon in the upper right corner of the Workstation Browser, as in the following example:

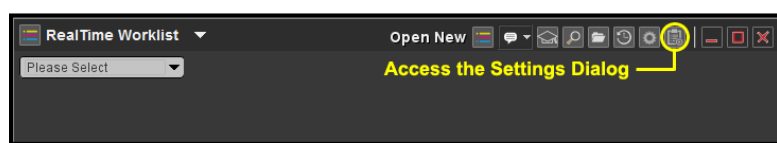


Figure 411: The Settings Icon

The **Settings** dialog is displayed with links to the various templates management pages displayed on the left-hand side, as in the following example:

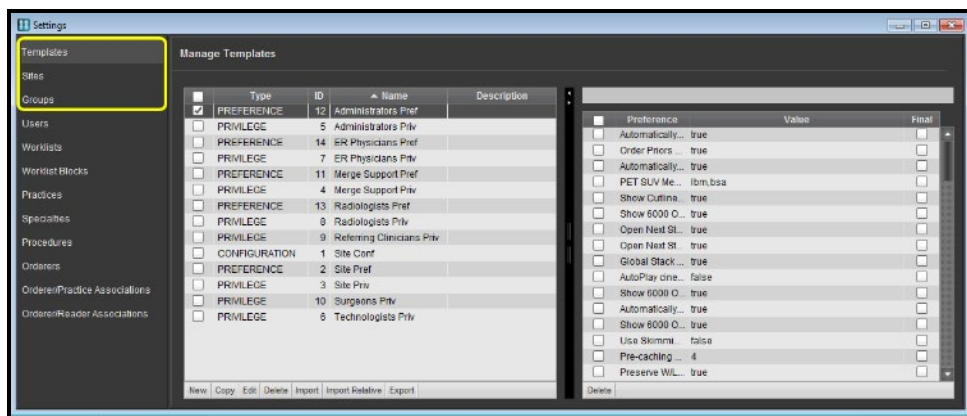


Figure 412: Template Management Pages on the Settings Dialog

11.2.2. Template Configuration

The **Manage Templates** page allows you to add, edit, copy, delete, export and import (complete or relative) templates that can be used at various levels within the system and is accessed by clicking on the **Templates** link on the main Settings dialog, as in the following example:

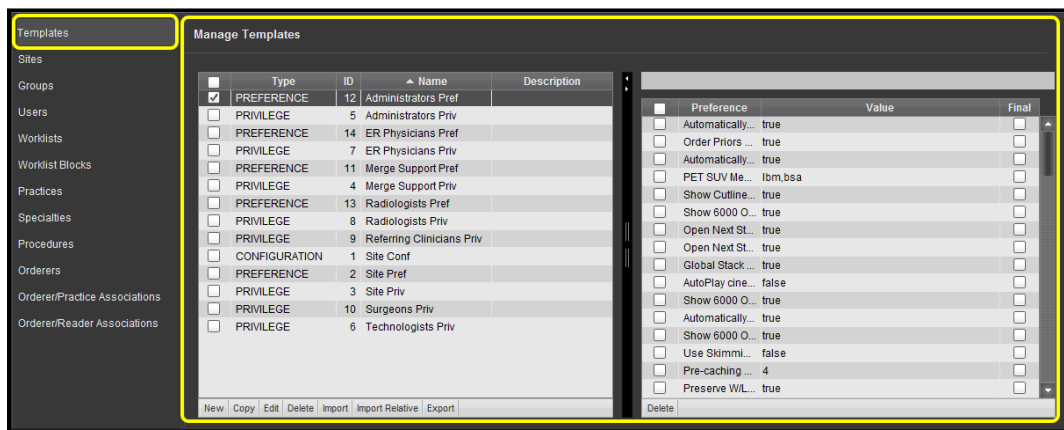


Figure 413: Manage Templates Page

NOTE: The Manage Templates page will be displayed by default when you first access the main Settings dialog.

The left panel of the Templates page displays a list of all templates currently existing within Merge PACS. The right panel of the Templates page displays the contents of the single template selected in the left panel (for privilege templates, this is the list of privileges enabled in the template, while for preference and configuration templates this is the list of preferences that have been set explicitly set and their values). Note that the right panel will only display information if a single template is selected in the left panel.

- To select a single template in the left panel, click anywhere on that template's row except for the check box.
- To select multiple templates (e.g., for copying, deleting or importing), click the checkbox for each template.
- By default, templates are listed alphabetically by name. To sort the list of existing templates, click on any of the column headers (Type, Name, etc.). Clicking the same header repeatedly will toggle between ascending and descending sort order.

a. Creating a New Template

At the manage Templates page you can create a new preference, privilege or configuration from scratch.

To create a new template from scratch:

1. Click on the **New** button at the bottom of the left panel, as in the following example:

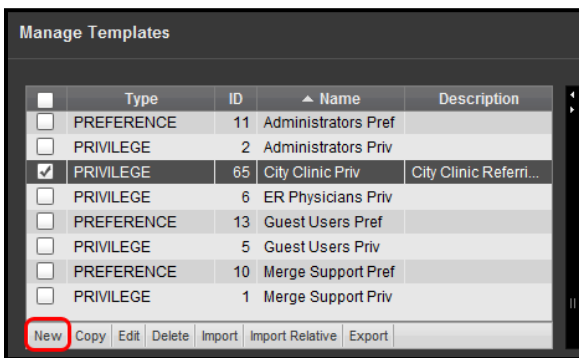


Figure 414: New Template Button

A new template is added to the list of existing templates, as in the following example:

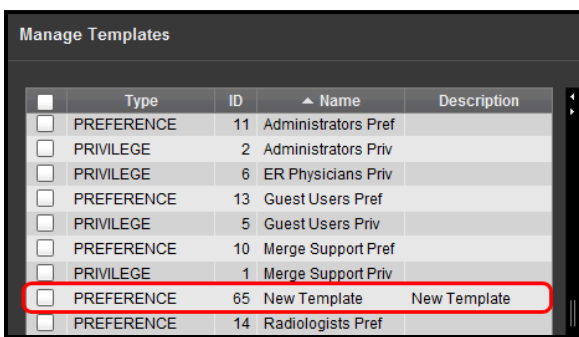


Figure 415: New Template

NOTE: The location of the new template within the list of templates will be based on the current sort order for the list, as described above.

2. Select the type of template you want to create by clicking on the **Type** column for the new template and selecting from the drop-down menu, as in the following example:

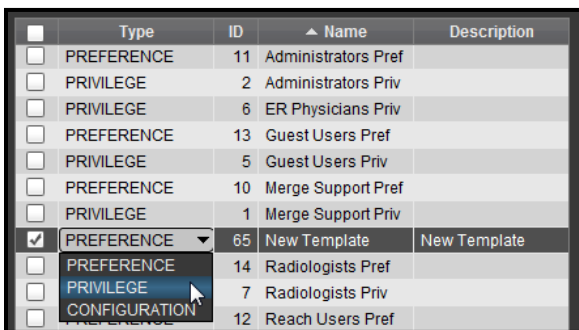


Figure 416: Selecting the Type of Template

NOTE: Once you select the template type and exit the Manage Templates page, you will not be able to change it later upon returning to the Manage Templates page. You can, however, delete it as described in Paragraph d below.

- Specify a name for this template by double-clicking on the **Name** column for the template to enable the Name field and entering the desired name, as in the following example:

☐	Type	ID	▲ Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☐	PRIVILEGE	6	ER Physicians Priv	
☐	PREFERENCE	13	Guest Users Pref	
☐	PRIVILEGE	5	Guest Users Priv	
☐	PREFERENCE	10	Merge Support Pref	
☐	PRIVILEGE	1	Merge Support Priv	
☒	PRIVILEGE	65	City Clinic Priv	New Template
☐	PREFERENCE	14	Radiologists Pref	

Figure 417: Entering the Name for this Template

NOTE: By convention, privilege templates include “Priv” in their name, preference templates include “Pref” and configuration templates include “Conf”, but this is not required.

NOTE: You can change the template name whenever you want, even after exiting the Manage Templates page and returning.

- If desired, you can optionally include a description for this template by double-clicking on the **Description** column for the template to enable the Description field and entering the desired description, as in the following example:

☐	Type	ID	▲ Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☐	PRIVILEGE	6	ER Physicians Priv	
☐	PREFERENCE	13	Guest Users Pref	
☐	PRIVILEGE	5	Guest Users Priv	
☐	PREFERENCE	10	Merge Support Pref	
☐	PRIVILEGE	1	Merge Support Priv	
☒	PRIVILEGE	65	City Clinic Priv	City Clinic Referring
☐	PREFERENCE	14	Radiologists Pref	

Figure 418: Entering the Description for this Template

NOTE: The Description field is just to provide additional detail on this page and is not used elsewhere in the system.

NOTE: You can change the template description whenever you want, even after exiting the Manage Templates page and returning.

- Follow the procedure described in Paragraph b below to edit the new template and specify its privileges or preferences.

b. Editing an Existing Template

To edit an existing template:

1. Select the desired template from the list of existing templates by clicking on it.

The current privileges or preference values for the selected, if any, are displayed in the right panel, as in the following example:

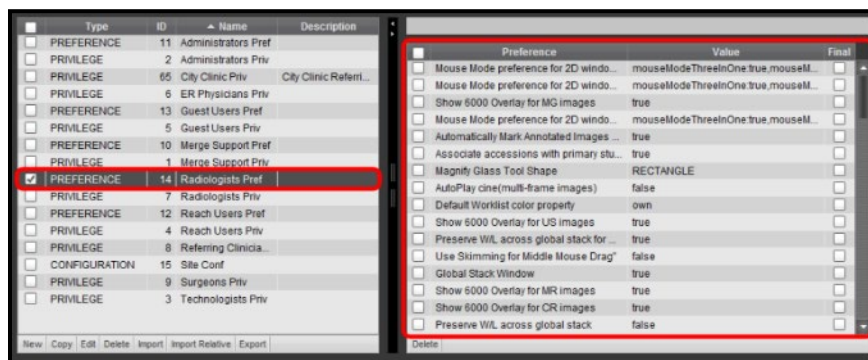


Figure 419: Viewing Current Privileges or Preferences for a Template

NOTE: The information displayed in the right panel can be sorted by clicking on any of the column headers.

NOTE: Only privileges or preferences that are different from the default values will be listed.

2. If desired, edit the name of the template by double-clicking on the **Name** column for the template and entering the desired name, as in the following example:

☐	Type	ID	Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☐	PRIVILEGE	65	City Clinic Priv	City Clinic Referrin...
☒	PRIVILEGE	67	City Clinic North Priv	City Clinic Referrin...
☐	PRIVILEGE	6	ER Physicians Priv	

Figure 420: Entering the Name for this Template

NOTE: By convention, privilege templates include “Priv” in their name, preference templates include “Pref” and configuration templates include “Conf”, but this is not required.

3. If desired, edit the description of the template by double-clicking on the **Description** column for the template and entering the desired description, as in the following example:

☐	Type	ID	Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☐	PRIVILEGE	65	City Clinic Priv	City Clinic Referrin...
☒	PRIVILEGE	67	City Clinic North Priv	City Clinic North Refe
☐	PRIVILEGE	6	ER Physicians Priv	

Figure 421: Entering the Description for this Template

NOTE: The Description field is just to provide additional detail on this page and is not used elsewhere in the system.

- If desired, you can delete one or more existing privileges or preferences entirely for this template by selecting the items you want to remove in the right panel and clicking on the **Delete** button at the bottom of the panel, as in the following example:

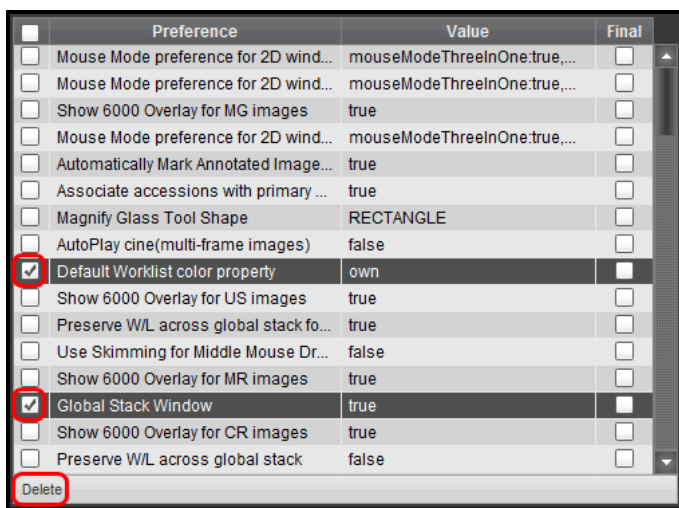


Figure 422: Deleting Existing Preferences from a Template

- If you have selected a **preference** template, you can mark one or more existing preferences as “final” by selecting the **Final** checkbox for those preferences, as in the following example:

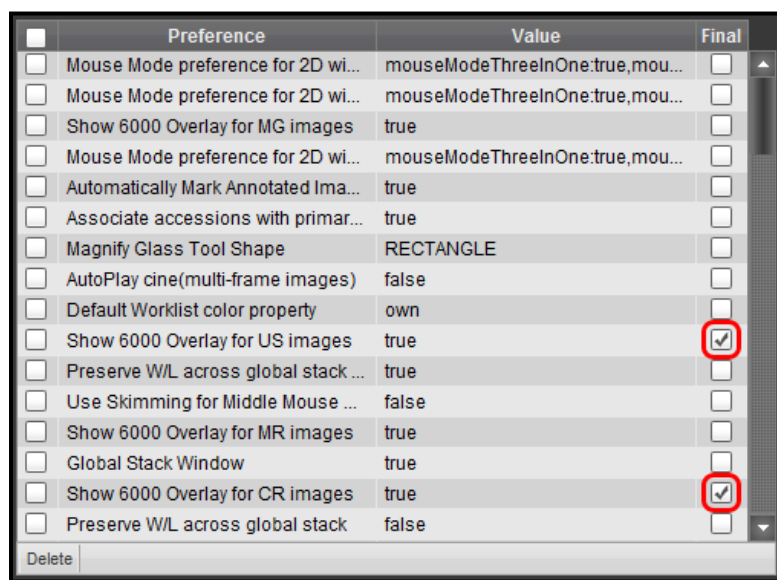


Figure 423: Deleting Existing Preferences from a Template

NOTE: Only preferences that have been changed from their default value will be listed here. If you want to mark a default preference value as final to prevent it from being overridden, follow the steps below to edit the template and change the value of that preference, save your changes at the Merge PACS Preferences dialog to add the preference to the template, re-edit the template to change the preference back to its original value, and then select the **Final** checkbox for that preference.

6. To configure the actual privileges or preferences for the selected template, click the **Edit** button at the bottom of the left panel, as in the following example:

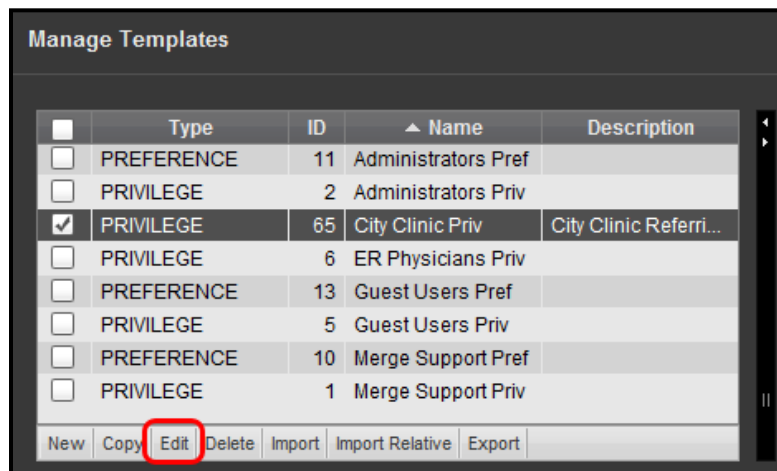


Figure 424: Editing an Existing Template

Depending on what type of template you are editing, one of the following three dialogs will be displayed:

- If you are editing a **privileges** template, the **Merge PACS Privileges** dialog will be displayed, as in the following example:

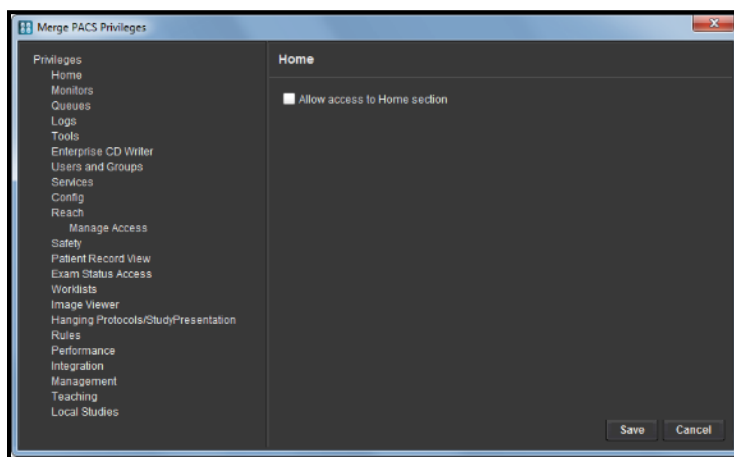


Figure 425: Merge PACS Privileges

NOTE: Refer to Appendix A.1 below for a description of all available privileges.

- If you are editing a **preferences** template, the **Merge PACS Preferences** dialog will be displayed with only user-level preferences showing, as in the following example:

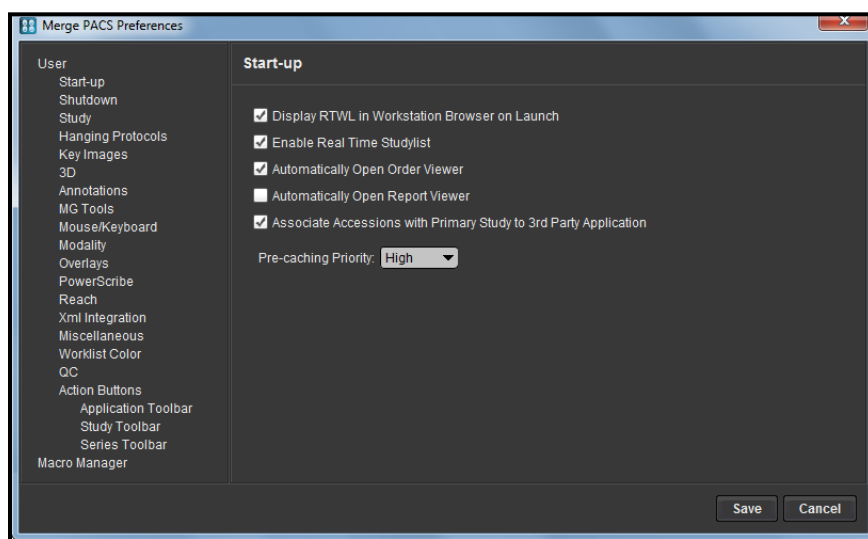


Figure 426: Merge PACS Preferences

NOTE: When creating new preference templates, you would typically only include preferences that are different from the default values instead of setting the value of every available preference.

NOTE: Refer to Appendix A.2 below for a description of all available user preferences.

- If you are editing a **configuration** template, the **Merge PACS Configuration** dialog will be displayed, as in the following example:

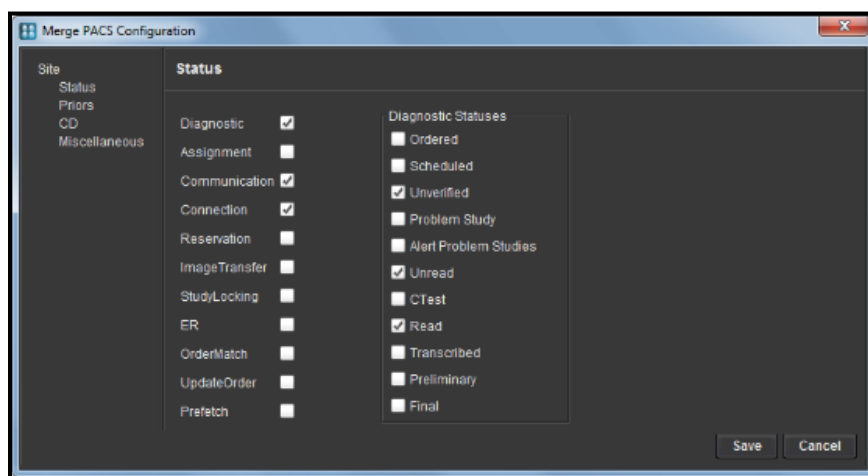


Figure 427: Merge PACS Configuration

7. Set the desired privileges or preferences from within the relevant dialog.

NOTE: Refer to Appendix A.3 below for a description of all available configuration preferences.

8. When finished, click the **Save** button at the bottom of the dialog to record your changes.

c. Copying an Existing Template

Instead of creating a new template entirely from scratch, you can also make a copy of an existing template of the same type and modify it as needed.

To make a copy of an existing template:

1. Select the template you want to copy in the left panel and click the **Copy** button at the bottom of the panel, as in the following example:

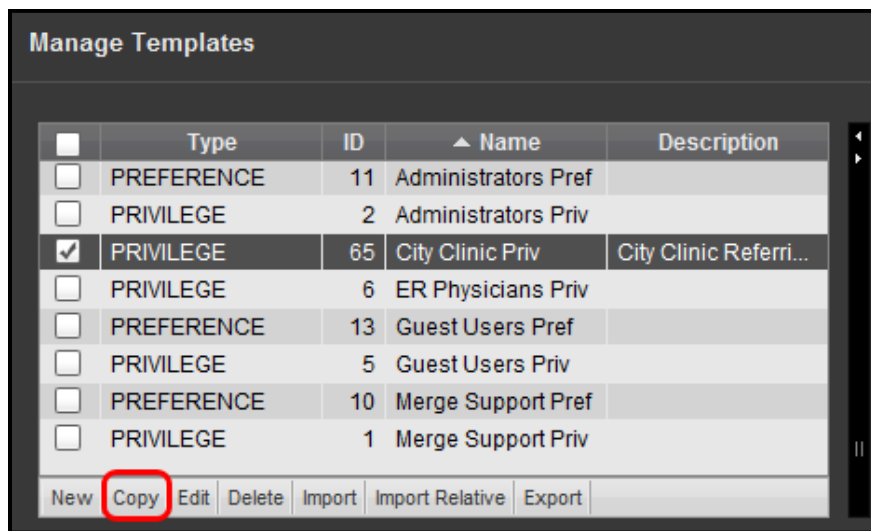


Figure 428: Copying an Existing Template

NOTE: The template copied must be the same type as the template you want to create.

A new template with a default name is created and displayed in the list of existing templates, as in the following example:

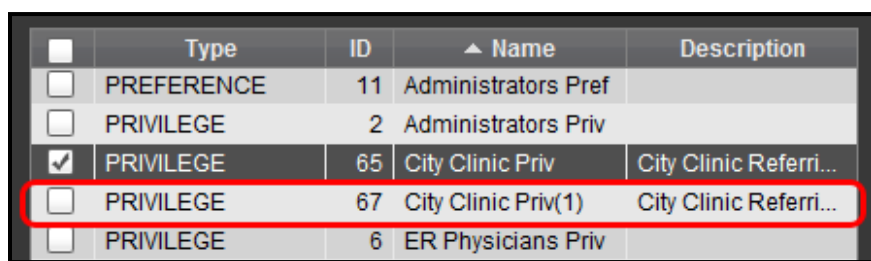


Figure 429: Newly Copied Template

2. Edit the name, description and/or settings and values of the template as necessary, following the procedures described in Paragraph b above.

d. Deleting an Existing Template

To delete one or more existing templates, click on the check box for each template you want to delete and then click the **Delete** button at the bottom of the left panel, as in the following example:

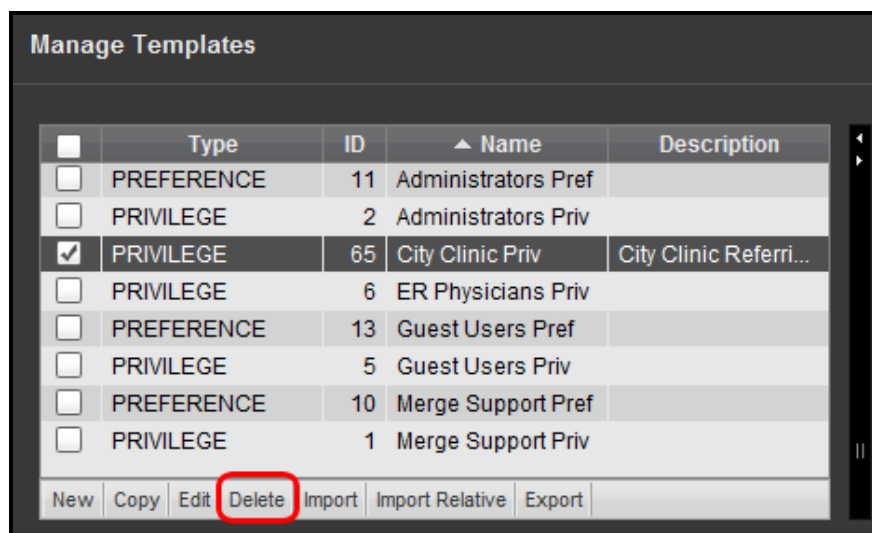


Figure 430: Deleting a Template

e. Exporting a Template

If desired, you can export one or more templates to files that can later be imported (e.g., to another Merge PACS Server).

To export templates:

1. Select one or more templates from the list of existing templates by clicking their checkboxes and click on the **Export** button, as in the following example:

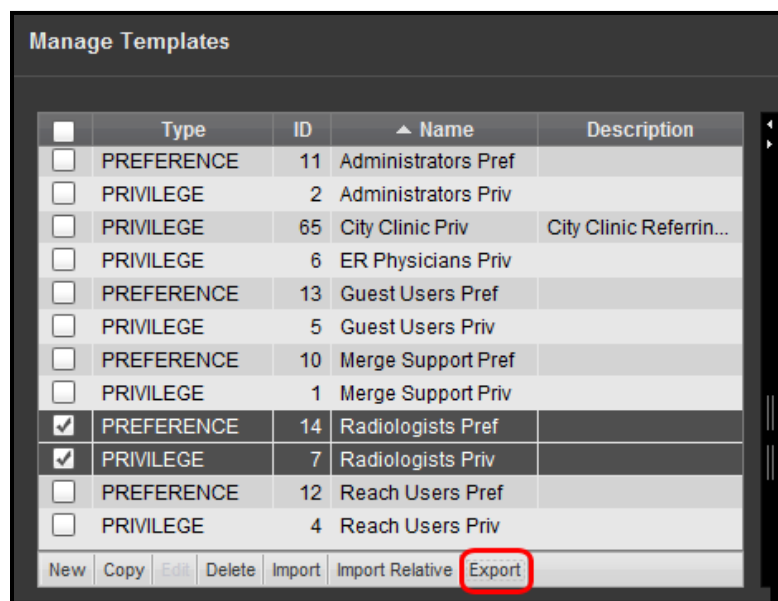


Figure 431: Exporting Templates

The **Directory Selection** dialog is displayed, as in the following example:

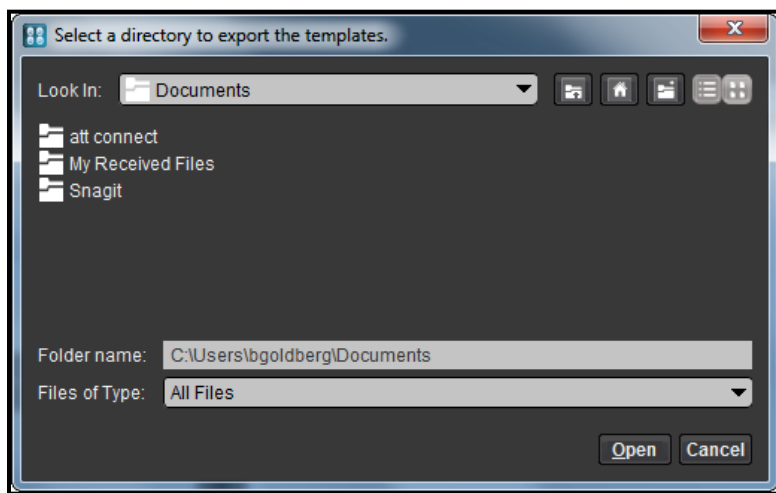


Figure 432: Select a Directory to Export the Template(s)

2. Browse to the location where you want to store the template files and click the **Open** button.

When the process is complete, a message similar to the following will be displayed:

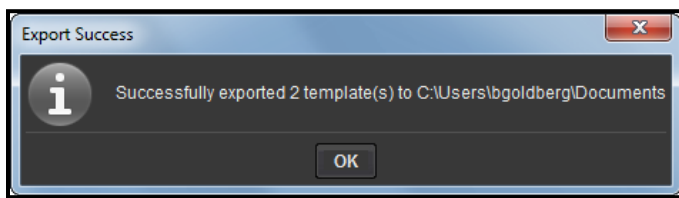


Figure 433: Export Success

NOTE: Templates are saved as text files in the format **<Template Name>#PREF.**

f. Importing a Complete Template

If desired, you can import a **complete** template into Merge PACS instead of creating one from scratch or copying one that already exists on the Server. This can be a preference, privilege or a configuration template and can either be a template exported from the Merge PACS Server (as described in Paragraph e above) or a set of User Preferences exported from the Workstation (as described in the *Merge PACS 8.2 Workstation Users Guide*).

Note the following:

- By default, an imported Server-based template will retain the name it had when exported, except that it will automatically be modified slightly if it has the same name as another template of the same type on the Server.
- Imported Workstation-based templates will be given the default name “PACS Prefs”, which will be slightly modified if there is already a template with the same name.
- Unlike importing relative templates, as described in Paragraph g below, the regular import process will preserve all the settings and values of the template, even if those match the default settings and values currently on the Server. As a result, the imported template may contain redundant or unnecessary information.

To import a complete template:

1. Click on the **Import** button at the bottom of the left panel, as in the following example:

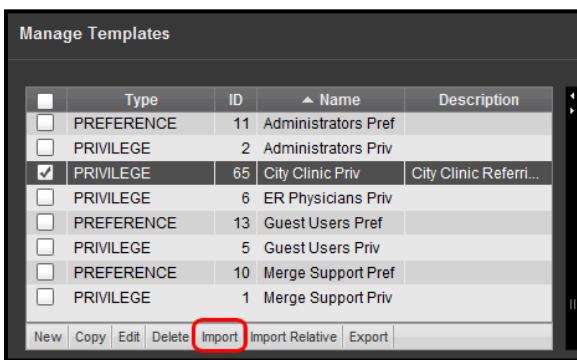


Figure 434: Importing a Template

NOTE: When importing a template, it doesn't matter what template is currently selected.

The **Select Template Files** dialog is displayed, as in the following example:

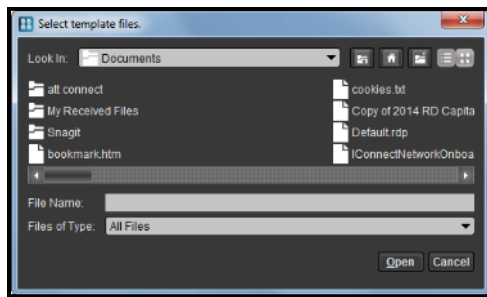


Figure 435: Select Template Files

2. Browse to and select the desired template file(s), and then click the **Open** button.

NOTE: Templates exported from the Server will be text files in the format **<Template Name>#PREF**. Templates exported from the Workstation will be standard text files with a **.txt** extension.

When the selected template or templates have finished being imported, a dialog such as the following will be displayed:

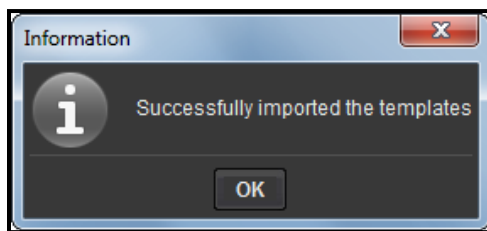


Figure 436: Import Successful

3. Click the **OK** button.

The list of templates in the left panel will refresh and the newly imported template(s) will be displayed, as in the following example:

☐	Type	ID	Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☐	PRIVILEGE	65	City Clinic Priv	City Clinic Referri...
☐	PRIVILEGE	6	ER Physicians Priv	
☐	PREFERENCE	13	Guest Users Pref	
☐	PRIVILEGE	5	Guest Users Priv	
☐	PREFERENCE	10	Merge Support Pref	
☐	PRIVILEGE	1	Merge Support Priv	
☐	PREFERENCE	74	PACS Prefs	
☐	PREFERENCE	14	Radiologists Pref	
☐	PRIVILEGE	7	Radiologists Priv	
☐	PRIVILEGE	75	Radiologists Priv(1)	
☐	PREFERENCE	12	Reach Users Pref	
☐	PRIVILEGE	4	Reach Users Priv	
☐	PRIVILEGE	8	Referring Clinica...	
☐	CONFIGURATION	15	Site Conf	
☐	PRIVILEGE	9	Surgeons Priv	
☐	PRIVILEGE	3	Technologists Priv	

New Copy Edit Delete Import Import Relative Export

Figure 437: Newly Imported Templates

- Edit the name, description and/or settings and values of the imported template(s) as necessary, following the procedures described in Paragraph b above.

g. Importing a Relative Template

As stated in Paragraph f above, the regular template import process preserves all the settings and values of the template, even if those match the default settings and values currently on the Server. Since this can result in redundant or unnecessary information, you can also import a **relative** template that can be linked to a specific site or group and will include only those settings and values that are different from those already defined for that site or group. The newly-imported template will be automatically linked to the specified site or group with a priority of one higher than the highest priority of a currently-linked template to the site or group.

To import a relative template:

- Click on the **Import Relative** button at the bottom of the left panel, as in the following example:

☐	Type	ID	Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☒	PRIVILEGE	65	City Clinic Priv	City Clinic Referri...
☐	PRIVILEGE	6	ER Physicians Priv	
☐	PREFERENCE	13	Guest Users Pref	
☐	PRIVILEGE	5	Guest Users Priv	
☐	PREFERENCE	10	Merge Support Pref	
☐	PRIVILEGE	1	Merge Support Priv	

New Copy Edit Delete Import Import Relative Export

Figure 438: Importing a Relative Template

NOTE: When importing a relative template, it doesn't matter what template is currently selected.

The **Import Preferences Relative To** dialog is displayed, as in the following example:

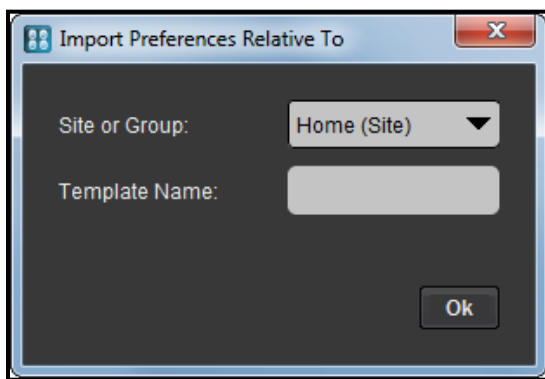


Figure 439: Specifying Where to Import Relative Preferences

2. Select the site or group you want to link the imported preference to from the drop-down **Site or Group** menu, as in the following example:

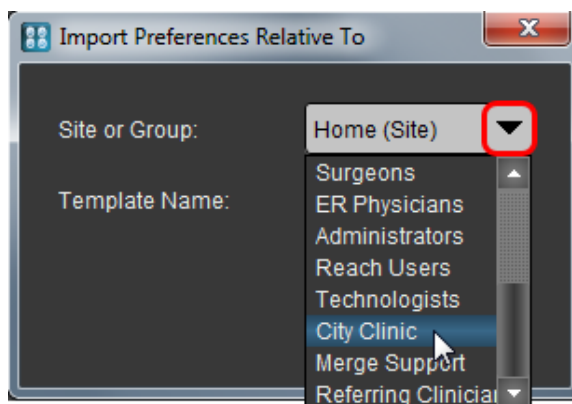


Figure 440: Specifying Where to Import Relative Preferences

3. Enter a name for this template in the **Template Name** field, as in the following example:

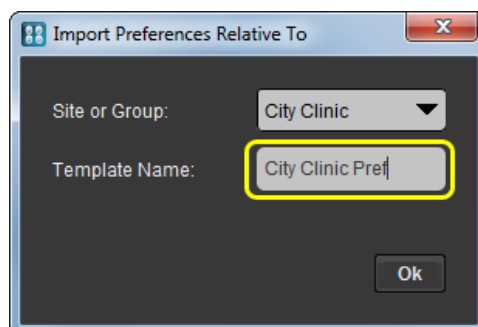


Figure 441: Enter the Template Name

NOTE: By convention, privilege templates include “Priv” in their name, preference templates include “Pref” and configuration templates include “Conf”, but this is not required.

4. Click the **OK** button.

The **Select Template Files** dialog is displayed, as in the following example:

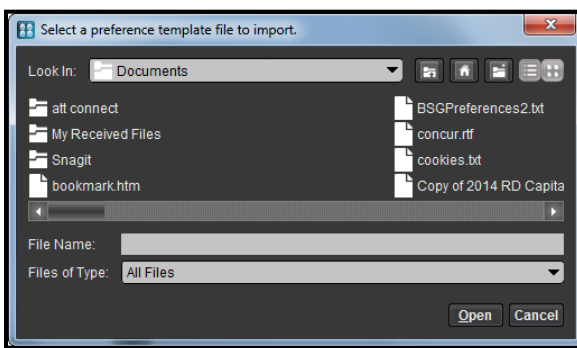


Figure 442: Select Template Files

- Browse to and select the desired template file, and then click the **Open** button.

NOTE: Templates exported from the Server will be text files in the format **<Template Name>#PREF**. Templates exported from the Workstation will be standard text files with a **.txt** extension.

When the selected template has finished being imported, a dialog such as the following will be displayed:

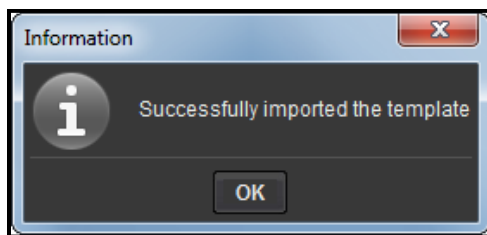


Figure 443: Import Successful

- Click the **OK** button.

The list of templates in the left panel will refresh and the newly imported template will be displayed, as in the following example:

☐	Type	ID	▲ Name	Description
☐	PREFERENCE	11	Administrators Pref	
☐	PRIVILEGE	2	Administrators Priv	
☐	PREFERENCE	76	City Clinic Pref	
☐	PRIVILEGE	65	City Clinic Priv	City Clinic Referri...
☐	PRIVILEGE	6	ER Physicians Priv	
☐	PREFERENCE	13	Guest Users Pref	
☐	PRIVILEGE	5	Guest Users Priv	

Figure 444: Newly Imported relative Template

- Edit the name, description and/or settings and values of the imported template(s) as necessary, following the procedures described in Paragraph b above.

11.2.3. Managing Site Templates

The **Manage Site Templates** page allows you to link an existing configuration, preference and/or privilege template to the site, as well as see the configurations, preferences and privileges currently set for the site. It is accessed by clicking on the **Sites** link on the main Settings dialog as in the following example:

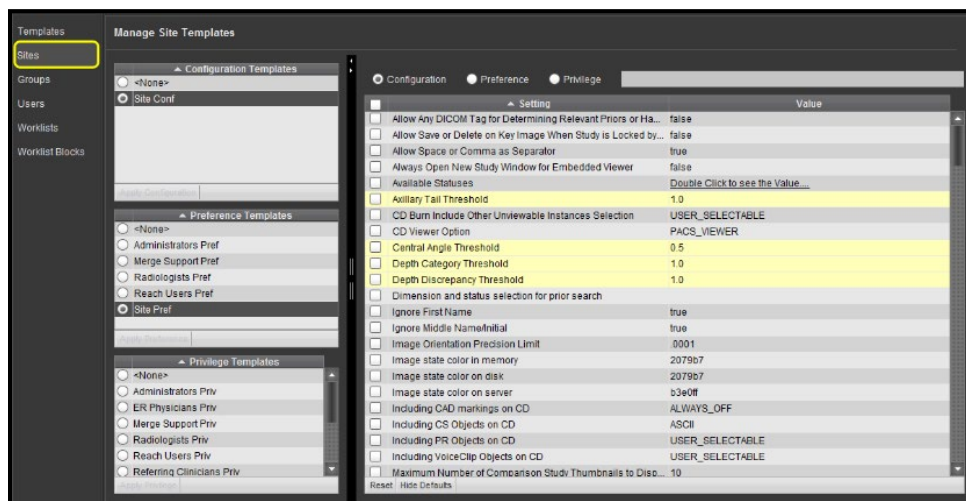


Figure 445: Manage Site Templates Page

The left panel includes separate windows to select the **Configuration Template**, **Preference Template** and **Privilege Template** for the site, as in the following example:

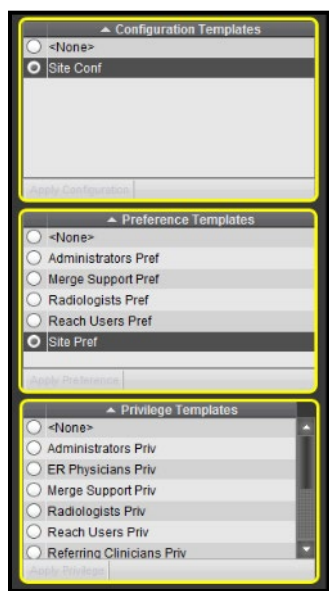
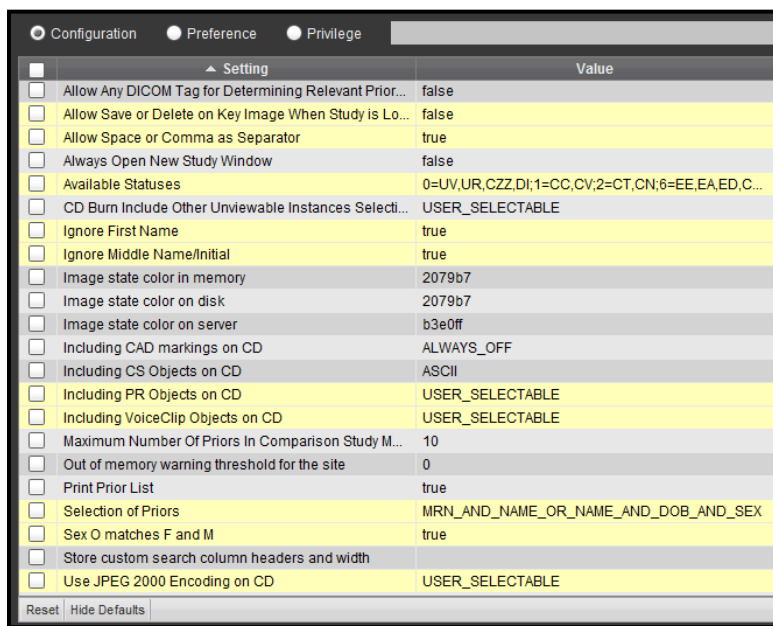


Figure 446: Template Selection Windows

- For new installations of Merge PACS 8.2, the default site templates will be as shown above.

- To select a different configuration, preference or privilege template (including “<None>”), select the desired template by clicking on the radio button for that template in the applicable window and then click on the applicable **Apply** button (which will be grayed out until you make a change).

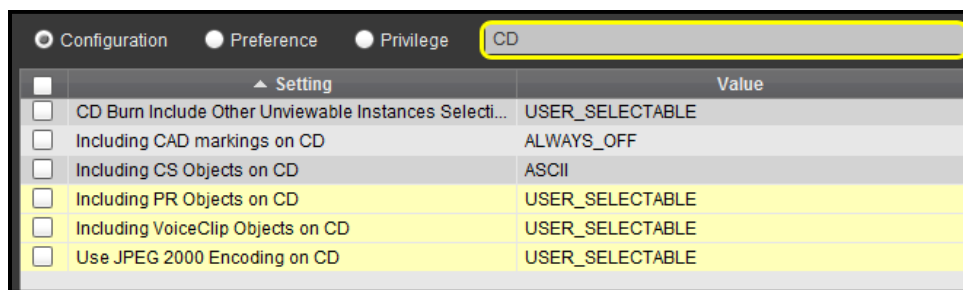
The right panel displays detailed information about all configurations, preferences and privileges currently set for the site, whether they come from a linked template or from the system defaults, as in the following example:



Setting	Value
☐ Allow Any DICOM Tag for Determining Relevant Prior...	false
☐ Allow Save or Delete on Key Image When Study is Lo...	false
☐ Allow Space or Comma as Separator	true
☐ Always Open New Study Window	false
☐ Available Statuses	0=UV,UR,CZZ,DI;1=CC,CV;2=CT,CN;6=EE,EA,ED,C...
☐ CD Burn Include Other Unviewable Instances Selecti...	USER_SELECTABLE
☐ Ignore First Name	true
☐ Ignore Middle Name/Initial	true
☐ Image state color in memory	2079b7
☐ Image state color on disk	2079b7
☐ Image state color on server	b3e0ff
☐ Including CAD markings on CD	ALWAYS_OFF
☐ Including CS Objects on CD	ASCII
☐ Including PR Objects on CD	USER_SELECTABLE
☐ Including VoiceClip Objects on CD	USER_SELECTABLE
☐ Maximum Number Of Priors In Comparison Study M...	10
☐ Out of memory warning threshold for the site	0
☐ Print Prior List	true
☐ Selection of Priors	MRN_AND_NAME_OR_NAME_AND_DOB_AND_SEX
☐ Sex O matches F and M	true
☐ Store custom search column headers and width	
☐ Use JPEG 2000 Encoding on CD	USER_SELECTABLE

Figure 447: Current Configurations, Preferences and Privileges

- Use the radio buttons at the top of the panel to select whether you want to view this site's **Configuration** settings, **Preference** values or **Privileges**.
- Sort the information displayed by clicking on a column header (clicking the same header repeatedly will toggle between ascending and descending sort order).
- Filter the information displayed by entering text to match in the text box at the top of the panel, as in the following example:



Setting	Value
☐ CD Burn Include Other Unviewable Instances Selecti...	USER_SELECTABLE
☐ Including CAD markings on CD	ALWAYS_OFF
☐ Including CS Objects on CD	ASCII
☐ Including PR Objects on CD	USER_SELECTABLE
☐ Including VoiceClip Objects on CD	USER_SELECTABLE
☐ Use JPEG 2000 Encoding on CD	USER_SELECTABLE

Figure 448: Filtering Information

- Hover your mouse over a setting, preference or privilege to see how it was set or enabled, as in the following example:

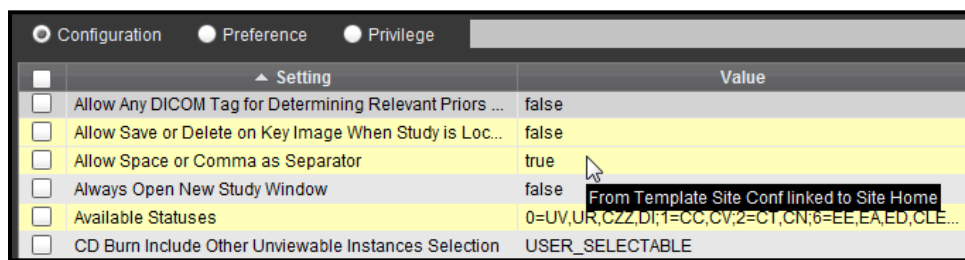


Figure 449: How this Setting Was Enabled

NOTE: For the **configurations** and **preferences**, you must hover over the **Value** column to see the pop-up information.

- When viewing **configurations** or **preferences**, settings that are different from the default site settings within the database are highlighted in yellow. To only show these settings, click on the **Hide Default** button at the bottom of the panel (the button will then change to **Show All** if you want to include the defaults again).
- When viewing **configurations** or **preferences**, you can restore the default value for one or more items by using the checkboxes to select the items and then clicking on the **Reset** button at the bottom of the panel.

11.2.4. Managing Group Templates

The **Manage Group Templates** page allows you to link one or more existing preference and/or privilege templates to specific groups, see the configurations, preferences and privileges currently set for those groups and set the priority of multiple linked preference templates. It also allows you to reset one or more preferences for a group to clear any user-level overrides for members of that group.

The Manage Group Templates page is accessed by clicking on the **Groups** link on the main Settings dialog as in the following example:

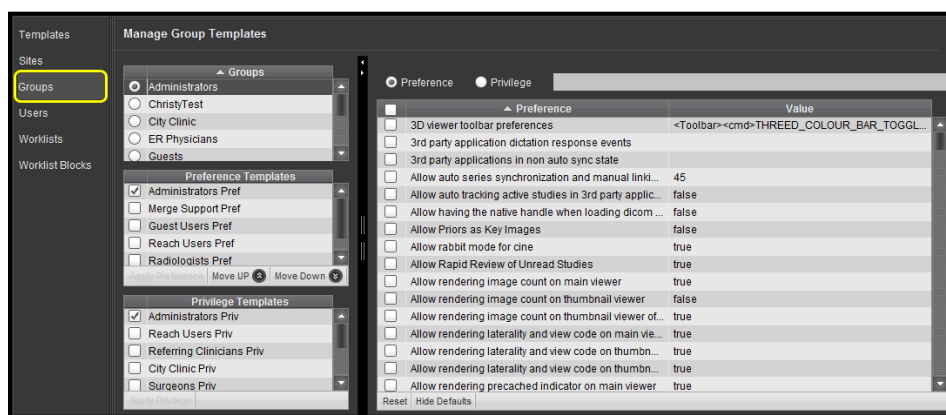


Figure 450: Manage Group Templates Page

The left panel includes separate windows to select the **Group** you want to configure as well as the **Preference Template(s)** and **Privilege Template(s)** you want to link to the selected group, as in the following example:

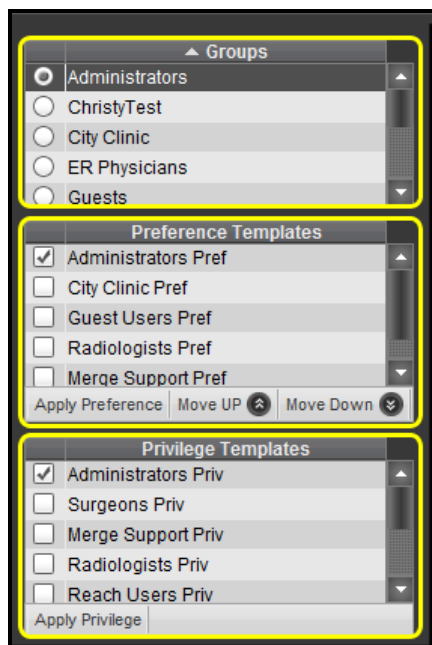


Figure 451: Group and Template Selection Windows

- Select the group whose templates you want to manage in the **Groups** window.
- In the **Preferences Templates** window, click on the checkboxes for the preference template(s) you want to link to the selected group.
 - The linked preference templates will have priority in the order they appear in this window, with the topmost preference having the highest priority. To change the relative priority of the linked templates, click on a linked template to select it and then click the **Move Up** or **Move Down** buttons as necessary.
 - Once you have made changes to the linked preference templates, click on the **Apply Preference** button to record the changes. Note that this button will be grayed out before you make any changes and may take a few second to activate once you have made changes.
- In the **Privilege Templates** window, click on the checkboxes for the privilege template(s) you want to link to the selected group and then click on the **Apply Privilege** button to record your changes. Note that the Apply Privilege button will be grayed out before you make any changes and may take a few second to activate once you have made changes.

NOTE: It is not possible to unlink the **Merge Support Priv** privilege template from the **Merge Support** group.

NOTE: You must log out and log back into the Merge PACS Workstation after adding or removing privilege templates for the changes to take effect for that group.

The right panel displays detailed information about all preferences and privileges currently set for the selected group, whether they come from a linked template, are inherited from a parent or come from the system defaults, as in the following example:

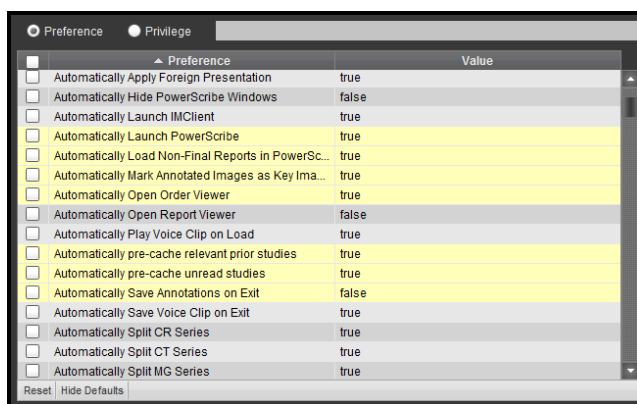


Figure 452: Current Preferences and Privileges

- Use the radio buttons at the top of the panel to select whether you want to view this group's **Preference** values or **Privileges**.
- Sort the information displayed by clicking on a column header (clicking the same header repeatedly will toggle between ascending and descending sort order).
- Filter the information displayed by entering text to match in the text box at the top of the panel, as in the following example:

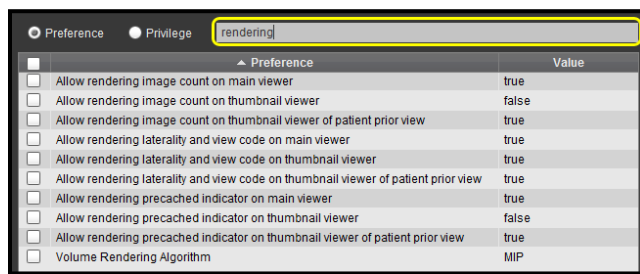


Figure 453: Filtering Information

- Hover your mouse over a preference or privilege to see how it was set or enabled, as in the following example:

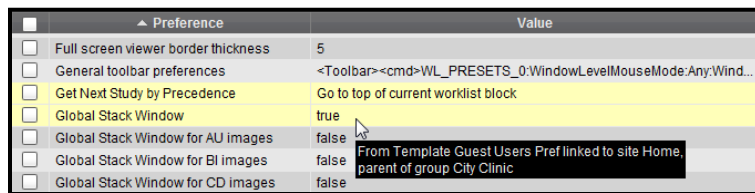


Figure 454: How this Setting Was Enabled

NOTE: For **preferences**, you must hover over the **Value** column to see the pop-up information.

- When viewing **preferences**, settings that are different from the default settings (*i.e.*, settings this group would have if no templates were linked to it) are highlighted in yellow. To only show these settings, click on the **Hide Default** button at the bottom of the panel (the button will then change to **Show All** if you want to include the defaults again).
- When viewing **preferences**, you can reset one or more items by using the checkboxes to select the items and then clicking on the **Reset** button at the bottom of the panel. This will clear any user-level overrides for the selected preferences set **by** all users in the group (via the Workstation Merge PACS Preferences dialog) or **for** all users in this group (via the Manage Users page described in Chapter 12 below).

NOTE: You can also reset the preferences for individual users from the Manage Users page, as described in Chapter 12 below.

Chapter 12. Managing Users

Users are created from the Users & Groups Management Page, as described in subsection 9.11.1 above, which includes providing a small set of basic demographic information such as name, address, e-mail, phone number, group membership and whether or not they are considered “providers”. Once a user is created, however, you can use the **Manage Users** page allows you to view the general privileges and preferences currently set for that user within the system, remove any user-level overrides for one or more preferences and/or edit a user’s preferences on the fly. You can also configure the Advanced Workflow information for the user, including their licensure regions, specialties, practice, reader location, etc.

12.1. Accessing the Manage User Page

The Manage User Page is accessed via the Merge PACS Workstation.

To access the Manage User page:

1. Log into the Merge PACS Workstation as an Admin user.
2. Click on the **Settings** icon in the upper right corner of the Workstation Browser, as in the following example:

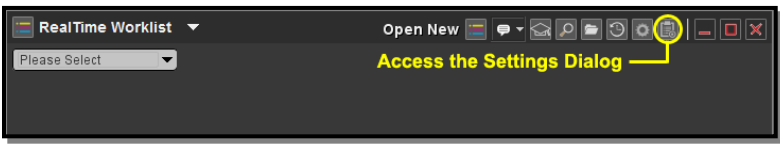


Figure 455: The Settings Icon

*The **Settings** dialog is displayed.*

3. Click on the **Users** link on the left-hand menu to access the Manage Users page, as in the following example:

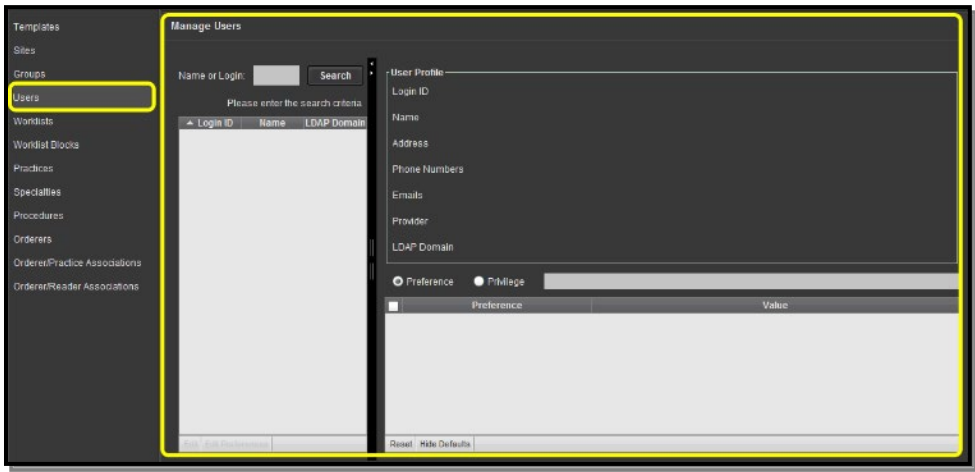


Figure 456: Manage Users Page

12.2. Selecting a User

The left-hand panel of the Manage Users page allows you to search for a particular user via the **Name or Login** search field and **Search** button at the top of the panel, as in the following example:

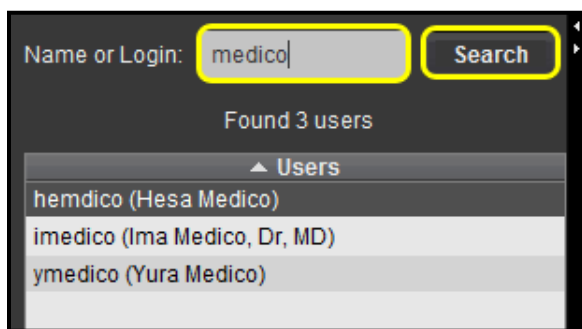


Figure 457: Searching for a User

A list of matching users is displayed. Clicking on a listed user will cause detailed information about that user to be displayed in the right-hand panel, as in the following example:

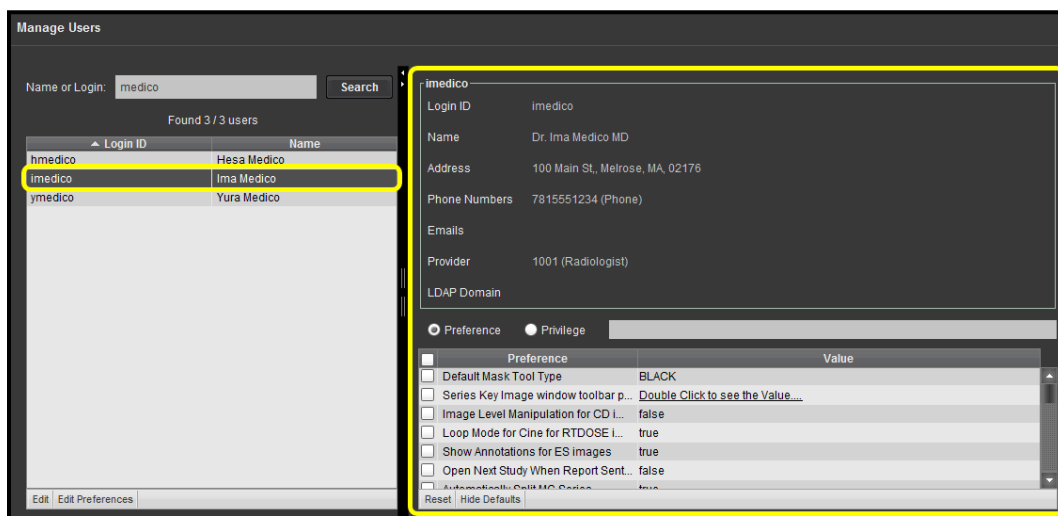


Figure 458: Searching for a User

- The top half of the right-hand panel displays personal information about the user, including Login ID, Name, Address, Provider Code, etc.
- The bottom half of the right-hand panel displays information about the preferences and privileges currently configured for this user.

12.3. General User Preferences and Privileges

12.3.1. Viewing Preferences and Privileges

By default, the bottom section of the right-hand panel displays detailed information about all preferences and privileges currently set for the selected user, whether they come from a template linked to one or more groups the user belongs to, are inherited from a group's parent or come from the system defaults, as in the following example:

Preference		Value
☐	Automatically Apply Foreign Presentation	true
☐	Automatically Hide PowerScribe Windows	false
☐	Automatically Launch IMClient	true
☐	Automatically Launch PowerScribe	true
☐	Automatically Load Non-Final Reports in PowerSc...	true
☐	Automatically Mark Annotated Images as Key Ima...	true
☐	Automatically Open Order Viewer	true
☐	Automatically Open Report Viewer	false
☐	Automatically Play Voice Clip on Load	true
☐	Automatically pre-cache relevant prior studies	true
☐	Automatically pre-cache unread studies	true
☐	Automatically Save Annotations on Exit	false
☐	Automatically Save Voice Clip on Exit	true
☐	Automatically Split CR Series	true
☐	Automatically Split CT Series	true
☐	Automatically Split MG Series	true

Figure 459: Current Preferences and Privileges

- Use the radio buttons at the top of the panel to select whether you want to view this user's **Preference** values or **Privileges**.
- Sort the information displayed by clicking on a column header (clicking the same header repeatedly will toggle between ascending and descending sort order).
- Filter the information displayed by entering text to match in the text box at the top of the panel, as in the following example:

Preference		Value
☐	Allow rendering image count on main viewer	true
☐	Allow rendering image count on thumbnail viewer	false
☐	Allow rendering image count on thumbnail viewer of patient prior view	true
☐	Allow rendering laterality and view code on main viewer	true
☐	Allow rendering laterality and view code on thumbnail viewer	true
☐	Allow rendering laterality and view code on thumbnail viewer of patient prior view	true
☐	Allow rendering precached indicator on main viewer	true
☐	Allow rendering precached indicator on thumbnail viewer	false
☐	Allow rendering precached indicator on thumbnail viewer of patient prior view	true
☐	Volume Rendering Algorithm	MIP

Figure 460: Filtering Information

- Hover your mouse over a preference or privilege to see how it was set or enabled, as in the following example:

☐	Preference	Value
☐	DICOM Overlays for LS images	<DICOMOverlay fontName="MS Sans Serif" fontStyle="0" f...
☒	DICOM Overlays for MG images	<DICOMOverlay fontName="MS Sans Serif" fontStyle="0" f...
☐	DICOM Overlays for MG tomo images	From Template Guest Users Pref linked to site Home, f...
☐	DICOM Overlays for MR images	parent of group Referring Clinicians, f...
☐	DICOM Overlays for NM images	parent of User imedico f...
☐	DICOM Overlays for OT images	<DICOMOverlay fontName="MS Sans Serif" fontStyle="0" f...

Figure 461: How this Setting Was Enabled

NOTE: For **preferences**, you must hover over the **Value** column to see the pop-up information.

- When viewing **preferences**, settings that are different from the default settings (*i.e.*, settings this user would have if no overrides were set at any level) are highlighted in yellow. To only show these settings, click on the **Hide Default** button at the bottom of the panel (the button will then change to **Show All** if you want to include the defaults again).
- When viewing **preferences**, you can reset any user-level overrides (*i.e.*, preferences set by the user via the Workstation Merge PACS Preferences dialog or set for the user from the Manage Users page) by using the checkboxes to select the preferences and then clicking on the **Reset** button at the bottom of the panel. This will restore the preference to whatever it would have been if no overrides were set at any level.

12.3.2.Editing User Preferences

If you want to edit a user's preferences on the fly, do the following:

- Make sure the desired user is selected and click on the **Edit Preferences** button at the bottom of the left panel, as in the following example:

The screenshot shows the 'Manage Users' window. On the left, a search bar contains 'imedico' and a table lists three users: 'hmedico', 'imedico', and 'ymedico'. The 'imedico' user is selected. At the bottom of this panel, the 'Edit Preferences' button is highlighted with a red rectangle. On the right, the details for the 'imedico' user are shown, including name, address, phone numbers, and emails. Below this, the 'Preference' tab is active, displaying a table of user preferences. The table has columns for 'Preference' and 'Value'. The first row, 'Default Mask Tool Type', is highlighted in yellow and has a value of 'BLACK'. Other preferences include 'Series Key Image window toolbar p...', 'Image Level Manipulation for CD L...', 'Loop Mode for Cine for RTDOSE I...', 'Show Annotations for ES Images', and 'Open Next Study When Report Sent...'. At the bottom of the preferences panel, there are buttons for 'Reset' and 'Hide Defaults'.

Figure 462: Edit this User's Preferences

The **Merge PACS Preferences** dialog will be displayed for the selected user with only user-level preferences showing, as in the following example:

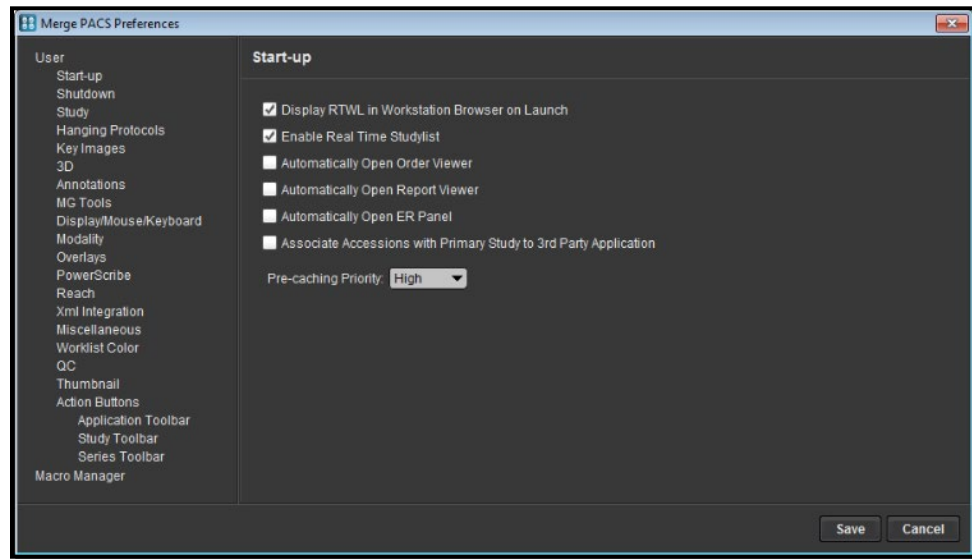


Figure 463: Merge PACS Preferences for this User

- Set the preferences for this user as desired.

NOTE: Refer to Appendix A.2 below for a description of all available user preferences.

NOTE: Only preferences that are not marked “final” can be edited, although all preferences will be displayed in the right panel, whether “final” or not.

12.4. Configuring Advanced Workflow Information

Once you have selected a user, you can define and/or edit the advanced workflow information for the user that is used in conjunction with the Automatic Assignment and Launch and Control features by clicking on the **Edit** button at the bottom of the left panel, as in the following example:

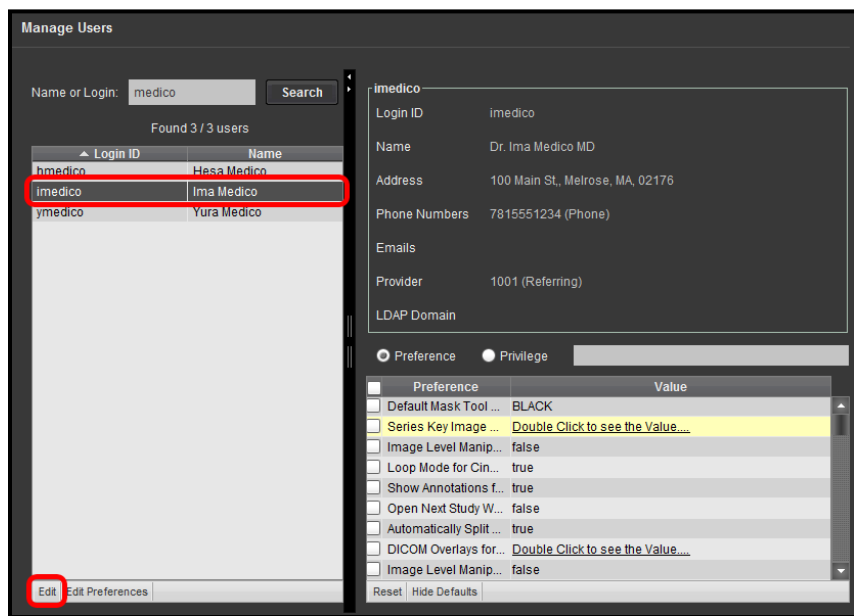


Figure 464: Edit User Button

When you click the Edit button, the **Edit User** page for this user will be displayed in a separate pop-up window, as in the following example:

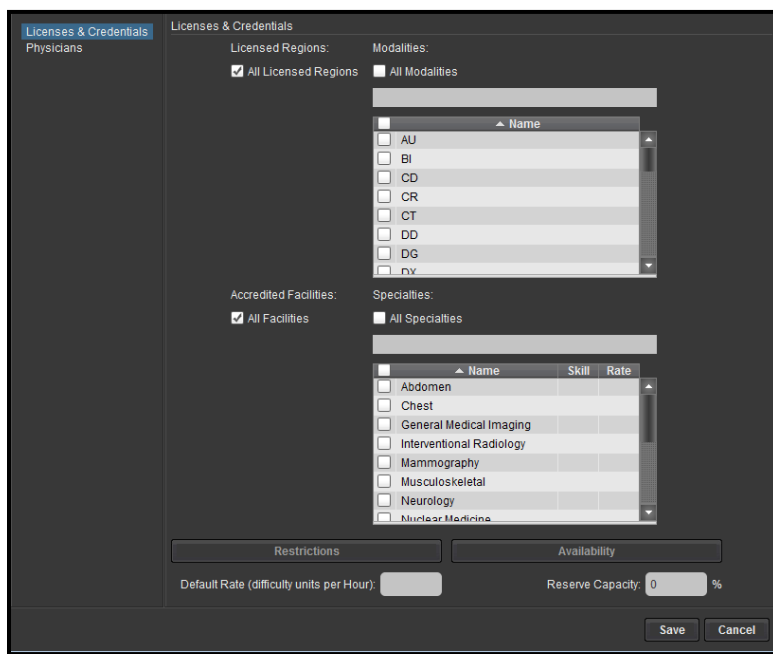


Figure 465: Edit User Page

12.4.1. License & Credential Information

The Licenses & Credentials section lets you define information about this user regarding the regions, facilities, modalities and specialties that this user is authorized to read studies from and is accessible by clicking on the **License & Credentials** link, as in the following example:

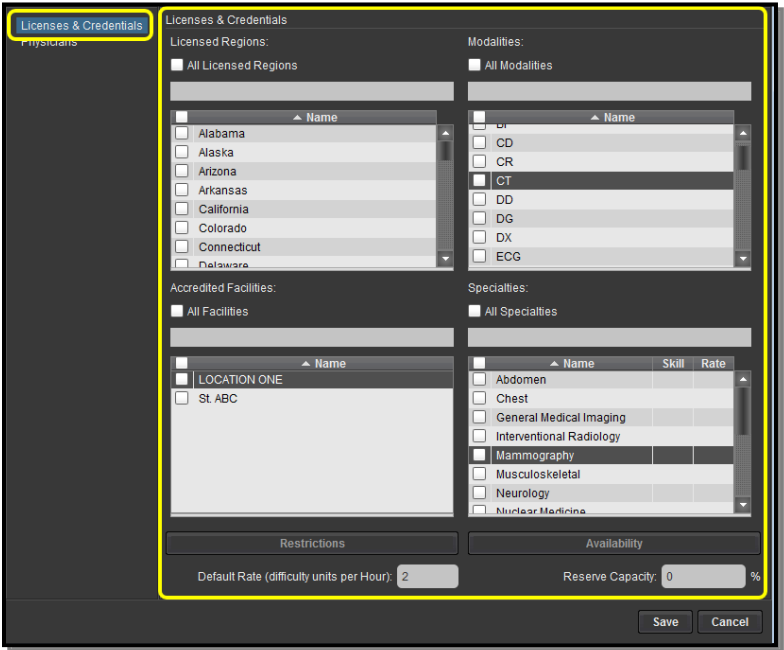


Figure 466: Licenses & Credentials

a. Licensed Regions

The **Licensed Regions** table determines the states (or other regions) where the user has privileges to view and read studies, as in the following example:

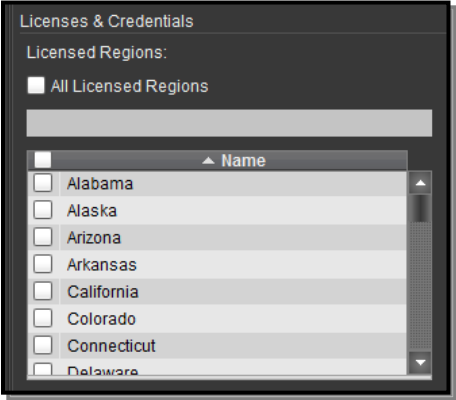


Figure 467: Licensed Regions

Licensed Regions can be used to restrict a user’s access to exams via a RTWL or Teaching worklist filter so that the user will only be shown exams that match the user’s assigned Licensure Regions and can also be used when creating Auto Assignment rules to determine whether a user is a candidate for a particular exam being assigned.

By default, each user is configured to be licensed in every available Licensed Region, meaning the “All Licensed Regions” checkbox will be selected and the Licensed Region table will be hidden, as in the following example:

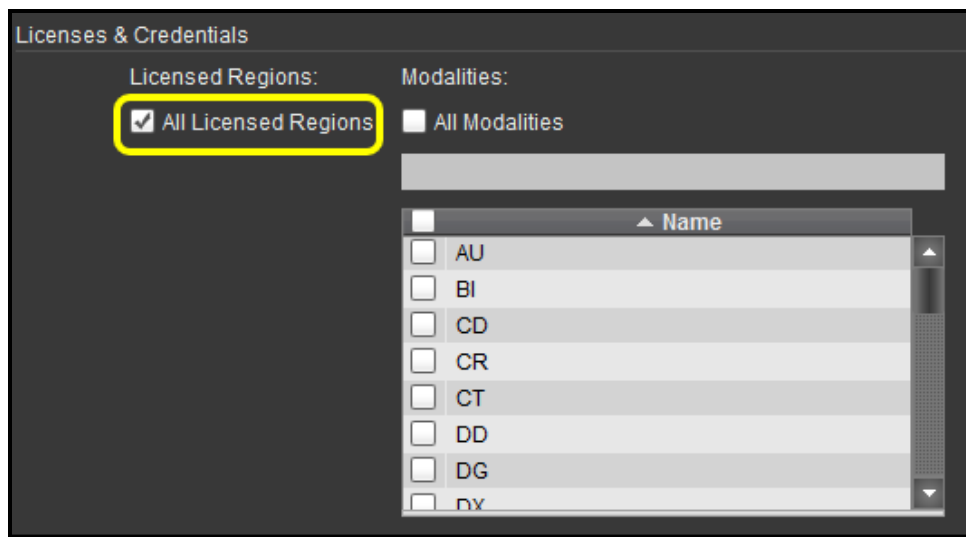


Figure 468: All Licensed Regions Selected

If desired, however, you can deselect the “All Licensed Regions” checkbox to display the Licensed Regions table and then manually select one or more regions in which this user is licensed to read. You can also use the text field at the top of the table to filter the number of regions displayed, as in the following example:

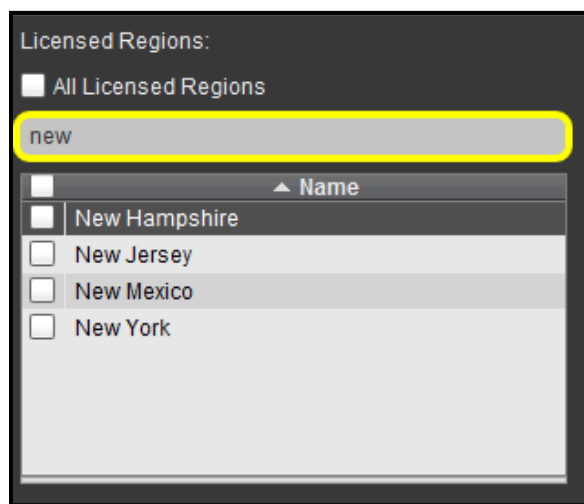


Figure 469: Filtering the List of Regions

NOTE: The available options in the Licensed Regions table are determined by the Licensure Regions that have been set up for Merge PACS, as described in subsection 9.13.10 above.

b. Accredited Facilities

The **Accredited Facilities** table determines the facilities where the user is accredited to view and read studies, as in the following example:

Accredited Facilities:

☐ All Facilities

☐	Name
☐	ACME Radiology
☐	LOCATION ONE
☐	Premiere Radiology
☐	St. ABC
☐	St. Nowhere

Figure 470: Accredited Facilities

Accredited Facilities can be used to restrict a user's access to exams via a RTWL or Teaching worklist filter so that the user will only be shown exams for the Facilities where they are accredited and can also be used when creating Auto Assignment rules to determine whether a user is a candidate for a particular exam being assigned.

By default, each user is configured to be accredited in every available facility, meaning the "All Facilities" checkbox will be selected and the Accredited Facilities table will be hidden, as in the following example:

Accredited Facilities:

☒ All Facilities

Specialties:

☐ All Specialties

☐	Name	Skill	Rate
☐	Abdomen		
☐	Chest		
☐	General Medical Imaging		
☐	Interventional Radiology		
☐	Mammography		
☐	Musculoskeletal		
☐	Neurology		
☐	Nuclear Medicine		

Figure 471: All Facilities Selected

If desired, however, you can deselect the “All Facilities” checkbox to display the Accredited Facilities table and then manually select one or more facilities in which this user is accredited to read. You can also use the text field at the top of the table to filter the number of facilities displayed, as in the following example:

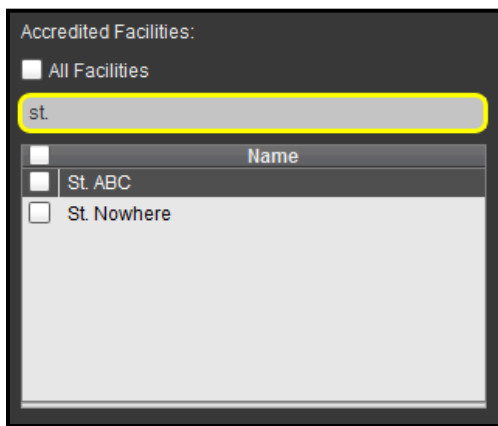


Figure 472: Filtering the List of Facilities

NOTE: The available options in the Accredited Facilities table are determined by the facilities that have been set up for Merge PACS, as described in subsection 10.5.2 above.

c. Modalities

The **Modalities** table determines the modalities the user is allowed to read, as in the following example:

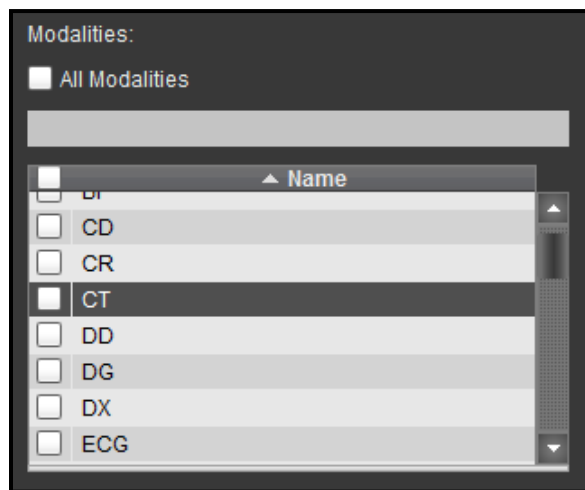


Figure 473: Allowed Modalities

Modalities can be used when creating Auto Assignment rules to determine whether a user is a candidate for a particular exam being assigned.

Click on the checkboxes next to one or more modalities to add them to the list of allowed modalities for this user. You can also use the text field at the top of the table to filter the number of modalities displayed, as in the following example:

Modalities:

☐ All Modalities

	Name
☐	CD
☐	DD
☐	DG
☐	DX
☐	RTDOSE
☐	RTRECORD

Figure 474: Filtering the List of Modalities

You can also click on the **All Modalities** checkbox to select all modalities and hide the Modalities table, as in the following example:

Licenses & Credentials

Licensed Regions:

☐ All Licensed Regions

Modalities:

☒ All Modalities

	Name
☐	Alabama
☐	Alaska
☐	Arizona
☐	Arkansas
☐	California
☐	Colorado
☐	Connecticut
☐	Delaware

Figure 475: All Modalities Selected

NOTE: The available options in the Modalities table are determined by the modalities that have been set up for Merge PACS.

d. Specialties

The **Specialties** table determines the specialties the user is allowed to read, as in the following example:



Specialties:

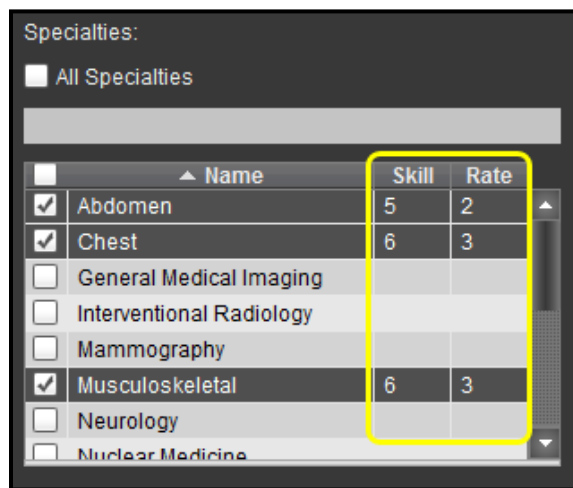
☐ All Specialties

☐	▲ Name	Skill	Rate
☐	Abdomen		
☐	Chest		
☐	General Medical Imaging		
☐	Interventional Radiology		
☐	Mammography		
☐	Musculoskeletal		
☐	Neurology		
☐	Nuclear Medicine		

Figure 476: Allowed Specialties

Specialties can be used when creating Auto Assignment rules to determine whether a user is a candidate for a particular exam being assigned.

Click on the checkboxes next to one or more specialties to add them to the list of allowed specialties for this user. For each specialty selected, you can also enter the user's skill level and reading rate per hour for that specialty in the fields provided, as in the following example:



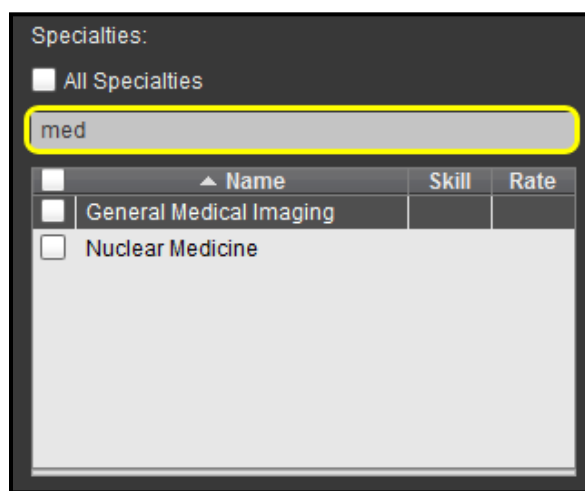
Specialties:

☐ All Specialties

☐	▲ Name	Skill	Rate
☒	Abdomen	5	2
☒	Chest	6	3
☐	General Medical Imaging		
☐	Interventional Radiology		
☐	Mammography		
☒	Musculoskeletal	6	3
☐	Neurology		
☐	Nuclear Medicine		

Figure 477: Selected Specialties with Skill and Rate Information

To filter the list of available specialties, enter text in the field at the top of the table, as in the following example:

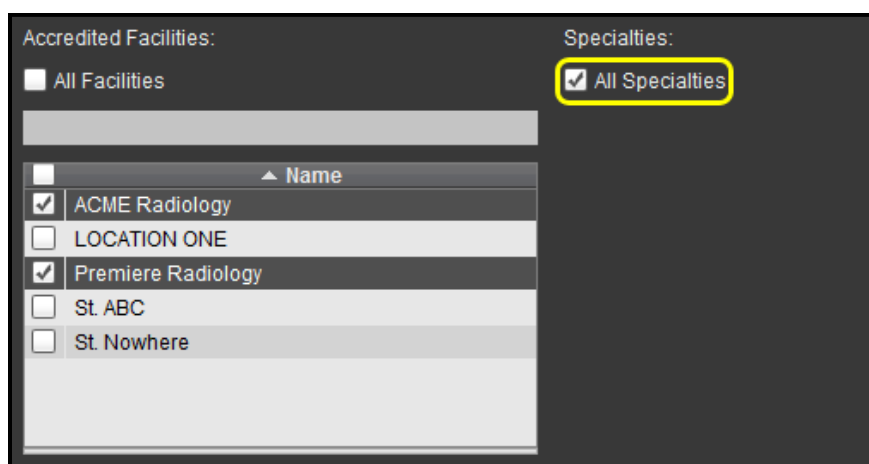


The screenshot shows a window titled "Specialties:". At the top, there is a checkbox labeled "All Specialties". Below it is a search input field containing the text "med". Underneath the search field is a table with three columns: "Name", "Skill", and "Rate". The table contains two rows of data:

	Name	Skill	Rate
☐	General Medical Imaging		
☐	Nuclear Medicine		

Figure 478: Filtering the List of Specialties

To select all specialties and hide the Specialties table, click on the **All Specialties** checkbox, as in the following example:



The screenshot shows a window with two sections. On the left, under "Accredited Facilities:", there is a checkbox labeled "All Facilities" and a table with a "Name" column. On the right, under "Specialties:", there is a checkbox labeled "All Specialties" which is checked and highlighted with a yellow box.

Accredited Facilities:

	Name
☒	ACME Radiology
☐	LOCATION ONE
☒	Premiere Radiology
☐	St. ABC
☐	St. Nowhere

Specialties:

☒ All Specialties

Figure 479: All Specialties Selected

NOTE: The available options in the Specialties table are determined by the specialties that have been set up for Merge PACS, as described in subsection 10.6.3 above.

e. Restrictions and Availability

Once you have specified at least one facility and one specialty for this user, you can define **restrictions** per accredited facility (e.g., for cases where a user is not allowed to read all of his specialties for all facilities) and **current availability** for reading specialties per accredited facility (e.g., for cases where a user may not be reading all of his specialties for all of his facilities, even though he is allowed to do so). This is done via the **Restrictions** and **Availability** buttons at the bottom of the page, as in the following example:

The screenshot shows a configuration window with two main sections: 'Accredited Facilities' and 'Specialties'. Both sections have a checkbox for 'All' (All Facilities, All Specialties). Below these are lists of facilities and specialties with checkboxes. At the bottom, there are two buttons: 'Restrictions' and 'Availability', both highlighted with yellow boxes. Below the buttons are input fields for 'Default Rate (difficulty units per Hour): 2' and 'Reserve Capacity: 20 %'.

Figure 480: Restrictions and Availability Buttons

NOTE: These buttons will be disabled (“grayed out”) unless at least one facility and one specialty is selected for this user.

To configure restrictions per accredited facility for this user:

1. Click on the **Restrictions** button.

The **Credential Restrictions** dialog for this user is displayed in a pop-up window, as in the following example:

The screenshot shows a dialog box titled 'Credential Restrictions for Ima'. It has two radio buttons: 'Do not customize Ima's credential restrictions' and 'Customize Ima's credential restrictions'. The 'Customize' option is selected. Below is a 'Facility' dropdown menu set to 'ACME Radiology'. Under the heading 'At ACME Radiology', there are two radio buttons: 'Ima can work on:' (selected) and 'Ima can work on everything but:'. Below these is a list of specialties with checkboxes: Abdomen, Chest, and Musculoskeletal. At the bottom are 'Save' and 'Cancel' buttons.

Figure 481: Credential Restrictions

2. Make sure that **Customize [USER NAME]'s credential restrictions** is selected.

NOTE: To disable credential restrictions for this user, select the **Do not customize [USER NAME]'s credential restrictions** option instead.

3. Select the desired Facility from the drop-down **Facility** menu.
4. Select whether the defined list of specialties is inclusive (“**can work on**”) or exclusive (“**can work on everything but**”).
5. Select one or more specialties from the list of specialties this user is allowed to read (click the check box at the top of the first column to select/deselect all specialties).
6. Click **Save**.

To configure current availability for reading specialties per accredited facility for this user:

1. Click on the **Availability** button.

The **Availability** dialog for this user is displayed in a pop-up window, as in the following example:

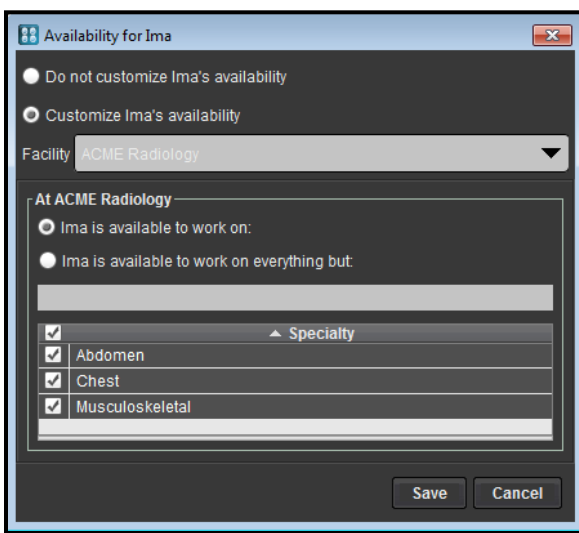


Figure 482: Availability

2. Make sure that **Customize [USER NAME]'s availability** is selected.

NOTE: To disable customized availability for this user, select the **Do not customize [USER NAME]'s availability** option instead.

3. Select the desired Facility from the drop-down **Facility** menu.
4. Select whether the defined list of specialties is inclusive (“**is available to work on**”) or exclusive (“**is available to work on everything but**”).
5. Select one or more specialties from the list of specialties this user is allowed to read (click the check box at the top of the first column to select/deselect all specialties).
6. Click **Save**.

f. Default Rate and Reserve Capacity

For exams whose specialty is not specified or for cases where there is no reading rate defined for a particular specialty for a user, you can set the default reading rate for the user by entering it in the **Default Rate** field at the bottom of the page, as in the following example:

The screenshot shows a user configuration window with two main sections: 'Accredited Facilities' and 'Specialties'. Both sections have a list of items with checkboxes. Below these sections are two buttons: 'Restrictions' and 'Availability'. At the bottom, there are two input fields: 'Default Rate (difficulty units per Hour):' with the value '2' and 'Reserve Capacity: 0 %'. The 'Default Rate' field is highlighted with a yellow box.

Figure 483: Default Rate

The default rate is specified as difficulty units per hour, which could be RVUs per hour or any other scale the site is using, and is intended to be used in with automatic assignment rules, for example, to determine a user's current load or remaining capacity.

When deciding the best assignee for a particular exam, the automatic assignment rules can take into account how many exams each user has left to read, how long it will likely take them to read each exam, and optionally, a reserve capacity for how much of their shift should be kept open (for example, for non-reading work such as consults or tumor boards). You can set the reserve capacity for this user by entering it as a percentage in the **Reserve Capacity** field at the bottom of the page, as in the following example:

The screenshot shows the same user configuration window as Figure 483. In this version, the 'Reserve Capacity: 0 %' field at the bottom right is highlighted with a yellow box.

Figure 484: Reserve Capacity

12.4.2. Physician Information

The Physician Information section lets you define information about this user regarding the user's availability to read studies that may be automatically assigned to the user (including that user's performance targets for both number of exams and difficulty units to read per hour, practice, reader location and rotation or shift) and is accessible by clicking on the **Physicians** link, as in the following example:

Figure 485: Physicians Information

NOTE: Users can override the defaults for reader location, current availability to read and current shift/rotation from within the Workstation.

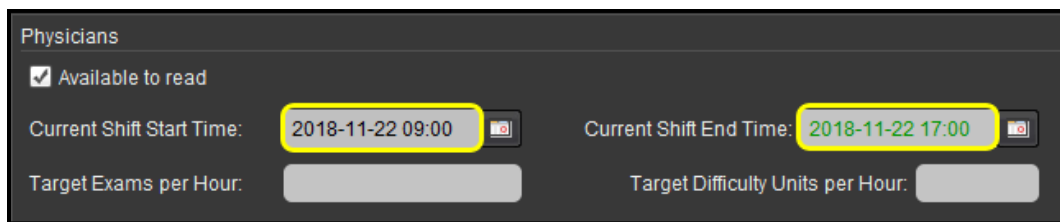
a. Availability to Read

If this user has the “Consider for Automatic Assignment” preference enabled, the user will generally be considered eligible to have studies assigned to him or her by the Advanced Distribution Engine, assuming they meet the requirements of the specific auto-assignment rules that have been defined and they are currently available to read. In addition, however, the user needs to be currently accepting automatic assignments, which can be set by selecting the **Available to read** checkbox, as in the following example:

Figure 486: Available to Read

b. Shift/Rotation Start and End Time

You can set the shift/rotation start and end time for this user by entering the date and time in the fields provided, as in the following example:



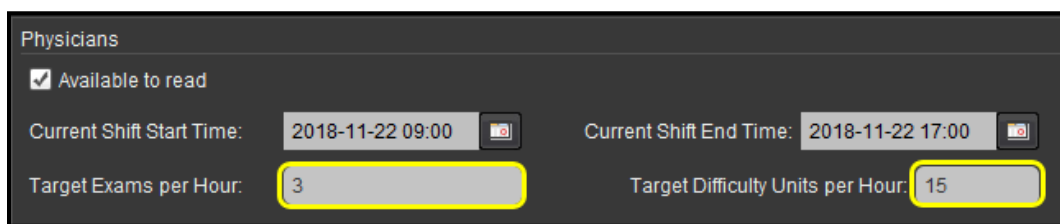
The screenshot shows a form titled "Physicians" with a checked checkbox "Available to read". Below this, there are two time fields: "Current Shift Start Time" and "Current Shift End Time". Both fields contain the date and time "2018-11-22 09:00" and "2018-11-22 17:00" respectively, with a small calendar icon to the right of each. Below these are two empty input fields: "Target Exams per Hour" and "Target Difficulty Units per Hour".

Figure 487: Current Shift Information

- The start and end time must include the date and be in the format **yyyy-mm-dd HH:mm**, as in the example above.
- When entering the times, the text will appear red until it is entered in the proper format, at which point it will display as green
- Time should be entered in 24-hour format (e.g., 5:00 PM should be entered as 17:00).
- You can click on the calendar icon next to each field to select the date from a pop-up calendar dialog. By default, the current time will be selected, which you can then manually edit as necessary in the fields themselves.

c. Target Exams and Difficulty Units per Hour

You can set the target exams per hour and target difficulty units per hour for this user in the fields provided, as in the following example:

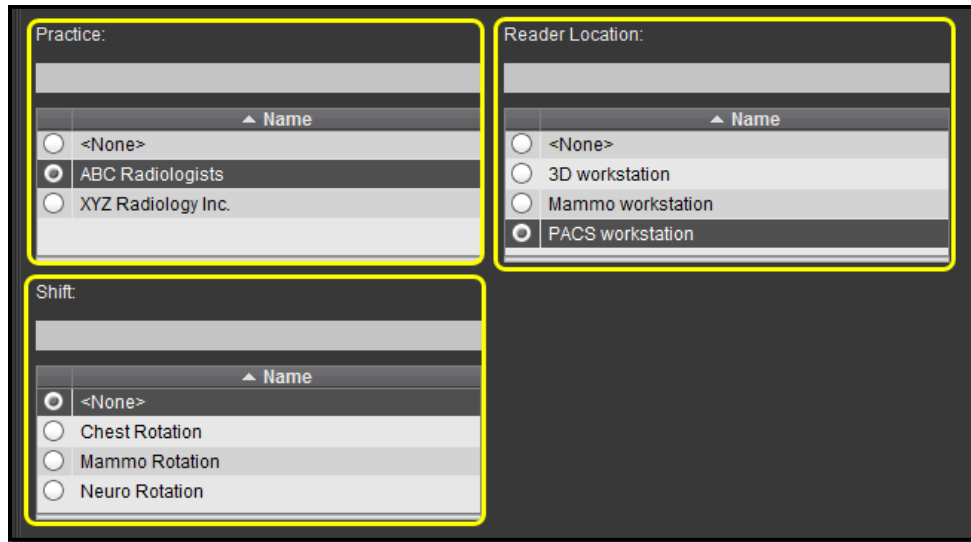


The screenshot shows the same "Physicians" form as Figure 487, but with the "Target Exams per Hour" field set to "3" and the "Target Difficulty Units per Hour" field set to "15". Both of these fields are highlighted with a yellow border. The time fields remain the same as in Figure 487.

Figure 488: Target Exams and Difficulty Units per Hour

d. Practice, Reader Location and Shift/Rotation

You can select the user's Practice, Reader Location and Shift/Rotation from the provided tables, as in the following example:



The screenshot shows a user configuration interface with three tables, each highlighted with a yellow border. Each table has a search field at the top and a list of options below, each with a radio button.

Practice:	
	▲ Name
☐	<None>
☒	ABC Radiologists
☐	XYZ Radiology Inc.

Reader Location:	
	▲ Name
☐	<None>
☐	3D workstation
☐	Mammo workstation
☒	PACS workstation

Shift:	
	▲ Name
☒	<None>
☐	Chest Rotation
☐	Mammo Rotation
☐	Neuro Rotation

Figure 489: Practice, Reader Location and Shift

- The available options in each table are determined by the practices, reader locations and reader availability custom tags currently configured for Merge PACS, as described in Sections 10.5 and 10.6 above.
- For each table, you can enter text in the field at the top to filter the list of options.
- The actual name of the “Shift” table is custom-configurable on a site-by-site basis.

Chapter 13. Creating and Managing Worklists

Once the RealTime Worklist component has been enabled and configured on the Merge PACS Server, you can create and manage individual worklists. Each worklist is comprised of one or more individual worklists blocks that can have their own individually configured behavior as to what studies should be displayed and how those studies should be displayed.

The steps involved in creating and managing worklists are as follows:

1. Populate the various worklist “dictionaries” with entries that can be used to provide exam filters for individual worklist blocks.
2. Create one or more specific worklists blocks with defined general block properties, hide criteria and exam filters.
3. Create one or more worklists with defined general worklist properties, columns, actions, blocks, caching options and associated user groups.

NOTE: Sites in a hub-spoke configuration or in a Production and Contingency Server configuration must always interact with the same hub worklist. The system is not designed for reading worklists being served from multiple servers at the same time but rather is designed for a single set of central shared worklists. Running worklists from multiple servers adds risk of study states not being in sync and “double reads”. Operation of the contingency server or spoke worklists when isolated (WAN outage or production server downtime) is still allowed, however.

13.1. Populating Worklist Dictionaries

Each worklist block includes a variety of exam filters that determine which studies should be made available to a particular user or group of users. Before you can create a worklist block, you must first configure the dictionary (i.e., define the available options) for each type of exam filter.

- The following exam filters can be configured from the Configuration section of the Merge PACS Management Pages, as described in Section 10.6 above:
 - **Modalities**
 - **Provider Types**
 - **Workflow Priorities**
 - **Workflow Status Dimensions**
 - **Workflow Status**
 - **HL7 Order Status Mapping**
 - **Licenture Regions**
 - **Locations**
 - **Patient Classes**
 - **IPID Configuration**

- The following exam filters can be configured from the Settings Dialog of the Merge PACS Workstation, as described in Section 10.6 above:
 - **Specialties**
 - **Procedures**
- In addition, **Facilities** can be configured from the Advanced Workflow Management Pages, as described in Section 10.5 above.

NOTE: Worklist blocks can also be filtered by **Station Names**. However, rather than requiring you to manually populate a separate worklist dictionary for each Station Name, the dictionary is automatically populated based on the entries in the DICOM Devices table. For information on adding devices to that table, refer to subsection 9.13.2 above.

13.2. Accessing the Worklist Configuration Pages

The Worklist Configuration Pages are accessed via the Merge PACS Workstation.

To access the Worklist Configuration Pages:

1. Log into the Merge PACS Workstation as a Super or Admin user.
2. Click on the **Settings** icon in the upper right corner of the Workstation Browser, as in the following example:

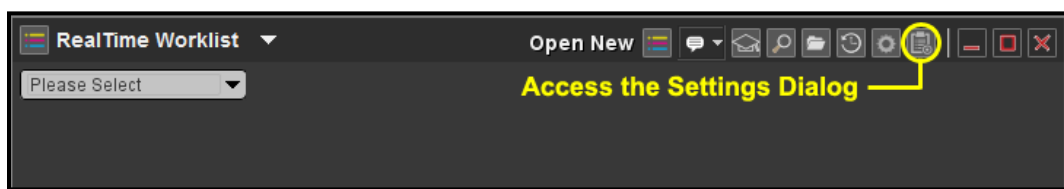


Figure 490: The Settings Icon

The **Settings** dialog is displayed with links to the two Worklist Configuration pages displayed on the left-hand side, as in the following example:

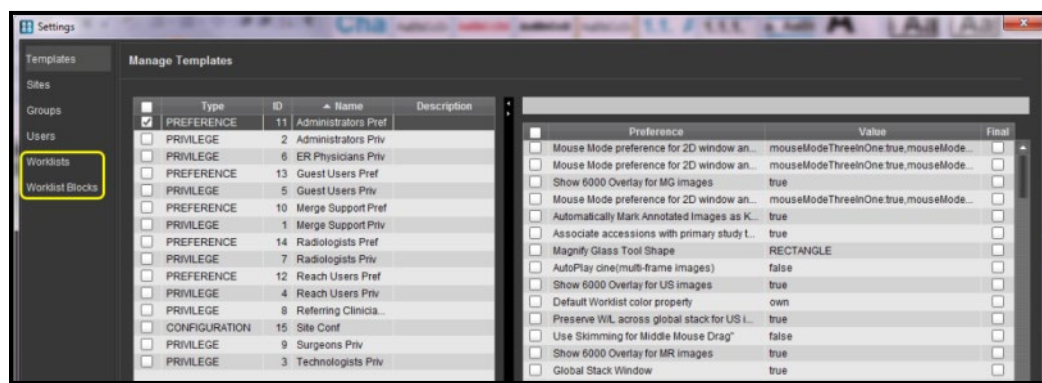


Figure 491: The Settings Dialog

13.3. Creating and Editing Worklist Blocks

Each worklist is comprised of one or more worklist blocks, and each worklist block can be configured to display exams according to specified display options and to include exams based on specified filters. Before you can create a worklist, as described in Section 13.4 below, you must first create one or more worklist blocks that can be included in that worklist.

NOTE: You can also create new worklist blocks “on the fly” while creating or editing a worklist, if desired, as described in subsection 13.4.10 below.

NOTE: Since exams might match the criteria set for multiple worklist blocks, it is possible for some exams to be displayed within multiple blocks on the same worklist.

13.3.1. Accessing the Worklist Blocks Management Page

From the **Settings** dialog described in Section 13.2 above, click on the **Worklist Blocks** link on the left side of the dialog, as in the following example:

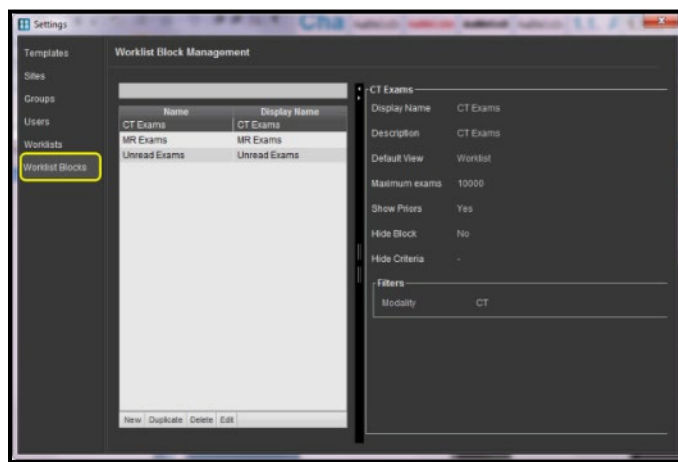


Figure 492: The Worklist Blocks Management Page

The left side of the Worklist Blocks Management Page displays a list of all existing worklist blocks, while the right side of the Worklist Blocks Management Page displays general information about the worklist block currently highlighted in the list.

NOTE: New installs of Merge PACS 8.2 have a default “Unread Exams” worklist block that will display studies that have a Diagnostic Status of “Unread.” If desired, this can be used as the starting point for new worklist blocks as described below.

CAUTION: It is strongly recommended that the Unread Exams worklist block be used (or created, if not there by default) by sites that use highly specific worklist filters to ensure studies are not missed.

13.3.2.Creating a New Worklist Block

To create a new worklist block from scratch, click on the **New** button at the bottom of the list of worklist blocks, as in the following example:

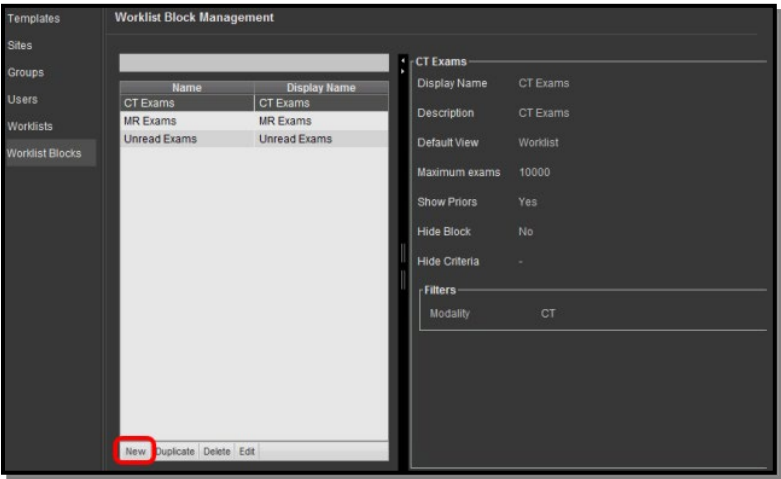


Figure 493: New Worklist Block Button

A blank **Worklist Block Editor** is displayed. Proceed to subsection 13.3.6 below.

13.3.3.Making a Copy of an Existing Worklist Block

To copy an existing workstation block (*i.e.*, to make modifications to it and save it with a new name), click on worklist block you want to copy in the list of existing worklist blocks and click on the **Duplicate** button at the bottom of the list, as in the following example:

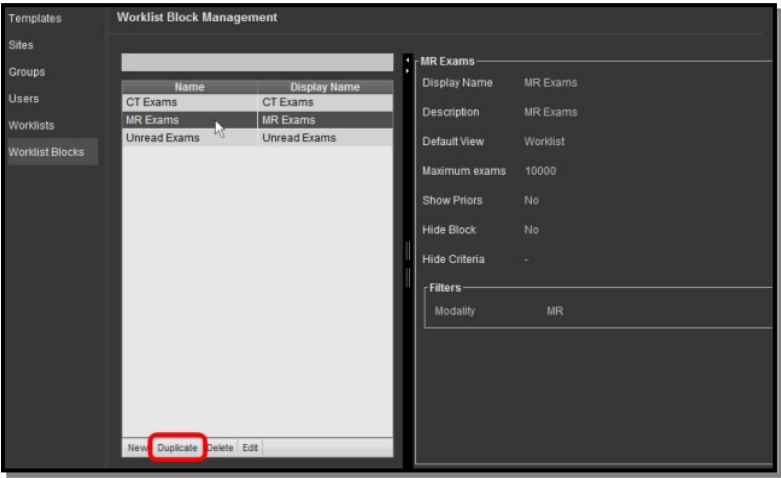


Figure 494: Duplictate Worklist Block Button

The **Worklist Block Editor** is displayed with information already entered. Proceed to subsection 13.3.6 below.

13.3.4.Editing an Existing Worklist Block

To edit an existing workstation block, double-click on worklist block you want to edit in the list of existing worklist blocks or click on the worklist block once to highlight it and then click on the **Edit** button at the bottom of the list, as in the following example:

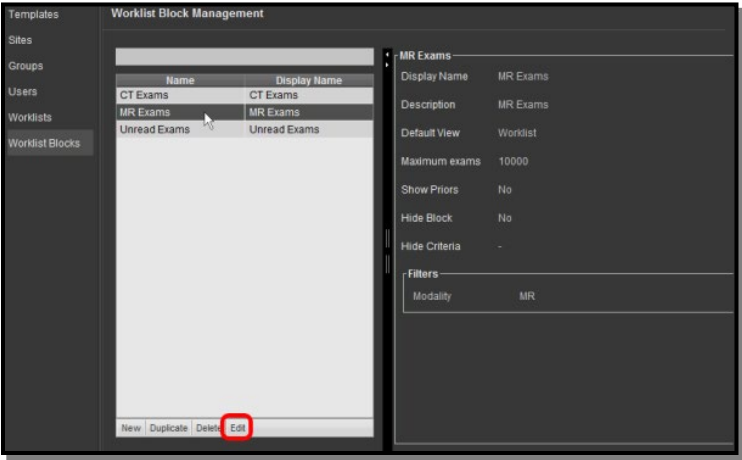


Figure 495: Edit Worklist Block Button

The **Worklist Block Editor** for the selected worklist block is displayed. Proceed to subsection 13.3.6 below.

13.3.5.Deleting an Existing Worklist Block

To delete an existing workstation block, click on worklist block you want to delete in the list of existing worklist blocks and click on the **Delete** button at the bottom of the list, as in the following example:

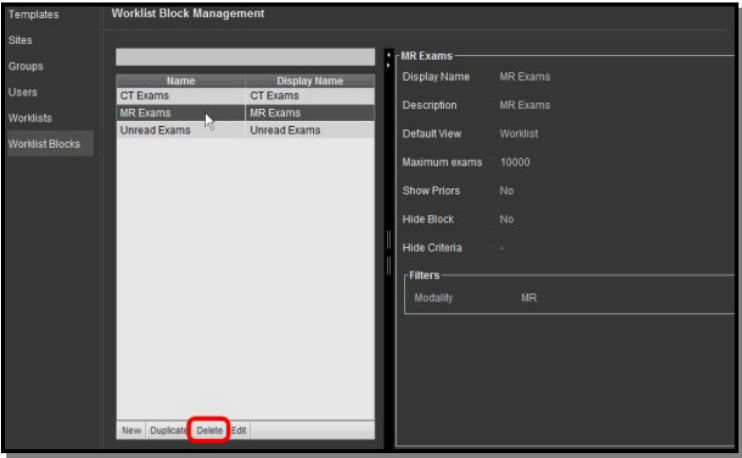


Figure 496: Delete Worklist Block Button

The selected worklist block is removed from the list of existing worklist blocks.

13.3.6.The Worklist Block Editor

When you click on the **New**, **Duplicate** or **Edit** button on the main Worklist Blocks Management page, the **Worklist Block Editor** will launch as a separate window, as in the following example:

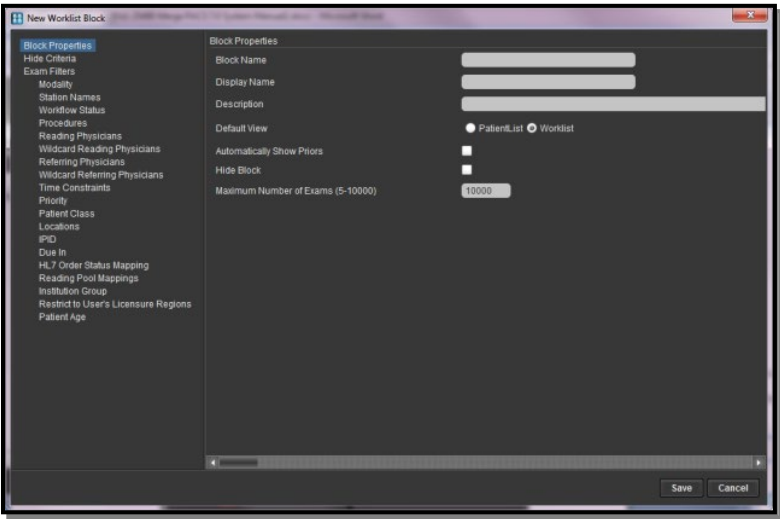


Figure 497: Worklist Block Editor

Depending on which button you clicked, the Worklist Block Editor will either be blank or already be populated with information. From the Worklist Block Editor you can enter or edit the information for this worklist block as described in the following subsections.

13.3.7.Block Properties

To configure the general properties for a worklist block, click on the **Block Properties** link on the left-hand side of the **Worklist Block Editor**, as in the following example:

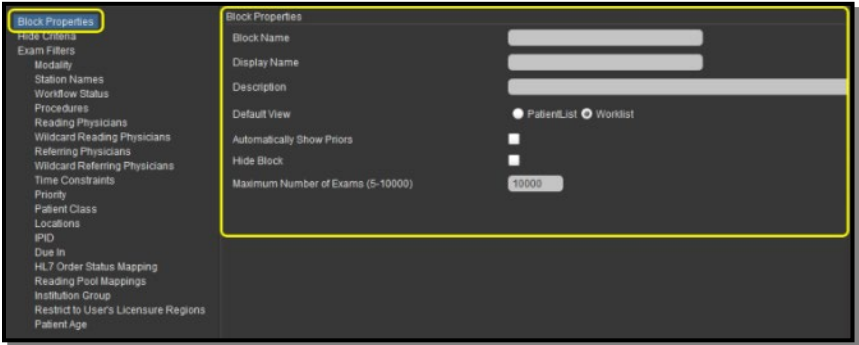


Figure 498: Block Properties

NOTE: The **Block Properties** link will be selected by default when you first access the Worklist Block Editor.

a. Block Name

In the **Block Name** field, enter a unique name for this worklist block.

b. Display Name

In the **Display Name** field, enter what you would like displayed for this worklist block as an option in the list of existing worklist blocks, if different than the Block Name. If left blank, the Block Name will automatically be filled in here.

c. Description

If desired, enter a more detailed description of the worklist block in the **Description** field. This field is optional and is only displayed within the Worklist Block Editor itself for the sake of clarification, if needed.

d. Default View

The Default View allows you to configure whether this worklist block will initially be displayed in **worklist** or **patient list** mode, as follows:

Option	Description
Worklist	Displays exams for this worklist block as a separate group for each date. When sorting by columns, sorting will be performed within each date group and not for the entire worklist block.
PatientList	Displays all exams for this worklist block in a single group, regardless of date. When sorting by columns, sorting will be performed on the entire group of exams within the worklist block.

NOTE: If **Consolidated View** is selected for the worklist this block appears in, as described in subsection 13.4.10 below, the entire worklist be displayed in **patient list** view regardless of the view defined for each worklist block.

e. Automatically Show Priors

Select **Automatically Show Priors** to specify that all prior studies for a study will be displayed in the worklist block when that study is opened in the Viewer.

NOTE: This does **not** mean that all prior exams will be displayed by default within the worklist block at all times, but only when a specific study is currently loaded within the Viewer.

f. Hide Block

As described in subsection 13.4.10 below, when defining which worklist blocks should be included in a particular worklist, you can select whether each worklist block should be displayed in an expanded (visible) or collapsed (hidden) state. By default, each worklist block will be set to be visible and you will need to manually select the “Hide” option for the worklist blocks you want to be initially hidden on the worklist.

If desired, however, you can click on the **Hide Block** option within the Block Properties for a worklist block so that the “Hide” option will automatically be selected whenever you choose to add that block to a worklist in the future.

NOTE: Selecting “Hide Block” will not affect any worklists that already contain this block.

g. Maximum Number of Exams

If desired, you can set an approximate maximum number of exams that can be displayed within this worklist block by entering an integer between 5-10,000 in the **Maximum Number of Exams** field. This is typically done to help control the load on the Server (*i.e.*, if large numbers of exams on a single worklist block are causing performance issues).

The actual maximum number of exams that can be displayed in this worklist block is determined by the integer entered in this field as well as the following:

- The **Worklist.Server.MaxExams** Server property. This is typically set to 10,000, but can be set lower for troubleshooting purposes and the maximum number of exams displayed on this worklist block cannot exceed the absolute maximum regardless of what is configured for the worklist block.
- The **Worklist.Server.HighWaterMark** Server property. This is set to 20 by default and this buffer will be automatically added to whatever number is configured for this worklist block. For example, if the Maximum Number of Exams for this block is set to 100 and the Worklist.Server.HighWaterMark is set to 20, the worklist block will actually be able to display up to 120 exams.

If the number of exams on the worklist block is actually truncated due to these settings, older exams will be omitted as necessary and a warning will be displayed to the user.

NOTE: For most worklist blocks, the recommended value is between 300 and 1000 exams.

13.3.8.Hide Criteria

To configure the hide criteria for a worklist block, click on the **Hide Criteria** link on the left-hand side of the **Worklist Block Editor**, as in the following example:

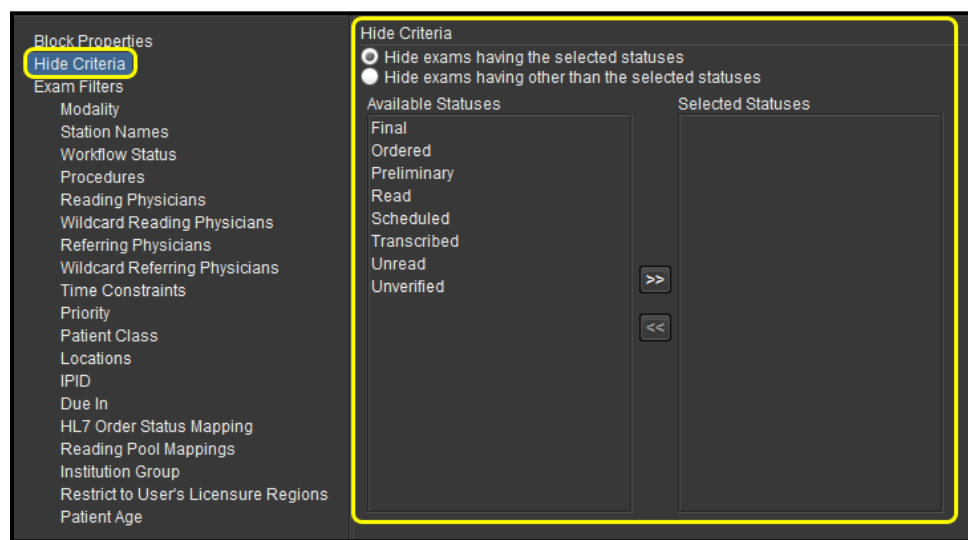


Figure 499: Hide Criteria

By default, all matching studies will be displayed when this worklist is first opened by the viewer, regardless of workflow status. If desired, however, you can configure the worklist to initially hide certain studies based on their statuses.

- Select whether you want to hide exams that have the selected statuses or hide exams that have **other than** the selected statuses by clicking on the appropriate radio button.
- Add status options to the list of selected statuses by highlighting the desired statuses in the **Available Statuses** column and clicking on the >> button in the center. If you make a mistake, highlight the incorrect item in the **Selected Statuses** column and click on the << button.

NOTE: Hide criteria only apply to worklists displayed in **Worklist** view and will not apply to worklists displayed in **Patient List** view.

13.3.9.Exam Filters

The **Exam Filters** section of the Worklist Block Editor allows you to view and edit the filters that determine whether a particular exam or order should be included in this worklist block or not. Click on the main **Exam Filters** link to view information about the filters currently defined for this worklist block, as in the following example:

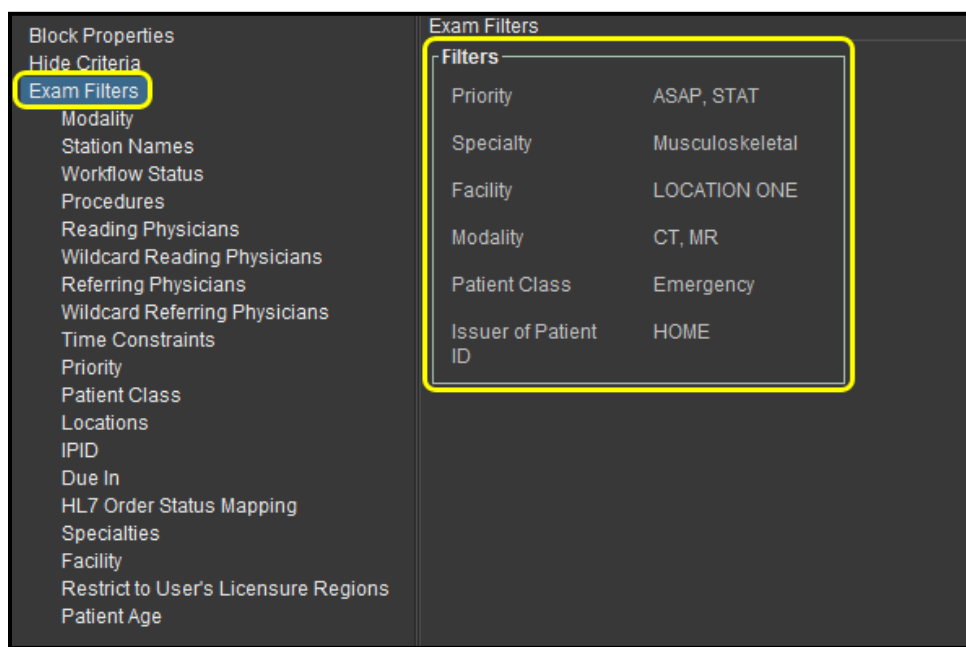


Figure 500: Exam Filters

The links below the main Exam Filters link will allow you to configure one or more individual filters as described below.

NOTE: For a complete description of how RTWL filters are mapped to database tables, HL7 fields and DICOM attributes, refer to Appendix D below.

a. Modality

By default, exams from all available modality types can be included in this worklist block. If desired, however, you can configure a Modality filter to have only exams from specified modality types included in the worklist block.

Click on the **Modality** link to display the **Modality Filter** section, as in the following example:

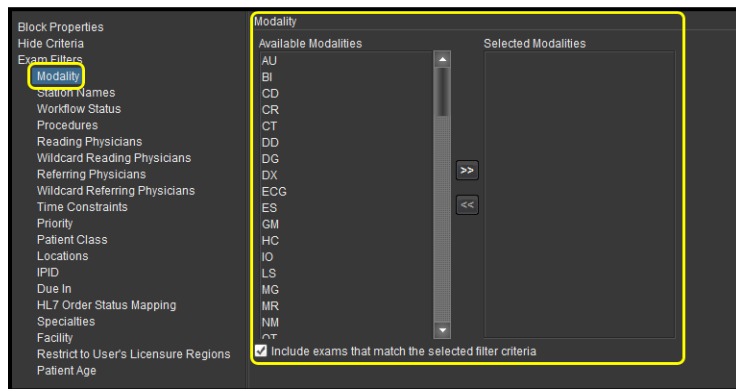


Figure 501: Modality Filter

- At the bottom of the section, select whether you want exams that match the chosen modalities to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add modalities to the list by highlighting the desired modalities in the **Available Modalities** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect modality in the **Selected Modalities** section and click on the **<<** button.

NOTE: A modality type must first be defined, as described in subsection 9.13.4 above, before it will appear in the Available Modalities section.

b. Station Names

By default, exams from all available station names (DICOM devices) can be included in this worklist block. If desired, however, you can configure a Station Name filter to have only exams from specified station names included in the worklist block.

NOTE: When Merge PACS is configured to run in **Standalone** mode, the list of available Station Names is based on the DICOM Devices that have been configured on the DICOM Devices Management Page, as described in subsection 9.13.2 above. The Station Name filter will apply to all exams with the matching Station Name in their DICOM header as well as all orders with an AE Title that has been associated with the matching Station Name in the DICOM Devices table.

NOTE: When Merge PACS is configured to run in **Integrated** mode, the list of available Station Names is taken directly from the list of Station Names that needs to be manually entered into the **AmicasServer.properties** file by Merge Healthcare Support.

Click on the **Station Names** link to display the **Station Names Filter** section, as in the following example:

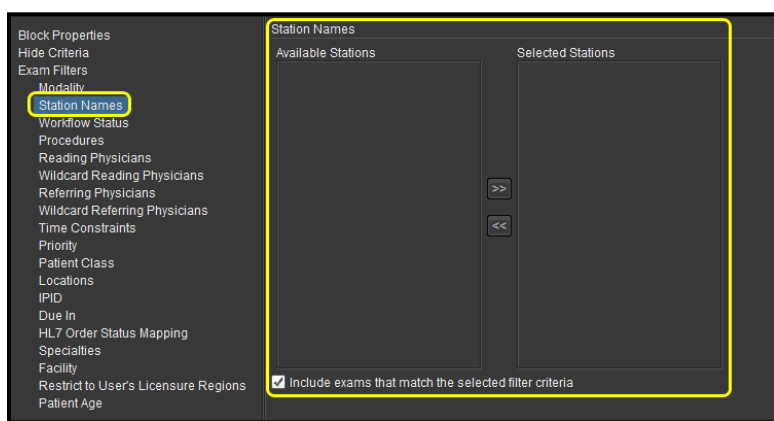


Figure 502: Station Names Filter

- At the bottom of the section, select whether you want exams that match the chosen station names to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add station names to the list by highlighting the desired modalities in the **Available Stations** section and clicking on the >> button in the center. If you make a mistake, highlight the incorrect station name in the **Selected Stations** section and click on the << button.

c. Workflow Status

By default, exams with any status can be included in this worklist block. If desired, however, you can configure a workflow status filter to have only exams with specified statuses be displayed. The workflow status filter can include one or more status expressions, and each expression is made up of one or more defined status dimensions, including any parameters.

Click on the **Workflow Status** link to display the **Workflow Status Filter** section, as in the following example:



Figure 503: Workflow Status Filter

- To **create** a new workflow status expression, do the following:
 - Click on the triangle to the left of the **New Filter** option, as in the following example:

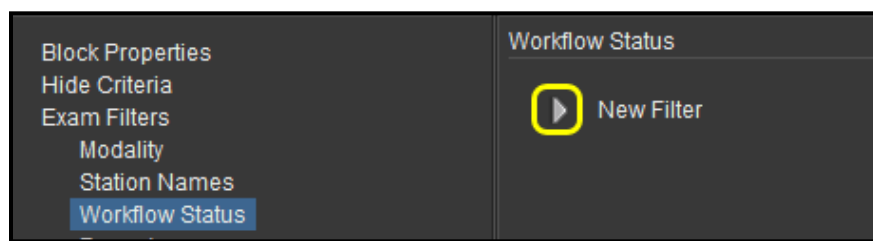


Figure 504: New Workflow Status Filter

A new numbered filter is created that displays a list of all available status options, as in the following example:

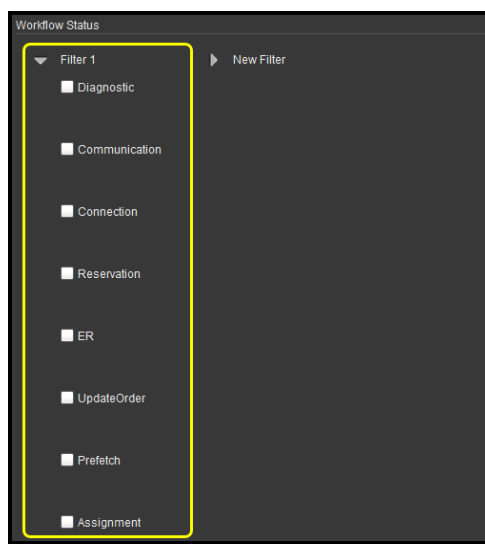


Figure 505: New Workflow Status Filter

- Select one or more statuses by clicking on the checkbox for those statuses.

The parameters associated with the selected status will appear directly below the status, as in the following example:

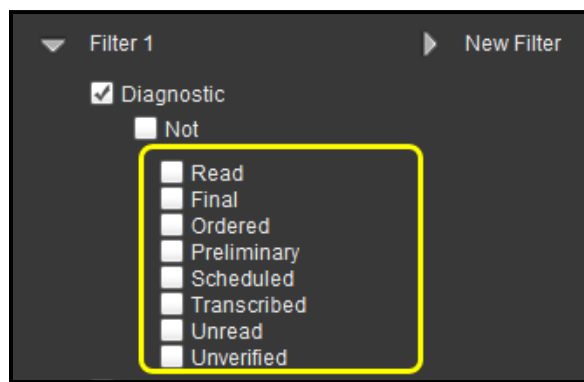


Figure 506: Available Parameters for this Status

- Select the parameters you want to be included in this filter. If you want to specify parameters that should **not** be included instead, select the **Not** option as well.

For example, if you wanted to define a filter that would include exams where the value of the Diagnostic status was neither Read, Final or Transcribed, you would select those three parameters as well as the Not option, as in the following example:

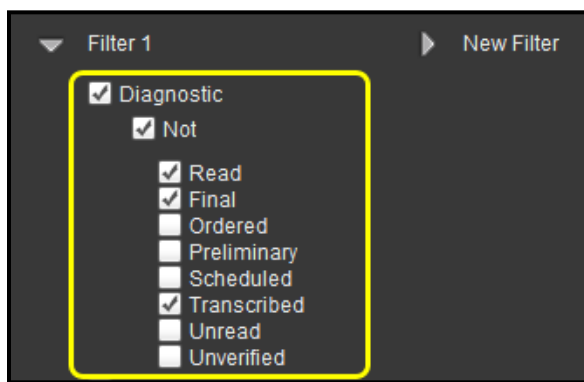


Figure 507: Only Include Exams where the Diagnostic Status is NOT Read, Final or Transcribed

Similarly, if you wanted to define a filter that would include exams where the value of the Assignment status was not assigned to the user (“ME”), you could do so as follows:

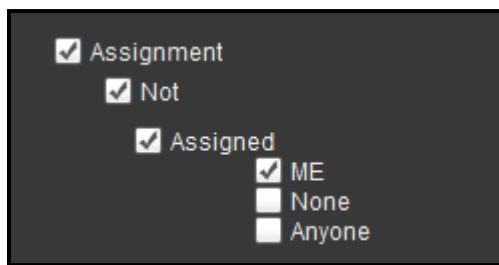


Figure 508: Only Include Exams where the Assignment Status is NOT Assigned to “ME”

- To **edit** an existing workflow status filter, use the checkboxes to change the selected values for that filter.
- To **remove** an existing filter, click on the triangle to the left of the filter’s name.

d. Procedures

By default, exams with any defined procedure can be included in this worklist. If desired, however, you can configure the worklist to only display exams that have specified procedures.

NOTE: For easier long-term maintenance of worklists, the use of Specialties, as described in subsection 10.6.3 above, is recommended instead of Procedures. Procedure codes should be mapped to Specialties, then Specialties can be used in the worklist block filters.

Click on the **Procedures** link to display the **Procedures Filter** section, as in the following example:

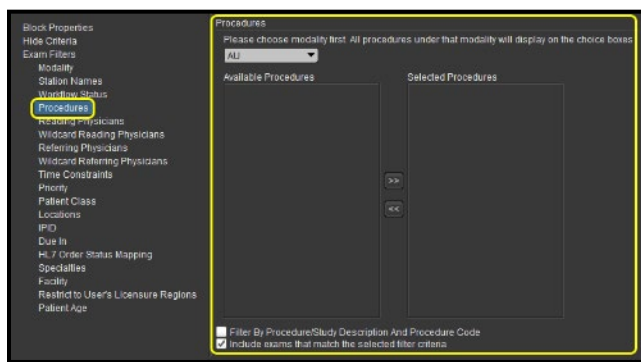


Figure 509: Procedures Filter

- Select a modality from the drop-down menu to populate the list of available procedures for that modality, as in the following example:

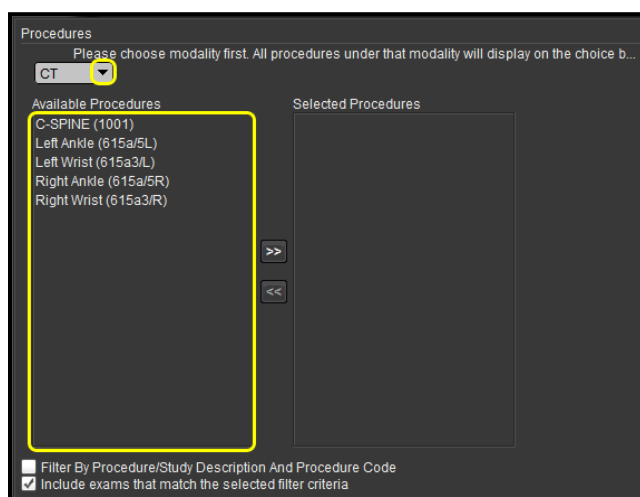


Figure 510: Selecting a Modality to Choose Procedures

- At the bottom of the section, select whether you want exams that match the chosen procedures to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add procedures for the chosen modality to the list by highlighting the desired procedures in the **Available Procedures** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect procedure in the **Selected Procedures** section and click on the **<<** button.

NOTE: A procedure must first be defined, as described in subsection 10.6.4 above, before it will appear in the Available Procedures section.

- By default, this worklist block will display exams with either a matching Procedure Description or Study Description, regardless of what the Procedure Code is. If you want to require that exams also have a matching Procedure Code in order to be displayed, select the **Filter By Procedure/Study Description and Procedure Code** checkbox at the bottom of the section.

e. Reading Physicians

By default, exams associated with any defined reading physician can be included in this worklist. If desired, however, you can configure the worklist to only display exams associated with specified reading physicians.

Click on the **Reading Physicians** link to display the **Reading Physicians Filter** section, as in the following example:

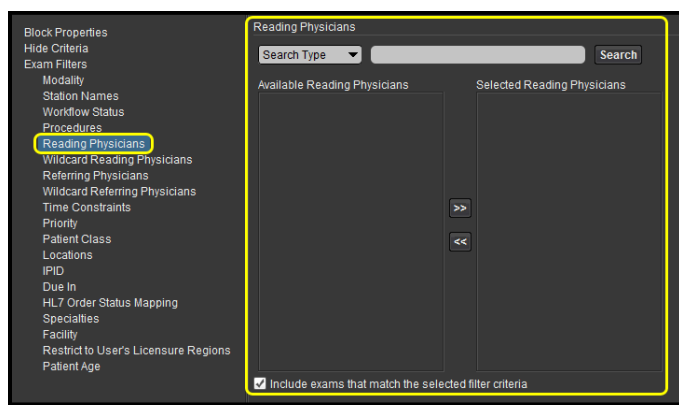


Figure 511: Reading Physicians Filter

- Use the query fields at the top of the section to narrow down the list of available reading physicians as follows:
 - From the **Search Type** drop-down menu, select “Provider Name,” “Provider Code” or “Provider Type”.
 - Enter a value to be matched in the field to the right of the drop-down menu.
 - Click on the **Search** button.

The list of physicians matching your search criteria is displayed in the **Available Reading Physicians** section, as in the following example:

Figure 512: Available Reading Physicians

- At the bottom of the section, select whether you want exams that match the chosen reading physicians to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add reading physicians to the list by highlighting the desired reading physicians in the **Available Reading Physicians** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect reading physician in the **Selected Reading Physicians** section and click on the **<<** button.

NOTE: A user must be designated as a Provider, as described in subsection 9.11.1 above, before that user will appear in the Available Reading Physicians section.

f. Wildcard Reading Physicians

The Wildcard Reading Physicians filter allows you to use wildcard characters to define a reading physician (*i.e.*, in cases where a physician's name may be spelled multiple ways within the RIS) that can be included in this worklist.

Click on the **Wildcard Reading Physicians** link to display the **Wildcard Reading Physicians Filter** section, as in the following example:

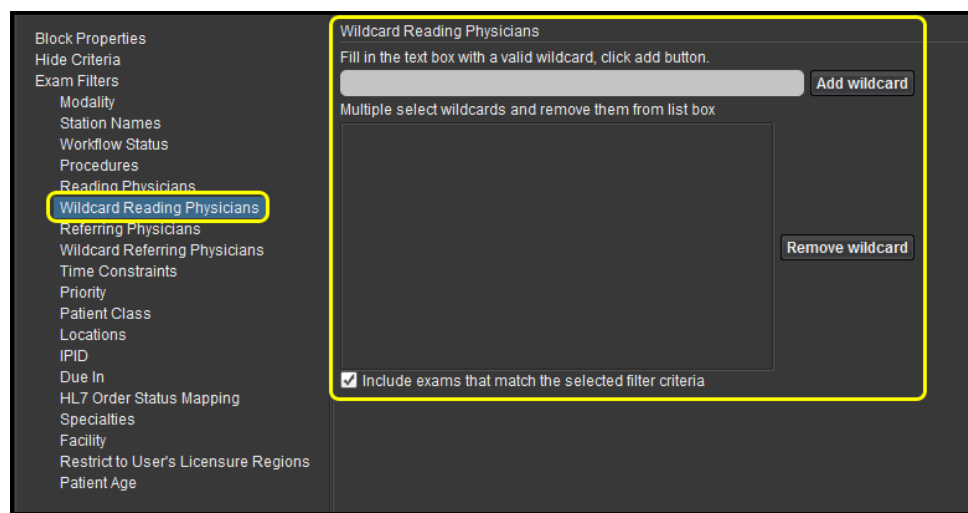


Figure 513: Wildcard Reading Physicians Filter

- At the bottom of the section, select whether you want the defined wildcard reading physicians to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- In the field provided at the top of the section, enter the name of the reading physicians you want included in this worklist, using * as a wildcard character as desired. For example, you can enter “**silver***” to include “Silver, Joshua”, “Silverman, Joan” and “Silverberg, Jessica” in the worklist. When finished, click on the **Add Wildcard** button to add this wildcard reading physician to the list.
- Once added to the list, you can remove a wildcard reading physician by clicking on that physician once to highlight it and then clicking on the **Remove wildcard** button. You can also select multiple wildcard reading physicians by holding down the **Shift** or **Ctrl** keys while clicking on them.

g. Referring Physicians

By default, exams associated with any defined referring physician can be included in this worklist. If desired, however, you can configure the worklist to only display exams associated with specified referring physicians.

Click on the **Referring Physicians** link to display the **Referring Physicians Filter** section, as in the following example:

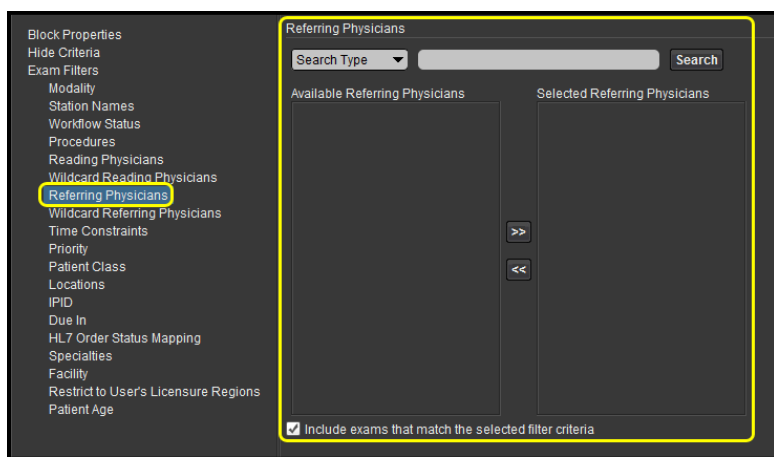
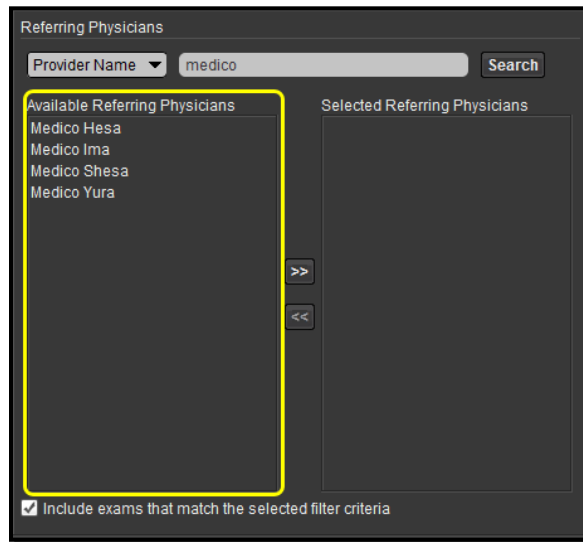


Figure 514: Referring Physicians Filter

- Use the query fields at the top of the section to narrow down the list of available referring physicians as follows:
 - From the **Search Type** drop-down menu, select “Provider Name,” “Provider Code” or “Provider Type”.
 - Enter a value to be matched in the field to the right of the drop-down menu.
 - Click on the **Search** button.

The list of physicians matching your search criteria is displayed in the **Available Referring Physicians** section, as in the following example:



The screenshot shows a window titled "Referring Physicians". At the top, there is a "Provider Name" dropdown menu and a search input field containing the text "medico". To the right of the input field is a "Search" button. Below the search bar, there are two main sections: "Available Referring Physicians" on the left and "Selected Referring Physicians" on the right. The "Available Referring Physicians" section contains a list of four names: "Medico Hesa", "Medico Ima", "Medico Shesa", and "Medico Yura". This list is enclosed in a yellow rectangular highlight. Between the two lists are two buttons: ">>" and "<<". At the bottom of the window, there is a checkbox labeled "Include exams that match the selected filter criteria", which is currently checked.

Figure 515: Available Referring Physicians

- At the bottom of the section, select whether you want exams that match the chosen referring physicians to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add referring physicians to the list by highlighting the desired reading physicians in the **Available Referring Physicians** section and clicking on the >> button in the center. If you make a mistake, highlight the incorrect referring physician in the **Selected Referring Physicians** section and click on the << button.

NOTE: A user must be designated as a Provider, as described in subsection 9.11.1 above, before that user will appear in the Available Referring Physicians section.

h. Wildcard Referring Physicians

The Wildcard Referring Physicians filter allows you to use wildcard characters to define a referring physician (i.e., in cases where a physician's name may be spelled multiple ways within the RIS) that can be included in this worklist.

Click on the **Wildcard Referring Physicians** link to display the **Wildcard Referring Physicians Filter** section, as in the following example:

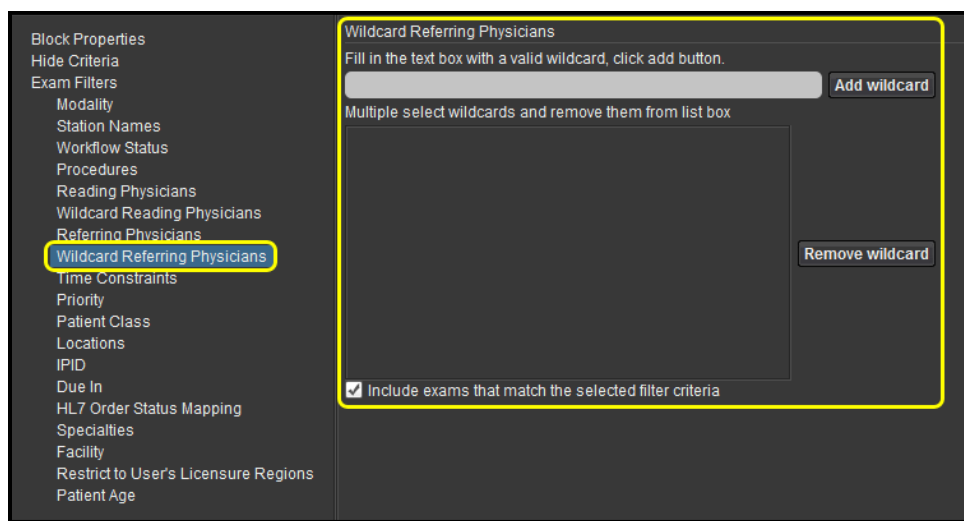


Figure 516: Wildcard Referring Physicians Filter

- At the bottom of the section, select whether you want the defined wildcard referring physicians to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- In the field provided at the top of the section, enter the name of the referring physicians you want included in this worklist, using * as a wildcard character as desired. For example, you can enter “**silver***” to include “Silver, Joshua”, “Silverman, Joan” and “Silverberg, Jessica” in the worklist. When finished, click on the **Add Wildcard** button to add this wildcard referring physician to the list.
- Once added to the list, you can remove a wildcard referring physician by clicking on that physician once to highlight it and then clicking on the **Remove wildcard** button. You can also select multiple wildcard referring physicians by holding down the **Shift** or **Ctrl** keys while clicking on them.

i. Time Constraints

By default, a worklist will display all exams that match the selected criteria for that worklist, regardless of when the orders associated with those exams are scheduled or when the exams themselves are actually performed. An individual worklist can be configured, however, to only display exams that were scheduled or performed in the past, the future, or both.

NOTE: If there is no order, the time constraints will be based solely on the Study Creation Date/Time of the study.

Click on the **Time Constraints** link to display the **Time Constraints Filter** section, as in the following example:

Time Constraints

Only the cases which are scheduled after the start time and before the end time will display on the RTWL. For example:
 If Start time = -1 Days, End Time = 0 Days, RTWL would show studies/orders from 1 day(s) ago to present time.
 If Start time = -2 Days, End Time = 2 Days, RTWL would show studies/orders from 2 day(s) ago to 2 day(s) in the future.
 If Start time = 0 Days, End Time =, RTWL would show studies/orders from present time.
 If Start time = -1 Days, End Time = 0 Days and 2 Hours, RTWL would show studies/orders from 1 day(s) ago to 0 day(s) 2 hour(s) in the future.

Start Time: Days Hours

End Time: Days Hours

Figure 517: Time Constraints Filter

Enter the start and end times for this time constraint, in hours and days, in the fields provided. Note that negative numbers can be used for days to indicate time periods in the past. For example:

- If Start time = -1 Days and End Time = 0 Days, RTWL would show studies/orders from 1 day(s) ago to the present time.
- If Start time = -2 Days and End Time = 2 Days, RTWL would show studies/orders from 2 day(s) ago to 2 day(s) in the future.
- If Start time = 0 Days and End Time = <NULL>, RTWL would show studies/orders from the present time.
- If Start time = -1 Days and End Time = 0 Days and 2 Hours, RTWL would show studies/orders from 1 day(s) ago to 0 day(s) 2 hour(s) in the future.

j. Priority

By default, exams for all available workflow priorities can be included in this worklist. If desired, however, you can configure the worklist to only display exams for specified priorities.

Click on the **Priority** link to display the **Priority Filter** section, as in the following example:

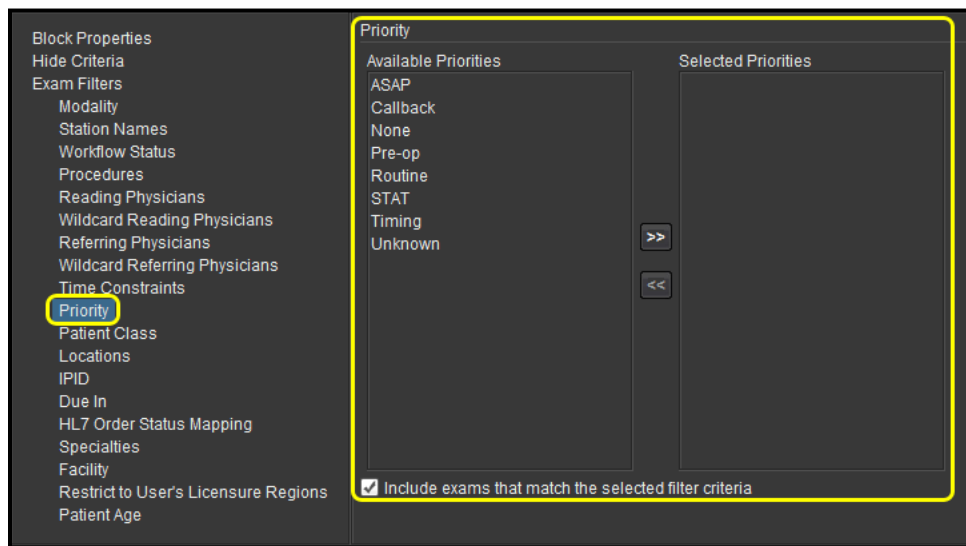


Figure 518: Priority Filter

- At the bottom of the section, select whether you want exams that match the chosen priorities to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add priorities to the list by highlighting the desired priorities in the **Available Priorities** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect priority in the **Selected Priorities** section and click on the **<<** button.

NOTE: A priority must first be defined, as described in subsection 9.13.6 above, before it will appear in the Available Priority section.

k. Patient Class

By default, exams for all available patient classes can be included in this worklist. If desired, however, you can configure the worklist to only display exams for specified patient classes.

NOTE: This is an HL7 specific filter and can only be used with a HL7 feed.

Click on the **Patient Class** link to display the **Patient Class Filter** section, as in the following example:

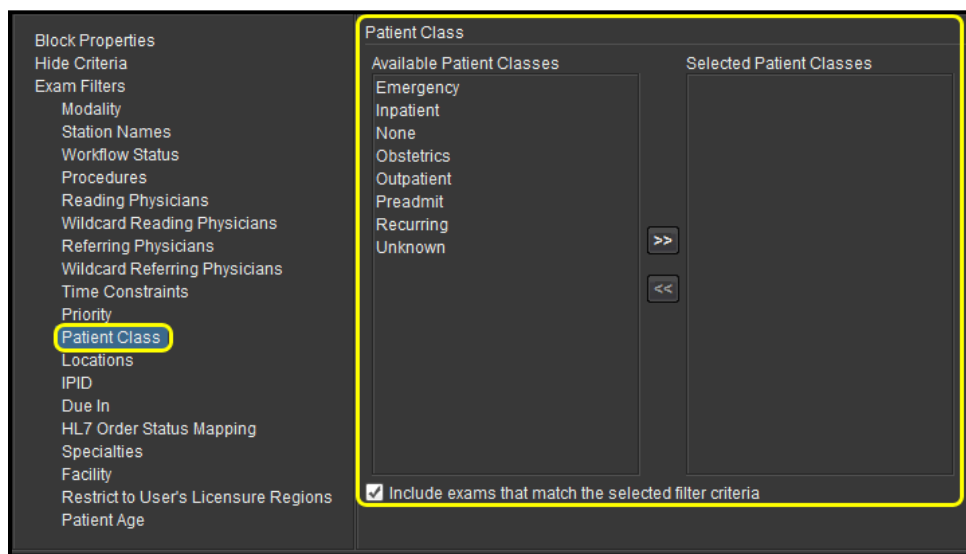


Figure 519: Patient Class Filter

- At the bottom of the section, select whether you want exams that match the chosen patient classes to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add patient classes to the list by highlighting the desired patient classes in the **Available Patient Classes** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect patient class in the **Selected Patient Classes** section and click on the **<<** button.

NOTE: A patient class must first be defined, as described in subsection 9.13.12 above, before it will appear in the Available Patient Classes section.

I. Locations

By default, exams from all defined locations can be included in this worklist. If desired, however, you can configure the worklist to only display exams from specified locations.

NOTE: This is an HL7 specific filter and can only be used with a HL7 feed.

Click on the **Locations** link to display the **Locations Filter** section, as in the following example:

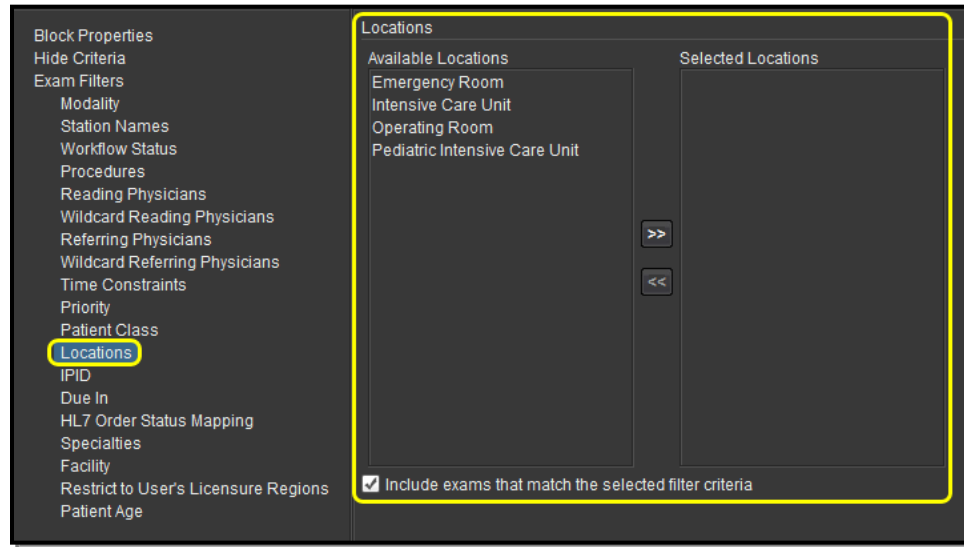


Figure 520: Locations Filter

- At the bottom of the section, select whether you want exams that match the chosen locations to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add locations to the list by highlighting the desired locations in the **Available Locations** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect location in the **Selected Locations** section and click on the **<<** button.

NOTE: A location must first be defined, as described in subsection 9.13.11 above, before it will appear in the Available Location Items section.

NOTE: The use of Patient Class (e.g. E for Emergency) is recommended rather than the location of the emergency department.

m. IPID

By default, exams for all available Issuer of Patient ID (IPID) values can be included in this worklist. If desired, however, you can configure the worklist to only display exams for specified IPID values.

Click on the **IPID** link to display the **IPID Filter** section, as in the following example:

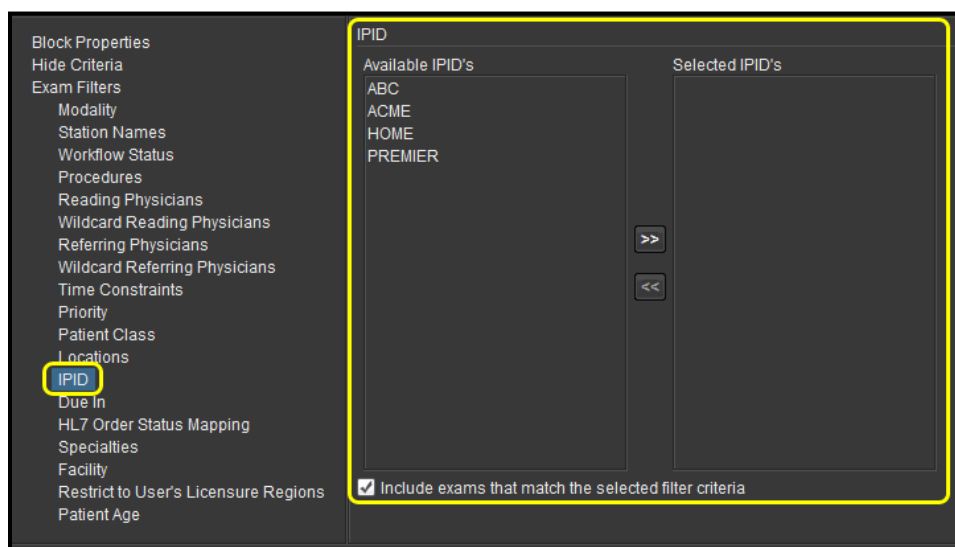


Figure 521: IPID Filter

- At the bottom of the section, select whether you want exams that match the chosen IPID values to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add IPID values to the list by highlighting the desired patient classes in the **Available IPID's** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect IPID value in the **Selected IPID's** section and click on the **<<** button.

NOTE: The list of available IPID values must first be defined, as described in subsection 9.13.13 above, before it will appear in the Available IPID section.

n. Due In

If the site's workflow includes deadline tracking due to **Service Level Agreements** (SLAs) and you have configured SLA values for various workflow priorities, as described in subsection 9.13.6 above, you can configure the worklist to only display exams that are due within a specified time.

Click on the **Due In** link to display the **Due In Filter** section, as in the following example:

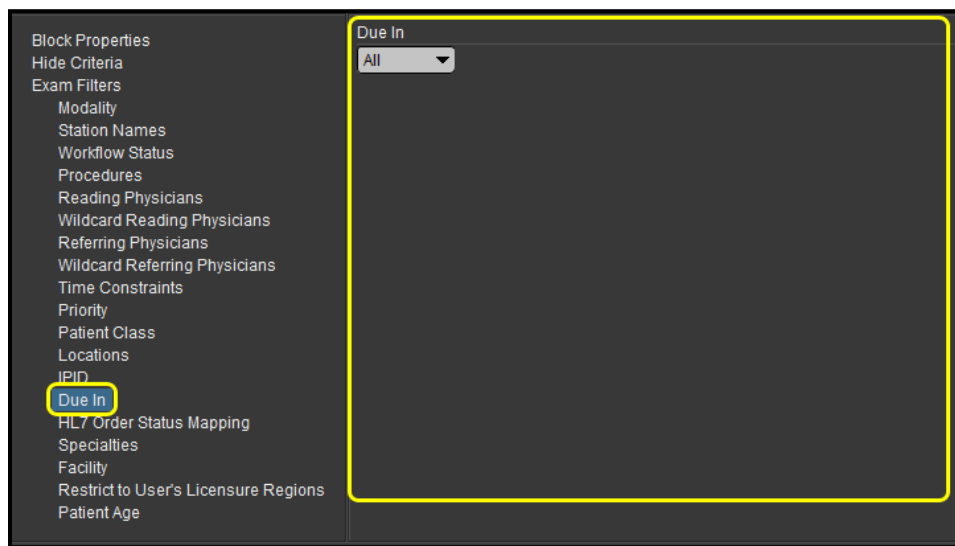


Figure 522: Due In Filter

To specify a Due In time to restrict which exams should be associated with this worklist block, select the desired value from the drop-down **Due In** menu, as in the following example:

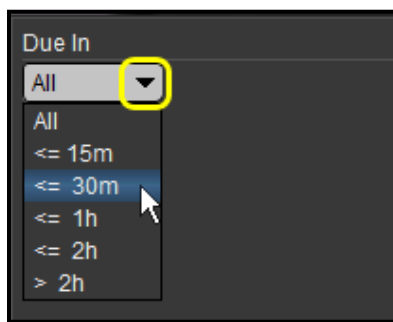


Figure 523: Selecting a Due In Filter

NOTE: The list of available options in the Due In menu is configurable, as described in subsection 1.2.80.y above.

o. HL7 Order Status Mapping

By default, exams can be included in this worklist regardless of their HL7 Order Status (as contained in ORC-5 of the incoming HL7 messages). If desired, however, you can configure the worklist to only display exams where the HL7 Order Status matches one or more specified HL7 Order Statuses that have been mapped within Merge PACS.

NOTE: This is an HL7 specific filter and can only be used with a HL7 feed.

Click on the **HL7 Order Status Mapping** link to display the **HL7 Order Status Mapping Filter** section, as in the following example:

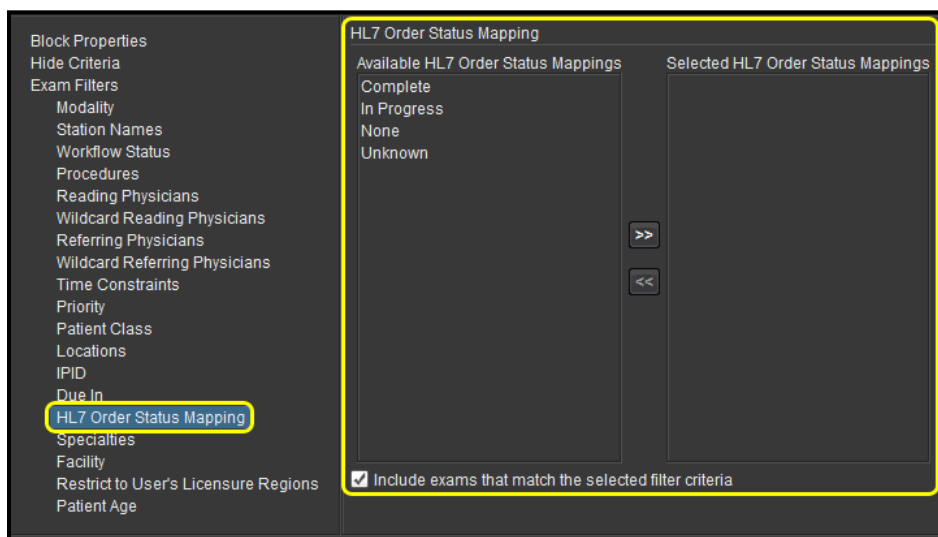


Figure 524: HL7 Order Status Mapping Filter

- At the bottom of the section, select whether you want exams that match the chosen HL7 Order Status Mapping values to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add HL7 Order Status Mapping items to the list by highlighting the desired HL7 Order Status Mapping items in the **Available HL7 Order Status Mappings** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect HL7 order status item in the **Selected HL7 Order Status Mappings** section and click on the **<<** button.

NOTE: An HL7 Order Status Mapping must first be defined, as described in subsection 9.13.9 above, before the associated HL7 order status will appear in the Available HL7 Order Status Mappings section.

p. Specialties

By default, exams can be included in this worklist regardless of their specialty value (as contained in OBR-18 of the incoming HL7 messages). If desired, however, you can configure the worklist to only display exams where the specialty value matches one or more specified specialty values that have been mapped within Merge PACS.

NOTE: This is an HL7 specific filter and can only be used with a HL7 feed.

NOTE: Specialties were referred to as “Reading Pools” in prior versions of Merge PACS.

Click on the **Specialties** link to display the **Specialties Filter** section, as in the following example:

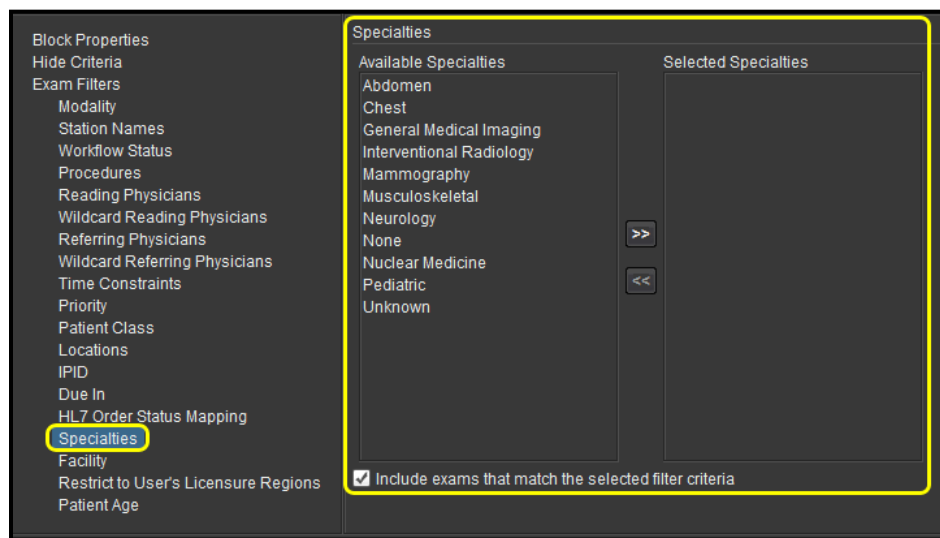


Figure 525: Specialties Filter

- At the bottom of the section, select whether you want the exams that match the chosen specialties to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add specialties to the list by highlighting the desired specialty in the **Available Specialties** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect specialty in the **Selected Specialties** section and click on the **<<** button.

NOTE: A specialty must first be mapped to a display name, as described in subsection 10.6.3 above, before it will appear in the Specialties section.

q. Facility

By default, exams from all defined facilities can be included in this worklist. If desired, however, you can configure the worklist to only display exams from specified facilities.

NOTE: The list of available facilities is automatically generated once a day by the system from the DICOM headers of exams that have been imported and are then automatically or manually mapped to facilities. If no exams have been imported for a system, and no facilities have been created, no facilities will appear on the list. Refer to subsection 10.5.2 above for information on manually mapping facilities.

Click on the **Facility** link to display the **Facility Filter** section, as in the following example:

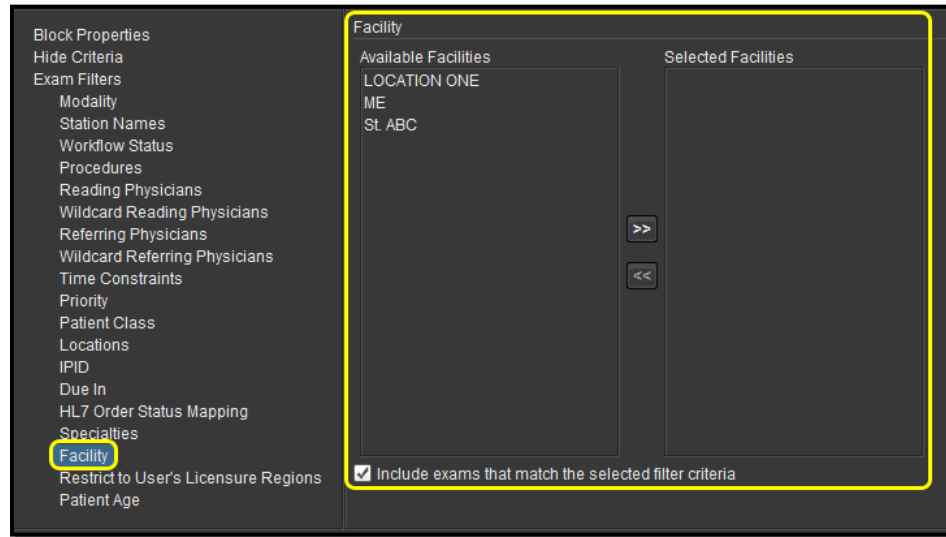


Figure 526: Facility Filter

- At the bottom of the section, select whether you want exams that match the chosen facilities to be **included** or **excluded** from the worklist block by selecting or deselecting the **Include exams that match the selected filter criteria** checkbox.
- Add facilities to the list by highlighting the desired facility in the **Available Facilities** section and clicking on the **>>** button in the center. If you make a mistake, highlight the incorrect facility in the **Selected Facilities** section and click on the **<<** button.

NOTE: Select the **ME** option to only display exams from facilities where the user is credentialed, as described in Section 12.4 above.

r. Restrict to User's Licensure Regions

By default, exams can be included in this worklist regardless of the Licensure Region (typically defined as a particular state) where the study was performed. If you have configured licensure regions for one or more facilities, as described in subsection 10.5.2 above, and users, as described in Section 12.4 above, however, you can choose to configure the worklist to only display exams that originate in Licensure Regions that have been associated with the current user.

Click on the **Restrict to User's Licensure Regions** link to display the **Restrict to User's Licensure Regions Filter** section, as in the following example:

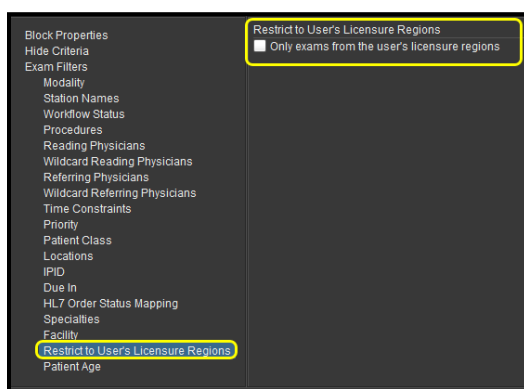


Figure 527: Restrict to User's Licensure Regions Filter

To restrict exams to the user's Licensure Regions, select the checkbox marked “**Only exams from the user's licensure regions.**”

s. Patient Age

By default, exams for patients of all ages can be included in this worklist. If desired, however, you can configure the worklist to only display exams for patients within a specified age range.

Click on the **Patient Age** link to display the **Patient Age Filter** section, as in the following example:

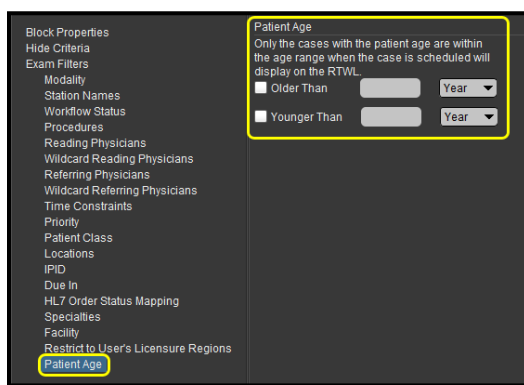


Figure 528: Patient Age Filter

- Select whether you want the range to include ages **Older Than** a specified age, **Younger Than** a specified age, or both.
- Select **Year** or **Month** from the appropriate drop-down menu(s) and enter the number of years or months in the fields provided.

13.3.10. Saving a Worklist Block

When you have finished making all the desired changes to this worklist block, click on the **Save** button at the bottom of the Worklist Block Editor, as in the following example:

The screenshot shows the 'Worklist Block Editor' window. On the left is a sidebar with a list of filter categories: Block Properties, Hide Criteria, Exam Filters, Modality, Station Names, Workflow Status, Procedures, Reading Physicians, Wildcard Reading Physicians, Referring Physicians, Wildcard Referring Physicians, Time Constraints, Priority, Patient Class, Locations, IPID, Due In, HL7 Order Status Mapping, Reading Pool Mappings, Institution Group, Restrict to User's Licensure Regions, and Patient Age. The 'Patient Age' category is currently selected. The main panel displays the configuration for 'Patient Age'. It includes a text description: 'Only the cases with the patient age are within the age range when the case is scheduled will display on the RTWL.' Below this, there are two filter options: 'Older Than' (which is checked) and 'Younger Than' (which is unchecked). The 'Older Than' filter has a text input field containing '65' and a dropdown menu set to 'Year'. The 'Younger Than' filter has an empty text input field and a dropdown menu set to 'Year'. At the bottom right of the window, there are two buttons: 'Save' and 'Cancel'. The 'Save' button is highlighted with a yellow rectangular border.

Figure 529: Worklist Block Save Button

NOTE: Clicking the **Save** button will store your information and exit the Worklist Block Editor, so you should only click it when you have finished making your changes or if you need to leave and finish making your changes later.

NOTE: You can also click the **Cancel** button to exit the Worklist Block Editor without saving any changes.

13.4. Creating and Editing Worklists

Once you have created one or more worklist blocks, as described in Section 13.3 above, you can create one or more worklists to display those blocks and associate them with specified user groups.

NOTE: You can also create new worklist blocks “on the fly” while creating or editing a worklist, if desired.

13.4.1. Accessing the Worklists Management Page

From the **Settings** dialog described in Section 13.2 above, click on the **Worklists** link on the left side of the dialog, as in the following example:

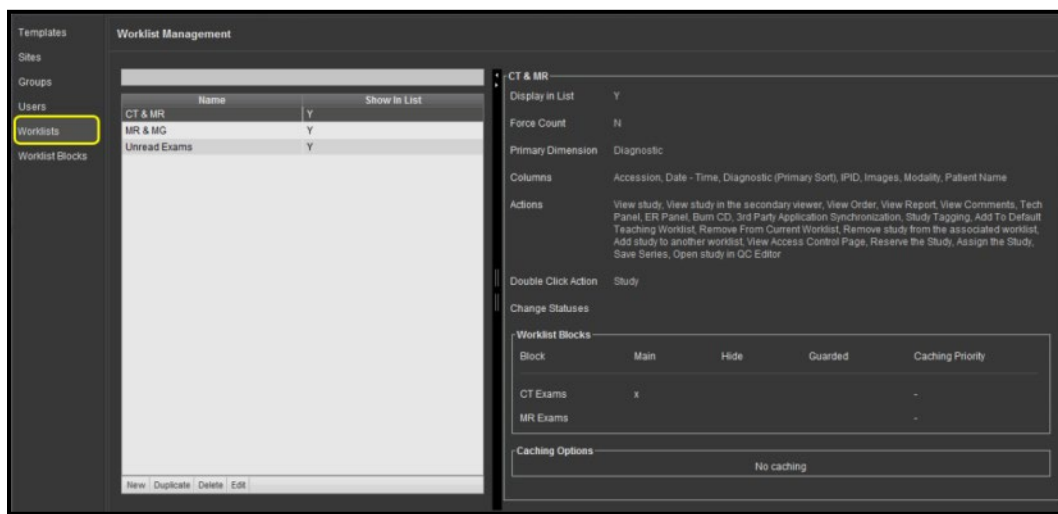


Figure 530: The Worklists Management Page

The left side of the Worklists Management Page displays a list of all existing worklists, while the right side of the Worklists Management Page displays general information about the worklist currently highlighted in the list.

NOTE: New installs of Merge PACS 8.2 have a default “Unread Exams” worklist that will display studies that have a Diagnostic Status of “Unread” and have a limited number of Action Icons and Display Columns set by default. If desired, this can be used as the starting point for new worklists.

CAUTION: It is strongly recommended that the Unread Exams worklist be used (or created, if not there by default) by sites that use highly specific worklist filters to ensure studies are not missed.

13.4.2. Creating a New Worklist

To create a new worklist from scratch, click on the **New** button at the bottom of the list of worklists, as in the following example:

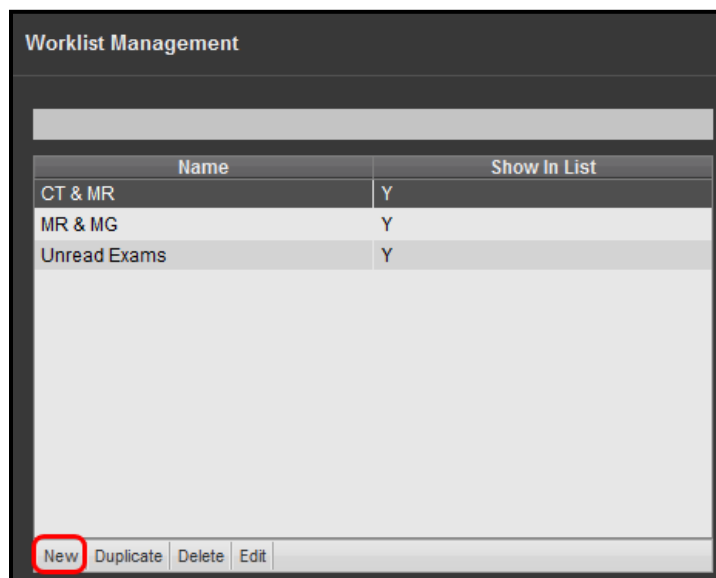


Figure 531: New Worklist Button

A blank **Worklist Editor** is displayed. Proceed to subsection 13.4.6 below.

13.4.3. Making a Copy of an Existing Worklist

To copy an existing workstation (*i.e.*, to make modifications to it and save it with a new name), click on worklist you want to copy in the list of existing worklists and click on the **Duplicate** button at the bottom of the list, as in the following example:

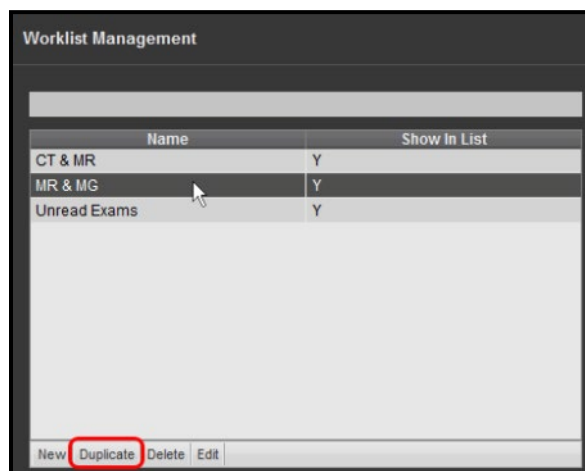


Figure 532: Duplciate Worklist Button

The **Worklist Editor** is displayed with information already entered. Proceed to subsection 13.4.6 below.

13.4.4.Editing an Existing Worklist

To edit an existing workstation, double-click on worklist you want to edit in the list of existing worklists or click on the worklist block once to highlight it and then click on the **Edit** button at the bottom of the list, as in the following example:

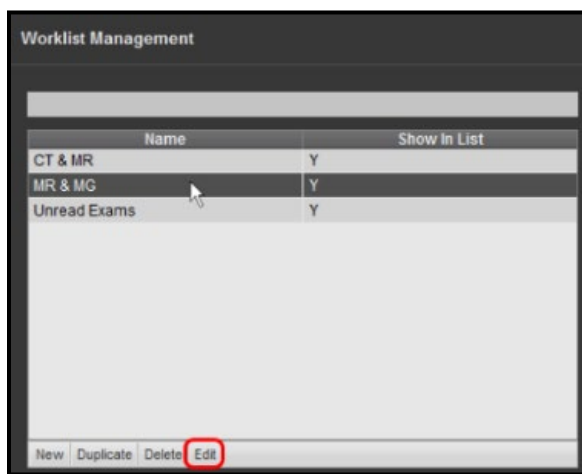


Figure 533: Edit Worklist Button

The **Worklist Editor** for the selected worklist block is displayed. Proceed to subsection 13.4.6 below.

NOTE: The Worklist Editor can also be accessed for specific worklists directly from within RTWL or RTSL, as described in Section 13.4.14 below.

13.4.5.Deleting an Existing Worklist

To delete an existing workstation, click on worklist you want to delete in the list of existing worklists and click on the **Delete** button at the bottom of the list, as in the following example:

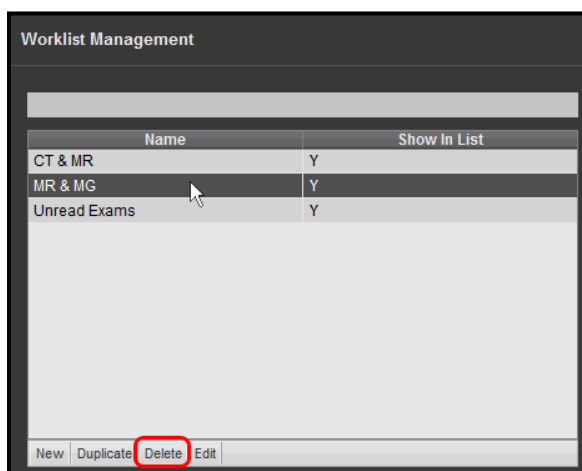


Figure 534: Delete Worklist Button

The selected worklist is removed from the list of existing worklist blocks.

13.4.6. The Worklist Editor

When you click on the **New**, **Duplicate** or **Edit** button on the main Worklists Management page, the **Worklist Editor** will launch as a separate window, as in the following example:

Figure 535: Worklist Editor

Depending on which button you clicked, the Worklist Editor will either be blank or already be populated with information. From the Worklist Editor you can enter or edit the information for this worklist as described in the following subsections.

13.4.7. Worklist Properties

To configure the general properties for this worklist, click on the **Properties** link on the left-hand side of the **Worklist Editor**, as in the following example:

Figure 536: Worklist Properties

NOTE: The **Properties** link will be selected by default when you first access the Worklist Editor.

a. Worklist Name

In the **Worklist Name** field, enter a unique name for this worklist that will be displayed as an option when users select a worklist.

NOTE: To prevent errors, the worklist name should contain no more than 64 characters.

b. Display in List

Select the **Display in List** option if you want this worklist to be displayed as an option when users select a worklist.

NOTE: This option will be selected by default, but you may want to deslect it to hide a worklist instead of deleting it entirely, *e.g.*, if you want to use it as a template for other worklists.

c. Worklist Force Count

Select the **Force Count** option if you want the study count for this worklist to be displayed within the RTSL Available Worklists panel at all times, even if the worklist is not currently being viewed by any user.

CAUTION: Forcing the study count to be displayed at all times can have serious memory consequences, especially if numerous large worklists are involved. Therefore, this option should be used only when necessary.

d. Primary Dimension

Use the **Primary Dimension** drop-down menu to specify which status dimension should be used as the main sort criteria for this worklist. Exams on this worklist will then be grouped by that dimension's status for each day.

e. Coloring of Rows in the Worklist

Select how you would like the rows and columns in this entire worklist to be colored as follows:

Option	Description						
Base Color	Select one of the following options to determine the base (<i>i.e.</i>, background) color of each row: • Grey • Determined by primary dimension status • Determined by Order Priority • Determined by Due In						
Columns Show Their Own Color	Select one of the following options: <table> <tr> <th>Option</th><th>Description</th></tr> <tr> <td>Yes</td><td> Individual columns will be colored according to the color defined for the status of those dimensions (if they are configured to allow the use of their status colors, as described in subsection 9.13.8 above), as in the following example:  </td></tr> <tr> <td>No</td><td> All columns will be colored the same as the selected Base Color, as in the following example:  </td></tr> </table>	Option	Description	Yes	Individual columns will be colored according to the color defined for the status of those dimensions (if they are configured to allow the use of their status colors, as described in subsection 9.13.8 above), as in the following example: 	No	All columns will be colored the same as the selected Base Color, as in the following example: 
Option	Description						
Yes	Individual columns will be colored according to the color defined for the status of those dimensions (if they are configured to allow the use of their status colors, as described in subsection 9.13.8 above), as in the following example: 						
No	All columns will be colored the same as the selected Base Color, as in the following example: 						

NOTE: Which coloring options are selected by default is determined by the “Worklist Color” preference set from the **Merge PACS Preferences** dialog, as described in Appendix A.2.1.6 below.

13.4.8. Display Columns

The Display Columns section allows you to set which of the available RealTime Worklist data columns will be displayed on this particular worklist, the order in which the selected columns will be displayed horizontally, and the order in which they will be sorted vertically by default.

To configure the display columns for this worklist, click on the **Columns** link on the left-hand side of the **Worklist Editor**, as in the following example:

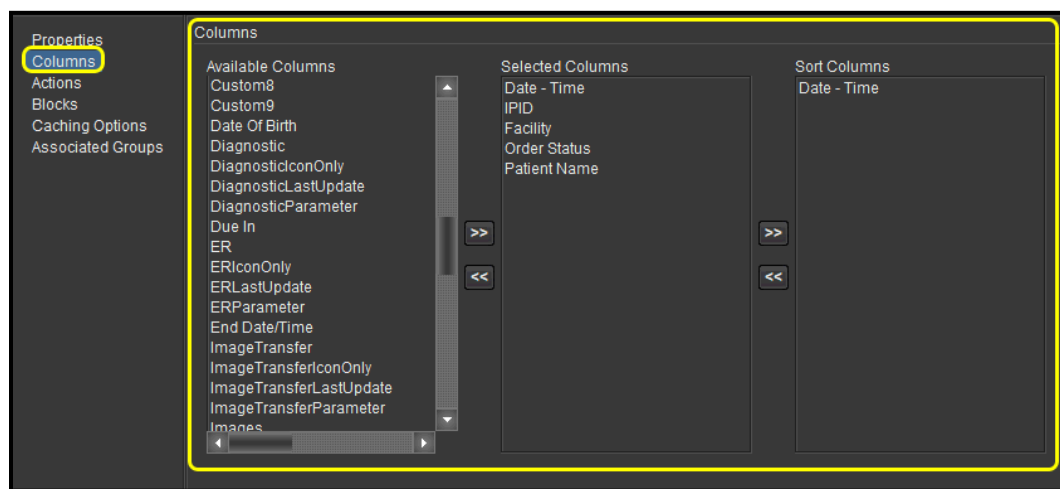


Figure 537: Columns

CAUTION: If the optional Merge RadStream component has been enabled, this worklist must have “Acuity” selected as both a display column and a sort column in order for exams to actually be prioritized by acuity. Enabling Merge RadStream will allow the system to set acuity scores, but the individual worklists must still be configured to display and sort by acuity.

In addition to the standard display columns for each status dimension, some status dimensions include an “IconOnly” column that will only display icon instead of text and a “Parameter” column that will display the currently selected parameter for the status. The IconOnly version cannot be used in conjunction with the standard version, but the Parameter version can be used in conjunction with either the standard or the IconOnly version.

The available “**LastUpdate**” columns for the various status dimensions can be included to display the last time a study was set in that dimension (e.g., selecting the **DiagnosticLastUpdate** column would display the date and time when the Diagnostic status was last updated for each study).

The available “**Parameter**” columns for the various status dimensions can be included to display the current value for a particular status (e.g., if the **Reservation Status** column indicates that a Study has a status of “Reserved,” the **Reservation Parameter** column would display the name of the user who has reserved the Study).

The following columns will display with different names within RTWL/RTWL than listed here (refer to Appendix D below for a full list of DICOM tags for each available column):

Column Name Selection	Display String in RTWL/RTSL
Patient ID	MRN
Procedure	Study Description
Date – Time	Date/Time
Institution Name	Institution
Licensure Region	License
UpdateOrder	Update Order

At the Display Columns section, you can do the following:

- To add a column to the list of columns that will be displayed in this worklist, drag and drop that column's name from the **Available Columns** section to the **Selected Columns** section.
- To change the display order for the columns in the Selected Columns section, click on the desired column names and drag them up and down until they are all in the desired order (columns at the top will be displayed to the left of those further down).
- To remove a column from the Selected Columns section so it does not show up in this worklist, drag and drop that column from the Selected Columns section back to the **Available Columns** column.
- To set the sort order for this worklist, drag and drop one or more columns from the Selected Columns section to the **Sort Columns** section and move them up and down to determine priority.

NOTE: Whatever status dimension was set as the “Primary Dimension” for this worklist, as described in subsection 13.4.7 above, will be the primary sort column for this worklist, and this cannot be changed.

- To remove a column from the Sort Columns section, drag and drop that column from Sort Columns section back to the **Selected Columns** section.

13.4.9.Display Actions

The Display Columns section allows you to set which of the available action icons will be displayed on this particular worklist.

To configure the display actions for this worklist, click on the **Actions** link on the left-hand side of the **Worklist Editor**, as in the following example:

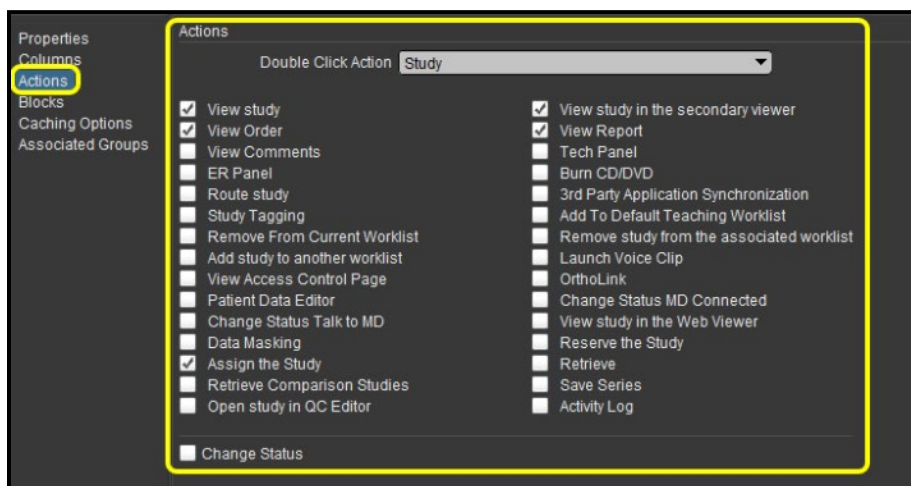


Figure 538: Actions

After selecting action icons to be displayed, make sure of the following:

- This icon has been associated with one or more workflow status options, as described in subsection 9.13.8 above.
- If the icon relates to an advanced Merge PACS Workstation privilege (such as 3D functionality), make sure the group associated with this worklist has the necessary privilege, as described in subsection 9.11.1.f above.

a. Available Actions

Select one or more of the following actions to be included in this worklist:

Icon	Name	General Description
	View Study	Load the selected study into the Merge PACS Viewer.
	View Study in Secondary Viewer	If you are already viewing a study in the Merge PACS Viewer and want to view another study without exiting the first one, click on the View Study in Separate Viewer icon to load the selected study into a separate Merge PACS Viewer.
	View Orders and/or Comments	Launch the Order Viewer as a separate pop-up window with any orders and/or comments displayed.

NOTE: If there is no order or comment currently associated with this study, the Orders and Comments icon will be disabled. If you need to add a comment, you can do so from the Technologist WorkPanel or the Comments Viewer, depending on your login privileges.

Icon	Name	General Description
	View Report	Launch the Report Viewer as a separate pop-up window.
	View Comments	Launch the Comment Viewer as a separate pop-up window with any comments displayed.
	Tech Panel	Launch the Technologist WorkPanel for the selected study in a separate pop-up window. The Technologist WorkPanel provides a number of different tools commonly used by Technologists in a single window.
	ER Panel	Launch the ER WorkPanel for the selected study in a separate pop-up window. The ER WorkPanel provides a number of different tools commonly used by Emergency Room Physicians in a single window.
	Burn CD/DVD	Add the selected study (together with any priors) to the Burn CD dialog's Study List. Note that multiple studies from multiple patients can be added to the same Burn CD Study List.
	Route Study	Send the selected study to another available Merge PACS location (<i>i.e.</i> , a different Merge PACS Server or a local DICOM device).
	Study Tagging	For users with the login privileges to access the optional Teaching Files feature, this will launch the Study Tagging window.
	Add to Default Teaching Worklist	For users with the login privileges to access the optional Teaching Files feature, add this study to whichever Teaching Worklist (including Briefcase) is currently configured as your default Teaching Worklist.
	Remove from Current Worklist	Remove a study that was manually associated with this worklist. Note that removal will be instantaneous and will not cause the Associate/Dissociate Study pop-up window to be displayed.
	Remove study from the associated worklist	Launch the Associate/Dissociate Study pop-up window to remove a study that was manually added to this RTWL worklist.
	Add study to another worklist	Launch the Associate/Dissociate Study pop-up window to add the selected study to another RTWL worklist to which you have access.
	Launch Voice Clip	Listen to the audio annotation, if any, that has been recorded for this study.
	View Access Control Page	Grant one or more users or groups access to this study/exam.
	OrthoLink	View the selected CR study within the Image Viewer and launch the orthopedic templating toolset installed on the user's workstation, such as Merge OrthoCase.
	Change Status	Change the workflow status and/or selected HL7 attributes for a particular study directly from the Worklist without opening the study within the Merge PACS Viewer. See paragraph b below for additional information.
	Patient Data Editor	Launch the Study Demographics window in a separate pop-up window that lets users view and edit patient and study demographic information for the selected study.

Icon	Name	General Description
	Communication Panel	If the optional Merge RadStream™ component has been enabled, launch the Communication WorkPanel for the selected study in a separate pop-up window. NOTE: The Communication Panel display action will only be available if the optional RadStream component has been enabled for Merge PACS. Note that, once this action has been added to the worklist, the Merge PACS Workstation will need to be restarted before it will actually appear on the worklist.
	Change Status MD Connected	Sets the connection dimension status of the selected study to MD Connected (or the equivalent status used by your site).
	Change Status Talk to MD	Sets the connection dimension status of the selected study to Talk to MD (or the equivalent status used by your site).
	View Study in the Web Viewer	If your system is configured to provide access to the iConnect® Access viewer, display the selected study within iConnect Access.
	Data Masking	For studies that are on a Teaching Worklist, creates a copy of the study with certain personally identifiable Protected Health Information (PHI) about the study (including Patient Name, MRN, Accession Number) replaced with randomized data.
	Reserve the Study	Flag this study as reserved to this user. If a worklist containing this study has been configured to include the Reservation display column, the Reservation status will be displayed in that column.
	Assign the Study	Flag this study to be assigned to another user. If that user has access to a worklist configured to display studies assigned to him, this study will appear on that worklist.
	Retrieve	If Merge PACS has an archive configured and this study is currently offline (<i>i.e.</i> , only available in the archive), submit a request to retrieve the study from the archive.
	Retrieve Comparison Studies	Submits a retrieval request for all comparison studies for the selected study so that they will be available for comparison when you load the primary study in the Viewer.
	Save Series	Save images from one or more Series for this study in a variety of available image formats for inclusion in a document or an e-mail message.
	Open Study in QC Editor	Open the study within the Quality Control Editor .
	Open Study in Patient Synopsis™	If your system is configured to provide access to Patient Synopsis™, display the selected Study within Patient Synopsis.
	Activity Log	Launch the Activity Log as a separate pop-up window with a list of events displayed.

b. Change Status Options

If you select the **Change Status** action icon, a number of additional options will be displayed to allow you to select one or more status items to which the user will be able to change status, as in the following example:

The 'Change Status' dialog box is displayed with the 'Change Status' checkbox selected. It contains several sections with checkboxes for different status options:

- Diagnostic:** Transcribed, Scheduled, CLEAR, Read, Unread, Final, Ordered, Unverified, Preliminary.
- Communication:** Conveyed, Convey Result, CLEAR.
- Connection:** Talk to MD, MD Connected, CLEAR.
- Reservation:** Reserved, CLEAR.
- ImageTransfer:** ImageTransferring, CLEAR.
- StudyLocking:** In Use, CLEAR.
- ER:** Rad Agree, ER Read, ER Acknowledge, CLEAR, Rad Disagree, ED Resolved.
- OrderMatch:** OrderMatching, CLEAR.
- UpdateOrder:** Arrived, Exam Started, Checked In, Exam Ended, CLEAR.
- Assignment:** Assigned, CLEAR.
- Prefetch:** In Progress, Needed, Complete, CLEAR, Failed, Partial Fail.
- QC:** QC Exception, QC Exception Pending, CLEAR.
- RETENTION:** REFERENCE, TELERAD, CLEAR.
- LOGY:** RADIOLOGY, CLEAR.

Figure 539: Defining Change Status Options

c. Setting the Double-Click Action

By default, double-clicking on an entry on this worklist will cause the applicable study to be loaded into the Merge PACS Viewer. This can be changed, however, by selecting a different option from the **Double Click Action** drop-down menu at the top of the page, as in the following example:

The 'Actions' panel is shown with the 'Double Click Action' dropdown menu open. The dropdown menu lists the following options: Study, Study2, Order, Report, CommentsViewer, Tech, ER, CD Burn, and Launch Voice Clip. The 'Study' option is currently selected. The 'Launch Voice Clip' checkbox is also visible at the bottom of the panel.

Figure 540: Configuring Double-Click Action

13.4.10. Worklist Blocks

The Blocks section lets define which worklist blocks will be displayed on this particular worklist, as well as the order in which they are displayed. From Blocks section you can also edit the information for any existing worklist block and create new worklist blocks on the fly.

To configure the worklist blocks for this worklist, click on the **Blocks** link on the left-hand side of the **Worklist Editor**, as in the following example:

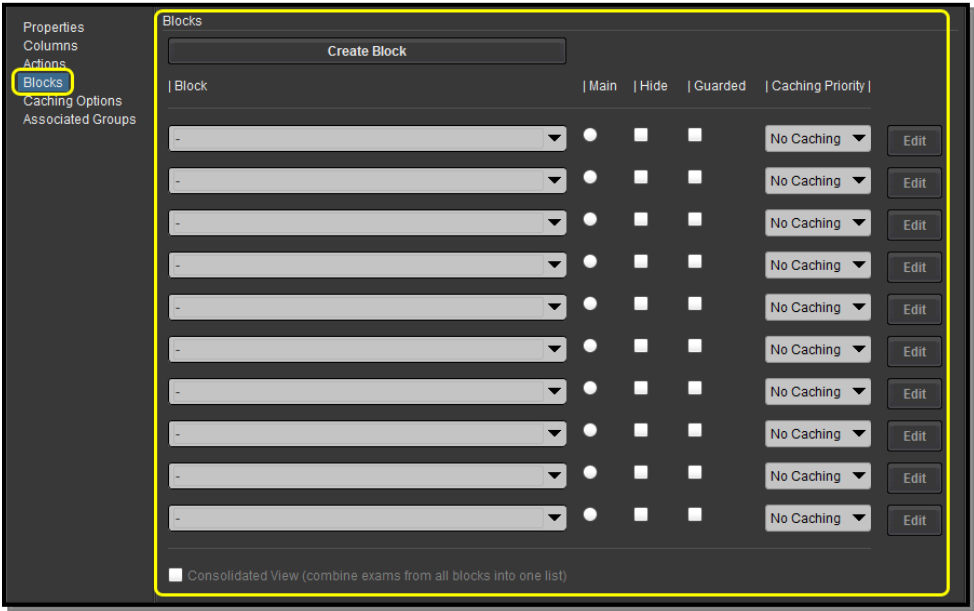


Figure 541: Blocks

a. Adding an Existing Worklist Block to this Worklist

To add an existing worklist block to this worklist:

1. Select the desired worklist block from one of the available drop-down menus in the **Block** column, as in the following example:

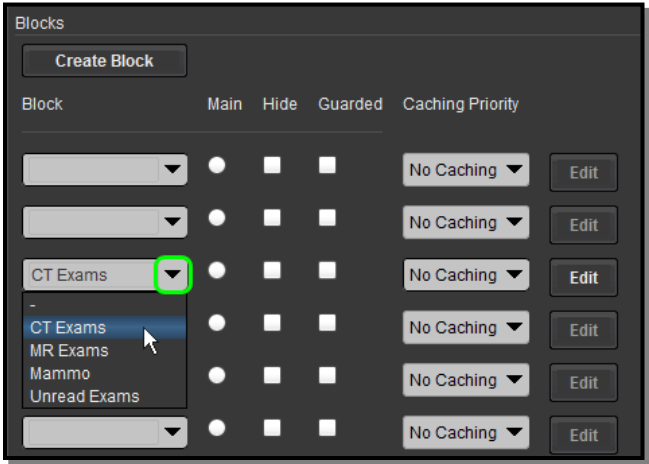


Figure 542: Selecting a Worklist Block to Add

- If you know the name of the desired block, you can also start typing its name in the text field to display a list of matching blocks, as in the following example:

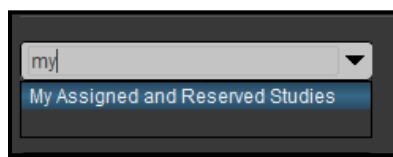


Figure 543: Filtering on Block Name

- You may add as many as ten worklist blocks to this worklist.
 - Worklist Blocks will be displayed within this worklist in the same top-down order they appear on this dialog.
 - To remove a worklist block that has previously been selected, choose the - option from the drop-down menu.
 - Leaving a dropdown menu for a worklist block blank will not cause any extra space to be displayed within this worklist. Therefore, you might want to leave the first one or two drop-down menus blank initially in case you later want to add additional worklist blocks to the top of this worklist.
2. If you want this worklist block to be the main worklist block for this worklist, click on the **Main** radio button for this worklist block.

NOTE: The main worklist block will be the one that exams are added to when users manually associate exams to this worklist.

3. If you want this worklist block to be hidden (*i.e.*, displayed in a collapsed state) by default on this worklist, select the **Hide** checkbox for this worklist block.
4. If you want this worklist block to be guarded (*i.e.*, skipped over when users click on the **Next Study** or **Previous Study** buttons within the PACS Viewer) on this worklist, select the **Guarded** checkbox for this worklist block.

NOTE: If a user is already viewing a study contained within a guarded worklist block, the user will still be able to navigate among other studies within that block using the **Next Study** and **Previous Study** buttons.

5. If desired, you can set the caching priority for this worklist block (relevant to other worklist blocks on this worklist) by selecting a number from 1-10 from the drop-down **Caching Priority** menu for this worklist block, as in the following example:

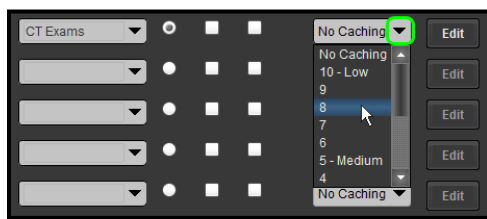


Figure 544: Setting the Caching Priority for a Worklist Block

6. Repeat steps 1-5 for any additional worklist blocks you want to add to this worklist.

b. Creating a New Worklist Block

If you want to add a worklist block that does not already exist, you can create it on the fly by clicking on the **Create Block** button at the top of the screen, as in the following example:

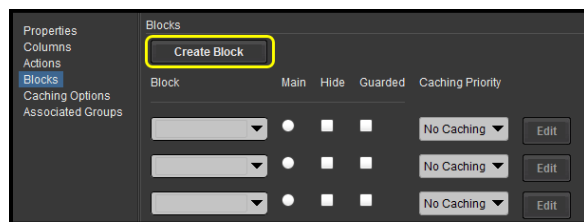


Figure 545: Create Block

A blank **Worklist Block Editor** is displayed. Refer to subsection 13.3.6 above for information on creating a new worklist block.

c. Editing an Existing Worklist Block

If you want to edit the configuration of a worklist block that you have added to this worklist, click on the **Edit** button associated with that worklist block, as in the following example:

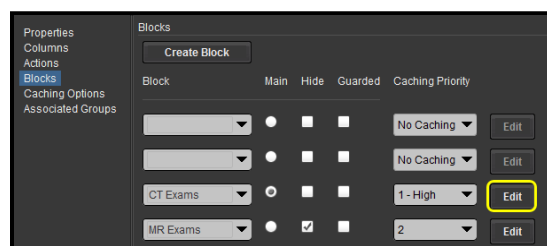


Figure 546: Create Block

The **Worklist Block Editor** for the selected worklist block is displayed. Refer to subsection 13.3.6 above for information on editing the worklist block.

d. Removing an Existing Worklist Block from a Worklist

To remove an existing worklist block from a worklist, choose the - option from the drop-down menu for that block, as in the following example:

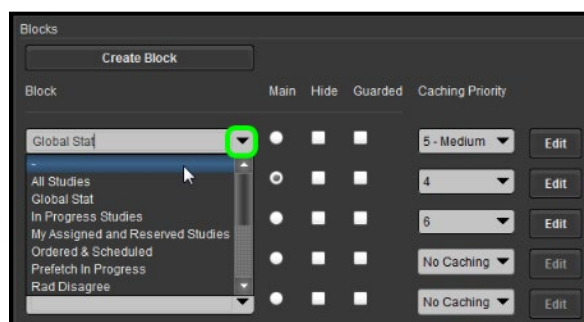


Figure 547: Removing an Existing Worklist Block

e. Enabling/Disabling Consolidated View

By default, each block on a worklist is displayed as a separate section and if users click on column headers to sort by a column the sorting will be done separately for each block. If desired, however, you can select the **Consolidated View** checkbox at the bottom of the page to have the entire worklist be displayed in “patient list” view (regardless of the view defined for each worklist block) so that exams from all blocks will be combined into one list that can all be sorted together, as in the following example:

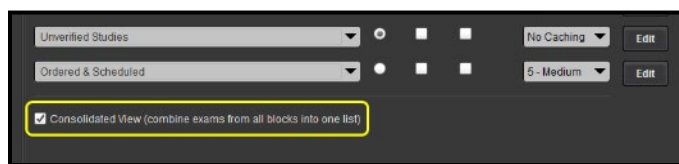


Figure 548: Enabling Consolidated View of Worklist Blocks

NOTE: The Consolidated View option sets the default state of the blocks that make up the worklist. Individual users can override Consolidated View for any given worklist, however, from the Worklist Right-click General Menu.

13.4.11. Caching Options

The Caching Options section allows you to enable and configure caching for this worklist.

To configure the caching options for this worklist, click on the **Caching Options** link on the left-hand side of the **Worklist Editor**, as in the following example:

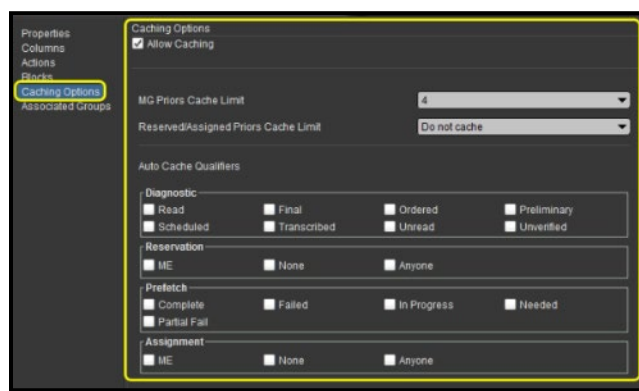


Figure 549: Caching Options

a. Allow Caching

Select the **Allow Caching** option to enable caching in general for studies on this worklist. Once this option is selected, the rest of the caching options described below will be displayed.

NOTE: The Allow Caching option is typically only used for worklists designed for reading physicians and sometimes technologists.

b. MG Priors Cache Limit

If you want this worklist to automatically cache prior mammography studies, select the maximum number of prior studies to be cached from the drop-down **MG Priors Cache Limit** menu, as in the following example:

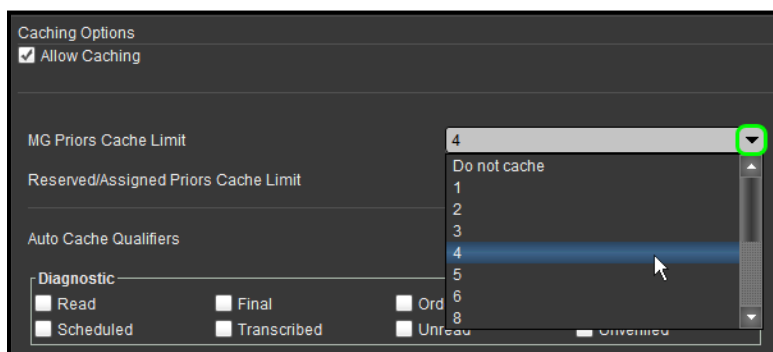


Figure 550: MG Priors Cache Limit Options

NOTE: Although there is a option to cache “All” MG Priors, this is not recommended for performance reasons.

c. Reserved/Assigned Priors Cache limit

If you want this worklist to automatically cache priors for studies that have been reserved by or assigned to users, select the maximum number of prior studies *per primary study* to be cached from the drop-down **Reserved/Assigned Priors Cache Limit** menu, as in the following example:

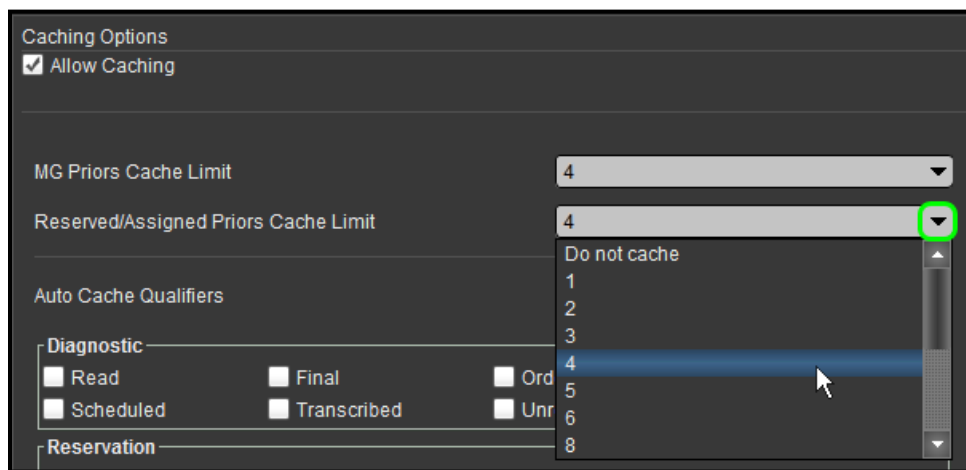


Figure 551: Reserved/Assigned Priors Cache Limit Options

When this option is enabled, prior studies will be cached when:

- the **primary** study is included in the worklist,
- the **primary** study is cacheable (*i.e.*, it matches the Auto Cache qualifiers defined in paragraph d below), and
- the **primary** study is either reserved or assigned to the current user.

NOTE: The prior studies themselves do not need to match the Auto Cache qualifiers or be reserved by or assigned to the user.

NOTE: In the case of MG priors, the maximum number of priors that will be cached is determined by the **MG Priors Auto Cache Limit** described in paragraph b above (assuming the **Auto Cache MG Priors** option has been enabled). If there are both MG and non-MG priors, the number of non-MG priors that can be cached will be determined separately (and in addition to the MG priors) according to the defined **Reserved/Assigned Priors Auto Cache Limit**.

NOTE: Although there is a option to cache “All” Reserved/Assigned Priors, this is not recommended for performance reasons.

d. Auto Cache Qualifiers

At the Auto Cache Qualifiers section you can configure the types of studies that will be automatically cached when auto-caching is invoked by the user. This can be done for each of the following four status dimensions:

- **Diagnostic**
- **Reservation**
- **Prefetch** (if available)
- **Assignment**

To set the auto cache qualifiers for a particular status dimension, click the checkboxes next to the desired status options for that dimension, as in the following example:

Auto Cache Qualifiers			
Diagnostic			
☒ Read	☒ Final	☒ Ordered	☐ Preliminary
☐ Scheduled	☐ Transcribed	☐ Unread	☐ Unverified
Reservation			
☒ ME	☐ None	☐ Anyone	
Prefetch			
☐ Complete	☐ Failed	☒ In Progress	☐ Needed
Assignment			
☐ ME	☒ None	☐ Anyone	

Figure 552: Auto Cache Qualifiers

The available status options will depend on the statuses currently defined for each status dimension, as described in subsection 9.13.7 above.

13.4.12. Associated Groups

As described in subsection 9.11.1.f above, worklists can be assigned to a group of users from the **Users & Groups** page. You can also assign worklists to groups directly from the **Worklist Editor**.

To configure the groups associated with this worklist, click on the **Associated Groups** link on the left-hand side of the **Worklist Editor**, as in the following example:

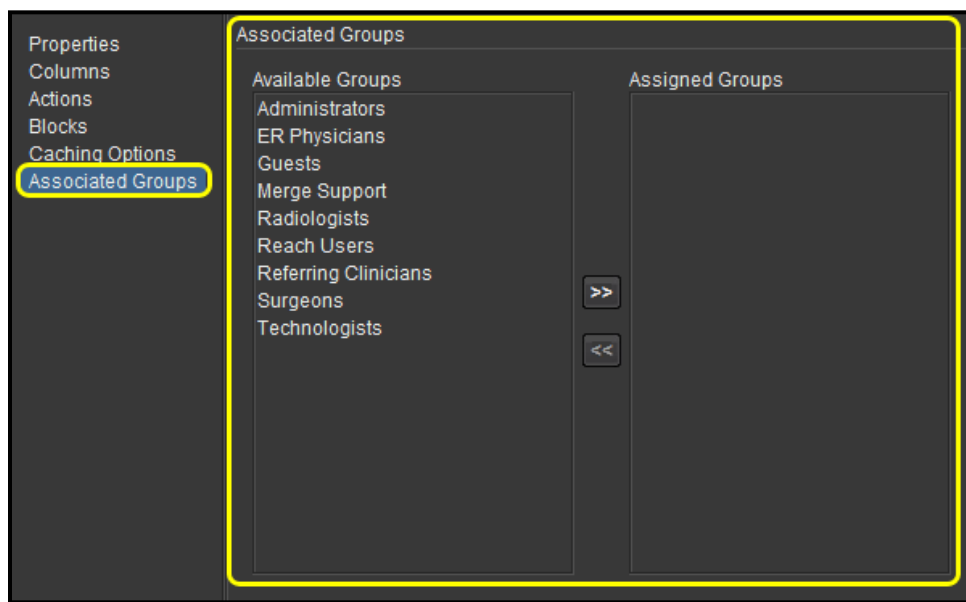


Figure 553: Associated Groups

NOTE: Before a group can be associated with a worklist, it must first be defined, as described in subsection 9.11.1.f above.

- To **associate** a group with this worklist, highlight the desired group in the **Available Groups** section and click on the >> button in the center to add it to the list of Assigned Groups.
- To **remove** (dissociate) a group from this worklist, highlight the desired group in the Assigned Groups section and click on the << button in the center to remove it from the list of Assigned Groups.

13.4.13. Saving a Worklist

Once you have finished defining all the properties for this worklist, click on the **Save** button at the bottom of the page to save all changes.

Alternatively, you can click on the Cancel Edits button to exit the Edit Worklist page without creating this worklist.

13.4.14. Modifying a Worklist from the PACS Workstation

If you have the **Allow Access to Worklist Definition** login privilege, you can also edit an existing worklist directly from RealTime Worklist or RealTime Study List itself without needing to go to the Settings page.

a. Modifying a Worklist from RTWL

At the RealTime Worklist user interface, select the worklist that needs to be modified, as in the following example:

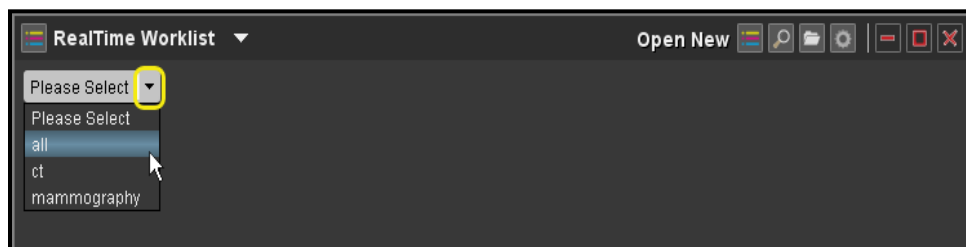


Figure 554: Selecting a Worklist to Modify

Once you have selected a worklist, the name of the worklist will be displayed near the top of the screen, together with a connection status light, as shown in the following example:

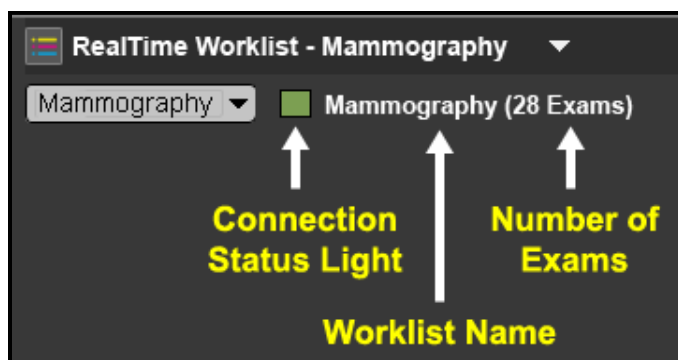


Figure 555: Worklist Connection Status and Name

To modify the worklist, click on the name of the worklist. This will bring up the Worklist Editor for that worklist.

Refer to subsection 13.4.6 above for information on editing the worklist.

b. Modifying a Worklist from RTSL

At the RealTime Study List user interface, select the worklist that needs to be modified in the Available Worklist panel, as in the following example:

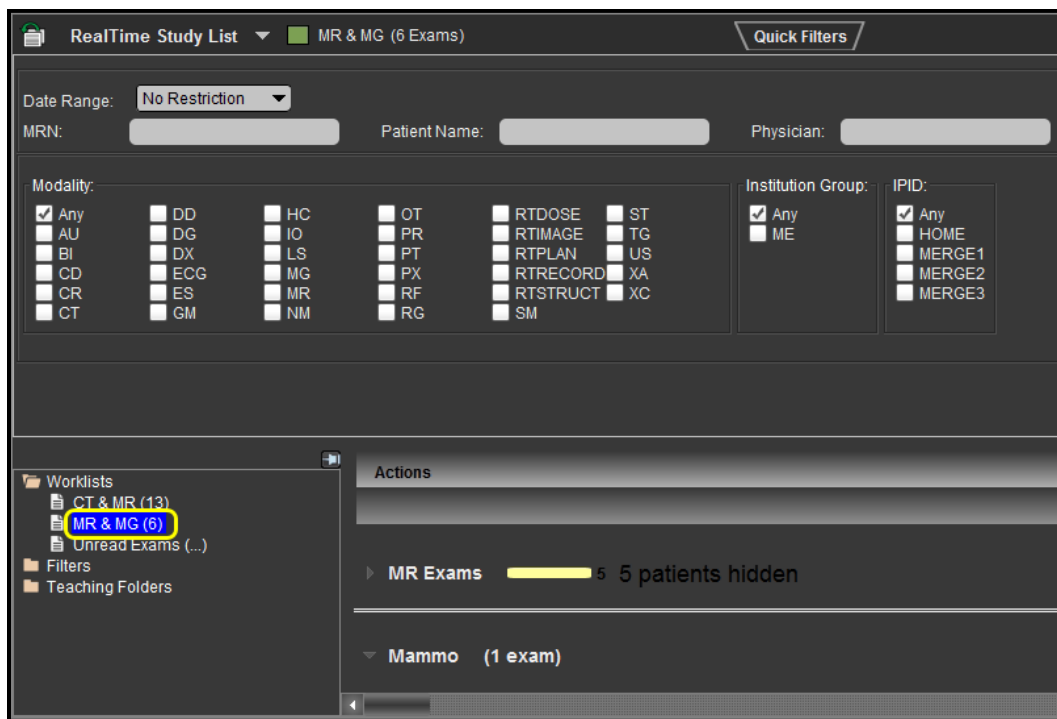


Figure 556: Selecting a Worklist to Modify

Once you have selected a worklist, the name of the worklist will be displayed near the top of the screen, together with a connection status light, as shown in the following example:

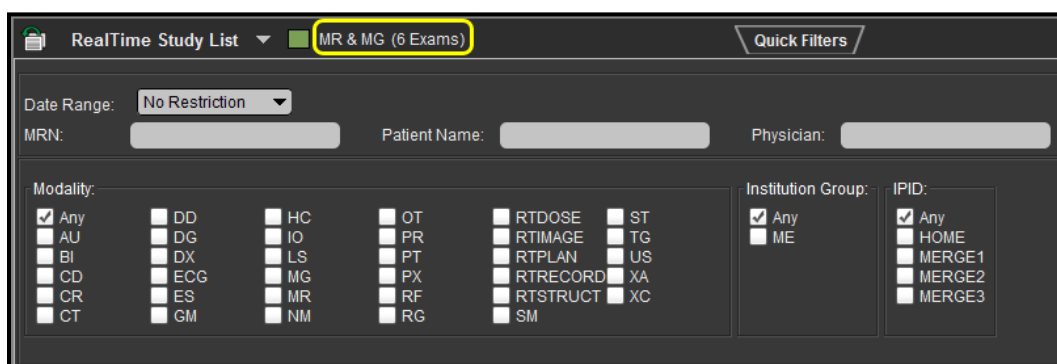


Figure 557: Worklist Connection Status and Name

To modify the worklist, click on the name of the worklist at the top of the screen. This will bring up the Worklist Editor for that worklist.

Refer to subsection 13.4.6 above for information on editing the worklist.

Chapter 14. Managing Hanging Protocols and User Rules

A **Hanging Protocol** is essentially a display template for a specified type of Study that controls the way a Study is displayed when it is first opened in the Merge PACS Viewer. When a Study is opened in the Merge PACS Viewer, the system checks to see if there is an existing Hanging Protocol associated with that type of Study and, if so, displays the Study according to settings contained within that Hanging Protocol. If there are multiple Hanging Protocols associated with a particular Study, the most recently created one will be used and the user will be given the ability to select a different protocol once the Study has been opened. Hanging protocols are associated with usernames, and which Hanging Protocols are available to a particular user will depend on that user's user account.

In addition, Merge PACS allows you to create and manage one or more user rules that affect the operation of the Merge PACS Workstation. The primary use case for these rules is to configure how Hanging Protocols are displayed within the Viewer, and you can create as many Hanging Protocol rules as desired. Default user rules are also available to manage how reports and VoiceClips are handled within the Workstation, as well as how scout images are determined, and these can be edited as necessary to meet the needs of a particular site. Custom user rules can also be used to configure when the ER Workpanel should be automatically launched, as describe in Appendix A.2.1 below.

NOTE: User rules are saved on a Server level, not per-user.

14.1. Managing Hanging Protocols

The creation, selection and management of Hanging Protocols and Study Presentations is done via the **Hanging Protocol Menu**, which is accessed by clicking on the **Hanging Protocol** button on the **Application Toolbar** within the Merge PACS Viewer. The appearance of the Hanging Protocol button will vary depending on whether there is Hanging Protocol or Study presentation currently loaded, as described below:



There is no Hanging Protocol or Study Presentation currently loaded for the current Study.



There is a **Hanging Protocol** currently loaded for the current Study.



There is a **Study Presentation** currently loaded for the current Study.

When you click on the Hanging Protocol button, the Hanging Protocol Menu will be displayed as in the following example:

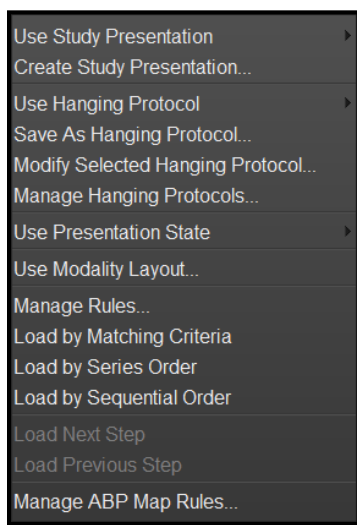


Figure 558: The Hanging Protocol Menu

NOTE: The availability of some of the options on the Hanging Protocol is determined by your login privileges and/or whether there is currently a hanging protocol applied to the Study.

14.1.1. Creating a New Hanging Protocol

Once you have manually adjusted the display of the Study currently being viewed, you can choose to save those display settings as a Hanging Protocol for other studies of the same type. This is done by then selecting **Save As Hanging Protocol** from the Hanging Protocol Menu, as in the following example:

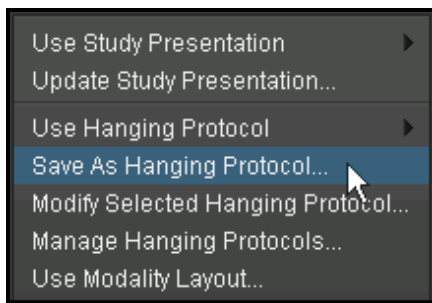


Figure 559: Creating a Hanging Protocol

Clicking on the **Save As Hanging Protocol** option will cause the **Create Hanging Protocol** window to be displayed, as in the following example:

Figure 560: The Create Hanging Protocol Window

The Create Hanging Protocol window is divided up into several sections, described below, each of which can be used to customize the hanging protocol.

a. General Information

At the top of the Create Hanging Protocol window is the General Information section, as in the following example:

Figure 561: The General Information Section

The following information can be entered:

Option	Description						
Edit for user	If you have the “Allow user to manage all Hanging Protocols” login privilege, this option will be displayed to allow you configure whether this hanging protocol should be available just to you or to other specified users as follows: <table><tr><th>Option</th><th>Description</th></tr><tr><td>Me</td><td>This hanging protocol will only be available to you. The drop-down menu to the right will be disabled, but your username will automatically be displayed there.</td></tr><tr><td>Users</td><td>The drop-down menu to the right will become active and you can select an existing user to have access to this Hanging Protocol. If this option is selected and you have selected a user other than yourself, this will be created as a private Hanging Protocol and the “Publish” checkbox described below will not be visible.</td></tr></table>	Option	Description	Me	This hanging protocol will only be available to you. The drop-down menu to the right will be disabled, but your username will automatically be displayed there.	Users	The drop-down menu to the right will become active and you can select an existing user to have access to this Hanging Protocol. If this option is selected and you have selected a user other than yourself, this will be created as a private Hanging Protocol and the “Publish” checkbox described below will not be visible.
Option	Description						
Me	This hanging protocol will only be available to you. The drop-down menu to the right will be disabled, but your username will automatically be displayed there.						
Users	The drop-down menu to the right will become active and you can select an existing user to have access to this Hanging Protocol. If this option is selected and you have selected a user other than yourself, this will be created as a private Hanging Protocol and the “Publish” checkbox described below will not be visible.						
Hanging Protocol Name	Enter a name for this Hanging Protocol so it can be easily identified.						
Hanging Protocol Description	If desired, enter additional descriptive information to help distinguish this protocol from other protocols for the same Study type.						
Publish	If you have the appropriate login privileges, you can click on the Publish box to allow other users to have access to this Hanging Protocol. Note that this option will be unavailable if you have selected to edit this Hanging Protocol for a user other than yourself, as described above.						

b. Matching Criteria

Below the General Information section is the Hanging Protocol Matching Criteria section, as in the following example:



Figure 562: The Hanging Protocol Matching Criteria Section

NOTE: You can choose to hide or display this section by clicking on the small triangle to the left of the section heading.

- By default, this section will already contain information to match the current Study, but you can edit this information as follows:
 - To **edit** an existing matching criterion, select the desired options from the two drop-down menus for that criterion and enter the desired text in the matching field on the right.
 - To **add** a new criterion, click on the **[+]** button to the right of the last criterion listed.
 - To **delete** an existing criterion, click on the **[-]** button to the right of that criterion.
 - To **reorder** the criteria and change the order in which they are applied, click on any criterion (any place other than on the drop-down menus, fields or buttons) and drag and drop it above or below another criterion.

- In general, the Hanging Protocol will match **all** the specified criteria. However, if you use the **same** criterion multiple times (e.g., “Study modality equals CR,” “Study modality equals DR”), the Hanging Protocol will match **any** of those criteria.
- When **Series number** is used as a matching criterion at the series level, the system will check for the matching series number. If there are no matching series, system will pick the first series as a matched record and it will apply the Hanging Protocol.
- For **mammography** images, if **Series Description** is used as a matching criterion it will match based on the Merge PACS internal image stack description that is used for splitting and is displayed in the navigation thumbnail header. For example, Breast Tomosynthesis images have a "(DBT)" suffix appended to their internal series description within Merge PACS, so if you wanted to configure the Hanging Protocol to load an R CC Breast Tomosynthesis image into a particular viewport, you could match based on the following:
 - View code equals “CC”
 - Laterality equals “R”
 - Series Description contains “(DBT)”
- If using the **equals** operator, the text must match exactly. Wildcard characters cannot be used.
- If using the **contains** operator, the following applies:
 - You can use an asterisk (*) as a wildcard to replace a single character or a string of characters to make the information more generic so as to apply to a larger number of studies. For example, “Study Description – contains – **Head CT * Contrast**” would apply to “Head CT **with** Contrast” , “Head CT **without** Contrast” and “Head CT **w/** Contrast.”
 - You do not need to put an asterisk at the beginning or ends of words. For example, “Study Description – contains – **spin**” would apply to all studies that have the word “spin” in the Study description (including “spine” and “spinal”).
 - You can use the vertical separator (|) [shift-\] as an “or” indicator. For example, “Study Description contains **spin|pelvis**” would apply to all studies that either have the word “spin” in the Study description **or** have the word “pelvis” in the description.
- You can use the **Regular Expression** operator to create flexible criteria, as in the following examples (note that all keywords should be lower case in the actual expressions):

Desired Criteria	Regular Expression
Starts with “Head”	head.*
Does not start with “Head”	(?!head.*).*
Ends with “Head”	.*head
Does not end with “Head”	(?!.*head).*
Contains “Head”	.*head.*
Does not contain “Head”	(?!.*head.*).*
Starts with “Head” and ends with “Neck”	head.*neck
Contains “Head” or “Neck”	.*(head neck).*
Contains “Head” and “Neck” in order	.*head.*neck.*
Contains “Neck” or “Head” but not “Brain”	(?!.*brain.*).*(neck head).*

- If your site has one or more custom study-level user rules created for use with Hanging Protocols, you can also select one of these rules instead of specifying detailed criteria by selecting **RULE** from the first criteria drop-down menu, as in the following example:

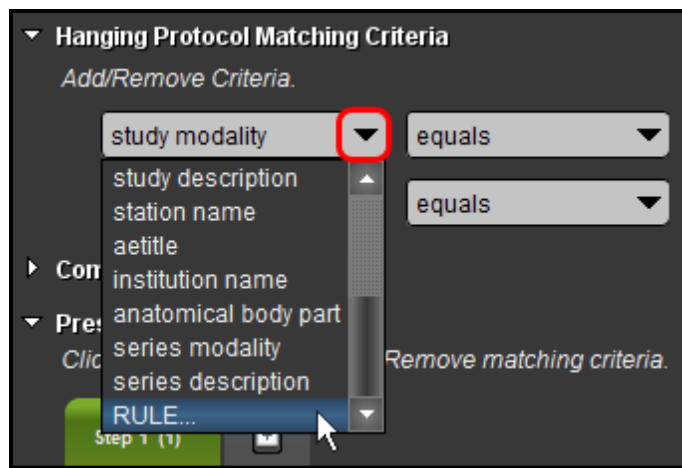


Figure 563: Using a Rule Instead of Matching Criteria

NOTE: Use of the HP Rules can dramatically simplify long term maintenance of the hanging protocols.

- Once you select the RULE option, the **Select a Rule** dialog will be displayed with a list of all available study-level Hanging Protocol rules, as in the following example:

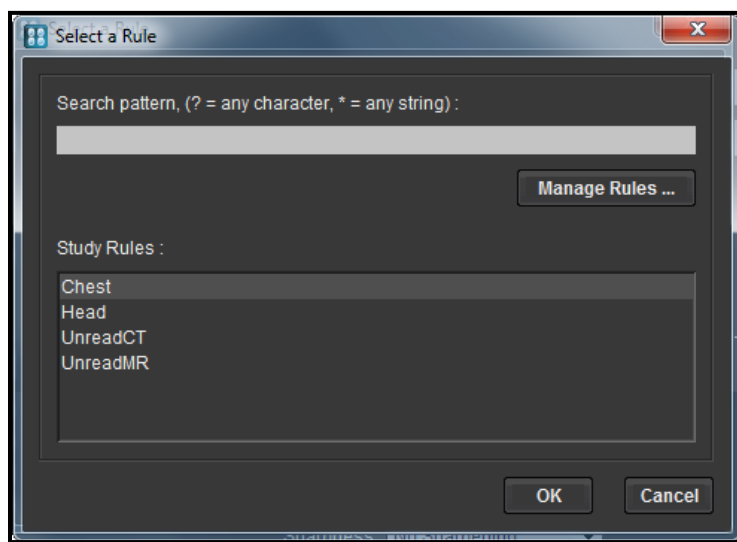


Figure 564: Selecting a Study Rule

- Click on the desired rule to select it and click the **OK** button.

NOTE: Filter the list of rules by entering text to match in the **Search pattern** field at the top

NOTE: If you have the login privilege to **add/edit/delete user rules**, you can edit the rules displayed by clicking on the **Manage Rules** button. Refer to Section 14.2 below for details on managing user rules.

Once you have selected a rule, the *Select a Rule* dialog will close and the chosen rule will be displayed as the only matching option, as in the following example:



Figure 565: Selected Rule

NOTE: Since only one rule may be selected for this Hanging Protocol, the **[+]** will re-launch the **Select a Rule** dialog instead of adding a new matching criteria.

c. Comparison Studies

Below the Hanging Protocol Matching Criteria section is the Comparison Studies section, as in the following example:



Figure 566: The Comparison Studies Section

NOTE: You can choose to hide or display this section by clicking on the small triangle to the left of the section heading.

- If one or more comparison studies are currently being displayed, this section will already contain information to match those studies. Otherwise, the entire section will be hidden except for the heading, and you can force it to be displayed by clicking on the small triangle next to the section header.
- Information in the Comparison Studies section can be edited as follows:
 - To **add** a listing for an additional **comparison Study**, click on the **[+]** button to the right of the last Study listed.
 - To **delete** a listing for an existing **comparison Study**, click on the **[-]** button to the right of that Study.
 - To **add** a new **matching criterion** for a Study, click on the **[+]** button to the right of the last criterion listed for that Study.
 - To **delete** an existing **matching criterion** for a Study, click on the **[-]** button to the right of that criterion.
 - To **reorder** the comparison **studies** listed, click on the description of a Study and drag and drop it above or below one of the other comparison studies.
 - To **reorder** the **criteria** for a Study, click on any criterion (any place other than on the drop-down menus, fields or buttons) and drag and drop it above or below another.

- In general, the Hanging Protocol will match **all** the specified criteria. However, if you use the **same** criterion multiple times (e.g., “Study modality equals CR,” “Study modality equals DR”), the Hanging Protocol will match **any** of those criteria.
- When Series number is used as a matching criterion at the series level, the system will check for the matching series number. If there are no matching series, system will pick the first series as a matched record and it will apply the Hanging Protocol.
- If using the **equals** operator, the text must match exactly.
- If using the **contains** operator, the following applies:
 - You can use an asterisk (*) as a wildcard to replace a single character or a string of characters to make the information more generic so as to apply to a larger number of studies. For example, “Study Description – contains – **Head CT * Contrast**” would apply to “Head CT **with** Contrast” , “Head CT **without** Contrast” and “Head CT **w/** Contrast.”
 - You do not need to put an asterisk at the beginning or ends of words. For example, “Study Description – contains – **spin**” would apply to all studies that have the word “spin” in the Study description (including “spine” and “spinal”).
 - You can use the vertical separator (|) [shift-\] as an “or” indicator. For example, “Study Description contains **spin|pelvis**” would apply to all studies that either have the word “spin” in the Study description **or** have the word “pelvis” in the description.
- You can use the **Regular Expression** operator to create flexible criteria, as in the following examples (note that all keywords should be lower case in the actual expressions):

Desired Criteria	Regular Expression
Starts with “Head”	head.*
Does not start with “Head”	(?!head.*).*
Ends with “Head”	.*head
Does not end with “Head”	(?!.*head).*
Contains “Head”	.*head.*
Does not contain “Head”	(?!.*head.*).*
Starts with “Head” and ends with “Neck”	head.*neck
Contains “Head” or “Neck”	.*(head neck).*
Contains “Head” and “Neck” in order	.*head.*neck.*
Contains “Neck” or “Head” but not “Brain”	(?!.*brain.*).*(neck head).*

- If your site has one or more custom study-level user rules created for use with Hanging Protocols, you can also select one of these rules instead of specifying detailed criteria by selecting **RULE** from the first criteria drop-down menu, as in the following example:

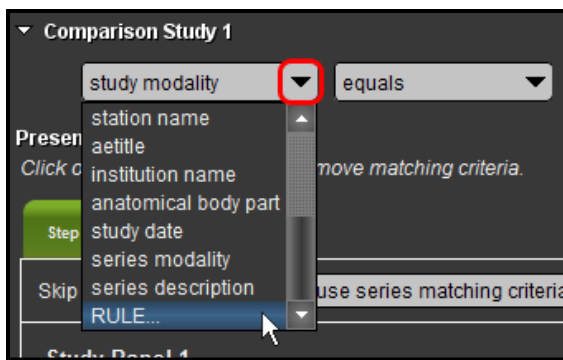


Figure 567: Using a Rule Instead of Matching Criteria

NOTE: Use of the HP Rules can dramatically simplify long term maintenance of the hanging protocols.

- Once you select the RULE option, the **Select a Rule** dialog will be displayed with a list of all available study-level Hanging Protocol rules, as in the following example:

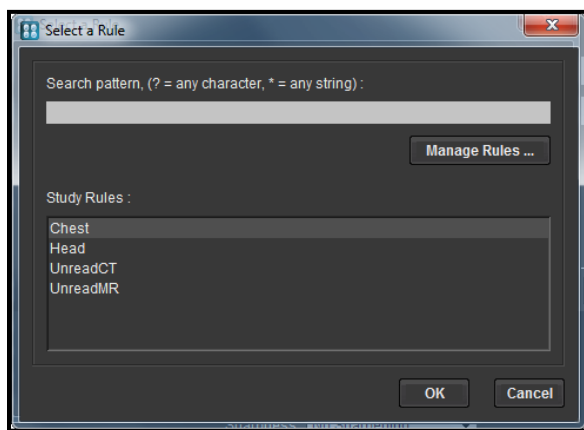


Figure 568: Selecting a Study Rule

- Click on the desired rule to select it and click the **OK** button.

NOTE: Filter the list of rules by entering text to match in the **Search pattern** field at the top

NOTE: If you have the login privilege to **add/edit/delete user rules**, you can edit the rules displayed by clicking on the **Manage Rules** button.

Once you have selected a rule, the **Select a Rule** dialog will close and the chosen rule will be displayed as the only matching option, as in the following example:



Figure 569: Selected Rule

NOTE: Since only one rule may be selected for this Hanging Protocol, the **[+]** will re-launch the **Select a Rule** dialog instead of adding a new matching criteria.

d. Presentation Layout

At the bottom of the Create Hanging Protocol window is the **Presentation Layout** section, as in the following example:

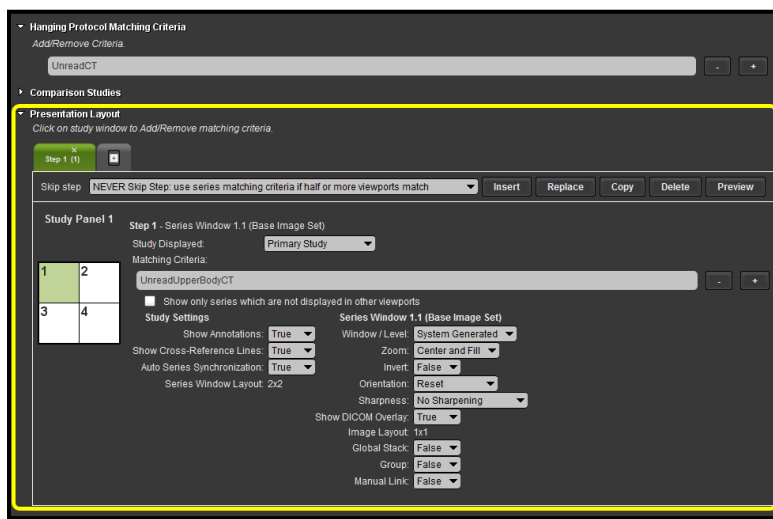


Figure 570: The Presentation Layout Section

If multiple studies are currently being displayed in separate Study panels, information for each panel will have a separate listing (you will need to use the scroll bar at the right of the window to view the information for these panels). In addition, the information for each Study panel will display the Series Layout currently being displayed within that panel.

NOTE: The Study and Series layout information will always reflect the Study and Series layout currently being displayed within the Merge PACS Viewer and cannot be edited from the Create Hanging Protocol Screen.

- The following information can be edited for each of the available Study panels:
 - To specify which **Study** is to be displayed within this Study Panel, select the desired Study from the drop-down **Study Displayed** menu, as in the following example:

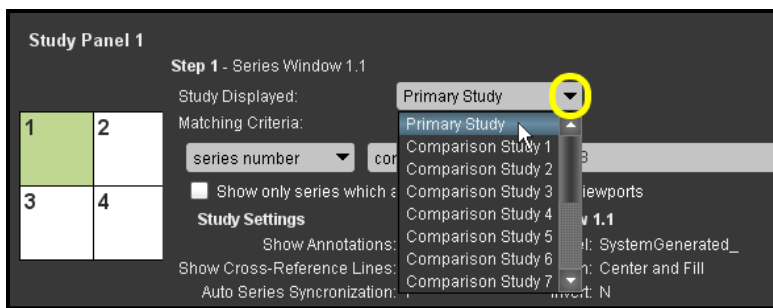


Figure 571: Specifying the Desired Study

- If the current Series Layout is displaying multiple Series Viewports, edit the information for the desired Viewport as follows:
 - If necessary, click on the Viewport you want to edit in the Series Layout diagram, as in the following example:

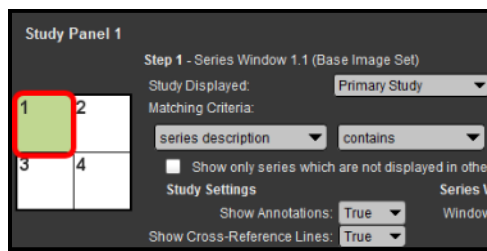


Figure 572: Specifying the Desired Study

- To **edit** an existing matching criterion for the selected Series Viewport, select the desired options from the two drop-down menus for that criterion and enter the desired text in the matching field on the right, as shown below:

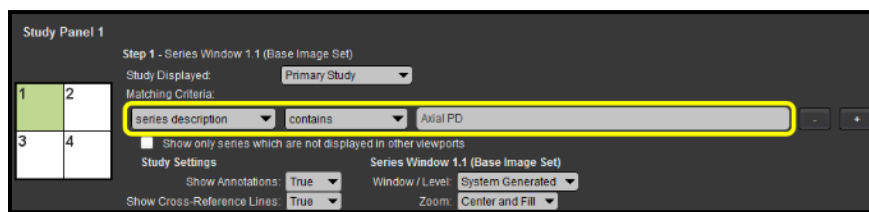


Figure 573: Editing Matching Criterion

- To **add** a new criterion, click on the **[+]** button to the right of the last criterion listed, as in the following example:

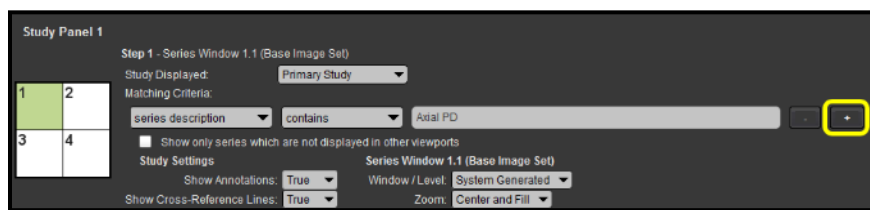


Figure 574: Adding a New Matching Criterion

- To **delete** an existing criterion, click on the **[-]** button to the right of that criterion, as in the following example:

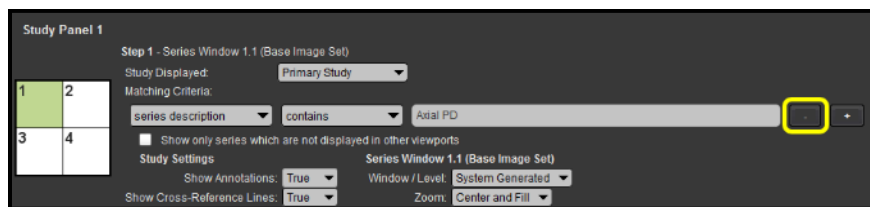


Figure 575: Deleting a Matching Criterion

- To **reorder** the criteria and change the order in which they are applied, click on any criterion (any place other than on the drop-down menus, fields or buttons) and drag and drop it above or below another criterion.
- To prevent Series that are already displayed in other viewports from being displayed in the current viewport, select the **Show only Series which are not displayed in other viewports** option, as in the following example:

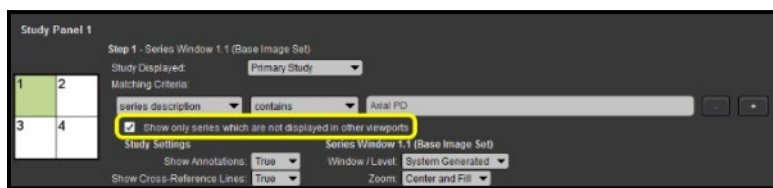


Figure 576: Show Only Series Which Are Not Displayed in Other Viewports

- If your site has one or more custom series-level user rules created for use with Hanging Protocols, you can also select one of these rules instead of specifying detailed criteria by selecting **RULE** from the first criteria drop-down menu, as in the following example:

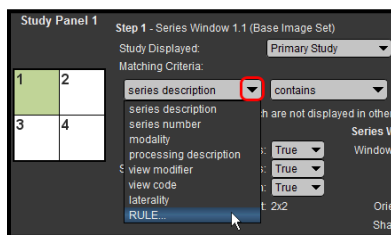


Figure 577: Using a Rule Instead of Matching Criteria

NOTE: Use of the HP Rules can dramatically simplify long term maintenance of the hanging protocols.

- Once you select the RULE option, the **Select a Rule** dialog will be displayed with a list of all available series-level Hanging Protocol rules, as in the following example:

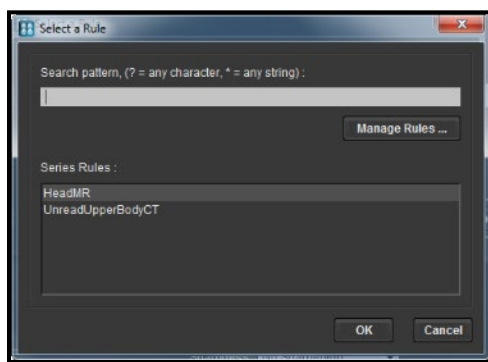


Figure 578: Selecting a Series Rule

- Click on the desired rule to select it and click the **OK** button.

NOTE: Filter the list of rules by entering text to match in the **Search pattern** field at the top

NOTE: If you have the login privilege to **add/edit/delete user rules**, you can edit the rules displayed by clicking on the **Manage Rules** button.

Once you have selected a rule, the Select a Rule dialog will close and the chosen rule will be displayed as the only matching option, as in the following example:

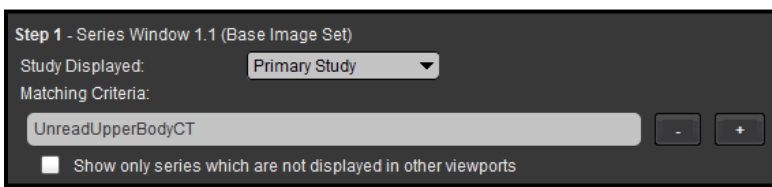


Figure 579: Selected Rule

NOTE: Since only one rule may be selected for this Presentation Step, the **[+]** will re-launch the **Select a Rule** dialog instead of adding a new matching criteria.

- If the current Series Layout includes any fused MPR Viewports, the graphic representing each fused MPR Viewport will be divided into a top and bottom section, as in the following example:

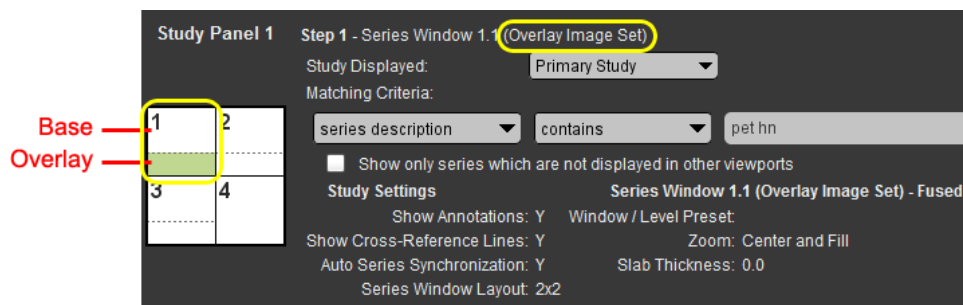


Figure 580: Fused MPR Viewport Representation

- Click on the top section of the graphic to change the matching criteria for the base Series.
 - Click on the bottom section of the graphic to change the matching criteria for the overlay Series.
- By default, the actual presentation information for this step will match the current presentation state of the study, but you can edit any of this information except for the Series Window Layout and Image Layout as shown in the following examples:

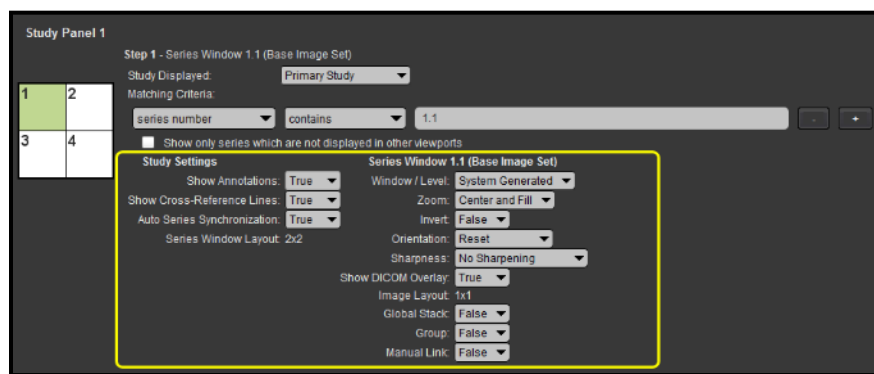


Figure 581: Editing Presentation State Information – Standard Viewport

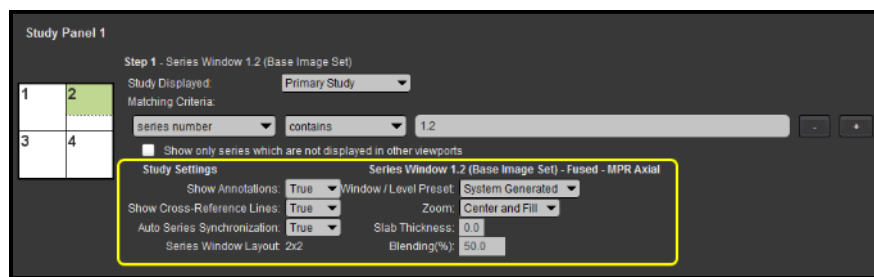


Figure 582: Editing Presentation State Information – MPR Viewport with Image Fusion

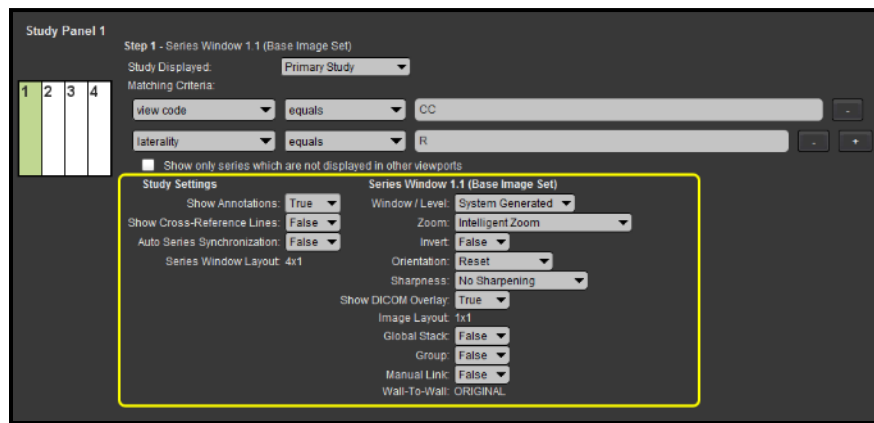


Figure 583: Editing Presentation State Information – Mammography Viewport

- You can also **add, insert, replace, edit, copy, move, delete** and **preview** steps as follows:
 - To **add** a new step, make whatever changes are desired to the display within the Merge PACS Viewer and then click on the **[+]** button at the top of the Presentation Layout Section, as in the following example:

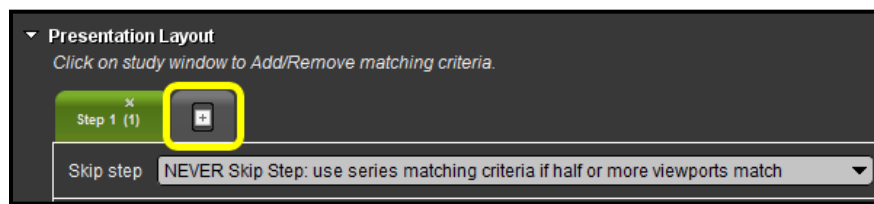


Figure 584: Adding a New Step

NOTE: You can also click on the **[+]** button without making any changes within the Viewer and then manually change the Presentation State settings for the new step within the Create Hanging Protocol window itself.

- To **Insert** a step **before** an existing step, click on the tab corresponding to the desired existing step at the top of the Presentation Layout Section and then click the **Insert** button, as in the following example:



Figure 585: Adding a New Step

You can also right-click on the existing step and select **Insert Step** from the Step Right-click Menu, as in the following example:

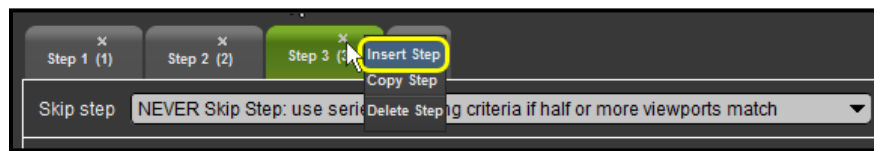


Figure 586: Inserting a Step from the Right-click Menu

- To **replace** an existing step, make whatever changes are desired to the display within the Merge PACS Viewer, click on the tab corresponding to that step at the top of the Presentation Layout Section and then click the **Replace** button, as in the following example:



Figure 587: Replacing a Step

- To **edit** a step instead of replacing it, select the desired step and then manually change the Presentation State settings for the step within the Create Hanging Protocol window itself.

- To **copy** a step (which can then be edited as necessary), click on the tab corresponding to that step at the top of the Presentation Layout Section and then click the **Copy** button, as in the following example:



Figure 588: Copying a Step

You can also right-click on the existing step and select **Copy Step** from the Step Right-click Menu, as in the following example:

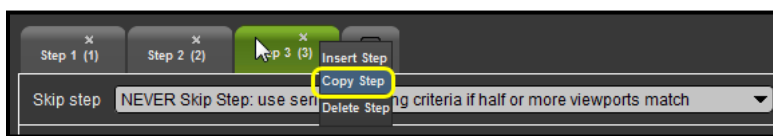


Figure 589: Copying a Step from the Right-click Menu

- To **move** a step (i.e., change the order the steps are performed), click on the header of the step you want to move and drag it to its new position, as in the following example:

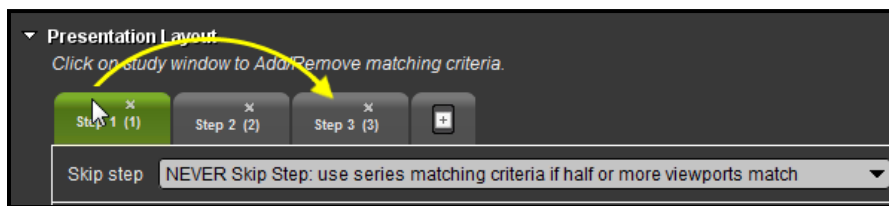


Figure 590: Moving a Step

- To **delete** a step, click on the tab corresponding to that step at the top of the Presentation Layout Section and then click the **Delete** button, as in the following example:

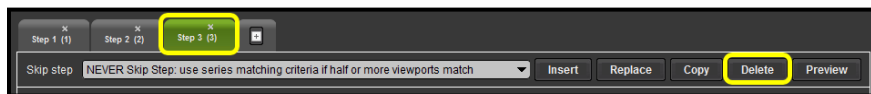


Figure 591: Deleting a Step

You can also delete a step by clicking on the small **x** in the step's tab or right-clicking on the step and selecting **Delete Step** from the Step Right-click Menu, as in the following example:

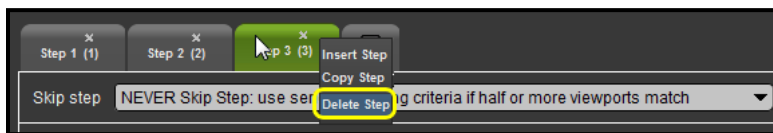


Figure 592: Deleting a Step from the Right-click Menu

- To **preview** a step, click on the tab corresponding to that step at the top of the Presentation Layout Section and then click the **Preview** button, as in the following example:

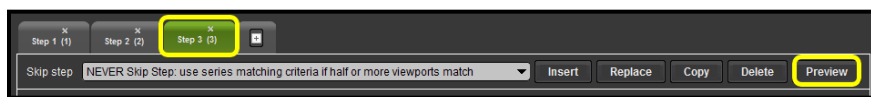


Figure 593: Previewing a Step

- For each step, you can also select an option to skip the step under specified circumstances from the drop-down **Skip step** menu, as in the following example:

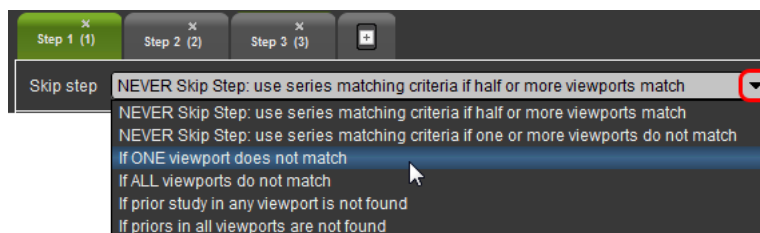


Figure 594: Skipping a Step

The following options are available:

- **NEVER Skip Step: use series matching criteria if half or more viewports match.**
- **NEVER Skip Step: use series matching criteria if one or more viewports do not match.**
- [Skip Step] **If ONE viewport does not match.**
- [Skip Step] **If ALL viewports do not match.**
- [Skip Step] **If prior Study in any viewport is not found.**
- [Skip Step] **If priors in all viewports are not found.**

e. Saving the Hanging Protocol

When you have finished making whatever edits are desired, as described in the preceding paragraphs, click on the **Save** button at the bottom of the Create Hanging Protocol window. Alternatively, you can click on the **Close** button to exit the window without saving your changes.

14.1.2.Editing an Existing Hanging Protocol

To edit an existing hanging protocol:

1. Click on the **Manage Hanging Protocols** option from the Hanging Protocol Menu.

The **Manage Hanging Protocols** window is displayed, as in the following example:

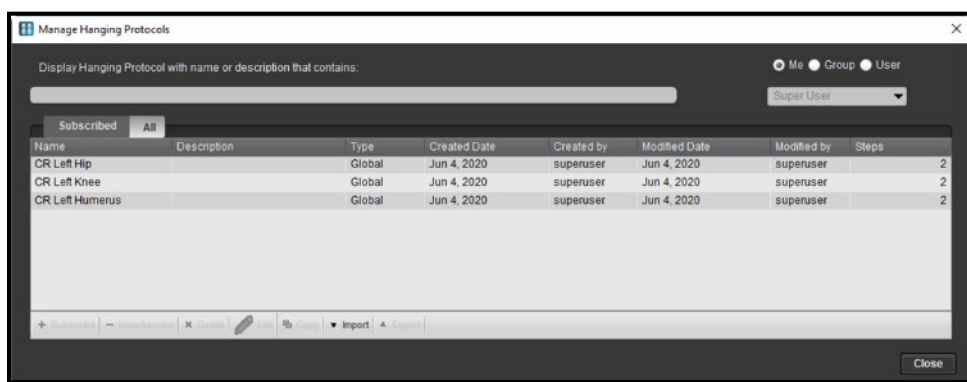


Figure 595: The Manage Hanging Protocols Window

2. Click on the **All** tab to view all the hanging protocols in the system.

NOTE: If the list of Hanging Protocols is lengthy, you can reduce the number of entries displayed by entering text in the “Display Hanging Protocol with name or description that contains” field at the top of the window. The window will then only display Hanging Protocols whose name or description contains that text.

3. Click on the desired hanging protocol once to select it and then click the **Edit** button at the bottom of the dialog.
 - When you click on the **Edit** button, a pop-up window will be displayed to indicate exactly which groups and/or users are currently subscribed to this Hanging Protocol, as in the following example:

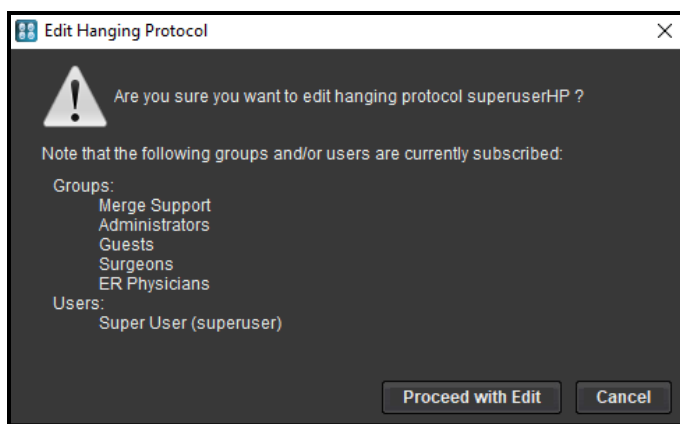


Figure 596: Currently Subscribed Users and Groups

- Click on **Proceed with Edit** to display the **Modify Hanging Protocol** window, which has the same options available as the Create Hanging Protocol window described in subsection 14.1.1.1 above.

NOTE: The **Edit** button will be enabled when the “Me” or “User” option selected, as described above, but will be disabled when the “Group” option is selected.

4. Make your edits as desired and click on the **Save** button at the bottom of the **Modify Hanging Protocol** window.

*The **Modify Hanging Protocol** window closes and the **Manage Hanging Protocols** window is displayed.*

14.1.3. Copying an Existing Hanging Protocol

To copy an existing hanging protocol:

1. Click on the **Manage Hanging Protocols** option from the Hanging Protocol Menu.

*The **Manage Hanging Protocols** window is displayed.*

2. Click on the **All** tab to view all the hanging protocols in the system.

NOTE: If the list of subscribed or all Hanging Protocols is lengthy, you can reduce the number of entries displayed by entering text in the field at the top of the window. The window will then only display Hanging Protocols whose name or description contains that text.

3. Click on the desired hanging protocol once to select it and then click the **Copy** button at the bottom of the dialog.
A copy of the selected hanging protocol is created and displayed with the name “Copy of <Name of Original Hanging Protocol>.”
4. Select the newly created hanging protocol and click on the **Edit** button at the bottom of the dialog to change the name and make any other desired changes, following the procedure described in subsection 14.1.2 above.

14.1.4.Deleting an Existing Hanging Protocol

To copy an existing hanging protocol:

1. Click on the **Manage Hanging Protocols** option from the Hanging Protocol Menu.
*The **Manage Hanging Protocols** window is displayed.*
2. Click on the **All** tab to view all the hanging protocols in the system.

NOTE: If the list of subscribed or all Hanging Protocols is lengthy, you can reduce the number of entries displayed by entering text in the field at the top of the window. The window will then only display Hanging Protocols whose name or description contains that text.

3. Click on the desired hanging protocol once to select it and then click the **Delete** button at the bottom of the dialog.
You will be prompted to confirm the deletion and then the selected hanging protocol will be removed from the list.
4. Select the newly created hanging protocol and click on the **Edit** button at the bottom of the dialog to change the name and make any other desired changes, following the procedure described in subsection 14.1.2 above.

14.1.5.Exporting Hanging Protocols

If desired, you can export some or all of the hanging protocols saved on the Server as **.xml** files that can later be imported onto another Merge PACS Server.

To export one or more hanging protocols:

1. Click on the **Manage Hanging Protocols** option from the Hanging Protocol Menu.
*The **Manage Hanging Protocols** window is displayed.*
2. Click on the **All** tab to view all the hanging protocols in the system.

NOTE: If the list of subscribed or all Hanging Protocols is lengthy, you can reduce the number of entries displayed by entering text in the field at the top of the window. The window will then only display Hanging Protocols whose name or description contains that text.

3. Click on the hanging protocols you want to export, using the **Shift** and **Ctrl** keys while clicking to select multiple hanging protocols, and click the **Export** button at the bottom of the dialog.

The **Directory Selection** dialog is displayed, as in the following example:

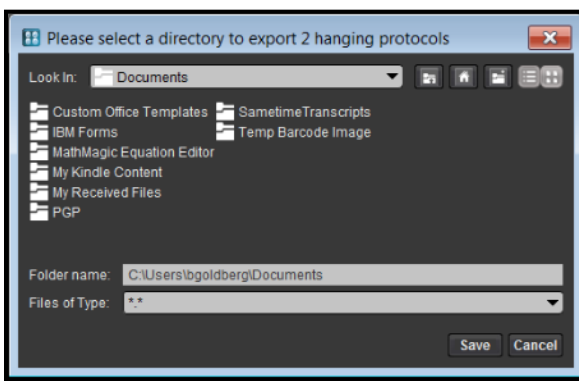


Figure 597: The Directory Selection Dialog

4. If necessary, browse to the directory where you want to temporarily store the exported hanging protocols or create a new directory by clicking on the **Create New Folder** button.
5. Click the **Save** button at the bottom of the dialog to export the selected hanging protocols to the designated directory.

14.1.6.Importing Hanging Protocols

If desired, you can import any hanging protocol .xml files that were exported from another Merge PACS Server onto this Server or are otherwise available for your use, as long as they are stored on a location you can access (the local workstation, a network drive, a USB drive, etc.).

NOTE: A number of hanging protocols are provided by default and are available for importation in the following directory on the installaiton drive of the Merge PACS Server:
<DriveLetter>:\amicas2\hangingprotocols.

To import a hanging protocol .xml file:

1. Click on the **Manage Hanging Protocols** option from the Hanging Protocol Menu.
*The **Manage Hanging Protocols** window is displayed.*
2. Click on the **Import** button at the bottom of the window.

*The **Directory Selection** dialog is displayed, as in the following example:*

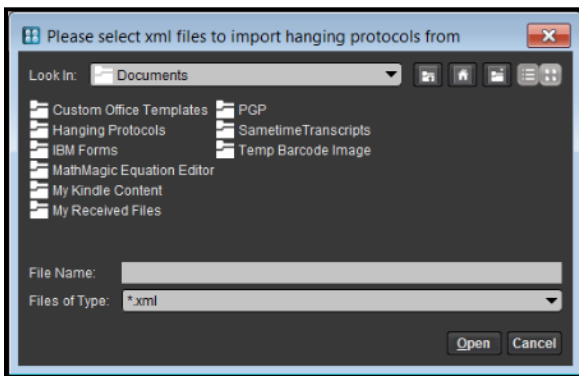


Figure 598: The Directory Selection Dialog

3. If necessary, browse to the directory where the hanging protocol .xml files are located.
4. Click on the hanging protocols you want to import, using the **Shift** and **Ctrl** keys while clicking to select multiple hanging protocols, and click the **Open** button at the bottom of the dialog.

A progress dialog will be displayed while the selected files are importing and a confirmation dialog will be displayed once the process is complete.

NOTE: If the newly imported files are not displayed, exit out of the **Manage Hanging Protocols** window and re-open it to refresh the information.

14.1.7.Subscribing and Unsubscribing Hanging Protocols

From the Manage Hanging Protocol window, you can subscribe and unsubscribe a Hanging Protocol to specific users and groups.

To subscribe and unsubscribe Hanging Protocols:

1. Click on the **Manage Hanging Protocols** option from the Hanging Protocol Menu.

The Manage Hanging Protocols window is displayed, as in the following example:

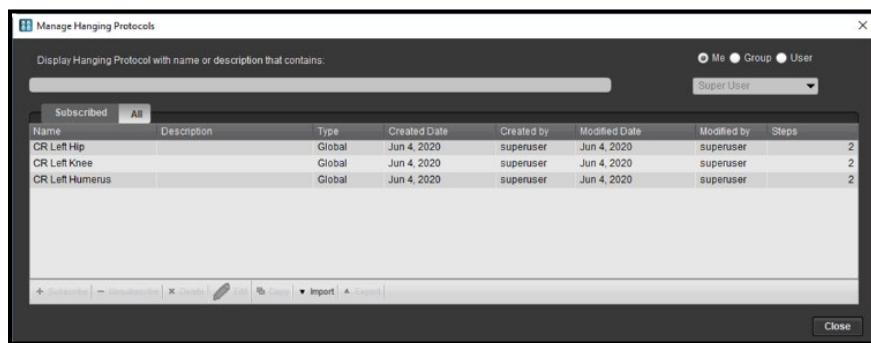
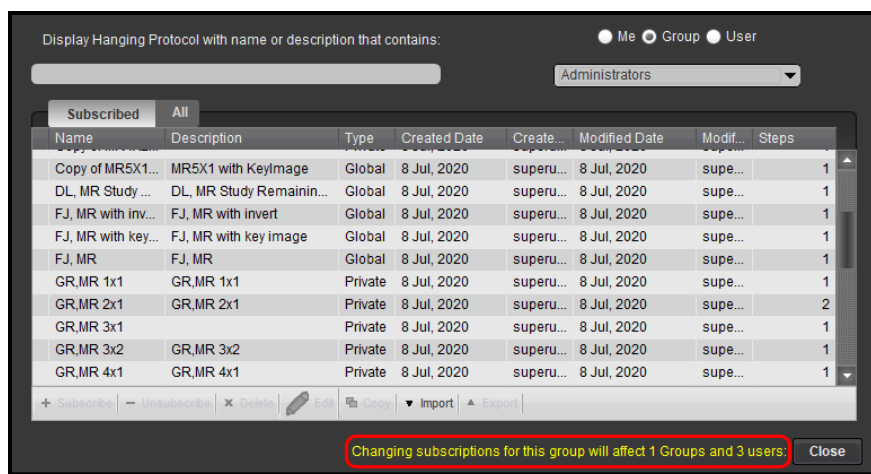


Figure 599: The Manage Hanging Protocols Window

2. Select whether you want to subscribe or unsubscribe Hanging Protocols for a “Group” or a particular “User” by selecting from the radio buttons at the top of the window and then selecting the desired group or use from the drop-down menu below those buttons.

If you selected the “Group” option, a warning message will be displayed on the **Subscribed** tab with a count of groups and users that will be affected if you change subscriptions for the selected group, as in the following example:



Currently Subscribed Users and Groups

3. Click either the **Subscribed** tab to view only Hanging Protocols that are currently subscribed to user or group or the **All** tab to view all defined Hanging Protocols.
4. Select one or more Hanging Protocols and click the **Subscribe** or **Unsubscribe** button as desired.

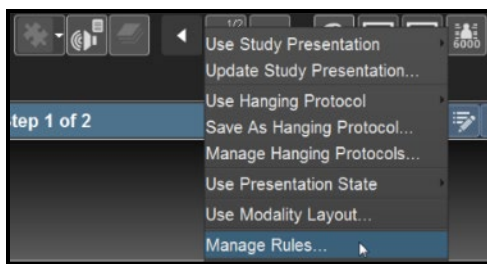
NOTE: Select multiple Hanging Protocols by holding down the Ctrl or Shift key while clicking on the desired Hanging Protocols.

14.2. Managing User Rules

Merge PACS allows you to create and manage any number of user rules for use with Hanging Protocols within the Workstation. These rules define various filters and matching conditions and can be applied when creating the Hanging Protocols themselves to determine which hanging protocols should be selected for a study and where series should be placed within the study layout. Custom user rules can also be used to configure when the ER Workpanel should be automatically launched, as described in Appendix A.2.1 below and as part of the Launch and Control feature that allows you to associate a custom macro with the “View Study” worklist action icon to control which application (including 3rd-party applications) will be launched under what circumstances.

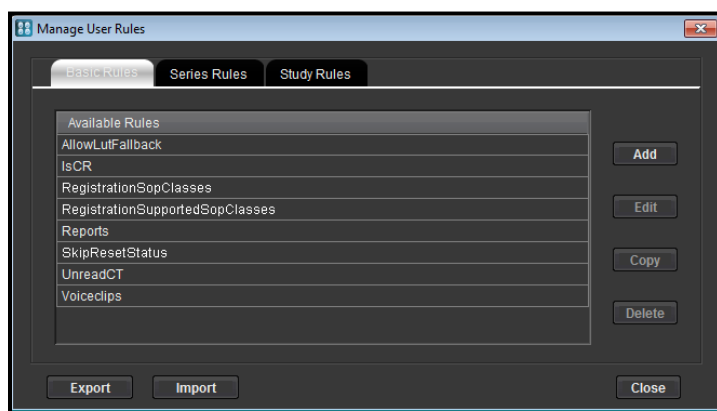
Use of the HP Rules can dramatically simplify long term maintenance of the hanging protocols. For example, conventional rules and HP criteria fall short for MR exams for a number of reasons including, but not limited to, series descriptions do not get updated on the scanner, text descriptions are inherently unreliable at many sites and sequence names can vary by manufacture. In Merge PACS 8.2, you can use the MR DICOM acquisition parameters like Repetition Time, Echo Time, Orientation and Flip Angle to drive HP layouts via the rules. For example, you could create a T1 Rule based on the number parameters that the TR is <800, TE <30, FA =90. This will allow the user the create MR HP using “by the numbers” in the acquisition instead of by the series description.

If you have the login privilege to **add/edit/delete user rules**, you can manage the rules governing the selection and implementation of Hanging Protocols by selecting **Manage Rules** from the **Hanging Protocol Menu** within the Workstation, as in the following example:



Managing User Rules

Clicking on the Manage Rules option will cause the **Manage User Rules** window to be displayed, as in the following example:



The Manage User Rules Window

At the Manage User Rules window you can add, edit, copy and delete the following types of rules:

Type of Rule	Description
Basic Rules	These rules exist solely as the building blocks for other rules and cannot themselves be selected when creating Hanging Protocols. They are typically simple rules, but can be as complicated as needed.
Series Rules	These rules can be selected when creating a Hanging Protocol to determine which Series should be displayed in which Viewports for a Hanging Protocol once the Hanging Protocol has been selected.
Study Rules	These rules can be selected when creating a Hanging Protocol to determine which Hanging Protocol should be selected for a study.

Each rule can, in turn, be composed of one or more of the following components:

Type of Segment	Description
DICOM Condition	Allows you to filter based on a specific DICOM tag and value.
Status Condition	Allows you to filter based on a specific status dimension as well as a possible value.
Other Exam Info Condition	Allows you to filter based on the facility, specialty and/or storing AE Title associated with specific exams.

User Profile	Allows you to filter based on specific information about the user reading the study such as the workstation the user is currently logged into.
Rule	Allows you to specify a previously created Basic rule to be included as a condition in this rule.
Condition Set	Allows you to specify a grouping of multiple conditions (Dicom Conditions, Status Conditions, additional Condition Sets and/or Basic Rules). In essence, this allows you to create another complex “on the fly” within the main rule without needing to create a new named rule from scratch.

NOTE: Conditions sets are available to preserve compatibility with rules imported from Merge RadSuite and would not typically be used for the creation of new rules.

NOTE: Merge PACS ships with a number of default rules that can be custom-configured.

14.2.1.Adding a New Rule

Adding a rule involves the following general process:

1. Creating and naming the rule
2. Adding one or more components to the rule
3. Configuring how the various components work together

NOTE: Since a condition set is essentially a complete rule in and of itself, with multiple components configured to work together in a certain way, the creation of condition sets will be discussed at the end of this subsection.

a. Creating and Naming a Rule

To create and name a new rule:

1. Select the type of rule you want to add from the tabs at the top of the Manage User Rules window, as in the following example:

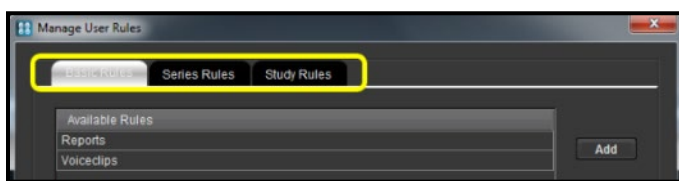


Figure 600: The Manage User Rules Window

NOTE: The procedure for adding any type of rule is the same, but basic rules are typically created first so they can be referenced within other rules.

2. Click the **Add** button.

The **Add Rule** window for the selected type of rule is displayed, as in the following example:

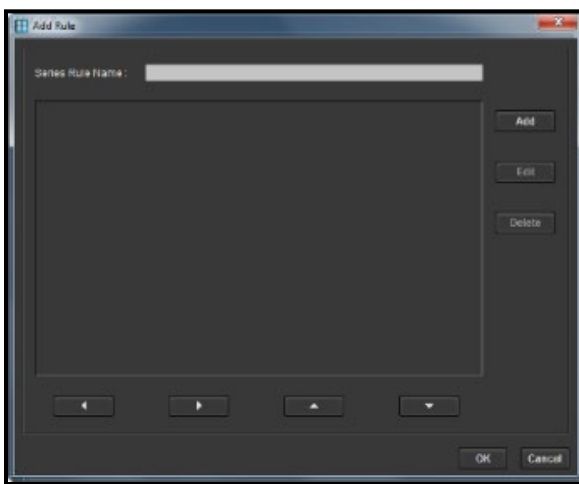


Figure 601: The Add Rule Window

3. Enter a name for this rule in the **Rule Name** field, as in the following example:

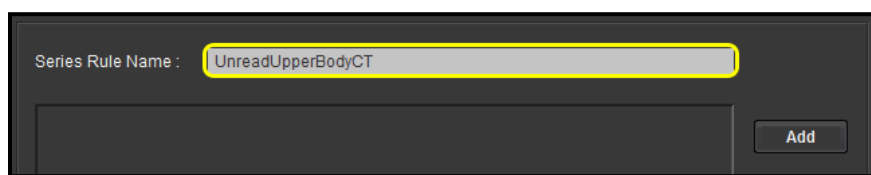


Figure 602: Entering the Rule Name

NOTE: In the example above, the rule is named **UnreadUpperBodyCT** to indicate that this rule will match on CT studies of the upper body that have a status of “Unread.”

b. Adding DICOM Conditions to the Rule

DICOM conditions allow you to filter a rule based on specific DICOM tags and values.

NOTE: Private DICOM tags are not supported.

To add one or more DICOM conditions to this rule:

1. Click on the **Add** button and select **Dicom Condition** from the menu of options, as in the following example:

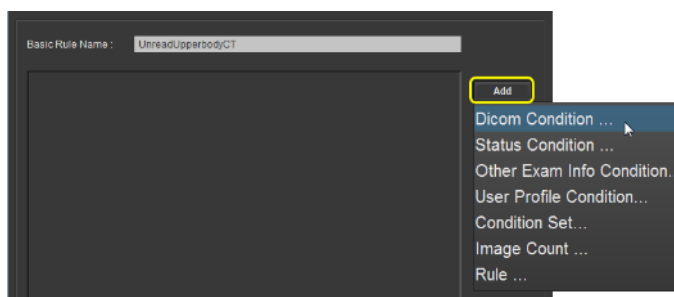


Figure 603: Adding a DICOM Condition

The **DICOM Tag Condition** dialog is displayed, as in the following example:

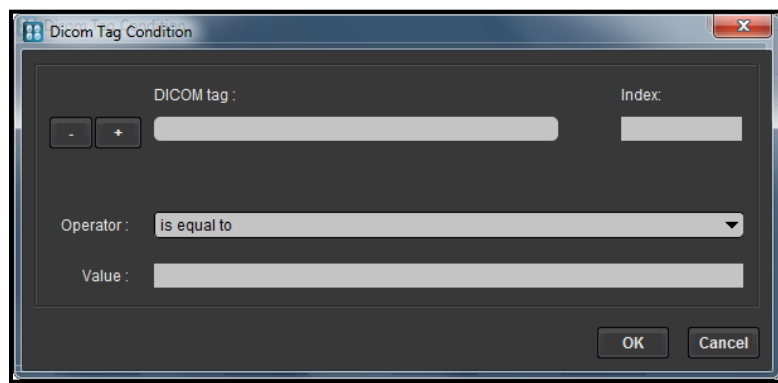


Figure 604: DICOM Tag Condition Dialog

2. Begin typing the name of the DICOM tag you want to match on in the **DICOM tag** field to display a filtered list menu of DICOM tags, as in the following example:

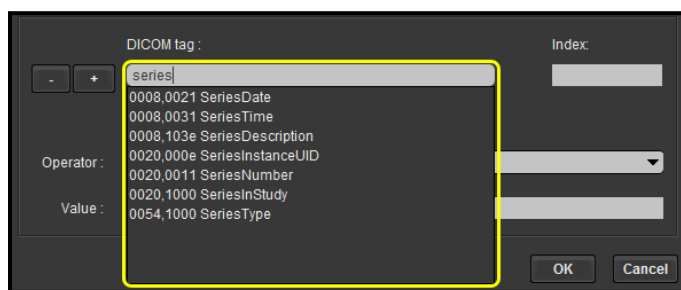


Figure 605: Matching DICOM Tags

3. Click on the desired tag in the list to select it.

The selected tag is displayed in the **DICOM tag** field, as in the following example:



Figure 606: Matching DICOM Tags

4. By default, this condition will match if the selected DICOM tag **is equal to** a specified value. If desired, however, you can select a different logical operator from the drop-down **Operator** menu, as in the following example:

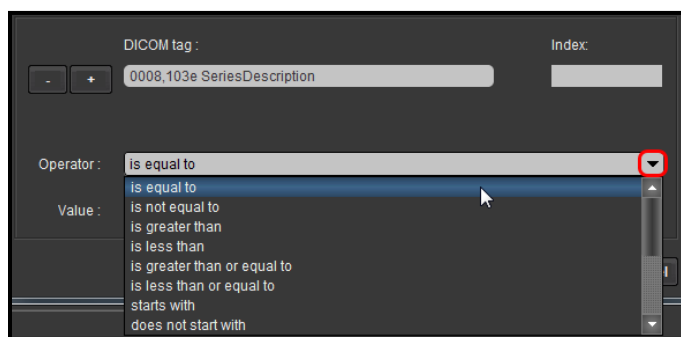


Figure 607: Selecting the Logical Operator

5. Enter the value to match on in the **Value** field, as in the following example:

The screenshot shows a configuration window for a DICOM tag. The 'DICOM tag' field contains '0008,103e SeriesDescription'. The 'Index' field is empty. The 'Operator' is set to 'is equal to'. The 'Value' field is highlighted with a yellow border and contains the text 'Chest'.

Figure 608: Entering the Value

6. For multi-valued DICOM attributes, you can specify the index number you want to match the value on in the **Index** field, as in the following example:

The screenshot shows a configuration window for a DICOM tag. The 'DICOM tag' field contains '0008,0008 ImageType'. The 'Index' field is highlighted with a yellow border and contains the number '2'. The 'Operator' is set to 'is equal to'. The 'Value' field contains the text 'SECONDARY'.

Figure 609: DICOM Tag Index

NOTE: In the example above, Image Type [0x0008, 0x0008] is a multi-valued attribute that can have a value of ORIGINAL or SECONDARY.

NOTE: If all the values needs to be considered while evaluating the selected logical operator for the DICOM Condition, you can enter a **negative** integer (e.g., “-1”) as the Index.

7. If desired, click the **[+]** button to add an additional DICOM tag that can be set to match against the same value with the same logical operator, as in the following example:

The screenshot shows a configuration window for a DICOM tag. The 'DICOM tag' field contains two entries: '0008,103E SeriesDescription' and '0008,1030 StudyDescription'. The 'Index' field is empty. The 'Operator' is set to 'is equal to'. The 'Value' field contains the text 'Chest'. The '+' button is highlighted with a yellow border.

Figure 610: Adding Another DICOM Tag

NOTE: In the example above, the condition would be met if **either** the Series Description **or** the Study Description were equal to “Chest.”

NOTE: Click the **[-]** button to remove an additional DICOM tag.

8. Click the **OK** button to save this DICOM condition and return to the main **Add Rule** window.

The newly added DICOM condition is displayed in the Add Rule window, as in the following example:

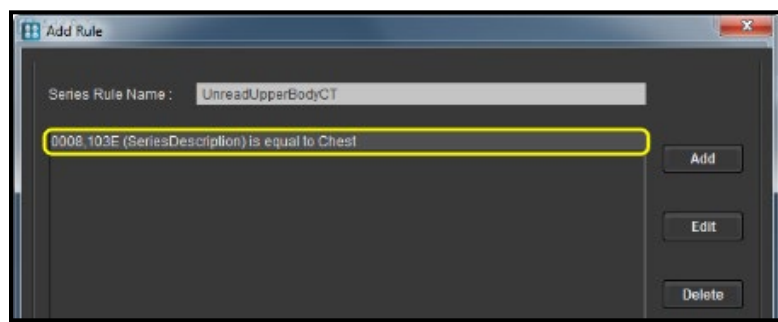


Figure 611: Newly Added DICOM Condition

9. Follow steps 1 through 8 above to create additional DICOM conditions as needed.
10. If necessary, click on a Dicom condition to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

c. Adding Status Conditions to the Rule

Status conditions allow you to filter a rule based on specific status dimensions and values.

NOTE: Status conditions are primarily intended for a future feature involving the launching of macros.

To add one or more status conditions to this rule:

1. Click on the **Add** button and select **Status Condition** from the menu of options, as in the following example:

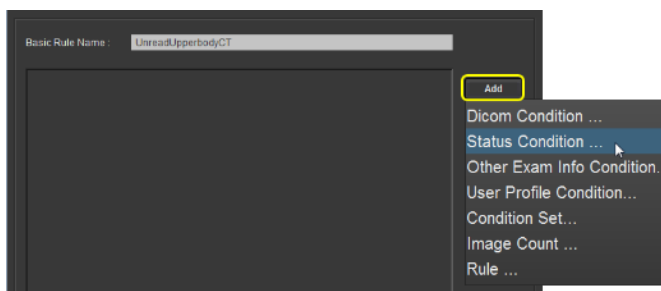


Figure 612: Adding a Status Condition

The **Status Condition** dialog is displayed, as in the following example:

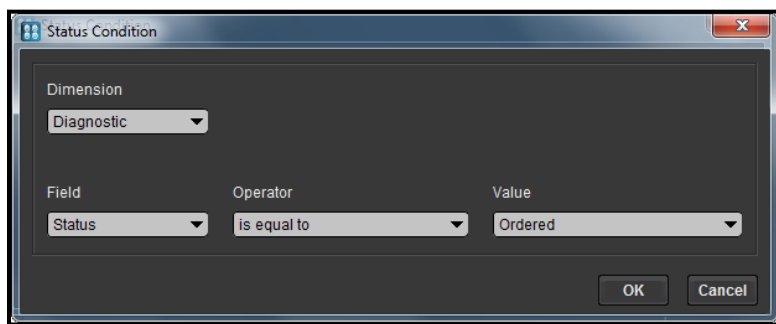


Figure 613: Status Condition Dialog

2. Select the desired status dimension from the drop-down **Dimension** menu.
3. If necessary, select the desired field (Status, Parameter or Notes) from the drop-down **Field** menu.
4. Select the desired logical operator from the drop-down **Operator** menu
5. Select the desired value to match on from the drop-down **Value** menu (or, if you are matching on a Parameter or Note field, enter the desired text in the Value field).

NOTE: The options in the Value menu are determined by the selected status dimension, and for some dimensions there may be only one option.

6. Click the **OK** button to save this Status condition and return to the main **Add Rule** window.

The newly added Status condition is displayed in the Add Rule window, as in the following example:

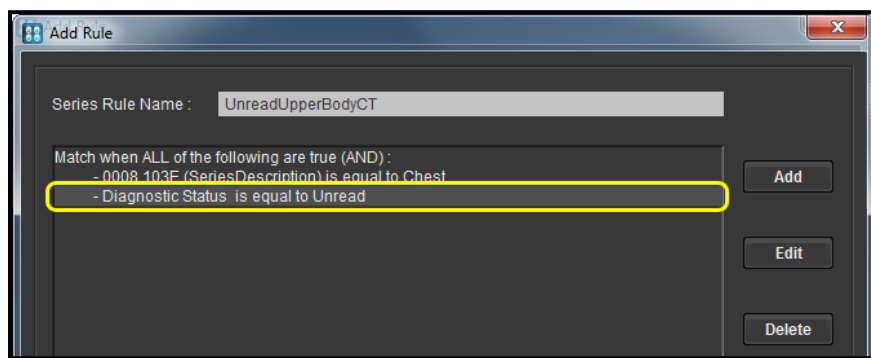


Figure 614: Newly Added Status Condition

NOTE: Once multiple conditions are added to a rule, they will automatically be grouped together with AND logic as in the example above. This logic can be changed, however, as described later in this procedure.

7. Follow steps 1 through 6 above to create additional status conditions as needed.
8. If necessary, click on a status condition to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

d. Adding Other Exam Information Conditions to the Rule

Other exam information conditions allow you to filter a rule based on the facility, specialty and/or storing AE Title associated with specific exams.

NOTE: Other exam information conditions are primarily intended for use with the **Launch and Control** feature that allows you to associate a custom macro with the “View Study” worklist action icon to control which application (including 3rd-party applications) will be launched under what circumstances.

To add one or more other exam information conditions to this rule:

1. Click on the **Add** button and select **Status Condition** from the menu of options, as in the following example:

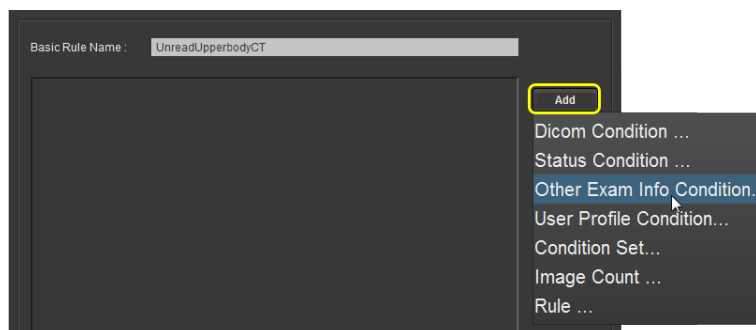


Figure 615: Adding an Other Exam Information Condition

The **Other Exam Information Condition** dialog is displayed, as in the following example:

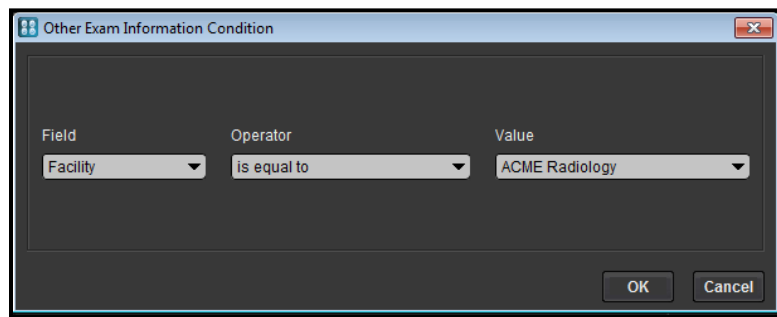


Figure 616: Other Exam Information Condition Dialog

2. Select the desired parameter from the drop-down **Field** menu. The following options are available:
 - **Facility**
 - **Specialty**
 - **Storing AE Title**
3. Select the desired logical operator from the drop-down **Operator** menu
4. Select or enter the desired value to match on from the drop-down **Value** menu as follows:
 - For **Facility**, the options in the Value drop-down menu are determined by the facilities that have been configured for this site, as described in subsection 10.5.2 above.

- For **Specialty**, the options in the Value drop-down menu are determined by the specialties that have been configured for this site, as described in subsection 10.6.3 above.
 - For **Storing AE Title**, manually enter the desired AE Title.
5. Click the **OK** button to save this other exam information condition and return to the main **Add Rule** window.

The newly added other exam information condition is displayed in the Add Rule window, as in the following example:

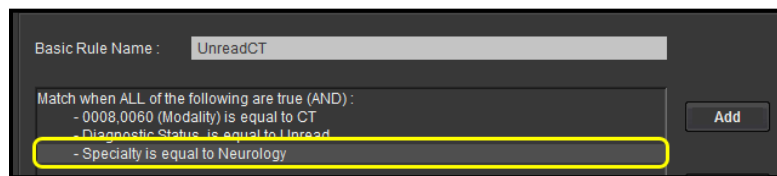


Figure 617: Newly Added Status Condition

NOTE: Once multiple conditions are added to a rule, they will automatically be grouped together with AND logic as in the example above. This logic can be changed, however, as described later in this procedure.

6. Follow steps 1 through 5 above to create additional status conditions as needed.
7. If necessary, click on an “other exam information” condition to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

e. Adding User Profile Conditions to the Rule

User profile conditions allow you to filter a rule based on the specific information about the user reading the study. In Merge PACS 8.2, this is limited to the location (*i.e.*, workstation) where the user is currently logged in.

NOTE: User profile conditions are primarily intended for use with the **Launch and Control** feature that allows you to associate a custom macro with the “View Study” worklist action icon to control which application (including 3rd-party applications) will be launched under what circumstances.

To add one or more user profile conditions to this rule:

1. Click on the **Add** button and select **User Profile Condition** from the menu of options, as in the following example:

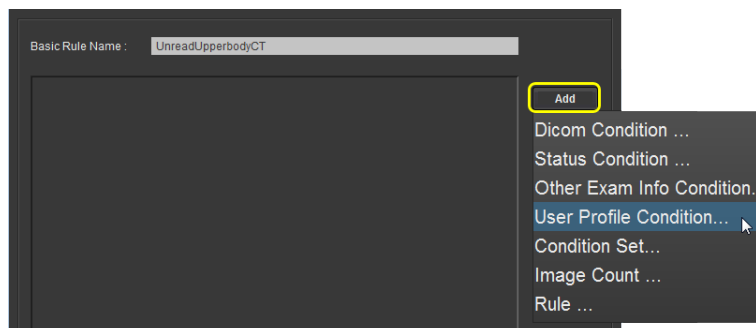


Figure 618: Adding a User Profile Condition

The **User Profile Condition** dialog is displayed, as in the following example:

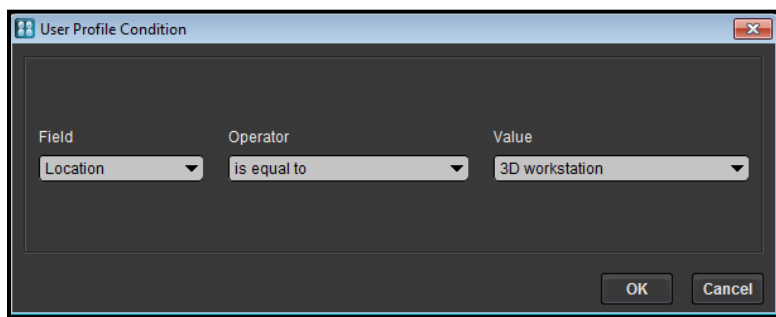


Figure 619: Other Exam Information Condition Dialog

2. The only option currently available for **Field** is location.
3. Select the desired logical operator from the drop-down **Operator** menu
4. Select the desired value to match on from the drop-down **Value** menu.

NOTE: The options in the Value drop-down menu are determined by the reader locations that have been configured for this site, as described in subsection 10.5.4 above.

5. Click the **OK** button to save this user profile condition and return to the main **Add Rule** window.

The newly added user profile condition is displayed in the Add Rule window, as in the following example:

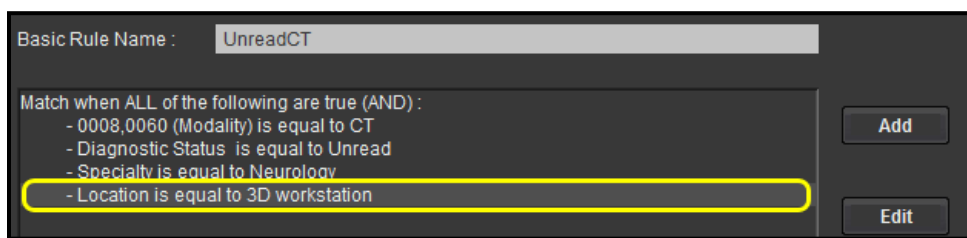


Figure 620: Newly Added User Profile Condition

NOTE: Once multiple conditions are added to a rule, they will automatically be grouped together with AND logic as in the example above. This logic can be changed, however, as described later in this procedure.

6. Follow steps 1 through 5 above to create additional status conditions as needed.
7. If necessary, click on a user profile condition to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

f. Adding Image Count Conditions to the Rule

Image Count allow you to filter a rule based on the number of images in a series or study.

To add one or more image count conditions to this rule:

1. Click on the **Add** button and select **Image Count** from the menu of options, as in the following example:

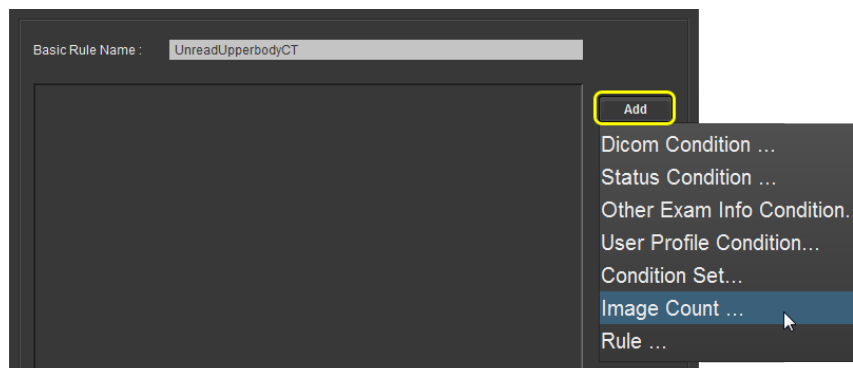


Figure 621: Adding an Image Count Condition

The **Image Count** dialog is displayed, as in the following example:

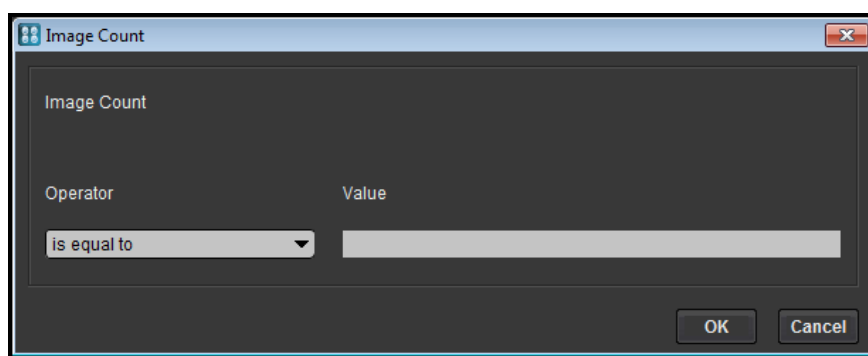


Figure 622: Image Count Dialog

2. Select the desired logical operator from the drop-down **Operator** menu.
3. Enter the desired number of images in the **Value** field.
4. Click the **OK** button to save this Image Count condition and return to the main **Add Rule** window.

The newly added Image Count condition is displayed in the Add Rule window, as in the following example:

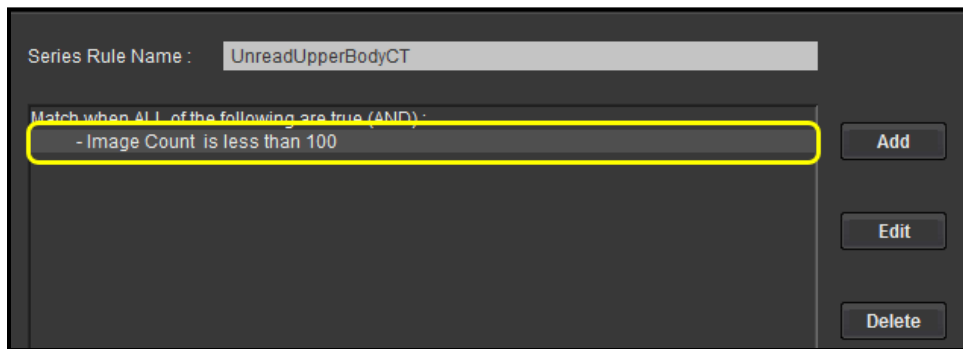


Figure 623: Newly Added Image Count Condition

NOTE: Once multiple conditions are added to a rule, they will automatically be grouped together with AND logic as in the example above. This logic can be changed, however, as described later in this procedure.

5. Follow steps 1 through 4 to create additional status conditions as needed.
6. If necessary, click on an Image Count condition to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

g. Adding Basic Rules to the Rule

If you have previously created one or more basic rules, you can add those rules as conditions to other rules, whether they be basic, study or series rules.

To add one or more existing basic rules as a condition to this rule:

1. Click on the **Add** button and select **Rule** from the menu of options, as in the following example:

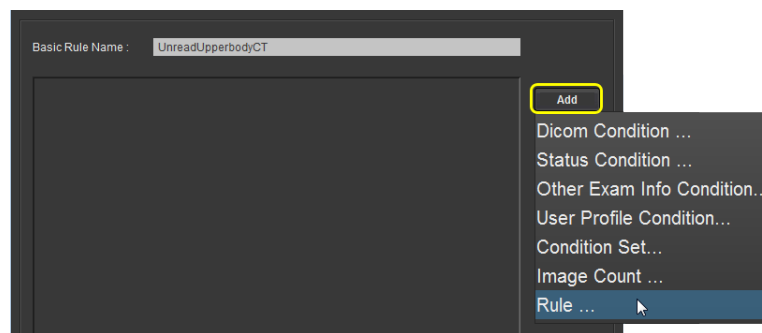


Figure 624: Adding a Basic Rule

The **Select a Rule** dialog is displayed, showing all currently existing basic rules, as in the following example:

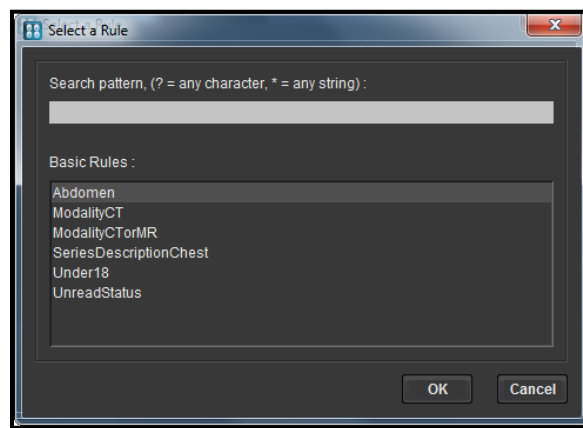


Figure 625: Select a Rule

- If desired, you can filter the list of basic rules by entering text to match in the **Search pattern** field at the top of the dialog, as in the following example:

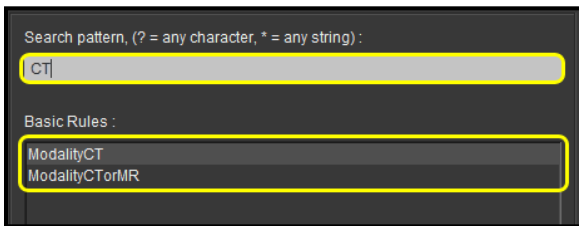


Figure 626: Filtering the List of Rules

- Click on the desired rule in the **Basic Rules** panel to select it and then click the **OK** button to save this basic rule as a condition and return to the main **Add Rule** window.

The newly added basic rule is displayed in the Add Rule window, as in the following example:

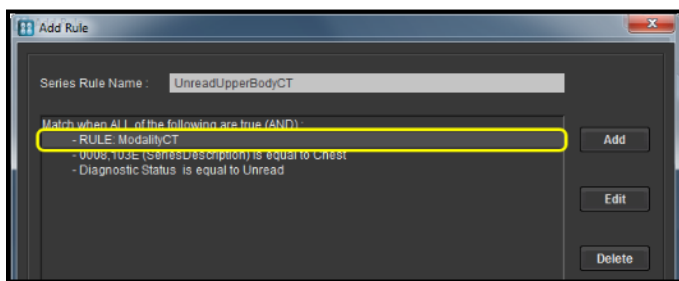


Figure 627: Newly Added Rule Condition

NOTE: Once multiple conditions are added to a rule, they will automatically be grouped together with AND logic as in the example above. This logic can be changed, however, as described later in this procedure.

- Follow steps 1 – 3 to add additional basic rules to this rule as needed.
- If necessary, click on a basic rule to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

h. Configuring Interaction of Rule Components

Once you have added multiple components to a rule, you can configure how the various components work together, including how they are grouped and the logical operators connecting them.

- Grouping of components is done using the buttons located at the bottom of the Add Rule window, as in the following example:

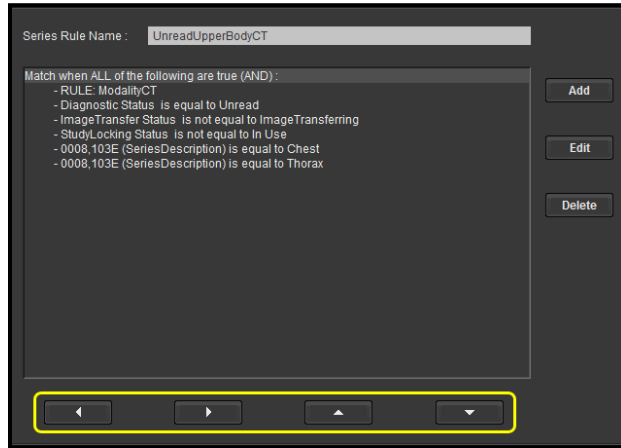


Figure 628: Component Integration Buttons

Click on a component to select it and then use the buttons as follows:

Button	Description
 	Click on the Up and Down arrows to position components next to each other in preparation for grouping them together.



Click the **Right** arrow to put the selected component into a sub-group with a default logical operator of AND, as in the following example:

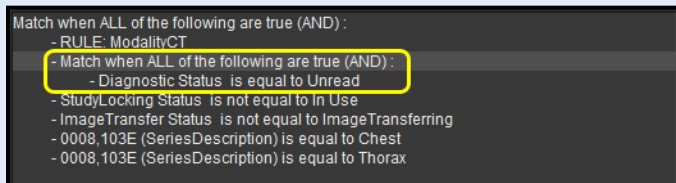


Figure 629: Component Sub-group

Selecting any component immediately below this group and clicking the **Right** arrow will add that component to the group, as in the following example:

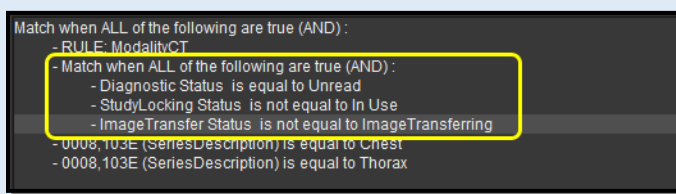


Figure 630: Multiple Component Sub-group

NOTE: If you want to add a component to a new sub-group instead of the one immediately above it, first move that component so that it is above the existing sub-group.



Click the **Left** arrow to remove a component from a sub-group.

- Change a logical operator at any level by clicking on the operator and then clicking the **Edit** button, as in the following example:

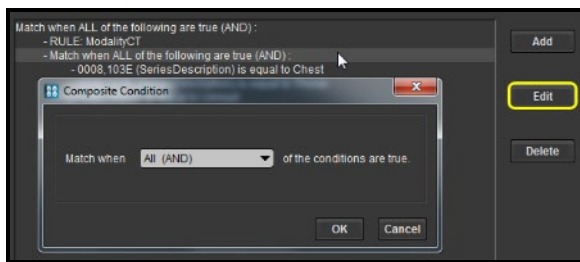


Figure 631: Editing a Logical Operator

Select the desired operator from the drop-down menu and then click the **OK** button to save your change and close the dialog.

- A sample series rule with fully integrated components might look similar to the following:

```
Match when ALL of the following are true (AND) :
- RULE: ModalityCT
- Match when ALL of the following are true (AND) :
  - Diagnostic Status is equal to Unread
  - StudyLocking Status is not equal to In Use
  - ImageTransfer Status is not equal to ImageTransferring
- Match when ANY of the following are true (OR) :
  - 0008,103E (SeriesDescription) is equal to Chest
  - 0008,103E (SeriesDescription) is equal to Thorax
```

Figure 632: Sample Completed Rule

In the example above, a series would match the rule if **ALL** of the following are true:

- The modality is equal to “CT” (as defined in the embedded “ModalityCT” rule)
- The Diagnostic status for the parent study is “Unread” **AND** the Study Locking status is not “In Use” **AND** the Image Transfer status is not “ImageTransferring”
- Either the Series Description is “Chest” **OR** the Series Description is “Thorax”.

i. Adding Condition Sets to the Rule

Condition sets allow you to essentially create an entire complex rule with multiple components (including dicom conditions, status conditions, basic rules and/or additional condition sets) “on the fly” and add it as a component of a rule without needing to create a new named rule.

NOTE: Conditions sets are available to preserve compatibility with rules imported from Merge RadSuite and would not typically be used for the creation of new rules.

To add a condition set to a rule:

1. At the **Add Rule** window, click on the **Add** button and select **Condition Set** from the menu of options, as in the following example:

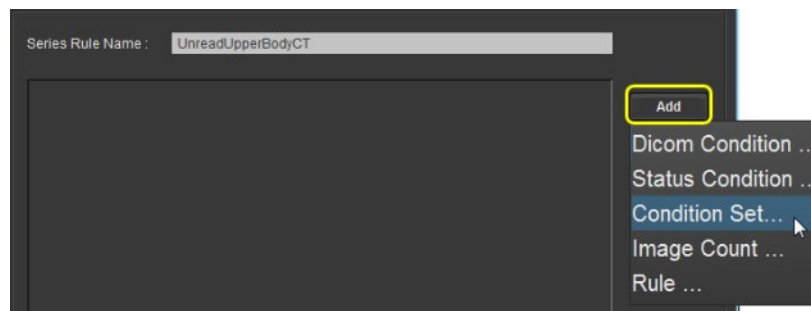


Figure 633: Adding a Condition Set

The **Add Condition Set** dialog is displayed, as in the following example:

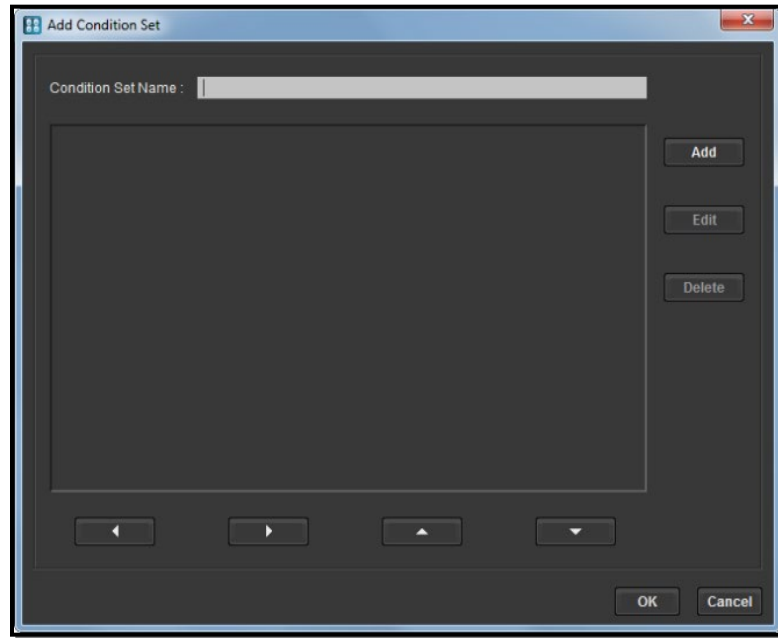


Figure 634: Add Condition Set

2. Enter a name for this condition set in the **Condition Set Name** field.
3. Set up this condition set as desired, adding one or more DICOM tag conditions, status conditions, basic rules and/or additional condition sets.
4. When finished, click the **OK** button to save this condition set and return to the main **Add Rule** window.

The newly added condition set is displayed in the Add Rule window, as in the following example:

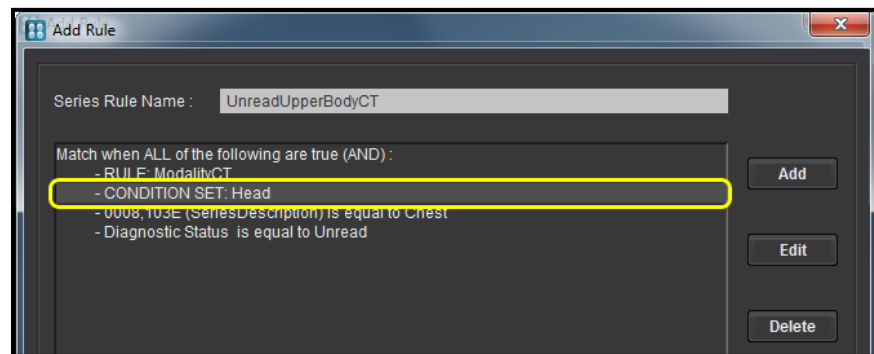


Figure 635: Newly Added Status Condition

5. Follow steps 1 through 4 above to add additional basic rules to this rule as needed.
6. If necessary, click on a basic rule to select it and then click the **Edit** button to make changes to it or the **Delete** button to remove it.

j. Sample Rule – Breast Tomosynthesis Series Matching

If you wanted to create a rule that would match for Breast Tomosynthesis (BTO) series with a view code of “CC” and laterality or “R”, you would perform the following steps:

1. Click on the **Series Rule** tab at the main **Manage User Rules** window and click on the **Add** button, as in the following example:

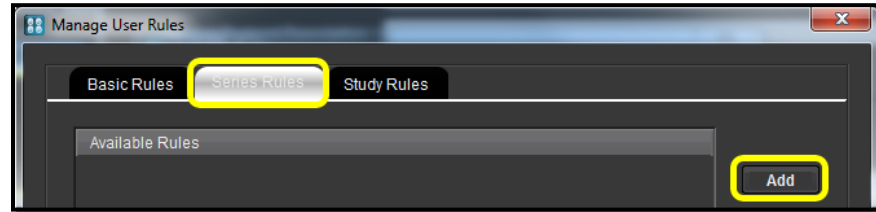


Figure 636: Series Rule Tab

2. At the **Add Rule** window, enter a name for this rule in the **Rule Name** field, as in the following example:



Figure 637: Entering the Rule Name

NOTE: In the example above, the rule is named **R CC BTO** to indicate that this rule will match on Breast Tomosynthesis series with a laterality code of “R” and a view code or “CC”.

3. Add a DICOM condition to match on studies with the standard SOP Class UID for BTO studies as follows:
 - a. Click on the **Add** button and select **Dicom Condition** from the menu of options, as in the following example:

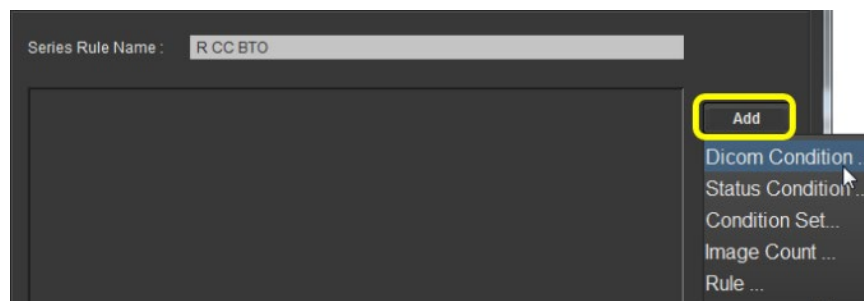


Figure 638: Adding a DICOM Condition

- b. At the **DICOM Tag Condition** dialog, define the first matching criteria as having a SOP Class UID equal to that of BTO images (**1.2.840.10008.5.1.4.1.1.13.1.3**), as in the following example:

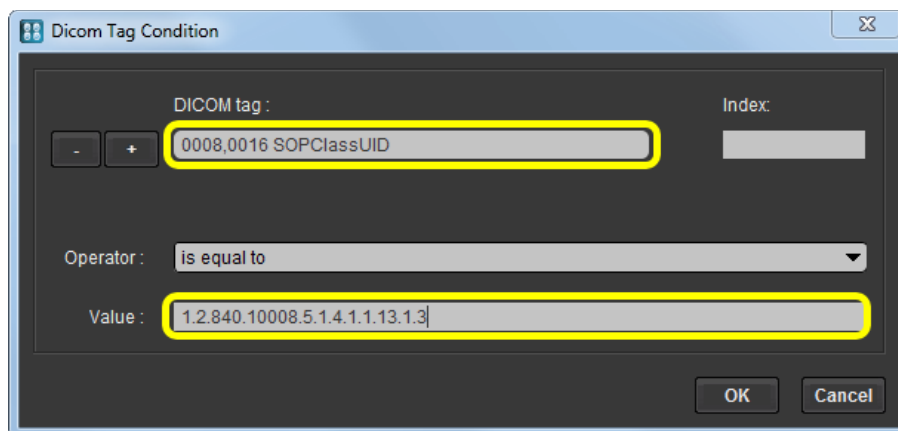


Figure 639: Defining the SOP Class UID

NOTE: To specify SOP Class UID as the DICOM tag, start entering “SOP” in the field and then select the correct option from the drop-down filtered list menu.

- c. Click the **OK** button to save this condition and return to the Add Rule window.
4. Add another DICOM condition to match on studies with the standard view code sequence/code value for “CC” views as follows:
 - a. Click on the **Add** button and select **Dicom Condition** from the menu of options, as in the following example:

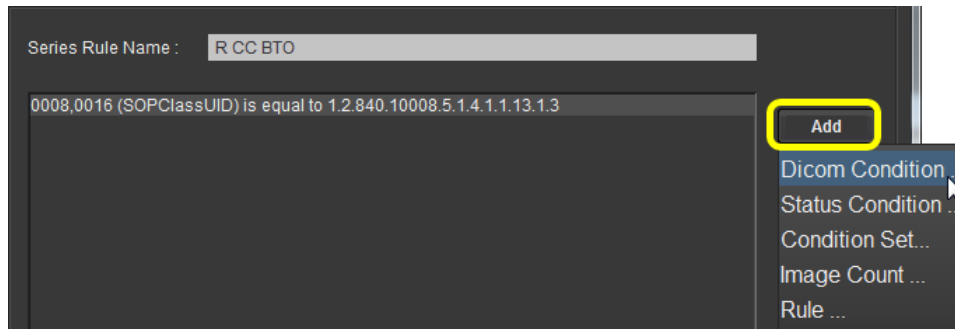


Figure 640: Adding a DICOM Condition

- b. At the **DICOM Tag Condition** dialog, define the second matching criteria as having a view code sequence and a view code value equal to the view code value for CC views (**R-10242**), as in the following example:

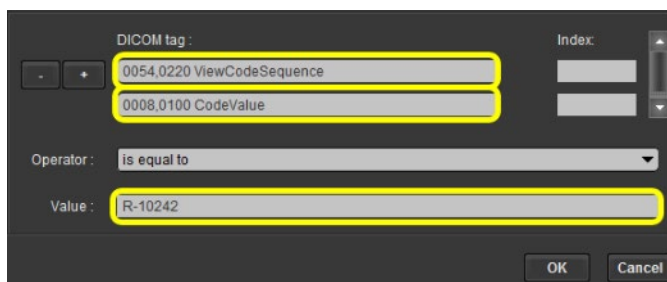


Figure 641: Defining the View Code

NOTE: After selecting ViewCodeSequence as the first DICOM tag, click on the **+** button to add a new field for CodeValue.

- c. Click the **OK** button to save this condition and return to the Add Rule window.
5. Add a final DICOM condition to match on studies with the shared functional group sequence/frame anatomy sequence value for “R” laterality as follows:
 - a. Click on the **Add** button and select **Dicom Condition** from the menu of options, as in the following example:

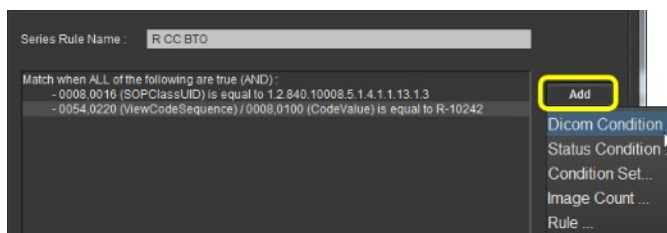


Figure 642: Adding a DICOM Condition

- b. At the **DICOM Tag Condition** dialog, define the third matching criteria as having a shared functional group sequence and a frame anatomy sequence equal to “R”, as in the following example:

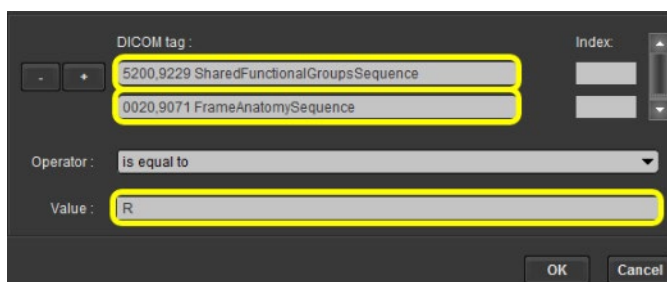


Figure 643: Defining the Laterality

- c. Click the **OK** button to save this condition and return to the Add Rule window.
6. At the Add Rule window, click on the **OK** button to complete the rule, as in the following example:

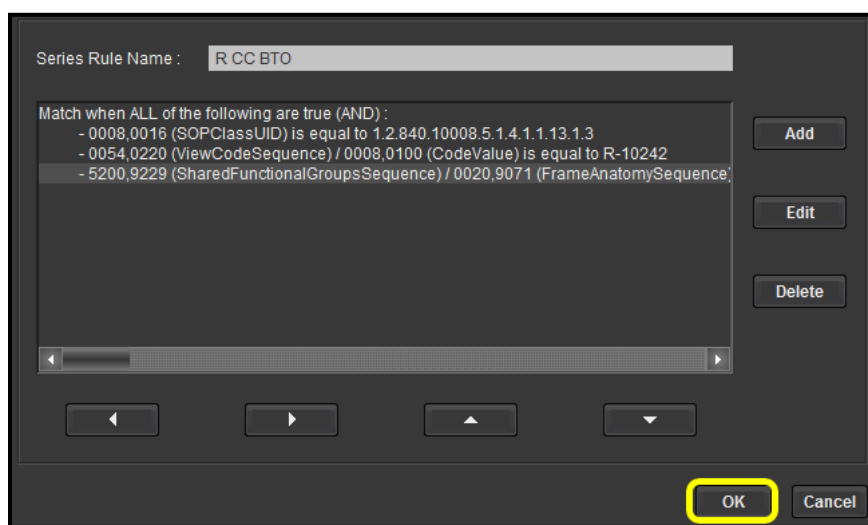


Figure 644: Completing the Rule

14.2.2.Editing an Existing Rule

To edit an existing rule:

1. Click on the desired rule in the appropriate tab on the main **Manage User Rules** window and click on the **Edit** button, as in the following example:

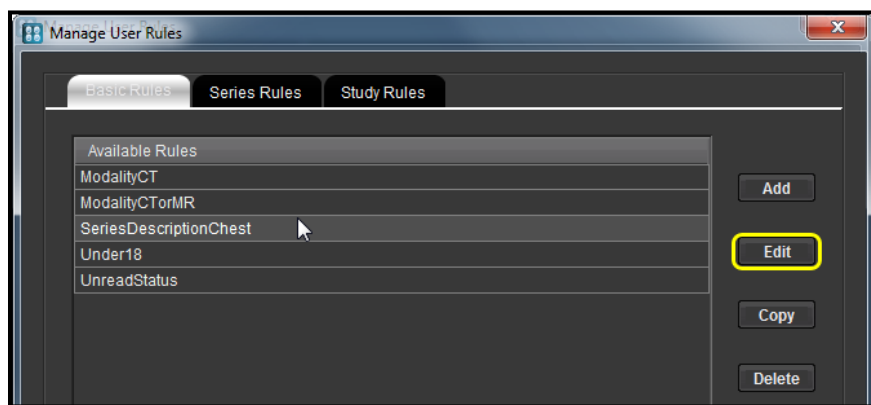


Figure 645: Edit Button

The **Edit Rule** window is displayed, as in the following example:

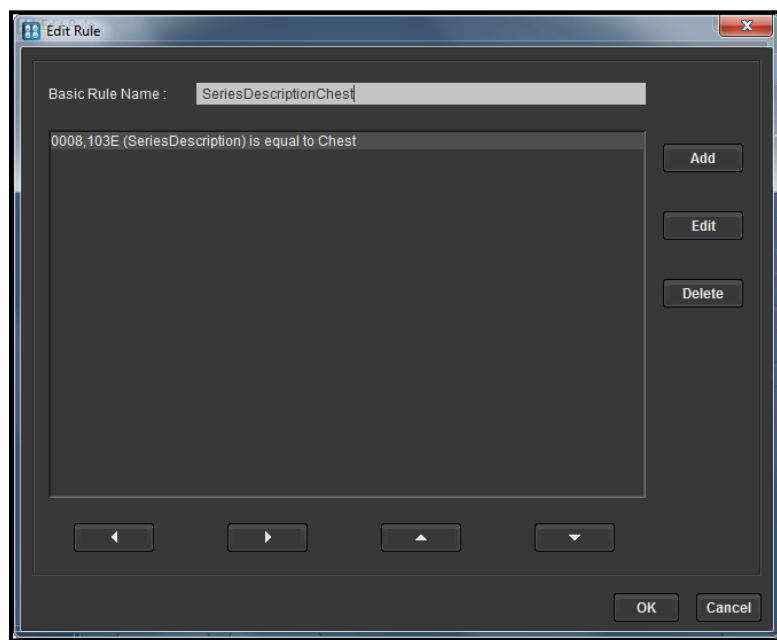


Figure 646: Edit Rule Window

2. Edit the rule as desired, following the procedures described in subsection 14.2.1 above.
3. When finished editing the rule, click the **OK** button at the bottom of the Edit Rule window.

14.2.3. Copying an Existing Rule

To copy an existing rule:

1. Click on the desired rule in the appropriate tab on the main **Manage User Rules** window and click on the **Copy** button, as in the following example:

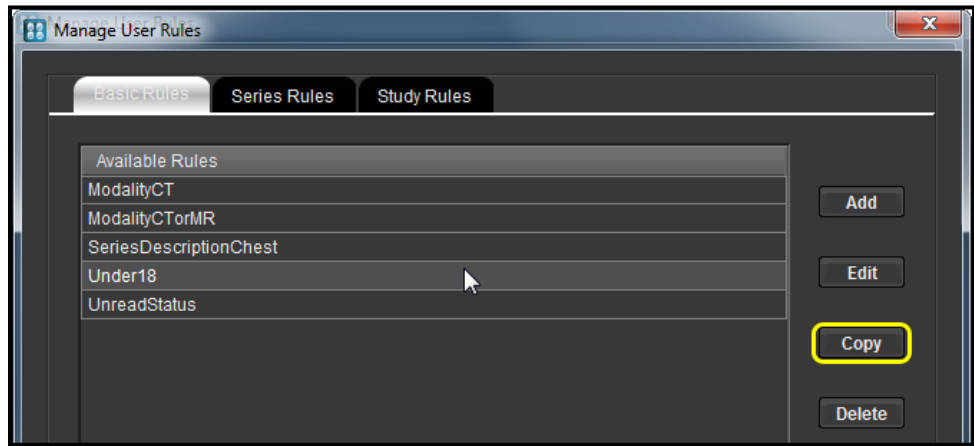


Figure 647: Copy Button

A copy of the rule is created and added to the list of available rules at the selected tab, as in the following example:

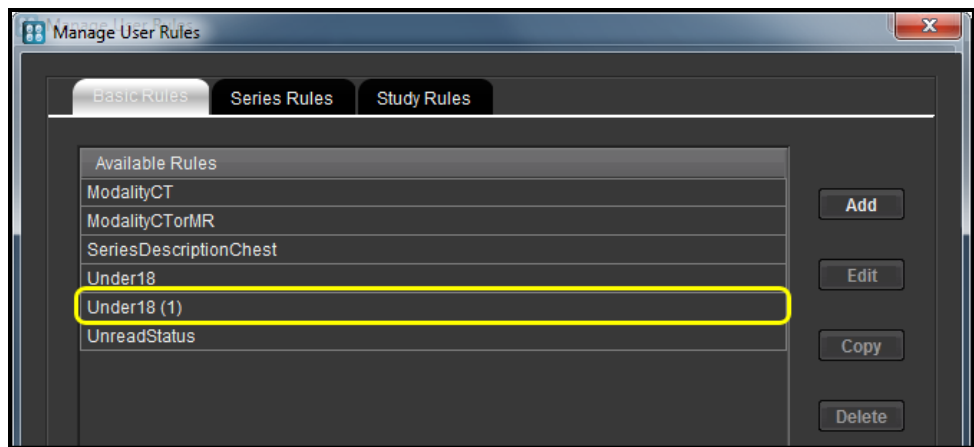


Figure 648: Copied Rule

2. Click on the new copy to select it and then click on the **Edit** button to make changes to it.

14.2.4. Deleting Existing Rules

To delete one or more existing rules:

1. Click on the rule or rules you want to remove in the appropriate tab on the main **Manage User Rules** window (pressing the **Shift** key to select multiple contiguous rules and the **Ctrl** key to select multiple non-contiguous rules) and then click on the **Delete** button, as in the following example:

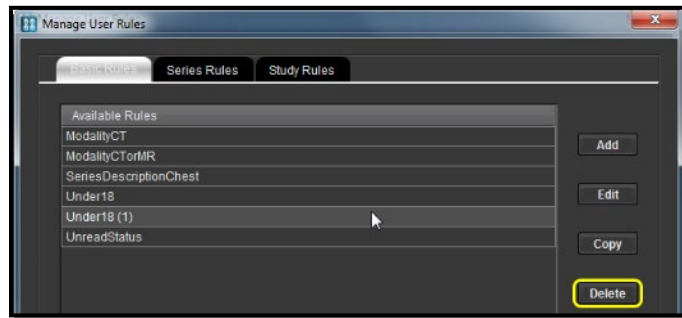


Figure 649: Delete Button

A warning message similar to the following is displayed:

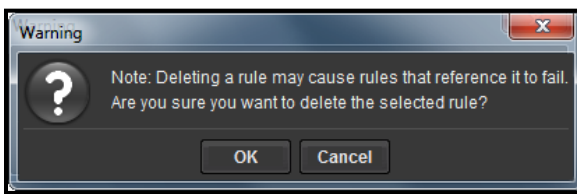


Figure 650: Deletion Warning

2. Click the **OK** button to proceed with the deletion.

14.2.5.Exporting Rules

If desired, you can export some or all of the rules saved on the Server to a **.rules** file that can later be imported onto another Merge PACS Server.

To export some or all rules:

1. At the main **Manage User Rules** window, click on the **Export** button at the bottom of the window.

The **Select Rules for Exporting** dialog is displayed, as in the following example:

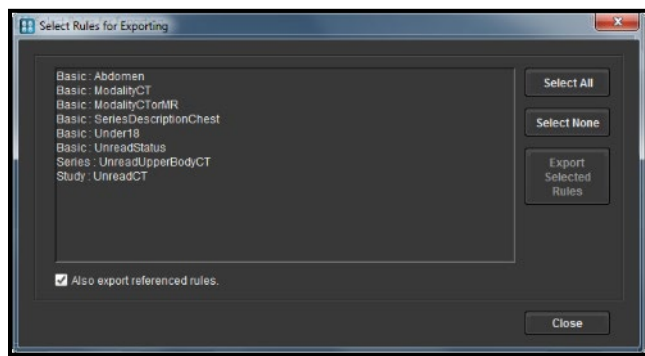


Figure 651: Selecting Rules for Exporting

2. Click on one or more rules to select them, pressing the **Shift** key to select multiple contiguous rules and the **Ctrl** key to select multiple non-contiguous rules.

NOTE: You can also click the **Select All** button to select all rules or the **Select None** button to clear your selections.

3. If you have not selected all rules and want to include any rules that are referenced within the selected rules, select the **Also export referenced rules** option.
4. When all desired rules have been selected, click the **Export Selected Rules** button.

The **Select File** dialog is displayed, as in the following example:

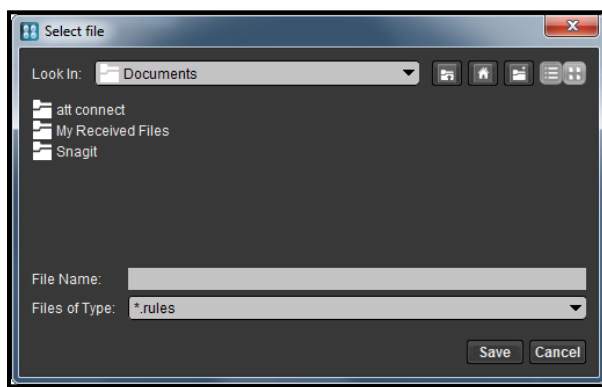


Figure 652: Selecting the Export File

5. Browse to the location where you would like to save the file, give it a name, and click on the **Save** button, as in the following example:

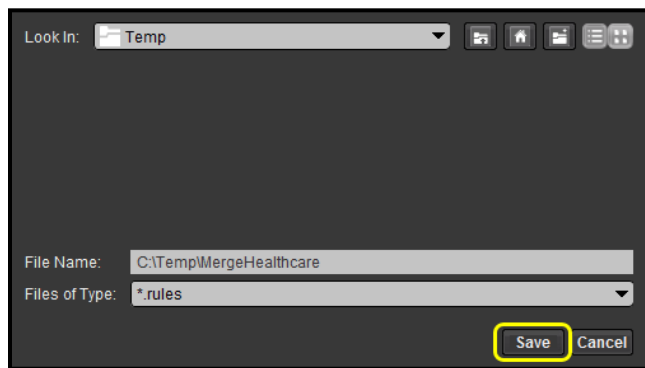


Figure 653: Saving the Export File

NOTE: In the example above, the file is called MergeHealthcare.rules and will be saved to C:\Temp.

Then the export process is complete, a confirmation message will be displayed, as in the following example:

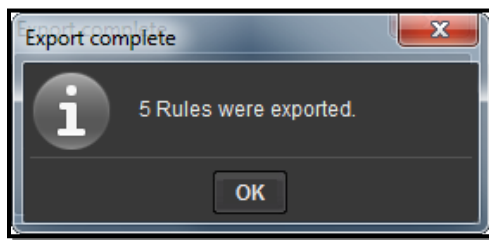


Figure 654: Export Complete

6. Click the **OK** button.

14.2.6.Importing Rules

NOTE: The Import Rules feature can only be used to import rules exported from a Merge PACS Server. If you need to import rules from a RadSuite Server, contact Merge Healthcare Support and ask them to run the Rules Conversion Tool.

If desired, you can import any **.rules files** that were exported from another Merge PACS Server onto this Server, as long as they are stored on a location you can access (the local workstation, a network drive, a USB drive, etc.).

NOTE: A number of user rules are provided by default and are available for importation in the following directory on the installation drive of the Merge PACS Server:
<DriveLetter>:\amicas2\rules.

To import a *.rules file:

1. At the main **Manage User Rules** window, click on the **Import** button at the bottom of the window.
*The **Select Files for Importing** dialog is displayed, as in the following example:*

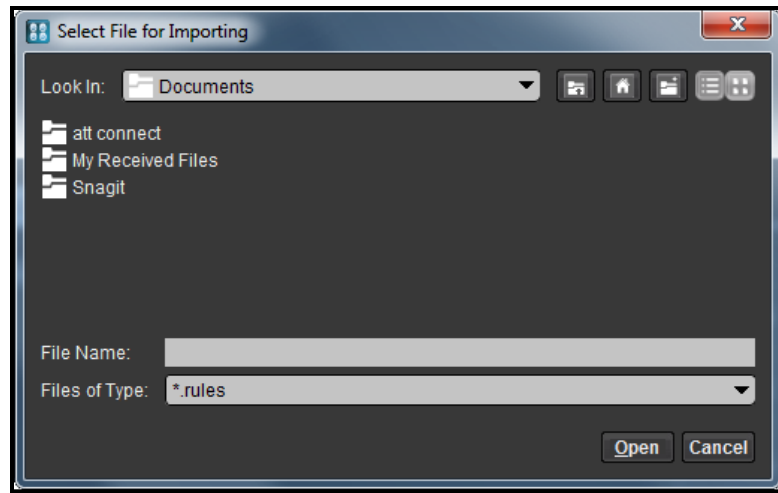


Figure 655: Selecting the File for Importing

2. Browse to the location where the *.rules files is located, select the file and then click on the **Open** button, as in the following example:

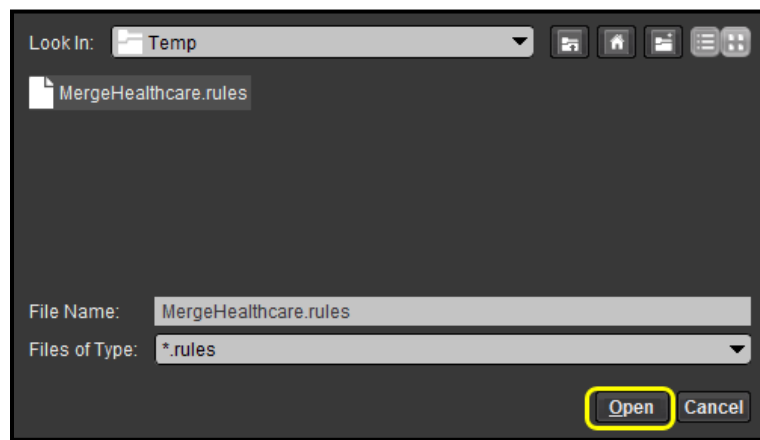


Figure 656: Selecting the File for Importing

The **Select Rules for importing** dialog is displayed, listing the rules contained in the selected *.rules file, as in the following example:

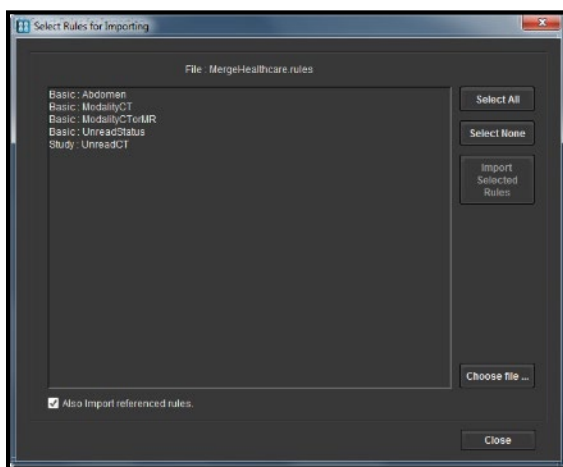


Figure 657: Selecting the Rules for Importing

3. Click on one or more rules to select them, pressing the **Shift** key to select multiple contiguous rules and the **Ctrl** key to select multiple non-contiguous rules.

NOTE: You can also click the **Select All** button to select all rules or the **Select None** button to clear your selections.

4. If you have not selected all rules and want to include any rules that are referenced within the selected rules, select the **Also export referenced rules** option.
5. When all desired rules have been selected, click the **Import Selected Rules** button.

If a rule exists on the Server with the same name as a rule being imported, a confirmation dialog will be displayed, as in the following example:

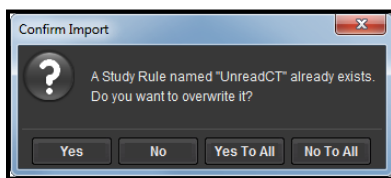


Figure 658: Confirm Import

When all rules have been imported, a confirmation message will be displayed, as in the following example:

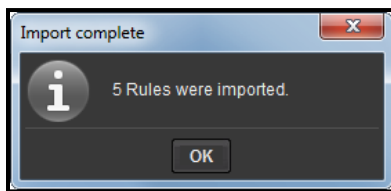


Figure 659: Import Complete

14.2.7.Editing the Reports User Rule

Standard HL7 reports are automatically viewable within the Workstation's Report Viewer. A special **Reports** user rule is provided by default to configure whether incoming DICOM objects such as DICOM Structured Reports or DICOM-wrapped encapsulated PDFs should also be viewable within the Report Viewer, and this rule can be edited as necessary to meet the needs of a specific site. In addition, if your site already has a rule created, Merge PACS can be configured by Merge Healthcare Support to use that rule as the default.

The default Reports rule provides a “match” (*i.e.*, the object will be considered a report) if **EITHER** of the following conditions is met:

- The SOPClassUID is equal to 1.2.840.10008.5.1.4.1.1.88.11 (*i.e.*, the object is a DICOM SR) **AND** the SeriesDescription does not start with “emageon study note”
- OR**
- The SOPClassUID is equal to 1.2.840.10008.5.1.4.1.1.104.1 (*i.e.*, the object is an encapsulated PDF)

To edit the Reports rule:

1. Select **Manage Rules** from the **Hanging Protocol Menu** within the Viewer.
*The **Manage User Rules** window is displayed.*
2. At the **Basic Rules** tab, click on the **Reports** rule in the list of Applicable Rules once to highlight it and then click on the **Edit** button, as in the following example:

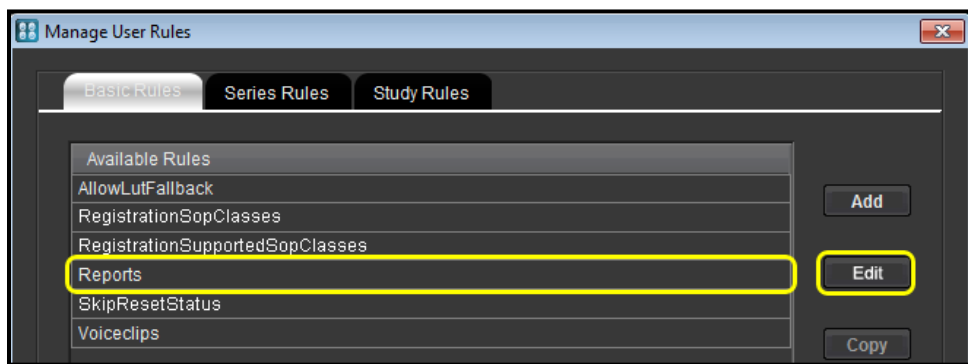


Figure 660: Editing the Reports Rule

The **Edit Rule** window is displayed, as in the following example:

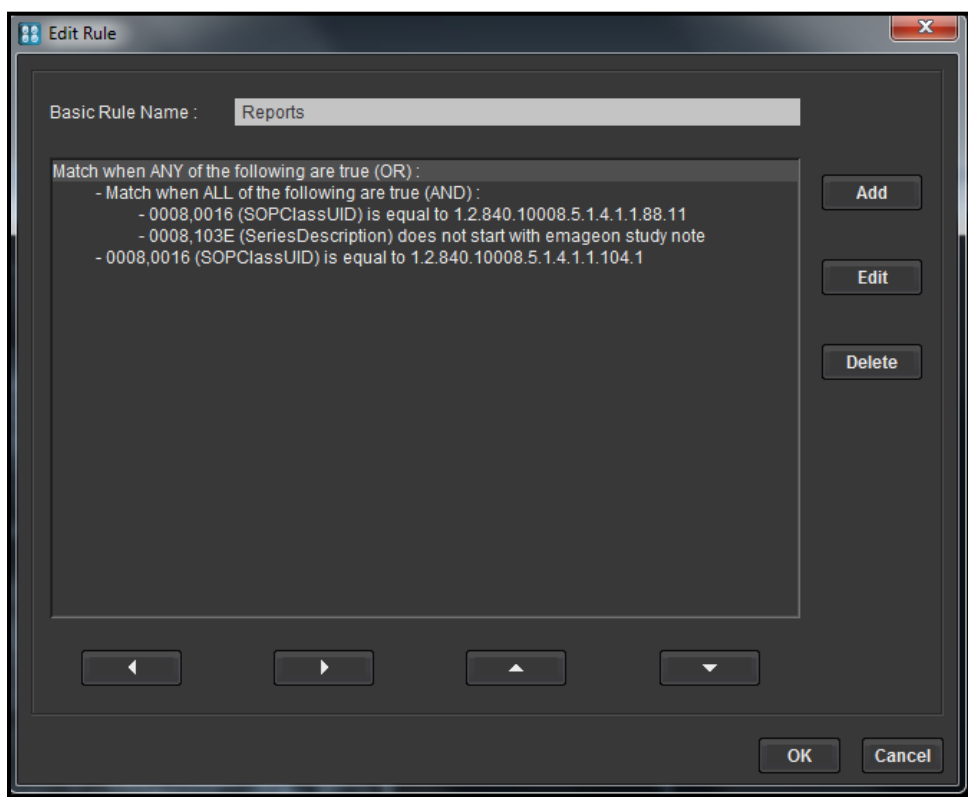


Figure 661: The Edit Rule Window

3. Edit the rule, referring to the detailed instructions in the Section 14.2.1 above as necessary, and click on the **OK** button at the bottom of the window.

14.2.8.Editing the Voiceclips User Rule

VoiceClips recorded and stored within Merge PACS are automatically accessible from within the Workstation's VoiceClip Player. A special **Voiceclips** user rule is provided by default to configure whether incoming DICOM objects should also be accessible within the VoicerClip Player, and this rule can be edited as necessary to meet the needs of a specific site. In addition, if your site already has a rule created, Merge PACS can be configured by Merge Healthcare Support to use that rule as the default.

The default Voiceclip rule provides a "match" (*i.e.*, the object will be considered a VoiceClip) if the the SOPClassUID is equal to 1.2.840.10008.5.1.4.1.1.9.4.1.

To edit the Voiceclips rule:

1. Select **Manage Rules** from the **Hanging Protocol Menu** within the Viewer.

The **Manage User Rules** window is displayed.

2. At the **Basic Rules** tab, click on the **Voiceclips** rule in the list of Applicable Rules once to highlight it and then click on the **Edit** button, as in the following example:

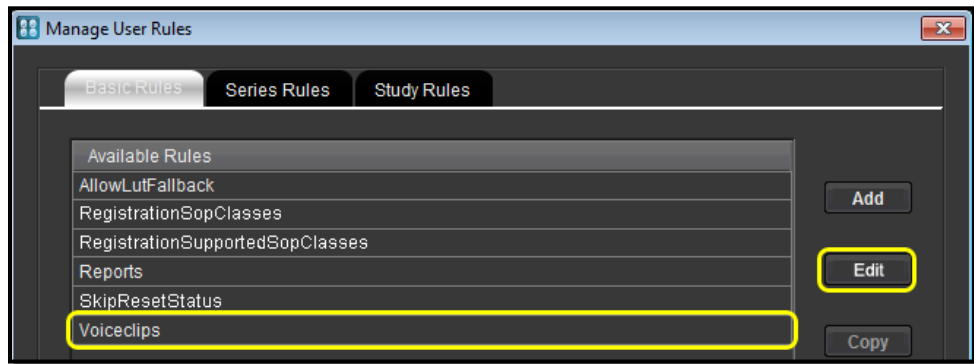


Figure 662: Editing the Voiceclips Rule

The **Edit Rule** window is displayed, as in the following example:

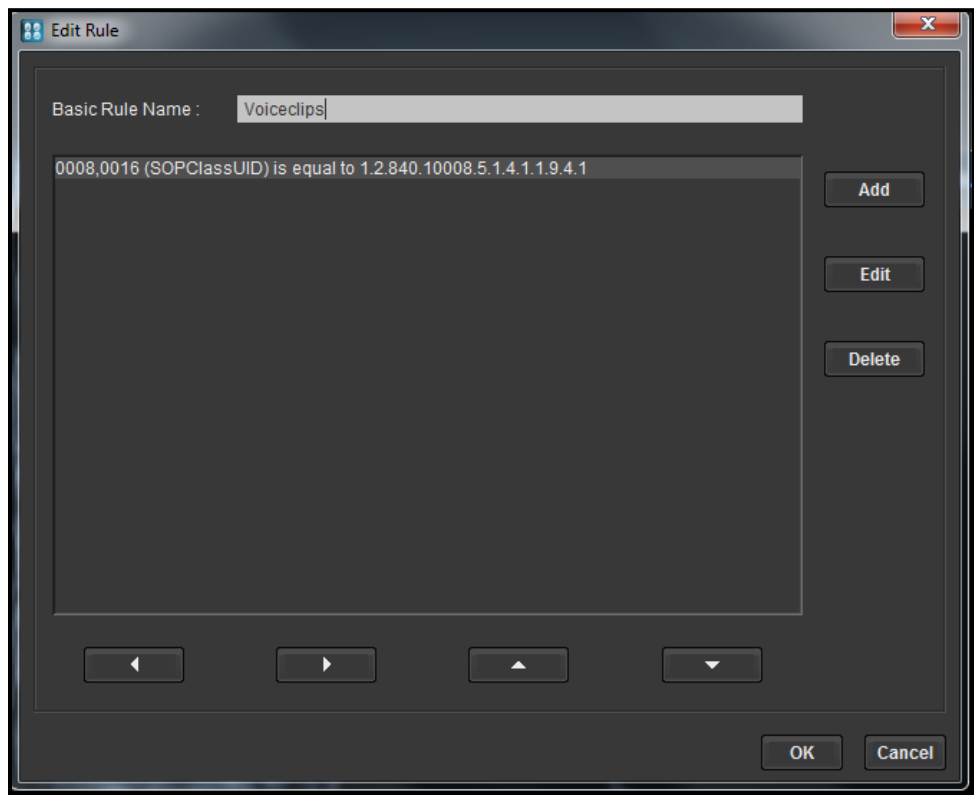


Figure 663: The Edit Rule Window

3. Edit the rule, referring to the detailed instructions in the Section 14.2.1 above as necessary, and click on the **OK** button at the bottom of the window.

14.2.9.Editing the Default SkipResetStatus User Rule

If a site is configured to use the Auto Registration feature, as described in 1.2.78 above, a special **SkipResetStatus** user rule is provided by default to check for the two registration SOP Class UIDs so that new registration objects will not reset the study status. In addition, if your site already has a rule created, Merge PACS can be configured by Merge Healthcare Support to use that rule as the default.

The default SkipResetStatus rule provides a “match” (*i.e.*, the object will be considered a registration object) if **EITHER** of the following conditions is met:

- The SOPClassUID is equal to 1.2.840.10008.5.1.4.1.1.66.1

OR

- The SOPClassUID is equal to 1.2.840.10008.5.1.4.1.1.66.3

To edit the SkipResetStatus rule:

1. Select **Manage Rules** from the **Hanging Protocol Menu** within the Viewer.

*The **Manage User Rules** window is displayed.*

2. At the **Basic Rules** tab, click on the **SkipResetStatus** rule in the list of Available Rules once to highlight it and then click on the **Edit** button, as in the following example:

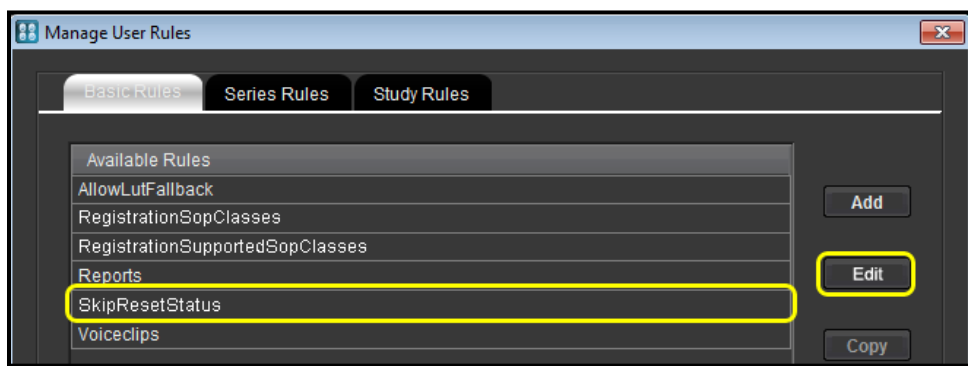


Figure 664: Editing the SkipResetStatus Rule

NOTE: Two other rules related to Auto Registration (**RegistrationSopClasses** and **RegistrationSupportedSopClasses**) are also included by default, but you should not need to edit them.

The **Edit Rule** window is displayed, as in the following example:

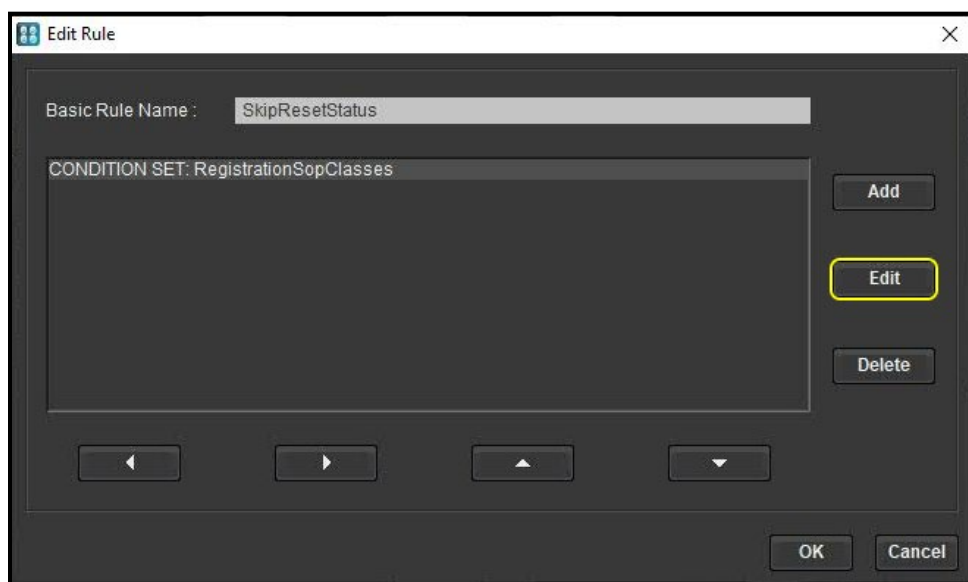


Figure 665: The Edit Rule Window

3. Click on the **Edit** button.

The **Edit Condition Set** window is displayed, as in the following example:

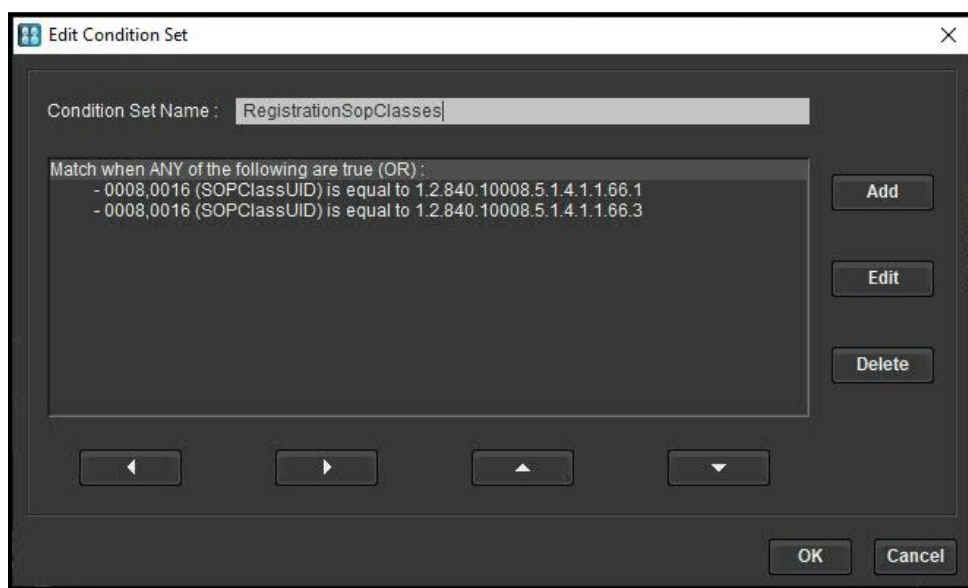


Figure 666: The Edit Condition Set Window

4. Edit the condition set as desired, referring to the detailed instructions in the Section 14.2.1 above as necessary, and click on the **OK** button at the bottom of the window.

14.2.10. Editing the ScoutSeriesRule User Rule

The **Displayed FOV Matching** feature within the Merge PACS Viewer links the magnification of all images, series and studies currently displayed in the Viewer to that of a “reference” image within the primary Study. The “reference” image will be the first non-key image in the primary Study window that is not a scout image.

The ScoutSeriesRule user rule determines what images are actually considered to be scout images for purposes of determining the “reference” image, and this rule can be customized to meet the needs of a particular site based on what their modalities do. By default, this rule provides a “match” (*i.e.*, the object will be considered a scout image) if any of the following are true:

- The **SeriesDescription** field contains “SCOUT”, “LOCAL”, “SCAN”, “SURV”, “TOPO” or “3-PL”.
- The **ScanOptions** field contains “SCOUT”, “LOCAL”, “SCAN”, “SURV”, or “TOPO”.
- The **ImageType** field contains “LOCALIZER”.

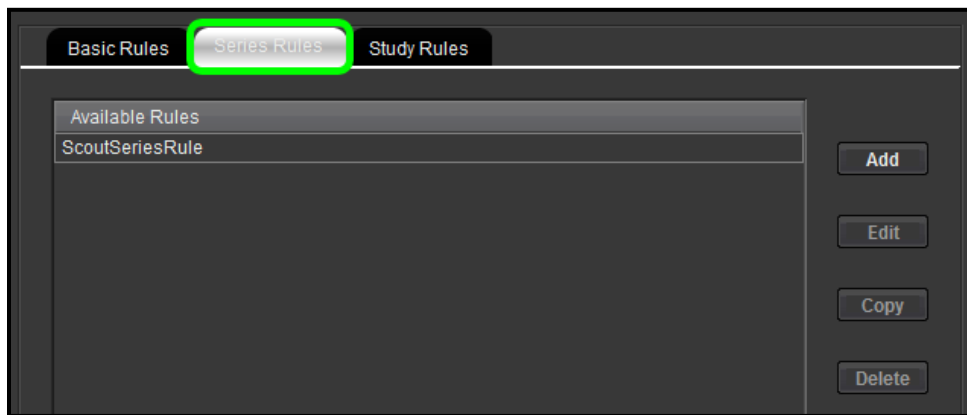
In addition, if a site already has a rule created that they want to use as the default, Merge PACS can be configured by Merge Healthcare Support to use that rule as the default.

To edit the ScoutSeriesRule rule:

1. Select **Manage Rules** from the **Hanging Protocol Menu** within the Viewer.

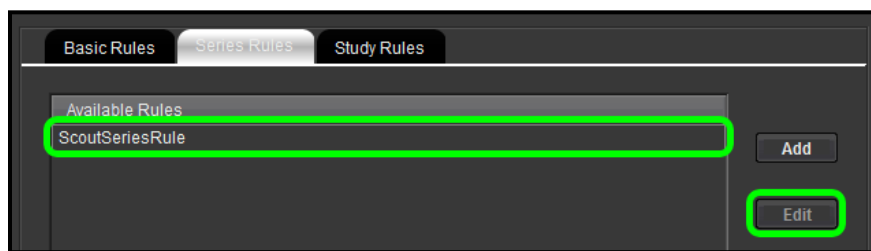
*The **Manage User Rules** window is displayed.*

2. Click on the **Series Rules** tab, as in the following example:



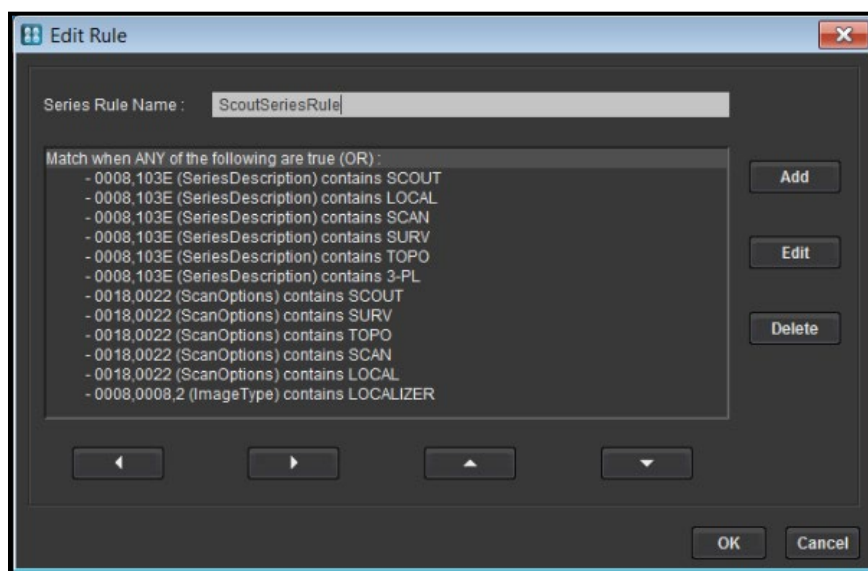
Viewing Series Rules

3. Click on the **ScoutSeriesRule** rule in the list of Available Rules once to highlight it and then click on the **Edit** button, as in the following example:



Editing the Voiceclips Rule

The **Edit Rule** window is displayed, as in the following example:



The Edit Rule Window

4. Edit the rule, referring to the detailed instructions in the Section 14.2.1 above as necessary, and click on the **OK** button at the bottom of the window.

14.2.11. Editing the Default AllowLutFallback User Rule

Prior to Merge PACS 8.2, if the Presentation State for an image did not have a Modality Look up Table (LUT), Merge PACS would use the Modality LUT found in the image itself. As of Merge PACS 8.2, however, Merge PACS will only use the Modality LUT found in the image itself for those presentation states which do not have a Modality LUT and which match the default

AllowLutFallback rule. For any presentation states that do not match the rule, Merge PACS will follow the DICOM standard and use the identity transformation for the image instead.

By default, the **AllowLutFallback** rule provides a “match” (*i.e.*, the image is eligible to have its Modality LUT used if the Presentation State associated with that image does not contain a Modality LUT) only if the Series Description does **not** contain the text “Generated by iConnect Access”.

In addition, if a site already has a rule created that they want to use as the default for this purpose, this can be set from within AmicasServer.properties.

To edit the default AllowLutFallback rule:

1. Select **Manage Rules** from the **Hanging Protocol Menu** within the Viewer.

The **Manage User Rules** window is displayed.

- At the **Basic Rules** tab, click on the **AllowLutFallback** rule in the list of Available Rules once to highlight it and then click on the **Edit** button, as in the following example:

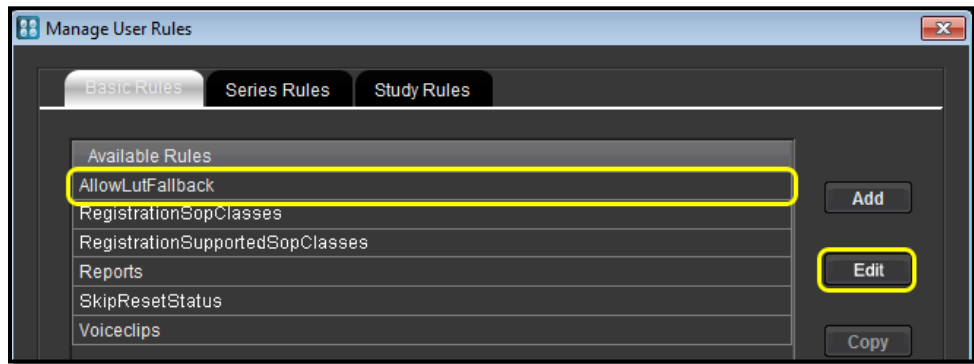


Figure 667: Editing the AllowLutFallback Rule

The **Edit Rule** window is displayed, as in the following example:

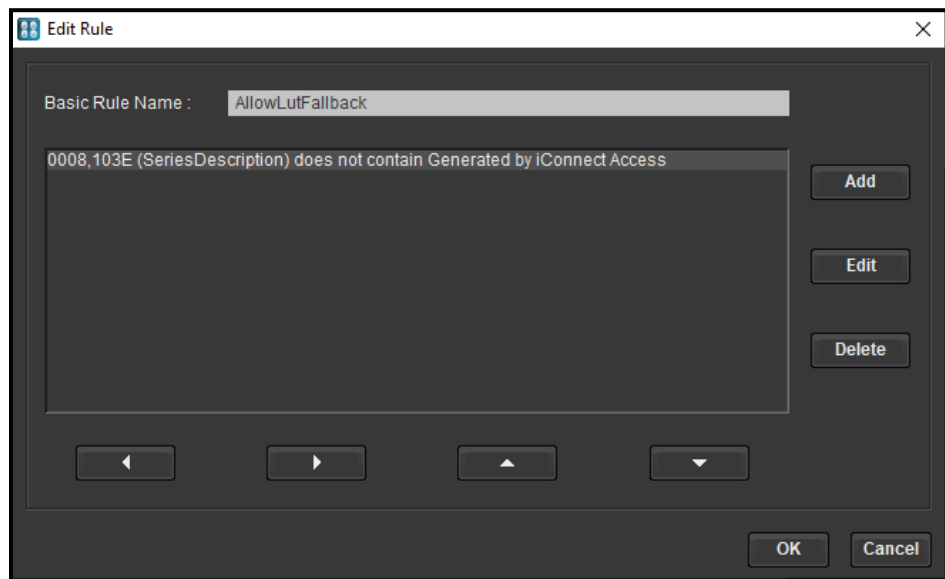
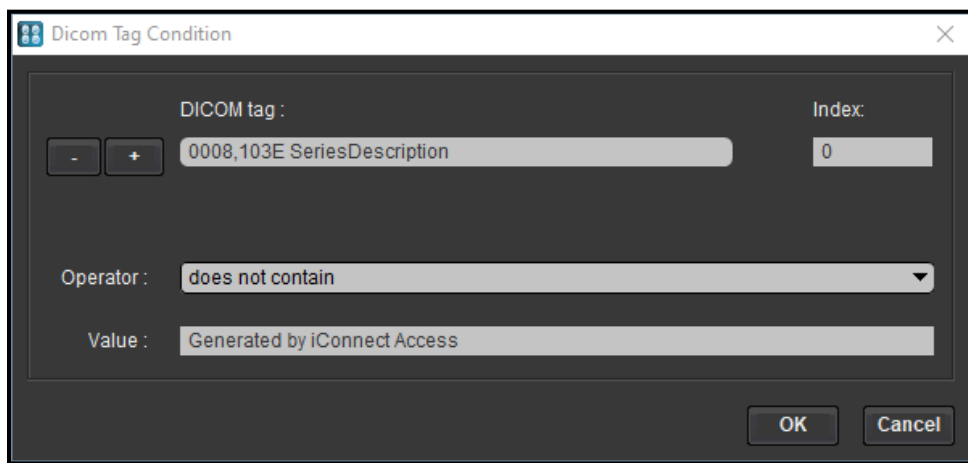


Figure 668: The Edit Rule Window

- Click on the **Edit** button.

The **DICOM Tag Condition** window is displayed, as in the following example:



DICOM tag :	Index:
- + 0008,103E SeriesDescription	0

Operator :

does not contain

Value :

Generated by iConnect Access

OK

Cancel

Figure 669: The DICOM Tag Condition Window

4. Edit the condition as desired, referring to the detailed instructions in the Section 14.2.1 above as necessary, and click on the **OK** button at the bottom of the window.

Chapter 15. Procedures for Configuring Launch and Control

The Launch and Control feature within Merge PACS allows you to set up a macro that can be used to determine which available applications (including the Merge PACS Viewer and any integrated 3rd-party viewers) should be launched when users click the “View Study” worklist action icon under various conditions (e.g., the type of study, the specific workstation or facility where the study is being read, etc.). When study is loaded into a 3rd-party viewer using this feature, the management of that study will stay with Merge PACS (e.g., the study will be locked and will be tracked for navigation in a go-to-next workflow).

All applications currently supported through **Viewer XML Integration** (M*Modal Fluency, Epic, etc.) are supported for use with the Launch and Control feature. Other applications that can make use of Merge PACS's XML interface may also be used.

Configuring Merge PACS to use Launch and Control feature involves the following steps:

- Associating existing **Viewer XML Integrations** with the **groups** whose members will be using this feature from the Merge PACS Users & Groups Management Pages.
- Optionally configuring **Facilities** and **Reader Locations** from the Advanced Workflow Management Pages in order to have those parameters considered.
- Optionally configuring **Specialties** from the Settings dialog of the Merge PACS Workstation in order to have that parameter considered.
- Configuring one or more **user rules** from the Hanging Protocol menu of the Merge PACS Viewer for use in setting the various conditions.
- Configuring a custom **macro** from the Preferences dialog of the Merge PACS Workstation to associate various rules with various actions.
- Assign the macro to the **Primary Exam Launch** user preference for each user who will be using this feature, either at the individual user level from the User Management page or at the group or site level using Templated User Management.

15.1. Associating Viewer XML Integrations with Groups

Viewer XML Integrations need to be associated with the group or groups whose members will be using the Launch and Control feature. This is done from the **Users & Groups Management Page**, as described in subsection 9.11.2 above.

NOTE: In order for these XML Integrations to be shown as available actions when creating the Launch and Control macro, as described in Section 15.5 below, you must also add them to whatever group you belong to (e.g., Administrator).

15.2. Configuring Facilities and Reader Locations

If you want the Launch and Control feature to consider the reader location (e.g., workstation) and facility where a study is being read when determining which application to launch, you will need to define the available facilities and reader locations within Merge PACS. This is done from the **Advanced Workflow Management Pages** as follows:

- Facilities are defined from the **Manage Facilities Page**, as described in subsection 10.5.2 above.
- Reader Locations are defined from the **Manage Reader Location Page**, as described in subsection 10.5.4 above.

15.3. Configuring Specialties

If you want the Launch and Control feature to consider the specialties associated with particular studies when determining which application to launch, you will need to define the available specialties within Merge PACS. This is done from the **Manage Specialties Page** on the Settings dialog of the Merge PACS Workstation, as described in subsection 10.6.3 above.

15.4. Configuring User Rules

Once you have configured all the necessary integrations, applications and parameters, you will need to configure one or more user rules that can be linked to specific actions within the custom Launch and Control macro to determine when certain actions (such as launching a particular application) should be performed. For example, if you want to have the macro launch one application for CR studies and a different application for MR studies, you will need to create separate user rules to determine if a study modality is “CR” and if a study modality is “MR”. Similarly, if you want the macro to launch a specific application only if the user is logged into a specific workstation or reading from a specific facility, you will need to create rules to determine if those conditions are met.

User rules are configured from the Hanging Protocol menu of the Merge PACS Viewer, as described in Section 14.2 above. The following, however, are some sample rules of the sort that you might want to configure for use with the Launch and Control macro.

15.4.1. Sample Rule – Study Modality is CR

To create a basic rule matching modality to “CR”:

1. From the Manage User Rules dialog, select the **Basic Rules** tab and click the **Add** button.

*The **Add Rule** dialog is displayed.*

2. At the Add Rule dialog, enter a short descriptive name for this rule (such as “IsCR”), click the **Add** button and select **Dicom Condition** from the drop-down list, as in the following example:

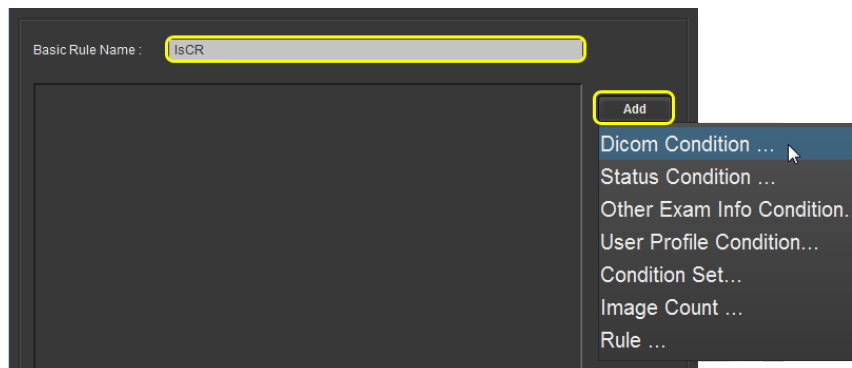


Figure 670: Adding a DICOM Condition

The **DICOM Tag Condition** dialog is displayed, as in the following example:

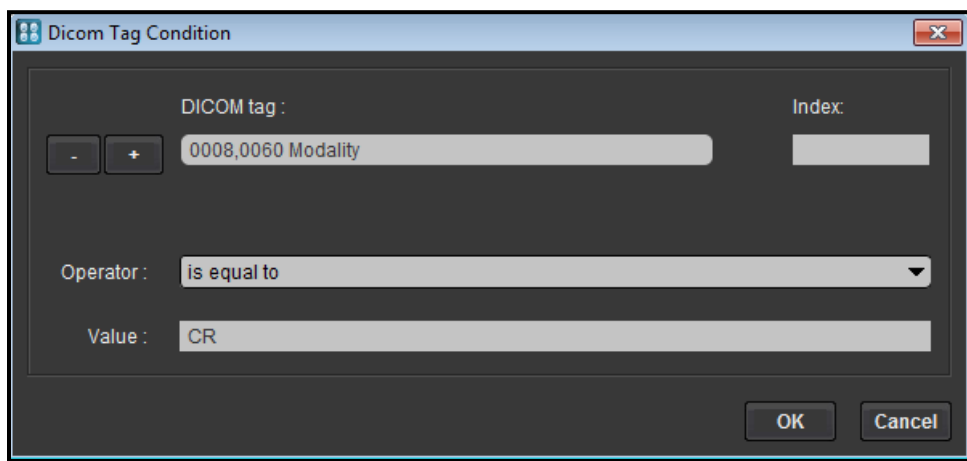


Figure 671: DICOM Tag Condition Dialog

3. At the DICOM Tag Condition dialog, do the following:
 - a. Enter **0008,0060 Modality** in the drop-down **DICOM Tag** field (note that you can simply enter “modality” in the field and then select the full name from the drop-down list).
 - b. Select **is equal to** from the **Operator** drop-down menu.
 - c. Enter **CR** in the **Value** field.
4. Click **OK** to complete the DICOM Condition and return to the Add Rule dialog.
5. At the Add Rule dialog, click **OK** to save the rule.

15.4.2. Sample Rule – Study Modality is MR and Specialty Is Neurology

To create a basic rule matching modality to “MR” and Specialty to “Neurology”:

1. At the Add Rule dialog, enter a short descriptive name for this rule (such as “MRandNeuro”), click the **Add** button and select **DICOM Condition** from the drop-down list, as in the following example:

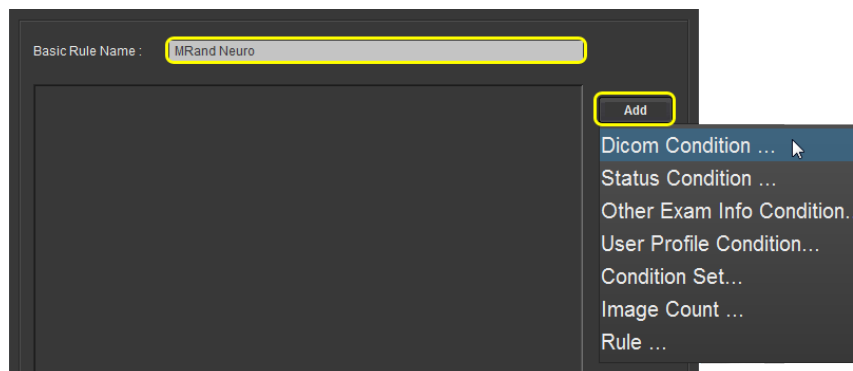


Figure 672: Adding a DICOM Condition

The **DICOM Tag Condition** dialog is displayed, as in the following example:

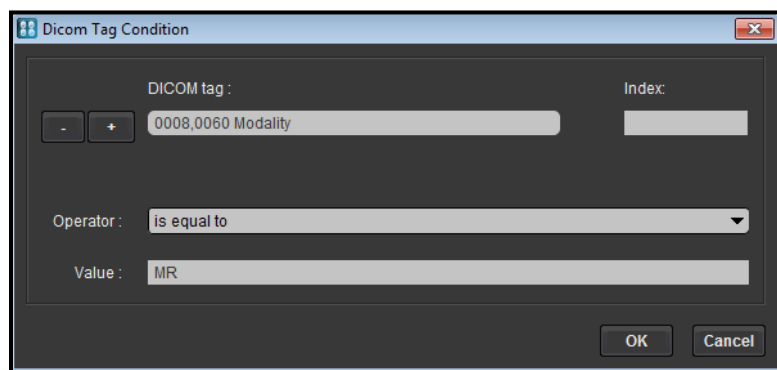


Figure 673: DICOM Tag Condition Dialog

2. At the DICOM Tag Condition dialog, do the following:
 - a. Enter **0008,0060 Modality** in the drop-down **DICOM Tag** field (note that you can simply enter “modality” in the field and then select the full name from the drop-down list).
 - b. Select **is equal to** from the **Operator** drop-down menu.
 - c. Enter **MR** in the **Value** field.
3. Click **OK** to complete the DICOM Condition and return to the Add Rule dialog.
4. At the Add Rule dialog, click the **Add** button and select **Other Exam Info Condition** from the drop-down list.

The **Other Exam Information Condition** dialog is displayed, as in the following example:

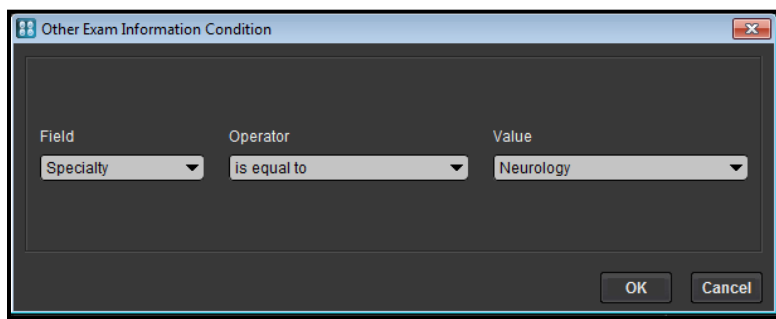


Figure 674: Other Exam Information Condition Dialog

5. At the Other Exam Information Condition dialog, do the following:
 - a. Select **Specialty** from the drop-down **Field** menu.
 - b. Select **is equal to** from the **Operator** drop-down menu.
 - c. Select **Neurology** from the drop-down **Value** field.

NOTE: The options in the Value drop-down menu are determined by the specialties that have been configured for this site, as described in subsection 10.6.3 above.

6. Click **OK** to complete the Other Exam Information Condition and return to the Add Rule dialog, as in the following example:

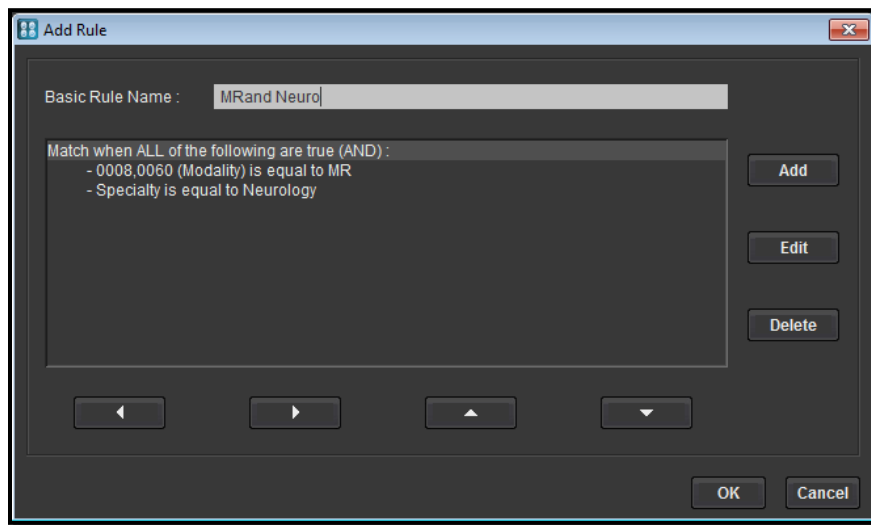


Figure 675: Completed Rule

7. At the Add Rule dialog, click **OK** to save the rule.

15.4.3. Sample Rule – Study Modality is CT and Location Is 3D Workstation

To create a basic rule matching modality to “CT” and Reader Location to “3D Workstation”:

1. At the Add Rule dialog, enter a short descriptive name for this rule (such as “CTand3DWorkstation”), click the **Add** button and select **DICOM Condition** from the drop-down list, as in the following example:

*The **DICOM Tag Condition** dialog is displayed.*

2. At the DICOM Tag Condition dialog, do the following:
 - a. Enter **0008,0060 Modality** in the drop-down **DICOM Tag** field (note that you can simply enter “modality” in the field and then select the full name from the drop-down list).
 - b. Select **is equal to** from the **Operator** drop-down menu.
 - c. Enter **CT** in the **Value** field.
3. Click **OK** to complete the DICOM Condition and return to the Add Rule dialog.
4. At the Add Rule dialog, click the **Add** button and select **User Profile Condition** from the drop-down list.

*The **User Profile Information Condition** dialog is displayed, as in the following example:*

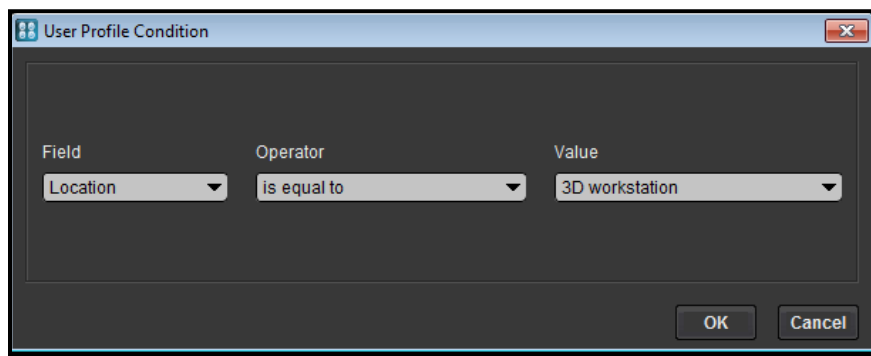


Figure 676: User Profile Condition Dialog

5. At the User Profile Condition dialog, do the following:
 - a. The only option currently available for **Field** is **location**.
 - b. Select **is equal to** from the **Operator** drop-down menu.
 - c. Select **3D Workstation** from the drop-down **Value** field.

NOTE: The options in the Value drop-down menu are determined by the reader locations that have been configured for this site, as described in subsection 10.5.4 above.

- Click **OK** to complete the User Profile Condition and return to the Add Rule dialog, as in the following example:

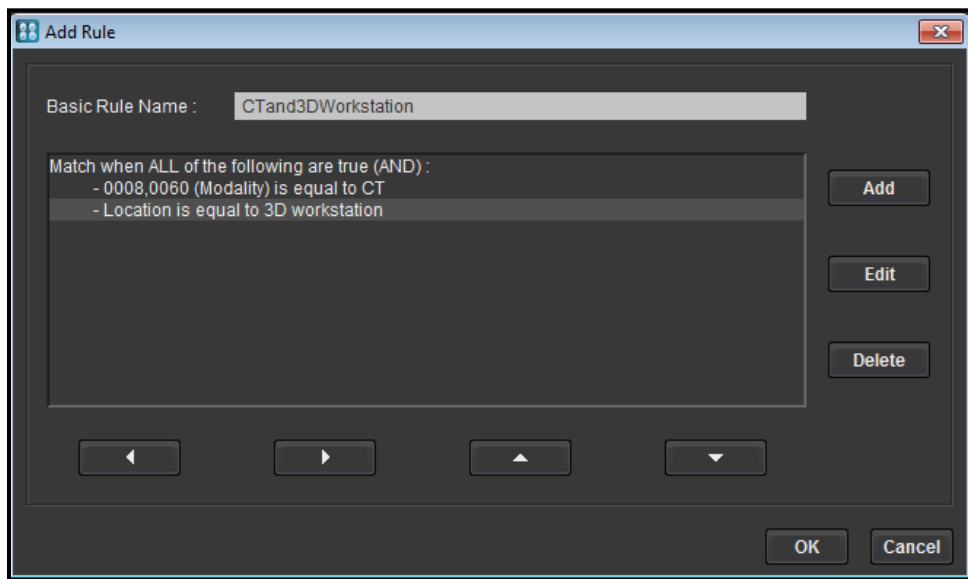


Figure 677: Completed Rule

- At the Add Rule dialog, click **OK** to save the rule.

15.5. Configuring a Custom Macro

Once you have configured all the necessary integrations, applications and parameters, as well as all the rules that you need, you will need to actually configure the macro to be used with the Launch and Control feature. For example, you might want to have a macro that launches the Fluency dictation application for all studies, one 3rd-party viewer (“Viewer A”) for MR Studies that are associated with the “Neurology” specialty, another 3rd-party viewer (“Viewer B”) for CT studies that are being read from the defined 3D Workstation, and the Merge PACS Viewer for all other studies.

Macros are configured from the Preferences dialog of the Merge PACS Workstation, as described in Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide*. The following, however, is a sample macro of the sort that you might want to configure for use with the Launch and Control feature.

To create a macro for the Launch and Control feature:

- From within the Merge PACS workstation, launch the Preferences dialog by clicking on the **Preferences** link in various right-click menus throughout the Merge PACS Workstation.

- At the Preferences dialog, click the **Macro Manager** option in the left-hand menu to display the Macro Manager, as in the following example:

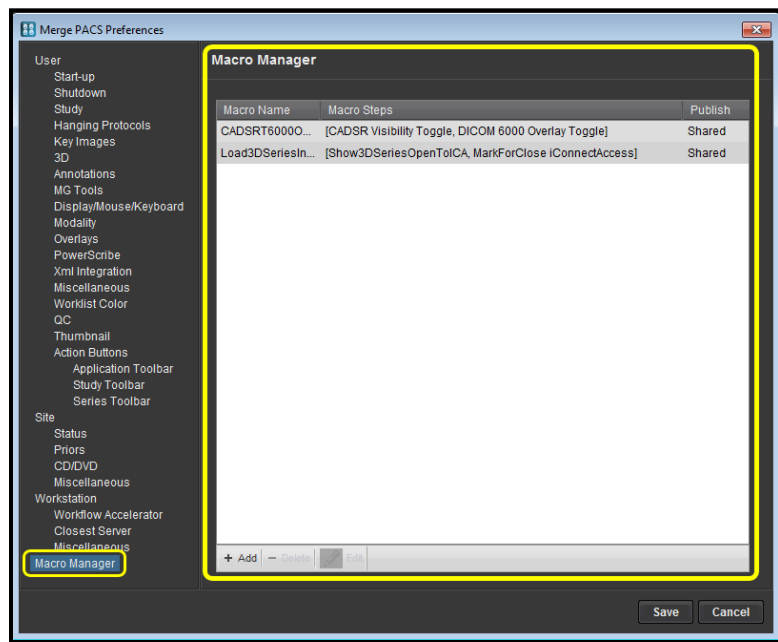


Figure 678: Macro Manager

- Click the **Add** button at the bottom of the Macro Manager screen.

The Macro Editor window is displayed, as in the following example:

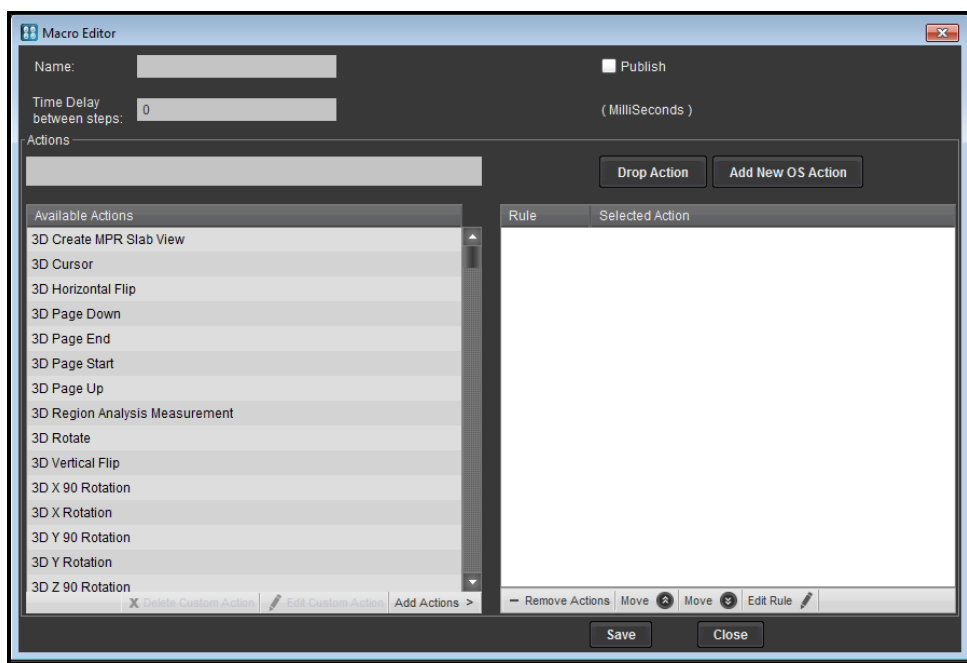


Figure 679: Macro Editor

NOTE: You may need to horizontally resize the Macro Editor window to display all the options at the bottom of each panel.

- In the **Name** field, enter a short, descriptive name for the macro, such as “LaunchPrimary”.

NOTE: The name cannot include any special characters such as .,:_-'"/>()!?\$&%#*#@^

- Locate the “**Deactivate Merge PACS Viewer for primary viewer launch**” action in the **Available Actions** panel (using the text field at the top of the panel to filter the list as necessary) and either double-click on it or click on it once to highlight it and then click on the **Add Actions** button at the bottom of the panel, as in the following example:

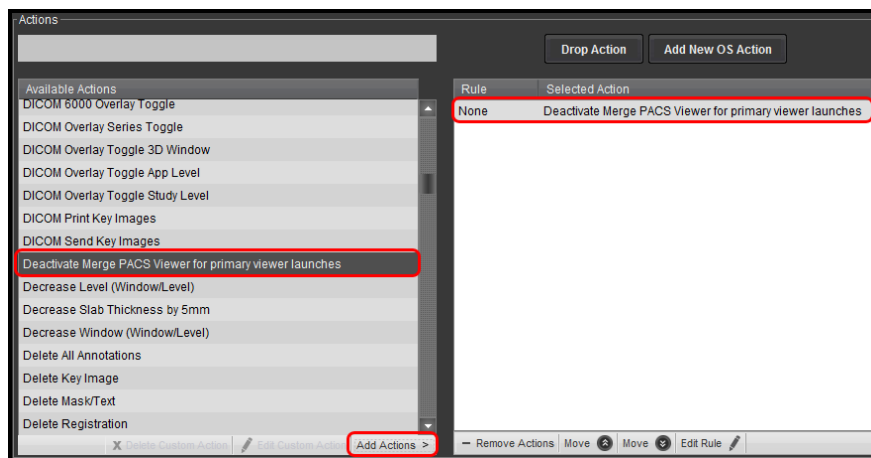


Figure 680: Adding “Deactivate Merge PACS Viewer for Primary Viewer Launch”

- Locate and add the “**Clear active XML integrations**” action, as in the following example:

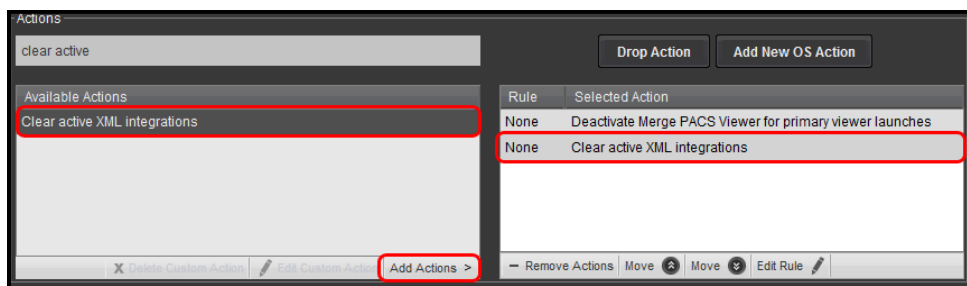


Figure 681: Adding “Clear Active XML Integrations”

- To have Fluency launched for all studies, locate and add the “**Add Fluency to the active XML integrations**” action, as in the following example:

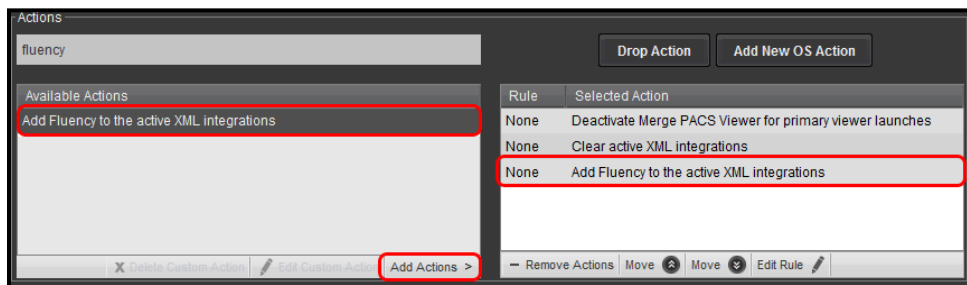


Figure 682: Adding Fluency

NOTE: Fluency must be configured as a Viewer XML Integrations and associated with the group you belong to in order for it to appear as an option in the Available Actions table.

8. To have “Viewer A” launch just for MR Studies that are associated with the “Neurology” specialty, perform the following steps:
 - a. Locate and add the “**Add Viewer A to the active XML integrations**” action, as in the following example:

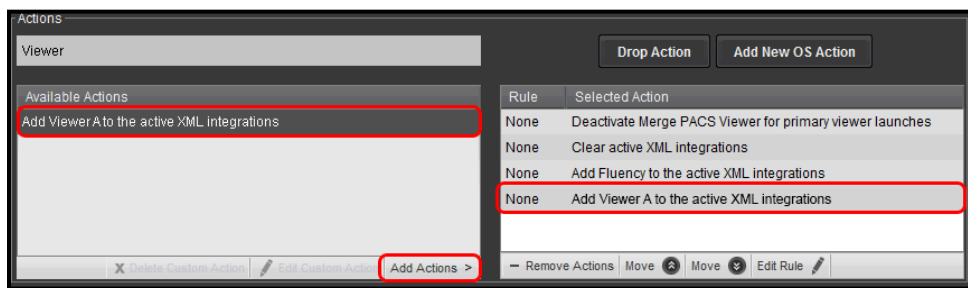


Figure 683: Adding “Viewer A”

NOTE: “Viewer A” must be associated with the group you belong to in order for it to appear as an option in the Available Actions table.

- b. Highlight the **Add Viewer A to the active XML integrations** action in the **Selected Action** panel and then click the **Edit Rule** button at the bottom of the panel.

The Rule Editor dialog is displayed as a separate pop-up window, as in the following example:

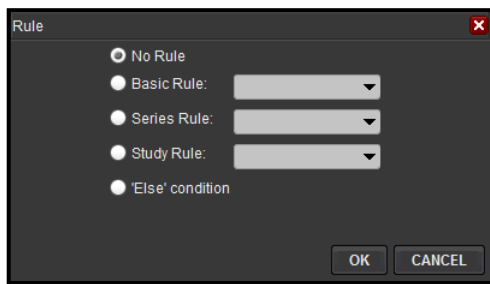


Figure 684: Rule Editor

- c. Click the **Basic Rule** radio button and then select the previously defined “MRand Neuro” rule from the associated drop-down menu, as in the following example:

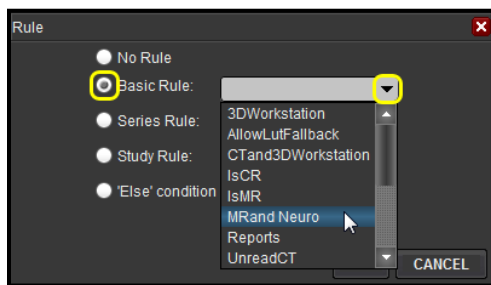


Figure 685: Selecting “MRandNeuro” Rule

NOTE: The rule must first be created, as described in Section 15.4 above, before it will appear as an option in the menu.

- d. Click **OK**.

The selected rule will be displayed in the Rule column of the Selected Action panel for this action, as in the following example:

Rule	Selected Action
None	Deactivate Merge PACS Viewer for primary viewer launches
None	Clear active XML integrations
None	Add Fluency to the active XML integrations
MRand Neuro	Add Viewer A to the active XML integrations

Figure 686: “MRandNeuro” Rule Applied to this Action

9. To have “Viewer B” launch just for MR Studies that are read from the pre-defined “3D Workstation” reader location, perform the following steps:
- Locate and add the “**Add Viewer B to the active XML integrations**” action, as in the following example:

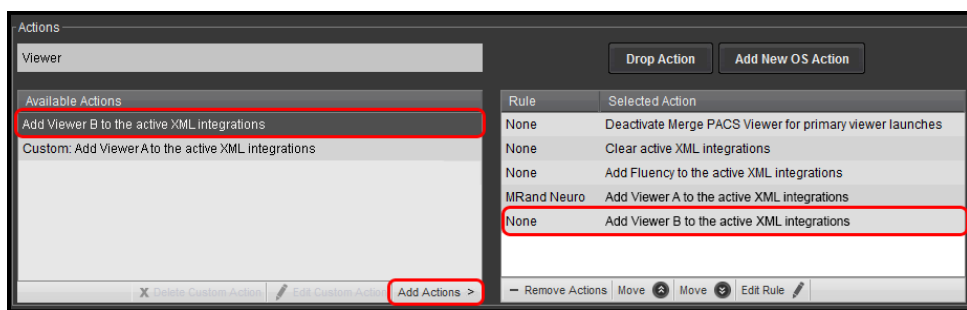


Figure 687: Adding “Viewer B”

NOTE: “Viewer B” must be configured as an XML Integration and associated with the group you belong to in order for it to appear as an option in the Available Actions table.

- Highlight the **Add Viewer B to the active XML integrations** action in the **Selected Action** panel and then click the **Edit Rule** button at the bottom of the panel.

The Rule Editor dialog is displayed as a separate pop-up window.

- Click the **Basic Rule** radio button and then select the previously defined “CTand3DWorkstation” rule from the associated drop-down menu, as in the following example:

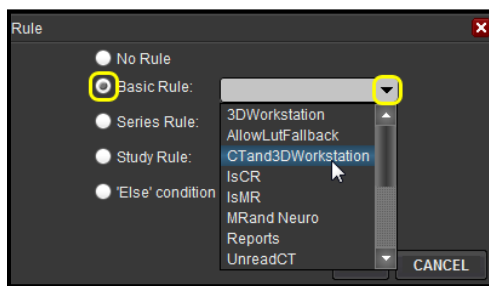


Figure 688: Selecting “CTand3DWorkstation” Rule

NOTE: The rule must first be created, as described in Section 15.4 above, before it will appear as an option in the menu.

- d. Click **OK**.

The selected rule will be displayed in the Rule column of the Selected Action panel for this action, as in the following example:

Rule	Selected Action
None	Deactivate Merge PACS Viewer for primary viewer launches
None	Clear active XML integrations
None	Add Fluency to the active XML integrations
MRand Neuro	Add Viewer A to the active XML integrations
CTand3DWorkstation	Add Viewer B to the active XML integrations

Figure 689: “CTand3DWorkstation” Rule Applied to this Action

10. To have the Merge PACS Viewer for all other studies not covered by the rules described above, perform the following steps:
- Locate and add the “**Activate Merge PACS Viewer for primary viewer launches**” action, as in the following example:

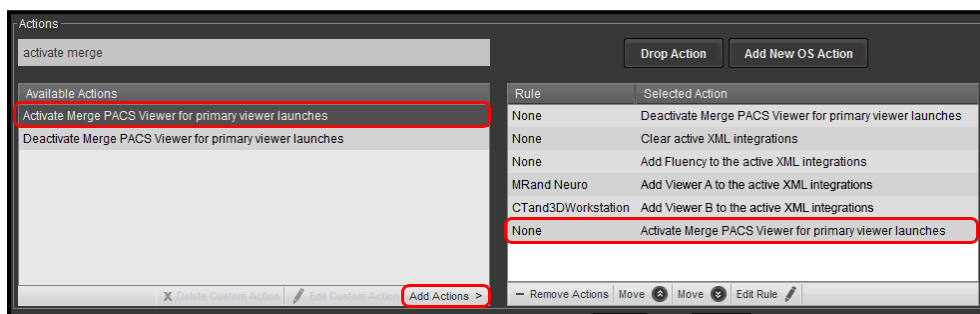


Figure 690: Activate Merge PACS Viewer for Primary Viewer Launches

- Highlight the **Activate Merge PACS Viewer for primary viewer launches** action in the **Selected Action** panel and then click the **Edit Rule** button at the bottom of the panel.

The Rule Editor dialog is displayed as a separate pop-up window.

- Click the ‘Else’ condition, as in the following example:

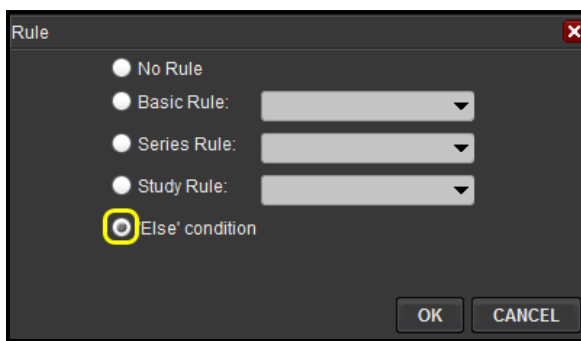


Figure 691: Selecting “CTand3DWorkstation” Rule

The “Else” Condition will be displayed in the Rule column of the Selected Action panel for this action, as in the following example:

Rule	Selected Action
None	Deactivate Merge PACS Viewer for primary viewer launches
None	Clear active XML integrations
None	Add Fluency to the active XML integrations
MRand Neuro	Add Viewer A to the active XML integrations
CTand3DWorkstation	Add Viewer B to the active XML integrations
Else	Activate Merge PACS Viewer for primary viewer launches

Figure 692: “Else” Condition Applied to this Action

NOTE: The “Else” Condition option will cause the associated action to be applied only if all other rule-based actions prior to this action in the macro (but after any earlier actions with an “Else” condition) are skipped because their rules evaluate to false.

- d. Click **OK**.
11. Click **Save** at the bottom of the **Macro Editor** dialog to save this macro.

The newly created macro will be displayed in the main Macro manager dialog, as in the following example:

Macro Manager		
Macro Name	Macro Steps	Publish
CADSRT60000...	[CADSR Visibility Toggle, DICOM 6000 Overlay Toggle]	Shared
LaunchPrimary	[Deactivate Merge PACS Viewer for primary viewer launches, Clear...	Private
Load3DSeriesIn...	[Show3DSeriesOpenToICA, MarkForClose IConnectAccess]	Shared

Figure 693: Newly Added Launch and Control Macro

12. Click **OK** at the bottom of the **Macro Manager** dialog.

15.6. Setting the Primary Exam Launch User Preference

Once you have created a custom macro for use with the Launch and Control feature, you need to assign that macro to the **Primary Exam Launch** user preference for each user who will be using this feature. This can either be done at the individual user level from the User Management page or at the group or site level using Templated User Management.

15.6.1. Setting the Primary Exam Launch User Preference for Individual Users

You can associate the Launch and Control macro with the “Primary Exam Launch” user preference for individual users from the Manage Users page, as described in Section 12.4 above.

To associate the Launch and Control macro with the “Primary Exam Launch” user preference for individual users:

1. Log into the Merge PACS Workstation as an Admin user.
2. Click on the **Settings** icon in the upper right corner of the Workstation Browser, as in the following example:

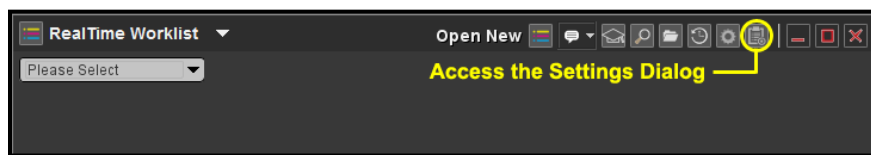


Figure 694: The Settings Icon

The **Settings** dialog is displayed.

3. Click on the **Users** link on the left-hand menu to access the Manage Users page, as in the following example:

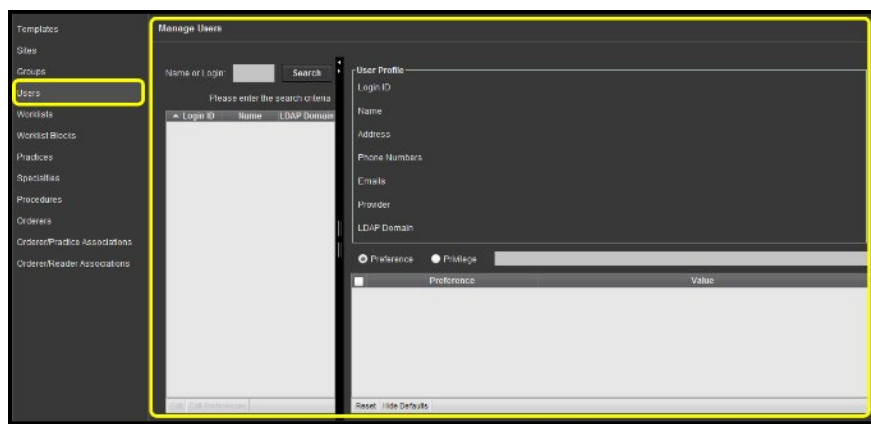


Figure 695: Manage Users Page

- At the top of the left-hand panel of the Manage Users page, locate the desired user via the **Name or Login** search field and **Search** button at the top of the panel, as in the following example:

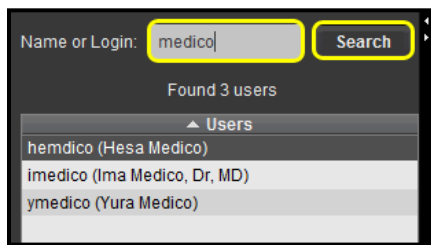


Figure 696: Searching for a User

A list of matching users is displayed

- Make sure the desired user is selected and click on the **Edit Preferences** button at the bottom of the left panel, as in the following example:

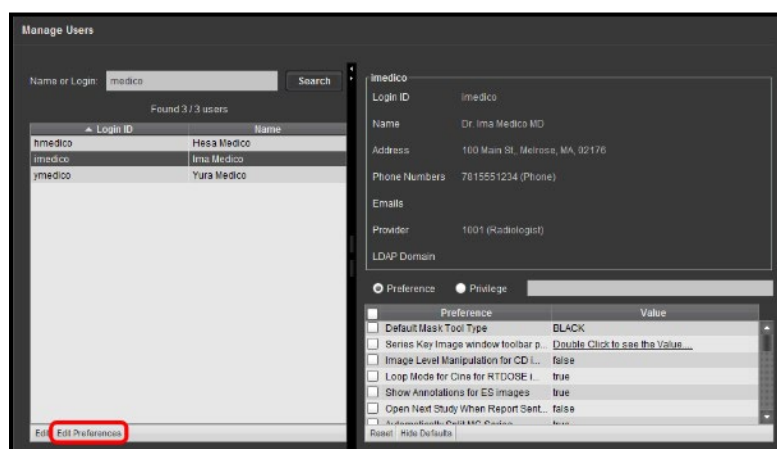


Figure 697: Edit this User's Preferences

*The **Merge PACS Preferences** dialog will be displayed for the selected user with only user-level preferences showing, as in the following example:*

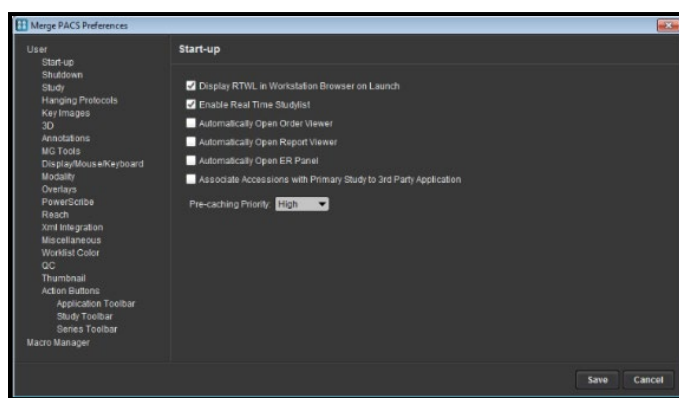


Figure 698: Merge PACS Preferences for this User

- Click on **Study** in the left-hand menu and then select the Launch and Control macro from the **Primary Exam Launch** Study Loading preference, as in the following example:

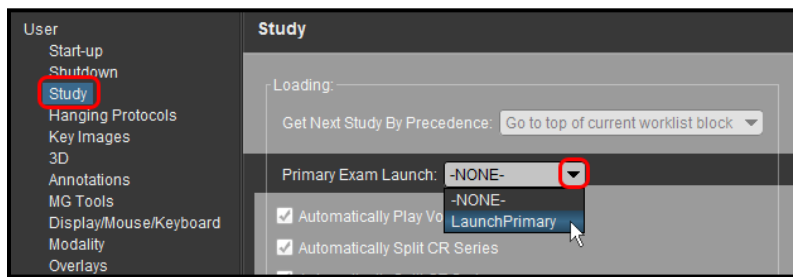


Figure 699: Selecting the Custom Macro Created for Launch and Control

- Click **Save** at the bottom of the Preferences dialog.

NOTE: This user preference can also be set by individual users from within the Preferences dialog of the Merge PACS Workstation, which will override anything set from the Manage Users page.

15.6.2. Setting the Primary Exam Launch User Preference via Templated User Management

You can associate the Launch and Control macro with the “Primary Exam launch” user preference for specified groups of users or for the entire site using Templated User Management, as described in Chapter 11 above.

To associate the Launch and Control macro with the “Primary Exam Launch” user preference for groups or the entire site:

- Log into the Merge PACS Workstation as a Merge Support or Admin user.
- Click on the **Settings** icon in the upper right corner of the Workstation Browser, as in the following example:

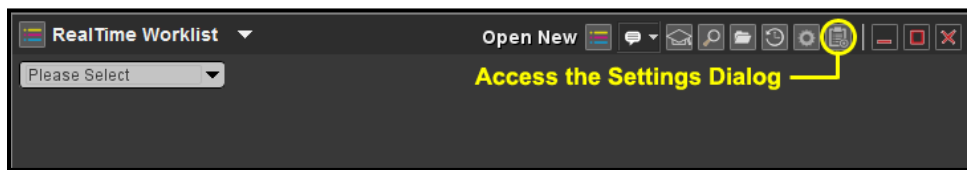


Figure 700: The Settings Icon

The **Settings** dialog is displayed with the **Manage Templates** page selected by default, as in the following example:

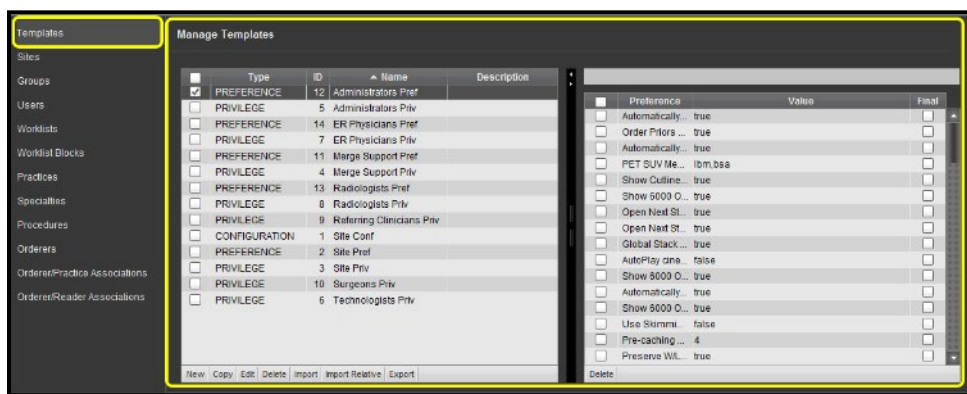


Figure 701: Manage Templates Page

3. Select the desired preference template from the list of existing templates by clicking on it. If you want to configure the preferences for a group, select the preference template associated with that group. If you want to configure the preference for the entire site, select the site preference template.
4. Click the **Edit** button at the bottom of the Existing Templates panel, as in the following example:

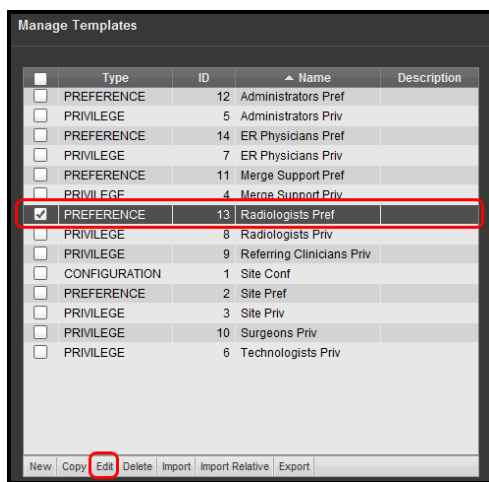


Figure 702: Manage Templates Page

The **Merge PACS Preferences** dialog will be displayed for the selected user with only user-level preferences showing, as in the following example:

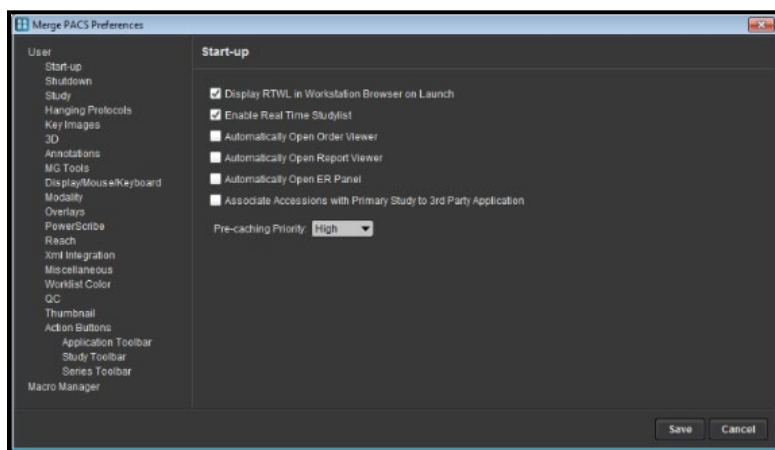


Figure 703: Merge PACS Preferences for this User

- Click on **Study** in the left-hand menu and then select the Launch and Control macro from the **Primary Exam Launch** Study Loading preference, as in the following example:

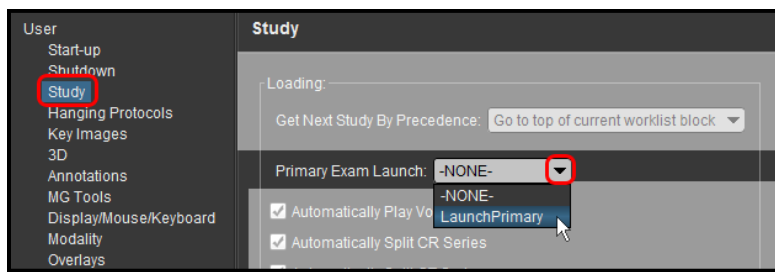


Figure 704: Selecting the Custom Macro Created for Launch and Control

- Click **Save** at the bottom of the Preferences dialog.

Chapter 16. Custom Configuring the Merge PACS Workstation

This chapter contains various procedures for custom configuring the Merge PACS Workstation to meet the needs of a particular site:

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16.1. Configuring Site-wide Workstation Preferences

The Merge PACS Preferences dialog, accessible from within the Merge PACS Workstation itself, allows you to configure a variety of site configuration preferences. These are the same preferences that can be configured in a Site Configuration Template, as described in Chapter 11 above, but any preferences set from the Preferences dialog will override the same preferences set in a linked Site Configuration Template.

To configure site-wide Workstation preferences:

1. Log into the Merge PACS Workstation as an Administrator.
2. From any right-click menu within the Workstation, select the **Preferences** option.

The **Merge PACS Preferences** dialog is displayed, as in the following example:

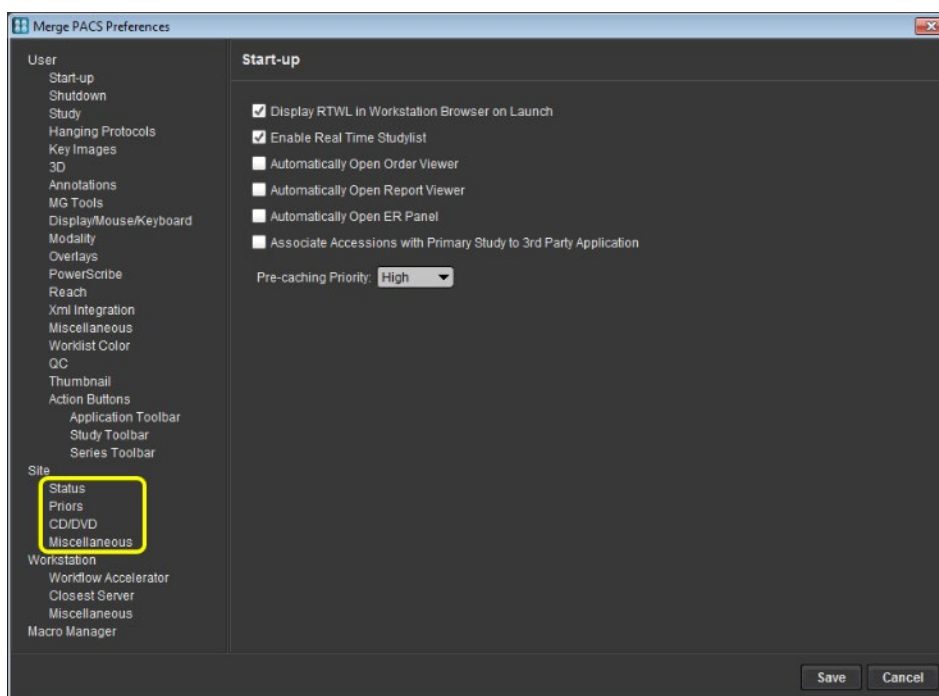


Figure 705: The Merge PACS Preferences Dialog

3. Click on the various Site links (highlighted in yellow in the example above) to set the preferences as desired.

NOTE: Refer to Appendix A.3 below for a description of all available Site Configuration preferences.

16.2. Saving and Restoring Quick Filter Column Defaults

Merge PACS includes a default set of data columns that are used for Quick Filter worklists within the RealTime Study List. If desired, you can configure the default list of data columns that are available for display directly from the Merge PACS Workstation itself and save your changes as a new system default, as well as restore the original default values. Customization can include the addition or removal of specific columns, as well as changes to the column order, sorting and size.

To save system-level Quick Filter column defaults:

1. Log into the Merge PACS Workstation as a user with Admin privileges, making sure your preferences are set to display RTSL instead of RTWL.
2. Use the Quick Filters section of RTSL to generate a sample Quick Filter worklist.
3. Use the column editing features, as described in subsection 3.6.5 of the *Merge PACS 8.2 Workstation Users Guide*, to customize the columns as desired.
4. In the **Available Worklist Panel**, right-click on the **Filters** folder and select **Save as System Column Defaults**, as in the following example:

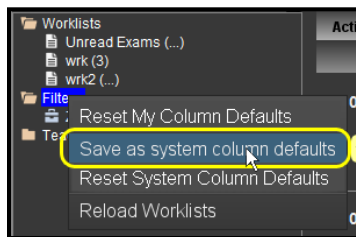


Figure 706: Saving as System Column Defaults

NOTE: Individual users can save their own defaults in their user preferences that will overwrite these saved system defaults. However, if a user decides to restore their column defaults (or if they have not saved any defaults of their own), these saved system defaults will be used.

To restore system-level Quick Filter column defaults:

1. Log into the Merge PACS Workstation as a user with Admin privileges, making sure your preferences are set to display RTSL instead of RTWL.
2. In the **Available Worklist Panel** of the RealTime Study List, right-click on the **Filters** folder and select **Reset System Column Defaults**, as in the following example:

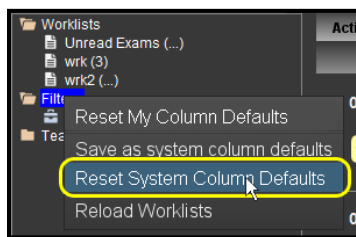


Figure 707: Resetting System Column Defaults

This will reset the system column defaults to those currently defined for your system.

16.3. Configuring Merge PACS Workstation to Work with Integrated Windows Authentication

If Integrated Windows Authentication has been enabled on the Merge PACS Server for the non-management web pages, Merge PACS Workstation users can have their client machines configured to make use of Integrated Windows Authentication's single login feature. This means that once the user has logged onto his or her computer, the user will not be prompted again for a username and password when accessing the Merge PACS Web Server.

CAUTION: Integrated Windows Authentication will only work with Internet Explorer.

To Configure Merge PACS Workstation to work with integrated Windows Authentication:

1. On the Client machine, launch Internet Explorer.
2. Go to **Tools** → **Internet Options** to display the Internet Options window.
3. At the **Internet Options** window, click on the **Security** tab and then click on the **Custom Level** button at the bottom of the screen to display the **Security Settings** window.
4. At the **Security Settings** window, scroll down to the bottom until you find the **User Authentication** options.
5. Click on the Automatic logon with current username and password option.
6. Click on the **OK** button to record your change and exit the **Security Settings** window and return to the **Internet Options** window, and then click on the **OK** button to exit the **Internet Options** window.

NOTE: If the user is connecting to the network remotely (e.g., via a VPN connection), the user will need to log onto his or her local machine with the same username and password that is required by the Merge PACS Web Server.

16.4. Launching the Merge PACS Workstation with Command Line Options

If desired, the Merge PACS Workstation can be run directly via the following file on the client machine:

C:\Program Files\Merge Healthcare\Merge PACS Workstation\Viewer_8.0.0.xxxxxx\bin\Viewer.exe

where xxxxxx represents the actual build number of the version currently running on the workstation.

NOTE: Previous versions of the Viewer may still be installed on the system so it is important to set the path to the current build.

This executable file does not require any command line options to run, but the following options are available:

Option	Description
"-J<XXX>"	Options that control JRE parameters (for example, "-J-Xmx500m" would change the Java memory setting for maximum Java heap size).
"-Pcache.path=<path>"	Overrides the default image cache location; may be useful if running multiple instances of the application. NOTE: Before using this command line option to start the Workstation, run the following command to shutdown the Study Cache Manager: taskkill /IM aStudyCacheMgr.exe CAUTION: This command line option should not be used in conjunction with the configurations specified in subsection 17.2.4.d below when launching the workstation.
"-Psecondary=true"	Causes a blue border to be displayed around the viewer; useful if running multiple instances to distinguish that it is a separate instance. By default, this option is set to "false."
"-PdefaultStudyUID=<SUID>"	Allows the viewer to automatically load a specific study found on the server with the specified study instance UID. If the value for this option is left blank ("-PdefaultStudyUID=") the currently loaded study will be cleared when the viewer is reused (and marked as "Read" depending on the user's privileges).
"-PlocalOnly"	Forces the application to run in local/stand-alone mode (no login dialog).
"-err" "<pathAndFilename>"	Specifies the path and filename to which to log error messages
"-out" "<pathAndFilename>"	Specifies the path and filename to which to log output messages
"-PserverRootURL=<theServerRootURL>", "-PdefaultUserName=<username>", "-PdefaultPassword=<password>"	When all three are set (for example, "-PserverRootURL= http://myhost/http://myhost " "-PdefaultUserName=jSmith" "-PdefaultPassword=password"), the specified criteria will be used and the login dialog will not be displayed
"-PviewerID=<UniqueViewerName>"	Specifies the name of the Viewer to reuse, if already running; else specifies the name to assign to the Viewer being started. If not specified the viewer will not be reusable by default. To reuse a viewer launched from the command line for URL pre-caching, PviewerID should not be specified, and Preusable (described below) should be set to "true."

Option	Description						
<code>"-PreuseViewer=true/trueNoReauth"</code>	Specifies what to do if the user attempts to launch the Viewer when another instance of the Viewer is already running, based on the defined value: <table> <tr> <th>Value</th><th>Description</th></tr> <tr> <td>true</td><td>The initial instance of the Viewer will be reused (<i>i.e.</i>, instead of launching a second instance that will run concurrently with the first instance) but the user will be reauthenticated if required.</td></tr> <tr> <td>trueNoReauth</td><td>The initial instance of the Viewer will be reused without prompting for user credentials or re-authenticating.</td></tr> </table> If viewer ID is specified, as described above, this option will have a default value of trueNoReauth and will not need to be included in the command line. You can still include this option to specify the actual value you would like, however. If viewerID is not specified, you will need to explicitly set this to either "true" or "trueNoReauth."	Value	Description	true	The initial instance of the Viewer will be reused (<i>i.e.</i> , instead of launching a second instance that will run concurrently with the first instance) but the user will be reauthenticated if required.	trueNoReauth	The initial instance of the Viewer will be reused without prompting for user credentials or re-authenticating.
Value	Description						
true	The initial instance of the Viewer will be reused (<i>i.e.</i> , instead of launching a second instance that will run concurrently with the first instance) but the user will be reauthenticated if required.						
trueNoReauth	The initial instance of the Viewer will be reused without prompting for user credentials or re-authenticating.						
<code>"-Preusable =true"</code>	Specifies that the Viewer should be reused for URL precaching purposes. By default this option is set to "false" if no viewerID is specified, as described above. Set this to "true" if you want to reuse this unnamed Viewer.						
<code>"-PsetVisible=false"</code>	Specifies that the Viewer should be hidden when launched. By default, this option is set to "true."						
<code>"-PshowSplashScreen=true"</code>	Specifies that the splash screen should be displayed when the Viewer is auto launched. By default, this option is set to "false."						
<code>"-PsetBrowserVisible=false"</code>	Will launch the Viewer directly without the Workstation Browser that displays RTWL, the Query Page, Patient Record, etc. By default, this option is set to "true."						

NOTE: Prior implementations of Merge PACS contained a **run.bat** file that ran aViewer.exe with certain command line options that were required at that time. Starting with v6 MR4, however, those command line options are no longer required and the run.bat file has been deprecated.

16.5. Enterprise-wide Distribution of Merge PACS Workstation

By default, the Merge PACS Workstation is installed on individual computers by having each user log onto the Merge PACS Server and letting the automatic installation process download the necessary files. If desired, however, sites can zip up the necessary files from one computer and then unzip the files onto other computers without the need for user involvement. The distribution and unzipping can be done manually or programmatically (e.g., via an MSI file).

In addition, a default MSI file is available that can be used.

16.5.1. Creating a Zip File of Merge PACS Workstation

To create a zip file of Merge PACS Workstation:

1. Install the Merge PACS Workstation on a single machine by accessing the Merge PACS Server from that machine and letting the automatic installation process download the necessary files.
2. Use any compression software to package the files found in the following directory:

`C:\Program Files\Merge Healthcare\Merge PACS Workstation`

NOTE: If a site wants to use a common (system-specific) cache override directory, this can be included in this package within the `C:\Program Files\Merge Healthcare\Merge PACS Workstation\Viewer_8.0.0.xxxxxx\config` directory.

CAUTION: If the Merge PACS Workstation is installed via this method, users will not be able to use PowerScribe.

16.5.2. Using the Default MSI File

A default MSI file is included with the PACS Server and can be found in the following location on the Merge PACS CD:

`<MergePACSCD>:\Amicas Distribution\Merge PACS Viewer
Software_8.2.0.xxxxxx.msi`

This MSI is intended to be used to install to the following default location on the client machine:

`C:\Program Files\Merge Healthcare\Merge PACS Workstation\Merge PACS Viewer
Software_8.2.0.xxxxxx`

The last folder in the installation path Merge PACS Viewer Software_7.3.0.xxxxxx is different for different versions of viewer. Users can install multiple versions of viewer on the same machine.

A variety of command line arguments may be included to customize how the Viewer is installed, as follows:

- If the viewer must be installed to a different location, this can be achieved using the TARGETDIR command-line argument, e.g.:

`msiexec /package Merge PACS Viewer Software_7.3.0.xxxxxx.msi /log
installLog.txt /quiet TARGETDIR="C:\MyDir"`

- A server address may optionally be specified during the installation. If specified, this server address will be used when the Merge PACS Workstation is first launched from the desktop shortcut or from aViewer.exe directly. The server address is specified using the SERVERADDRESS command-line argument, e.g.:

```
msiexec /package ViewerMSI.msi /log installLog.txt /quiet  
SERVERADDRESS="myservername"
```

- If you belong to the Administrator group and have installed the latest version of the Viewer using the "Just Me" option, running MSIEXEC without the FORCE=1 parameter, the Viewer will be installed as if the "Just Me" option were chosen. To force the Viewer to be installed into the Program Files directory, add ALLUSERS=1 and FORCE=1 parameter, e.g.:

```
msiexec /package ViewerMSI.msi /log installLog.txt /quiet ALLUSERS=1  
FORCE=1 SERVERADDRESS="myservername"
```

- If you do not belong to the Administrator group and want to use MSIEXEC to install the viewer as "Just Me", add the parameters MSIINSTALLPERUSER=1 ALLUSERS=2, e.g.:

```
msiexec /package ViewerMSI.msi /log installLog.txt /quiet  
MSIINSTALLPERUSER=1 ALLUSERS=2 SERVERADDRESS="myservername"
```

NOTE: Merge PACS supports the display of DICOM-encapsulated reports within the Merge PACS Workstation, but this requires that Adobe Reader needs to be installed on each Workstation that will be using this functionality. Therefore, you might want to push out Adobe Reader to these Workstations.

Chapter 17. Procedures for Maintaining Merge PACS

17.1. Overview

This chapter contains procedures for the ongoing maintenance of Merge PACS and for the resolution of problems in the operation of the system.

17.2. General Merge PACS Maintenance and Troubleshooting

This section contains procedures for:

- Monitoring the processing of studies by Merge PACS.
- Troubleshooting Merge PACS.

17.2.1. Monitoring the Processing of Studies

You can use the Merge PACS Management Web page to monitor the processing of studies. Point your browser at:

`http://<your-server>/merge-management`

Bookmark this page and examine it periodically over the course of the day in order to determine how the server is performing.

NOTE: Before you can view the Merge PACS Administration Web pages, Merge PACS challenges you for a username/password – contact Merge Healthcare Support if you need this information.

With appropriate access rights, you can keep track of the following:

Monitor	Purpose
Import Monitor	Are studies currently being imported?
Scheduler Monitor	Are periodic/explicit commands being executed?
Image Import Queue	Is there a backlog of studies to be processed?
Command Queue	Are explicit commands waiting to be executed?
Auditor	Were any studies not successfully imported?
Listener	Have DICOM Q/R requests been performed successfully?

17.2.2.Using the Import Monitor

On the Merge PACS Management Web page, click on **Monitors** → **Importer** to see the current activity of the **Importer** process, as shown in the following example:

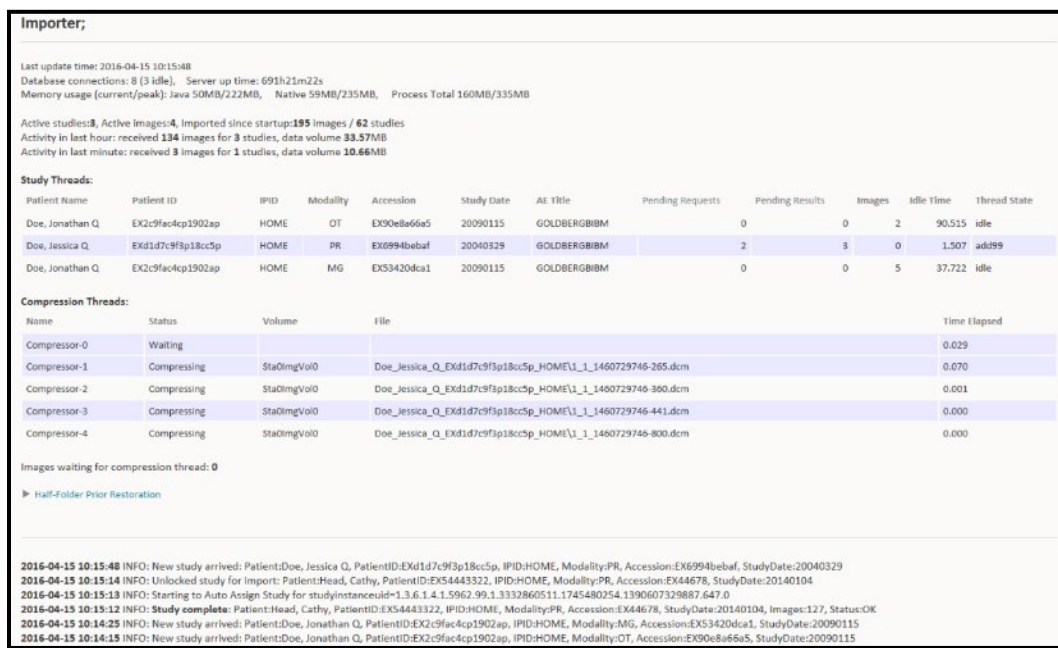


Figure 708: Sample Importer Status

The **Import Server** status window displays:

- Current Date and Time
- Server Version Information
- Images handled in the last 60 seconds
- Data Volume handled in the last 60 seconds
- Date, Time and Demographics of studies being imported
- Images imported since startup
- Length of time the Importer has been running (“up time”)

17.2.3.Using the Scheduler Monitor

On the Merge PACS Management Web page, click on **Monitors** → **Scheduler** to see the current activity of the **CommandScheduler** process, as shown in the following example:

```
CommandScheduler; Version: 6.6.2

Last update time: 2014-07-25 09:14:49
Database connections: Demographic(3) Preference(-14), Memory Usage: Current(22MB) Peak(22MB)
2014-07-25 09:12:53 Status: Scheduler: Command: [AnetCommunicator - Local ANETDB] completed.
2014-07-25 09:12:53 Status: AnetCommunicator - Total Sent=0 Total Not Sent=0 (New=0 InProcess=0 Errors=0) Total Duplicates Sent=0
2014-07-25 09:11:38 Status: Found 0 candidate studies to delete.
2014-07-25 09:11:38 Status: DICOM Deletion Qualifier query:select AmicasStudy.StudyInstanceUID from AmicasStudy where AmicasState in ('IMPORTED', 'FAILED', 'PARTIALR') and
AmicasStudy.ModifiedDate < ? and not exists (select StudyInstanceUID from InImageQueue where InImageQueue.StudyInstanceUID = AmicasStudy.StudyInstanceUID) and not
exists (select StudyInstanceUID from ForwardingQueue where ForwardingQueue.StudyInstanceUID = AmicasStudy.StudyInstanceUID) and not exists (select StudyInstanceUID from
StudyLock where StudyLock.StudyInstanceUID = AmicasStudy.StudyInstanceUID) order by ModifiedDate asc fetch first 200 rows only for read only optimize for 200 rows
NOTE That param='2014-07-25 08:11:38.563'
2014-07-25 09:11:38 Status: Running CheckResources Command [DeleteLLBasedOnWatermarksOrderByStudydateandImportdate: false,
DeleteCmprBasedOnWatermarksOrderByStudydateandImportdate: false, DeleteDicomBasedOnTime: true, MaxDeleteCompressed: 200, DeleteVoiceClipsBasedOnTime: false,
DeleteVoiceClipsBasedOnWatermarks: false, MaxDeleteMisc: 200, MaxDeleteLossless: 200, DeleteLosslessCompressedOlderThanHours: 720, DeleteLosslessBasedOnWatermarks:
true, MaxDeleteDICOM: 200, DeleteCompressedBasedOnWatermarks: false, DeleteDicomBasedOnWatermarks: true]
2014-07-25 09:07:50 Status: Scheduler: Command: [AnetCommunicator - Local ANETDB] completed.
```

Figure 709: Sample Command Scheduler Status Window

The **CommandScheduler** status window displays the periodic and explicit commands that the CommandScheduler is currently executing:

- **Periodic commands** are scheduled for repeated execution at certain time intervals. The following are common periodic commands.

Command	Description
CleanupUncompressed	Uncompressed DICOM images are deleted every hour.
SystemMonitor	The Import Queue is checked every 15 minutes. If it exceeds 1000 images an email alert can be sent out via Remote Notification.
UpdateStatus	Email updates with Importer statistics and the current state of the database can be automatically sent out via Remote Notification.

- **Explicit commands** are commands that are either initiated manually by a Merge PACS administrator or else performed automatically by a process running within Merge PACS. For example, the following are common explicit commands:

Command	Description
Validate	Editing of study demographics
Update	Editing of patient demographics
Delete	Deleting patient studies
Merge	Merging two patients together
CleanupDB	Cleaning up database log tables
SendMail	Sending general e-mail messages

17.2.4. Troubleshooting the Merge PACS Workstation

The following paragraphs describe solutions to various known issues with Merge PACS Workstation.

a. User Session Is not Closed Even After Closing the Viewer (IE8 and Later Only)

Internet Explorer 8 introduced an enhanced “Session Merging” feature that lets multiple browser windows and tabs share login credentials. As a result, if a user has any component of the Merge PACS Workstation open in one window or tab and then opens another window or tab, the user’s login credentials will be preserved in the new window or tab even after the original window or tab (actually running the Merge PACS Workstation) is closed. As a result, the user will be able to launch the Merge PACS Workstation in the second window or tab without being prompted to login.

If this constitutes a security concern, there are two possible workarounds:

- Instruct users to open new browser windows by selecting **New Session** from the File menu within Internet Explorer instead of **New Window**.
- Create a shortcut to Internet Explorer for each user with **-nomerge** included as a parameter.

b. Merge PACS Workstation Runs Slowly Due to Virus Scanning

Although it is important to have virus scanning software running on the client workstation, performance can be adversely affected if the virus scanning software has to scan large numbers of Merge PACS files, including image data, which may be on the workstation’s hard drive. Refer to subsection 1.6.1 above for a list of files and directories that should be excluded from virus scanning.

c. Studies with Extremely Large Images Cannot Be Viewed

By default, the Merge PACS Viewer allocates 15% of physical memory for Java when running under a 32-bit (“small”) OS and 25% of physical memory for Java when running under a 64-bit (“large”) OS. In addition, it will use a maximum of 50% of the remaining memory for the image cache and a maximum of 5% of the remaining memory for the image header cache. Studies with extremely large images, however (especially Mammography studies with large film scans), may cause “Out of Memory” errors to display with these default settings.

These default settings can be changed to reserve a specified amount of memory for Java as well as to reserve a specified amount or percentage of memory for image caches and/or image headers. This can be done on a Workstation-by-Workstation basis for users who work with these types of studies or on the PACS Server (by Merge Support) to be used as a default for all Workstations that don’t override the settings locally.

Note the following:

- The “Advanced” tab of the Merge PACS Workstation “About” box can be used to determine the current memory usage of an individual Workstation, as described in the *Merge PACS 8.2 Workstation Users Guide*.

- Depending on the administrative privileges of the user who downloaded the Workstation software, the necessary configuration file will be stored in one of the following locations on the Workstation:
 - C:\Program Files\Merge Healthcare\Merge PACS Workstation\Viewer_8.0.0.xxxxxx\config
 - C:\Users\dicom\AppData\LocalLow\AMICAS\config\
- If a configuration file defines both a percentage of memory to be reserved and an absolute value of memory to be reserved for the image cache and/or image header cache, the percentage settings will take precedence over the absolute values.

To specify the memory settings for a particular Workstation (Web Launch):

1. On the desired Workstation, locate the following file as appropriate:
 - C:\Program Files\Merge Healthcare\Merge PACS Workstation\Viewer_8.2.0.xxxxxx\config\AmicasViewerLaunchConfig.txt
 - C:\Users\dicom\AppData\LocalLow\AMICAS\config\AmicasViewerLaunchConfig.txt
2. Copy and rename the file to the following location as appropriate:
 - C:\Program Files\Merge Healthcare\AmicasViewerLaunchConfig_override.txt
 - C:\Users\<loginID>\AppData\Local\AMICAS\AmicasViewerLaunchConfig_override.txt
4. Open the newly copied file in a text editor and add (or modify if already present) one or more of the following properties as desired:
 - To set the amount of memory (in megabytes) to be reserved for **Java**, use the following lines of code:


```
-J-Xmx<MEGABYTES>m (e.g., -J-Xmx192m)
-J-Xms<MEGABYTES>m (e.g., -J-Xmx192m)
```
 - To set an **absolute** amount of memory (in megabytes) to be reserved for the **Image Cache**, use one of the following:
 - For 32-bit operating systems:


```
com.amicas.viewer.util.MemoryUtils.ImageCacheSizeInMbSmall=
<MEGABYTES>
```
 - For 64-bit operating systems:


```
com.amicas.viewer.util.MemoryUtils.ImageCacheSizeInMbLarge=
<MEGABYTES>
```
 - To set a **percentage** of memory (in megabytes) to be reserved for the **Image Cache**, use one of the following:
 - For 32-bit operating systems:


```
com.amicas.viewer.util.MemoryUtils.ImageCacheSizeInPercentSmall=<PERCENT>
```
 - For 64-bit operating systems:


```
com.amicas.viewer.util.MemoryUtils.ImageCacheSizeInPercentLarge=<PERCENT>
```

NOTE: Prior to 7.1, this percentage reserved for the Image Cache was controlled by **-J-Dcom.amicas.study.model.ImageFrameCache.MaxMemPercentSmall** and **-J-Dcom.amicas.study.model.ImageFrameCache.MaxMemPercentLarge**. Those settings will still be used if they exist (uncommented) in the configuration file and the new **com.amicas.viewer.util.MemoryUtils.ImageCacheSizeInPercentSmall** and **com.amicas.viewer.util.MemoryUtils.ImageCacheSizeInPercentLarge** settings are not entered.

- To set an **absolute** amount of memory (in megabytes) to be reserved for the **Image Header Cache**, use one of the following:

- For 32-bit operating systems:

```
com.amicas.viewer.util.MemoryUtils.HeaderCacheSizeInMbSmall=
<MEGABYTES>
```

- For 64-bit operating systems:

```
com.amicas.viewer.util.MemoryUtils.HeaderCacheSizeInMbLarge=
<MEGABYTES>
```

- To set a **percentage** of memory (in megabytes) to be reserved for the **Image Header Cache**, use one of the following:

- For 32-bit operating systems:

```
com.amicas.viewer.util.MemoryUtils.HeaderCacheSizeInPercentSmall=<PERCENT>
```

- For 64-bit operating systems:

```
com.amicas.viewer.util.MemoryUtils.HeaderCacheSizeInPercentLarge=<PERCENT>
```

5. Save and exit the file. The changes will take effect the next time the Merge PACS Viewer is launched.

To change the memory settings for a Workstation (aViewer.exe launch):

One or more of the following command line arguments can to be added when launching aViewer.exe:

- For 64-bit Operating Systems:

```
"-J-Xmx<MEGABYTES>m"
```

```
"-J-Xms<MEGABYTES>m"
```

```
"-J-Dcom.amicas.viewer.util.MemoryUtils.ImageCacheSizeInPercentLarge=
<PERCENTAGE>"
```

```
"-J-Dcom.amicas.viewer.util.MemoryUtils.HeaderCacheSizeInMbLarge=
<PERCENTAGE>"
```

- For 32-bit Operating Systems:

```
"-J-Xmx<MEGABYTES>m"
```

```
"-J-Xms<MEGABYTES>m"
```

```
"-J-Dcom.amicas.viewer.util.MemoryUtils.ImageCacheSizeInPercentSmall=
<PERCENTAGE>"
```

```
"-J-Dcom.amicas.viewer.util.MemoryUtils.HeaderCacheSizeInMbSmall=
<PERCENTAGE>"
```

d. C: Drive on Workstation Is Not Optimal for Image Cache Directory

By default, the PACS Workstation image cache directory is located on the C: drive. For some workstations, however (especially those with reduced space on the C: drive), system performance can be enhanced by having the image cache directory located on a different drive.

The default setting can be changed on a Workstation-by-Workstation basis for users who have reported performance issues. This is done by creating a special text file that specifies where the image cache directory should be stored.

NOTE: An alternative method for specifying the image cache directory is described in Section 16.4 above using the “-pcache.path” command line option. That command line option should not be used in conjunction with the instructions specified below.

To change the location of the image cache directory:

1. Open a text editor such as Windows Notepad.
2. Enter the desired cache directory in this text file. This directory can have any name, but it needs to be accessible to all users (*i.e.*, it should not depend on login or require administration rights). For example, if you want the image cache to be located on the D: drive, you might type the following:

```
d:\amicas_cache
```

NOTE: If the directory does not exist, it will automatically be created.

3. Save this text file to the following location:

```
C:\Users\<user>\AppData\LocalLow\AMICAS\config\cacheoverridedirectory.txt
```

4. Log off the Windows user account and log back in for the current user in order for the change to take effect.

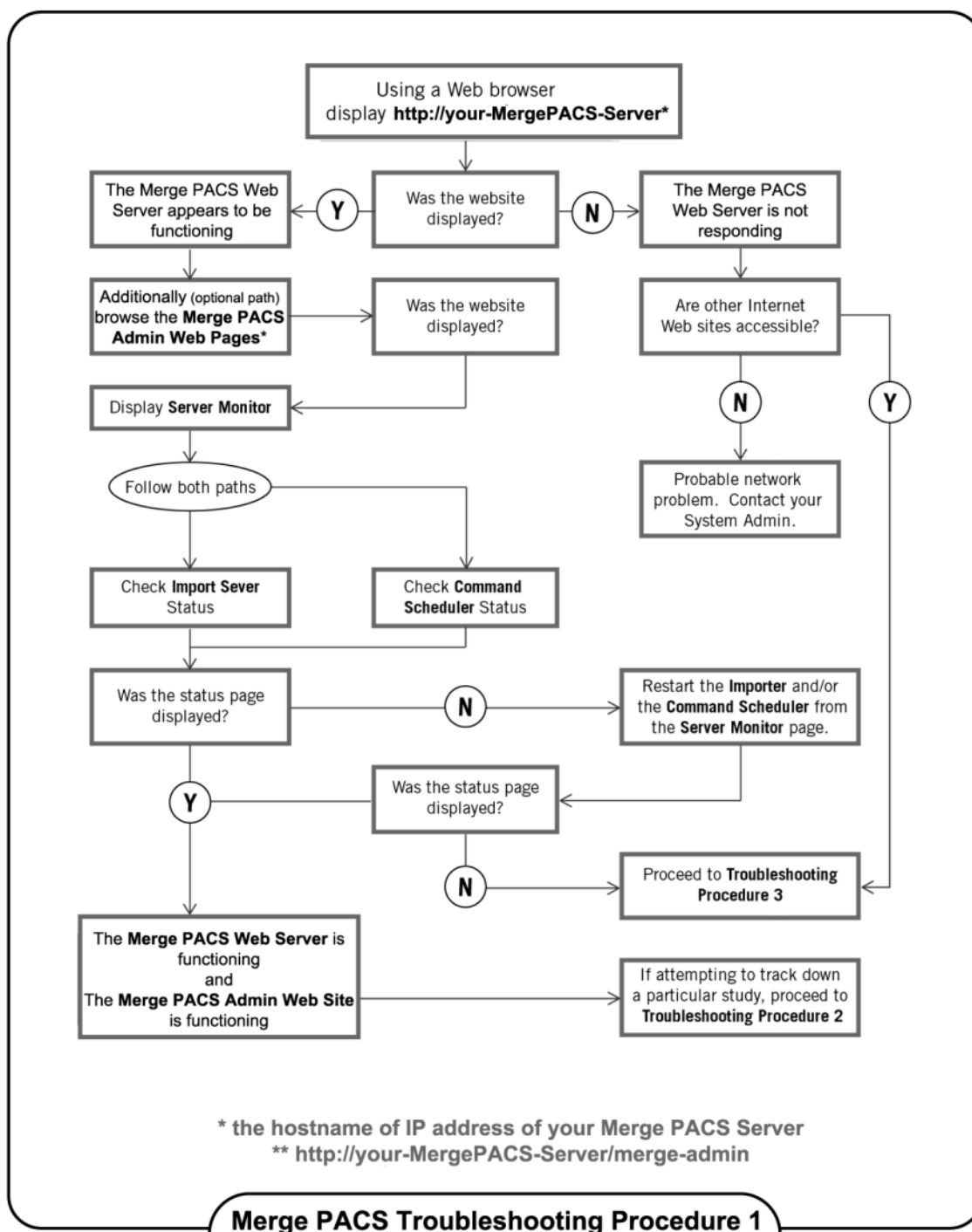
17.2.5. Troubleshooting Merge PACS

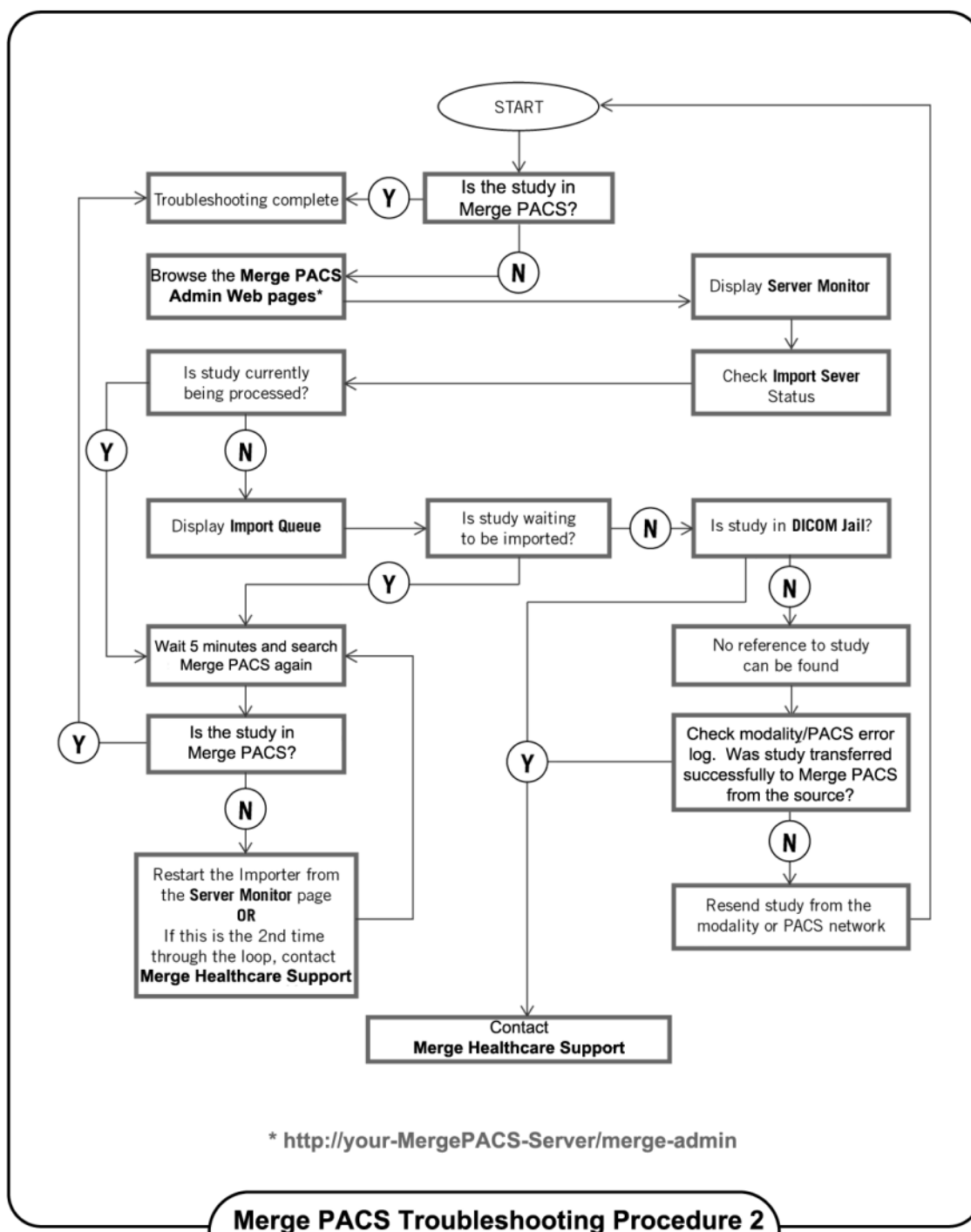
This section lists several potential problems you might encounter with Merge PACS and, for each problem, provides you with a troubleshooting flow chart that addresses the problem.

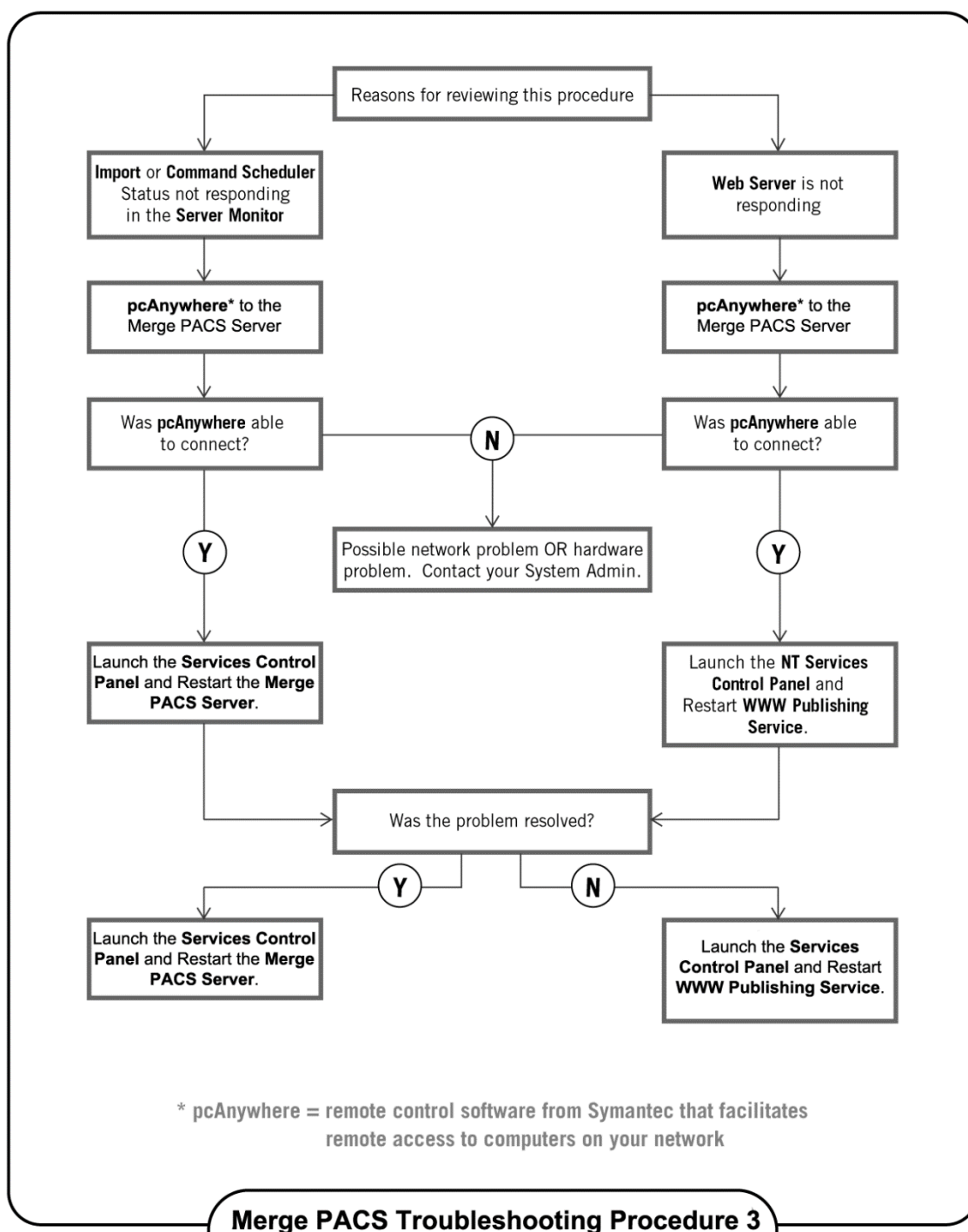
Problem	Procedure
Merge PACS not responding over the Web?	Troubleshooting Procedure 1
Patient study not found in Merge PACS?	Troubleshooting Procedure 2
Server Monitor not responding?	Troubleshooting Procedure 3
Customer unable to view reports	Report Troubleshooting
Viewer - image quality issues?	Collect patient demographics* and call Merge Healthcare Technical Support
Viewer - installation problems?	Collect system information** and call Merge Healthcare Technical Support

Table 9-1: Troubleshooting Approaches

*** Patient Demographics:****Medical Record Number or Patient ID****Patient Name****Accession Number (if applicable)****Study Date and Time****Modality****** System Information Required:****Customer Name****Contact Person****Telephone No.****Pager No.****Hostname/Computer Name****License Number****Station ID of Merge PACS Server****IP Address of Merge PACS Server****Modem No. (if applicable)**







17.3. Shutting Down and Starting Up the Merge PACS Server

17.3.1.Merge PACS Server shutdown procedure

Depending on how long the Merge PACS Server is going to be down, it may be necessary to interrupt the feed of images to the Merge PACS Server. If the downtime is expected to be more than half an hour then it is recommended (though not essential) to turn off the image feed.

If this is not done, and the Merge PACS Server is down for an extended period of time, then the devices that are sending images to Merge PACS will start generating repeated errors while trying to connect to Merge PACS. These errors could impact the performance of the sending device.

To shut down the Merge PACS Server:

1. Shutdown the Amicas service using the Services Control Panel.
2. Shutdown the Apache service using the Services Control Panel.
3. Shutdown the computer.

17.3.2.Merge PACS Server startup procedure

To start up the Merge PACS Server:

1. Start the computer. The Amicas service, Apache service and Database services will start automatically.
2. Browse to <http://<your-server>/merge-management>. If you can see this web page, the WWW Publishing Service is running. Click on the Server Monitor button and verify that the Merge PACS processes are running.
3. View images as per usual and verify that they are accessible.
4. Restart the image feed to the Merge PACS Server and determine if new images are being imported and stored successfully on the system. If the image feed was not stopped at the outset of this procedure then new studies should start coming through automatically.

17.4. Automatic Assignment of Exams

By default, all users who have access to the Merge PACS Server can view all exams (including studies, reports and orders) on the Server. As described in Section 9.11 above, however, it is possible to give particular users and/or groups restricted access to exams by removing the default privilege that grants them access to all exams. Once that privilege has been removed, a user can only view exams that have been specifically assigned to the user or group to which the user belongs. This is true whether the user runs queries from within the Viewer or from any of the Management pages.

If the Site has a RIS, the Merge PACS Server can be configured to automatically assign exams to a user if the user's Provider Code matches any of following fields of an order generated by the RIS:

- **Attending Doctor**
- **Referring Doctor**
- **Consulting Doctor**
- **Admitting Doctor**
- **Order Doctor**
- **Other Health Care Provider**

This will work with users who are manually entered into Merge PACS via the Users & Groups Management Pages, as described in Section 9.11 above.

In addition, however, user information can be imported into Merge PACS in bulk from an Excel comma separated values (*.csv) file with specific information for each referring physician.

17.4.1. Creating an Excel CSV File with Referring Physician Information

The CSV file for the users must follow the CSV file format definition with the first row as the header. The header must contain the following as the column names (case-sensitive and spelled exactly). But the order does NOT matter.

Description	Required?	Field Length	Comments
Login Name	YES	64	This is the login name for the user. Although the maximum length of the field is 64, it is recommended that the login name is between 4 and 16 characters.
Provider Code	NO	64	This is the Provider Code that makes up part of the Referring Physician field of the order. Multiple codes must be separated by commas, and must all be grouped together within a set of double quotes (<i>e.g.</i> , "10003,20003"). Each provider code cannot be longer than 64 characters.
Provider Type	NO	64	If specified Provider Type does not already exist in the dictionary of Provider Types (see subsection B.1.3 of Appendix B), a new Provider Type will be created and added to the dictionary.
Name Prefix	NO	16	The title of the user's name, such as Mr., Mrs., or Ms., etc.
Last Name	YES*	64	The last name of the user.
First Name	YES*	64	The first name of the user.
Middle Name	NO	64	The user's middle name.
Name Suffix	NO	32	The name suffix, such as Jr., III, etc.
Name Degree	NO	32	The user's degree as part of the name, such as MD., PhD, ESQ., etc.

Description	Required?	Field	
		Length	Comments
Address Line1	YES*	128	The first line of the user's address.
Address Line2	NO	128	The second line of the user's address.
City/Town	YES*	64	The city or town of the user's address.
State	YES*	64	The 2-letter code representing the state of the user's address.
Zip Code	YES*	16	The zip code of the user's address.
Phone Number	YES*	16	The user's phone number.
Fax Number	NO	16	The user's fax number.
Email	YES*	128	The user's primary email address.
Email2	NO	128	The user's additional email address.
Email3	NO	128	The user's additional email address
Notify	NO	1	Y or N. This indicates if the user should be notified of any exam status changes.
Groups	NO	128	Multiple group names must be separated by commas, and must all be grouped together within a set of double quotes (e.g., "East,West"). The users with the same group name will be placed in the same group. If the group with the name does not exist, the group will be automatically created. Each group name cannot exceed 128 characters.
Password	YES	64	Enter the user's password (note that passwords must be between 6 and 12 characters long and are case sensitive).

NOTE: * If the tool is executed in the "relaxed" mode, these fields are not required.

The following is a sample CSV file containing user information:

```

Login Name,Provider Code,Provider Type,Name Prefix,Last Name,First Name,Middle
Name,Name Suffix,Name Degree,Address Line1,Address Line2,City/Town,State,Zip
Code,Phone Number,Fax Number,Email,Email2,Email3,Notify,Groups>Password
d0001,10001,,Mr,Abott,Zada,A,,MD,1 Infinite Loop,,Easton,MA,12345,617-779-
7878,617-779-7879,az@xyz.com,,N,,111111
d0002,10002,,Ms,Baker,Yana,B,,MD,2 Computer Drive,,Weston,MA,23456,617-779-
7878,617-779-7879,by@xyz.com,,N,,111111
d0003,10003,,Mr,Carter,Xavier,C,,MD,3 Microsoft Way,,Redmond,WA,
34567,617-779-7878,617-779-7879,cx@xyz.com,,N,"East,West",111111
d0004,10004,,Ms,Decker,Wendy,D,,MD,4 Dell Circle,,Houston,TX,45678,617-779-
7878,617-779-7879,dw@xyz.com,,N,"East,South",111111
d0005,"10005,10006",,Mr,Evans,Victor,E,,MD,5 Apple Rd,,Boston,MA,
56789,617-779-7878,617-779-7879,ev@xyz.com,,N,West,111111

```

CAUTION: Commas may not be used in any of the fields except for Provider Code and Groups.

17.4.2.Using the ProviderDataSynchronizationTool

The program used to import user information is called **ProviderDataSynchronizationTool** and it is run from a DOS command prompt. The program has two separate modes:

Mode	Description
Insert Mode	Only creates a new account if the user's login name does not already exist in the system. If the user's login name exists in the system, the user in the CSV file is skipped for processing. If you are not ready to allow all the users to have access to the system, you can create them using a special disabled option.
Update Mode	Only creates a new account if the user's login name does not already exist in the system. However, if the user's login name does already exist in the system, the user's information (except the user's password) will be updated based on the values defined in the CSV file. Note that the password can also be force-updated with the <code>-p</code> option for the program.

The program runs through the CSV in two passes. In the first pass, the program validates the CSV file. The validation process checks these conditions:

- The required fields are not empty. The required fields are First Name, Last Name, Address Line 1, City, State, Zip Code, Phone Number, Email 1, Login Name, and Password.

NOTE: The program can be run in an optional **relaxed** mode where the only required fields are Login Name and Password.

- The Login Names within the CSV file are unique.
- The Provider Codes within the CSV file are unique when they are present.
- The Login Name/Provider Code does not conflict with the existing users in the system.

In the second pass, the program will insert or update the users depending on the mode.

The batch import tool is located on the Merge PACS Server at the following location:

```
C:\amcas2\scripts\run-ProviderSynchronizationTool.bat
```

The program is run from a DOS command line and takes the following parameters:

Parameter	Description
-f file	specify the file path of the CSV file
-u	Operate in Update mode (update the user if user already exists except for the password)
-p	reset the password (only works in Update mode)
-d	Create the user in disabled mode.
-r	Run the program in relaxed mode where the only required columns are Login Name and Password
-h	Print the help message

For example, if the CSV file was located at **c:\temp\users.csv** and you wanted to run the program in **Update** mode, you would enter the following at a DOS command prompt:

```
C:\amcas2\scripts\run-ProviderSynchronizationTool.bat -f c:\temp\users.csv -u
```

17.5. Using the Problem Studies Tool

As described in subsection 9.9.4 above, the Problem Studies Tool Management page allows users to fix various problems that have caused one or more of a study's images to be placed in the DICOM Jail. There are five separate types of problem that can be monitored and addressed with the Problem Studies Tool:

Problem	Description
DICOM Study Attribute Mismatch	One or more images in a study have one or more study level DICOM attributes that do not match the attributes for the first image of that study that was received by Merge PACS.
DICOM Study Attribute Missing	One or more of the images in a study are missing one or more study level DICOM attributes that are required by Merge PACS.
DICOM Series Attribute Mismatch	One or more images in a series have one or more series level DICOM attributes that do not match the attributes for the first image of that series that was received by Merge PACS.
DICOM Series Attribute Missing	One or more of the images in a study are missing one or more series level DICOM attributes that are required by Merge PACS.
DICOM Image Attribute Mismatch	The same study has been sent to Merge PACS twice, but one or more images have one or more image-specific attributes that are not the same for both studies.
Patient Identity Mismatch	A study that has been imported into Merge PACS and watermarked is later pushed to Merge PACS or restored from Enterprise Archive with the same Study Instance UID but a different patient identity (normalized MRN, normalized Patient Name, Issuer of Patient ID). Note that, with regard to Enterprise Archive in particular, studies will only be jailed if Merge PACS has been configured so that the information received from EA will be given precedence over the information stored within Merge PACS, as described in subsection 1.2.61 above.

CAUTION: Great care should be exercised when changing attributes via PDE operations. Mistakes could result in problems such as delayed reading of studies, studies read out of order, etc.

To access the Problem Studies Tool:

1. Using a web browser, go to the following URL:
`http://<your-server>/merge-management`

- Click on the **Tools** tab and then select the **Problem Studies** link to display the Problem Studies Tool main window, as shown in the following example

Problem Studies Tool

Station Name

Patient ID

Patient Name

Date Search By 2014 25

Problem Studies Found:

Study Date	Patient Name	Patient ID	IPID	DOB	Sex	Station Name	Problems	Action
2004-10-04 11:52:09	Doe, Jonatham	123456789	HOME	2012-12-20	O		DICOM study attribute mismatch	Fix

Figure 710: The Problem Studies Tool

- By default, the Problem Studies Tool page displays all problem studies for the **current day**, regardless of the originating source (the “Station”), Patient ID, and Patient Name.
- You can limit or expand the list of studies shown by selecting a particular **Station Name** from the pull-down menu provided (or selecting “All stations”), entering a custom **date** or **date range**, or entering a specific **Patient ID** or **Patient Name**, and then clicking on the appropriate **Search** button.
 - If searching by **Patient Name**, you can enter the name you are searching for in the format “**last name, first name**” or simply enter the **last name** to find all people with that last name.
 - You can also use **wildcard searches** for **Patient Name**, as described in Section 9.2 above.

NOTE: Each search criterion is independent and cannot be combined with other search criteria. Thus, if you search for problem studies from a specific station, the search will automatically include all Patient IDs, all Patient Names, and all dates.

- Click on the **Fix** button next to the study whose problem you want to address to bring up a separate “**Fix a Problem Study**” page.
- Once you have fixed the problem or problems for the study, click on the **Apply All Changes** button at the bottom of the page to save the changes and return to the main Problem Studies Tool page.

How you fix a particular problem depends on the type of problem. The following sections detail the various types of problems that can be fixed with the Problem Studies Tool. Note that a particular study may have more than one problem associated with it, and all the problems will be listed on a single “Fix a problem Study” page.

NOTE: If a particular image has been jailed for more than one reason (*i.e.*, it has multiple attributes that are different from what is already listed in the Merge PACS database), you will have to use the Problem Studies Tool multiple times to address each problem.

17.5.1.DICOM Study Attribute Mismatch

DICOM study attribute mismatch					
Attribute Name	# Images	AMICAS Value	Incoming Values	Actions	
Patient Birth Date	1	19910227	19190227	Delete Jailed: ☐	New Value: 19190227 19190227 v

Figure 711: DICOM Study Attribute Mismatch

A DICOM study attribute mismatch typically occurs when one or more study-specific attributes are changed partway through a study. The list of study-specific DICOM attributes includes the following:

- **Study ID**
- **Accession Number**
- **Patient Name**
- **Patient ID**
- **Patient Birth Date**
- **Patient Sex**

For each mismatched attribute, the “AMICAS Value” represents the information that was entered for the first image in the study and the “Incoming Value” represents the information that was entered for the jailed image or images.

The following actions may be taken to fix the problems with this study:

- To enter a new attribute value for an image or set of images, either enter the correct value in the field provided in the “Actions” column or else select a value from the associated pull-down menu. There are three possible outcomes:
 - If you select the **AMICAS Value** from the pull-down menu, the jailed images will be given the same attribute as the non-jailed images in the study.
 - If you select the **Incoming Value** from the pull-down menu, all the non-jailed images for this study will be given the same attribute as the jailed images. Note that this will only affect the study in question and not other studies for the same patient.
 - If you enter a **new value**, both the jailed and non-jailed images for this study will be given this new attribute. Note that this will only affect the study in question and not other studies for the same patient.

NOTE: This option is not available if the mismatched attribute for the study is **Study Instance UID** or **Series Instance UID**. Images with either of those attributes mismatched can only be deleted, as described below.

- To delete the jailed image or images with a particular attribute mismatch, click on the **Delete Jailed** box in the “Actions” column for that attribute.
- To delete all jailed images for this study, click on the **Delete All Jailed Images** button at the bottom of the screen.

When finished, click on the **Apply All Changes** button at the bottom of the screen (you do not need to do this if you used the **Delete All Jailed Images** button).

17.5.2.DICOM Study Attribute Missing

DICOM study attribute missing				
Attribute Name	# Images	AMICAS Value	Incoming Values	Actions
Patient's ID	1			Delete Jailed: ☐ New Value:
Apply All Changes Delete All Jailed Images				

Figure 712: DICOM Study Attribute Missing

A missing DICOM study attribute typically occurs when the **Study ID** or **Patient ID** is omitted at the time the study is sent.

The following actions may be taken to fix the problems with this study:

- To enter a new attribute value for an image or set of images, enter the correct value in the field provided in the “Actions” column.
- To delete the jailed image or images with a particular missing attribute, click on the **Delete Jailed** box in the “Actions” column for that attribute.
- To delete all jailed images for this study, click on the **Delete All Jailed Images** button at the bottom of the screen.

When finished, click on the **Apply All Changes** button at the bottom of the screen (you do not need to do this if you used the **Delete All Jailed Images** button).

17.5.3.DICOM Series Attribute Mismatch

DICOM series attribute mismatch						
Attribute Name	Series Number	Series Date	# Images	AMICAS Value	Incoming Values	Actions
Series Number	23007	2006-10-03	1	10002345	23007	Delete Jailed: ☐ New Value: 10002345 10002345
Modality	23007	2006-10-03	1	BN	CR	Delete Jailed: ☐ New Value: BN BN
Station Name	23007	2006-10-03	1	pr08-elchest	pr10-elchest	Delete Jailed: ☐ New Value: pr08-elchest pr08-elchest
Apply All Changes Delete All Jailed Images						

Figure 713: DICOM Series Attribute Mismatch

A DICOM series attribute mismatch typically occurs when one or more series-specific attributes are changed partway through a series. The list of series-specific DICOM attributes includes the following:

- **Series Number**
- **Modality**
- **Station Name**

For each mismatched attribute, the “AMICAS Value” represents the information that was entered for the first image in the series and the “Incoming Value” represents the information that was entered for the jailed image or images.

The following actions may be taken to fix the problems with this study:

- To enter a new attribute value for an image or set of images, either enter the correct value in the field provided in the “Actions” column or else select a value from the associated pull-down menu. There are three possible outcomes:
 - If you select the **AMICAS Value** from the pull-down menu, the jailed images will be given the same attribute as the non-jailed images in the series.
 - If you select the **Incoming Value** from the pull-down menu, all the non-jailed images for this series will be given the same attribute as the jailed images. Note that this will only affect the series in question and not other series or studies for the same patient.
 - If you enter a **new value**, both the jailed and non-jailed images for this series will be given this new attribute. Note that this will only affect the series in question and not other series or studies for the same patient.
- To delete the jailed image or images with a particular attribute mismatch, click on the **Delete Jailed** box in the “Actions” column for that attribute.
- To delete all jailed images for this study, click on the **Delete All Jailed Images** button at the bottom of the screen (you do not need to do this if you used the **Delete All Jailed Images** button).

When finished, click on the **Apply All Changes** button at the bottom of the screen.

17.5.4.DICOM Series Attribute Missing

Attribute Name	Series Number	Series Date	# Images	AMICAS Value	Incoming Values	Actions
Modality	23007	2006-10-03	1			Delete Jailed: ☐ New Value:

Figure 714: DICOM Series Attribute Missing

A missing DICOM series attribute typically occurs when the **Modality** is omitted at the time the study is sent.

The following actions may be taken to fix the problems with this study:

- To enter a new attribute value for an image or set of images, enter the correct value in the field provided in the “Actions” column.
- To delete the jailed image or images with a particular missing attribute, click on the **Delete Jailed** box in the “Actions” column for that attribute.
- To delete all jailed images for this study, click on the **Delete All Jailed Images** button at the bottom of the screen.

When finished, click on the **Apply All Changes** button at the bottom of the screen (you do not need to do this if you used the **Delete All Jailed Images** button).

17.5.5.DICOM Image Attribute Mismatch

DICOM image attribute mismatch
Note: You cannot select incoming values for image level attributes

Attribute Name	Series #	Image #	Image Date	AMICAS Value	Incoming Values	Actions
Image Type	6	1	2006-10-03 09:33:58	PPR	SINGLE PLANE	Delete Jailed: ☐ New Value: PPR PPR
Rows	6	1	2006-10-03 09:33:58	1024	512	Delete Jailed: ☐ New Value: 1024 1024
Columns	6	1	2006-10-03 09:33:58	1024	512	Delete Jailed: ☐ New Value: 1024 1024

Figure 715: DICOM Image Attribute Mismatch

A DICOM image attribute mismatch occurs when the same study is sent to Merge PACS twice, but one or more images have different image-specific attributes. The list of image-specific DICOM attributes includes the following:

- **Image Type**
- **Content Date**
- **Image Orientation**
- **Number of Rows**
- **SOP Class UID**
- **Content Time**
- **Slice location**
- **Number of Columns**
- **Image Number**
- **Image Position**
- **Pixel Spacing**
- **Lossy Compression**

For each mismatched attribute, the “AMICAS Value” represents the information that was entered for the image in the previous study and the “Incoming Value” represents the information that was entered for the jailed image or images.

The following actions may be taken to fix the problems with this study:

- To enter a new attribute value for an image or set of images, either enter the correct value in the field provided in the “Actions” column or else select a value from the associated pull-down menu. Note that you cannot enter or select the “Incoming Value.”
- To delete the jailed image or images with a particular attribute mismatch, click on the **Delete Jailed** box in the “Actions” column for that attribute.
- To delete all jailed images for this study, click on the **Delete All Jailed Images** button at the bottom of the screen.

When finished, click on the **Apply All Changes** button at the bottom of the screen (you do not need to do this if you used the **Delete All Jailed Images** button).

17.5.6. Patient Identity Mismatch

Problem Studies Tool

Station Name: LOGIQ7 (dropdown) [Search]

Patient ID: [Search]

Patient Name: [Search]

Date Search By: Date (dropdown) Today Yesterday 2013 04 25 [Search]

Problem Studies Found:

Study Date	Patient Name	Patient ID	IPID	DOB	Sex	Station Name	Problems	Action
2000-05-03 10:42:30	DOE JANE Q	9999999	Home	1957-06-16	F	LOGIQ7	Patient identity mismatch	[Fix]

Figure 716: Patient Identity Mismatch

A patient identity mismatch occurs when a study that has been imported into Merge PACS and watermarked is later pushed to Merge PACS or restored from Enterprise Archive with the same Study Instance UID but a different patient identity (normalized MRN, normalized Patient Name, Issuer of Patient ID).

Clicking on the 'Fix' button will give you the option to enter a new demographic value or select an existing demographic value, as in the following example:

Fix a Problem Study

Study Information

Study Date	Patient Name	Patient ID	IPID	DOB	Sex	Accession #	Modality
2000-05-03 10:42:30	DOE JANE S	99999999	Home	1957-06-16	F	5732512	US

Study Description

275W2

Patient identity mismatch

Attribute Name	# Images	AMICAS Value	Incoming Values	Actions
Patient's Name	1	DOE^JANE^Q	DOE^JOANNE^A	☐ Delete Jailed: ☐ New Value: DOE^JANE^S^^ DOE^JANE^S^^ DOE^JOANNE^A

[Apply All Changes] [Delete All Jailed Images]

Figure 717: Fixing Patient Identity Mismatch

The following actions may be taken to fix the problems with this study:

- To enter a new attribute value for an image or set of images, either enter the correct value in the **New Value** field or else select a value from the drop-down menu. When finished, click on the **Apply All Changes** button at the bottom of the screen
- To delete all jailed images for this study, do one of the following:
 - Click on the **Delete Jailed** checkbox or any listed attribute mismatch and then click on the **Apply All Changes** button at the bottom of the screen.
 - Click on the **Delete All Jailed Images** button at the bottom of the screen.

17.6. Monitoring EMRLink Exceptions

If your site has Merge OrthoEMR™ installed and sharing of images is required between Merge PACS and OrthoEMR, a feature called EMRLink can be enabled on the Merge PACS Server. When EMRLink is enabled, a JPEG copy of each image will be created and exported to the local file system on import to the Merge PACS database. A service on the system will then send these images to the OrthoEMR database and they will become eligible for inclusion in an EMR report.

When EMRLink is running, any images that cannot be transferred to the OrthoEMR Server will be copied to the **C:\failed_jpegs** directory on the Merge PACS Server. Similarly, any images that cannot be assigned (e.g., because the target patient does not exist within the Merge PACS database) will be copied to the **C:\unassigned_jpegs** directory on the Merge PACS Server.

It is important to monitor both of these directories on a regular basis and delete any images that are placed there, both to prevent the directories from getting too full and because the images may contain Protected Health Information.

If a significant number of images are failing to be transferred, you should also contact Merge Healthcare Technical Support and provide a copy of all log files located in the **C:\helpers\EMRLink** directory on the Merge PACS Server.

NOTE: Log files are automatically deleted after a configurable amount of time (the default is 30 days, but this can be changed by Merge Healthcare Technical Support) .

17.7. Restoring the System

In the event of a server failure, it is the responsibility of the site to return the server to the condition it was in at the time of the original Merge PACS installation. This includes the reloading of the operating system and PCAnywhere as needed. The system backups and registry settings can be used to bring the system back to its most recent status just prior to the server failure. From this point, the database may be restored by Merge Healthcare Technical Support.

17.8. Restoring Production Server from a Contingency Server

If Merge PACS has been configured to failover from the Production Server to a Contingency Server, the following procedures should be followed to restore the system from Contingency to Production after the problem causing the failover has been resolved.

Before you begin the procedure to restore the Production Server, keep the following in mind during downtime.

- If you are using a single IP switch then the transition from PACS Production to Contingency should be seamless as no user should be forced to change their Web Address. Users just need to login and out of the PACS Production Server.
- Since Contingency is configured to send all images, status updates to Production by default items will be queued until the Production Server is back up.
- Telmed imports should be monitored during downtime to make sure the last set of changes imported successfully.

- Remember that no items will be archived until the Production Server is back up and operational.

17.8.1.Preliminary Steps

- Make sure Production Server has been brought up to Windows OS state but with no PACS processes running.
- Make sure modalities have temporarily stopped sending to Contingency
- Make sure HL7 Feeders should temporarily stop sending (HL7 Orders not sent to Production will be resent once Production has been restored).
- Prior to restoring production, verify that all items in the Telmed Queue are queued and no items have failed status.
- Check and update jobs that need to be sent to the Production Server by doing the following:
- Go to db2cmd
- Switch to the ContingencyServerScripts directory using the following command:

```
CD <DRIVE>:\amcas2\scripts\FailoverDBScripts\
ContingencyServerScripts
```
- Connect to the IMG Database using the following command:

```
db2 connect to IMGXXX user dicom using password
```
- While connected to the IMGXXX database run the following command

```
Db2 -vf TelmedQueue.sql
```
- Once complete look at the output results, as in the following example:

```
Administrator: DB2 CLP - DB2COPY1
C:\>db2 -vf C:\PROD_CONTINGENCY\TelmedQueue.sql
select count(*), JobType, CURRENTSTATUS, LASTSTATUS, NRETRIES, NRETRIED from TelmedJob where JobType like '%OUT' group by JobType, CURRENTSTATUS, LASTSTATUS, NRETRIES, NRETRIED
1
-----
1 IMAGE_OUT ER IP 6 6
11 STATUS_OUT ER IP 6 6
2 record(s) selected.
```

Figure 718: Output Results

- Compare them with the below table as per statuses:

DB Column Value	Description
NE	NEW
WT	WAITING
SE	SELECTED
IP	IN PROGRESS
ER	ERROR
OK	COMPLETED
OH	ONHOLD

12. If any items have a CURRENTSTATUS of ER and the number of Retries is greater than the maximum allowed (15), perform image status and non-image status updates to the database as follows:
 - **Image Status Update** – from the db2cmd window connected to the IMGXXXX database, run the following command:
`Db2 -vf ImageUpdate.sql`
 - **Non-Image Status Update** – from the db2cmd window connected to the IMGXXXX database, run the following command:
`Db2 -vf StatusUpdate.sql`
13. Rerun the Db2 -vf TelmedQueue.sql (Step c) to confirm the job status updates are successful.

17.8.2. Bringing Production Back Up

1. Initiate the Merge PACS Production Server Services back to operational.
2. Have users log out and log back in to get reconnected to Production.
3. Have the modalities begin resending to PACS Production.
4. Have HL7 Feeds begin resending to both PACS Production and Contingency.
5. Monitor Telmed Export jobs until completion of all queued jobs. Once the jobs are completed, run a status and image check to verify all items are updated.

17.8.3. Production vs. Contingency Status Check

1. On the Contingency Server, run the following commands in order when connected to the IMGXXX Database in the db2cmd window:
`CD <DRIVE>:\AMICAS2\scripts\FailoverDBScripts\ContingencyServerScripts
ProductionAndContingencyStatusCheck.bat`
2. When prompted, enter the number of hours the production server has been down. (Note could be this can be within the nearest hour).

NOTE: The **st.ModifiedDate** parameter is the time of the Production Server failure. Since Modalities have returned to sending, you may decide to change the ModifiedDate parameter to be (**st.ModifiedDate between FailureStartDateTime and ProdRestartDateTime**).

3. When batch file is finished, enter the following command:
`db2 "DROP TABLE CONTG_STATUS"`
4. Copy the exported **ContingencyServerStatus.txt (from the C Drive)** file over to the Production Server's C drive.
5. While connected to the IMGXXX database of the Production Server, run the following commands at the db2cmd window:
`CD <DRIVE>:\AMICAS2\scripts\FailoverDBScripts\ProductionServerScripts
CreateStatusSet.bat`
6. Once completed, run the following commands to check study status, order status and series/image counts:

7. **Study Status Check** – To validate all status changes have successfully taken place within the StudyStatus table run the below command in the db2cmd window connected to the IMG XXXX database on the Production Server:

```
Db2 -vf StudyStatusCheck.sql
```
8. **Order Status Check** – To validate all status changes have successfully taken place within the OrderStatus table run the below command in the db2cmd window connected to the IMG XXXX database on the Production Server:

```
Db2 -vf CheckOrderStatus.sql
```
9. **Series and Images Check** – To validate all images were successfully transferred to the Production Server, run the below command in the db2cmd window connected to the IMG XXXX database on the Production Server:

```
Db2 -vf CheckSeries_ImagesCounts.sql
```

NOTE: To output items to a text file you can use the **-z** parameter in the db2cmd window.

NOTE: It is the site's responsibility to determine which is the correct status and image count. There exists a possibility of annotations and key objects stored prior to production going offline.

10. Update items where status is not in sync between Production and Contingency through the RealTime Worklist.
11. For Images and Series that are different, delete and resend studies.
12. Rerun status and image checks to verify all items are now reflected correctly.
13. Once completed, run the following command from the db2cmd window connected to the IMGXXXX database to remove the added table for comparison:

```
Db2 DROP TABLE PRODSTATUS
```

Production and Contingency should now be operational.

17.9. Configuring a VServer of a NetScaler Appliance to Accept Traffic for a Specific Set of Ports

If your site has a Virtual Server (VServer) on a NetScaler appliance, you might want to configure a virtual server only to accept TCP traffic on certain ports as described below. In such a case, you can create multiple VServers, approximately 10000, to accept traffic for specific ports, or create a VServer with the protocol type ANY. A VServer with the protocol type ANY accepts all the traffic.

The following are the default ports you would need to include for failover (HL7 is not included as they will always go to the direct Server):

Item	Port(s)
DICOM Modality Worklist	104
HTTP	80, 8080
HTTPS	443, 8443 (if TLS is enabled)
Messenger	5222
Service Monitors	11095-11111

NOTE: These ports are default and are subject to be changed by your PACS Support representative.

To configure a VServer of a NetScaler appliance to accept traffic for a specific set of ports:

1. Expand the Load Balancing node of the Navigation pane on the appliance.
2. Select the Virtual Servers node.
3. Click **Add** on the **Load Balancing Virtual Servers** page.
4. Enter ***** in the **Port** field.
5. Select **ANY** from the **Protocol** list.
6. Enter the other required details to create the VServer.
7. Click **Create**.
8. Click **Close**.

The VServer with protocol ANY is created.

9. Select the VServer with protocol **ANY**.
10. Click **Open** on the **Load Balancing Virtual Servers** page.
11. Click on the **Advanced** tab.
12. Expand the **Listen Policy** section.
13. Enter **1** in the **Listen Priority** field.
14. Click **Configure**.
15. Enter the following in the **Expression** field:
`CLIENT.TCP.DSTPORT.EQ(80) || CLIENT.TCP.DSTPORT.BETWEEN(11095,11111)`
16. Click **Create**.
17. Click **OK**.

Appendix A. Privileges, Preferences and Configurations

As described in Chapter 11 above, privileges determine what features a user has access to (including access to specific management pages and specific features within the Merge PACS Workstation), while preferences set default values for various optional configurations within the Merge PACS Workstation. Individual users inherit their privileges and default preferences from the groups they belong to as well as from the site as a whole, and these privileges and preferences are configured via privilege and preference templates that can be linked to groups and to the site, as well as a special configuration template that can be linked to the site to deal with site-wide preferences.

This appendix describes the various privileges and preferences that can be included in templates.

A.1. Privileges

Privileges for each template are set from the **Merge PACS Privileges** dialog, as in the following example:

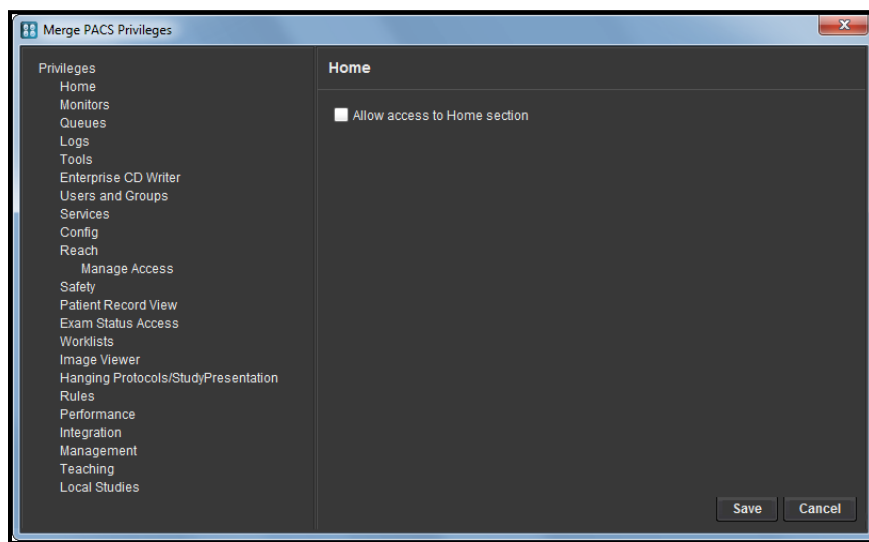


Figure 719: Merge PACS Privileges

NOTE: You must log out and log back into the Merge PACS Workstation after adding or removing privileges for the changes to take effect.

A.1.1. In General

Some or all of the following access privileges can be included in a privilege template that can be linked to a site or to a group:

NOTE: Many of these privileges can be set under multiple subheadings for convenience.

a. Home

Select **Allow access to Home section** to provide access to the **Home** tab of the Merge PACS Management Pages, as described in Section 9.5 above

b. Monitors

Select one or more of the following to provide access to specific monitor pages at the **Monitors** tab of the Merge PACS Management Pages, as described in Section 9.6 above:

The following privileges are available with regard to the **Monitors** section of the Merge PACS Management Pages, as described in Section 9.6 above:

- Allow access to the Monitors - **Archiver** section
- Allow access to the Monitors - **Auditor** section
- Allow access to the Monitors - **ECDR Manager** section
- Allow access to the Monitors - **Importer** section
- Allow access to the Monitors - **Listener** section
- Allow access to the Monitors - **Scheduler** section
- Allow access to the Monitors - **Client Notification Server** section
- Allow access to the Monitors - **Sender** section
- Allow access to the Monitors - **Worklist Server** section

c. Queues

Select one or more of the following to provide access to, and special privileges at, specific queue pages at the **Queues** tab of the Merge PACS Management Pages, as described in Section 9.7 above:

- Allow access to Queues - **Archive** section
- Allow modification of jobs in **Archive** queue
- Allow access to Queues - **Auto-Routing** section
- Allow modification of jobs in **Auto-Routing** queue
- Allow access to Queues - **Command** section
- Allow modification of jobs in **Command** queue

- Allow access to Queues - **Forwarding** section
- Allow modification of jobs in **Forwarding** queue
- Allow access to Queues - **HL7 Outbound Interface** section
- Allow modification of jobs in **HL7 Outbound Interface** queue
- Allow access to Queues - **Image Import** section
- Allow modification of jobs in **Image Import** queue
- Allow access to Queues - **Insert File** section
- Allow modification of jobs in **Insert File** queue
- Allow access to Queues - **Telmed Export** section
- Allow modification of jobs in **Telmed Export** queue
- Allow access to Queues - **Telmed Import** section
- Allow modification of jobs in **Telmed Import** queue
- Allow access to Queues - **DICOM Print** section
- Allow modification of jobs in **DICOM Print** queue
- Allow management of items in **DataStorage**
- Allow modification of jobs in **Sender Job** queue

d. Logs

Select one or more of the following to provide access to specific log pages at the **Logs** tab of the Merge PACS Management Pages, as described in Section 9.8 above:

- Allow access to Logs – **Access Log** section
- Allow access to Logs – **Event Log** section
- Allow access to Logs – **Performance Log** section
- Allow access to Logs – **Study Log** section
- Allow access to Logs – **User Login/Logout** section
- Allow access to Logs – **User Modifications** section

In addition, the **Allow access to Activity Log** privilege will provide access to the pop-up Activity Log when a worklist is configured to include the Activity Log action icon.

e. Tools

Select one or more of the following to provide access to, and special privileges at, specific tool pages at the **Tools** tab of the Merge PACS Management Pages, as described in Section 9.9 above:

- Allow access to Tools - **Access Control** section
- Allow access to Tools - **Archive Tools** section
- Allow access to Tools - **Archive Info** section

- Allow testing of **study archive**
- Allow access to Tools - **Patient Info** section
- Allow access to Tools - **Problem Studies** section
- Allow deleting jailed images from the Problem Studies tool
- Allow access to Tools - **Send Studies** section
- Allow validation of invalid studies
- Allow access to Tools - **Quality Control**
- Allow access to Tools - **Disaster** Recovery section

f. Enterprise CD Writer

Select one or more of the following to provide access to, and special privileges at, specific pages at the **Enterprise CR Writer** tab of the Merge PACS Management Pages, as described in Section 9.10 above:

- Allow access to Enterprise CD Writer - **Create CDs Manually** section
- Allow access to Enterprise CD Writer - **CD Queue** section
- Allow modification of **CD Queue** jobs
- Allow access to Enterprise CD Writer - **CD Writers** section
- Allow access to Enterprise CD Writer - **Auto Writing Rules** section

g. Users and Groups

The following privileges control access to various Management Pages and Manage Templates Pages associated with Users, Groups and Site:

Privilege	Description
Allow access to Users, Group & Site – Users Section	Provides access to the Users Management Page, as described in subsection 9.11.1 above, the Credentialing and Licensure Management page, as described in subsection 9.13.10 above, and the Manage Users page, as described in Chapter 12 above.
Allow access to Users, Group & Site – Groups Section	Provides access to the Groups Management Page, as described in subsection 9.11.1.f above, and the Manage Group Templates page, as described in subsection 11.2.4 above.
Allow access to Users, Group & Site – Site Section	Provides access to the Sites Management Page, as described in subsection 9.11.3 above, and the Manage Site Templates page, as described in subsection 11.2.3 above.
Allow access to superusers	Provides the rights to add/remove a user from the Merge Support or Administrator groups.
Allow access to site administrator users	Provides the rights to add/remove a user from the Administrator group.

Privilege	Description
Allow access to group worklist association	Provides access to the "Worklist" tab on the Manage Groups page, as described in subsection 9.11.2 above, and the "Associated Groups" tab at the Worklist Editor dialog, as described in subsection 13.4.2 above.
Allow user to manage user preferences	Provides access to the Merge PACS Preferences dialog within the Workstation.

h. Services

Select **Allow access to Services section** to provide access to the **Services** tab of the Merge PACS Management Pages, as described in Section 9.12 above

i. Config

Select one or more of the following to provide access to specific configuration pages at the **Configuration** tab of the Merge PACS Management Pages, as described in Section 9.13 above:

- Allow access to Configuration - **Merge PACS Servers** section
- Allow access to Configuration - **Application Links** section and creation of links to **External Applications**
- Allow access to Configuration - **Archive Config** section
- Allow access to Configuration - **Compression Prefs** section
- Allow access to Configuration - **DICOM Devices** section
- Allow access to Configuration - **External Applications** section
- Allow access to Configuration - **Image Volumes** section
- Allow access to Configuration - **Import Devices** section
- Allow access to Configuration - **Patient Location** section
- Allow access to Configuration - **Modality Dictionary** section
- Allow access to Configuration - **Patient Class Dictionary** section
- Allow access to Configuration - **Preferences** section
- Allow access to Configuration - **Procedure Dictionary** section
- Allow access to Configuration - **Property Files** section
- Allow access to Configuration - **Providers** section
- Allow access to Configuration - **Provider Type Dictionary** section
- Allow access to Configuration - **Refresh Rate** section
- Allow access to Configuration - **Schedule DB Backup** section
- Allow access to Configuration - **Server Config** section
- Allow access to Configuration - **Transfer Rules** section

- Allow access to Configuration - **Workflow Priorities** section
- Allow access to Configuration - **Workflow Statuses** section [also provides access to **Workflow Status Dimensions**]
- Allow access to Configuration - **Viewer XML Integration** section [also provides access to **Viewer XML Event Configuration**]
- Allow access to Configuration - **LDAP Servers** section
- Allow access to Configuration - **Deletion Rules** section
- Allow access to Configuration - **DICOM Printers** section
- Allow access to **Specialty Mappings Configuration**
- Allow access to **HL7 Order Status Mapping Configuration**
- Allow access to **IPID Configuration**
- Allow access to **Facility Configuration**
- Allow access to Configuration - **External Volumes** section
- Allow access to **Practice Configuration**
- Allow access to **Order/Provider Preferences**
- Allow access to **Manage Passwords**

In addition, you can click on the **Manage Access** sub-option under the main **Config** option to provide one or more of the following access-related privileges

- Allow access to **order information**
- Allow access to **final reports**
- Allow access to **preliminary reports**
- Allow **granting exam access to others**.

j. Safety

Provide one or more of the following safety-related privileges within the Workstation:

- Allow **Series-level comparison**
- Allow **moving primary study window**
- **Warn user when images are not viewed prior to setting of status**

NOTE: The "Not all images reviewed" dialog will always be displayed for mammography studies regardless of whether this privilege is selected or not.

- Allow access to **preliminary reports**
- Allow user to **add, edit, or delete annotations after study is read**
- Allow user to **add, edit, or delete annotations after Study is final**
- Allow user to **add, edit, or delete key images after study is read** [requires Allow setting of Key Images privilege]

- Allow user to **add, edit, or delete key images after study is final** [requires Allow setting of Key Images privilege]

k. Patient Record View

Provide one or more of the following privileges relating to the **Patient Record, Query Page, Local Study Open** and the **Merge PACS Viewer**:

- Allow access to **Patient Record**
- Allow access to **Local Studies**
- Allow access to **worklist**
- Allow access to **Query Viewer**
- Allow access to **Order Viewer**
- Allow access to **Report Viewer**
- Allow access to preliminary reports
- Allow access to **Notes Viewer**
- Allow access to **Route Studies**
- Allow access to **DICOM Q/R**
- Allow access to **Technologist Work Panel**
- Allow **granting exam access to others**
- Allow access to **ER Work Panel**
- Allow access to **Operator Work Panel**
- Allow creation of **notes**
- Allow **all users to delete both text comments and audio comments**
- Allow **editing of patient and study demographic data**
- Allow **changing workflow status**
- Allow user access to **Management Pages** [from within the Workstation]
- Allow user to create **DICOM CD/DVDs**

l. Exam Status Access

Provide one or more of the following privileges relating to exam and status access control:

- Allow user to **lock the study while reading it**
- Allow **changing workflow status**
- **Group and lock all studies belonging to same exam together**
- Allow users to **lock and set status of multiple unread studies in one session**
- Allow **access to all exams in the system**

- Allow User to **Assign Exams**
- Allow access to **update HL7 attributes**

NOTE: Select the **Allow access to all exams in the system** to provide this user with unrestricted access to all exams within Merge PACS. If this option is not selected, the user will only have access to those exams that have been specifically assigned to him, either manually or automatically, as described in subsection 9.9.8 above.

CAUTION: The default Radiologist Privilege Template that is shipped with Merge PACS includes the “Allow access to all exams in the system” privilege and this privilege should be removed from the template if you do not want all Radiologists using this template to have access to all exams.

m. Worklists

If the optional **RealTime Worklist** component has been enabled, you can give this user access to RTWL as well as provide him or her with any of the following advanced Worklist Configuration privileges:

Privilege	Description
Allow access to Worklist	Provides access to RealTime Worklist.
Allow user to view all worklists	Provides the ability to view all worklists, whether or not assigned to the user.
Allow user to view own worklists only	Provides the ability to only view worklists associated with the group(s) the user is a member of.
Allow worklist to download/cache images for viewer	Provides the right to use the Workflow Accelerator feature with worklists that have this feature enabled.
Allow creation of worklists that include any region or institution	Provides the right to create a filter worklist that displays exams from all regions or Institutions, even if the user has been credentialed for specific Facilities or been associated with specific Licensure Regions.
Allow access to worklist definition	Provides access to the Worklist Configuration Pages, as described in Section 13.4 above.
Allow access to worklist block definition	Provide access to the Worklist Blocks configuration pages, as described in Section 13.3 above.
NOTE: Users with the Allow access to worklist definition privilege, but not the Allow access to worklist block definition privilege, will be able to define the existing blocks they want to be part of a particular worklist but will not be able to actually edit or create new blocks.	
Allow access to Quick Filters	Provides access to the Quick Filters section of the RealTime Study List. If not selected, user will only be able to access existing worklists from the RTSL and/or create new teaching worklists.
Allow Access to Dashboard	If Merge PACS is configured to integrate with an Analytics / Dashboard Server, provides access to the Mini Dashboard within the Merge PACS Workstation.

n. Image Viewer

Provide one or more of the following privileges relating to the **Merge PACS Viewer** (refer to the *Merge PACS 8.2 Workstation Users Guide* for additional information):

- Allow access to **Image Viewer**
- Allow access to **3D Viewer**
- Allow user to **listen to Voice Clips**
- Allow user to **record Voice Clips**
- Allow **deleting of text and audio comment**
- Allow user to **view annotations**
- Allow user to **view Spine Labels**
- Allow user to **add, edit, or delete annotations** [requires Allow setting of Study Presentations privilege, as described below]
- Allow user to **add, edit, or delete annotations after study is read**
- Allow user to **add, edit, or delete annotations after Study is final**
- Allow user to **view key images**
- Allow **setting of Key Images**
- Allow user to **add, edit, or delete key images after study is read** [requires Allow setting of Key Images privilege]
- Allow user to **add, edit, or delete key images after study is final** [requires Allow setting of Key Images privilege]
- Allow user to **overwrite autologout timeout**
- Allow user to **customize the user interface** [allows user to create/edit keyboard shortcuts and manage macros from the Merge **PACS Preferences dialog**]
- Allow user to **print to DICOM printers**
- Allow access to **lossy images**
- **Hide patient/study/series information** [provides “**blinded reading**” capability]
- Allow user to **access other series/studies**
- Allow access to **QC special tools** [e.g., Pixel Masking, Burnt-in Text and Save Masks / Text]
- Allow access to **MG workflow** [provides access to all mammography-specific tools and features (Intelligent Zoom, Quadrant Zoom, Binocular Tool, Hot Light Tool, CAD Summary, mammo-specific hanging protocols, etc.) as well as the Mammography Thumbnail Viewer]
- Allow user to **publish a macro**
- Allow user to **add, edit, or delete Spine Labels**
- Allow user to **add an image to a study** [*i.e.*, from an orthopedic templating toolkit application when OrthoLink is enabled]
- Allow access to **3D Targeted Review**

- **Allow Viewer Annotations to be saved using Button** [provides access to “Save Annotations” button on Application Toolbar]

CAUTION: If users have the “Allow Viewer Annotations to be saved using button” privilege, the only way for the user to save annotations will be to use the “Save Annotations” button. Also, if the user/group previously had the “Automatically Save Annotations on exit” preference enabled, that function will be disabled if the user/group has access to the “Allow Viewer Annotations to be saved using button” privilege.

CAUTION: See Appendix A.1.3 below for important information regarding blinded reading.

CAUTION: “Allow access to 3D Targeted Review” is for internal use only and should not be changed.

NOTE: Refer to Appendix A.1.1.v below for detailed information on the interaction between lossy and lossless image privileges.

NOTE: If “Allow user to access other series/studies” is **not** selected, the user will not be able to access any series other than the series launched via URL. This includes access to Patient Record, Load Comparison Study, Go to Next, creating/modifying HPs and SPs, etc.

NOTE: Not having the “Allow access to lossy images” privilege may impact performance when loading tomosynthesis or other images.

o. Hanging Protocols/Study Presentations

Provide one or more of the following privileges relating to Hanging Protocols and Study Presentations within the Merge PACS Workstation:

- Allow access to **Hanging Protocols**
- Allow user to **create hanging protocols**
- Allow **sharing of Hanging Protocols**
- Allow user to **manage all Hanging Protocols**
- Allow access to **study presentation**
- Allow **setting of Study Presentations**
- Allow access to **viewing of anatomical body part rules**
- Allow user to **manage anatomical body part rules**
- Allow user to **manage anatomical body part map**
- Allow user to **apply presentation states** (*i.e.*, externally created Presentation States available for a Series)

NOTE: If **Allow user to apply presentation states** is selected, the most recent externally created Presentation States available for a Series will be automatically applied on top of the currently selected Hanging Protocol or Study Presentation, if any, when the Series is opened.

p. Rules

Provide one or more of the following privileges relating to rules management, as described in Chapter 14 above:

- Allow user to **manage** rules
- Allow user to **export** rules
- Allow user to **import** rules

q. Performance

Provide one or more of the following privileges relating to performance and optimization within the Merge PACS Workstation:

- Allow auto background caching
- Allow access to **Lossless Images**
- Allow access to Lossless Images by default
- Allow MG Pre-caching of Priors
- Allow pre-caching of priors for XML and URL pre-caching requests

NOTE: Refer to Appendix A.1.1.v below for detailed information on the interaction between lossy and lossless image privileges.

r. Integration

Provide one or more of the following privileges relating to the integration with third-party applications within the Merge PACS Workstation:

- Allow user to use the **Viewer XML Integration**
- Allow user to use **OrthoLink**
- Allow access to **VoiceLink** (*i.e.*, direct API-based integration with dictation/report applications)
- Allow user to launch **Patient Synopsis** (includes the ability to launch it as a separate window via a worklist action icon or have it embedded within the Patient Record).

s. Management

Provide one or more of the following privileges relating to the management of Merge PACS:

- Allow user access to **Management Pages**
- Allow user to perform **QC Delete operations**
- Allow user to **manage site preferences**
- Allow user to **edit Workstation Preferences**

- Allow user to access **QC Editor and perform basic QC operations**
- Allow user to perform **QC Merge, Move, and Split Operations**
- Allow user to perform **QC Merge Operations between patients**
- Allow user to perform **QC Move Operations between patients**
- Allow full access to **Watch**
- Allow user to perform **Advanced QC Tag Editing**
- Allow editing of **patient demographic data**
- Allow editing of **study demographic data**

NOTE: The “**Allow user to edit Workstation Preferences**” privilege currently only controls access to the Closest Server Preferences section of the Workstation Preferences.

NOTE: The “**Allow full access to Watch**” privilege provides full administrative login privileges to access the optional Merge PACS Watch (“Watch”) module, including the Watch Management tab that is used to configure Watch. Note that Watch can optionally be configured to allow any user with a valid Merge PACS account to access the main Watch application, even without this privilege, but those users will not be able to access the options on the Watch Management tab. For information on using Watch, refer to the *Merge PACS 8.2 Watch User Guide*.

NOTE: If the “**Allow editing of patient demographic data**” privilege is **disabled**, the edit button will be disabled when a patient node is selected within the QC Editor, patient demographics will not be editable at the DICOM Tag Editor within the QC Editor (users can still see the information, but these cannot be edited) and the patient demographics icon on the Patient Record within the Workstation will not be visible.

NOTE: If the privilege “**Allow editing of study demographic data**” is **disabled**, the edit button will be disabled when a study node is selected within the QC Editor, study demographics will not be editable at the DICOM Tag Editor within the QC Editor (users can still see the information, but these cannot be edited) and the pencil icon used to launch the Patient Data Editor from the Study Details dialog within the Workstation will not be visible.

NOTE: By default, when the “**Allow editing of study demographic data**” privilege is changed (enabled or disabled), the change will not be picked up by the Scheduler until it is restarted or until the default cached value expires, whichever comes first. The expiry time for cached privileges is 30 minutes by default, but this can be changed by Merge Support if desired. We do not recommend adjusting this setting lower than 10 minutes.

t. Teaching

Provide one or more of the following privileges relating to the optional Teaching Files feature:

Privilege	Description
Allow user access to Teaching Files tools	Provides the right to view Teaching Worklists, tag studies and query on teaching tags.
Allow user to Manage Tags	Provides access to the Teaching Tags management page described in subsection 9.13.17 above.
Allow user to add tag metadata	Provides the right to add values to the various picklists at the Study tagging window within the Merge PACS Workstation.
Allow user to create teaching worklists	Provides the right to create new teaching worklists in addition to adding studies to existing worklists

u. Local Studies

Select “Allow user to import local studies” to provide the ability to import studies from a local source (such as a CD or a directory on the Workstation) into Merge PACS from the **Local Studies** page of the Merge PACS Workstation.

v. Advanced Workflow

Select “Allow access to Advanced Workflow Configuration” to provides access to the Advanced Workflow Configuration Management Pages.

A.1.2. Lossy and Lossless Privileges

As described in Appendix A.1.1 above, templates can include one or more of the following privileges with regard to lossy and lossless images:

- Allow access to **lossy images**
- Allow access to **Lossless Images**
- Allow access to **Lossless Images by default**

The interaction of these privileges can affect the following behaviors (note that users may get their privileges from multiple templates linked to their groups and/or to the site):

- Whether studies can be opened in the PACS Viewer
- Whether lossless studies will be loaded as lossless or lossy.
- Whether studies can be burned to CDs from the Workstation
- Whether non-image instances (such as structured reports and audio comments) will be displayed in the Viewer

- Whether the “Show Lossy” checkbox will be available on the Study Right-click Menu within the Viewer

NOTE: When Merge PACS is configured to run in **Standalone** mode, Lossy and Lossless privileges determine whether users will have access to images in the Lossy and Lossless image **volumes**, regardless of how the images were compressed by Merge PACS (if at all). For example, images that come in as lossy get stored in the Lossy volume only, and a user would need the “Allow access to lossy images” privilege to see them.

NOTE: When Merge PACS is configured to run in **Integrated** mode, images are stored on the paired EA and not in Lossy and Lossless image volumes on the Merge PACS Server. As a result, the Lossy and Lossless privileges generally control how the images are loaded instead of controlling access to them, although users will still be unable to access any images if neither privilege is enabled.

NOTE: Not having the “Allow access to lossy images” privilege may impact performance when loading tomosynthesis or other images.

a. User Has No Lossy/Lossless Privileges

If a user does not have any of the three privileges described above, the following will be true for that user whether Merge PACS is configured to run in **Standalone** or **Integrated** mode:

- The user will be **unable** to open studies within the PACS Viewer (an “Insufficient Privileges” warning message will be shown and the study will not be opened).
- The user will be **unable** to burn studies to a CD (an “Insufficient Privileges” warning message will be shown and the CD Burn dialog will not be displayed).

b. User Has Lossy Privilege Only

If a user has the “**Allow access to lossy images**” privileges, but does not have either the “**Allow access to Lossless Images by default**” or the “**Allow access to Lossless Images**” privileges, the following will be true for that user:

- If Merge PACS is configured to run in **Standalone** mode:
 - The user **will** be able to open studies within the PACS Viewer and only **lossy** images will be displayed.
 - The user **will** be able to burn studies to a CD.
 - Non-image instances **will** be displayed.
 - The “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
- If Merge PACS is configured to run in **Integrated** mode:
 - Lossless images will load as lossy.
 - Lossy images that were originally ingested as lossy will load as lossy.
 - Lossy images that were converted from lossless to lossy by EA will load as lossy.

c. User Has “Access to Lossless” Privilege Only

If a user has the "**Allow access to Lossless Images**" privilege, but does not have either the "**Allow access to Lossless Images by default**" or the "**Allow access to lossy images**" privileges, the following will be true for that user:

- If Merge PACS is configured to run in **Standalone** mode:
 - The user **will** be able to open studies within the PACS Viewer if **lossless** images are available (if only lossy images are available for a study, the original lossy images will be loaded in the viewer).
 - The user **will** be able to burn studies to a CD if **lossless** images are available (if only lossy images are available for a study, CD Burn dialog will be displayed).
 - Non-image instances **will** be displayed.
 - The “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
- If Merge PACS is configured to run in **Integrated** mode:
 - Lossless images will load as lossless and the “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
 - Lossy images that were originally ingested as lossy will load as lossy.
 - Lossy images that were converted from lossless to lossy by EA will load as lossy.

d. User Has “Access to Lossless By Default” Privilege Only

If a user has the "**Allow access to Lossless Images by default**" privilege, but does not have either the "**Allow access to Lossless Images**" or the "**Allow access to lossy images**" privileges, the following will be true for that user whether Merge PACS is configured to run in **Standalone** or **Integrated** mode:

- The user will be **unable** to open studies within the PACS Viewer (an “Insufficient Privileges” warning message will be shown and the study will not be opened).
- The user will be **unable** to burn studies to a CD (an “Insufficient Privileges” warning message will be shown and the CD Burn dialog will not be displayed).

e. User Has “Access to Lossy” and “Access to Lossless By Default” Privileges Only

If a user has the “**Allow access to lossy images**” and the "**Allow access to Lossless Images by default**" privilege, but does not have the "**Allow access to Lossless Images**" privilege, the following will be true for that user:

- If Merge PACS is configured to run in **Standalone** mode:
 - The user **will** be able to open studies within the PACS Viewer and only **lossy** images will be displayed.
 - The user **will** be able to burn studies to a CD.

- Non-image instances **will** be displayed.
- The “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
- If Merge PACS is configured to run in **Integrated** mode:
 - Lossless images will load as lossy and the “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
 - Lossy images that were originally ingested as lossy will load as lossy.
 - Lossy images that were converted from lossless to lossy by EA will load as lossy.

f. User Has “Access to Lossless” and “Access to Lossless By Default” Privileges Only

If a user has the “**Allow access to Lossless images**” and the “**Allow access to Lossless Images by default**” privilege, but does not have the “**Allow access to lossy images**” privilege, the following will be true for that user:

- If Merge PACS is configured to run in **Standalone** mode:
 - The user **will** be able to open studies within the PACS Viewer if **lossless** images are available (if only lossy images are available for a study, the original lossy images will be loaded in the viewer).
 - The user **will** be able to burn studies to a CD if **lossless** images are available (if only lossy images are available for a study, CD Burn dialog will be displayed).
 - Non-image instances **will** be displayed.
 - The “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
- If Merge PACS is configured to run in **Integrated** mode:
 - Lossless images will load as lossless and the “Show Lossy” checkbox will **not** be available on the Study Right-click Menu.
 - Lossy images that were originally ingested as lossy will load as lossy.
 - Lossy images that were converted from lossless to lossy by EA will load as lossy.

g. User Has “Access to Lossy” and “Access to Lossless” Privileges Only

If a user has the “**Allow access to lossy images**” and the “**Allow access to lossy images**” privilege, but does not have the “**Allow access to Lossless Images by default**” privilege, the following will be true for that user:

- If Merge PACS is configured to run in **Standalone** mode:
 - The user **will** be able to open studies within the PACS Viewer and **lossy** images will be displayed by default.
 - The user **will** be able to burn studies to a CD.
 - Non-image instances **will** be displayed.

- The “Show Lossy” checkbox **will** be available on the Study Right-click Menu and the user will be able to toggle between lossy and lossless images if both are present.
- If Merge PACS is configured to run in **Integrated** mode:
 - Lossless images will load as lossless, but the user can choose to display them as lossy via the “Show Lossy” checkbox.
 - Lossy images that were originally ingested as lossy will load as lossy.
 - Lossy images that were converted from lossless to lossy by EA will load as lossy.

h. User Has All Lossy/Lossless Privileges

If a user has all three of the privileges described above, the following will be true for that user:

- If Merge PACS is configured to run in **Standalone** mode:
 - The user **will** be able to open studies within the PACS Viewer and **lossless** images will be displayed by default if available.
 - The user **will** be able to burn studies to a CD.
 - Non-image instances **will** be displayed.
 - The “Show Lossy” checkbox **will** be available on the Study Right-click Menu and the user will be able to toggle between lossy and lossless images if both are present.
- If Merge PACS is configured to run in **Integrated** mode:
 - Lossless images will load as lossless, but the user can choose to display them as lossy via the “Show Lossy” checkbox.
 - Lossy images that were originally ingested as lossy will load as lossy.
 - Lossy images that were converted from lossless to lossy by EA will load as lossy.

A.1.3. Blinded Reading Considerations

CAUTION: The “Hide patient/study/series information” privilege **MUST NOT** be selected for normal users, since turning off the demographics raises patient safety concerns and the intent of this privilege is to **ONLY** be used for clinical trials and for presentations (e.g. Tumor Board) under controlled conditions.

In order to use this feature, the users **MUST** each have to have a second login for this purpose and **MUST NOT** have Mark as Read or most other status change privileges. Furthermore, these users should have the bare minimum of admin tools available to avoid incidental Patient Health Information (PHI) display.

A single shared login for this purpose would violate HIPAA auditing.

When used for presentations, the PC/Laptop must be configured for dual display operation (**NOT** cloning). The worklist and all windows that could have PHI must be used on the primary display (as arranged by the user). The image display with any desktop sharing (e.g. WebEx) or projector must be used on the secondary display.

There is no way to Guarantee that PHI may not have accidental / incidental display of demographics (e.g. something pops up on the "public monitor"). It is the responsibility of the user to rehearse their use of this for public presentations prior to the public presentation.

As described in Appendix A.1.1 above, disabling the “Hide patient/study/series information” privilege provides a “blind reading” capability whereby the user will not be able to view patient, series and study-specific information in the following locations within the Merge PACS Workstation:

- **The Viewer Titlebar**
- **All DICOM Overlays**
- **Series Descriptions** (both within navigation thumbnails and individual Viewports).

In addition, access to the DICOM Attributes Viewer, the Order Viewer and the Report Viewer will all be disabled from within the Viewer.

The primary use case for this is to provide for “blinded reading” for use with clinical trials where a user may be required to review/read a study without seeing the identifying information for that study/patient.

NOTE: The information is only hidden from within the Viewer. It will still be visible in other locations such as RTWL, the Query Page, etc.

NOTE: The actual information that is hidden is configurable on a site-by-site basis, as described in subsection 1.2.80.ii above.

A.2. Preferences

Some or all of the following preferences can be included in a preference template that can be linked to a site or to a group.

NOTE: These preferences can also be set (as overrides) for individual users from the “User” section of the **Merge PACS Preferences Dialog** within the Workstation.

NOTE: Refer to the *Merge PACS 8.2 Workstation Users Guide* for a detailed description of each of these preferences.

A.2.1. Start-up Preferences

The following preferences relate to starting up the Workstation and/or the PACS Viewer:

Preference	Description
Display RTWL in Workstation Browser on Launch	If selected, RealTime Worklist (or RealTime Study List, if applicable) will be displayed by default in the Workstation Browser when it is launched.
Enable RealTime Study List	If you have login privileges to access the RealTime Study list, this option will be selected by default. If deselected, you will see the RealTime Worklist within the Workstation Browser instead.
Automatically Open Dashboard	If your site is configured to provide automatic assignments of exams, the user is eligible to have exams assigned to him or her, and the user has the “Allow Access to Dashboard” user privilege, the Mini Dashboard will be automatically launched when the user first accesses the Workstation.
Automatically Open Order Viewer	If selected, the Order Viewer will automatically be launched when a study is first opened within the Merge PACS Viewer.
Automatically Open Report Viewer	If selected, the Report Viewer will automatically be launched when a study is first opened within the Merge PACS Viewer.

Preference	Description						
Automatically Open ER Panel	If selected, the ER Workpanel will automatically be launched when a study is first opened within the Merge PACS Viewer. If the “conditional” sub-option is enabled, you can further select one of the following: <table> <tr> <th>Option</th><th>General Description</th></tr> <tr> <td>ER Statuses</td><td>Allows you to select one or more ER statuses, and the ER Workpanel will only be automatically launched when a study’s status is among the selected ER statuses.</td></tr> <tr> <td>Rule</td><td>Displays a Select Rule button that will let you search for and select a previously defined study rule, and the ER Workpanel will only be automatically launched when the conditions defined by the selected rule are met. When searching for a rule, you can also click on the Manage Rules button to launch the Manage User Rules Window to create a new study rule, as described in Section 14.2 above, which can then be selected.</td></tr> </table>	Option	General Description	ER Statuses	Allows you to select one or more ER statuses, and the ER Workpanel will only be automatically launched when a study’s status is among the selected ER statuses.	Rule	Displays a Select Rule button that will let you search for and select a previously defined study rule, and the ER Workpanel will only be automatically launched when the conditions defined by the selected rule are met. When searching for a rule, you can also click on the Manage Rules button to launch the Manage User Rules Window to create a new study rule, as described in Section 14.2 above, which can then be selected.
Option	General Description						
ER Statuses	Allows you to select one or more ER statuses, and the ER Workpanel will only be automatically launched when a study’s status is among the selected ER statuses.						
Rule	Displays a Select Rule button that will let you search for and select a previously defined study rule, and the ER Workpanel will only be automatically launched when the conditions defined by the selected rule are met. When searching for a rule, you can also click on the Manage Rules button to launch the Manage User Rules Window to create a new study rule, as described in Section 14.2 above, which can then be selected.						
Mark Studies as Viewed when Launched in Web Viewer	If selected, studies will be given a status of “Viewed” as soon as you click on the View Study in iConnect Access Viewer button on a worklist. NOTE: This option will only be available if Merge PACS is configured for use with the iConnect Access Viewer. CAUTION: When this option is enabled, the study status will be set to “Viewed” prior to actually launching the iConnect Access Viewer. As a result, the status will be set to “Viewed” even if the Viewer fails to launch or the study cannot be displayed.						
Associate accessions with primary study to 3rd Party Application	If selected, and your system is configured to integrate with a third-party dictation application, users will be prompted to select multiple accession numbers if there are multiple unread studies when they click the 3rd party Application Synchronization button.						
Display Availability Dialog on Launch	If selected, the Reader Location and Availability Confirmation dialog will automatically be displayed when the user first logs into the Workstation.						

Preference	Description
Pre-caching Priority	If you have background pre-caching enabled, you can select from one of the following options to determine what priority the pre-caching process should be given by your Workstation: • Low • Medium • High • Highest With regard to precaching of multi-frame images in particular, selecting High or Highest will allow the Workstation to download four multi-frame images in parallel, Medium will download two images in parallel, and low will only download one at a time.

A.2.2. Shutdown

The following preferences relate to shutting down the Workstation and/or the PACS Viewer:

Option	General Description
Automatically Save Annotations on Exit	If selected and the user has login privileges to save annotations, any text annotations the user adds to a Study's images will automatically be saved when the user exits the Study. In addition, the user will not be prompted to save annotation changes when creating or editing a Study Presentation. If not selected, the user will be prompted to save annotations when the user exits and will also be prompted to save annotation changes when creating or editing Study Presentations. NOTE: If a user has the Allow Viewer Annotations to be saved using button privilege, as described in Appendix A.1.1 above, he or she will not be able to use the Automatically Save Annotations on Exit user preference. NOTE: QC special annotations created by Pixel-Masking and Burnt-in Text tools will not be saved automatically.
Automatically Save Voice Clip on Exit	If selected, any recorded voice clips will automatically be saved when the user exits the study.
Warn if Annotations Cannot Be Saved	If selected, a warning message will be displayed when you attempt to create an annotation if annotations cannot be saved for this study (e.g., due to your login privileges, the status of the study, etc.) NOTE: To avoid unnecessary user annoyance, this preference should not be selected unless the users are truly concerned about being able to save annotations and measurements for prior/comparison studies.

Option	General Description
Delete Image Cache On Exit	If you have background pre-caching enabled, selecting this option will cause all images to be deleted from your workstation's hard drive when you exit the Merge PACS Workstation.
Keep Viewer Running in Background upon Exit	If selected, the Merge PACS Viewer will be kept running in the background when you exit the Workstation so that it will load faster the next time it is launched. To exit the Viewer completely, right-click on the Merge PACS Workstation icon on the Windows Taskbar and select "Exit Merge PACS Diagnostic Workstation" from the pop-up menu. NOTE: This option will only be available if you are running The Merge PACS Viewer from within another application (i.e., as an "embedded viewer") or from a command-line with <code>setBrowserVisible=false</code>.
Log off Windows On Exit	If selected, you will be logged out of Windows when you exit the Merge PACS Workstation.

A.2.3. Study Preferences

The following preferences relate to how studies are loaded and locked:

a. Study Loading

The following preferences relate to the loading of studies:

Option	General Description
Get Next Study by Precedence	This preference controls the behavior of the Open Next Study , Mark Study Read and Skip Study worklist navigation buttons on the Viewer Toolbar and the Open Next Study and Mark Study Read buttons on the Patient Record Toolbar. The following options are available:
Option	General Description
Go to top of worklist	Opens the first candidate study in the first unguarded block of the worklist
Go to top of current worklist block	Opens the first candidate study in the current worklist block or, if there are no candidate studies in this block, opens the first candidate study in the first unguarded block of the worklist.
Go to next	The behavior depends on whether the worklist navigation button is clicked while viewing a study from a “guarded” or an “unguarded” worklist block, as described in subsection 13.4.10 above: - From an unguarded block: - Opens the next remaining candidate study in the current block, if any. - If there is no remaining candidate Study in the current block, opens the next candidate Study in next unguarded block in the worklist (based on display order). - If there is no remaining candidate study in the worklist, opens the first candidate study in the first unguarded block of the worklist. - From a guarded block: - Opens the next remaining candidate study in the current block, if any. - If there is no remaining candidate study in the current block, opens the the first candidate study in the current worklist block.
None	Disables this preference if the Load Next Study Based on Acuity Score (described below) is enabled. Otherwise, behaves the same as the “Go to next” option

NOTE: A “candidate” Study is one that has viewable images, is not currently hidden (either by itself, as part of a hidden status group or as part of a hidden date group, but not just when within a hidden worklist block) and is neither locked nor reserved by another user.

Option	General Description
	NOTE: The “Go to top of worklist” and “Go to top of current worklist block” options are recommended for use with worklists configured so that studies may be removed when their statuses are changed (e.g., a worklist that is configured to only show studies in “Unread” or “Unverified” status will remove a Study once it is manually marked as “Read” or the user clicks on the Mark Study Read button). The “Go to next” option should be used with worklists that do not remove studies to prevent getting stuck on the first few studies in the worklist.
Load Next Study Based on Acuity Score	If selected and the optional Merge RadStream component is enabled, the Open Next Study and Mark Study Read buttons on the Patient Record Toolbar and the Merge PACS Viewer Toolbar will select the next study based on acuity score instead of position on the worklist. NOTE: If this option is selected, you must set the Get Next Study by Precedence preference described above to “None.”
Automatically Play Voice Clip on Load	If selected and the user has login privileges to listen to audio annotations via VoiceClip, the first audio annotations that has been recorded for a study will be played automatically when the study is first opened.
Automatically Split CR Series	If selected, CR Series with multiple images will automatically be split into separate Series for display purposes within the Merge PACS Viewer.
Automatically Split CT Series	If selected, CT series will be split into multiple series for display purposes within the Merge PACS Viewer in the following cases: • When the origin of the images change • When the orientation of the images change (except for short runs, i.e., of 5 slices or less) • When the gap between slices exceeds 5 times the current slice thickness
Automatically Split MR Series	If selected, MR Series with multiple images will automatically be split into separate Series for display purposes within the Merge PACS Viewer based on: • Echo Number • Temporal Position Identifier (TPI) • Positional Sequence • Stack Axis/Positional Break
Automatically Split MG Series	If selected, MG Series with multiple images will automatically be split so that each image will be its own Series and then grouped based on the following rules: • 2D images will be grouped based on laterality, view code and view code modifier. Conventional images and synthetic images representing the same laterality and view will be grouped together. • 3D breast tomosynthesis images will be separated from 2D images or other 3D images even if they represent the same laterality and view. • 3D breast tomosynthesis images (thin slices) will be labeled with (DBT) suffix. • GE thick slices (slabs) are labeled with (DBT Sb) suffix. • Siemens Insight 3D rotational images are labeled with (Insight). CAUTION: Splitting is recommended for all Mammography studies. Deselection of this option will cause multiple issues during the viewing of Mammography studies.

Option	General Description
Automatically Split DX Series	If selected, DX Series with multiple images will automatically be split into separate Series for display purposes within the Merge PACS Viewer.
Preserve Presentation State within Viewport	If selected, the current presentation state (W/L, zoom, etc.) within each Series Viewport will be maintained when a new Series is opened within that Series Viewport.

b. Locking

The following preferences relate to the locking of studies:

Option	General Description										
Lock and Update Study Status in Secondary Viewer	If selected, studies opened in a Secondary Merge PACS Viewer window will be locked and users will be prompted to set the status upon exiting. If not selected, studies opened in the Secondary Viewer window will be read-only (they will not be locked, you will be unable to change the workflow status and the context of any dictation application will remain on the study in the Primary Viewer). NOTE: When this option is enabled, the Secondary Viewer will behave like a Primary Viewer. If the application is configured to sync automatically, the study will be automatically synced with the 3rd-party application when it is opened in the Primary Viewer. If a second study is then opened in the Secondary Viewer without closing the primary study, the primary study will lose the sync and the secondary study will automatically sync with the third party application instead.										
Unread Study Lock	Select one of the following options to configure how unread studies are locked: <table> <tr> <th>Option</th><th>General Description</th></tr> <tr> <td>All</td><td>If selected, all studies related to the study currently being viewed (primary as well as priors) will be locked from within RTWL.</td></tr> <tr> <td>Same Modality</td><td>If selected, only studies of the same modality type as the study currently being viewed will be locked from within RTWL.</td></tr> <tr> <td>Only Relevant</td><td>If selected, only relevant prior studies related to the study currently being viewed will be locked from within RTWL.</td></tr> <tr> <td>Only Primary Study</td><td>If selected, only the primary study currently being viewed will be locked from within RTWL.</td></tr> </table>	Option	General Description	All	If selected, all studies related to the study currently being viewed (primary as well as priors) will be locked from within RTWL.	Same Modality	If selected, only studies of the same modality type as the study currently being viewed will be locked from within RTWL.	Only Relevant	If selected, only relevant prior studies related to the study currently being viewed will be locked from within RTWL.	Only Primary Study	If selected, only the primary study currently being viewed will be locked from within RTWL.
Option	General Description										
All	If selected, all studies related to the study currently being viewed (primary as well as priors) will be locked from within RTWL.										
Same Modality	If selected, only studies of the same modality type as the study currently being viewed will be locked from within RTWL.										
Only Relevant	If selected, only relevant prior studies related to the study currently being viewed will be locked from within RTWL.										
Only Primary Study	If selected, only the primary study currently being viewed will be locked from within RTWL.										

c. Comparison Studies

The following preferences relate to the loading of comparison studies:

Option	General Description
Display Warning Indicator on All Priors	If selected, all prior studies loaded in comparison mode will bear the warning icon indicating that they are prior studies.

Option	General Description						
Allow Rapid Review of Unread Studies	If selected, studies with a status of UNREAD will be included for use with the Viewer's Rapid Review feature.						
Consider All Unverified as Unread Studies	If selected, studies with a status of UNVERIFIED will be treated the same as UNREAD studies while in comparison mode.						
Automatically Pre-cache Relevant Prior Studies	If selected, images for relevant prior studies will be pre-cached as soon as the primary study is loaded into the Merge PACS Viewer. If not selected, images for relevant prior studies will not be cached until the prior study is itself loaded into the viewer.						
Automatically pre-cache unread studies	If selected, images for unread prior studies will be pre-cached as soon as the primary study is loaded into the Merge PACS Viewer. If not selected, images for unread prior studies will not be cached until the prior study is itself loaded into the viewer.						
Order Priors	Select one of the following options to configure how comparison studies are ordered (including the display of Navigation Thumbnail Images for comparison studies that are manually dragged and dropped into the Viewer): <table> <tr> <th>Option</th><th>General Description</th></tr> <tr> <td>Chronologically</td><td>If selected, the list of prior studies will be displayed in reverse chronological order instead of by relevance.</td></tr> <tr> <td>By Relevance</td><td>If selected, the list of prior studies will be sorted by relevance. In the case of Navigation Thumbnail Images for comparison studies, this would only apply to non-relevant studies that are dragged and dropped into the viewer.</td></tr> </table>	Option	General Description	Chronologically	If selected, the list of prior studies will be displayed in reverse chronological order instead of by relevance.	By Relevance	If selected, the list of prior studies will be sorted by relevance. In the case of Navigation Thumbnail Images for comparison studies, this would only apply to non-relevant studies that are dragged and dropped into the viewer.
Option	General Description						
Chronologically	If selected, the list of prior studies will be displayed in reverse chronological order instead of by relevance.						
By Relevance	If selected, the list of prior studies will be sorted by relevance. In the case of Navigation Thumbnail Images for comparison studies, this would only apply to non-relevant studies that are dragged and dropped into the viewer.						

d. Presentation State

Option	General Description
Automatically Apply Presentation State	If selected, if there are any externally created ("foreign") Presentation States available for a Series, the most recent one will be automatically applied when the Series is opened, whether as part of the primary study or as part of a comparison study. Note that the externally created Presentation State will be applied on top of the currently selected Hanging Protocol or Study Presentation, if any.

A.2.4. Hanging Protocols

The following preferences relate to Hanging Protocols:

Option	General Description
Override DICOM Overlay Setting	If selected, any DICOM Overlay settings contained within a Hanging Protocol or Study Presentation will be overridden by the DICOM Overlay button and the DICOM Overlay Series Toggle option.
Include UNREAD Studies as Comparison Candidates	If selected, studies with a status of UNREAD will be considered valid comparison candidates for Hanging Protocols.
Include Studies Newer than Primary as Comparison Candidates	If selected, studies that have a time/date stamp newer than the primary study will be considered valid comparison candidates for Hanging Protocols.
Show Unviewed Images after Last Hanging Protocol Step (MG Only)	If selected, when a Hanging Protocol is applied to an Mammography study, all unviewed images will automatically be displayed in a 2x1 layout after the last step has been applied. The images will be displayed using modality preferences and in the same sequence as they are present in a study.

A.2.5. Key Images

The following preferences relate to Key Images:

Option	General Description
Automatically Mark Annotated Images as Key Images	If selected, any images that have annotations added to them will automatically be flagged as key images.
Open Key Image Viewport on Primary Study Load	If selected, the Key Image Viewport will automatically be displayed when the primary study is loaded, if applicable.
Open Key Image Viewport on Comparison Study Load	If selected, the Key Image Viewport will automatically be displayed when a prior study is loaded, if applicable.
Open Key Image Viewport When Image Added	If selected, the Key Image Viewport will automatically be displayed whenever an image is flagged as a key image.
Show Key Image Series for Comparison Study	If selected, key images for comparison studies will be displayed when the primary study is being viewed.
Allow Priors as Key Images	If selected, images that are part of a prior study will be able to be flagged as key images.
Auto Layout Key Image Window	If selected, the system will attempt to optimize the layout of the images within the Key Image Viewport (as opposed to displaying them in the order they were flagged).

A.2.6. 3D

The following preferences relate to The Merge PACS Viewer's inherent 3D functionality:

Option	General Description
Default Single View Orientation	Allows you to select one of the following options as the default single view orientation: • Axial • Coronal • Sagittal
Grouped 3D View Orientation	Allows you to select one of the following options as the default grouped view orientation: • Axial • Coronal • Sagittal
Volume Rendering Algorithm	Allows you to select one of the following options as the default volume rendering algorithm: • MIP • MinP • Average • CVR • Faded MIP
Initial Slab Thickness (mm)	Allows you to select one of the following options as the default initial slab thickness in millimeters: • 2.5 • 5.0 • 10.0 • No Slab
Thin Slice Interpolation	Allows you to select one of the following options as the default thin slice interpolation for MPR images to improve the image quality: • NONE • TRILINEAR • TRICUBIC
Thick Slice Interpolation	Allows you to select one of the following options as the default thick slice interpolation for MPR images to improve the image quality: • NONE • TRILINEAR • TRICUBIC NOTE: This should be set to Tricubic or Trilinear unless there are large numbers of lossy thick slice images that are used. Use of the "None" option will result in sub-optimal image quality for PET images.

Option	General Description						
Thick / Thin Boundary (mm)	Allows you to set the default thick / thin boundary (in millimeters) for MPR images, which will determine whether the image quality should be in thin slice interpolation or thick slice interpolation.						
Enable Skimming for Volume Navigation Mouse Mode	When selected, the standard “page” mouse tool will act as a “skimming” tool for 3D volumes, except that no warning will be displayed that images are being skipped. CAUTION: This option should NOT be enabled for any reading physician (radiologist, ER physician, orthopedic surgeon, etc.). It is only intended as a convenience for non-reading/ non-interpreting referring physicians.						
Enable Background Loading for Tomosynthesis	When enabled, 3D processing will be performed for multi-frame breast tomosynthesis images in the background as soon as they are loaded into the Viewer in order to speed up the process of generating 3D volumes such as MIP or MPR. This will be done only if the following conditions are met, however: • The Workstation is 64-bit. • The image is already pre-cached before it is loaded into the Viewer. • The image is not a thick-slice (“slab”) image. • There is sufficient memory to perform the operation. NOTE: If a study contains multiple tomosynthesis images, the 3D volumes will be processed in sequential order instead of simultaneously. If a user manually selects to generate a 3D volume for one image while a volume is being processed in the background for another image, the background process will be cancelled and the user selected volume will be processed instead.						
Integrate with Vitrea	If Merge PACS has been configured to use Vitrea® to provide 3D functionality instead of the Merge PACS Viewer’s inherent 3D rendering tools, select this option and enter the following information in the fields provided: <table> <tr> <th>Parameter</th><th>Value</th></tr> <tr> <td>Load Command</td><td>visend load STUDYUID SERIESUID IMAGEUID</td></tr> <tr> <td>Unload Command</td><td>visend unload STUDYUID SERIESUID IMAGEUID</td></tr> </table>	Parameter	Value	Load Command	visend load STUDYUID SERIESUID IMAGEUID	Unload Command	visend unload STUDYUID SERIESUID IMAGEUID
Parameter	Value						
Load Command	visend load STUDYUID SERIESUID IMAGEUID						
Unload Command	visend unload STUDYUID SERIESUID IMAGEUID						

A.2.7. Annotations

The following preferences relate to the use of annotations within the Viewer:

Option	General Description
Line Thickness	Allows you to select the thickness of all annotation measurement lines. The following options are available: • Thin • Medium • Thick
Show Cursor While Annotating	By default, the mouse cursor will be displayed on the screen while you use any annotation tool. If desired, you can choose to hide the cursor while annotating by deselecting this option.
Allow Xref Update on Mouse Hover	By default, the “active” Viewport for the purpose of cross-reference lines is whichever Viewport the mouse cursor is currently hovering over, regardless of whether you have actually clicked on that Viewport. If this option is not selected, you must actually click on a Viewport to make it “active” for the purpose of cross-reference lines.
Allow Small Line Annotations	By default, very small line measurements (<i>i.e.</i>, of only a few pixels) are not recognized. If this option is selected, line measurements of any size (<i>i.e.</i>, if you have moved the mouse cursor to any new location after clicking), will remain after releasing the mouse button.
Render Annotations in Grayscale Mode with Shadow Effect	If selected, any text annotations will be displayed as gray text with a shadow, so as to be more visible against white backgrounds. NOTE: This preference will only be effective if Global Stack has not been enabled for the Series Viewport.
PET SUV Measurements	Allows you to select one or more of the following Standardized Uptake Values (SUVs) to be displayed when using the Probe, ROI, Region Analysis Area and Region Analysis Volume tools with PET images: • Body Weight (g/ml) • Lean Body Mass (g/ml) • Ideal Body Weight (g/ml) • Body Surface Area (cm³/ml) Refer to Appendix B below for details on how SUVs are calculated.

A.2.8. MG Tools

The following preferences relate to the use of Mammography-specific tools within the Viewer:

Option	General Description
Magnifying Glass	Allows you to select the size and shape of the Magnify tool's viewing area as follows: • Size: ○ Smallest ○ Small ○ Medium ○ Large ○ Largest ○ Custom • Shape: ○ Rectangle ○ Circle
Binoculars	Allows you to select the size and shape of the Binocular tool's viewing area as follows: • Size: ○ Smallest ○ Small ○ Medium ○ Large ○ Largest ○ Custom • Shape: ○ Rectangle ○ Circle
Hot Light	Allows you to select the size and intensity of the Binocular tool's viewing area as follows: • Size: ○ Smallest ○ Small ○ Medium ○ Large ○ Largest ○ Custom • Intensity: ○ 10% ○ 15% ○ 20% ○ 25% ○ Custom

A.2.9. Merge Messenger

The following preferences relate to configuring your Merge Messenger preferences and will be available if this site has been configured to use Merge Messenger:

Option	General Description
Messenger Client always on top	If selected, the main Merge Messenger window will always be on top of other application windows unless minimized.
Automatically Launch Messenger Client	If selected, the Merge Messenger application will be automatically launched when you log into the Merge PACS Workstation (<i>i.e.</i>, the Merge Messenger icon will be displayed on the Windows System Tray as “Online” and will begin listening for messages). If not selected, the Merge Messenger icon will display as “Not Connected” on the System Tray as well as in the Workstation Browser and the Merge PACS Viewer. To launch Merge Messenger, click on the Merge Messenger icon in the Workstation Browser or the Merge PACS Viewer once to launch the application and then, once the icon changes to “online” status, click it again to open the Merge Messenger window as usual.
Chat window always on top	If selected, the Merge Messenger Chat window will always be on top of other application windows unless minimized.
Show chat message in system tray	If selected, the content of any messages received will be displayed in an alert balloon in the Windows System Tray / Task Bar.
Report Idle Time	Allows you to select one of the following options: • Never – Remain “Online” until you manually select another status or log off entirely. • Based on mouse or keyboard use – Automatically change your status to “Away” after a set period of keyboard and mouse inactivity.
Minutes before becoming Idle	If the Based on mouse or keyboard use option has been selected for “Report Idle Time”, this allows you to specify how many minutes of inactivity should trigger a change to “Away” status. Note that you can either select a number from the drop-down menu or enter a number manually.

A.2.10. Display/Mouse/Keyboard

The following preferences relate to the use of your mouse and keyboard, as well as the overlay font height:

a. Mouse

The following preferences are related to your mouse usage:

Option	General Description
Interaction	Allows you to configure whether you need to hold down the left mouse button in order to use the various Mammography-specific tools (Binocular , Hot Light , Dual Link Magnifier). The following options are available: Operate on Click-drag – The left mouse button must be held down. Operate on Click – Click once to activate the tool, click again to turn off.
Use Skimming for Middle Mouse Drag	If selected, the Skimming tool will be available by pressing and holding the center wheel on your mouse as you drag your cursor across a Series Viewport.
Delayed Right Click	If selected, the Series Right-click Menu will only appear if you click on hold the mouse button for a brief period of time and single clicking with the right-mouse button will toggle among mouse modes. If not selected, single-clicking with the right mouse button will immediately display the menu.
Ignore Horizontal Mouse Movement when Paging	If not selected, you will be able to page through images forwards and backwards by moving the mouse to the right and left, in addition to moving the mouse up and down.
Assign Mouse Button Actions	Select this option and then click on the Customize button to launch the Mouse Configuration dialog.
Customize Mouse-Cycle Modes/Toolbox	Click on the Customize button to launch the Customize Mouse-Cycle Modes dialog.

NOTE: Refer to Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide* for a detailed description of each of these preferences.

b. Keyboard

Most of the various actions (whether functions or tools) within the Workstation are associated with keyboard shortcuts that let users perform the function or select the tool by pressing one or more keys on their keyboard. New installations of Merge PACS come with a set of default keyboard shortcuts, but these can be customized at the site or group level via a template or at the user level via the Preferences menu within the Workstation.

Note the following:

- Not all keys or keystroke combinations are allowed (a warning message will be displayed if you attempt to use an invalid key).
- Some actions have shortcuts that are not configurable.
- Any macros that have been created, as described in Appendix A.2.20, can also be assigned to keyboard shortcuts.

Click on the **Display** button to launch the **Keyboard Shortcut Assignments** dialog in a separate pop-up window, which allows you to view a list of all currently assigned keyboard shortcuts that can be sorted, printed out or exported to a Comma Separated Value (.csv) file that can be opened in spreadsheet applications such as Microsoft® Excel, as in the following example:

NOTE: You can also view all assigned keyboard shortcuts by clicking on the **View All Assigned** button at the **Customize Keyboard Shortcuts** dialog.

NOTE: Refer to Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide* for a detailed description of how to create custom keyboard shortcuts.

c. Display Height

As described in Appendix A.2.12 below, you can customize the display of the Lossy Compression overlay text and each DICOM field included in the DICOM overlay text to appear in a bold and/or color highlighted font as well as to appear in a small, medium or large font size. In addition, you can configure a different “base font size” to be used for the all overlay text with different screen resolutions.

In order to configure a different base font size for different screen solutions, however, you must first define what constitutes a “small” and “large” screen resolution in the **Display Height** section, as shown below:

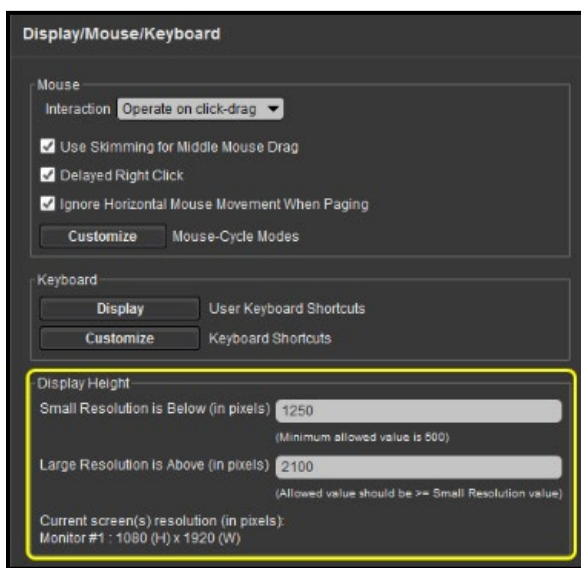


Figure 720: Viewing All Assigned Keyboard Shortcuts

You can manually define “small” resolution as being below a certain number of pixels in height and define “large” resolution as being above a certain number of pixels in height. The system will then automatically calculate “medium” resolution as being anything between those two numbers:

Option	General Description
Small Resolution is Below (in pixels)	Enter the maximum height (in pixels) of a monitor to be considered “small resolution”
Large Resolution is Above (in pixels)	Enter the minimum height (in pixels) of a monitor to be considered “large resolution”

For example, a user might define “Small Resolution” as any screen with a resolution of anything below 1200 (pixels) in height and “Large Resolution” as any screen with a resolution of anything above 2048 (pixels) in height. The system would then automatically calculate “Medium Resolution” as any screen with a resolution between 1200 and 2048 pixels in height.

A.2.11. Modality

The Modality preferences section allows you to set a variety of default preferences for different modalities, as in the following example:

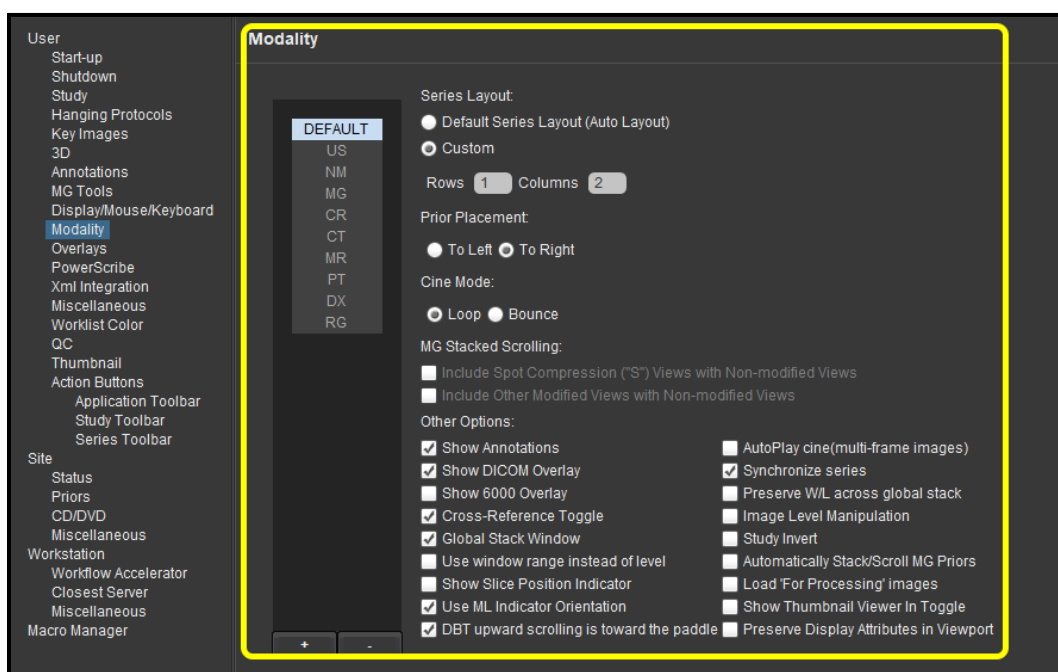


Figure 721: Modality Preferences

The left-hand side of the screen has a menu displaying various specific image modality types that can be selected as well as a **DEFAULT** option for every other type of modality as well as modalities for which specific settings have not been defined. You can also use the + and – buttons at the bottom of the menu to add new modality types or remove modality types that you have previously added (you cannot remove any of the four predefined modality types).

For each modality type, you can set the default series layout as well as various miscellaneous preferences, as described below:

Option	General Description						
Series Layout	For each modality type you can select one of the following two options with regard to Series Layout: <table> <tr> <th>Option</th><th>General Description</th></tr> <tr> <td>Default Series Layout (Auto Layout)</td><td>This allows the Merge PACS Viewer to automatically determine the optimal series layout, based on the number of series in a given study.</td></tr> <tr> <td>Custom</td><td>This allows you to manually enter the number of rows and columns that should be displayed.</td></tr> </table>	Option	General Description	Default Series Layout (Auto Layout)	This allows the Merge PACS Viewer to automatically determine the optimal series layout, based on the number of series in a given study.	Custom	This allows you to manually enter the number of rows and columns that should be displayed.
Option	General Description						
Default Series Layout (Auto Layout)	This allows the Merge PACS Viewer to automatically determine the optimal series layout, based on the number of series in a given study.						
Custom	This allows you to manually enter the number of rows and columns that should be displayed.						
Prior Placement	By default, when users view Primary Mammography and Prior Mammography studies together in Compare Studies mode or Stack Scrolling mode, the Prior study will be displayed to the right of the Primary study. If desired, however, you can configure the Viewer to always display Prior Mammography studies to the left of the Primary study by selecting the To Left option. NOTE: This option will only be available when setting the MG (or the DEFAULT) modality.						
Cine Mode	Depending on the modality, the Cine mode will operate in one of the following two modes by default: <table> <tr> <th>Mode</th><th>Description</th></tr> <tr> <td>Loop</td><td>After the last image in the series is displayed the paging cycle will start over again with the first image</td></tr> <tr> <td>Bounce</td><td>The paging cycle will keep reversing directions when the last and first images are displayed. This is the default for MG, PT and NM modalities.</td></tr> </table> If desired, however, you can change the default Cine mode for one or more modalities by selecting the desired Cine Mode option.	Mode	Description	Loop	After the last image in the series is displayed the paging cycle will start over again with the first image	Bounce	The paging cycle will keep reversing directions when the last and first images are displayed. This is the default for MG, PT and NM modalities.
Mode	Description						
Loop	After the last image in the series is displayed the paging cycle will start over again with the first image						
Bounce	The paging cycle will keep reversing directions when the last and first images are displayed. This is the default for MG, PT and NM modalities.						
MG Stacked Scrolling	For Modality type MG, you can configure the Viewer to include modified views along with non-modified views in the stack as follows: <table> <tr> <th>Mode</th><th>Description</th></tr> <tr> <td>Include Spot Compression ("S") Views with Non-modified Views</td><td>If selected, the spot compression ("S") view will be included in the stack of non-spot compression views (e.g., "RCCS" will be displayed in the "RCC" stack).</td></tr> <tr> <td>Include Other Modified Views with Non-modified Views</td><td>If selected, the modified views (other than spot compression view) will be included in the stack of non-modified views (e.g., "RML" will be displayed in the "RML0" stack).</td></tr> </table> NOTE: These options will only be available when setting the MG (or the DEFAULT) modality and only if Automatically Stack/Scroll MG Priors is enabled, as described below.	Mode	Description	Include Spot Compression ("S") Views with Non-modified Views	If selected, the spot compression ("S") view will be included in the stack of non-spot compression views (e.g., "RCCS" will be displayed in the "RCC" stack).	Include Other Modified Views with Non-modified Views	If selected, the modified views (other than spot compression view) will be included in the stack of non-modified views (e.g., "RML" will be displayed in the "RML0" stack).
Mode	Description						
Include Spot Compression ("S") Views with Non-modified Views	If selected, the spot compression ("S") view will be included in the stack of non-spot compression views (e.g., "RCCS" will be displayed in the "RCC" stack).						
Include Other Modified Views with Non-modified Views	If selected, the modified views (other than spot compression view) will be included in the stack of non-modified views (e.g., "RML" will be displayed in the "RML0" stack).						

Option**General Description****Other Options**

For each modality type, you can set the following miscellaneous display preferences:

Option	General Description
Show Annotations	If selected, the display of annotations will be automatically enabled whenever a study of this particular modality type is loaded into the Viewer.
AutoPlay Cine (multi-frame images)	If selected, multi-frame images will automatically be loaded in Cine mode whenever a study of this particular modality type is loaded into the Viewer.
Show DICOM Overlay	If selected, the DICOM Overlay option will be automatically enabled whenever a study of this particular modality type is loaded into the Viewer.
Synchronize Series	If selected, the Automatic Series Synchronization option will be automatically enabled whenever a study of this particular modality type is loaded into the Viewer.
Show 6000 Overlay	If selected, the DICOM 6000 Overlay option will be automatically enabled (if applicable) whenever a study of this particular modality type is loaded into the Viewer.
Preserve W/L Across Global Stack	If selected, window and level settings will be carried over from one series to the next when using the Global Stack feature.
Show Cross Reference	If selected, the Reference Lines button on the Toolbar will be automatically enabled whenever a study of this particular modality type is loaded into the Viewer.
Image Level Manipulation	If selected, the Image Level Manipulation option will automatically be enabled whenever a study of this particular modality type is loaded into the Viewer.
Global Stack Window	If selected, the Global Stack option will be automatically enabled whenever a study of this particular modality type is loaded into the Viewer.
Study Invert	If selected, the Invert option will be automatically enabled for all images whenever a study of this particular modality type is loaded into the Viewer.
Use Window Range Instead of Level	If selected, the Window/Level tool will appear as a special Window Range cursor whenever a study of this particular modality type is loaded into the Viewer. Note that this option will be checked by default for NM and PT modalities.

Option	General Description
Automatically Stack/Scroll MG Priors	If selected, any relevant prior mammography studies will automatically be loaded into the Viewer along with the primary study and the Navigation Thumbnails for those prior studies will be displayed alongside the primary study's Navigation Thumbnails in both the main Viewer and the Mammography Thumbnail Viewer. In addition, the Scroll MG Priors tool in the Viewer Toolbar will become active. If the Show Thumbnails for Comparison Studies preference is also enabled, as described in Appendix A.2.18 below, thumbnails for relevant priors that are not specifically mammography images (such as MR breast or US breast images) will also be displayed. NOTE: If this preference is not selected, Navigation Thumbnails for relevant prior mammography and non-mammography studies will still be displayed if the Show Thumbnails for Comparison Studies preference is selected, although the Scroll MG Priors tool will not be active. NOTE: This option is only available when setting the preferences for the Default and Mammography (MG) modalities.
Show Slice Position Indicator	If selected, the Slice Position Indicator will be displayed for Breast Tomosynthesis images (only available when setting the MG or the DEFAULT modality)
Load “For Processing” Images	By default, MG images that are classified as “for processing” (i.e., that are intended to be used for processing before the study is reviewed, but may not be intended to be reviewed as part of a normal read) will not be loaded into the Viewer or cached in RTWL. You can change this, however, by selecting the Load “For Processing” Images preference. This option is only available when setting the preferences for the Default and Mammography (MG) modalities.

Option	General Description
	NOTE: The number of images for a given study listed on a worklist or the Query Page is based on the images stored on the PACS Server, not those loaded into the Viewer. Therefore, if this option is not selected and a study has any “for processing” images available, the number of images shown as available for a study will be less than the actual number of studies loaded.
Use ML Indicator Orientation	By default, the slice position indicator for multi-frame breast tomosynthesis images indicates Medial-Lateral (M-L) for MLO views, as opposed to Right-Left (R-L), and Superior-Inferior (S-I) for CC views, as opposed to Head-Foot (H-F). You can change this to display R-L and H-F instead, however, by deselecting the Use ML Indicator Orientation option. This option is only available when setting the preferences for the Default and Mammography (MG) modalities.
Show Thumbnail Viewer in Toggle	By default, clicking on the Patient Record icon on the Application Toolbar within the Viewer will toggle the display of the Patient Record on and off, but will not affect the display of the Mammography Thumbnail Viewer (when viewing Mammography Images). Selecting the Show Thumbnail Viewer in Toggle option, however, will cause the Patient Record icon to toggle the display of both the Patient Record and the Mammography Thumbnail Viewer. Note that this option is only available when setting the preferences for the Default and Mammography (MG) modalities.
DBT upward scrolling is toward the paddle	By default, when viewing mammography images, scrolling “upward” on DBT series will move the images toward the paddle. To change this behavior, this option should be deselected. This option is only available when setting the preferences for the Default and Mammography (MG) modalities.

Option	General Description
Preserve Display Attributes in Viewport	If selected, any changes applied to orientation, zoom, or invert on an image in a viewport will be applied to all images in the viewport (and the paired viewport, if MG stacked scrolling is enabled and Paired Stacking mode is selected) and will generally be persisted after the viewport content is changed. The one exception is that these display attributes will not be persisted in cases where the viewport content is changed to an image with a different laterality. This option is only available when setting the preferences for the Default and Mammography (MG) modalities. Note the following: • This preference is applicable whether using modality layout or if a Hanging protocol is applied. • This preference acts independently from the Preserve Presentation State Within Viewport preference, described in Appendix A.2.3 above. • This preference also acts independently from the Image Level Manipulation preference, described above, since that preference controls the display of images <i>within</i> a single series, whereas this preference controls the display of images <i>between</i> different series.

A.2.12. Overlays

The Overlays Preferences section allows you to configure how DICOM overlay and Lossy Compression overlay text is displayed on images within Series Viewports, as in the following example:

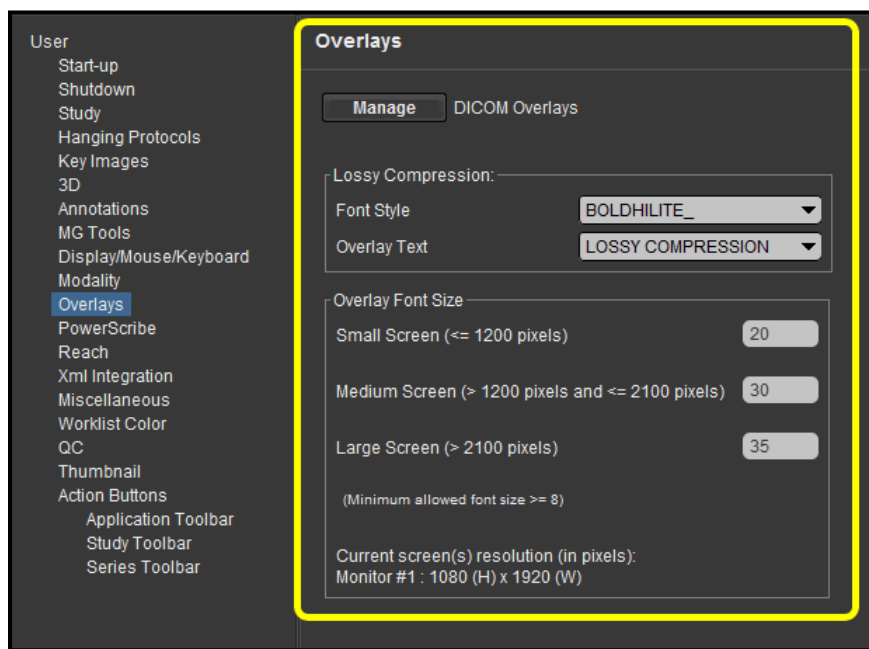


Figure 722: Overlay Preferences

The following preferences are available:

Option	General Description
Manage DICOM Overlays	The DICOM Overlay that is displayed within the Merge PACS Viewer can be custom configured to display differently for specific modalities. For each modality, the following things can be customized: • Which DICOM fields or Viewer keywords should be displayed • Where (which quadrant) each DICOM field or Viewer keyword should be displayed • Which, if any, DICOM fields or Viewer keywords should be considered “minimal” (<i>i.e.</i>, to be displayed when the DICOM Overlay feature is turned off). NOTE: Viewer keywords let you display information derived from current settings within the Viewer, such as Window/Level and Zoom, as opposed to hard-coded information sent from the modalities. In addition, a default can be customized that will apply to all modalities that don’t have specific configurations.

Option	General Description
Lossy Compression	By default, the text that is displayed on an image to indicate that it is being displayed with lossy compression is labeled “Lossy Compression” and is displayed in a large bold yellow font. If desired, however, the label and font type can be changed.
Overlay Font Size	Once you have defined what constitutes a “small” and “large” screen resolution, as described in Appendix A.2.10 above, you can set the base overlay font size to be used with each of those resolutions. For each screen resolution, the “Overlay Font Size” defines the base font that is used for any “medium” font style selected for the Lossy Compression overlay text and individual DICOM overlay fields. The “small” and “large” font styles are then calculated based on that base font by subtracting or adding an offset of five pixels. NOTE: The recommended lower base font size is 14.

NOTE: Refer to Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide* for a detailed description of each of these preferences.

A.2.13. PowerScribe

NOTE: These preference will only work with versions of PowerScribe prior to PowerScribe 360.

If your site is configured to integrate with PowerScribe, one or more of the following PowerScribe-related preferences may be selected:

Option	General Description
Automatically Launch PowerScribe	If selected, PowerScribe will automatically be launched in the background when the Merge PACS Workstation is first opened.
Automatically Hide PowerScribe Windows	If selected, all PowerScribe windows will be hidden when not in use.
Automatically Load Non-Final Reports	If selected, reports that are not marked as final will automatically be loaded into PowerScribe in addition to reports that are marked as final.
Automatically Add Addendum to Final Reports	If selected, any addendums to final reports will automatically be added within PowerScribe.

In addition, you may define what PowerScribe should do when specific Report events occur as follows:

Event	General Description
Report is Saved	Select whether PowerScribe should: • Close Study • Open Next Study

Event	General Description
Report is Signed	Select whether PowerScribe should: • Close Study • Open Next Study
Report is Approved (V5.0 only)	Select whether PowerScribe should: • Close Study • Open Next Study
Report is Sent to Editor (V5.0 only)	Select whether PowerScribe should: • Close Study • Open Next Study

A.2.14. XML Integration

If your site has been configured to use Bi-directional XML integration, you can configure the actions performed by the Workstation when various responses are sent from the integrated third-party application. You can also configure what notifications are sent to the third-party application (via xml drop) in certain circumstances. This is done by selecting the **Configure** button under **XML Integration**, as in the following example:

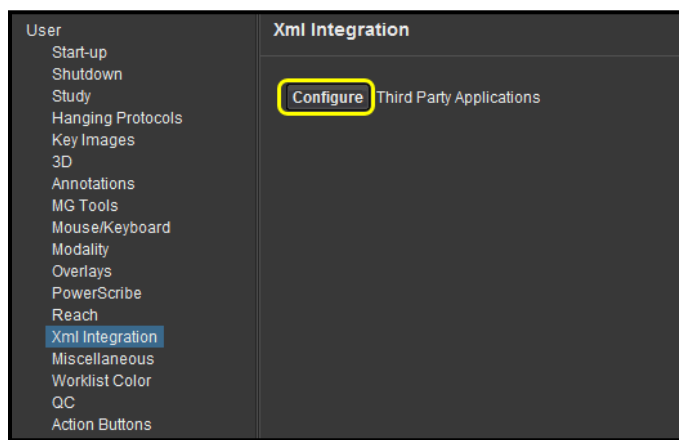


Figure 723: XML Integration Preferences

NOTE: The Configure button will only be available if bi-directional XML integration has been enabled.

When you select this option, the **Third Party Application Configuration** window will be displayed as a separate pop-up window, as in the following example:

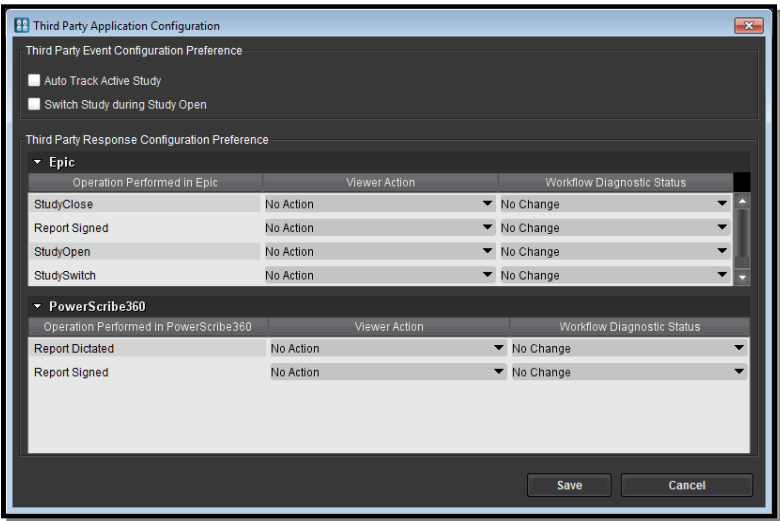


Figure 724: The Third Party Application Configuration Window

The following preferences relate to XML Integration within the Viewer:

Option	General Description
Third-party Event Configuration Preferences	If Merge PACS has been configured to notify a 3 rd -party application whenever a user switches between studies in general, you can configure the following options:
Auto Track Active Study	If selected, the 3 rd -party application will automatically be notified of a study switch whenever you change study focus (e.g., when you switch from the Primary Viewer to the Secondary Viewer). This will cause the context within the 3 rd -party application to switch to the study that was switched to.
Switch Study during Study Open	If selected, the 3 rd -party application will automatically be notified of a study switch whenever you open a study for viewing in the Primary Viewer.

Option	General Description
Third-party Response Configuration Preferences	Allows you to configure response configuration preferences for one or more individual third-party applications. Each available third-party application will be listed separately, and you can expand/collapse the information for each application by clicking on the triangle to the left of that application's name. Each application will list one or more "response events" that describe the operation performed by the application (<i>e.g.</i>, "Report Dictated" or "Report Signed"). The list of actual response events may vary depending on the application and is also customizable on a site-by-site basis. For each response event, you can specify what action you want the Workstation to perform and what the workflow diagnostic status for the study should be set to when the response event is received. This is done by selecting from the drop-down Viewer Action and Workflow Diagnostic Status menus for each response event.

NOTE: Refer to Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide* for a detailed description of each of these preferences.

A.2.15. Miscellaneous

The following miscellaneous options relate to the operation of the Workstation:

Option	General Description
Report Viewer Font Size	Allows you to select one the default font size used in the Report Viewer. The following options are available: • 15 [pt] • 18 [pt] • 22 [pt]
UI Color Model	Allows you to select the color scheme for the Merge PACS Viewer. The following options are available: • Color • Grayscale
List Priors without Timestamps	Allows you to specify where prior studies without timestamps should appear in relation to other prior studies. The following options are available: • Top – Above all other prior studies • Bottom – Below all other prior studies
DICOM Attribute Viewer Font Size	Allows you to select one the default font size used in the DICOM Attribute Viewer. The following options are available: • SMALL • MEDIUM • LARGE

Option	General Description
Order Viewer Default Visible Section	By default, when you access the Order Viewer and there is too much information to be displayed without vertical scroll bars, the top section of the Order Viewer (typically the Allergies section) will be displayed and you will need to scroll down to view the other sections (Diagnosis, Orders or Questionnaire). If desired, however, you can select to have the scroll bar automatically jump one of the following sections whenever you open the Order Viewer. • Allergies • Diagnosis • Orders • Questionnaire NOTE: These options will have no effect if the display of the Allergies and/or Diagnosis sections in the Order Viewer has been turned off completely for your site.
Series Layout Configuration	Allows you to set how many rows and columns of options will be displayed when users click on the the Series Layout Menu icon.
Consider for Automatic Assignment	Allows user to be considered eligible to have exams automatically assigned to him or her.
Show only latest report for each study in the Report Viewer	Allows you to configure the Report Viewer to display only one report (the most recent) for each study in the Report Viewer.
Select Priors Automatically when Adding to CD/DVD Burning list	If selected, when adding studies to the CD Burning dialog's study list, the prior studies will be automatically selected.
Use Toolbox Series Right-click Menu	If selected, the Series Right-Click Menu will contain only the active tools that are present in the mouse cycle mode for the modality from which the menu was invoked instead of the complete set of options.
Show status as check box	If selected, any status dimension that has only one available status option will be displayed as a checkbox instead of a drop-down menu on the Update Study Status dialog.
Wrap When Paging Images	If selected, images will “wrap” when you are paging through them. In other words, when you reach the last image in a Series while paging forward, the first image will be displayed next. Similarly, when you reach the first image in a Series while paging backwards, the last image will be displayed next.
Show Application Toolbar	If not selected, the main Viewer Toolbar will be hidden from view.
Skip Update Status for Close Viewer	If selected, clicking on the Close Viewer button on the Viewer Toolbar will exit the application without prompting you to update the study status.
Enable Alternative CAD Marker Association	The Merge PACS Viewer uses a standard scheme to automatically associate structured reports with DICOM images. If the reports are not being correctly associated with the DICOM images for a given site, select this option to use an available alternative scheme.

Option	General Description
Ignore private DICOM Tags	By default, the Merge PACS Workstation will process all DICOM tags associated with images, including private DICOM tags. With datasets that contain a lot of private data, however, this can cause unacceptable amounts of memory usage. If this is a problem, select this option to prevent private DICOM tags from being processed.
Enable Column Filtering	By default, you can click on the column headers of the various worklists and enter text below the header, and the worklist will be filtered to only display studies where the value in the column matches the text entered. Deselecting this option will disable column filtering.
Enable Window/Level Clamping	If a Hanging Protocol includes custom-defined Window/Level presets, those custom W/L settings may not be appropriate for all images in other studies to which the Hanging Protocol is applied (causing some images to be displayed as completely black, for example). If this option is selected, the system will automatically override custom-defined presets and change the W/L to “auto” for images the system determines would not display correctly with the specified W/L setting.
Viewer Defaults to use Displayed FOV Match	Select this to enable the Displayed FOV Match Feature by default. If this option is selected, clicking on the Displayed FOV Match icon on the Application Toolbar will toggle the feature off instead of on.
Show Cutlines Options in Menu	Select this to display the optional Cutlines feature in the Series Right-click Menu. Cutlines are be used to display all the reference lines of one or more series on a single image of another series. As with standard Reference Lines, this feature is used primarily for CT and MR images where there are sequential "slices". With CT studies, there is typically a master or "scout" image against which the images of a second series are referenced. With MR studies, on the other hand, there are typically multiple viewing angles of the same set of images, and the references lines will show the relationship between the linked series.

Option	General Description
Include Header for Print	When images are printed by selecting Print from the Series Right-click Menu, a header is displayed at the top of the printed image that reads, “Not for Diagnostic Use” and will also display the following additional information by default: • Patient Name • MRN • Patient DOB • Patient Code • Study Date/Time • Referring Physician • Series Number • Accession Number • Image Number This extra information will be included by default for standard Merge PACS implementations and disabled for OrthoPACS installations, but can be changed as desired on a site-by-site basis by selecting or deselecting this option Note that, regardless of whether this option is selected or not, each printed image will also include a footer that includes the following information: • Patient Name • MRN • Study Date • Study Description • Series Description • Image Number • Frame Number • Date Printed
Show Patient Synopsis in Patient Record	If Merge PACS is configured to integrate with Patient Synopsis and the user has the privilege to access it, selecting this option will cause Patient Synopsis information for the currently open exam to be displayed within the Patient Record.
Import User Preferences	Click the Import button to import user preferences from another Workstation or from a different user on the same Workstation, as described in Appendix A.2.22 below
Export User Preferences	Click the Export button to export your user preferences to a file that can be used to import the preferences to another Workstation or to a different user on the same Workstation, as described in Appendix A.2.21 below.
Restore User Preference Defaults	Click the Restore button to restore your user preferences to their default values. NOTE: Selecting this option will cause the Merge PACS Workstation to close and you will need to manually restart the Workstation in order to continue.

A.2.16. Worklist Color

The following options allow you to set the default color scheme for worklists that will be applied to new worklists the user creates:

Option	Description						
Base Color	Select one of the following options to determine the base (<i>i.e.</i>, background) color of each row: • Grey • Determined by primary dimension status • Determined by Order Priority • Determined by Due In						
Columns Show Their Own Color	Select one of the following options: <table> <tr> <th>Option</th><th>Description</th></tr> <tr> <td>Yes</td><td> Individual columns will be colored according to the color defined for the status of those dimensions (if they are configured to allow the use of their status colors, as described in subsection 9.13.7 above), as in the following example:  </td></tr> <tr> <td>No</td><td> All columns will be colored the same as the selected Base Color, as in the following example:  </td></tr> </table>	Option	Description	Yes	Individual columns will be colored according to the color defined for the status of those dimensions (if they are configured to allow the use of their status colors, as described in subsection 9.13.7 above), as in the following example: 	No	All columns will be colored the same as the selected Base Color, as in the following example: 
Option	Description						
Yes	Individual columns will be colored according to the color defined for the status of those dimensions (if they are configured to allow the use of their status colors, as described in subsection 9.13.7 above), as in the following example: 						
No	All columns will be colored the same as the selected Base Color, as in the following example: 						

NOTE: The selected option will only be applied to new worklists the user creates and will not apply to all worklists that you view.

NOTE: The selected color scheme can be overridden for specific worklists as described subsection 13.4.7 above

A.2.17. QC

The QC Preferences section allows you to customize the default display of columns within the Selection Panel of the Quality Control Editor for different types of studies, series and images on a site-wide basis, as in the following example:

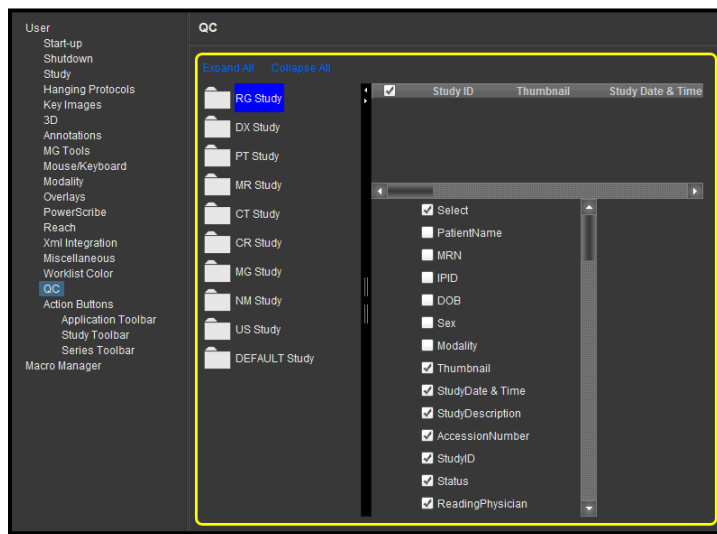


Figure 725: Column Editing Panel

For each available modality type, you can individually customize the display of the entire study, the series and individual images within the Selection panel as follows:

- Select the display whose default column values you want to customize by clicking on it in the **Display Selection** panel, as in the following example:

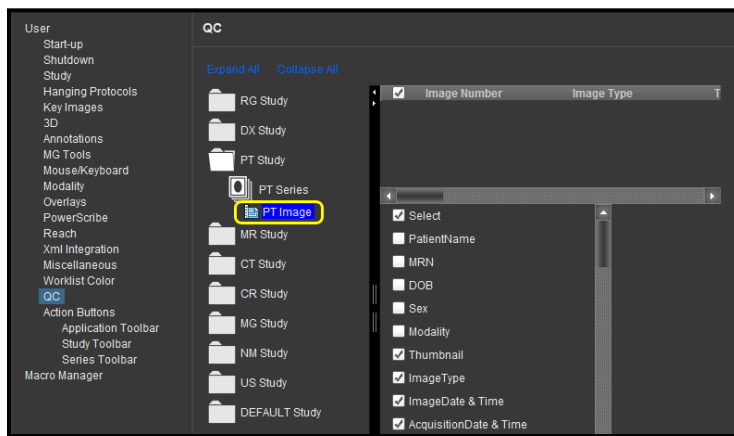


Figure 726: Selecting Display to Customize

NOTE: Double-click on a **Study** folder to display the **Series** folder and double-click the **Series** folder to display the **Image** instance. You can also click on the **Expand All** and **Collapse All** links at the top of the screen.

- Once you have selected a display to customize, use the **Column Selection** panel to select or deselect individual columns to be displayed, as in the following example:

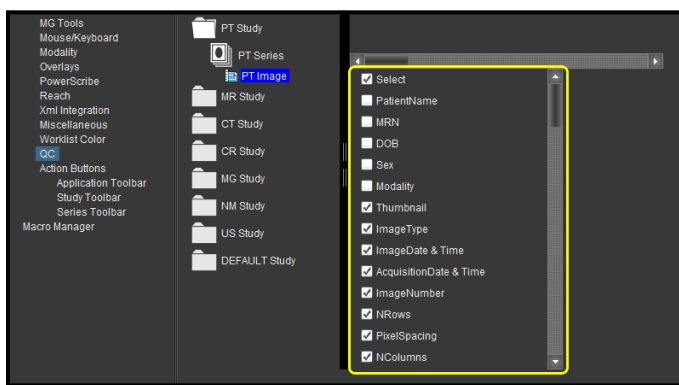


Figure 727: Selecting Columns to Be Displayed

NOTE: The actual columns displayed in the Column Selection panel will depend on whether you are customizing the display of a study, series or image.

- Resize any column by placing your mouse cursor between that column's header and the column header to its right until a double-sided arrow is displayed, as in the following example, and then clicking and dragging the border between the columns to the left or right as desired:



Figure 728: Resizing a Column

NOTE: Use the scroll bar below the column headers to view all columns.

- Change the order of the columns by clicking on any column header and dragging it to a new location.
- Change the column sort order by clicking on the column header you want to sort on. Note the following:
 - Clicking once on a column header will cause the rows to be sorted in ascending order. Clicking again will reverse the sort order to descending.
 - You can sort on multiple columns by holding down the **CTRL** key as you click. Columns will sort in the order that they were selected. Immediately after selecting a column for sorting, you can click on it again to reverse the sort order for that particular column as long as you are still pressing the **CTRL** key.

For more information on the QC Editor, refer to the *Merge PACS 8.2 Quality Control Editor Users Guide*.

A.2.18. Thumbnail

The Thumbnail Preferences section allows you to configure how the Series Navigation Thumbnail icons are displayed within the Viewer. The following options are available:

Option	Description
Show Thumbnails for Comparison Studies	If selected, Series Navigation Thumbnails for all available (“online”) relevant comparison studies will automatically be displayed beside the primary study’s Series Navigation Thumbnails when the primary study is loaded into the Viewer. If the Automatically Stack/Scroll MG Priors preference is enabled, as described in Appendix A.2.11 above, thumbnails for relevant priors that are not specifically mammography images (such as an MR breast image) will also be displayed in both the main Viewer and the Mammography Thumbnail Viewer when viewing mammography images. NOTE: The order in which the comparison studies are displayed is controlled by the Show Unread Studies First preference described below. NOTE: Changes to this preference will not affect studies currently displayed in the Viewer until those studies are reloaded.
Show Unread Studies First	If selected, navigation thumbnails for unread relevant comparison studies will be displayed first, followed by other relevant comparison studies and then foreign studies. If not selected, all comparison studies will be sorted based on the configured Order Priors Comparison Studies preference, as described in Appendix A.2.3 above. NOTE: This option will only be available if Show Thumbnails for Comparison Studies is enabled.
Show Thumbnails	If not selected, the Series Navigation Thumbnails within the Merge PACS Viewer will be hidden from view.
Thumbnail Location in the Viewer	Let’s you configure the location of the Series Navigation Thumbnails within the Viewer. The following options are available: • Top • Bottom • Left • Right NOTE: When displaying multiple studies, the system will prevent Series Navigation Thumbnails from displaying between two study panels so as to avoid confusion, e.g., displaying them to the left and right in a two-panel layout instead of both to the left or both to the right. The actual location will depend on the study layout selected.

A.2.19. Action Buttons

The Action Buttons Preferences section allows you to custom configure which of the available tools should be displayed on the various toolbars within the Viewer and customize the specific tools to a limited degree, as in the following example:

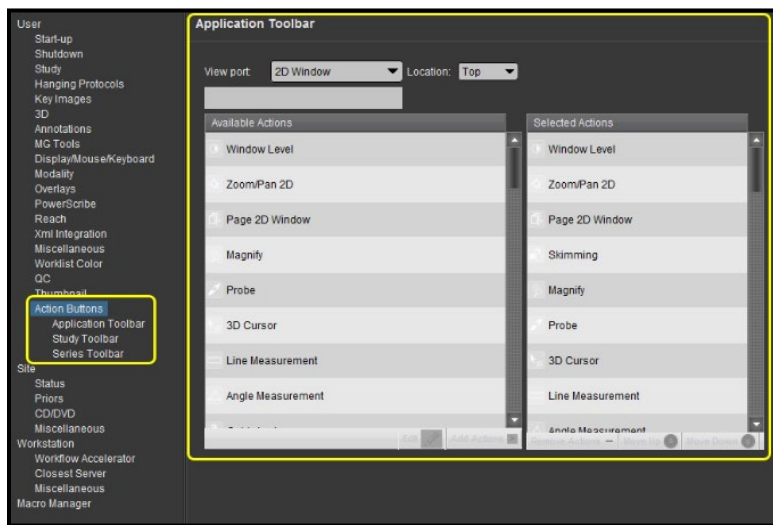


Figure 729: Action Buttons Preferences

NOTE: For the Study and Series toolbars, you can also configure when you want the tools to be displayed or hidden.

NOTE: You can also configure the Application toolbar for the separate Quality Control Editor application from this section.

The following items can be added to individual toolbars:

- Any default tool available for the toolbar
- Any macros that have been created and assigned to a tool icon

NOTE: Refer to Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide* for a detailed description of each of these preferences.

A.2.20. Managing Macros

A macro is a sequence of defined actions that can be performed via a single keystroke or toolbar button. The actions that can be included in a macro can be any of the following:

- a **default system action** such as rotate, toggle cine, show cross-reference lines, etc.
- a **custom drop command** (to drop a JPEG and/or XML event for the current study)
- a **custom operating system command** (for example, to launch an external application via command line)
- another **macro**

NOTE: Once you have created a macro, you need to assign it to a keyboard shortcut, as described in Appendix A.2.10 above, or add it as an Action Button on a Toolbar, as described in Appendix A.2.19 above.

NOTE: If a macro requires user action (such as entering a comment) and the user cancels the action, the rest of the macro will continue to run as configured.

The Macro Manager allows you to create and manage one or more macros, as in the following example:

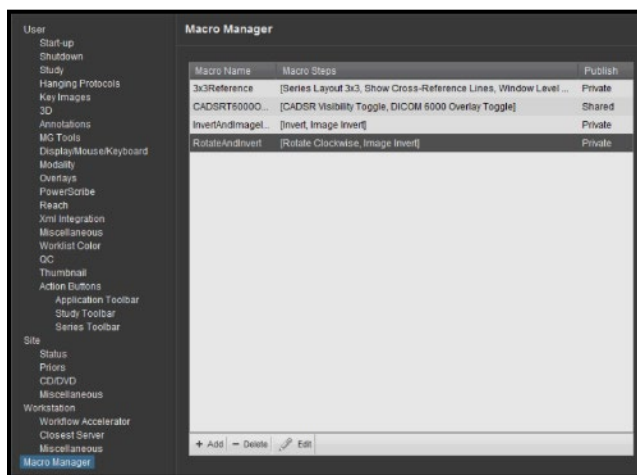


Figure 730: Macro Manager

NOTE: Merge PACS includes a default macro called **CADSRT60000OVERLAY** that toggles the display of both the DICOM 6000 overlay and the CADSRT overlay with a single click and that can be added to the Application, Study and/or Series toolbars. It cannot be edited or deleted from the Macro Manager directly, but you can configure how it appears on individual toolbars (tooltip, icon, placement, etc.) as described in Appendix A.2.19 above.

NOTE: Refer to Chapter 24 of the *Merge PACS 8.2 Workstation Users Guide* for detailed instructions for creating, editing and deleting macros.

A.2.21. Exporting User Preferences

As described in Appendix Note: above, the **Export User Preferences** option allows you to export all user preferences to an external **.xml** or **.txt** file that can then be used to import the preferences to another Workstation or to a different user on the same Workstation.

When you select this option, the **Export User Preferences** window will be displayed as a separate pop-up window, as in the following example:

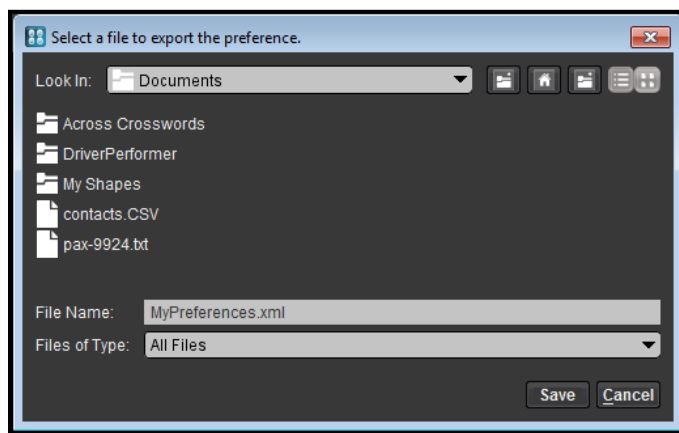


Figure 731: The Export User Preferences Window

By default, the file will be saved to your individual Windows user directory. If desired, however, you can select a different location for the file from the **Look In** drop-down menu, as in the following example:

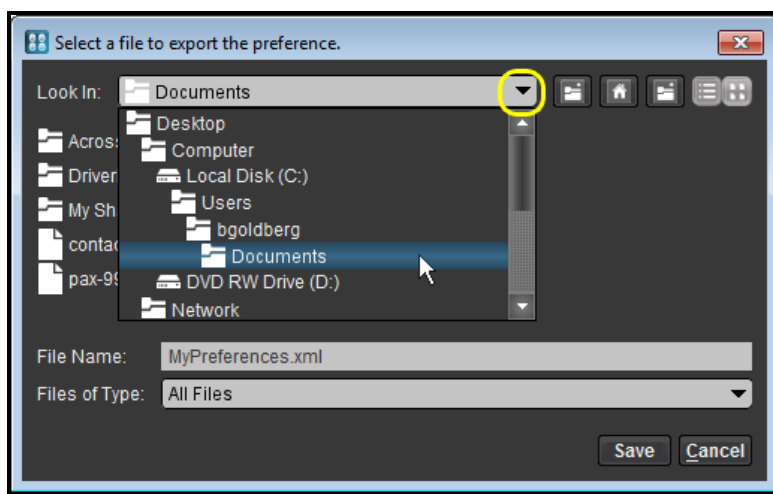


Figure 732: Selecting a Location for the Exported File

If desired, you can change the name of the file in the **File Name** text field, as in the following example:

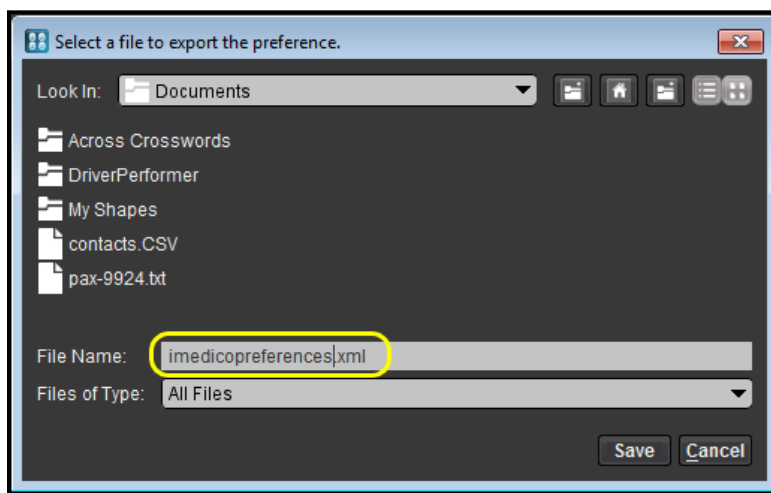


Figure 733: Selecting a Location for the Exported File

When finished, click on the **Save** button at the bottom of the window.

A.2.22. Importing User Preferences

As described in Appendix Note: above, the **Import User Preferences** allows you to import all user preferences from another Workstation or from a different user on the same Workstation. Note that the preferences are stored in an external **.xml** or **.txt** file that must first be created, as described in Appendix A.2.21 above.

When you select this option, the **Import User Preferences** window will be displayed as a separate pop-up window, as in the following example:

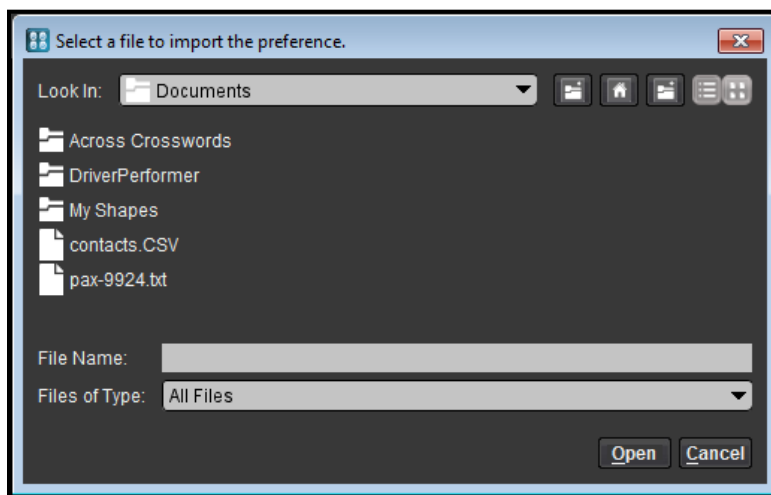


Figure 734: The Import User Preferences Window

By default, Import Preferences Dialog will display the contents of your individual Windows user directory. If the file containing the preferences is displayed there, click on it to select it. If the file is located somewhere else, you can browse to the file's location from the **Look In** drop-down menu, as in the following example:

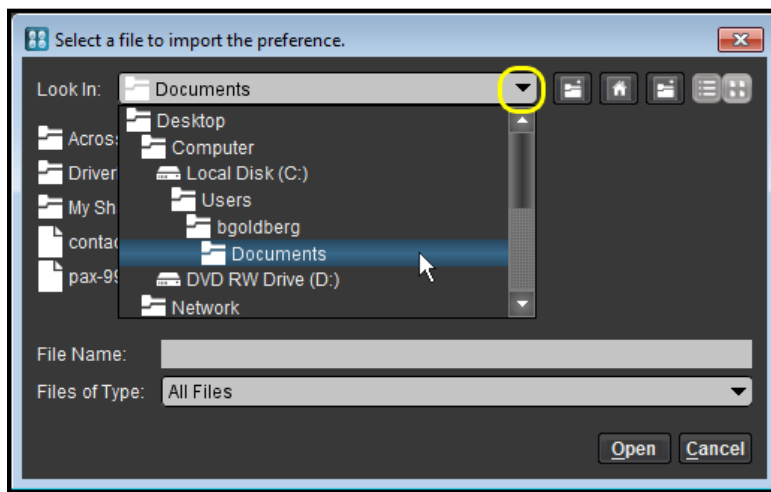


Figure 735: Selecting the File to Be Imported

Once you have located the directory where the preferences file is stored, click on the name of the file once to select it and then click on the **Open** button, as in the following example:

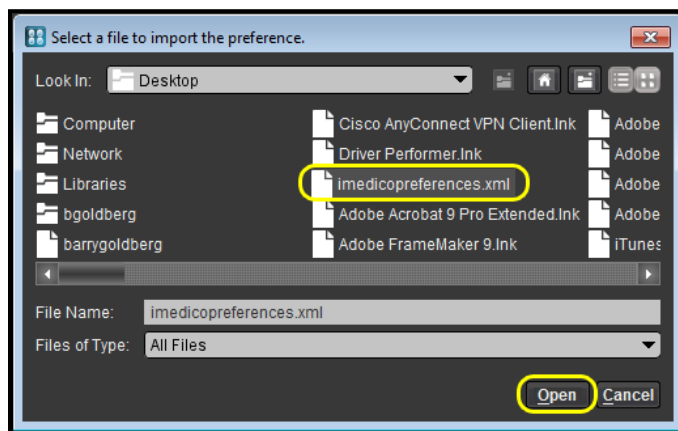


Figure 736: Selecting the File to Be Imported

NOTE: You can also double-click on the name of the file to open it.

NOTE: Selecting this option will cause the Merge PACS Workstation to close and you will need to manually restart the Workstation in order to continue.

A.3. Configurations

The **Merge PACS Configuration** dialog allows you to set a variety of site-level preferences that can provide default settings for all users, as in the following example:

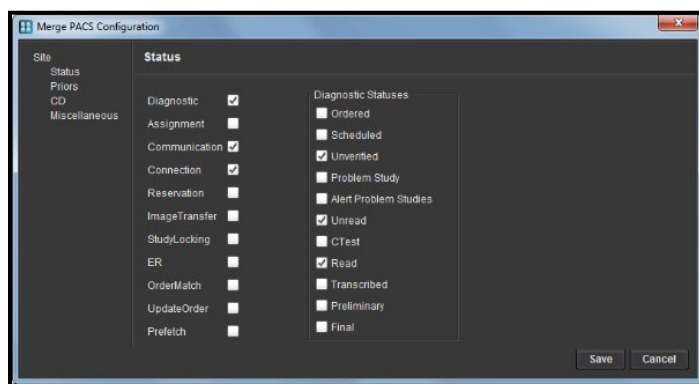


Figure 737: Merge PACS Configuration

NOTE: These preferences can also be set (as overrides) from the “Site” section of the **Merge PACS Preferences Dialog**, as described in subsection 16.1 above.

A.3.1. Status Preferences

By default, the Update Study Status dialog within the Merge PACS Workstation presents the user with a complete set of available status dimensions and status options that can be assigned to a particular study. If desired, however, you can limit the dimensions and options that appear in this dialog by selecting only certain statuses and options, as in the following example:

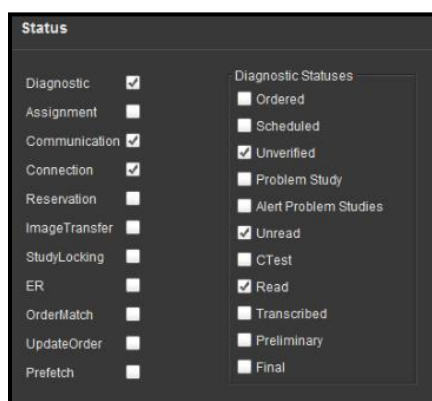


Figure 738: Statuses and Options

NOTE: When you select a Status in the first column, the applicable options for that status will be displayed in the second column.

NOTE: These preferences only affect the Update Study Status dialog that is presented automatically when a user exits a study. The Change Workflow Status dialog, accessed by clicking the Change Status icon, will always show all available dimensions and statuses.

A.3.2. Priors Preferences

The Priors Preferences section allows you to set default preferences related to how prior studies are determined and treated within the Workstation, as in the following example:

Priors

Criteria for Selection of Priors (MRN and Name) or Name/DOB/Sex

☒ Ignore First Name

☒ Ignore Middle Name/Initial

☒ Sex O matches F and M

☒ Allow Space or Comma as Separator

☐ Filter on RETENTION

☒ Filter on 'ology

☐ RADIOLOGY

☐ RADIOLOGY

☐ Same as primary study

☐ None

☐ Allow any DICOM Tag for Determining Relevant Priors or Hanging Protocol

Maximum Number of Priors in Comparison Study Menu: 10

Maximum Number of Comparison Study Thumbnails to Display: 10

Maximum Number of Relevant Prior Studies to Automatically Pre-cache: 4

☒ Print Prior Report List

☒ Perform extended search when searching by IPID=Any

Figure 739: Priors Preferences

The following preferences can be configured:

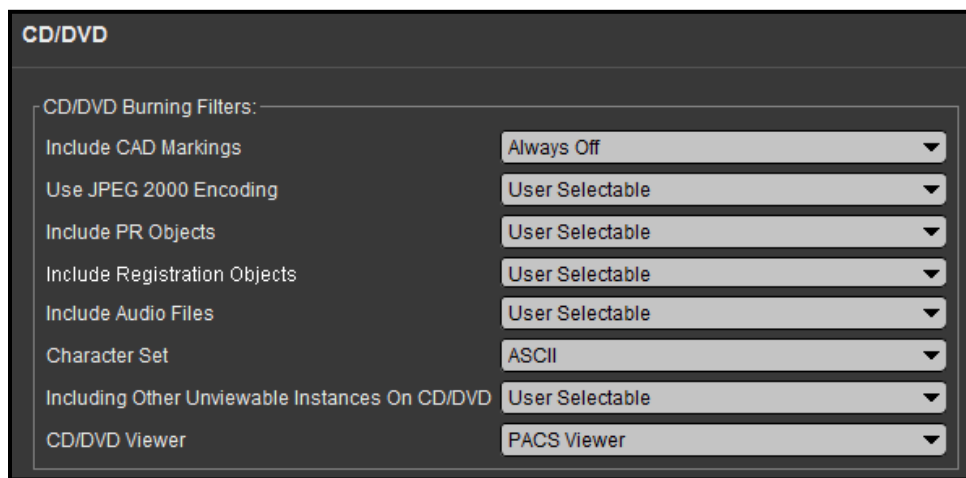
Preference	Description								
Criteria for Selection of Priors	Select one of the following options to determine how prior studies are determined: <table><tr><th>Preference</th><th>Description</th></tr><tr><td>MRN</td><td>Determine priors based solely on MRN</td></tr><tr><td>IPID and MRN</td><td>Determine priors based on a combination of Issuer of Patient ID (IPID) and MRN (i.e., both must match to be considered a prior).</td></tr><tr><td>MRN and Name</td><td>Determine priors based on a combination of MRN and Patient Name (i.e., both must match to be considered a prior).</td></tr></table>	Preference	Description	MRN	Determine priors based solely on MRN	IPID and MRN	Determine priors based on a combination of Issuer of Patient ID (IPID) and MRN (i.e., both must match to be considered a prior).	MRN and Name	Determine priors based on a combination of MRN and Patient Name (i.e., both must match to be considered a prior).
Preference	Description								
MRN	Determine priors based solely on MRN								
IPID and MRN	Determine priors based on a combination of Issuer of Patient ID (IPID) and MRN (i.e., both must match to be considered a prior).								
MRN and Name	Determine priors based on a combination of MRN and Patient Name (i.e., both must match to be considered a prior).								

Preference	Description
IPID and MRN and Name	Determine priors based on a combination of IPID, MRN and Patient Name (<i>i.e.</i> , all must match to be considered a prior).
MRN or Name/DOB/Sex	Determine priors based on either MRN or a combination of Patient Name, Date of Birth and Sex.
(MRN and Name) or Name/DOB/Sex	Determine priors based on either a combination of MRN and Patient Name or a combination of Patient Name, Date of Birth and Sex.
Name/DOB/Sex	Determine priors based on a combination of Patient Name, Date of Birth and Sex.
NOTE: For all selection of priors options that involve IPID and MRN, matching is performed based on exact values (e.g., “ABC” will be considered different from “abc” and “123” will be considered different from “1.2.3”). NOTE: “MRN” is NOT recommended and is only provided for legacy system compatibility. “MRN and Name” or “(MRN and Name) or Name/DOB/Sex” coupled with “Ignore First Name” “Sex O matches F or M” (described below) is the recommended default. Please review in light of your local ethnic population realities.	
Ignore First Name	When performing any type of match involving Patient Name, the first name and middle name/initial will be ignored.
Ignore Middle Name/Initial	When performing any type of match involving Patient Name, the middle name or initial will be ignored. Note that this option will be selected by default for both new installs and upgrades, but will become grayed out if Ignore First Name is selected since Ignore First Name also ignores Middle Name/Initial automatically.
Sex O Matches F and M	When performing any type of match involving Sex, allow a value of “O” to match a value of “M” or “F”.
Allow Space or Comma as Separator	When performing any type of match, allow a space or comma to be used as a separator and not just a ^.
Filter on <Study Type Group>	When Merge PACS is configured to run in Integrated mode, the attached iCEA Server can be set up to tag studies according to Study Type (e.g., Cardiology, Radiology, Neurology, etc.) and multiple Study Types can be placed together into a Study Type Group (e.g., an “OLOGY” group that would include the aforementioned Cardiology, Radiology and Neurology study types). Merge PACS can be configured to permit filtering of priors based on specific study types (organized by study type groups). Each study type group so configured will appear as an option, with each associated study type listed as a sub-option (together with a “Same as primary study” option).

Preference	Description
Allow Any DICOM Tag for Determining Relevant Priors or Hanging Protocol	By default, when users create Anatomical Body Parts (ABP) mapping rules from within the Workstation, they can only select to match on the Modality and Study Description DICOM tags. If desired, however, you can configure a site to provide a wider list of available DICOM tags for use when setting up ABP mapping rules, such as Procedure Code Sequence, Body part Examined, Protocol Name and Study Comment. To expand the list of available DICOM tags for use when creating ABP mapping rules, select the Allow Any DICOM Tag for Determining Relevant Priors or Hanging Protocol option from the Site Preferences Menu. Note that this preference only controls the list of available DICOM tags present when a new rule is created. If a rule has been created that references one or more additional DICOM tags, it will not be affected if this preference is later disabled.
Maximum Number of Priors in Comparison Study Menu	By default, the drop-down menu used to select a comparison study within the Viewer will display a maximum of ten studies. If desired, you can change the maximum number.
Maximum Number of Comparison Study Thumbnails to Display	By default, when the Show Thumbnails for Comparison Studies option workstation preference is selected, as described in Appendix A.2.18 above, navigation thumbnails will be displayed for maximum of ten comparison/prior studies. If desired, you can change the maximum number.
Maximum Number of Relevant Prior Studies to Automatically Pre-cache	By default, if the Automatically Pre-cache Relevant Prior Studies workstation preference is selected, as described in subsection A.2.3 above, a maximum of 4 relevant priors will be pre-cached. If desired, however, this can be changed to a value between 0 and 12.
Print Prior List	When reports are viewed within the Report Viewer of the Merge PACS Workstation for a primary study, a list of links to any prior reports is included at the bottom of the report. By default, when the report is printed from the Report Viewer, the list of prior reports is included in the printed report. If desired, the lists of prior reports can be suppressed on a site-wide basis so that they will still be available as clickable links within the Report Viewer but will not appear on the printed report by deselecting the Print Prior List option.
Perform extended node search when IPID=Any	If selected, when the IPID matching criterion for a DICOM device is set to “any,” the system will perform extended searches on all DICOM Devices which are configured as extended nodes. If not selected, the system will not perform an extended search when the IPID matching criterion for a DICOM device is set to “any.” NOTE: By default, this preference will be enabled.

A.3.3. CD/DVD Preferences

The CD/DVD Preferences section allows you to set default preferences related to studies are burned onto CDs or DVDs from within the Workstation, as in the following example:



CD/DVD Burning Filters:	
Include CAD Markings	Always Off
Use JPEG 2000 Encoding	User Selectable
Include PR Objects	User Selectable
Include Registration Objects	User Selectable
Include Audio Files	User Selectable
Character Set	ASCII
Including Other Unviewable Instances On CD/DVD	User Selectable
CD/DVD Viewer	PACS Viewer

Figure 740: CD/DVD Burning Preferences

In general, the Merge PACS Workstation lets users burn patient studies onto CDs or DVDs if their workstations have an attached CD or DVD burner. The CD/DVD Burning user interface includes a number of checkboxes that lets users determine whether to include various optional items as well as to determine how the study images should be encoded. Some of these checkboxes can be disabled on a site-by-site basis, however, if a site wants to have a default setting that cannot be overwritten by individual users.

In addition, you can configure the character-set attribute that Merge PACS will set in the DICOM directory (DICOMDIR) that it creates on the CD/DVD.

The following preferences can be configured:

Preference	Description						
Include CAD markings	By default, CAD markings (SR objects) associated with studies are not available to be included on the CD/DVD. Users can be given the option to include them, however. The following system-wide options are available: <table border="1"> <tbody> <tr> <td>Always Off</td><td>CAD markings will never be burned onto CDs/DVDs.</td></tr> <tr> <td>Always On</td><td>CAD markings will always be burned onto CDs/DVDs.</td></tr> <tr> <td>User Selectable</td><td>Users will be prompted for each study whether to burn the associated CAD markings onto the CD/DVD or not.</td></tr> </tbody> </table>	Always Off	CAD markings will never be burned onto CDs/DVDs.	Always On	CAD markings will always be burned onto CDs/DVDs.	User Selectable	Users will be prompted for each study whether to burn the associated CAD markings onto the CD/DVD or not.
Always Off	CAD markings will never be burned onto CDs/DVDs.						
Always On	CAD markings will always be burned onto CDs/DVDs.						
User Selectable	Users will be prompted for each study whether to burn the associated CAD markings onto the CD/DVD or not.						

Preference	Description						
Use JPEG 2000 Encoding	By default, users can select for each study burned onto a CD/DVD whether the images for that study should be compressed using JPEG2000 encoding (which results in smaller images that may not be viewable by non-Merge PACS viewers) or instead burn uncompressed DICOM images (which results in larger images that are viewable by all Viewers). This can be changed on a site-wide basis as follows, however: <table> <tr> <td>Always Off</td><td>Images will always be burned in uncompressed DICOM format.</td></tr> <tr> <td>Always On</td><td>Images will always be burned in compressed JPEG 2000 format</td></tr> <tr> <td>User Selectable</td><td>Users will be prompted for each study whether or not to use JPEG 2000 encoding.</td></tr> </table>	Always Off	Images will always be burned in uncompressed DICOM format.	Always On	Images will always be burned in compressed JPEG 2000 format	User Selectable	Users will be prompted for each study whether or not to use JPEG 2000 encoding.
Always Off	Images will always be burned in uncompressed DICOM format.						
Always On	Images will always be burned in compressed JPEG 2000 format						
User Selectable	Users will be prompted for each study whether or not to use JPEG 2000 encoding.						
Include PR Objects	By default, users can select whether Presentation State (PR) objects (including annotations) associated with studies should be included on the CD/DVD. This can be changed on a site-wide basis as follows, however: <table> <tr> <td>Always Off</td><td>PR objects will never be burned onto CDs/DVDs.</td></tr> <tr> <td>Always On</td><td>PR objects will always be burned onto CDs/DVDs.</td></tr> <tr> <td>User Selectable</td><td>Users will be prompted for each study whether to burn the associated PR objects onto the CD/DVD or not.</td></tr> </table>	Always Off	PR objects will never be burned onto CDs/DVDs.	Always On	PR objects will always be burned onto CDs/DVDs.	User Selectable	Users will be prompted for each study whether to burn the associated PR objects onto the CD/DVD or not.
Always Off	PR objects will never be burned onto CDs/DVDs.						
Always On	PR objects will always be burned onto CDs/DVDs.						
User Selectable	Users will be prompted for each study whether to burn the associated PR objects onto the CD/DVD or not.						
Include Registration Objects	By default, users can select whether DICOM registration objects associated with studies should be included on the CD/DVD. This can be changed on a site-wide basis as follows, however: <table> <tr> <td>Always Off</td><td>Registration objects will never be burned onto CDs/DVDs.</td></tr> <tr> <td>Always On</td><td>Registration objects objects will always be burned onto CDs/DVDs.</td></tr> <tr> <td>User Selectable</td><td>Users will be prompted for each study whether to burn the associated Registration objects onto the CD/DVD or not.</td></tr> </table>	Always Off	Registration objects will never be burned onto CDs/DVDs.	Always On	Registration objects objects will always be burned onto CDs/DVDs.	User Selectable	Users will be prompted for each study whether to burn the associated Registration objects onto the CD/DVD or not.
Always Off	Registration objects will never be burned onto CDs/DVDs.						
Always On	Registration objects objects will always be burned onto CDs/DVDs.						
User Selectable	Users will be prompted for each study whether to burn the associated Registration objects onto the CD/DVD or not.						
Include Audio Files	By default, users can select whether audio comments (VoiceClips) associated with studies should be included on the CD/DVD. This can be changed on a site-wide basis as follows, however: <table> <tr> <td>Always Off</td><td>Audio comments will never be burned onto CDs/DVDs.</td></tr> <tr> <td>Always On</td><td>Audio comments will always be burned onto CDs/DVDs.</td></tr> <tr> <td>User Selectable</td><td>Users will be prompted for each study whether to burn the associated Audio comments onto the CD/DVD or not.</td></tr> </table>	Always Off	Audio comments will never be burned onto CDs/DVDs.	Always On	Audio comments will always be burned onto CDs/DVDs.	User Selectable	Users will be prompted for each study whether to burn the associated Audio comments onto the CD/DVD or not.
Always Off	Audio comments will never be burned onto CDs/DVDs.						
Always On	Audio comments will always be burned onto CDs/DVDs.						
User Selectable	Users will be prompted for each study whether to burn the associated Audio comments onto the CD/DVD or not.						

Preference	Description
Character Set	Select the character-set attribute(s) that Merge PACS will set in the DICOMDIR that it creates on the CD/DVD as well as which character set you would like to be used by default. The following options are available: ASCII PACS will not set the character set attribute (0008,0005) in the DICOMDIR that it creates on the CD/DVD. LATIN1 PACS will set the character set attribute (0008,0005) to "ISO_IR 100" in the DICOMDIR that it creates on the CD/DVD. AUTO PACS will set the character set attribute (0008,0005) to the union of the character sets in the patient hierarchy that is being burned to CD/DVD. Note the following: • If all character set options are disabled, the ASCII option shall be used. • If less than two options are enabled, the character-set selector on the Burn CD/DVD dialog shall not be shown.
Include Other Unviewable instances on CD	By default, users can select whether other unviewable instances associated with studies should be included on the CD. This can be changed on a site-wide basis as follows, however: Always Off Unviewable instances will never be burned onto CDs/DVDs. Always On Unviewable instances will always be burned onto CDs/DVDs. User Selectable Users will be prompted for each study whether to burn the unviewable instances onto the CD/DVD or not.
CD/DVD Viewer	By default, CDs and DVDs burned from within the Merge PACS Workstation will have the option to include the Merge PACS CD Viewer. If desired, however, you can change this so that users will instead have an option to include Merge eFilm Lite viewer if you have eFilm available on your system. NOTE: The eFilm Lite Viewer does not currently support display of Key Images and PR Objects, although those items will still be burned onto the CD/DVD if selected. NOTE: Please contact Merge Healthcare Support to have eFilm made available on your system.

NOTE: In cases where you are having exchange issues with old or "dumb" PACS systems, turn off JPEG2000 compression and do not include CAD. SR, PR, KO, voice clips on the CD. Instead, you should use the PDI (browser) images.

A.3.4. Miscellaneous Preferences

The Miscellaneous Preferences section allows you to set various miscellaneous preferences that don't fit into any of the other configuration sections, as in the following example:

Figure 741: Miscellaneous Site Preferences

The following miscellaneous site-preferences can be set:

Preference	Description
Allow Save/Delete of Key Image When Study Is Locked	By default, authorized users can save and delete key images from a study even when that study is currently locked by another user. Deselect this option to prevent this from happening.
Always Open New Study Window for Embedded Viewer	By default, when a user is viewing a study in the Embedded Viewer and selects another study to view, the second study will be displayed in the same Embedded Viewer window. Select this option to force the second study to be displayed in a new Embedded Viewer window instead.
3D Image Orientation Precision Limit	By default, Merge PACS determines if consecutive slices have valid DICOM image orientation to generate a 3D volume based on whether various calculations match their applicable constraints within a .0001 tolerance. For some oblique slice orientations, however, these constraints may need to be relaxed (e.g., to a tolerance of .00001) to account for rounding errors that would prevent MPR from working with series containing those images. This setting allows you to change the tolerance as needed.
Area Measurements	Determines whether the area of ROI or Region Analysis Area measurements should be displayed in square centimeters (cm²) or square millimeters (mm²).
Restore Site Preference Defaults	Click the Restore button to reset all site preferences to their default values.

Appendix B. Standardized Uptake Values (SUV)

When using the **Probe**, **ROI**, **Region Analysis Area** and **Region Analysis Volume** tools with PET images, values can be displayed with any of the following units, based on the data provided by the modality:

Unit	Description	Unit	Description
NONE	unitless	MLG	milliliter/gram
CNTS	counts	1CM	1/centimeter
CM2	centimeter ²	UMOLML	micromole/milliliter
CM2ML	centimeter ² / milliliter	PROPCNTS	proportional to counts
PCNT	percent	PROPCPS	proportional to counts/sec
CPS	counts/second	MLMINML	milliliter/minute/milliliter
BQML	Becquerels/milliliter	MLML	milliliter/milliliter
MGMINML	milligram/minute/milliliter	GML	grams/milliliter
UMOLMINML	micromole/minute/milliliter	STDDEV	standard deviations
MLMING	milliliter/minute/gram		

To aid in the interpretation of PET images, if the units are in **BQML**, **GML**, or **CM2ML**, the **Probe**, **ROI**, **Region Analysis Area** and **Region Analysis Volume** tools will display the absorption of an injected isotope in body tissue based on **Standardized Uptake Values (SUV)**. SUV statistics are indicated based on the DICOM data present in the image and one or more of the following user preferences for SUV display:

- **Body Weight (BW)**
- **Lean Body Mass (LBM)**
- **Body Surface Area (BSA)**
- **Ideal Body Weight (IBW)**

CAUTION: Standardized Uptake Values are not absolute values. For details on the equations used to calculate these values in the application, see below.

CAUTION: Standardized Uptake Values are displayed only if the image contains values for the required DICOM tags.

B.1. Requirements for Displaying Standardized Uptake Values

If the units are **GML** or **CM2ML** than the modality pixel values represent the SUV, no correction has to be applied.

If the units are **CM2ML**, the SUV will be considered BSA; all other SUV types will display N/A

If the units are **GML**, the SUV can be either BW or LBM. There is an extra tag SUV Type (0054, 1006) that can distinguish between BW and LBM; if not present, the SUV will be considered implicitly BW; all other SUVs will display N/A

If the units are **BQML**, a correction has to be applied to the modality pixel values. In order to calculate the SUV, the following DICOM tags are required to be present in the dataset (refer to Appendix B.1 for details how these DICOM tags are used in SUV calculations):

DICOM Tag Name	Group, Element	Required Value	BW	LBM	IBW	BSA
Units	(0054,1001)	BQML(Becquerels per milliliter)	X	X	X	X
Decay Correction	(0054,1102)	START or ADMIN	X	X	X	X
Series Time	(0008,0031)	Specified (Expressed in hhmmss.frac)	X	X	X	X
Radionuclide Total Dose	(0018,1074)	Specified in BQ (becquerels) and > 0	X	X	X	X
Radiopharmaceutical Start Time	(0018,1072)	Specified (Expressed in hhmmss.frac)	X	X	X	X
Radionuclide Half Life	(0018,1075)	Specified in seconds and > 0	X	X	X	X
Patient Weight	(0010,1030)	Specified and > 0.0 (Expressed in kg)	X	X	X	X
Patient Size	(0010,1020)	Specified and > 0.0 (Expressed in meters)			X	X

NOTE: If one of the required DICOM tags listed above is not present in the dataset, the SUV will display N/A.

B.2. Basic Calculation for Standardized Uptake Value (SUV)

The general equation for SUV is as follows:

$$\text{SUV} = \text{Modality Pixel Value} * (\text{Body Parameter}) / (\text{Corrected Dose})$$

- **Body Parameter** has a different value for each type of SUV:

SUV	Body Parameter Value
BW (g)	Patient Weight * 1000
LBM (g)	• Male (Patient Sex is 'M'): (1.10 * Patient Weight - 128.0 * POWER ((Patient Weight / (Patient Size * 100)), 2)) * 1000 • Female (Patient Sex is 'F'): (1.07 * Patient Weight - 148.0 * POWER ((Patient Weight / (Patient Size * 100)), 2)) * 1000 Lean Body Weight refers to the sum of the weight of your bones, muscles and organs (everything other than fat in your body)¹
IBW (g)	• Male (Patient Sex is 'M'): (48.0 + 1.06 * (Patient Size - 152)) * 1000 • Female (Patient Sex is 'F'): (45.5 + 0.91 * (Patient Size - 152)) * 1000²
BSA (cm2)	POWER(Patient Weight, 0.425) * POWER(Patient Size * 100, 0.725) * 0.007184 * 10000 ³

- **Corrected Dose** has different values depending on the Decay Correction value:
 - If the Decay Correction is **ADMIN** then the **corrected dose** = Radionuclide Total Dose
 - If the Decay Correction is **START** then the **corrected dose** = Radionuclide Total Dose * EXP(-0.693147 * TIME_DIFF / Radionuclide Half Life), where TIME_DIFF is the difference in seconds between Radiopharmaceutical Start Time (the time the patient activity was measured) and the Series Time (start scan time); if the two times are equal then the corrected dose = Radionuclide Total Dose. Note that Radionuclide Half Life is usually 6588 seconds for F18 isotope.

NOTE: The Merge PACS Viewer uses only tags defined by the DICOM standard to calculate the SUVs; it does not take into account any private tags used by some modality vendors.

¹ James WPT. *Research on obesity*. London. Her Majesty's Stationery Office

² Hamwi GJ. *Therapy: changing dietary concepts*. In: *Diabetes Mellitus: Diagnosis and Treatment (vol. 1)*. Danowski TS (ed). American Diabetes Association. New York. 1964, pp73-8.

³ DuBois D; DuBois EF: *A formula to estimate the approximate surface area if height and weight be known*. Arch Int Med 1916 17:863-71.

Appendix C. Creating a COM+ Application

On 64-bit Operating Systems, the auto-login into PS360 feature does not work properly because of an incompatibility between a 64-bit Viewer and 32-bit PowerScribe 360 clients. In order to address this issue, you must create a COM+ application that will facilitate the communication between the Viewer and PS360 client installation.

To create a COM+ application:

1. On the Workstation running both Merge PACS and Powerscribe 360, go to **Control Panel → Administrative Tools → Component Services**.
2. In the left-hand panel, expand **Computers → My Computer**, right-click on **COM+ Application** and select **New → Application** from the pop-up menu, as in the following example:

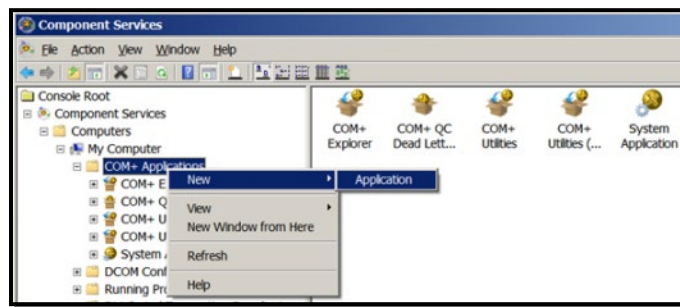


Figure 742: Adding a New COM+ Application

The **COM + Application Install Wizard** is launched.

3. Click on **Create an empty application**.

The **Create Empty Application** screen is displayed, as in the following example:

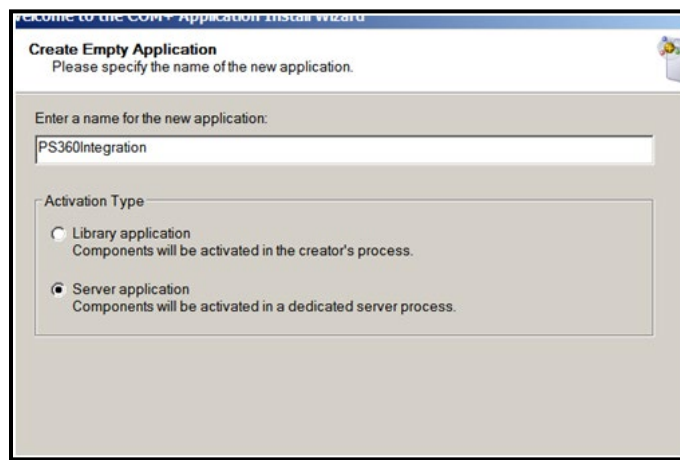


Figure 743: Creating an Empty Application

4. Provide a name for the application (as an example PS360Integration), make sure that the **Server application** option is selected, and click on the **Next** button.

The **Set Application Identity** screen is displayed, as in the following example:

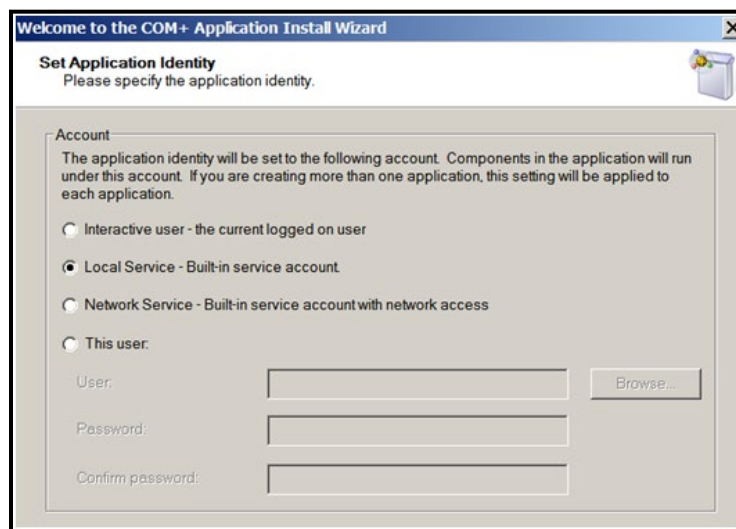


Figure 744: Setting Application Identity

5. Make sure that the **Local Service** option is selected and click on the **Next** button.

The **Add Application Roles** screen is displayed.

6. Click the **Next** button.

The **Add Users to Roles** screen is displayed.

7. Click the **Next** button, followed by the **Finish** button.

The new application will be displayed in the list of COM+ applications.

8. Expand the new COM+ application, right-click on **Components** and select **New → Component** from the pop-up menu, as in the following example:

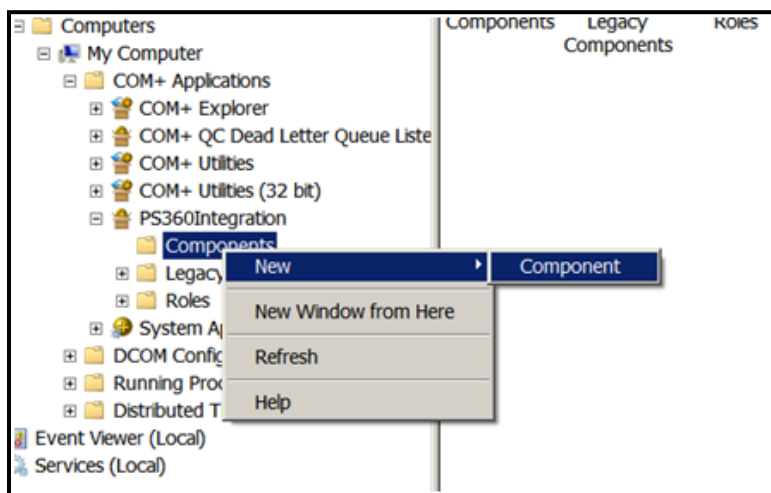


Figure 745: Adding a New Component

The **COM+ Component Install Wizard** is launched

9. Click on Import component(s) that are already registered.

The **Choose Components to Import** screen will be displayed, with a list of all the registered COM Components, as in the following example:

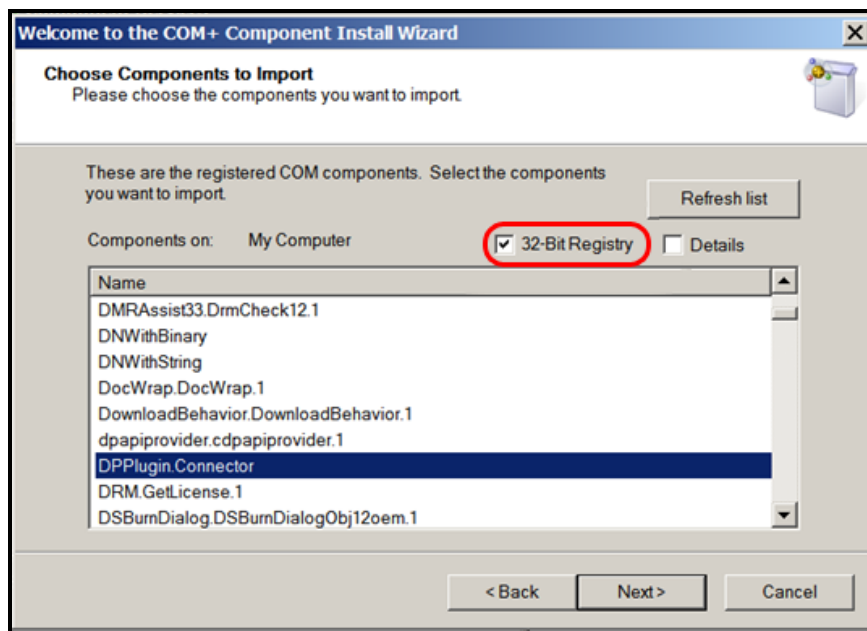


Figure 746: Choosing Components to Import

10. Make sure that **Enable 32-bit Registry** is enabled, select **DPPlugin.Connector** and click on the **Next** button.
11. Click the **Finish** button.
12. In the right-hand panel, right-click on **PS360Integration** application and select **Properties** from the pop-up menu, as in the following example:

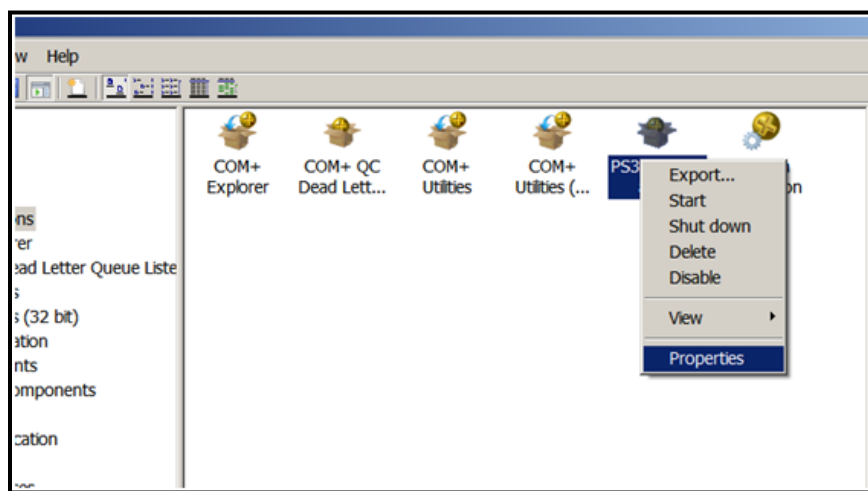


Figure 747: Accessing Properties of New COM+ Application

The **PS360Integration Properties** window is displayed.

13. Click on the **Security** tab and make sure that **Enforce access checks for this application** is **not** selected, as in the following example:

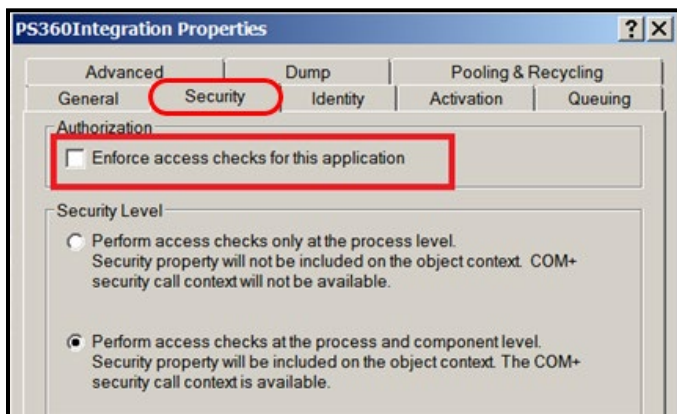


Figure 748: Disabling Enforced Access Checks

14. Click on the **Advanced** tab and make sure that **Leave running when idle** is selected, as in the following example:

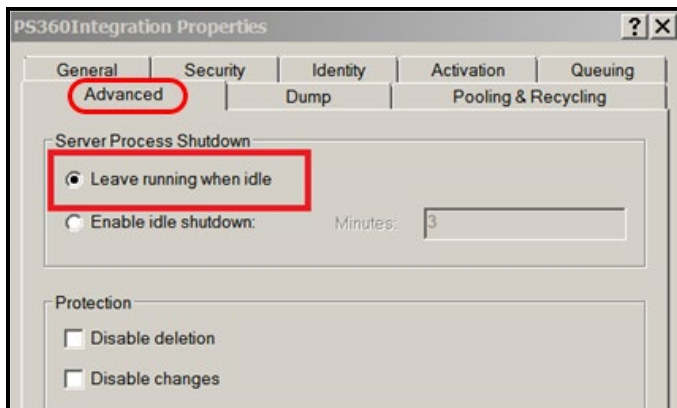


Figure 749: Leave Running When Idle

15. Close the Properties dialog.
16. Right-click on **PS360Integration** application and select **Start** from the pop-up menu, as in the following example:

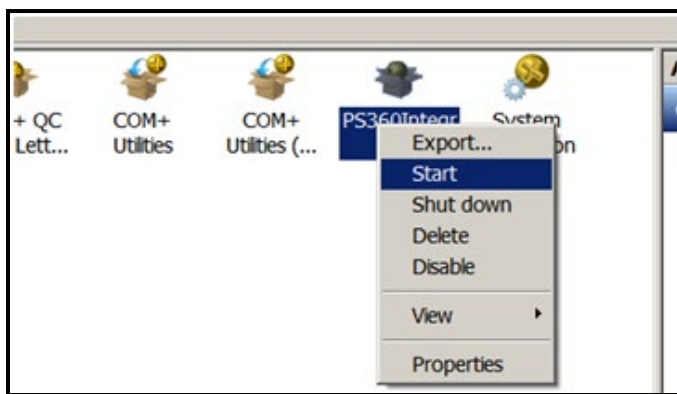


Figure 750: Starting the New COM+ Application

The COM+ application will start and the icon associated with it will flicker to indicate that is running.

17. Log into the PS360 client directly using powerscribe/powerscribe credentials or any active credentials and verify that the authentication works.
18. Log out and log into PS360 from within The Merge PACS Viewer to verify that the authentication works.

NOTE: For troubleshooting purposes, the PS360 client process should be displayed in Task Manager as in the following example:

explorer.exe	5140	dragomir	00	91,928 K	C:\Windows\Explorer.EXE
FF_Protection.exe *32	5548	dragomir	00	1,820 K	"C:\Program Files (x86)\ST
GoogleToolbarNotifier.exe *32	5888	dragomir	00	1,132 K	"C:\Program Files (x86)\Gc
hidfind.exe	5804	dragomir	00	1,976 K	"C:\Program Files\DellITPa
hkcmd.exe	5436	dragomir	00	3,212 K	"C:\Windows\System32\hk
IFrmewrk.exe	5664	dragomir	00	21,712 K	"C:\Program Files\Commor
igfxext.exe	1172	dragomir	00	2,476 K	C:\Windows\system32\igfx
igfxpers.exe	5500	dragomir	00	4,452 K	"C:\Windows\System32\igl
igfxsrvc.exe	5392	dragomir	00	3,172 K	C:\Windows\system32\igfx
igfbtray.exe	1592	dragomir	00	3,768 K	"C:\Windows\System32\igl
ipoint.exe	5836	dragomir	00	9,336 K	"C:\Program Files\Microsol
mmc.exe	5572	dragomir	00	5,496 K	"C:\Windows\system32\mi
MSASCui.exe	5788	dragomir	00	7,372 K	"C:\Program Files\Microsof
Nuance.PowerScribe360.exe *32	4112	dragomir	00	69,932 K	"C:\Users\dragomir\AppData
nvwsvc.exe	4856	SYSTEM	00	10,000 K	C:\Windows\system32\nvw
NvXDSync.exe	3532	SYSTEM	00	10,096 K	"C:\Program Files\NVIDIA
OUTLOOK.EXE *32	6824	dragomir	00	123,288 K	"C:\Program Files (x86)\Mi
rundll32.exe *32	5592	dragomir	00	1,172 K	C:\Windows\SysWOW64\R
splwow64.exe	4432	dragomir	00	2,600 K	C:\Windows\splwow64.exe
stray64.exe	5420	dragomir	00	10,536 K	"C:\Program Files\IDT\WD

Figure 751: Starting the New COM+ Application

Appendix D. Mapping of RTWL Filters to Database Tables, HL7 Field and DICOM Attribute

The following table describes mapping of RTWL filters to database tables, HL7 fields and DICOM attributes:

Filter	Definition	Order	Study or Synthetic Order
Modalities	This filter is applied against the type of modality that will be performed (e.g. CR, CT, MR) Modalities used in defining the filter are defined in the Modality Dictionary	Sent in HL7 ORM-001 Diagnostic Order message in OBR:24 . Stored in database table SchedProcStep.Modality	(0008, 0060) Stored in: AmicasSeries.Modality
Station Names	This filter is applied against the StationName where the study was performed.	We get the Scheduled Station AE Title in HL7 ORM-001 Diagnostic Order OBR:21 and now lookup into the DicomDevice Dictionary for the StationName that is associated with the AE Title. Stored in SchedProcStep.StationAETitle	StationName (0040, 1010) Stored in: AmicasSeries.StationName
Status	This filter is applied against the Workflow status (e.g. UV, UR,DI) as they are defined in the Workflow Status Dictionary. There may be customer defined statuses as well as system generated statuses.	The Order Status is set to Scheduled if the HL7 ORM-001 Diagnostic Order message has a Scheduled Date and Time in OBR:7 The Order Status is stored in the table OrderStatus.Status with key fields of MRN and ACCNUM Accession Number	A Study Status will be set once there is a performed study associated. This is a non-DICOM Merge PACS defined element. The Study Status is stored in StudyStatus.Status with key identifier field of the StudyInstanceUID. A Transient Status may also be set (e.g. LO – in Use), this status is stored in StudyStatusTran.Status with StudyInstanceUID as the key identifier.
Procedures	This filter is applied against the scheduled/performed procedure meaning. Procedure names used in the filter are defined in the Procedure Dictionary.	The procedure is sent in the HL7 ORM-001 Diagnostic Order message as OBR:4 component 2 (the text portion). This gets stored in ReqProc.ReqProcMeaning	The Study Description (0008, 1030) Stored in: AmicasStudy.Description

Filter	Definition	Order	Study or Synthetic Order
Reading Physicians	This filter is applied against the Reading Physician that appears in the Dicom Header of the Study. Physicians used in the filter are defined in the Provider Dictionary. Note that only the text portion of the name (or wildcard) is filtered against, since we do not get a reading physician code in the Dicom Header	Not derived from Order. It would be rare to know the reading physician at the time of the order.	The ReadingPhysician: (0008, 1060) Stored in: AmicasStudy.ReadingPhysician
Referring Physicians	This filter is applied against the Referring Physician associated with the order. Physicians used in the filter are defined in the Provider Dictionary. The filter uses only the code or Id if defined in the dictionary against the RefPhysID field in the database, or if not defined will attempt a match on the name components.	This field is sent in HL7 ORM-001 Diagnostic Order message in PV1:8 (we use PV1:7 if PV1:8 is empty) The Referring Physician components are stored in ReqProc.RefPhysician and component 1 (code or id) of PV1:8 is stored in ReqProc.RefPhysID to facilitate more efficient filter matching.	Referring Physician (0008, 0090) Stored in: AmicasStudy.ReferPhysician Note that the referring physician code is not available as a DICOM attribute and therefore is not stored in the AmicasStudy table and is not available for study filtering.
Time Constraint	This filter allows matching on orders scheduled for a date for x days in the future or x days in the past or both. The filter is applied either against the Schedule date/time OR the PerformedStudy Study date/time	This field is sent in the HL7 ORM-001 Diagnostic Order message in OBR:7 and stored in the SchedProcStep.ProcStartDate and SchedProcStep.ProcStartTime as well as SchedProcStep.SchStartTimeStamp	Study Date and Time (0008, 0020) (0008, 0030) Stored in: AmicasStudy.StudyDate and AmicasStudy.StudyTime
Priority	This filter is applied against the Order's priority level. It is either enabled or disabled ("Y" or "N"). If enabled, it will only match with STAT priority Orders. There is no study level priority.	The field is sent in the HL7 ORM-001 Diagnostic Order message in OBR:27 component 6. The HL7 Priority is stored (ReqProc.HL7Priority) and also mapped to a DICOM priority which is also stored (ReqProc.ReqProcPriority)	No Study or Synthetic Order representation.
Patient Class	This filter allows matching against the PatientClass (e.g. I, O, E [Inpatient, Outpatient, Emergency]). Patient Classes used for filtering are defined in the Patient Class dictionary.	The field is sent in the HL7 ORM-001 Diagnostic Order message in PV1:2 and stored in SrvReq.PatientClass	No Study or Synthetic Order representation.
Locations	This filter allows matching against the Current PatientLocation (Point of Care) at the time of the Order. Locations used for filtering are defined in the Location Dictionary.	The field is sent in the HL7 ORM-001 Diagnostic Order message in PV1:3 and stored in SrvReq.CurLocation (matching is applied only to component 1 – Point of Care)	No Study or Synthetic Order representation.
IPID	This filter allows matching against the PatientIssuerID issued by the institution.	The field is sent in the HL7 ORM-001 Diagnostic Order message in PID Segment, Patient ID Internal, CX component 4 subcomponent 1~and stored in REQPROC.PTIDISSUER.	"Issuer of Patient ID" field (0010,0021).
Due In	This filter is applied against the due in time configured based on the Order's priority level.	Field not sent in order, however, based on Priority in OBR:27 component 6 and stored in WFPRIORITYD.SLA	No Study or Synthetic Order representation.
HL7 Order Status Mapping	This filter is applied against the Order's Status.	The field is sent in the HL7 ORMs and ORUs, message segment ORC 5 (order status) and stored in RISORDERSTATUS.STATUS	No Study or Synthetic Order representation.

Filter	Definition	Order	Study or Synthetic Order
Specialty Mapping	This filter is applied against the Order's priority level. It is either enabled or disabled ("Y" or "N"). If enabled, it will only match with STAT priority Orders. There is no study level priority.	The field is sent in the HL7 ORMs, OBR-18 segment and is stored in RISORDERSTATUS>READINGPOOL.	No Study or Synthetic Order representation.
Facility	This filter allows matching against the institution that is recorded in the DICOM Header for the performed study. Institutions are placed into the InstitutionGroup table by a periodic command (MiscellaneousJobs) in CommandScheduler	Field not sent in order.	InstitutionName (0008, 0080) Stored in: AmicasStudy.Institution (used in filtering)
Patient Age	This filter if enabled allows matching against the patient age at the time of the exam. Values accepted are patient > age, patient < age, or patient > age and patient < age (range).	Computed field based on patient date of birth as compared to the exam/order date in terms of days. HL7 field = PID:7 Patient.PatientBirthDate	PatientBirthdate (0010, 0030) Stored in: AmicasStudy.PatientBirthDate

Appendix E. Creating New Users for OpenLDAP

The following procedure describes how to create new users within OpenLDAP for use with OrthoPACS.

NOTE: New users should only be created at the hub server in a hub/spoke setup.

NOTE: To create users you will first need access to the LDAP Admin software. LDAP Admin can be accessed from the OrthoPACS hub server, or can be installed on another workstation on the network.

E.1. Connecting to OpenLDAP Using LDAP Admin

To connect to Open LDAP using LDAP Admin:

1. The first time you access LDAP Admin, you will need to add information for connecting to OpenLDAP. To do this, open LDAP Admin and select **New connection**, as in the following example:

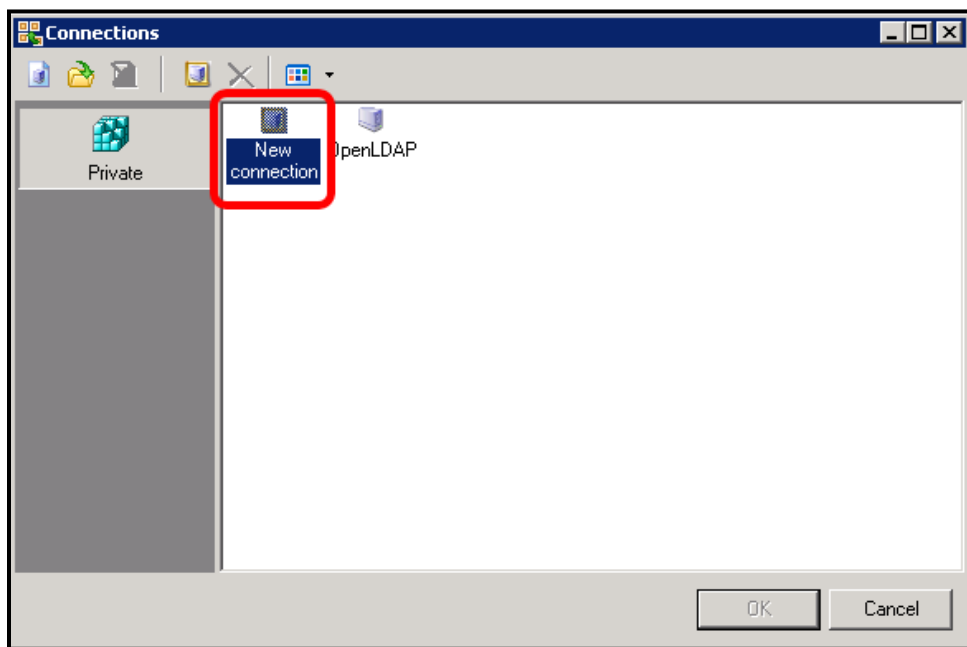


Figure 752: Selecting New Connection

The **Connection Properties** dialog is displayed, as in the following example:

Figure 753: Connection Properties

2. Add the connection information for your OpenLDAP instance as follows:

Parameter	Value
Connection name	OpenLDAP
Host	If LDAP Admin is installed on the hub server, use 127.0.0.1. If LDAP Admin is installed on a workstation, use the IP address of the Server.
Port	389
Base	ou=<Organizational Unit>,dc=maxcrc,dc=com
NOTE: <Organizational Unit> is typically Merge, but may vary depending on how OpenLDAP was originally configured on your Server.	
Username	cn=manager,dc=maxcrc,dc=com
Password	Password set during the OpenLDAP install

3. You can now use LDAP Admin to connect to and manage OpenLDAP by launching LDAP Admin, selecting **Connect** from the **Start** menu, and selecting **OrthoPACS**, as in the following example:

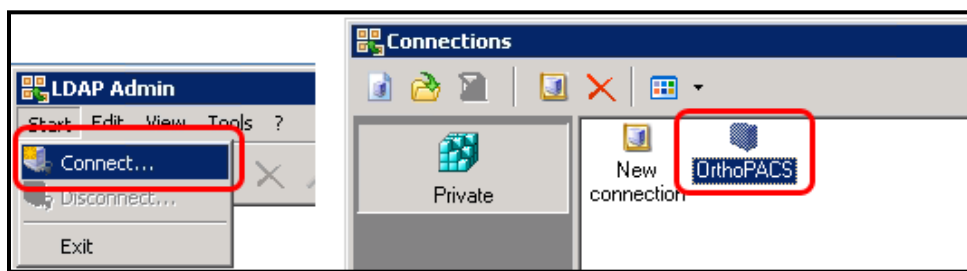


Figure 754: Connecting to OpenLDAP

E.2. Creating OpenLDAP Users

To create OpenLDAP users:

1. Connect to **OpenLDAP** using LDAP Admin.

NOTE: If you use the default username/password supplied in the registry file you will not need to select a server or login again.

2. Right-click the root of your OrthoPACS domain in the tree and select **New → User** from the pop-up menu, as in the following example:

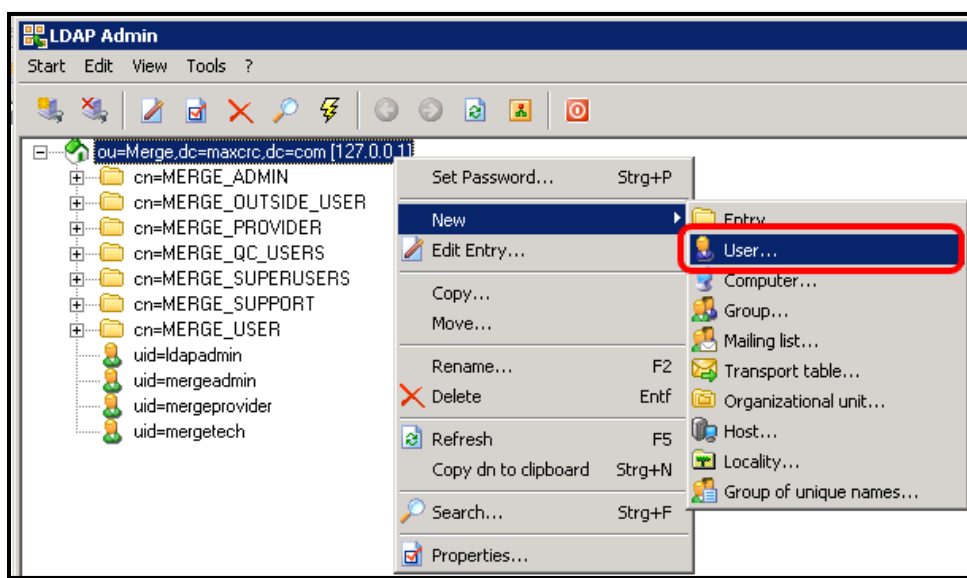


Figure 755: Selecting New User

NOTE: Users are created in the root, not in the individual groups.

The **Create User** dialog displays, as in the following example:

Figure 756: Create User

3. Enter the desired information in the following fields:
 - **First Name**
 - **Second Name** (Last Name)
 - **Display Name**
 - **Username** (please use lower case letters only)
 - **Home Directory** (use "C:\")
4. Click on the **Membership** tab and then click the **Add** button at the bottom of the screen.

The **Choose Groups** dialog is displayed:

Figure 757: Choose Groups

5. At the Choose Groups dialog, select the desired group and click the **OK** button.

NOTE: Each user should only belong to one group.

The **Choose Groups** dialog closes.

6. At the Create User dialog, click the **OK** button.

The Create User dialog closes

7. Create a password for the new user as follows:

- a. Right-click on the new user's user name and select "Set Password" from the pop-up menu.

*The **Set Password** dialog is displayed, as in the following example:*

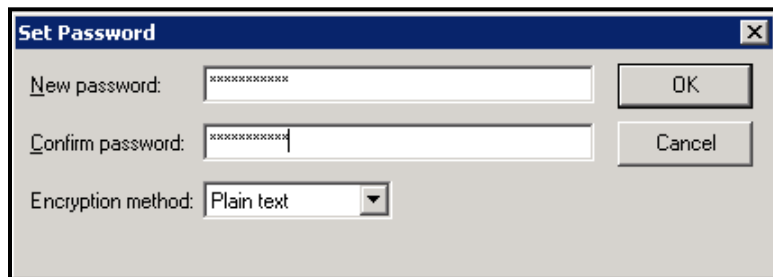


Figure 758: Set Password

- b. Enter and confirm the desired password, select "**Plain text**" for the Encryption method, and click **OK**.
8. At the main LDAP Admin screen, right-click on the Common Name (**cn=**) of the group to which you added the new user and select **Copy dn to clipboard** from the pop-up menu, as in the following example:

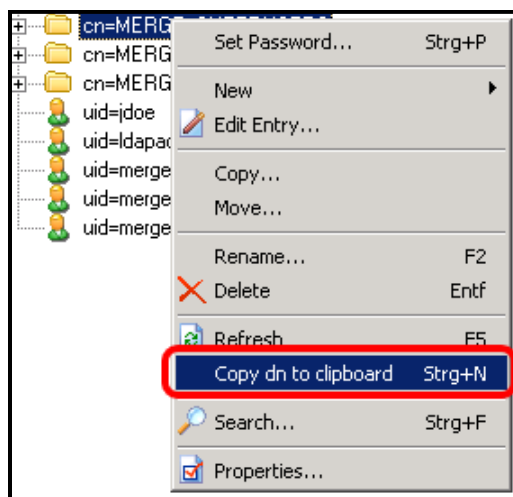


Figure 759: Copy dn to clipboard

9. Right-click on the new user and select **Edit Entry** from the pop-up menu, as in the following example:

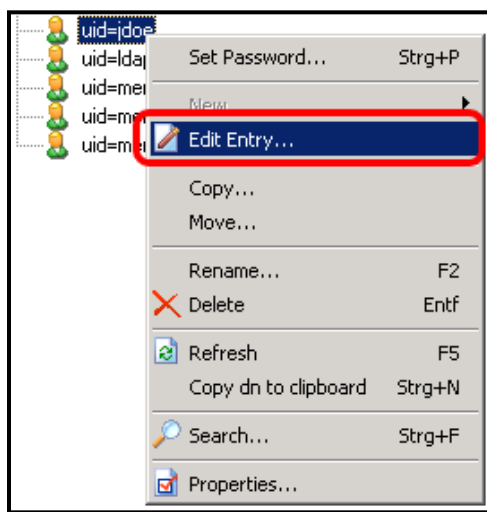


Figure 760: Edit Entry

The **Edit Entry** screen is displayed, as in the following example:

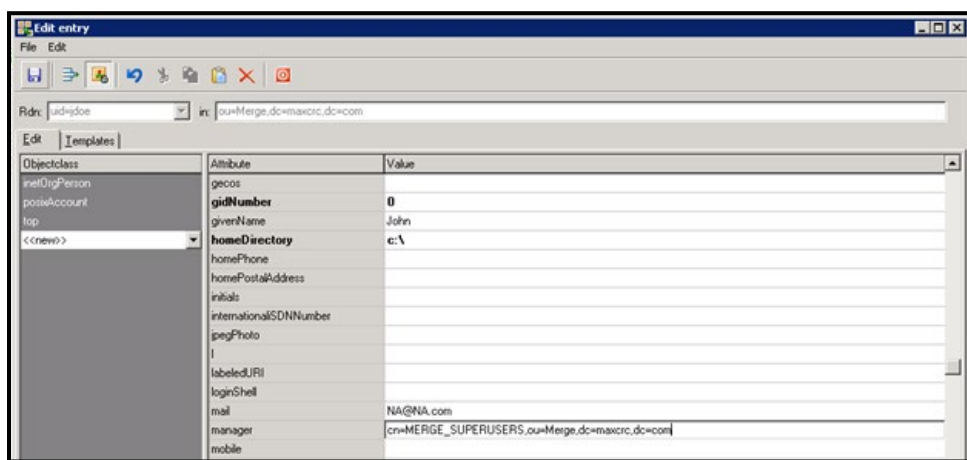


Figure 761: Edit Entry

10. At the Edit Entry dialog, do the following:
 - a. For the value of **mail**, enter the user's email address (used by OrthoPACS Web Viewer).
 - b. For the value of the **manager** attribute, paste in the distinguished name (dn) of group that was copied to the clipboard earlier.
 - c. Click the **Save** icon.

The **Edit Entry** dialog closes.

11. After all users have been added, make sure to backup your Open LDAP domain and users as follows:
 - a. At the LDAP Admin window, select **Export** from the Tools menu.
 - b. Choose a filename and a secure location, then click **OK**, as in the following example:

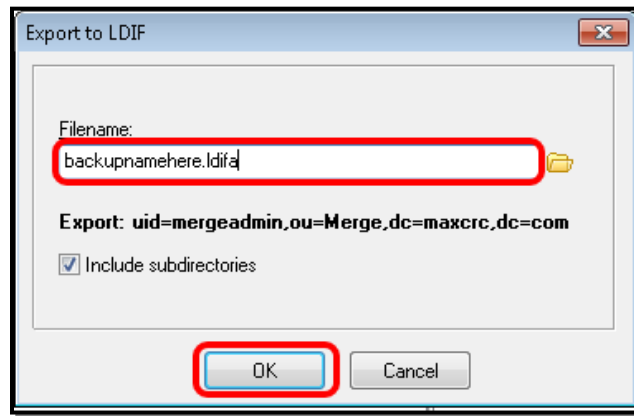


Figure 762: Edit Entry

This will allow you to restore your domain, groups, and users if you run into a problem.

12. Test that the user can login to PACS and Web Viewer.

Appendix F. Creating an X-keys Macro to Simulate Localized Bounce

For Merge PACS workstations that use the X-keys XK-24 USB Keypad, it is possible to create a custom X-keys macro on those Workstations to provide a simulated “localized bounce” function for use with multi-frame breast tomosynthesis images. The standard “bounce” mode of the default cine feature within the Workstation causes all frames of an image to be displayed, bouncing back and forth between the first and last frames. The localized bounce macro, on the other hand, allows users to select a specific frame and then have the bounce mode centered on that frame.

F.1. Keyboard

The keyboard used for the steps described in this Appendix is the X-keys® **XK-24** USB Keypad, as can be found described at the following location:

<http://xkeys.com/XkeysKeyboards/index.php>

F.2. Initial Set-up Instructions

NOTE: If no installation disk is included in the package from X-keys, the installation software can be downloaded from the X-keys site.

1. Add the appropriate key legends to the keys on the keypads
2. Go to <http://xkeys.com/PISupport/Software.php>
3. Install MacroWorks
4. Reboot the system

F.3. Adding a Macro to the Keyboard

1. Plug-in the X-keys keyboard and wait for windows to recognize it
2. Press the programming switch (located on the top edge of the keyboard) to open the MacroWorks 3.1 (MW3) programming window or select the MW3 desktop icon.
3. Select a key from the keypad or double-click on a button on the MW3 dialog to open the MW3 Macro window for that button.
4. Type the keystrokes or select text, shortcut, function to define the macro for that key.

5. In order to define the localized bounce macro, add left and right keystrokes with a delay between each in order to give time to view the image.
 - a. To add keystrokes, select the "Keystrokes" tab and press the appropriate key. The table which actually add keys for both keydown and keyup.
 - b. To add a delay, switch to the "Function" tab, select "Show Functions" and double click on "Delay".

The following sequence creates the localized bounce feature:

- Keys: Left
- Func: Delay[500]
- Keys: Left
- Func: Delay[500]
- Keys: Left
- Func: Delay[500]
- Keys: Right
- Func: Delay[500]
- Keys: Right
- Func: Delay[500]
- Keys: Right
- Func: Delay[500]
- Keys: Right
- Func: Delay[500]
- Keys: Left
- Func: Delay[500]
- Keys: Left
- Func: Delay[500]
- Keys: Left
- Func: Delay[500]

As can be seen in the following example:

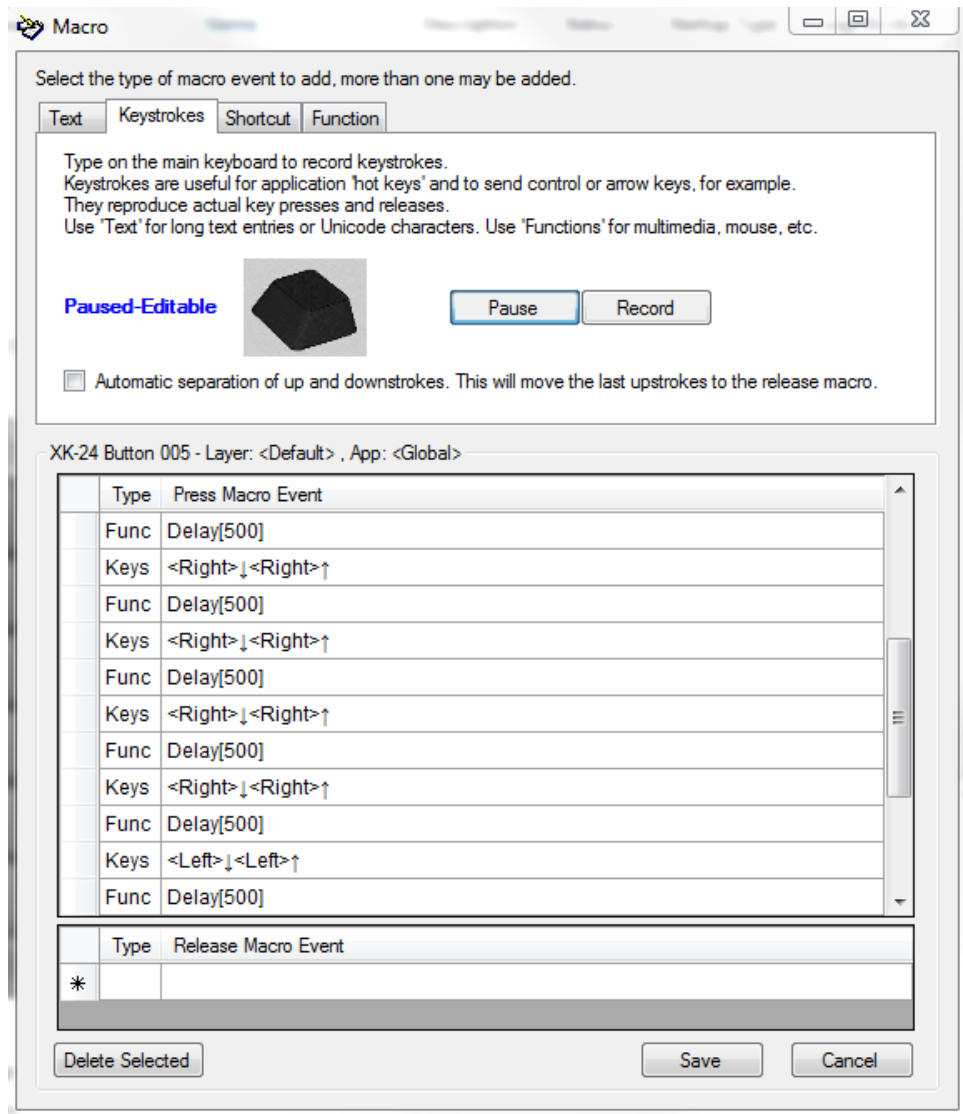


Figure 763: Localized Bounce Macro

- 6. Click the **Save** button to save the macro.

On the MW3 programming window, the button with the macro should have a new icon to indicate that it has a macro associated with it, as in the following example:

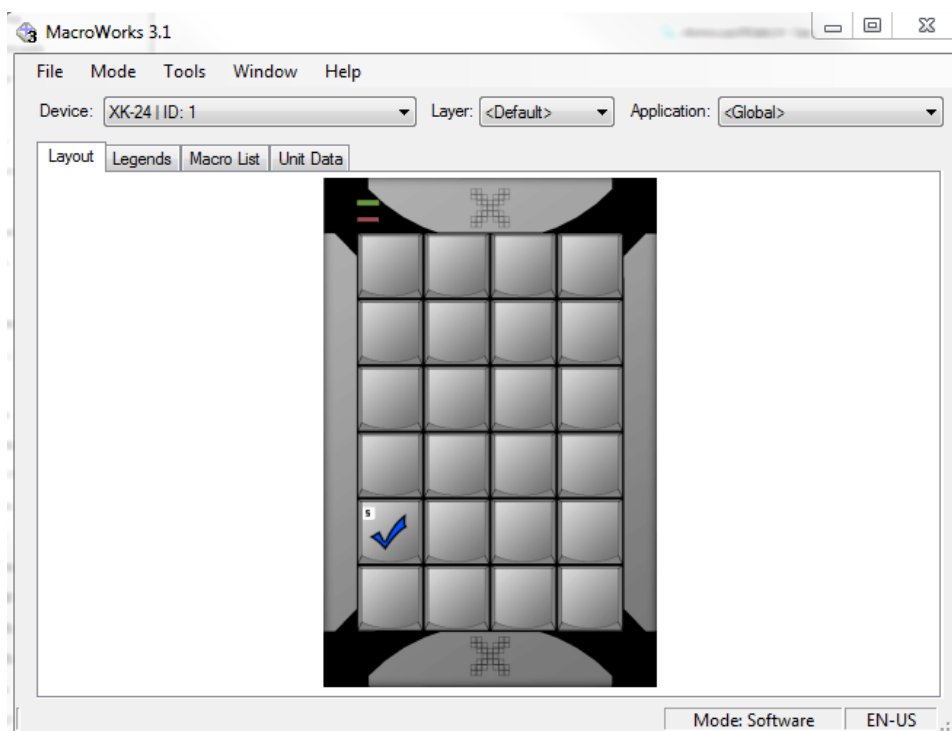


Figure 764: New Icon for Localized Bounce Macro

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